



**GOVERNMENT OF ANDHRA PRADESH**  
(ANDHRA PRADESH ROAD DEVELOPMENT CORPORATION)

ANDHRA PRADESH ROAD SECTOR PROJECT  
Feasibility Study, Design and Detailed Engineering

**Detailed Project Report**  
**UPGRADING OF KAKINADA - RAJAHMUNDY ROAD**

**VOLUME – VII**  
**Environmental Impact Assessment**  
**And Environmental Management Plan**

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## ABBREVIATIONS

|       |                                     |
|-------|-------------------------------------|
| SOS   | Strategic Option Study              |
| R&B   | Road & Building Department          |
| APRDC | A.P. Road Development Corporation   |
| APRSP | Andhra Pradesh Road Sector Project  |
| GOI   | Government of India                 |
| GOAP  | Government of Andhra Pradesh        |
| SH    | State Highway                       |
| MDR   | Major District Road                 |
| ODR   | Other District Road                 |
| EIA   | Environmental Impact Assessment     |
| EMP   | Environmental Management Plan       |
| DoF   | Department of Forest                |
| MoEF  | Ministry of Environment and Forest  |
| FHWA  | Federal Highways Authority          |
| CPCB  | Central Pollution Control Board     |
| IMD   | India Meteorological Department     |
| COI   | Corridor of Impact                  |
| PIA   | Project Influence Area              |
| PD    | Project District                    |
| ROW   | Right of Way                        |
| AAQ   | Ambient Air Quality                 |
| SPM   | Suspended Particulate Matter        |
| RPM   | Respirable Particulate Matter       |
| AADT  | Annual Average Daily Traffic        |
| PUC   | Pollution under Control Certificate |
| ASI   | Archaeological Survey of India.     |
| NOC   | No Objection Certificates           |
| RH    | Relative Humidity                   |
| PAP   | Project Affected Person             |
| RAP   | Resettlement Action Plan            |
| OMC   | Optimum Moisture Content            |
| CSC   | Construction Supervision Consultant |
| EMU   | Environment Management Unit         |
| EO    | Environment Officer                 |

PIU  
PU

Project Implementation Unit  
Package Unit



## EXECUTIVE SUMMARY

### A.1 BACKGROUND

APRDC has planned to improve the State road network by providing better quality and safer roads to the users in sustainable manner with loan assistance from World Bank. Improved quality of roads, better institutional operation and management system of APRDC and safe roads are important features of the project component. In this connection RDC has selected 38 corridors (cumulative length of 2002 km) of high-density (traffic) roads through Strategic Option Study (SOS) to be developed in phased manner.

### A.2 OBJECTIVES OF THE ASSIGNMENT

The main objective would be to alleviate the current unsafe and congested conditions of the road network connecting the villages and towns by providing better quality and safe roads to the users in a sustainable and environment friendly manner. Government of India, GoI through Ministry of Environment and Forest (MoEF) enforces **Environment (Protection) Rules, 1986** for environmental protection because of intervention of new projects or activities, or on expansion and modernization of existing projects or activity based on their environmental impacts.

### A.3 SCOPE OF ENVIRONMENTAL ASSESSMENT (EA)

The environmental assessment scope includes screening and scoping, environmental assessment and environmental management plans for the individual project roads as required. The EA process also envisages to develop a comprehensive environmental management frame work for the entire project which will adopted as part of the corporate environmental policy for AP Road Development Corporation.

### A.4 DESCRIPTION OF PROJECT ROAD

The project road Kakinada – Rajahmundry (KR) road, is a part of SH-78 in the Andhra region of Andhra Pradesh. The KR road starts at km 0/0 in Kakinada, in East Godavari district and runs north-west direction to Samalkota (project road has an overlap for a length of 3.3km with SH-148) and again from Samalkota runs south west direction to terminate at km61/6 in Rajahmundry (Junction with NH5 at km 60/1) near Dawleswaram barrage. Cumulative length of KR road is 61.6km and serves as an important link between Kakinada Port and NH5. **The project road is very close to the Samalkota canal and it appears that the PR serves like a inspection road to Samalkota canal.**

The project road is proposed for improvement from existing 2L to 4 L in HS-01 (km 0.0 to km 16.0) and remaining section HS-02 (km 16.0 to km 60.1) is proposed for improvement from existing 2L to 2 L+PS, with no major realignments. Sufficient land width is available in the HS -01 section for four laning, except some additional land is required for the bypass (length 3.1km) and minor realignments. Similarly for HS-02 NO additional land width is required as the proposed improvement is limited to 1.5 m paved shoulder and 1 m earthen shoulder on either side of the existing pavement. Hence the total land acquisition required for the project road upgradation is 14.25 Hectares, on an average 2.5m width all along the project road, well below 20m (Above 20 m the project requires the MoEF Clearance), hence NO MoEF clearance is required.

### A.5 KEY ENVIRONMENTAL LAWS AND REGULATIONS

Table 1 presents the environmental regulations and legislations relevant to AP state road project.

**Table 1: Environmental Regulations and Legislations**

| SR. No | Act / Rules                     | Purpose                                    | Applicable Yes/ No | Reason for Applicability                                                           | Authority                             |
|--------|---------------------------------|--------------------------------------------|--------------------|------------------------------------------------------------------------------------|---------------------------------------|
| 1      | Environment Protection Act-1986 | To protect and improve overall environment | Yes                | As all environmental notifications, rules and schedules are issued under this act. | MoEF, GoI; DoE, State Gov. CPCB; SPCB |

| SR. No | Act / Rules                                                                                                                                                                  | Purpose                                                                                                                     | Applicable Yes/ No | Reason for Applicability                                                                                                                                               | Authority                                                                                                                           |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 2      | Environmental Impact Assessment Notification 4 <sup>th</sup> April, 2011                                                                                                     | To provide environmental clearance to new development activities following environmental impact assessment                  | NO                 | This notification is NOT applicable to Project road, as this is an up-gradation of existing state highway.                                                             | MoEF, EIAA                                                                                                                          |
| 3      | Notification for use of fly ash                                                                                                                                              | Reuse large quantity of fly ash discharged from thermal power plant to minimize land use for disposal                       | NO                 | No thermal plant within 100km reaches.                                                                                                                                 |                                                                                                                                     |
| 4      | Coastal Regulation Zone(CRZ) Notification 1991 (2002)                                                                                                                        | Protection of <b>fragile coastal</b> belt                                                                                   | NO                 | Road is not located along coastal belt                                                                                                                                 |                                                                                                                                     |
| 5      | National Environment Appellate Authority Act (NEAA) 1997                                                                                                                     | Address Grievances regarding the process of environmental clearance.                                                        | Yes                | Grievances if any will be dealt with, within this act.                                                                                                                 | NEAA                                                                                                                                |
| 6      | The Land Acquisition Act 1894 & 1989                                                                                                                                         | Set out rule for acquisition. of land by government                                                                         | Yes                | This act will be applicable to as there will be acquisition of land for widening, geometric improvements and realignments.                                             | Revenue Department State Government.                                                                                                |
| 7      | MoEF Circular on Marginal Land Acquisition and Bypasses 1999                                                                                                                 | Defining "marginal land" acquisition relating to the 1997 Notification                                                      | NO                 | Not applicable as per Environmental Impact Assessment Notification 14th Sep-2006                                                                                       | MoEF                                                                                                                                |
| 8      | The Forest (Conservation) Act 1927<br>The Forest (Conservation) Act. 1980<br>forest (conversion ) Rules 1981                                                                 | To check deforestation by restricting conversion of forested areas into non- forested areas                                 | NO                 | NO forest area.                                                                                                                                                        | Forest Department, GoAP                                                                                                             |
| 9      | MoEF circular (1998) on linear Plantation on roadside, canals and railway lines modifying the applicability of provisions of forest (Conversation) Act, to linear Plantation | Protection / planting roadside strip as avenue/strip plantations as these are declared protected forest areas.              | NO                 | Roadside tree plantation in Andhra Pradesh does not come under the forest act.                                                                                         | MoEF                                                                                                                                |
| 10     | Wild Life Protection Act 1972                                                                                                                                                | To protect wildlife through certain of National Parks and Sanctuaries                                                       | NO                 | This act is not applicable to as there are NO points of wildlife crossing along project corridor.                                                                      | Chief Conservator Wildlife, Wildlife Wing, Forest Department, GoAP                                                                  |
| 11     | Air (Prevention and Control of Pollution) Act, 1981                                                                                                                          | To control air pollution by & Transport controlling emission of air Department. Pollutants as per the prescribed standards. | Yes                | This act will be applicable during construction; for obtaining NOC for establishment of hot mix plant, workers' camp, construction camp, etc.                          | SPCB                                                                                                                                |
| 12     | Water Prevention and Control of Pollution) Act1974                                                                                                                           | To control water pollution by controlling discharge of pollutants as per the prescribed standards                           | Yes                | This act will be applicable during construction for (establishments of hot mix plant, construction camp, workers' camp, etc.                                           | SPCB                                                                                                                                |
| 13     | Noise Pollution (Regulation and Control Act) 1990                                                                                                                            | The standards for noise for day and night have been promulgated by the MoEF for various land uses.                          | Yes                | This act will be applicable as vehicular noise on project routes required to assess for future years and necessary protection measure need to be considered in design. | SPCB                                                                                                                                |
| 14     | Ancient Monuments and Archaeological Sites and Remains Act1958                                                                                                               | Conservation of cultural and historical remains found in India                                                              | NO                 | This act not applicable as the project route is not close to any Ancient Monument, declared protected under the act.                                                   | Archaeological Dept<br>Gol, Indian<br>Heritage Society and<br>Indian National<br>Trust for Art and<br>Culture Heritage<br>(INTACH). |
| 15     | Public Liability and Insurance Act 1991                                                                                                                                      | Protection form hazardous materials and accidents.                                                                          | Yes                | Contractor need to stock hazardous material like diesel, Bitumen, Emulsions etc.                                                                                       | SPCB                                                                                                                                |
| 16     | Explosive Act 1984                                                                                                                                                           | Safe transportation, storage and <b>use of explosive</b> material                                                           | Yes                | For transporting and storing diesel, bitumen etc.                                                                                                                      | Chief Controller of Explosives                                                                                                      |

| SR. No | Act / Rules                                                          | Purpose                                                                                        | Applicable Yes/ No | Reason for Applicability                                                                                                    | Authority                       |
|--------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 17     | Minor Mineral and concession Rules                                   | For opening new quarry.                                                                        | Yes                | Regulate use of minor minerals like stone, soil, river sand etc.                                                            | District Collector              |
| 18     | Central Motor Vehicle Act 1988 and Central Motor Vehicle Rules 1989  | To check vehicular air and noise pollution.                                                    | Yes                | This rule will be applicable to road users and construction Machinery.                                                      | Motor Vehicle Department        |
| 19     | National Forest Policy 1952<br>National Forest Policy (Revised) 1988 | To maintain ecological stability through preservation and restoration of biological diversity. | NO                 | This policy will not be applicable as NO eco sensitive feature exists along the project corridor.                           | Forest Department, GoA and GoAP |
| 20     | The Mining Act                                                       | The mining act has been notified for safe and sound mining activity.                           | Yes                | The construction of project road will require aggregates. These will be procured through mining from riverbeds and quarries | Department of mining, GoAP      |

On the basis of data and information collected during field survey and discussion with local expert and visualise potential associated impact, consultant has categorised this project as **category-B**, which requires a lesser level of environmental investigation.

#### A.6 BASE LINE ENVIRONMENT

Data was collected from secondary sources for the macro-environmental setting like climate, physiography (Geology and slope), biological and socio-economic environment within Project Influence Area, PIU/ Project District, PD. First hand information have been collected to record the micro-environmental features within Corridor of Impact, CoI. Collection of first hand (Primary) information includes preparation of base maps, extrapolating environmental features on proposed road design, tree enumeration, location and measurement of socio-cultural features abutting project road.

#### A.7 STAKEHOLDER CONSULTATION

Consultative procedure, since the inception of the project, has been continued during pre-feasibility, feasibility, Environmental Assessment and Management Plan preparation stage. Considering the fact that involving local communities in the project planning is basis of the participatory planning, often suggestion and option given by the people improves technical and economic efficiency of the project and suggested improvements proposals (if adopted by the project) of the people also generates sense of ownership within communities thus eases implementation process.

#### A.8 INFORMATION DISSEMINATION

- While conducting tree enumeration and inventory of road side environmental features (**hot spots**), the consultant has also conducted information dissemination by one to one campaigning about proposed improvement,
- Potential project affected families were consulted to inform them about proposed road improvement program and possible environmental conflict such as tree cutting, relocation of hand pumps,
- Pictorial Methods were adopted to explain proposed improvement and possible environmental impacts in the concerned villages,
- Information dissemination through pumhlet / *Khadapatralu* among the villagers explaining proposed road improvement.

#### A.9 ANALYSIS OF ALTERNATIVES

The entire length of the project road has very typical kind of restrictions from both sides, LHS and RHS. On LHS, all through canal runs parallel to project road and on RHS railway track runs parallel to road for around 30% length of it. Since on LHS canal is there, all built-up sections have been developed on the RHS of the Road, rather between railway track and project road.

There are four locations where project road passes through congested area:

1. Samalkota (km 14/0 to km 15/0)
2. Anaparthi (km 41/0 to km 42/0)
3. Kesavaram (km 49/0 to km 50/0)
4. Kadiyam (km 56/5 to km 57/0)

At all four locations, project road is located between canal and railway track. Owing to this lateral confinement, no bypass or major realignment is proposed except at Samalkota town.

#### A.10 POTENTIAL IMPACT

Table 2 below presents the general environmental impacts expected due to the proposed upgradation of the project road. Impacts have been assessed based on the first hand information collected from the screening & scoping of environmental attributes. The quanta of all the impacts on Natural Environment are analyzed in detail.

**Table 2: General Impacts on Natural Environment**

| Project Activity        | Planning and Design Phase | Pre-construction Phase             |                                                                          | Construction Phase                                                                                 |                                         |                                                 |                                          |                                    | Road Operation                                                                 | Indirect effects of operation or Induced development |
|-------------------------|---------------------------|------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------|------------------------------------------|------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------|
| Env. component Affected | Land acquisition          | Removal of Structures              | Removal of trees and vegetation                                          | Earth works including quarrying                                                                    | Laying of pavement                      | Vehicle & Machine operation & maintenance       | Asphalt & crusher plants                 | Sanitation & Waste (labour campus) | Vehicle operation                                                              |                                                      |
| <b>Air</b>              |                           | Dust generation during dismantling | Reduced buffering of air and noise pollution, Hotter, drier microclimate | Dust generation                                                                                    | Asphalt odour                           | Noise, dust, pollution                          | Noise, soot, odour, dust, pollution      | Odour / smoke                      | Noise, dust, pollution                                                         | other pollution                                      |
| <b>Land</b>             | Loss of productive Land   | Generation of debris               | Erosion and loss of top soil                                             | Erosion and loss of top soil                                                                       |                                         | Contamination by fuel and lubricants Compaction | Contamination Compaction of soil         | Contamination from wastes          | Spill from accidents Deposition of lead                                        | Change in cropping pattern                           |
| <b>Water</b>            | Loss of water sources     | Siltation due to loose earth       | Siltation due to loose earth                                             | Alteration of drainage Break in continuity of ditches Siltation, Stagnant water pools in quarries. | Reduction of ground water recharge area | Contamination by fuel and lubricants            | Contamination by asphalt leakage or fuel | Contamination from wastes Overuse  | Spill Contamination by fuel, lubricants and washing of vehicles                | Increased contamination of ground water              |
| <b>Noise</b>            |                           | Noise Pollution                    | Noise Pollution due to machinery                                         | Noise Pollution                                                                                    |                                         | Noise pollution                                 | Noise Pollution                          |                                    | Noise Pollution                                                                | Noise pollution                                      |
| <b>Flora</b>            |                           | Loss of Biomass                    |                                                                          | Lowered productivity Loss of ground for vegetation                                                 |                                         | Removal of vegetation                           | Lower productivity Use as fuel wood      | Felling trees for fuel             | Impact of pollution on vegetation Lowered productivity Toxicity of vegetation. |                                                      |
| <b>Fauna</b>            |                           |                                    | Disturbance Habitat loss                                                 | Disturbance                                                                                        |                                         | Disturbance                                     | Disturbance                              | Poaching                           | Collision with traffic                                                         | Distorted habitat                                    |

#### A.11 AVOIDANCE, MITIGATION & ENHANCEMENT

Prevention or avoidance of impact is better than mitigation of impact. Hence avoidance and reduction of adverse impacts approaches were adopted during the design stage through continued interaction between the design and environmental teams. This is reflected in the designs of the horizontal & vertical alignment, cross sections adopted, construction methods and construction materials. In-depth site investigations have been carried out so that sensitive environmental resources are effectively avoided, leading to the environmentally best-fit alignment option. As a

result many of the trees, cultural properties, water bodies etc. have been avoided at the design stage itself, as presented in Table 3 below.

**Table 3: Environmental features saved through Avoidance measure at design stage**

| Environmental Features        | Potential Impact | Under Direct Impact | Saved through alignment design |
|-------------------------------|------------------|---------------------|--------------------------------|
| Trees (nos.)                  | 5732             | 4162                | 1570                           |
| Surface Water source          | 4                | 2                   | 2                              |
| Ground Water source           | 8                | 3                   | 5                              |
| Schools and Hospitals         | 14               | 2                   | 12                             |
| Sensitive Cultural Properties | 67               | 29                  | 38                             |
| Bus Shelters                  | 19               | 5                   | 14                             |

#### A.12 ENVIRONMENTAL MANAGEMENT PLAN

Environmental Management Plan (EMP) deals with the implementation procedure of the guidelines and measures recommended to avoid, minimize and mitigate environmental impacts of the project. It also includes management of measures suggested for enhancement of the environmental quality along the highways.

The institutional arrangement made under project will look into the implementation of project as well as EMP and the various legal settings applicable to the project are briefly stated in chapter 3.

The avoidance, mitigation & enhancement measures for protection of the environment along highways have been discussed in detail in previous chapter. Although the social environmental impacts, its mitigation and management is an essential component of the EMP, this chapter excludes it for the purpose of clarity and procedural requirements. Social environmental elements have been separately dealt in separate volume namely, Resettlement and Rehabilitation Action Plan (RAP).

#### A.13 OBJECTIVE OF EMP

The EMP is a plan of action for mitigation / management / avoidance of the negative impacts of the project and enhancement of the project corridor. For each measure to be taken, its location, timeframe, implementation and overseeing / supervision responsibilities are listed. These components of the EMP have been given in Chapter 9 which explains the environmental issues and the avoidance/ mitigation/ minimization or enhancement measures adopted and/or to be adopted during different phases of the project. It also provide the references for the suggested measures, responsible agency for its implementation/ management as well as its timeframe.

#### A.14 COST ESTIMATES FOR ENVIRONMENTAL MANAGEMENT

Mitigation measures proposed in the EMP will be implemented by the Contractor. The works to be undertaken by the Contractor have been quantified and the quantities included in the respective BOQ items such as earth works, slope protection, noise barriers, road safety features, and shrub plantation.

Provisional quantities have also been included for additional measures that may be identified during construction and for silt fencing which will depend on the Contractors work methods and site locations. Items and quantities have also been included for enhancement measures.

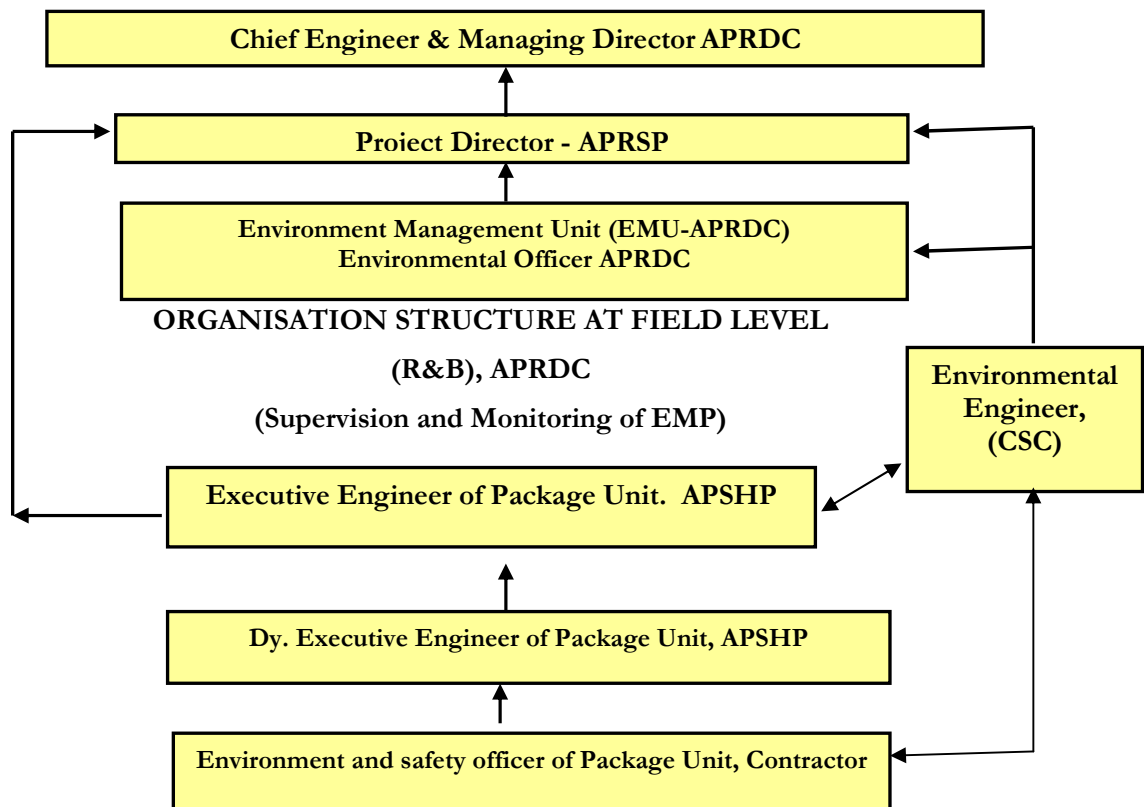
More general environmental management measures to be followed by the contractor have been included in the specifications and this EMP. These cannot be quantified and are to be included in the contract rates. A total of **Rs. 74,15,000 lacks** has been allocated for the environmental management for the Project road.

**A.15 ORGANIZATIONAL FRAMEWORK**

The proposed project will be implemented by APRDC through its Environmental Management Unit (EMU). The EMU comprises of officers from Department of Forest, GoAP, and other environmental Engineers. The EMU will be coordinating with the field level implementing agencies such as Engineer (Supervision Consultant), Contractor and field level APSHP officials. Role and responsibilities of important officials is detailed in chapter 9.

**ORGANISATIONAL FRAMEWORK OF PIU - APRSP**

(Supervision and Monitoring of EMP)

**Fig. 1: Organization Framework**

## ***CHAPTER - 1: INTRODUCTION***

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## CHAPTER 1: INTRODUCTION

### 1.1 BACKGROUND

Andhra Pradesh is a progressive state with a population of 76 millions. Around 75% of the total population is living in the rural areas. Improving infrastructure facilities will promote agricultural and industrial growth in the state. The Road Policy (2005-15) also aims at providing an efficient, safe and environmentally sustainable road network in the state. In order to improve the quality of the corridors, Road and Building Department (RBD) of Government of Andhra Pradesh (GoAP) has entrusted the responsibility of maintenance and improvement of key State roads (core network) to Andhra Pradesh Road Development Corporation (APRDC<sup>1</sup>).

APRDC has planned to improve the State road network by providing better quality and safer roads to the users in sustainable manner with loan assistance from World Bank. Improved quality of roads, better institutional operation and management system of APRDC and safe roads are important features of the project component. In this connection RDC has selected 38 corridors (cumulative length of 2002 km) of high-density (traffic) roads through Strategic Option Study (SOS) to be developed in phased manner.

The screening and scoping exercise has been carried out as a part of the feasibility study and to list out the priority roads in the state. The environmental assessment and analysis has been documented in this report.

### 1.2 THE PROJECT

#### 1.2.1 Project Description

Feasibility study was carried out on 38 roads (cumulative length 2002 km) selected through Strategic Option Study of 10,040km of key road network of state roads. Considering the economic, social and environmental criteria, 21 roads are selected for implementation in two phases, eight in phase I and thirteen roads in phase II. Further these phase I corridors have been agreed for implementation in two years. Four corridors each in year-1 and year-2 of Phase-I.

Corridors selected for implementation in Phase-I, year-1 are

1. Kandi-Shadnagar (65.60 Km)
2. Chittoor-Puttur (60.80 Km)
3. Kurnool-Devanakonda (60.60 km)
4. Mydukuru – Jammalamadugu (41 Km)

Corridors selected for implementation in Phase-I, year-2 are

5. Pedna-Nuzvid-Vissannapeta

#### **6. Kakinada-Rajahmundry**

7. Jagityal-Peddapalli

The present report on Environmental Impact Assessment (EIA) deals with the environmental analysis of Kakinada - Rajahmundry Road in accordance with the World Bank's guidelines on Environmental Assessment. Figure 1.1 Index Map showing location & elevation of Kakinada – Rajahmundry Road.

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<sup>1</sup> GoAP is in the process of operationalizing RDC as per RDC act to manage core network assigned to APRDC (about 10,000km of roads).

### 1.2.2 Objectives of the Assignment

The main objective would be to alleviate the current unsafe and congested conditions of the road network connecting the villages and towns by providing better quality and safe roads to the users in a sustainable and environment friendly manner. Government of India (GoI) through Ministry of Environment and Forest (MoEF) enforces **Environment (Protection) Rules, 1986** for environmental protection because of intervention of new projects or activities, or on expansion and modernization of existing projects or activity based on their environmental impacts.

The report in hand is prepared in accordance with the World Bank's operational policies<sup>2</sup> guidelines on Environmental Assessment and to meet the statutory requirement of Ministry of Environmental and Forest (MoEF), State Pollution Control Board, State Forest Department, etc. The objectives of this study are stated below:

- To present to decision makers a clear assessment of potential impact associated with the proposed project intervention,
- To apply a methodology which assesses and predict potential impacts and provides a) the means for impact prevention and mitigation, b) the enhancement of project benefits, and c) the minimization of long-term impacts;
- To provide a specific forum in which consultation is systematically undertaken in a manner that allows stakeholders to have direct input to the environmental management process.
- To assess the analysis of alternatives to bring environmental considerations into the upstream stages of development planning as well as the later stage of site selection, design and implementation, and
- To recommend the environmental management Measures to reduce adverse impacts.

In order to achieve these objectives, detailed surveys and other studies have been carried out along the project roads to identify Valued Ecosystem Components (VEC) and corridor specific significant environmental issues (SEI). For investigation/monitoring purpose the study area has been defined as under.

- **Corridor of Impact, CoI:** is the 20m wide strip on either side, beyond Right of Way, RoW.
- **Project Influence Area, PIA:** is the 10km area on either side along the alignment.
- **Project District, PD:** is/are the district/districts through which project road is passing.

### 1.2.3 Scope of Environmental Analysis (EA)

The environmental assessment scope includes screening and scoping, environmental assessment and environmental management plans for the individual project roads as required. The EA process also envisages to develop a comprehensive environmental management frame work for the entire project which will adopted as part of the corporate environmental policy for AP Road Development Corporation.

#### Environmental Screening and Scoping

Environmental screening exercise of the project roads were undertaken to facilitate inputs on environmental considerations, apart from social, economic, and traffic & transport considerations in selection of project roads out of 2056kms of roads, identified

<sup>2</sup> Applicable safeguards instruments are prepared based on Bank guidelines like environmental assessment (O.P. 4.01), Natural Habitat(O.P. 4.04), Forests(O.P. 4.06), Involuntary Resettlement(O.P.4.10) and Indigenous Peoples(O.P. 4.12)

through strategic options study. Further, this report also provides scoping inputs in determining the major environmental issues and defines the scope of work for conducting environmental assessment. As per the recommendation of the Environmental Screening report, detailed Environmental Assessment has been carried out for the project roads. The scoping exercise defines geographical Boundaries for the project roads for impact assessment as well as defining the project influence area to assess the impacts due to project activities.

### **Environmental Impact Assessment (EIA)**

The EIA for selected project roads includes establishing environmental baseline in the study area, identify the range of environmental impacts, specify the measures to avoid, minimize, and mitigate negative impacts and maximize positive impacts and integrate possible environmental enhancement measures. The proposed measures will be formulated in the form of an environmental management plan with necessary budget and institutional roles for effective implementation. The EMPs for individual projects and integration of the same in to project implementation agreements, including construction contract documents.

### **Environmental Management Framework**

An Environmental Management Framework will be designed for the implementation of the project. The environmental management framework shall consists of over all framework which will be developed as a guidance document providing environmental planning and design criteria for the current as well as future project roads, generic environmental management measures, institutional mechanism for implementation, capacity building and training process, and resource material for RDC to function adequately to mainstream the environmental management.

#### **1.2.4 Project Benefits**

The implementation of the project will have the following direct benefits:

- (i) Improved quality of life for the population in the project area. Economic boost to the local population by facilitate easy transportation of materials and having better connectivity for the commercial centres.
- (ii) Provides employment facility for the local population.

#### **1.2.5 Structure of the Report**

**Chapter-2:** Project Description, a brief description of the project corridor is envisaged focusing on various proposed improvements of corridor with a mention on right of way, roadway improvements, cross drainage structures, community facilities, traffic projections etc.

**Chapter-3:** Environment Regulatory Framework presents the legal and administrative framework of World Bank, Government of India and Government of Andhra Pradesh. This section underlines various clearances involved for the project corridor at the State level and at the Central level.

**Chapter-4** Baseline Environmental Status, the existing environmental conditions along the corridor was ascertained by conducting a recognizance survey along with collection of secondary information pertaining to the corridor. Primary data for various environmental parameters was generated using suitable monitoring devises. The methodology was strictly adhered to the Central Pollution Control Board's stipulated guidelines.

**Chapter-5** Public Consultation was carried out in order to know the reactions of local population and the project affected people, PAP. Meetings were held with the stake holders

to record their views on the impacts caused and the suggested remedies to be adopted for the proposed project corridor.

**Chapter-6** Analysis of Alternatives was carried out during feasibility study, covered in Environmental Screening and Scooping report, and the approved alternative is discussed in detail in this chapter.

**Chapter-7** Environmental Impacts, likely impacts caused on various environmental and social parameters by the various activities proposed for the project corridor was recorded in this chapter.

**Chapter-8** Mitigation & Enhancement Measures, various mitigation & enhancement measures were suggested for the impacts caused due to various activities.

**Chapter-9** Based on potential impact and proposed mitigation measure Environmental Management Plan, is prepared. Organizational framework for execution and monitoring of EMP is also prepared along with cost estimate for Environment management.



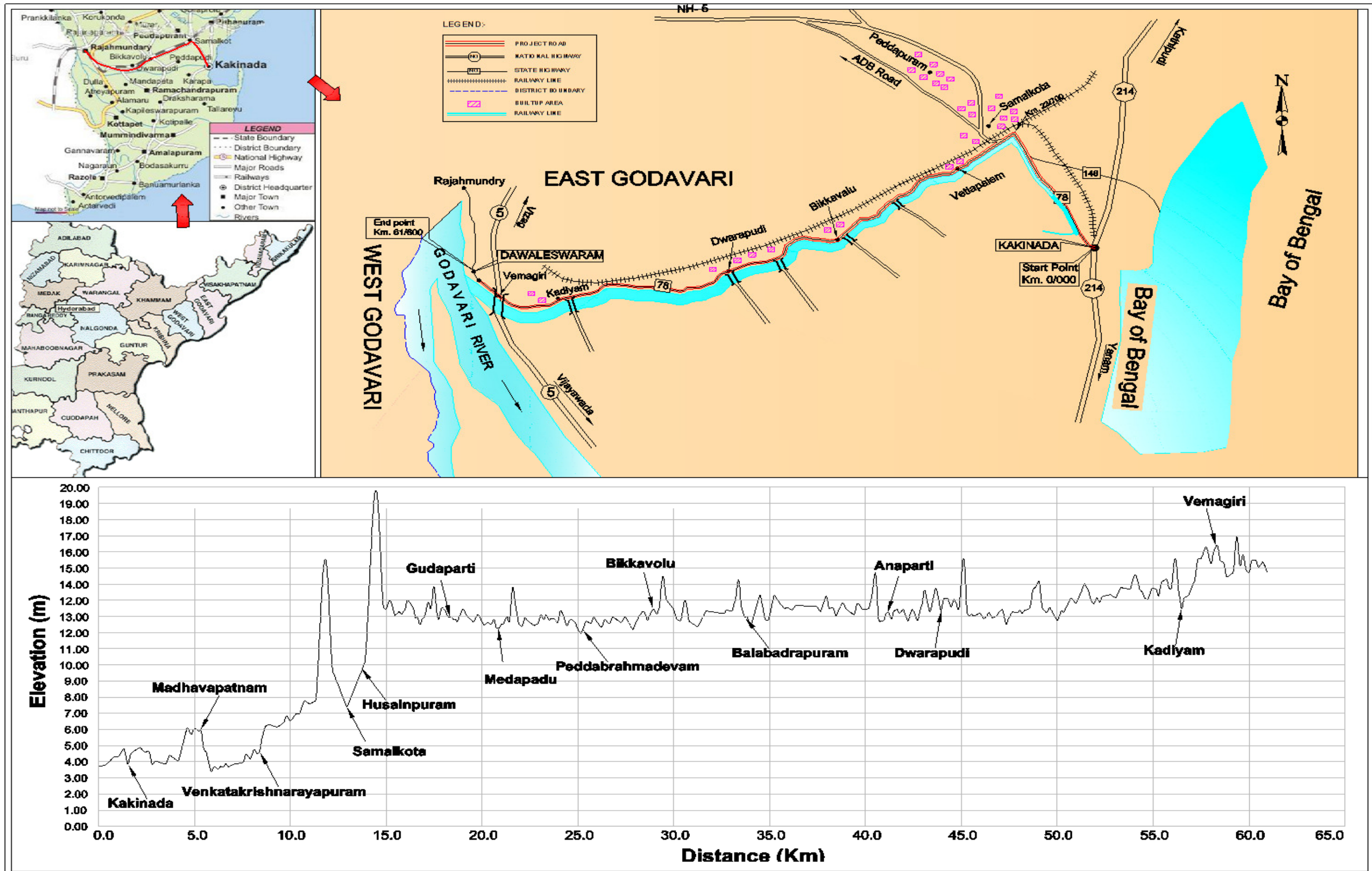


Fig. 1.1: Index Map showing Location and Elevation of Kakinada - Rajahmundry Road

## ***CHAPTER - 2: DESCRIPTION OF THE PROJECT***

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## CHAPTER - 2: DESCRIPTION OF THE PROJECT

### 2.1 GENERAL

The project road Kakinada – Rajahmundry (KR) road, is a part of SH-78 in the Andhra region of Andhra Pradesh. The KR road starts at km 0/0 in Kakinada, in East Godavari district and runs north-west direction to Samalkota (project road has an overlap for a length of 3.3km with SH-148) and again from Samalkota runs south west direction to terminate at km61/7 in Rajahmundry (Junction with NH5 at km 60/1) near Dawleswaram barrage. Cumulative length of KR road is 61.7km and serves as an important link between Kakinada Port and NH5. The location map of the project road is shown in Fig. 2.1.

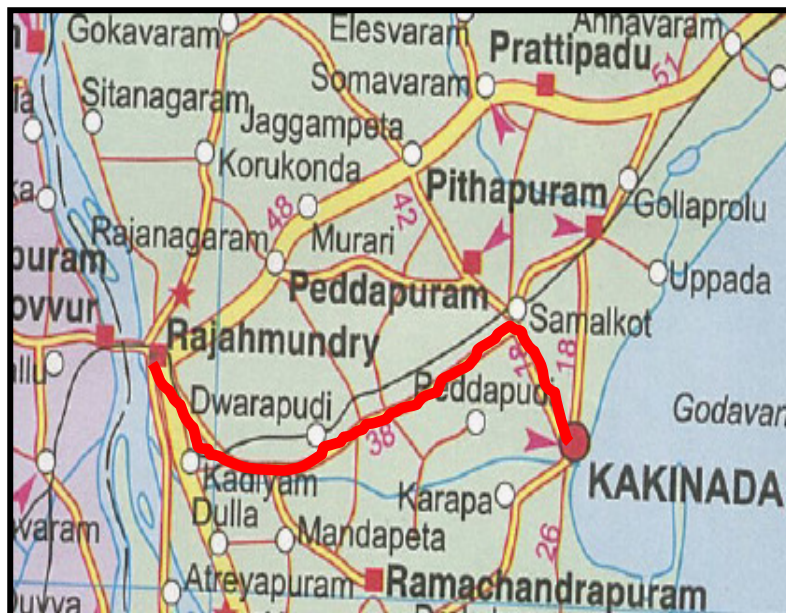


Fig 2.1: Location Map of Kakinada - Rajahmundry

The latitude and longitude of Kakinada - Rajahmundry are (17° 00'N, 78° 26'E) and (15° 05'N, 81° 35' E) respectively. Topographic survey sheets of Survey of India (Scale 1:50,000) No: 65H/13, 65K/4, and 65L/1 encompass the project road.

The KR road is divided into two homogenous sections based on the present traffic trend and the details are presented in table 2.1 below:

Table 2.1: List of Homogeneous Sections

| Section ID                       | Project Road Section    | Existing Chainage | Length (km) |
|----------------------------------|-------------------------|-------------------|-------------|
| KR-01                            | Kakinada – Samalkota    | km 0/0 - 16/0     | 16.0        |
| KR-02                            | Samalkota – Rajahmundry | km 16/0 - 61/7    | 45.7        |
| Total Length of the Project Road |                         |                   | 61.7        |

### 2.2 PRESENT CHARACTERISTICS

#### 2.2.1 Right of Way (ROW)

Width of ROW is not uniform along the project road. The following table gives the distribution of length of Project road and %age length with respect to ROW width.

Table 2.2: Distribution of Road Width

| Width of Existing ROW |           |           |           |       | No ROW Data Available | Total Length |
|-----------------------|-----------|-----------|-----------|-------|-----------------------|--------------|
| <16m                  | 16m – 20m | 20m – 25m | 25m – 30m | > 30m |                       |              |
| 42.0                  | 3.2       | 8.5       | 7.0       | -     | 1.0                   | 61.7         |
| 68%                   | 5%        | 14%       | 11%       | -     | 2%                    | 100%         |

From the above table it has been found that the KR road passes through a narrow corridor with 70% length having ROW < 16m, less than the minimum requirement for two lane standard Highway. This calls for major land acquisition requirement all along the PR.

## 2.2.2 Traffic Scenario

The vehicle-wise average daily traffic (ADT) figures were estimated by classified count survey. Table 2.3 presents a summary of the vehicle-wise AADT on the project road, for each homogenous section.

**Table 2.3: AADT along Homogeneous Sections of Kakinada - Rajahmundry Road**

| Sec    | Description | FAST MOVING VEHICLES |      |                       |      |      |     |       |      |             |                  |               | SLOW MOVING VEHICLES |          |                 |                |
|--------|-------------|----------------------|------|-----------------------|------|------|-----|-------|------|-------------|------------------|---------------|----------------------|----------|-----------------|----------------|
|        |             | 2 W                  | 3 W  | Car/<br>Jeep/<br>Taxi | Bus  |      | LCV | Truck |      |             | Agri.<br>Tractor |               | Cycle                | Rickshaw | Animal<br>Drawn |                |
|        |             |                      |      |                       | Mini | Full |     | 2-X   | M-X  | Articulated | With<br>Trailer  | No<br>Trailer |                      |          | Bullock<br>Cart | Horse<br>Drawn |
| Factor |             | 0.5                  | 1.0  | 1.0                   | 1.5  | 3.0  | 1.5 | 3.0   | 3.0  | 4.5         | 4.5              | 1.5           | 0.5                  | 1.5      | 6.0             | 6.0            |
| KR-01  | AADT        | 5846                 | 3194 | 2223                  | 121  | 699  | 563 | 818   | 669  | 30          | 149              | 101           | 3100                 | 116      | 34              | 0              |
|        | PCU         | 2923                 | 3194 | 2223                  | 182  | 2097 | 563 | 2454  | 2007 | 135         | 671              | 152           | 1550                 | 174      | 204             | 0              |
|        | %           | 33.1                 | 18.1 | 12.6                  | 0.7  | 4.0  | 3.2 | 4.6   | 3.8  | 0.2         | 0.8              | 0.6           | 17.6                 | 0.7      | 0.2             | 0.0            |
|        | Total PCU   | 16884                |      |                       |      |      |     |       |      |             |                  |               | 1928                 |          |                 |                |
|        | Total PCU   | 18812                |      |                       |      |      |     |       |      |             |                  |               |                      |          |                 |                |
| KR-02  | AADT        | 3666                 | 847  | 966                   | 19   | 316  | 500 | 1108  | 1117 | 24          | 131              | 16            | 483                  | 6        | 7               | 0              |
|        | PCU         | 1833                 | 847  | 966                   | 29   | 948  | 500 | 3324  | 3351 | 108         | 590              | 24            | 242                  | 9        | 42              | 0              |
|        | %           | 39.8                 | 9.2  | 10.5                  | 0.2  | 3.4  | 5.4 | 12.0  | 12.1 | 0.3         | 1.4              | 0.2           | 5.2                  | 0.1      | 0.1             | 0.0            |
|        | Total PCU   | 12766                |      |                       |      |      |     |       |      |             |                  |               | 293                  |          |                 |                |
|        | Total PCU   | 13059                |      |                       |      |      |     |       |      |             |                  |               |                      |          |                 |                |

Based on the base year traffic as presented above, traffic projections have been done to derive the traffic demand at horizon years which are important inputs in proposing the improvement options. Based on the traffic projections and capacity analysis it is concluded to upgrade the existing facility to 2/4-lane configuration.

## 2.2.3 Road Width

The carriageway width / roadway width of the project road is not uniform. The following table gives the distribution of length of Project road and %age length with respect to carriageway width (Single lane, Intermediate Lane, Double Lane, Double lane with Paved Shoulder and Four Lane)

**Table 2.4: Length Distribution of Road Width**

| Type of Carriageway Configuration and Length |      |      |      |     | Total Length |
|----------------------------------------------|------|------|------|-----|--------------|
| SL                                           | IL   | 2L   | 2LPS | 4L  |              |
| -                                            | 1.40 | 55.7 | 3.8  | 0.8 | 61.7         |
| -                                            | 3%   | 90%  | 6%   | 1%  | 100%         |

From the road inventory it can be summarised that 90% of the KR road is 2-lane and the carriageway at Kakinada town is widened to 2LPS and 4L to cater to the local and slow moving traffic. The width of earthen shoulder is varying from 1.0m to 1.5m against the standard width of 2.5m.

## 2.2.4 Land Use and Roadside Environments

The land use all along the KR road on the LHS is Samalkota Irrigation canal which runs parallel to the project road and as such all the settlements/builtup areas and agricultural lands are located on the RHS. For around a total length of 5 km, railway line runs parallel to the KR road in the close proximity on the RHS.

The following table indicates the distribution of length of Project road on RHS and %age Length in terms of land use and roadside environment.

**Table 2.5: Summary of Land Use**

| <b>Land Use pattern</b> |                       |                     |               |                   | <b>Total Length</b> |
|-------------------------|-----------------------|---------------------|---------------|-------------------|---------------------|
| <b>Urban Built-up</b>   | <b>Rural Built-up</b> | <b>Agricultural</b> | <b>Barren</b> | <b>Industrial</b> |                     |
| 7.0                     | 11.1                  | 43.2                | -             | 0.4               | <b>61.7</b>         |
| 11%                     | 18%                   | 70%                 | -             | 1%                | <b>100%</b>         |

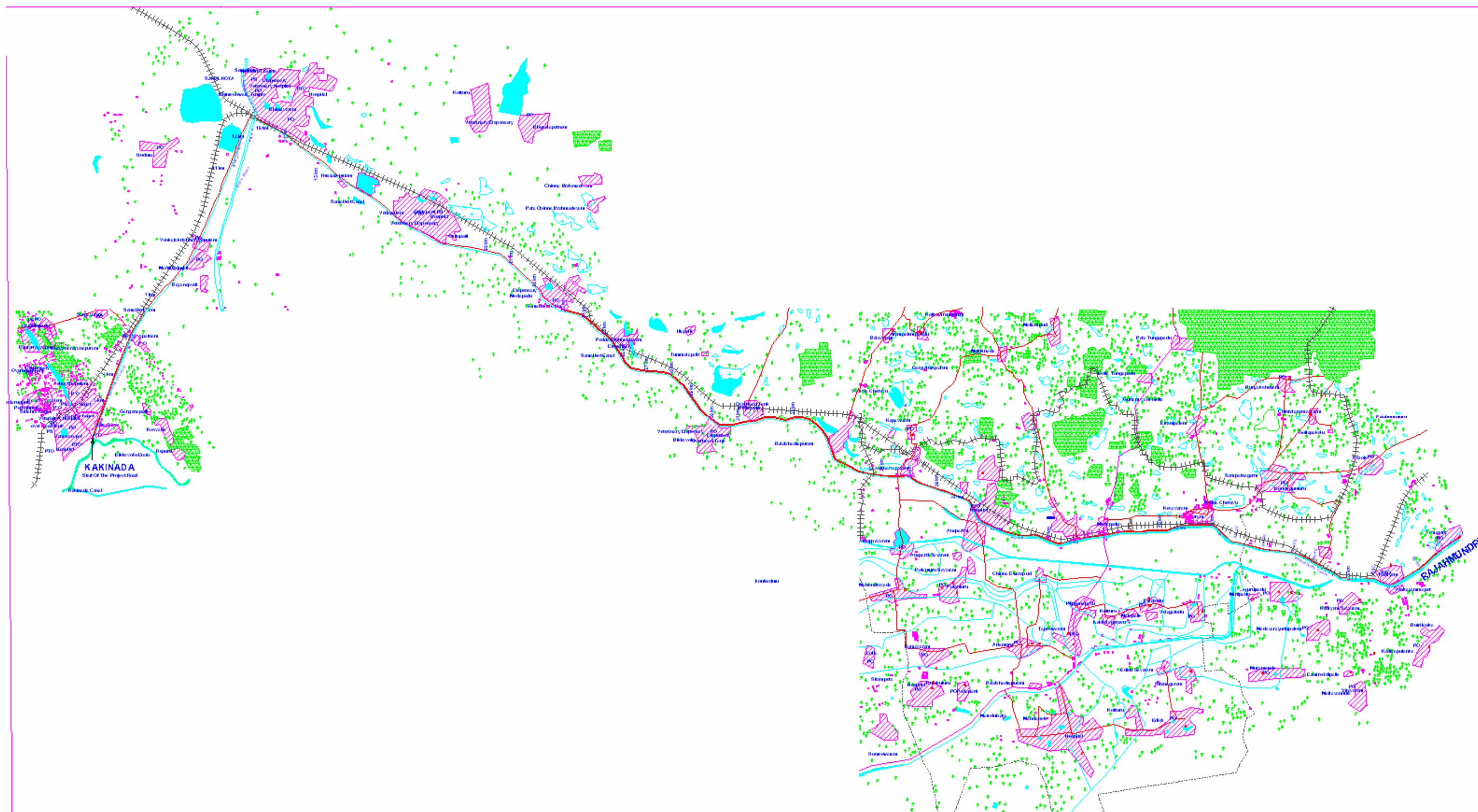


Fig 2.2: Land Use Plan of Kakinada - Rajamundry Road

## 2.2.5 Villages and Urban-Built Up Sections

Around 30% length of the project road passes through urban and rural builtup area. Along the project road section there are a number of major built up sections to name some Kakinada, Samalkota, Biccavolu, Anaparthi, Dwarapudi, Kadiyam & Vemagiri which need a special attention while finalizing the cross-section and improvement to geometric if any.

The total number of villages/settlements along the KR road is presented in table 2.6 below:

**Table 2.6: Inhabitation along the Project Road**

| Section | Length (km) | No. of villages |
|---------|-------------|-----------------|
| KR-01   | 16.0        | 4               |
| KR-02   | 45.7        | 10              |

## 2.3 PROPOSED IMPROVEMENTS

### 2.3.1 Proposed CW Configuration and Cross Sections

The existing KR road is proposed to be upgraded to 2-lane paved shoulder and 4-lane configuration, the details of which are presented below:

**Table 2.7 Section wise Proposed Improvements**

| Section ID | Section Name           | Design Chainage |        | Design Length (km) | Upgrading Proposal |
|------------|------------------------|-----------------|--------|--------------------|--------------------|
|            |                        | From            | To     |                    |                    |
| KR-01      | Kakinada – Samalkota   | 0/000           | 15/000 | 15.0               | 4 Lane + ES        |
| KR-02      | Samalkota – Rajamundry | 15/000          | 60/900 | 45.9               | 2 Lane + PS        |

### 2.3.2 Identification of Realignment and Bypass Provisions

The entire length of Kakinada – Rajahmundry has very typical kind of lateral confinement with Samalkota Irrigation and Navigational Canal running parallel on the LHS. On the RHS, Railway line is running parallel to the project road at around four locations at very close proximity.

Samalkota is an important town well connected with road and railway line in East Godavari district along the project road. Following branches of SH, MDR and Rural roads converge/diverge at Samalkota:

**Table 2.8: Important Road links at Samalkota**

| Sl. No. | Highway Type | Description                                            |
|---------|--------------|--------------------------------------------------------|
| 1       | SH-148       | Kakinada-Rajahmundry ADB road leading to Kakinada Port |
| 2       | SH-148       | Kakinada-Rajahmundry ADB road leading to Rajahmundry   |
| 3       | SH-78        | Kakinada- Rajahmundry road leading to Kakinada City    |
| 4       | SH-78        | Kakinada- Rajahmundry road leading to Rajahmundry City |
| 5       | MDR          | Samalkota-Gokavaram road                               |
| 6       | MDR          | Samalkota-Prattipadu                                   |
| 7       | MDR          | Samalkota-Uppada                                       |
| 8       | RR           | Samalkota-Unduru                                       |

Due to the above links converging and diverging at Samalkota, the traffic in and around Samalkota is very high. Further, Chennai – Howrah railway line is crossing Samalkota and the goods transport activity at Samalkota railway junction is very high due to its close proximity to Kakinada port.

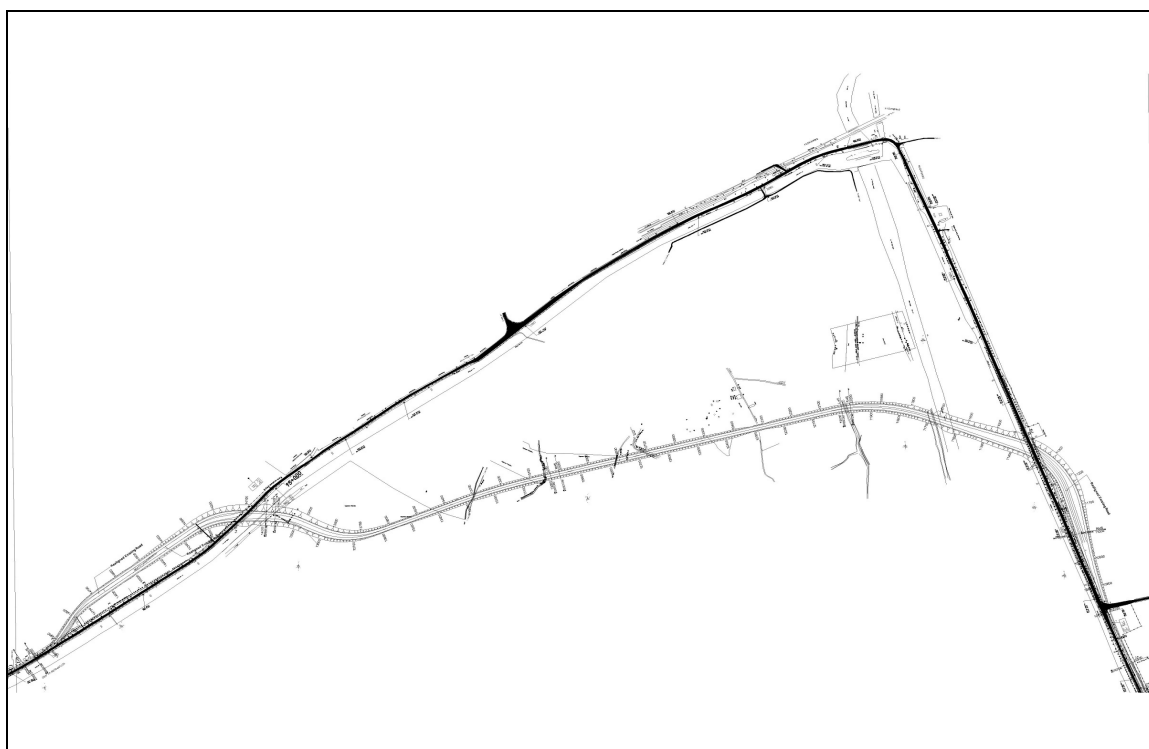
The existing alignment between km 12/4 and km 15/0 at Samalkot is very critical and upgrading of this portion along the existing alignment is almost impossible. At Samalkot, for around 2.6 km length from km 12/4 to 15/0, road has many constraints like bridge over Yeleru river, siphon structure between Yeleru river and Samalkot canal, railway bridge (ROB), level crossing and road intersections and project road is sandwiched between Samalkot canal and railway track. Table 2.9 summarises the details of critical locations along this section.

**Table 2.9: Description of Critical locations between km 12/4 and 15/0 at Samalkota**

| Location | Chainage | Description                                                                                                                                                                                                                                                                                         |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A        | Km 12/6  | Very sharp curve (90°), bridge and Aqueduct. Widening at this location is not possible due to the existence of Yeleru drain and Aqueduct on LHS and Railway line on RHS.                                                                                                                            |
| B        | Km 13/0  | Major canal on LHS and Samalkota railway station on RHS (Railway line located at 3-5 m from edge of the CW). Widening on both sides not possible.                                                                                                                                                   |
| C        | Km 14/0  | The intersection leading to Peddapuram/Samalkota and ADB road is located on ROB and the ramps of this intersection are just on the edge of the carriageway of KR road. Apart from the above Sugar factory is located close to the road. Hence widening of KR road at this location is not possible. |

Due to the above constraints at Samalkota, the existing bottleneck shall become more severe as the 4 lane traffic converges to 2 lane road at location which is already very critical and congested. Hence unless the problem at Samalkota is eased out, the upgrading of project road to 4 lane at both the ends of Samalkota shall be very much underutilized.

Due to the above mentioned critical locations and high intensity of traffic along KR road, the Consultants have proposed 4 Lane bypass. The schematic of the existing alignment and the proposed bypass is presented in Fig 2.3



**Fig 2.3: Schematic of existing alignment and bypass at Samalkota**

The proposed bypass is designed as 4Lane divided carriageway and takes off with a ramp at km 11/5 and crosses the Yeleru River and travels for a total length of 3.2 km to end at km 15/5 on existing alignment. The junctions will be developed at both the ends of bypass to facilitate easy entry/exit to/from the bypass.

## 2.4 CULVERTS AND BRIDGES

### 2.4.1 Surface and Longitudinal Drainage

The existence of Irrigation canal on LHS all along the KR road is functioning as drain to collect surface runoff and hence the provision of longitudinal drains has been considered on RHS in builtup locations. The road side drains are provided in built-up area to drain out the surface water.

### 2.4.2 Inventory of Cross Drainage Structures

The Bridge and culvert inventory was carried out from January to April 2007 and information has been collected in the format recommended by IRC. The detailed information on all the structural components, HFL, LWL dimensions of all the components, linear water way, vertical clearances, drainage spouts, handrails etc. are all given in the tabular form, and submitted in the Feasibility Report. The inventory also contains the recommendations whether the bridge is to be retained/repaired/or dismantled. There are 70 culverts and 14 bridges on the project road. A summary of cross-drainage structures are given in Table 2.10.

**Table 2.10: Summary of Existing Bridges and Cross-Drainage Structures**

| Road Name              | Length (km) | No of Culverts (Length <6m) |      |         |       | No. of Bridges |       |       |
|------------------------|-------------|-----------------------------|------|---------|-------|----------------|-------|-------|
|                        |             | Pipe                        | Slab | Others* | Total | Minor          | Major | Total |
| Kakinada – Rajahmundry | 60.9        | 6                           | 34   | 9       | 49    | 13             | 1     | 14    |

*Others\*: Includes Arch or Cut Stone or Mix/extended type*

#### Superstructure

All slab culverts are of Reinforced Cement Concrete (RCC). Arch culverts are made of stone masonry and the superstructure is made of cut-stone slab. Some slab culverts show spalling of concrete, mostly at the ends of the slabs.

#### Sub-structure

Abutments, wing walls of slab culverts and Head walls of Hume pipe / Arch culverts are in stone Masonry.

#### Parapet wall

Parapet walls of most of the culverts are partially or fully damaged and will require repair. Quite a few culverts do not have the parapet wall.

### 2.4.3 Formation Width of New Culverts and Bridges

The formation width of new bridges has been considered as follows in the feasibility studies-

- For Two lane -12.0 m
- For Four lane – 2 x 10.25 m

### 2.4.4 Sighting of Bridge and its approaches

In order to avoid land acquisition problem, it is assumed to dismantle the existing dilapidated culverts/bridges and construct the new high level bridges at the same location except in cases where it is not possible due to improvement in horizontal geometry. A temporary diversion road shall be constructed during the construction of new culverts/bridges.



**A) Length and Span Arrangement**

The overall length of the proposed culverts/bridges is kept higher the existing length based on the discharge and other criteria.

**B) Type of Superstructure**

The following types of super-structure have been assumed while working out the cost of bridges for feasibility studies-

- RCC solid slabs for spans up to 10.0m.
- RCC T-beam girder type super-structure has been assumed for spans varying from 10.0m to 25.0m in straight bridges.

**C) Type of Substructure**

The substructure consists of abutments and piers.

- **Abutments**

Plain Cement Concrete (PCC) solid wall type abutments have been assumed for heights up-to 12.0m and counterfort type abutments have been assumed for larger heights for calculating the cost of sub-structure for feasibility studies.

- **Piers**

Plain Cement Concrete (PCC) Piers are proposed for multi span bridges. A combination of rectangular and circular sections has been proposed for the piers.

**D) Type of Foundations**

Open type foundations are provided for all the bridges.

**2.5 IDENTIFICATION OF BORROW AREAS**

Extensive survey was conducted to identify borrow areas for locating suitable soil/soil aggregates mixes to be used in the construction of embankment and sub-grade. Efforts were made to locate borrow areas near the project road to avoid long haulage of the materials. Similarly, detailed survey was carried out to identify stone metal quarries of hard stone for use in Granular Sub-base (GSB), Wet Mix Macadam (WMM), bituminous and cement concrete works.

The following conclusions have been made based on the exploration of potential borrow sources:

- Construction materials are available with in an economical lead;
- Construction materials can be obtained/exploited without any hassles during the implementation of the project;
- The soil/crushed stone aggregates/ sand etc are exhibiting higher strength properties than recommended.
- To procure good quality crushed stone aggregate in large quantity, the contractor is required to install his own crushing plant thus enhancing the quality of the materials; and
- Two locations of borrow areas have been identified and tested for soil to be used in embankment, shoulder and subgrade construction.
- Out of two locations LL, (PI <12), dry density and (CBR> 12) two falls within the permissible limits as per MoST Specifications.

**2.6 ROAD SAFETY REVIEW**

Road safety audit is a formal procedure for assessing accident potential and safety performance in the provision of new road schemes, the improvement and the rehabilitation of existing roads and in the maintenance of existing roads. The road safety Audit is an important aspect of the project preparation. It is carried out at every stage of the project preparation. i.e. Feasibility Stage and



Detailed Project Report Stage to ensure that the safety is not compromised at any stage. The basic aim for road safety audit is to identify areas of major concern, including black spots and accident-prone stretches on individual project roads and to propose measure to be taken for improving the engineering design with respect to road safety aspects.

Consultants have carried out a detailed reconnaissance along the project roads and identified areas of major concern, including black spots and accident-prone stretches on each project road. Based on critical analysis of accident-prone stretches, the observed main causes for accidents are as list below:

1. Inadequate width, of shoulders, varies from 1.0m to 1.5m, which is not sufficient for parking of disabled vehicles and space for emergency stops for vehicles away from the carriageway.
2. Majority of horizontal curves are not provided with transition curve and do not have sufficient setback distance and safe stopping distance.
3. The vertical alignment of project roads traverses through plain to hilly terrain and do not have sufficient safe stopping distance and compatibility between horizontal and vertical alignment is missing.
4. Lack of segregation of traffic in both direction and lack of access control in built-up area
5. Non-provision of acceleration and deceleration lanes and intersection/ junction.

Consultants approached respective Police Station for providing the accident data. Police people were reluctant to part with the information on accident data. However, the reasons of the accidents were discussed with them to have a better understanding of the problem. The locations where two or more accidents have occurred in a year have been thoroughly analyzed/ studied, by the Consultants, from the technical and engineering aspect. The Table 2.11 contains the deficient geometric locations along the Project Road:

**Table 2.11: Deficient Geometric Location from Safety Aspects**

| Locations (km)                                                                                                                                                                                   | Deficiency                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 1+543-1+567, 26+765-26+836, 40+505-40+569, 49+003-49+050, 60+677-60+791, 60+812-60+893                                                                                                           | Radiuses of Horizontal curves are substandard             |
| 3+550-3+860, 11+200-12+000, 13+750-14+600, 17+320-17+550, 19+390-19+730, 21+450-21+700, 29+220-29+560, 30+160-30+720, 34+350-34+650, 40+350-40+650, 44+810-45+250, 48+850-49+180, 59+180-59+470. | Vertical curves /Alignment at approaches to bridges       |
| 0+000, 0+250, 0+600, 1+000, 1+480, 10+720, 15+200, 40+420, 59+320                                                                                                                                | Major Junctions, need proper design as per traffic demand |
| 10+670-15+000                                                                                                                                                                                    | Samalkota bypass                                          |

The data given in the Table reveals that the horizontal and vertical geometric needs to be improved at some locations to give a safe passage to the commuters. At built-up locations, there is no provision of railing or fencing to separate the movement of pedestrians and vehicles. No intersection has any segregated islands. All along the project road, there is no bus and truck lay-bye.

From safety of commuters and safeguard of irrigation canal all along the LHS of the project road, retaining wall and safety barrier/fence (W-beam crash barrier) needs to be provided to protect the side slopes/ embankment and safety.

Following measures have been taken up for improve the traffic safety:

1. Geometric Design Aspects
2. Design of Intersections

3. Traffic Control and Road Safety Features
4. Roadside facilities
5. Traffic Calming

#### A) Geometric Design Aspects

All geometric design elements have been carried out as per Design standards stipulated for Project in consonance with IRC codal provisions. Comprehensive design standards, which link individual design elements to best estimates of actual speed have been utilised. The objective is that drivers must not be presented with the unexpected. The emphasis has been given on maintaining continuity or giving adequate warning where it could not be made.

Following realignment locations have been identified to improve the horizontal geometrics of the project road.

**Table 2.12: Realignment sections.**

| Realignment Locations |        | Length (m) | Remarks           |
|-----------------------|--------|------------|-------------------|
| From                  | To     |            |                   |
| 11+590                | 11+780 | 190        | Minor Realignment |
| 11+780                | 14+400 | 2620       | Samalkota Bypass  |
| 14+400                | 14+700 | 300        | Minor Realignment |
| 31+700                | 32+000 | 300        | Minor Realignment |
| 34+390                | 34+550 | 160        | Minor Realignment |
| 35+600                | 35+700 | 100        | Minor Realignment |
| 57+280                | 57+410 | 130        | Minor Realignment |
| Total(m)              |        | 3800       |                   |

**Adequate measures have been taken so that the raising if any in settlement areas should be minimum possible and in no case exceed 0.5m.**

The design speed has been kept quite consistent, and speed difference between two consecutive curves is not exceeded. All horizontal curves are designed for 100kmph or 80kmph or the lowest one is 50ph, as detailed in table 2.13 below:

**Table 2.13: Summary of Proposed Horizontal Alignment**

| Total No of Curves | Radius of Curve (m) |         |         |         |            |       | Design speed (kmph) |    |    |     |
|--------------------|---------------------|---------|---------|---------|------------|-------|---------------------|----|----|-----|
|                    | <150                | 150-200 | 200-400 | 400-800 | 800 - 2000 | >2000 | 40                  | 65 | 80 | 100 |
| 148                | 0                   | 5       | 34      | 36      | 37         | 31    | 4                   | 6  | 96 | 37  |

All horizontal curves have been designed with proper transition curves and super-elevation runoff. All vertical curves have been design for minimum stopping sight distance. Due has been taken to avoid the raising of profile at urban/built-p locations and in any case the raising has been restricted to 0.5m.

In built-up areas, the cross-section 1 and 3 (provision of drain and footpath) has been adopted to segregate the local pedestrians from through motorised traffic.

#### B) Design of Intersections

Accident data reveals number of accidents at intersection account for almost 30%-40% of all reported road accidents in India. During the detailed design proposal, all major and minor junctions have been studied thoroughly with respect to traffic volume and geometric. The important minor junctions leading to villages and major settlements have been identified and proper junction layouts (including road markings and traffic signs) have been applied as per IRC-SP: 41-1994. Design of

major junctions has been carried out based on peak hour traffic data. List of major and minor junctions is given below:

**Table 2.14: Summary of Proposed Minor and Major Junctions**

| Sl.no | Type of Junction | Numbers/Chainage           | Remarks                    |
|-------|------------------|----------------------------|----------------------------|
| 1     | Minor - T/ Y/ X  | 101 Nos                    | Typical Design             |
| 1     | Major            | 0+000 (Design ch: 0+000)   | Proposed <b>T</b> Junction |
| 2     | Major            | 0+250 (Design ch: 0+250)   | Proposed <b>X</b> Junction |
| 3     | Major            | 0+600 (Design ch: 0+600)   | Proposed <b>T</b> Junction |
| 4     | Major            | 1+000 (Design ch: 1+000)   | Proposed <b>X</b> Junction |
| 5     | Major            | 1+480 (Design ch: 1+480)   | Proposed <b>X</b> Junction |
| 6     | Major            | 10+670 (Design ch: 10+720) | Proposed <b>Y</b> Junction |
| 7     | Major            | 16+000 (Design ch: 15+000) | Proposed <b>Y</b> Junction |
| 8     | Major            | 30+200 (Design ch: 29+400) | Proposed <b>X</b> Junction |
| 9     | Major            | 35+400 (Design ch: 34+450) | Proposed <b>X</b> Junction |
| 10    | Major            | 41+600 (Design ch: 40+420) | Proposed <b>T</b> Junction |
| 11    | Major            | 45+900 (Design ch: 45+100) | Proposed <b>X</b> Junction |
| 12    | Major            | 49+600 (Design ch: 48+750) | Proposed <b>X</b> Junction |
| 13    | Major            | 56+900 (Design ch: 56+150) | Proposed <b>X</b> Junction |
| 14    | Major            | 60+100 (Design ch: 59+320) | Proposed <b>X</b> Junction |

### C) Traffic Control and Road Safety Features

Traffic control devices and road safety features, comprising of following, plays a key role in influencing driver behaviour, orientation and information:

- Traffic Signs
- Road Markings
- Delineators
- Road lighting
- Crash Barrier

#### (i) Traffic Signs

Traffic signs are divided into three broad categories; Warning signs, Regulatory signs and informatory signs. Warning sign play a crucial role in terms of road safety, in giving advance information about hazards ahead. Typical examples of warning signs are Curve ahead, Pedestrian crossings, gap in median etc. Regulatory/ Mandatory signs regulates the side road or through traffic in order to have a safe movement. STOP, speed limit, NO parking etc, are some of the regulatory traffic signs. Informatory signs are provided to give information and guidance about the facilities available to the road users. Village sign boards, Fuel stations, Hospitals etc are some of the information sign boards.

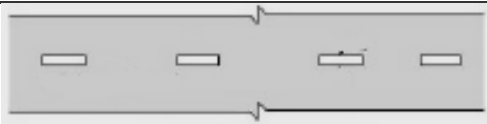
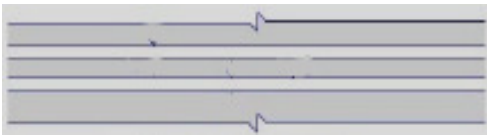
All critical locations were identified after the proposed geometric improvements along the project road and cross roads, where warning, Regulatory and Informatory signs are required.

Traffic signs have been designed to convey clear and unambiguous messages to road users so that they can be understood quickly and easily. IRC-67:1977, code for Practice of Road Sign, has been followed for finalisation of sign installation. On unkerbed roads segments, the extreme edge of the sign would be 2 to 3m from the edge of the carriageway. On kerbed roads it is proposed as not less than 60cm away from the edge of the kerb.

## (ii) Road Markings

Road markings play a very important role in guiding the driver and providing him/ her with the information necessary to negotiate conflict points on the road network. It is given a high priority for improving safety. IRC-35:1997, Code of Practice for Road Markings, provides the recommended practice for use of road markings. Road markings are lines, words and Symbols attached to the carriageway or adjacent to the carriageway, for controlling, warning, guiding and informing the users. Yellow, white and black colours are the standard colours used for marking. Road marking are classified as longitudinal marking, marking at intersections, marking at hazardous locations, marking for parking, word messages. The longitudinal marking proposed along the project road as given below:

**Table 2.15: Longitudinal Markings Proposed for the Project Road**

| Type of line                          | Figure                                                                                                                     | Description                                                                                                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Broken line                           | <br>SINGLE BROKEN LINE                    | Broken lines are permissive in character and may be crossed with discretion, if traffic permits.                                                                                                                                   |
| Solid lines                           | <br>SINGLE SOLID LINE                    | Solid lines are restrictive in character and indicate that crossing is not permitted except for entry or exit from a side road.                                                                                                    |
| Double solid lines                    | <br>PAIR OF SOLID LINES                 | Double solid lines indicate maximum restrictions and are not to be crossed except in emergent usage.                                                                                                                               |
| Combination of broken and solid lines | <br>COMBINATION OF BROKEN & SOLID LINES | In a combination of broken and solid lines, a solid line may be crossed, with discretion, if the broken line is nearer to the direction of travel. Vehicle from the opposite directions are not permitted to cross the solid line. |

Thermoplastic road markings have been proposed considering their long-lasting effect.

## (iii) Delineators

The delineators are most likely to be effective on dangerous bends, on approaches to intersections and on embankments. Though all horizontal curves are design to the IRC standards, delineators have been proposed for additional guidance and information to the commuter at curves (Radius less than 1000m). Locations along the high embankment have also been provided with delineators. Delineators with reflectorised panels have been considered.

## (iv) Road lighting

Night-time accidents in urban areas can be substantially reduced by the implementation of adequate street lighting. The stretches of the project road traversing through the built-up/urban areas are proposed with adequate lighting. Road lighting has also been proposed at all major junctions. One/ Two/ Three arm bracket Sodium vapour lamps have been considered

**(v) Crash Barrier**

In addition to the adequate provisions of roadway width and roadside design, crash barrier/guard rails have been proposed to be installed along the roadway edge on either side based on the sections of the project road having:

- At all the bridge approaches
- Embankment with >3 m height
- Horizontal curves with deflection angle >30 degrees and Radius  $\leq 240$ .

For the embankments upto 3 m height, reasonably flatter side slope of 1 vertical: 3 horizontal or more would enable erring vehicles to return to the traffic stream. The locations of provision of crash barrier are given below:

Table 2.16: Locations of proposed Crash Barrier

| Sl. No    | Chainage |        | Length (km) | Remarks                | Reason                         |
|-----------|----------|--------|-------------|------------------------|--------------------------------|
|           | From     | To     |             |                        |                                |
| 1         | 1.543    | 1.567  | 0.024       | On outer side of Curve | Def Angle > 30° & Radius ≤ 240 |
| 2         | 26.765   | 26.836 | 0.071       | On outer side of Curve |                                |
| 3         | 40.505   | 40.569 | 0.064       | On outer side of Curve |                                |
| 4         | 49.003   | 49.050 | 0.047       | On outer side of Curve |                                |
| 5         | 60.677   | 60.791 | 0.114       | On outer side of Curve |                                |
| 6         | 60.812   | 60.893 | 0.081       | On outer side of Curve |                                |
| Total (m) |          |        | 0.401       |                        |                                |
| 1         | 1.510    | 1.535  | 0.025       | On either side         | High Embankment                |
| 2         | 11.465   | 12.410 | 0.945       | On either side         |                                |
| 3         | 13.980   | 14.720 | 0.740       | On either side         |                                |
| 4         | 37.900   | 37.920 | 0.020       | On either side         |                                |
| 5         | 54.070   | 54.080 | 0.010       | On either side         |                                |
| Total (m) |          |        | 2.141       |                        |                                |
| 1         | 1.640    | 1.820  | 0.180       | On left Side           | Critical Locations             |
| 2         | 2.040    | 2.070  | 0.030       | On left Side           |                                |
| 3         | 2.620    | 2.850  | 0.230       | On left Side           |                                |
| 4         | 17.150   | 17.250 | 0.100       | On left Side           |                                |
| 5         | 17.325   | 17.450 | 0.125       | On left Side           |                                |
| 6         | 19.350   | 19.675 | 0.325       | On left Side           |                                |
| 7         | 21.010   | 21.200 | 0.190       | On left Side           |                                |
| 8         | 21.975   | 22.100 | 0.125       | On left Side           |                                |
| 9         | 22.875   | 23.100 | 0.225       | On left Side           |                                |
| 10        | 23.580   | 23.800 | 0.220       | On left Side           |                                |
| 11        | 24.190   | 24.225 | 0.035       | On left Side           |                                |
| 12        | 24.275   | 24.320 | 0.045       | On left Side           |                                |
| 13        | 25.850   | 25.990 | 0.140       | On left Side           |                                |
| 14        | 30.280   | 30.330 | 0.050       | On left Side           |                                |
| 15        | 32.050   | 32.850 | 0.800       | On left Side           |                                |
| 16        | 34.200   | 34.240 | 0.040       | On left Side           |                                |
| 17        | 34.300   | 34.350 | 0.050       | On left Side           |                                |
| 18        | 34.550   | 35.000 | 0.450       | On left Side           |                                |
| 19        | 34.600   | 34.700 | 0.100       | On left Side           |                                |
| 20        | 34.820   | 34.980 | 0.160       | On left Side           |                                |
| 21        | 37.200   | 37.230 | 0.030       | On left Side           |                                |
| 22        | 37.280   | 37.400 | 0.120       | On left Side           |                                |
| 23        | 37.800   | 37.900 | 0.100       | On Right Side          |                                |
| 24        | 38.800   | 40.150 | 1.350       | On left Side           |                                |
| 25        | 44.900   | 45.100 | 0.200       | On left Side           |                                |
| 26        | 48.600   | 48.800 | 0.200       | On left Side           |                                |
| 27        | 48.900   | 49.000 | 0.100       | On left Side           |                                |
| 28        | 54.950   | 55.100 | 0.150       | On left Side           |                                |
| 29        | 55.800   | 56.000 | 0.200       | On left Side           |                                |
| Total (m) |          |        | 6.070       |                        |                                |
| 1         | 10.950   | 11.090 | 0.140       | LCW -Innerside         | Bypass                         |

| Sl. No    | Chainage |        | Length (km) | Remarks        | Reason |
|-----------|----------|--------|-------------|----------------|--------|
|           | From     | To     |             |                |        |
| 2         | 11.320   | 11.590 | 0.270       | LCW -Innerside |        |
| 3         | 11.700   | 11.950 | 0.250       | LCW -Innerside |        |
| 4         | 13.700   | 14.040 | 0.340       | LCW -Innerside |        |
| 5         | 14.350   | 14.850 | 0.500       | LCW -Outerside |        |
| 6         | 11.440   | 11.630 | 0.190       | RCW -Innerside |        |
| 7         | 11.740   | 11.950 | 0.210       | RCW -Innerside |        |
| 8         | 13.650   | 14.030 | 0.380       | RCW -Innerside |        |
| 9         | 14.600   | 14.850 | 0.250       | RCW -Outerside |        |
| Total (m) |          |        | 8.600       |                |        |

#### D) Roadside Facilities

Following roadside facilities have been provided on Kakinada - Rajahmundry road:

- On-street Parking
- Bus Lay-byes and Shelters

##### (i) On-street Parking

At quite a few locations, project road traverses through built-up areas where small commercial activities are carried out on both side of the road. The proposed 1.5m paved shoulder may be utilised for short time parking of the vehicles. This kerb side parking is permitted only if stationary vehicles do not unduly interfere with free and safe movement of vehicles. Besides, bus laybys and truck laybys are proposed for off-street parking,

##### (ii) Bus Lay-byes and Shelters

Consultants have identified the location/site for the proposed bus-lay bye for each village or built-up locations based on the following criteria:

- Bus shelters are proposed near to residences to minimise walking distance and major intersection/ junction should have direct pedestrian links segregated from motorised traffic.
- It should be positioned in straight and level sections of road and should be visible from a long distance in both directions.
- Bus bays are proposed as per the recommendations of IRC:80-1981. The typical bus bay consists of deceleration and acceleration lanes of 45m length with stopping lane of 3.5m wide, 30m long. Adequate arrangements have also been made to drain off surface water.
- The location details of bus laybys and shelters are presented in table 2.17.

**Table 2.17: Location of Proposed Bus Bays and Shelters**

| Sl.No | Description          | Location |        | Remarks                     |
|-------|----------------------|----------|--------|-----------------------------|
|       |                      | LHS      | RHS    |                             |
| 1     | Bus Shelter& Bus Bay | 0+230    |        | Kakinada                    |
| 2     | Bus Shelter& Bus Bay | 0+280    |        | Kakinada                    |
| 3     | Bus Shelter& Bus Bay |          | 0+280  | Kakinada                    |
| 4     | Bus Shelter& Bus Bay | 5+440    |        | Madhavapatnam               |
| 5     | Bus Shelter& Bus Bay |          | 5+440  | Madhavapatnam               |
| 6     | Bus Shelter& Bus Bay | 8+490    |        | Venkatakrishnarayapuram     |
| 7     | Bus Shelter& Bus Bay |          | 8+490  | Venkatakrishnarayapuram     |
| 8     | Bus Shelter& Bus Bay | 10+400   |        | Samarlakota Bypass starting |
| 9     | Bus Shelter& Bus Bay |          | 10+400 | Samarlakota Bypass starting |

| Sl.No | Description          | Location |        | Remarks                   |
|-------|----------------------|----------|--------|---------------------------|
|       |                      | LHS      | RHS    |                           |
| 10    | Bus Shelter& Bus Bay | 15+380   |        | Samarlakota Bypass Ending |
| 11    | Bus Shelter& Bus Bay |          | 15+500 | Vetlapalem                |
| 12    | Bus Shelter& Bus Bay |          | 17+730 | Gudaparti                 |
| 13    | Bus Shelter& Bus Bay |          | 21+250 | Medapadu                  |
| 14    | Bus Shelter& Bus Bay |          | 24+740 | Peddabrahmadevam          |
| 15    | Bus Shelter& Bus Bay |          | 29+150 | Bikkavolu                 |
| 16    | Bus Shelter& Bus Bay |          | 30+520 | Bikkavolu                 |
| 17    | Bus Shelter& Bus Bay |          | 30+990 | Bikkavolu                 |
| 18    | Bus Shelter& Bus Bay |          | 33+550 | Balabhadrapuram           |
| 19    | Bus Shelter& Bus Bay | 34+460   |        | Balabhadrapuram           |
| 20    | Bus Shelter& Bus Bay | 40+530   |        | Anaparthi                 |
| 21    | Bus Shelter& Bus Bay |          | 44+700 | Dwarapudi                 |
| 22    | Bus Shelter& Bus Bay | 48+580   |        | Kesavaram Rly Stn         |
| 23    | Bus Shelter& Bus Bay | 53+330   |        | Andhrapradesh paper mill  |
| 24    | Bus Shelter& Bus Bay | 56+500   |        | Kadiyam                   |
| 25    | Bus Shelter& Bus Bay | 58+800   |        | Vemageri                  |

### E) Traffic Calming

Speed is arguably a factor in every accident. Lower speeds reduce both the likelihood of the accident happening and the severity, if it does occur. Speed reduction benefits have received much attention in recent years and an international review of speed and accidents concluded that, on an average, each **5km reduction in speed would result in a 22% reduction in fatal accidents**. (1 mile leads to 7%)

Traffic calming measures need to be applied in a formal structured manner. All features should only be constructed where approach speeds are such that all road users can perceive the calming feature and traverse it safely. They should not be used in isolation but as part of a strategy covering a stretch of road or an area. Accordingly following two types of Traffic calming measures, i.e. Speed humps and rumble strips have been proposed along the project road.

- Road Humps
- Rumble Strips

#### (i) Road Humps

Road Humps or Speed breakers are formed by providing a rounded hump of 3.7m width (17m radius) and 100mm height for the preferred advisory crossing speed of 25kph for general traffic as per the **IRC: 99-1988**. The basic material for construction is bituminous concrete formed to required shape. Road humps have been proposed on minor roads at junctions /intersections with major roads, School and Hospital zones. Proper signboards and markings are provided to advise the drivers in advance of the situation. Road humps are extended across carriageway up to the edge of paved shoulder. Proper signboards and marking are proposed to advise the drivers in advance of the Road humps.

**Table 2.18: Location of Proposed Road Humps**

| Sl.No | Chainage | Description                      |
|-------|----------|----------------------------------|
| 1     | 0+200    | PRG College                      |
| 2     | 0+600    | Hospital                         |
| 3     | 1+600    | Ideal College Of Arts & Sciences |
| 4     | 5+700    | Saraswathi Vidyapita             |
| 5     | 17+950   | School                           |
| 6     | 30+200   | School                           |



| Sl.No | Chainage | Description         |
|-------|----------|---------------------|
| 7     | 41+500   | Private Hospital    |
| 8     | 57+300   | Veterinary Hospital |
| 9     | 57+850   | School              |
| 10    | 60+050   | School              |

## (ii) Rumble Strips

Rumble Strips are formed by a sequence of transverse strips laid across a carriageway with maximum permitted height of 20mm. These rumble devices produce audible and vibratory effects to alert drivers to take greater care and do not normally reduce traffic speeds in themselves. The typical design details of rumble strips proposed are transverse strips of Pre-mix bituminous concrete 500mm wide and overall thickness 20mm laid across a carriageway up to the end of paved shoulder. There will be 6 such transverse strips spaced at 0.5 m c/c. Rumble strips are proposed at:

- Sharp curves with radius less than 170m.
- Transition zones (speed limit zones).
- Village/built-up approaches.
- Sensitive receptors (Schools and hospitals)

**Table 2.19: Location of Proposed Rumble Strips**

| Sl. No | Chainage | Description      | Village Name              |
|--------|----------|------------------|---------------------------|
| 1      | 0+000    | Village Approach | Kakinada                  |
| 2      | 5+100    | Village Approach | Madhavapatnam             |
| 3      | 8+300    | Village Approach | Venkata Krishna Rayapuram |
| 4      | 14+700   | Village Approach | Husainpuram               |
| 5      | 17+300   | Village Approach | Gudaparti                 |
| 6      | 21+000   | Village Approach | Medapad                   |
| 7      | 24+600   | Village Approach | Peddabrahmadevam          |
| 8      | 29+100   | Village Approach | Bikkavolu                 |
| 9      | 33+700   | Village Approach | Balbadpuram               |
| 10     | 40+900   | Village Approach | Anaparthi                 |
| 11     | 43+500   | Village Approach | Dwarapudi                 |
| 12     | 55+800   | Village Approach | Kadiyam                   |
| 13     | 58+200   | Village Approach | Vamageri                  |

Proper signboards and marking are proposed to caution the drivers in advance of the situation.

## 2.7 ROAD CONSTRUCTION STANDARDS, NORMS AND GUIDELINES

Following the road construction standards and norms and management procedure has been adopted to keep the standards and guidelines maintained by the Indian Roads Congress (IRC):

- Guidelines for Environmental Impact Assessment of Highway Projects, IRC: 104- 1988.
- Recommended Practice for Treatment of Embankment slopes for erosion control, IRC: 36-1974.
- Recommended Practice for Borrow pits for Road Embankment for Road manual operation, IRC: 10-1961.
- Recommended Practice for the construction of Earth Embankments for Road Works, IRC: 36-1970.

- v) Highway Safety Code, IRC, special publication no. 44.
- vi) Guidelines on Bulk Bitumen Transportation and Storage Equipment, IRC, special publication 39.
- vii) Recommended Practice for Tools Equipment and Appliances for Concrete Pavement Construction, IRC: 43-1972.
- viii) Recommended Practice for use and Upkeep of Equipment, Tools and Appliances for Bituminous Pavement Construction, IRC: 72-1978. Road Accident Forms A-1 and 4, IRC: 33-1982.
- ix) The factories act 1956 for hygiene and safety requirements of construction workers.
- x) Other relevant codes of BIS and National Building Codes.

## ***CHAPTER - 3: ENVIRONMENT REGULATORY FRAMEWORK***

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## CHAPTER-3: ENVIRONMENT REGULATORY FRAMEWORK

### 3.1 APPLICABLE REGULATIONS

The chapter presents a review of the existing institutions and legislations relevant to the project at the National and State level. The various statutory clearances from various state and central government authorities and the institutional and legal frameworks are discussed in the following paragraphs.

#### 3.1.1 Legal Framework

The Government of India has laid out various policy guidelines, acts and regulations pertaining to environment. The Environment (Protection) Act, 1986 provides umbrella legislation for the protection of environment. As per this Act, the responsibility to administer the legislation has been jointly entrusted to the Ministry of Environment and Forests (MoEF) and the Central Pollution Control Board (CPCB)/Andhra Pradesh State Pollution Control Board (APSPCB) in the present context. More details on the legal framework of Government of India and State Government regulations and clearance procedures are envisaged in the following paragraphs.

#### 3.1.2 Key Environmental Laws and Regulations

The project road is proposed for improvement from existing 2L to 4 L in HS-01 (km 0.0 to km 16.0) and remaining section HS-02 (km 16.0 to km 60.1) is proposed for improvement from existing 2L to 2 L+PS, with no major realignments. Sufficient land width is available in the HS -01 section for four laning, except some additional land is required for the bypass (length 3.1km) and minor realignments. Similarly for HS-02 NO additional land width is required as the proposed improvement is limited to 1.5 m paved shoulder and 1 m earthen shoulder on either side of the existing pavement. Hence the total land acquisition required for the project road upgradation is 14.25 Hectares, on an average 2.5m width all along the project road, well below 20m (Above 20 m the project requires the MoEF Clearance), hence NO MoEF clearance is required.

Table 3.1 presents the environmental regulations and legislations relevant to AP state road project.

**Table 3.1: Environmental Regulations and Legislations**

| SR. No | Act / Rules                                                              | Purpose                                                                                                    | Applicable Yes/ No | Reason for Applicability                                                                        | Authority                             |
|--------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------|---------------------------------------|
| 1      | Environment Protection Act-1986                                          | To protect and improve overall environment                                                                 | Yes                | As all environmental notifications, rules and schedules are issued under this act.              | MoEF. Gol; DoE, State Gov. CPCB; SPCB |
| 2      | Environmental Impact Assessment Notification 4 <sup>th</sup> April, 2011 | To provide environmental clearance to new development activities following environmental impact assessment | NO                 | This notification is NOT applicable to Project road, as this is upgradation of existing highway | MoEF. EIAA                            |
| 3      | Notification for use of fly ash                                          | Reuse large quantity of fly ash discharged from thermal power plant to minimize land use for disposal      | NO                 | No thermal plant within 100km reaches.                                                          |                                       |
| 4      | Coastal Regulation Zone(CRZ) Notification 1991 (2002)                    | Protection of <b>fragile coastal belt</b>                                                                  | NO                 | Road is not located along coastal belt                                                          |                                       |
| 5      | National Environment Appellate Authority Act (NEAA) 1997                 | Address Grievances regarding the process of environmental clearance.                                       | Yes                | Grievances if any will be dealt with, within this act.                                          | NEAA                                  |

| SR. No | Act / Rules                                                                                                                                                                  | Purpose                                                                                                                     | Applicable Yes/ No | Reason for Applicability                                                                                                                                               | Authority                                                                                                         |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 6      | The Land Acquisition Act 1894 & 1989                                                                                                                                         | Set out rule for acquisition. of land by government                                                                         | Yes                | This act will be applicable to as there will be acquisition of land for widening, geometric improvements and realignments.                                             | Revenue Department State Government.                                                                              |
| 7      | MoEF Circular on Marginal Land Acquisition and Bypasses 1999                                                                                                                 | Defining “marginal land” acquisition relating to the 1997 Notification                                                      | NO                 | Not applicable as per Environmental Impact Assessment Notification 14th Sep-2006                                                                                       | MoEF                                                                                                              |
| 8      | The Forest (Conservation) Act 1927<br>The Forest (Conservation) Act. 1980 forest (conversion) Rules 1981                                                                     | To check deforestation by restricting conversion of forested areas into non- forested areas                                 | NO                 | NO forest area.                                                                                                                                                        | Forest Department, GoAP                                                                                           |
| 9      | MoEF circular (1998) on linear Plantation on roadside, canals and railway lines modifying the applicability of provisions of forest (Conversation) Act, to linear Plantation | Protection / planting roadside strip as avenue/strip plantations as these are declared protected forest areas.              | NO                 | Roadside tree plantation in Andhra Pradesh does not come under the forest act.                                                                                         | MoEF                                                                                                              |
| 10     | Wild Life Protection Act 1972                                                                                                                                                | To protect wildlife through certain of National Parks and Sanctuaries                                                       | NO                 | This act is not applicable to as there are NO points of wildlife crossing along project corridor.                                                                      | Chief Conservator Wildlife, Wildlife Wing, Forest Department, GoAP                                                |
| 11     | Air (Prevention and Control of Pollution) Act, 1981                                                                                                                          | To control air pollution by & Transport controlling emission of air Department. Pollutants as per the prescribed standards. | Yes                | This act will be applicable during construction; for obtaining NOC for establishment of hot mix plant, workers' camp, construction camp, etc.                          | SPCB                                                                                                              |
| 12     | Water Prevention and Control of Pollution) Act1974                                                                                                                           | To control water pollution by controlling discharge of pollutants as per the prescribed standards                           | Yes                | This act will be applicable during construction for (establishments of hot mix plant, construction camp, workers' camp, etc.                                           | SPCB                                                                                                              |
| 13     | Noise Pollution (Regulation and Control Act) 1990                                                                                                                            | The standards for noise for day and night have been promulgated by the MoEF for various land uses.                          | Yes                | This act will be applicable as vehicular noise on project routes required to assess for future years and necessary protection measure need to be considered in design. | SPCB                                                                                                              |
| 14     | Ancient Monuments and Archaeological Sites and Remains Act1958                                                                                                               | Conservation of cultural and historical remains found in India                                                              | NO                 | This act not applicable as the project route is not close to any Ancient Monument, declared protected under the act.                                                   | Archaeological Dept Gol, Indian Heritage Society and Indian National Trust for Art and Culture Heritage (INTACH). |
| 15     | Public Liability and Insurance Act 1991                                                                                                                                      | Protection form hazardous materials and accidents.                                                                          | Yes                | Contractor need to stock hazardous material like diesel, Bitumen, Emulsions etc.                                                                                       | SPCB                                                                                                              |
| 16     | Explosive Act 1984                                                                                                                                                           | Safe transportation, storage and <b>use of explosive</b> material                                                           | Yes                | For transporting and storing diesel, bitumen etc.                                                                                                                      | Chief Controller of Explosives                                                                                    |

| SR. No | Act / Rules                                                          | Purpose                                                                                        | Applicable Yes/ No | Reason for Applicability                                                                                                    | Authority                       |
|--------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 17     | Minor Mineral and concession Rules                                   | For opening new quarry.                                                                        | Yes                | Regulate use of minor minerals like stone, soil, river sand etc.                                                            | District Collector              |
| 18     | Central Motor Vehicle Act 1988 and Central Motor Vehicle Rules 1989  | To check vehicular air and noise pollution.                                                    | Yes                | This rule will be applicable to road users and construction Machinery.                                                      | Motor Vehicle Department        |
| 19     | National Forest Policy 1952<br>National Forest Policy (Revised) 1988 | To maintain ecological stability through preservation and restoration of biological diversity. | NO                 | This policy will not be applicable as NO eco sensitive feature exists along the project corridor.                           | Forest Department, GoI and GoAP |
| 20     | The Mining Act                                                       | The mining act has been notified for safe and sound mining activity.                           | Yes                | The construction of project road will require aggregates. These will be procured through mining from riverbeds and quarries | Department of mining, GoAP      |

### 3.1.3 Environmental Requirements of the State

Unlike other states, in A.P. the road side plantation within the right of way comes under the authority of R&B Department. But for felling and transportation of Trees, permission has to be taken from forest department at DFO level, under A.P. WALTA Act 2002.

#### Andhra Pradesh Water, Land Trees Act, (WALTA) 2002

The Act came into force on April 19, 2002 with an objective ***“to promote water conservation, and tree cover and regulate the exploitation and use of ground and surface water for protection and conservation of water sources, land and environment and matters, connected therewith or incidental thereto”***.

| SR. No | Act / Rules    | Purpose                                       | Applicable Yes/ No | Reason for Applicability                                                | Authority          |
|--------|----------------|-----------------------------------------------|--------------------|-------------------------------------------------------------------------|--------------------|
| 1      | WALTA Act 2002 | To promote water conservation, and tree cover | Yes                | Tree cutting and transit permission is required from forest department. | Forest Department. |

### 3.1.4 Other Legislation Applicable to Road Construction Projects

Environmental issues during road construction stage generally involve equity, safety and public health issues. The road construction agencies require complying with laws of the land, which include *inter alia*, the following:

- ✚ **Workmen's Compensation Act 1923** (the Act provides for compensation in case of injury by accident arising out of and during the course of employment);
- ✚ **Payment of Gratuity Act, 1972** (gratuity is payable to an employee under the Act on satisfaction of certain conditions on separation if an employee has completed 5 years);
- ✚ **Employees PF and Miscellaneous Provision Act 1952** (the Act provides for monthly contributions by the employer plus workers);
- ✚ **Maternity Benefit Act, 1951** (the Act provides for leave and some other benefits to women employees in case of confinement or miscarriage, etc.);
- ✚ **Contact Labor (Regulation and Abolition) Act, 1970** (the Act provides for certain welfare measures to be provided by the contractor to contract labour);
- ✚ **Minimum Wages Act, 1948** (the employer is supposed to pay not less than the Minimum Wages fixed by appropriate Government as per provisions of the);

- ✚ ***Payment of Wages Act, 1936*** (it lays down as to by what date the wages are to be paid, when it will be paid and what deductions can be made from the wages of the workers);
- ✚ ***Equal Remuneration Act, 1979*** (the Act provides for payment of equal wages for work of equal nature to Male and Female workers and not for making discrimination against Female employees);
- ✚ ***Payment of Bonus Act, 1965*** (the Act provides for payments of annual bonus subject to a minimum of 83.3% of wages and maximum of 20% of wages);
- ✚ ***Industrial Disputes Act, 1947*** (the Act lays down the machinery and procedure for resolution of industrial disputes, in what situations a strike or lock-out becomes illegal and what are the requirements for laying off or retrenching the employees or closing down the establishment);
- ✚ ***Industrial Employment (Standing Orders) Act; 1946*** (the Act provides for laying down rules governing the conditions of employment);
- ✚ ***Trade Unions Act, 1926*** (the Act lays down the procedure for registration of trade unions of workers and employers. The trade unions registered under the Act have been given certain immunities from civil and criminal liabilities);
- ✚ ***Child Labour (Prohibition and Regulation) A; 1986*** (the Act prohibits employment of children below 14 years of age in certain occupations and processes and provides for regulation of employment of children in all other occupations and processes. Employment of child labour is prohibited in Building and Construction Industry);
- ✚ ***Inter-State Migrant Workmen's (Regulation of Employment and Conditions of Service) Act, 1979*** (the inter-state migrant workers, in an establishment to which this Act becomes applicable, are required to be provided certain facilities such as housing, medical aid, traveling expenses from home to the establishment and back, etc.);
- ✚ ***The Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996 and the Cess Act of 1996*** (all the establishments who carry on any building or other construction work and employs 10 or more workers are covered under this Act; the employer of the establishment is required to provide safety measures at the building or construction work and other welfare measures, such as canteens, first-aid facilities, ambulance, housing accommodation for Workers near the workplace, etc.);
- ✚ ***The Factories Act, 1948*** (the Act lays down the procedure for approval of plans before setting up a factory, health and safety provisions, welfare provisions, working hours and rendering information-regarding accidents or dangerous occurrences to designated authorities);
- ✚ **Hazardous Wastes (Management and Handling) Rules, 1989;**
- ✚ **Chemical Accidents (Emergency Planning, Preparedness and Response) Rules, 1996.**

### 3.1.5 World Bank Environmental Requirements

The various environmental requirements of World Bank, GoI for the project roads are described below.

#### (i) Applicability of various WB safe guard Policies

The World Bank has ten safeguard policies; the details and applicability of the safe guard policies to the Project road are provided in the **Table 3.2**.

Environmental requirements of the World Bank are specified in detail in its Operational Policy (OP) 4.01 and other related Operation Policies. In instances in which the procedural and regulatory requirements differ, the more stringent applies. The World Bank environmental requirements are based on a three-part classification system.

- *Category A* - requires a full Environmental Assessment (EA).
- *Category B* - projects require a lesser level of environmental investigation.



- *Category C* - projects require no environmental analysis.

On the basis of data and information collected during field survey and discussion with local expert and visualise potential associated impact, consultant has categorised this project as **category-B**, which requires a lesser level of environmental investigation.

**Table 3.2: Applicability of WB Safe Guard Policies**

| WB Safe Guard Policy | Subject Category         | Triggered Or Not | Reason For Its Applicability                                                                                           | Mitigation Measures                             | Documentation  |
|----------------------|--------------------------|------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------|
| OP 4.01              | Environmental Assessment | Triggered        | Umbrella policy                                                                                                        | All necessary mitigation measures incorporated. | EIA & EMP.     |
| OP 4.04              | Natural Habitats         | Not Triggered    | Eco-sensitive-Forestry and wildlife related issues                                                                     | Not Applicable                                  | Not Applicable |
| OP 4.36              | Forestry                 | Not Triggered    | No forest land.                                                                                                        | Not Applicable                                  | Not Applicable |
| OP 4.09              | Pest Management          | Not Triggered    | Not Applicable                                                                                                         | Not Applicable                                  | Not Applicable |
| OP 4.30              | Involuntary Resettlement | Not Triggered    | Road widening will lead to loss of livelihoods, loss of land and Buildings etc                                         | Not Applicable                                  | Not Applicable |
| OP 4.20              | Indigenous people        | Not Triggered    | No separate Indigenous people development Plan is required for the Project.                                            | Not Applicable                                  | Not Applicable |
| OP 4.11 (draft)      | Cultural Property        | Not Triggered    | A number of temples, shrines, churches etc are located adjacent to road ROW. But NO declared cultural property exists. | Not Applicable                                  | Not Applicable |

### 3.1.6 Summary of Clearance Requirements

The summary table showing time requirements for agency responsible for obtaining clearance, and a stage at which clearance will be required is given below:

| Sl. No. | Type of Clearance                                                                                               | Applicability                                                      | Project Stage                                        | Responsibility | Time Required  |
|---------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------|----------------|----------------|
| 1       | Environmental Clearance from MoEF/SIAA                                                                          | Not Applicable                                                     | Not Applicable                                       | Not Applicable | Not Applicable |
| 2       | Forest Clearance for land diversion                                                                             | Not Applicable                                                     | Not Applicable                                       | Not Applicable | Not Applicable |
| 3       | Tree felling permission                                                                                         | For roadside tree cutting                                          | Pre construction                                     | APRDC          | 15 days        |
| 4       | NOC and consents under Air, Water & Environment Act and noise rules from SPCB                                   | For establishment of construction camp.                            | Construction stage (Prior to initiation of any work) | The Contractor | 2-3 Months     |
| 5       | NOC and consents under Air, Water & Environment Act and noise rules from SPCB                                   | For operating construction plant, crusher, batching plant etc.     | Construction stage (Prior to initiation of any work) | The Contractor | 1-2 Months     |
| 6       | Explosive License from Chief Controller of Explosives,                                                          | For storing fuel oil, lubricants, diesel etc. at construction camp | Construction stage (Prior to initiation of any work) | The Contractor | 2-3 Months     |
| 7       | Permission for storage of hazardous chemical from CPCB                                                          | Manufacture storage and Import of Hazardous Chemical               | Construction stage (Prior to initiation of any work) | The Contractor | 2-3 Months     |
| 8       | Quarry Lease Deed and Quarry License from State Department of Mines and Geology                                 | Quarry operation (for new quarry)                                  | Construction stage (Prior to initiation of any work) | The Contractor | 2-3 Months     |
| 9       | Permission for extraction of ground water for use in road construction activities from State Ground Water board | Extraction of ground water                                         | Construction stage (Prior to initiation of any work) | The Contractor | 2-3 Months     |
| 10      | Permission for use of water for construction purpose from irrigation department                                 | Use of surface water for construction                              | Construction stage (Prior to initiation of any work) | The Contractor | 2-3 Months     |
| 11      | Labor license from labor commissioner office                                                                    | Engagement of Labor                                                | Construction stage (Prior to initiation of any work) | The Contractor | 2-3 Months     |

As mentioned in above table pre -construction regulatory clearance is required for tree felling and forest land acquisition from forest department. In Andhra Pradesh tree-felling permission is obtained from forest department under WALTA Act 2002. For this project, the designated officer is Divisional Forest Officer, East Godavari. An application along with list of trees has already been processed and submitted to concern officials.

## ***CHAPTER - 4: BASELINE ENVIRONMENT***

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## CHAPTER 4: BASELINE ENVIRONMENT

### 4.1 BACKGROUND

Previous chapters have highlighted scope of environmental assessment, existing features of the road and proposed improvement, methodology and regulations applicable to environmental assessment. In this chapter an attempt has been made to prepare a baseline environmental setting so as applicability of Government of India (GoI) regulatory requirements as well as environmental management practices of operational procedure of World Bank could be envisioned. Based on the existing environmental scenario potential impacts of road improvement will be identified and accordingly management plan will be proposed in forthcoming sections. The baseline environmental conditions will help in comparing and to monitor the predicted negative and positive impacts resulting from the project during construction and operation phases.

Data was collected from secondary sources for the macro-environmental setting like climate, physiography (Geology and slope), biological and socio-economic environment. First hand information have been collected to record the micro-environmental features within and adjacent to the project corridor. Collection of first hand (Primary) information includes preparation of base maps, extrapolating environmental features on proposed road design, tree enumeration, location and measurement of socio-cultural features abutting project road.

Data was also recorded at sites used for extraction of materials for construction but generally outside the project corridor e.g. borrows areas and quarries. Ambient Air, Noise and water quality samples were collected at important locations in terms of environment quality to prepare a baseline database. Consultation was another source of information to explain local conditions like submergence, recent floods etc. However these consultation results were largely based on short term memories like information on floods, submergence but still it was helpful in comparing secondary information.

Following section describes the nature, type and characteristics of the physical, biological, cultural and socioeconomic components along the project roads.

### 4.2 AIR ENVIRONMENT - BASELINE

Air quality is relatively free from pollution along the project road traversing through agrarian rural and small semi urban areas. As a part of the Environmental Impact Assessment (EIA), climatic component has been studied to establish the benchmarks to understand air quality in the project influence area.

#### 4.2.1 Metrological factors and Climate

**Climate:** Among all other physical factors, Climate is the most important factor-influencing environment because it plays a vital role in determining the evolution of landforms (erosion, soil characteristics), types of flora and fauna (ecological diversity), the productivity of ecosystems as well it has an influence on the pollution loads on the environment. Rainfall, temperature, and winds are the principal climatic components that serve to transport, disperse various forms of pollution into the atmosphere and on the ground.

The characteristics of these climatic components (temperature, rainfall and winds) are averaged for comparatively longer period (about 50 years) to establish a general pattern in the project area.

In general, the climate of project influence area (PIA) of APRSP is classified as tropical (winterless) but for the purpose of Environmental Assessment, the climate of the PIA has been divided into following distinct seasons:

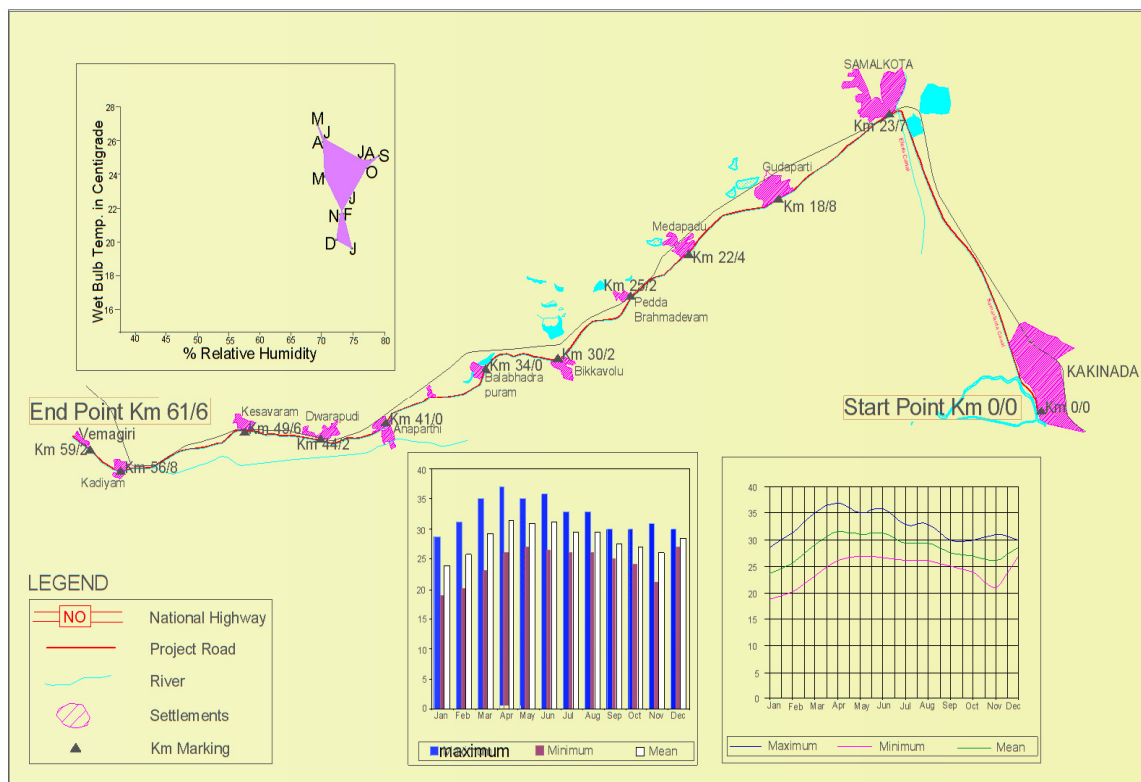
1. Rainy – October to January
2. Summer – March to June (hot and sunny)
3. Summer – End July to September (Hot and cloudy)

However for the baseline environment of the project, road under study has only two distinct seasons. (i) Hot season and (ii) Rainy season. During rainy season the project area is more or less

similar to monsoon and called as (Am), whereas in dry season it is more like {savannah (As)} with distinct hot season.

The various climatic factors such as temperature, humidity and rainfall pattern in the project area have been discussed in detail in the following sections. Following section deals about.

**Temperature:** May is the hottest month with mean monthly temperature is about 43°C in this area. The mean daily temperature during summer is as high as 43°C during March to May (**Table 4.1**). The temperature pattern of the project area will be useful as monitoring indicator especially during construction period near hot mix plants and stone crushers. December and January are comparatively the coolest months with average mean monthly temperature less than 18°C. But no months, the mean monthly temperature goes below 18 degree centigrade.



(Source: Statistical Hand Book- 2006)

**Fig. 4.1: Temperature Pattern within Project Influence Area/District**

**Humidity:** The nature and characteristics of the pollutants will vary with change of the humidity in the atmosphere. Fog provides possibility for suspended particles to coalesce and also enhances chemical reaction of the gaseous pollutants. The relative humidity is high during North East monsoon season as well as during south west monsoon season and in remaining period of the year the air is generally dry. The monthly variation is Temperature, Humidity and wind speed within project influence area/ project district is presented in Table 4.1 below. The relative humidity is the lowest during summers when it reaches as low as 50% and during monsoon it goes to as high as 86%.

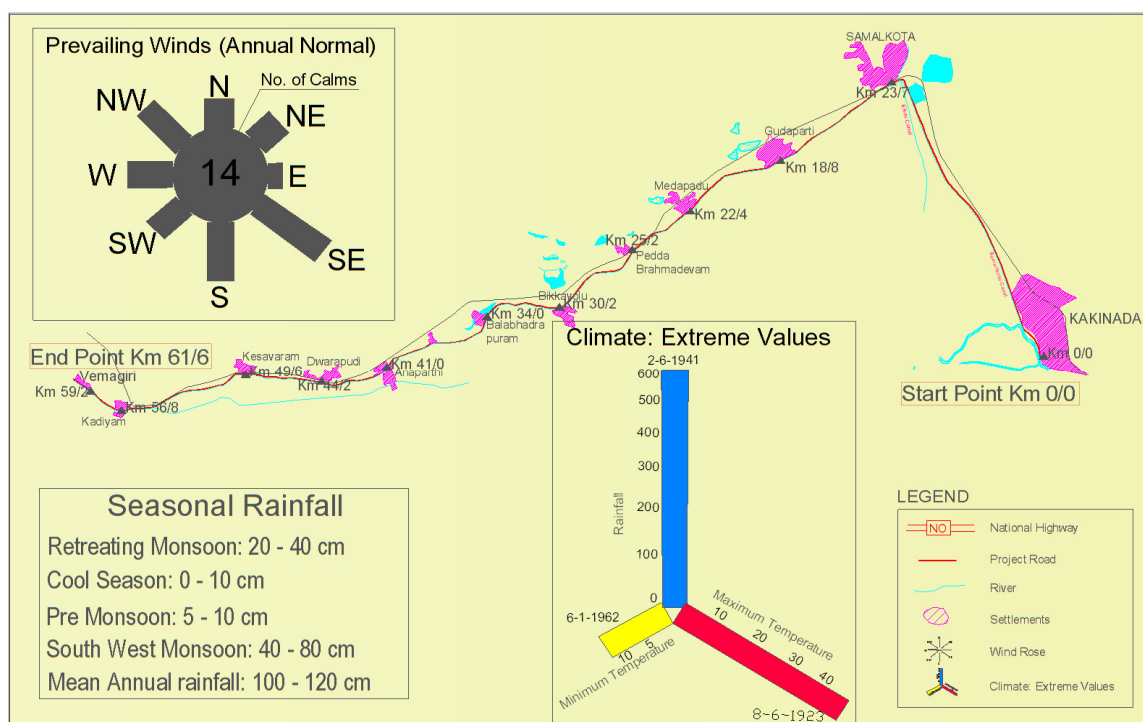
**Table 4.1 Important Climatic Condition within Project Influence Area/Project District**

| Sr. No. | Month | Temperature(°C) |      | Relative Humidity (%) |       | Wind Speed       |
|---------|-------|-----------------|------|-----------------------|-------|------------------|
|         |       | Max             | Min  | 8.30                  | 17.30 | M.W.S/km(24 hrs) |
| 1       | Jan   | 28.6            | 18.8 | 82.5                  | 65    | 4                |
| 2       | Feb   | 31.2            | 20.1 | 76                    | 59    | 3                |
| 3       | March | 35.2            | 23.1 | 73.3                  | 47    | 3.3              |
| 4       | Apr   | 37              | 26   | 72.3                  | 62    | 5                |
| 5       | May   | 35              | 27   | 69                    | 60.6  | 4.6              |

| Sr. No. | Month | Temperature(°C) |      | Relative Humidity (%) |       | Wind Speed       |
|---------|-------|-----------------|------|-----------------------|-------|------------------|
|         |       | Max             | Min  | 8.30                  | 17.30 | M.W.S/km(24 hrs) |
| 6       | June  | 35.9            | 26.6 | 70.3                  | 59.6  | 5.6              |
| 7       | July  | 33              | 26   | 80.6                  | 71.3  | 5                |
| 8       | Aug   | 33              | 26   | 78                    | 65    | 6.6              |
| 9       | Sep   | 30              | 25   | 81.6                  | 74.3  | 6.6              |
| 10      | Oct   | 30              | 24   | 81.6                  | 77    | 3.3              |
| 11      | Nov   | 31              | 21   | 73.3                  | 66.3  | 2.6              |
| 12      | Dec   | 30              | 27   | 79.3                  | 64    | 2.6              |

(Source: Statistical Hand Book- 2006)

**Precipitation:** October to January is the rainfall season. During the rainy months, the area receives almost 90% of the total rainfall. March to June is dry period for the project area. No rainy days is found during these months. However extreme climatic event like cyclones/depression do form occasionally. Fig. 4.2 below shows the wind rose diagram, maximum, minimum temperature and rainfall recorder within the project district in last 70 years.



(Source: Statistical Hand Book- 2006)

**Fig. 4.2: Extreme Climatic Conditions within Project Influence Area/District**

The average annual rainfall in the area is 873 mm. Rainfall mainly occurs during the months of SW monsoon, from June and ends with receding SW monsoon up ending of September. Rainfall during SW monsoon is generally caused by trade winds from Arabian sea called retreating monsoon. The rainfall pattern suggests that most of pre-construction activities (mainly earthwork) should be completed before monsoon season to avoid excessive consumption of water.

**Wind:** Wind speed and wind direction have a significant role on the dispersion of atmospheric pollutants and therefore, the air quality of the area. Ground level concentrations for the pollutants are inversely proportional to the wind speed in the down wind direction, while in upwind direction no effect will be observed and in cross wind direction partial effect due to the emission sources is observed.

The wind flow pattern in the project area is governed by continental location of project area, rain shadow region of western Ghat thus a relatively drier monsoon winds from SW monsoon. However, weaker depression from Bay of Bengal also influences wind direction. During October and November, wind blows from all directions. With the onset NE monsoon, the eastern winds blow

right up to the end of March followed by a full for a fortnight in April, then the waterlines commence to blow followed by SW monsoon up to the middle of September. There is a gap of 15 to 20 days before the northeastern period commence to blow, accompanied by showers. (See fig. 4.2).

**Cloud Cover:** It can be observed that clear sky will prevail from March to May and more cloud cover will prevail during the month of July and August without rainfall. Monsoon weather condition and cloud covers is guided by monsoon winds (retreating monsoon). Cloud in these period do not prolong for longer period.

Ambient air quality refers to the background air quality levels in a region, characterised by concentrations of various pollutants in the atmosphere. The presence of air pollutants and their concentrations depends on the type of polluting sources, and other factors that influence their flow and dispersion. In most cases vehicular emissions are the predominant source of air pollution. Existing ambient air quality data on various sections of the project corridors was collected to establish a baseline database. The aim was to identify areas that already have high pollution levels or are expected to experience so, on account of the road project, and to design adequate mitigation measures, as applicable.

The activities, which generate modify atmospheric air quality, are transportation (i.e., motor vehicle emissions, which are addressed in this study); industry; domestic and construction. The principal sources of air pollution due to road projects are hot mix plants and machineries used during construction phase and the vehicles that ply over it during the operation phase. The major pollutants of significance to roadside air quality, on account of vehicular emissions, are Suspended Particulate Matter (SPM), Respirable Particulate Matter (RPM), Sulphur Dioxide (SO<sub>2</sub>), Nitrogen Oxides (NO<sub>x</sub>), Hydrocarbons (HC), Carbon-Monoxide (CO).

Dispersal of pollutants depends upon factors like prevailing wind direction and other weather conditions, height of the source, and characteristics of roadside plantation and presence of other sinks along the project corridor.

**Monitoring locations and Results:** Factors that modify generation of these pollutants are traffic characteristics (volume, speed, composition and mode), vehicle maintenance levels, road geometry and areas of congestion. To assess the ambient air quality along the project corridors, ambient air quality monitoring was carried out at identified locations. The monitoring locations were selected based on the sensitivity of the receptors to vehicular traffic and to obtain baseline concentrations of the various representative land uses along the corridors. Locations and results of the monitoring are presented in **Table 4.2**. 24 hourly continuous air quality monitoring was carried out for pollutants.

**Table 4.2: Air Quality Monitoring Results**

| Sl. No.                                              |                 |                                                   | 1              | 2                   | 3              | 4            | 5               | 6              |
|------------------------------------------------------|-----------------|---------------------------------------------------|----------------|---------------------|----------------|--------------|-----------------|----------------|
| Name of location                                     |                 | National Ambient Air Quality Standards (24 Hours) | Kakinada       | Venkatakrishnapuram | Samarlakota    | Medapeta     | Balabhadrapuram | Anaparthi      |
| Chainage                                             |                 |                                                   | Km 1/250       | Km 8/550            | Km 15/000      | km 22/300    | Km 34/600       | Km 42/000      |
| Land use                                             |                 |                                                   | Junction point | Built Up            | Junction point | Built Up     | Built Up        | Junction point |
| Location code                                        |                 |                                                   | AQ-1           | AQ-2                | AQ-3           | AQ-4         | AQ-5            | AQ-6           |
| Date of monitoring                                   |                 |                                                   | 12.01.2008     | 12.01.2008          | 13.01.2008     | 13.01.2008   | 14.01.2008      | 14.01.2008     |
| Pollutants concentration in $\mu\text{g}/\text{m}^3$ | RPM             | 100                                               | <b>132.5</b>   | <b>130.4</b>        | <b>133.2</b>   | <b>120.0</b> | <b>138.6</b>    | <b>123.5</b>   |
|                                                      | SPM             | 200                                               | <b>356.2</b>   | <b>327.5</b>        | <b>365.3</b>   | <b>321.3</b> | <b>225.5</b>    | <b>372.41</b>  |
|                                                      | SO <sub>2</sub> | 80                                                | 21             | 18                  | 29             | 21           | 22              | 24             |
|                                                      | NO <sub>2</sub> | 80                                                | 31.2           | 28.3                | 36.55          | 31.90        | 38.44           | 32.56          |
|                                                      | Pb              | 1                                                 | 0.68           | 0.61                | 0.66           | 0.54         | 0.51            | 0.54           |
| mg/m <sup>3</sup>                                    | CO              | 2                                                 | 1.5            | 1.2                 | 1.6            | 1.7          | 1.2             | 1.7            |
| PPM                                                  | HC              | Not Established                                   | 49.2           | 39.8                | 53.4           | 48.5         | 43.5            | 51.2           |

(Source: Field investigation by consultant's EA team-Sep 2008)



From the above table it may be concluded that the RPM and SPM are above the limit set by National Ambient Air Quality Standards, however all other parameters are well within permissible limit.

#### 4.3 LAND ENVIRONMENT- BASELINE

The land environment describes the baseline aspects of the nature and geomorphic features, soil conditions and quality, borrow and material resources and land use Characteristics. Portion of this section has analyzed role of these landscape features in road improvement.

**Geo-morphological aspects :** The landscape falls in the contour of 150-600 meter of peninsular plateau. The project area is an old pen plain, made of deeply dissected plateau of archaean gneiss and granite for most of the part with hillocks of small relief's. Weathering of rocks has been resulted into senile topography with dissected plateau and remnant of hillocks. The project road through out pass at the foothills section in rolling terrain.

**Geology:** The project region (PIA of APRSP) contains a wide variety of geological formation ranging from oldest Dharwar Schist to the recent alluvium of coastal colluviums. The older rocks have been subjected to erosion, which resulted in different type of topography in the project area. Thus these geological formations have undergone process of evolution and are evident from its footprints in varied topography in the region.

**Table 4.3: Geological Formations of Project Influence Area**

| Formations            | :: | Age                                                        |
|-----------------------|----|------------------------------------------------------------|
| The Dharwars          | :: | Lower Pre-cambrian-around 2500 million years.              |
| The Archaean Gneisses | :: | Lower Pre-cambrian-more than 500 million years.            |
| The Charnockites      | :: | Pre-cambrian-mote than 500 million years.                  |
| The Khondalites       | :: | Pre-cambrian-more than 500 million years.                  |
| The cuddapahs         | :: | Upper Pre-Cambrian-more than 500 million years.            |
| The Kurnools          | :: | Lower Paleozoic-500 to 300 million years.                  |
| The Lower Gondwanas   | :: | Upper Carboniferous-250 to 190 millions years.             |
| The Upper Gondwanas   | :: | Middle Triassic-Lower Cretaceous-170 to 100 million years. |
| The Deccan Traps      | :: | Mesozoic-Lower Tertiary-150 to 50 million years.           |
| Tertiary Formations   | :: | Miocene-Pliocene-35 to 15 million years.                   |
| Laterites             | :: | Pleistocene-1 million to 25000 years.                      |
| Alluvium              | :: | Recent-25000 years to the present.                         |

As stated earlier, the partly metamorphosed archaeans or peninsular gneisses dominate the rock formation in the project area (proposed road). Some of the topography is of sedimentary formation also. Most of the length of the road is passing through rolling terrain. **Borrow Areas:** All along the project road borrow / material sources have been identified. There are Five borrow areas identified along the proposed project road. The samples were collected from these sources to carry out the necessary tests to study the adherence to the requirements of the MORTH specifications and to establish the suitability and availability of borrow areas for embankment, sub grade and shoulder construction. In addition to these details, the quantum of the materials available and the lead distances from the project road were also collected which would be useful while arriving at the detailed rate analysis.

**Table 4.4: Locations of the Soil Sample Analysis**

| S. N. | Location Code | Description of the Monitoring location | Chainage  | Type of Land |
|-------|---------------|----------------------------------------|-----------|--------------|
| 1.    | S1            | Venkatakrishnapuram                    | Km 8/550  | Open land    |
| 2.    | S2            | Vetlapalem                             | Km 17/800 | Open land    |
| 3.    | S3            | Bikkavolu                              | Km 34/150 | Open land    |
| 4.    | S4            | Dwarapudi                              | Km 44/800 | Open land    |
| 5.    | S5            | Vemagiri                               | Km 59/400 | Open land    |

(Source: Field investigation by consultant's EA team-Sep 2008)

| S. N. | Location Code | Description of the Monitoring location | Chainage  | Type of Land |
|-------|---------------|----------------------------------------|-----------|--------------|
| 3.    | S3            | Bikkavolu                              | Km 34/150 | Open land    |
| 4.    | S4            | Dwarapudi                              | Km 44/800 | Open land    |
| 5.    | S5            | Vemagiri                               | Km 59/400 | Open land    |

(Source: Field investigation by consultant's EA team-Sep 2008)

The proposed project may cause soil erosion and variant compaction of soil (due to movement of heavy equipment) during the construction phase. A management plan to minimize the impact and mitigate the problems will be required.

**Table 4.5: Available Nutrients in the Soil Samples**

| S. No | Parameter              | SQ-1        | SQ-2  | SQ-3  | SQ-4  | SQ-5  |
|-------|------------------------|-------------|-------|-------|-------|-------|
| 1     | Colour                 | Light Brown | Red   | Brown | Red   | Red   |
| 2     | Textural Class         | SCL         | SL    | SCL   | SL    | SL    |
| 3     | Sand%                  | 55          | 48    | 58    | 54    | 52    |
| 4     | Silt%                  | 29          | 32    | 26    | 25    | 29    |
| 5     | clay%                  | 18          | 20    | 17    | 28    | 19    |
| 6     | pH                     | 6.9         | 7.5   | 7.8   | 7.5   | 7.8   |
| 7     | Alkalinity meq/100gm   | 0.25        | 0.82  | 1.4   | 0.5   | 1.1   |
| 8     | Nitrogen (kg/ha)       | 234.5       | 255.1 | 200.6 | 225.8 | 264.3 |
| 8     | Phosphorus (kg/ha)     | 47.5        | 19.6  | 28.4  | 52.3  | 41.2  |
| 9     | Potassium (kg/ha)      | 118.6       | 148.5 | 190.5 | 110.5 | 155.5 |
| 10    | Total organic carbon % | 0.34        | 0.35  | .28   | 0.32  | .33   |
| 11    | Lead (mg/kg)           | ND          | 0.001 | ND    | ND    | ND    |
| 12    | Arsenic (mg/kg)        | ND          | ND    | ND    | ND    | ND    |
| 13    | Oil & Grease (mg/kg)   | 0.05        | ND    | 0.02  | 0.07  | 0.03  |

(Source: Field investigation by consultant's EA team-Sep 2008)

#### 4.4 WATER ENVIRONMENT- BASELINE

As mentioned in earlier section also, the project area is part of relatively drier region of the State. Many groundwater sources such as wells (bore well and open well), and hand pumps are located on both sides of the proposed road. The proposed project may contaminate the surface waters during the construction, operation as well as maintenance phases. A plan for monitoring and mitigation will, therefore, be required to avoid the pollution or deterioration of the water sources.

Six locations were selected along the proposed project length for monitoring of existing water quality in the area. One surface water body and five are ground water sources. The details of the monitoring points are given in **Table 4.6**.

**Table 4.6: Water Quality Monitoring Locations**

| Location Code | Location             | Chainage  | Name of Water body    | Type of Water Body |
|---------------|----------------------|-----------|-----------------------|--------------------|
| <b>GW1</b>    | Kakinada             | Km 1/250  | Ground Water          | Hand Pump          |
| <b>GW2</b>    | Venkatakrishtnapuram | Km 8/550  | Ground Water          | Hand Pump          |
| <b>SW 1</b>   | Samalakota Canal     | Km 15/000 | Surface Water         | Canal              |
| <b>GW3</b>    | Medapeta             | km 22/300 | Ground Water          | Hand Pump          |
| <b>SW 2</b>   | Dwarapudi            | Km 51/500 | Drinking Water source | Water Tank         |
| <b>GW4</b>    | Kadiyam              | Km 57/750 | Ground Water          | Hand Pump          |

(Source: Field investigation by consultant's EA team-Sep 2008)

#### Water Quality Standard

The canal, nallahs and ponds located along the proposed project road are a source of water for the local inhabitants, for domestic as well as agricultural uses. An analysis of their present status and the quality of the water has been done. They have been classified as per the CPCB classification norms for best use as shown in **Table 4.7**.

**Table 4.7: CPCB Best Use Classification for Surface Water Bodies**

| SNo. | Parameters                                       | Class A | Class B | Class C | Class D | Class E |
|------|--------------------------------------------------|---------|---------|---------|---------|---------|
| 1    | pH                                               | 6.5–8.5 | 6.5–8.5 | 6.0-9.0 | 6.5–8.5 | 6.5–8.5 |
| 2    | Dissolved oxygen (as O <sub>2</sub> ), mg/l, min | 6       | 5       | 4       | 4       | -       |
| 3    | BOD, 5 days at 20° C, max                        | 2       | 3       | 3       | -       | -       |
| 4    | Total coliform organism, MPN/100 ml, max         | 50      | 500     | 5000    | -       | -       |
| 5    | Free ammonia (as N), mg/l, max                   | -       | -       | -       | 1.2     | -       |
| 6    | Electrical conductivity, $\mu$ mhos/cm, max      | -       | -       | -       | -       | 2250    |
| 7    | Sodium absorption ratio, max.                    | -       | -       | -       | -       | 26      |
| 8    | Boron (as B), mg/l, max.                         | -       | -       | -       | -       | 2       |

Class A: Drinking water source without conventional treatment but after disinfection

Class B: Outdoor bathing (organized)

Class C: Drinking water source after conventional treatment and after disinfections

Class D: Propagation of Wild life and Fisheries

Class E: Irrigation, Industrial Cooling, and Controlled Waste Disposal

Below E: Not meeting A, B, C, D & E Criteria

#### **Water Quality Variation along the Project Road**

**Water Quality:** In general, water quality along the Corridors is good and complies with CPCB Surface Water Quality norms. Ground water is a major source of domestic as well as agricultural water supply in the area. Details of water quality features are mentioned in table below.

**Table 4.8: Monitoring Results for selected Ground Water Sources**

| S. No. | Location                                   | Hand Pump (GW1) | Hand Pump (GW2) | Samalkota Canal (SW1) | Hand Pump (GW3) | Water Tank (SW2) | Hand Pump (GW4) |
|--------|--------------------------------------------|-----------------|-----------------|-----------------------|-----------------|------------------|-----------------|
|        | Parameter                                  | Concentration   |                 |                       |                 |                  |                 |
| 1      | Temperature °C                             | 21              | 19.5            | 20                    | 21              | -                | 19.5            |
| 2      | colour                                     | Colourless      | colourless      | colourless            | Colourless      | Hazel            | colourless      |
| 3      | Taste                                      | Agreeable       | Agreeable       | Agreeable             | Agreeable       | Agreeable        | Agreeable       |
| 4      | Odour                                      | Odourless       | Odourless       | Odourless             | Odourless       | Agreeable        | Odourless       |
| 5      | Turbidity NTU                              | 0.4             | 1.1             | 0.5                   | 0.6             | 0.5              | 1.1             |
| 6      | Conductivity ( $\mu$ mhos/cm)              | 399             | 465             | 425                   | 398             | 432              | 465             |
| 7      | pH                                         | 7.3             | 6.8             | 7.2                   | 7.3             | 6.8              | 6.8             |
| 8      | Total Hardness as CaCO <sub>3</sub> (mg/l) | 100             | 128             | 150                   | 102             | 140              | 128             |
| 9      | Total Alkalinity                           | 37              | 45              | 50                    | 41              | 50               | 37              |
| 10     | Nitrate as NO <sub>3</sub> (mg/l)          | 0.36            | 0.62            | 0.72                  | 0.46            | 0.72             | 0.36            |
| 11     | Chloride as Cl (mg/l)                      | 27.5            | 28.6            | 20.2                  | 28.56           | 31.5             | 27.5            |
| 12     | Iron as Fe (mg/l)                          | 0.026           | 0.025           | 0.030                 | 0.021           | 0.013            | 0.013           |
| 13     | Fluoride as F (mg/l)                       | 0.38            | 0.48            | 0.55                  | 0.44            | 0.63             | 0.63            |
| 14     | Total Coliform MPN/100ml                   | <3              | Absent          | Absent                | Absent          | <3               | <3              |

(Source: Field investigation by consultant's EA team-Sep 2008)

#### 4.5 AMBIENT NOISE - BASELINE

Road construction results in increase in noise levels due to movement and operation of machinery, heavy vehicles, loading and unloading of construction materials, apart from high noise levels at the asphalt plants (90 - 100 dB(A)). These activities are intermittent and localised. During the operation phase, noise is generated from vehicle movement in three ways, namely from the vehicle body parts, from the tyre-roadway system (also known as the rolling noise) and from the driver behaviour, such as use of horns.

Noise from the vehicle body parts includes engine, inlet, exhaust, transmission, suspension, gearbox, cooling fan, during acceleration and chassis, etc. Vehicle condition is very important to this noise source. The rolling noise/frictional noise (noise from the tyre-roadway system) includes aerodynamic noise, noise from tyre-road interaction, brakes, etc. The noise level depends upon on the type and condition of tyres and pavement. At higher speed, these types of noise increase at same rate. At lower speeds in urban areas, where lower gears are used, noise from the vehicle body parts tends to be independent of vehicle speed whereas noise from the tyre-roadway system becomes less important. Driver behaviour contributes to road noise by using vehicle's horns, sudden breaking on vehicle speed, depending on the road surface and whether the surface is wet or dry.

##### Factors and parameters

The noise from a traffic stream is not constant but varies from moment to moment and it is necessary to use an index to arrive at a single figure estimate of the overall noise level for assessment purposes. Variation in traffic volume, vehicle composition and surrounding commercial and industrial activities are the dominating factors that influence the propagation of noise. Energy equivalent noise level  $L_{eq}$  (in dB(A)) was the primary parameter selected.

##### Ambient Noise Standards

Ambient noise standards were established as per the CPCB/MoEF Gazette Notification dated 26<sup>th</sup> December 1989. It is based on the 'A' weighted equivalent noise level,  $L_{eq}$ .

**Table 4.9: National Ambient Noise Standards**

| Area Code | Category of Zones | Day* limits of $L_{eq}$<br>(dB(A)) | Night* Limits of $L_{eq}$<br>(dB(A)) |
|-----------|-------------------|------------------------------------|--------------------------------------|
| A         | Industrial        | 75                                 | 70                                   |
| B         | Commercial        | 65                                 | 55                                   |
| C         | Residential       | 55                                 | 45                                   |
| D         | Silence Zone **   | 50                                 | 40                                   |

*Source: GoI, CPCB, 1989*

##### Monitoring locations & results

The noise measurement has been carried out continuously for a period of 24 hours. At each site 2880 data were recorded, each taken at an interval of 30 seconds of which 1800 data recorded at day time (07:00 AM to 10:00 PM) and 1080 at night time (10:00 PM – 07:00 AM). (Table 4.10)

**Table 4.10: Equivalent Noise Level Data**

| Sl. No. | Location code | Location (Junction Points) | Chainage  | Leq value dB (A) |            |
|---------|---------------|----------------------------|-----------|------------------|------------|
|         |               |                            |           | Day time         | Night time |
| 1.      | NQ - 1        | Kakinada                   | Km 1/250  | 65               | 55         |
| 2.      | NQ - 2        | Venkatakrishnapuram        | Km 8/550  | 56               | 50         |
| 3.      | NQ - 3        | Samarlakota                | Km 15/000 | 72               | 60         |
| 4.      | NQ - 4        | Medapeta                   | km 22/300 | 68               | 58         |
| 5.      | NQ - 5        | Bikkavolu                  | Km 34/050 | 64               | 52         |
| 6.      | NQ - 6        | Anaparthi                  | Km 42/000 | 70               | 58         |
| 7.      | NQ - 7        | Dwarapudi                  | Km 51/500 | 74               | 61         |
| 8.      | NQ - 8        | Kadiyam                    | Km 57/750 | 54               | 45         |
| 9.      | NQ - 9        | Vemagiri                   | Km 59/400 | 65               | 52         |

(Source: Field investigation by consultant's EA team-Sep 2008)

While comparing the MoEF Ambient Noise Norms for different categories, Leq noise levels at all locations during day time & during night time, were found to be exceeding their respective limits. This is because of regular movement of diverse vehicles.

It may be noted that as the noise survey was carried out at sensitive receptors i.e. School Zone and Hospital Zone respectively, noise standards is found above the limits at almost all the location.

#### 4.6 FLORA AND FAUNA - BASELINE

Andhra Pradesh contains a great diversity of natural ecosystems ranging from desert, semi-arid lands, mangroves and coral reefs in the west, to dry deciduous forests in the central and southern zone and moist deciduous and evergreen forests in the north and coastal areas. Irrespective of such diversity, forest cover in Andhra Pradesh(23%) is not upto ecological standard (33%). To stop rapid depletion of the residual forests, the GoAP has initiated a number of afforestation and conservation programmes in recent years including community based forest management (World Bank funded).

##### 4.6.1 Forest Area

**No forest with in corridor of impact and No National Park or Wild life Sanctuary within the project influence area, that is with in 10km from the project road.**

##### 4.6.2 Flora

The tree cover along the project area is dry deciduous interspersed with moist deciduous type near the water source or near the villages, they normally yield species belonging to the species Neem, Bagi, Giriship, Tamrind, Kanuga. A detailed Tree enumeration was carried out within the Corridor of Impact and is submitted to Forest department for clearance/tree felling permission.

The roadside trees are continuous throughout the corridor except some stretches where agricultural land or undulating topography are observed. Environmental impact survey establishes that average distance of trees from the centerline of road is 8m-10m on each side. The trees are generally having good canopy during environmental impact survey such canopy building stretches were identified. Many of these stretches are even very close to present carriageways. Every effort has been made by engineering and environmental team to save these trees.

**Timber resource along the road:** Generally strip plantations have limited value and uses to local communities. In the entire project corridors trees have not been planted in succession or as per requisite planning. Species selected for plantation is random often do not have timber value. The different species of trees that are presented above however no endangered floral species have been identified along the corridor of Impact.

##### 4.6.3 Fauna

Domestic animals, common Crows, Eagles, and Myna have been spotted during the reconnaissance visit along the corridor of impact. However NO endangered wild species have been noticed. No wildlife sanctuaries and wild life crossings exists in the Project Influence Area.

##### 4.6.4 Cultural Properties Resources

There are important community structures along the project road, which has cultural and sociological importance. As presented in table 4.11 these CPRs are mostly very near to proposed CoI thus having direct impact at many locations.

**Table 4.11: Cultural Property Resources in Kakinada - Rajahmundry Road**

| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) |
|---------|-------------------------|----------------|-------------------------|----------------------------|
| 1       | Venkatesh Temple        | 1+800          | Right                   | 4.5                        |
| 2       | Church                  | 2+000          | Right                   | 11                         |
| 3       | Temple                  | 3+000          | Left                    | 9.5                        |

| Sl. No. | Name of Common Property         | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) |
|---------|---------------------------------|----------------|-------------------------|----------------------------|
| 4       | Church                          | 3+300          | Right                   | 7                          |
| 5       | Temple                          | 4+100          | Right                   | 4.5                        |
| 6       | Temple                          | 4+825          | Right                   | 10.5                       |
| 7       | Temple/Pipaltree                | 5+525          | Right                   | 5.5                        |
| 8       | Temple                          | 8+700          | Right                   | 5.5                        |
| 9       | Church                          | 9+175          | Right                   | 13.5                       |
| 10      | Temple(Banyan&Neem)             | 10+575         | Right                   | 4.5                        |
| 11      | Temple                          | 16+000         | Left                    | 6.5                        |
| 12      | Temple                          | 16+150         | Right                   | 9.5                        |
| 13      | Temple (BW)                     | 16+200         | Left                    | 5                          |
| 14      | Temple                          | 16+250         | Left                    | 6.5                        |
| 15      | Temple                          | 16+450         | Right                   | 4.5                        |
| 16      | Temple                          | 17+550         | Left                    | 6.5                        |
| 17      | Temple                          | 18+000         | Right                   | 4.5                        |
| 18      | GaneshaTemple                   | 18+050         | Left                    | 6.5                        |
| 19      | Temple                          | 18+600         | Right                   | 5.5                        |
| 20      | Temple                          | 22+100         | Left                    | 6.5                        |
| 21      | Temple& Lands                   | 22+650         | Left                    | 8.5                        |
| 22      | Temple                          | 22+700         | Left                    | 5.5                        |
| 23      | Temple                          | 25+200         | Left                    | 6.5                        |
| 24      | Temple                          | 25+200         | Left                    | 4.5                        |
| 25      | Temple                          | 25+200         | Right                   | 4.5                        |
| 26      | Temple                          | 25+350         | Left                    | 4                          |
| 27      | Temple                          | 28+000         | Right                   | 4                          |
| 28      | Temple                          | 30+300         | Left                    | 6.5                        |
| 29      | Temple                          | 31+800         | Left                    | 6.5                        |
| 30      | Temple/Ghat                     | 34+670         | Left                    | 5.5                        |
| 31      | Sai Temple                      | 35+100         | Right                   | 28.5                       |
| 32      | Temple                          | 35+100         | Right                   | 8.5                        |
| 33      | Temple                          | 35+100         | Right                   | 5.5                        |
| 34      | Statue Of Lion In Mouth Goddess | 35+200         | Right                   | 5.5                        |
| 35      | Temple                          | 38+100         | Left                    | 6                          |
| 36      | Temple                          | 40+050         | Right                   | 5                          |
| 37      | Temple                          | 40+150         | Right                   | 4.5                        |
| 38      | Temple                          | 40+400         | Right                   | 5                          |
| 39      | Temple                          | 40+400         | Right                   | 5                          |
| 40      | Temple                          | 41+200         | Left                    | 6.5                        |
| 41      | Temple                          | 41+350         | Right                   | 5.5                        |
| 42      | Temple                          | 41+550         | Left                    | 4                          |
| 43      | Temple                          | 41+700         | Left                    | 5.5                        |
| 44      | Church (BW)                     | 42+100         | Right                   | 8.5                        |
| 45      | Temple                          | 44+150         | Right                   | 6.5                        |
| 46      | Big Temple (BW)                 | 44+600         | Right                   | 8.5                        |
| 47      | Temple                          | 44+700         | Right                   | 8.5                        |
| 48      | Temple                          | 45+000         | Left                    | 7.5                        |

| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) |
|---------|-------------------------|----------------|-------------------------|----------------------------|
| 49      | Temple                  | 45+350         | Left                    | 10.5                       |
| 50      | Temple                  | 45+900         | Left                    | 6.5                        |
| 51      | Temple Saibaba          | 46+100         | Right                   | 5.5                        |
| 52      | Temple                  | 48+800         | Right                   | 5.5                        |
| 53      | Temple                  | 49+550         | Right                   | 4.5                        |
| 54      | Temple                  | 52+300         | Right                   | 15                         |
| 55      | Sai Temple              | 54+200         | Right                   | 10.5                       |
| 56      | Temple                  | 55+500         | Left                    | 5.5                        |
| 57      | Temple                  | 56+800         | Left                    | 5.5                        |
| 58      | Temple                  | 57+050         | Right                   |                            |
| 59      | Temple                  | 57+200         | Left                    | 6.5                        |
| 60      | Temple (BW)             | 57+300         | Left                    | 8.5                        |
| 61      | Temple                  | 57+300         | Left                    | 5                          |
| 62      | Temple                  | 58+950         | Right                   | 6.5                        |
| 63      | Ganesh Temple           | 59+550         | Right                   | 3.5                        |
| 64      | Temple                  | 60+200         | Right                   | 4.5                        |
| 65      | Mary Statue             | 60+750         | Right                   | 7                          |
| 66      | Church                  | 60+750         | Right                   | 28.5                       |
| 67      | Park & Hanuman Temple   | 61+800         | Right                   | 6.5                        |

(Source: Field investigation by consultant's EA team-Sep 2008)

## ***CHAPTER - 5: STAKEHOLDERS CONSULTATION***



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## CHAPTER – 5: STAKEHOLDERS CONSULTATION

### 5.1 INTRODUCTION

Consultative procedure since the inception of the project has been continued during pre-feasibility, feasibility, Environmental Assessment and Environmental Management Plan preparation stage considering the fact that involving local communities in the project planning is basis of the participatory planning. Because, often suggestion and option given by the people improves technical and economic efficiency of the project and suggested improvements proposals (if adopted by the project) of the people also generates sense of ownership within communities thus eases implementation process.

Following section highlights level of consultative procedure adopted at various stages, strategies to participatory and continued consultation, lessons learnt from the stakeholder's consultation in project planning.

### 5.2 CONSULTATION DURING PRE-FEASIBILITY STAGE

#### 5.2.1 Identification of Stakeholders

Identification of important Stakeholders for this project is done taking into consideration of their expected roles in the planning and implementation of the project. Primary stakeholders are the main stakeholders with whom the project will have direct interaction.

| Primary Stakeholders<br>(Main stakeholders)    | Potential PAPs, Forest and RDC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Secondary Stakeholders<br>(Other stakeholders) | Groups of affected persons;<br>Village representatives like Sarpanch and members, PRIs, Village level health workers, Patwaris<br>Tribal groups<br>Women Groups such as DWACRA groups Officials at Mandal (now Tahsil level)<br>Local voluntary organizations like CBOs and NGOs;<br>Field level R&B Engineers, PWD, Government of Andhra Pradesh,<br>AP State AIDS Control Society representatives,<br>Commissioner office of Social Welfare, Tribal Development,<br>Other project stakeholders such as official of line Department |

#### 5.2.2 Planning for Consultation

The consultation mechanism has been planned in stages at each level of project preparation. While village level and Block level consultation was planned during environmental assessment continue till the operation stage, District level consultation, key informant interview and other focused consultative procedure was planned during detailed environmental impact assessment stage. One of the features of present consultation program is to have continued involvement of local engineers of R&B and **APSHIP** in planning and preparation of environmental management framework for the implementation of project largely based on input from concern stakeholders. The consultation strategies and phases have been discussed below.

**Table 5.1: Planning and Present Status of Consultation**

| Stakeholders                                        | Level of Consultation | Start Stages                               | Future course         |
|-----------------------------------------------------|-----------------------|--------------------------------------------|-----------------------|
| PPAPs, community, Women groups, Tribal groups, NGOs | Village               | Pre-feasibility, feasibility and DPR stage | Started Contd         |
| APRDC Officials, R&B officials at Field,            | PIU and Field Offices | Pre-feasibility, feasibility and DPR stage | First round completed |
| RDO, MRO, Forest Official                           | Tahsil/Block          | Pre-feasibility, feasibility and DPR stage | Ongoing               |

| Stakeholders                                                      | Level of Consultation | Start Stages | Future course |
|-------------------------------------------------------------------|-----------------------|--------------|---------------|
| District Officials of line Department, such as DRDA, Forest, SLAO | District              | DPR stage    | Continued     |
| Pollution Control Board                                           | State                 | DPR stage    | Continued     |
| District and State Forest Officials                               | State                 | DPR stage    | Continued     |

### 5.3 CONSULTATION DURING FEASIBILITY STAGE

At the feasibility stage of the environmental assessment, consultation at individual level, in groups and focused group discussions at strategic locations such as bypasses, locations of affected community and other **eco-sensitive** receptors are conducted. The consultation mechanism devised to ensure that people are consulted; their project related decision/opinions are gathered and agreement reached on their suggestion/preferences is shared with the community. For this purpose, the consultation mechanism was initiated with the information dissemination, followed by structured consultation and FGDs.

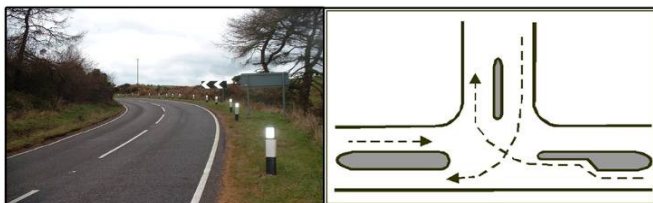
#### 5.3.1 Information Dissemination

- While conducting tree enumeration and inventory of road side other important environmental features (**hot spots**), the consultant has conducted information dissemination along the project route by one to one canvassing about proposed improvement,
- Potential project affected families were consulted to inform them about proposed road improvement program and possible environmental conflict such as tree cutting, relocation of hand pumps,
- Pictorial Methods were adopted to explain proposed improvement and possible environmental impacts in the concerned villages,
- Information dissemination through pumhlet/ *Khadapatralu* among the villagers explaining proposed road improvement (Fig 5.1)
- People are requested to gather at common places such as places of panchayat, Gramsabha and resource person of consultant explaining about the project using board and markers (**Rachabanda**).

## ఆంధ్రప్రదేశ్ రోడ్డు సెక్టార్ ప్రాజెక్ట్ (ఎ.పి.ఆర్.ఎస్.పి)

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ఆంధ్రప్రదేశ్ రోడ్డు మరియు భవనాల శాఖ వారిచే నియమించబడ్డ ఆంధ్రప్రదేశ్, రోడ్డు అభివృద్ధి సంస్థ (ఎ.పి.ఆర్.డి.సి) వరల్డ్ బ్యాంక్ సహకారంతో రాష్ట్రంలో వున్న రహదారులను సుమారుగా 900 కి.మీ వరకు ఉన్న రోడ్లను నాణ్యత కల్గిన రోడ్లుగా అభివృద్ధి పరిచే విధంగా ప్రణాళిక తయారు చేసినది.



### రోడ్డు అభివృద్ధిలో కొన్ని ముఖ్యాంశాలు

1. ఈ ప్రాజెక్టు - పోటోగ్రాఫ్లో చూపించిన విధంగా మంచి నాణ్యత గల రోడ్లుగా తయారు చేయడం.
2. రోడ్ల వెడల్పు గురించి ప్రజల అభిప్రాయాలను తెలుసుకోవడం.
3. ప్రజల అభిప్రాయాలను సేకరించుట, ఉరి పెద్దలను కలుసుకొనుట మరియు విషయాలను తెలియపరుచుట
4. రోడ్డు భద్రత ఈ ప్రాజెక్టు యొక్క ముఖ్య ఉద్దేశ్యము.
5. నవ్వు పరివారం, స్టేట్ రి సెటిల్మెంట్ మరియు రి హేబిటిషన్ పాలసీ 2005 చట్టం ఆధారంగా ఇవ్వబడును.
6. భూ ఆక్రమణలు తగ్గించడానికి అన్ని రకాల ప్రయత్నాలు తీసుకుంటాము.
7. హెచ్ ఐ వి ఎయిడ్స్ గురించి అవగాహన కల్పించటం

### రోడ్డు ప్రణాళికలో ప్రజల భాగస్వామ్యం

1. మునిసిపాలిటీ రోడ్డు వివరములు తెలియజేయడం.
2. బైపాస్ రోడ్డు స్థలం
3. ప్రమాదాలు ఎక్కువగా జరిగేటటువంటి ప్రదేశాలు
4. ప్రజల యొక్క ముఖ్య జీవనాధారాలు ఏవి
5. నవ్వుపరివారం ఏ రూపంలో ఇవ్వాలి (డబ్బు లేక స్థలానికి స్థలం)



Fig 5.1: Information dissemination through pamphlet Distribution

### 5.3.2 Structured Consultation

To provide better and qualitative planning options; consultations with specific objectives, agenda were conducted in structured manner. For this purpose, much advance date and venue of consultation were fixed in coordination with the PRIs representatives at villages level and Tahsildar, Forest officials (range official), Mandal Samakhya Group at Mandal level.

Such planned consultation has brought emergent issues of proposed road improvement in the agenda for discussion, some of the suggestion given by the community really highlighted crux of local concern of proposed improvement as well as management plan needs to adopted for there local concern. For example, the proposed Samalkota bypass bisected congested location and sharp curve at many locations. Daily newspapers also appreciated many of these consultations. Some of these newspaper cutting is attached along with the consultation findings.

One of the basic feature of the present consultation strategy to involve local R&B engineers, revenue administration in project planning simultaneously so that implementation (particularly options of regulatory clearance such as Forest, pollution board and utility shifting<sup>1</sup>) could be faster and focused.

### 5.3.3 Role and Responsibilities Identified during Consultation

Based on consultation during feasibility stage of environmental assessment a framework for LA, R&R and Environmental impact mitigation have been envisaged in the following table along with responsibilities of officials and expected benefits from the project.

**Table 5.2: Role and Responsibilities Identified after Consultation for APRSP**

| Stakeholders                                                                                                        | Roles and Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Expected Benefit for the Project                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Potential Project Affected Persons,<br>Project affected groups,<br>Project Affected Communities,<br>Host population | <ul style="list-style-type: none"> <li>Participate in formal and informal public meeting,</li> <li>Raise critical issues relevant to the environment,</li> <li>Suggest alternative alignments, environmental impact management,</li> <li>Options of widening to save trees and other environmental features,</li> <li>Methodologies for agreement on compensation and assistance,</li> <li>Suggest methodologies for continued participation in project cycle</li> </ul> | <ul style="list-style-type: none"> <li>Easing implementation.</li> <li>Incorporation of good practices (From long term memories of the people) of the past in project design.</li> <li>Planning for road safety issues.</li> <li>Community Capacity building and sense of ownership of the project,</li> </ul> |
| Engineers – APRDC and PWD, R&B                                                                                      | <ul style="list-style-type: none"> <li>Land Acquisition</li> <li>Forest Clearance</li> <li>Ensure continued consultation</li> <li>Participate in Block and District Level Meeting</li> </ul>                                                                                                                                                                                                                                                                             | <ul style="list-style-type: none"> <li>Ease implementation</li> <li>People oriented planning</li> <li>Ensured public cooperation</li> <li>Determination of value</li> </ul>                                                                                                                                    |
| Forest Official                                                                                                     | <ul style="list-style-type: none"> <li>Enumeration of trees</li> <li>Identification of eco sensitive hot spots</li> <li>Scrutiny of application form for forest clearance</li> <li>Permission for tree cutting</li> <li>Salvaging/ Auctioning of trees</li> </ul>                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>Faster Forest Clearance for the project</li> <li>Maintenance and growth of sapling at nursery</li> <li>Transplanting trees along the corridor</li> <li>Avenue Plantation</li> </ul>                                                                                     |
| GIS and Informatics Department, PWD, R&B Government of AP                                                           | <ul style="list-style-type: none"> <li>Land use classification</li> <li>Extent of Forest (Reserved and open), boundaries of sanctuaries and other eco-sensitive receptors along the project road</li> </ul>                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"> <li>To demarcate forest boundaries, major irrigation and other structures and boundaries</li> </ul>                                                                                                                                                                         |
| NGOs/CBOs<br>Veluru Mandal Samakhya Group, Water User Association<br>PRIs, Community forest management groups       | <ul style="list-style-type: none"> <li>Ensure public participation in project preparation and implementation,</li> <li>Assist APRDC and Government for dovetailing Government schemes for income generation schemes,</li> <li>Suggest RDC about afforestation schemes</li> </ul>                                                                                                                                                                                         | <ul style="list-style-type: none"> <li>Public are informed</li> <li>Opinion and preferences of people are known</li> <li>Easy to develop community capacity development plan.</li> </ul>                                                                                                                       |

## 5.4 CONSULTATION DURING DETAILED PROJECT PREPERATION STAGE

Consultation during environmental management plan preparation stage has been conducted at local level during survey stage, District level and State level. Local level consultation has emphasized individual level consultation during socio-economic baseline (25%), environmental impact and Census Survey, consultation with groups of PAFs at pre-decided dates and venue with representatives of PRIs, tribal groups, velugu groups (DWACRA groups), women groups, and social forestry organization. District Level Consultation (DLC) was focused to involving stakeholders from line department such as Forest, irrigation and revenue other than project-affected persons. These DLC have been framed within the preview of role and responsibilities identified during feasibility

<sup>1</sup> Implementation Completion report (Report No.29883-IN, 2004) observed that these activities were responsible for delay completion of APSHP.

stage as mentioned in the table above. State level consultation is conducted to have policy related decisions for environmental impact management.

#### 5.4.1 Local Level Consultation

Detailed village level consultations were conducted during environmental impact survey stage. Critical stretches along the proposed corridors were identified and meeting was held. At locations FGDs were also organized. Following table summarizes highlights of some of the consultation held along the project road.

**Table 5.3: Issues Raised and Agreement reached during consultation**

| Location                                                                                             | Major Issues/concern/suggestion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Agreement reached                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Venue:</b><br><b>ANAPARTHI</b><br><b>Dist. : East</b><br><b>Godavari</b><br><b>Date: 19/07/07</b> | <ul style="list-style-type: none"> <li>• Loss of residential, commercial and religious structures.</li> <li>• Drainage, water logging &amp; flooding problem.</li> <li>• Brief discussion of the project and the funding agency.</li> <li>• ROW Deficient stretches.</li> <li>• Land acquisition plan and the compensation to the affecting people.</li> <li>• Loss of common property resources.</li> <li>• Highway related diseases.</li> </ul>                                                                                                                    | <ul style="list-style-type: none"> <li>• People living here are very poor.</li> <li>• Issues related with dismantling of religious structures should be solved with the joint meetings of people and the executing agency.</li> <li>• Demanding for acquiring less land so that the displacement can be minimized to the extent possible.</li> <li>• Compensation at the current market rate should be given to the effected people.</li> </ul>                                                                                           | <ul style="list-style-type: none"> <li>• 5 Temples and a Church is going to affect in the village</li> <li>• Loss of business structures and business, should be compensated</li> <li>• Compensation for homestead land should be given</li> <li>• Income will be enhanced through this Project</li> <li>• Facilities for public toilets and tube wells should be provided</li> <li>• Temples should be re-constructed</li> </ul> |
| <b>Venue:</b><br><b>KADIYAM</b><br><b>Dist. : East</b><br><b>Godavari</b><br><b>Date:21/07/07</b>    | <ul style="list-style-type: none"> <li>• Brief discussion of the project and the funding agency.</li> <li>• Approximate land needed for the project</li> <li>• ROW Deficient stretches.</li> <li>• Land acquisition plan and the compensation to the affecting people.</li> <li>• Loss of common property resources.</li> <li>• Highway related diseases</li> <li>• Drinking water facility.</li> <li>• Drainage, water logging &amp; flooding problem.</li> <li>• Road accidents and safety issues.</li> <li>• Accident-prone area and dangerous curves.</li> </ul> | <ul style="list-style-type: none"> <li>• Accidents occur frequently in the village.</li> <li>• Road widening is the only way for this problem.</li> <li>• Speed breakers should be planned at regular intervals.</li> <li>• Income enhancement through road widening is a positive outcome of this Project</li> <li>• Business structures and homestead land, if affected, should be compensated for loss.</li> <li>• Temples, tube wells should be constructed</li> <li>• Demanding for saving age old temple in the village.</li> </ul> | <ul style="list-style-type: none"> <li>• People are aware of this project.</li> <li>• Speed breakers should be planned at regular intervals.</li> <li>• There are many people residing along the road side</li> <li>• Land acquisition strategy should be devised</li> </ul>                                                                                                                                                      |
| <b>Venue:</b><br><b>VEMAGIRI</b><br><b>Dist. : East</b><br><b>Godavari</b><br><b>Date: 19/07/07</b>  | <ul style="list-style-type: none"> <li>• Proposed way of the road.</li> <li>• Drinking water facility.</li> <li>• Drainage, water logging &amp; flooding problem.</li> <li>• Safety issues.</li> <li>• ROW Deficient stretches.</li> <li>• Land acquisition plan and the compensation to the affecting people.</li> <li>• Resettlement and Rehabilitation of the displaced population</li> </ul>                                                                                                                                                                     | <ul style="list-style-type: none"> <li>• Road is very narrow and accidents occur frequently.</li> <li>• Traffic is more and road widening is best solution.</li> <li>• Durga temple should not be dismantled.</li> <li>• Compensation should be given to affected people at the current market rate.</li> <li>• Speed breakers should be planned at regular intervals.</li> </ul>                                                                                                                                                         | <ul style="list-style-type: none"> <li>• People are aware of the benefits of road widening.</li> <li>• No speed breakers observed.</li> <li>• There are many people residing along the road side.</li> </ul>                                                                                                                                                                                                                      |



#### 5.4.2 District Level Consultation

The proposed environmental planning envisages District level consultation at two stages.

- (i) Consultation with the officials of Revenue, Forests and line Department
- (ii) Consultation with the PAPs, elected representatives and other stakeholders
- (i) Consultation with the District Officials:** Consultation with officials of the District administration, APSHP as well as RDC key persons<sup>2</sup> are organized. Issues likely to be discussed in the meeting will be regulatory clearances such as Permission of tree cutting, Land acquisition, issues related to 4(1) notification under LA act 1894, Utility shifting, planning for relocation site under APRRP 2005 etc.
- (ii) Consultation with Forest Officials:** Since the proposed project will require many trees to be felled, hence tree felling permission will be required from forest office, under WALTA act.
- (iii) Formal and structured Consultation with PAPs, elected representatives and District Administration:** This consultation is indicative of successfulness of project, intervention measures required for the environmental management, steps to be followed for consent award etc. Issues raised during village and Mandal level consultation will be discussed and endorsed, concerns of the people is given due attention during the meetings, major bottlenecks of the project will also be addressed to the elected representatives to resolve at the Government level.

#### 5.4.3 State Level Consultation

State level consultation was held at East Gadavari involving State Principle Secretary, Collector, Local R&B Engineers, and APRDC officials along with the consultant. The prime concern of the State level consultation was to formalize procedure and mechanism of regulatory clearances, utility shifting, land acquisition etc. various other issues are discussed regularly with state level officers.

### 5.5 SUGGESTED FRAMEWORK FOR ENVIRONMENTAL MANAGEMENT PLAN

#### 5.5.1 Issues of Tree Cutting

Kakinada – Rajahmundry Road (part of SH-78) is very typical because of irrigation canal is running parallel on left side all along the project road. At many places the clearance (open land) for widening is about 14 meters. While discussing strategies to save trees in these kind of stretches people suggested that these trees are very old and widening from one end to other end (Toe-Toe) with adequate safety measures should be taken up during design. During local level consultation, people also raised concern over one side tree felling especially where on rolling terrain and sharp curves, because it will cause flash floods and soil erosion. Therefore it was decided that tree felling would be restricted with maximum compromise (within MoRTH limits) in alignment design. And adequate plantation measures, not limiting to compulsory plantation, rather avenue plantation will be proposed.

#### 5.5.2 Compensatory Afforestation

As per provision of Andhra Pradesh Forest WALTA Act compensatory afforestation will be done in the ratio 1:2. Community during consultation has suggested to plant local species, which provide shelter to the people (Canopy building trees) and fruit bearing trees.

#### 5.5.3 Implementation Framework

It is learnt during the consultation that with key stakeholders that APSH-I the project has afforestation provision by Department of Forest through contractor involving Engineer. The provision has created total confusion and recently GoAP has decided not involving contract with the private parties for such practices. Therefore the project proposes to award afforestation work directly to Department of Forest with monitoring provision through APRDC. For this purpose Ministry of Works and Ministry of Forest will initiate MoU to activate operational mechanism of such procedure.

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<sup>2</sup> Environment officer, RDO as special LAO for APRSP, are key concern person of RDC.

**5.5.4 Monitoring Plan and Training**

Information collected during environment (air quality, water quality and other parameters) survey and consultation suggests periodic monitoring plan should be gauged by considering specific but limited number of environmental parameters. Therefore, Monitoring strategies for the project should devise a specific plan. Training and capacity building component of environmental team should be part of consolidated training program of RDC and budgeted in training and institutional component of the project.

**5.5.5 Cultural Properties Resources (CPRs) Enhancement**

Regarding cultural properties enhancement, village community is willing to come forward to cooperate with RDC in the enhancement of religious properties. This is learnt from consultation that generally CPRs do not receive due attention during construction as a result actual enhancement do not take place. Therefore project authority should plan for alternative implementation arrangement or strengthen its periodic monitoring of physical and financial progress of such enhancement.



## ***CHAPTER - 6: ANALYSIS OF ALTERNATIVES***

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## CHAPTER-6: ANALYSIS OF ALTERNATIVES

This chapter presents a comparative analysis of various alternatives considered to avoid or minimise impacts that would be inevitable if technically (based on design speed and geometrics) best-fit alignment is followed. Cross-sections adopted for the upgradation component as presented in Chapter -1 are flexible in design to avoid most of the impacts within RoW. An analysis of various alternatives is attempted to arrive at the technically and environmentally best-fit alternative.

There are large settlements as seen in the baseline environmental scenario along the corridors, where there is constricted RoW and traffic is higher leading to congestion, delay as well as various environmental impacts. Such locations are identified for bypassing the through traffic. Several alternatives are analyzed for avoiding localized environmental impacts & arriving at the best-fit alignment. Screening procedure for road links & alignments are presented in the following sections.

### 6.1 WITH OR WITHOUT PROJECT SCENARIO

The ‘with’ and ‘without’ project scenarios are analysed with respect to the development of the state by the backdrop of requirement of reliable quality infrastructure for sustained growth economy and consequent well-being of its citizens.

Providing better connectivity will ensure that goods and people from areas covered by the road can move in and out of the areas quicker and save time. Increased trade and commerce activity are expected. Accounting just for the savings in the Vehicle Operating Costs makes the project viable. The project roads have been designed to connect the important settlements with shorter & better road network. It provides greater impetus to religious and commercial establishments. The industrial activity will be getting a good boost due to this road to government plan of developing industrial in Andhra region. The industrial activity will give rise to employment potential for people in and around the region. However, there would be an increase in the vehicular pollution-air and noise, in the vicinity of the road. Some agricultural land will have to be diverted for road use to construct bypasses at Samalkota locations. In other settlements, some people will lose their properties close to the road to accommodate the proposed widening.

If the project is not implemented, there is every likelihood that the project roads will deteriorate further. Only certain roads may be maintained regularly. There is every likelihood of deterioration of the existing pavements. In the absence of the project, the highway department will also find it extremely difficult to generate funds for such a massive improvement of the road infrastructure from its own resources. Increased air pollution, due to slow moving traffic and congestion, will follow. Noise levels will rise due to deterioration of the pavement as well as increased honking. Without the bypass, the traffic would continue passing through the Samalkota town and continue to pose a safety risk for the residents in the already congested town.

Therefore, the “with” project scenario, with its minor adverse impacts is more acceptable than the “without” project scenario which would mean an aggravation of the existing problems. Potential benefits of the proposed road improvements are substantial and far-reaching both in terms of the geographical spread and time. Hence, it is clear that the implementation of the project will be a definite advantage to project road & East Godavari district in order to achieve all-round development of its economy and progress for its people.

### 6.2 ROUTE ALTERNATIVES

#### ANALYSIS OF ALTERNATIVES: KAKINADA - RAJAHMUNDRY (KR)

The entire length of the project road has very typical kind of restrictions from both sides, LHS and RHS. On LHS, all through canal runs parallel to project road and on RHS railway track runs parallel to road for around 30% length of it. Since on LHS canal is there, all built-up sections have been developed on the RHS of the Road, rather between railway track and project road.

There are four locations where project road passes through congested area:

#### 1. Samalkota (km 14/0 to km 15/0)

2. Anaparthi (km 41/0 to km 42/0)
3. Kesavaram (km 49/0 to km 50/0)
4. Kadiyam (km 56/5 to km 57/0)

At all four locations, project road is located between canal and railway track. Owing to this lateral confinement, no bypass or major realignment is proposed. It is proposed to improve the geometric, by acquiring some land from canal side, by placing a retaining wall.

#### **AREA OF SPECIAL PROBLEM: KAKINADA - RAJAHMUNDRY (KR)**

The Kakinada –Samalkota – Rajahmundry road has very high intensity of traffic, around 17,500.pcu per day. The traffic volume has already exceeded the Design Service Volume of 2-lane road and calls for immediate widening to four lane. The project road has been divided into two traffic homogenous links as given below:

1. Kakinada (km 0/0) to Samalkota km 16/00 (17499 PCU, 2007)
2. Samalkota to Rajahmundry (11995. PCU, 2007)

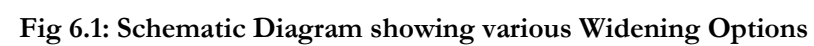
The project road is having 4-lane configuration from 0/00 to 1/600. The entire length of the project road has very typical kind of restrictions (lateral confinement) for widening from two lane to four lane, from both sides, LHS and RHS. On LHS, all through Samalkota Irrigation canal runs parallel to project road and on RHS railway track runs parallel to road for around 30% length of it. Since on LHS canal is there, all built-up sections have been developed on the RHS of the Road, or rather between railway track and project road. Shown by Schematic Diagram below. Typical Cross section is presented in Figure 4.2 A

#### **Alternative 1: Acquiring land from Canal RoW.**

Existing ROW along project road is around 14m as per R&B records. The canal on LHS is having around 40m ROW. Consultants has developed a scheme to widen the first segment, having higher traffic intensity, as shown in Figure below to widen it to four lane by acquiring 8m strip all along the canal. The transfer or acquisition of land from Irrigation department to R&B department has to be decided by Government of Andhra Pradesh. The first segment has been proposed for four lane widening as part of Phase-1 Project. Typical Cross Section is presented in Figure 4.2 B

#### **Alternative 2: Widening away from Canal, No land Acquisition from canal RoW.**

Existing ROW along project road is around 24m as per R&B records. Land acquisition in necessary to widen the project road to 4-lane configuration. Considering that no land acquisition has to be done towards canal side, consultant has made a provision of Protection works/ Retaining Wall so as to minimise land acquisition towards builtup sections. Typical Cross Section is presented in Figure 4.2C



**DETAILED INVESTIGATION - SAMALKOTA TOWN**

The entire length of Kakinada – Rajahmundry has very typical kind of lateral confinement with Samalkota Irrigation and Navigational Canal running parallel on the LHS. On the RHS, Railway line is running parallel to the project road at around four locations at very close proximity.

Samalkota is an important town well connected with road and railway line in East Godavari district along the project road. Following branches of SH, MDR and Rural roads converge/diverge at Samalkota:

**Table 6.1: Important Road links at Samalkota**

| Sl. No. | Highway Type | Description                                            |
|---------|--------------|--------------------------------------------------------|
| 1       | SH-148       | Kakinada-Rajahmundry ADB road leading to Kakinada Port |
| 2       | SH-148       | Kakinada-Rajahmundry ADB road leading to Rajahmundry   |
| 3       | SH-78        | Kakinada- Rajahmundry road leading to Kakinada City    |
| 4       | SH-78        | Kakinada- Rajahmundry road leading to Rajahmundry City |
| 5       | MDR          | Samalkota-Gokavaram road                               |
| 6       | MDR          | Samalkota-Prattipadu                                   |
| 7       | MDR          | Samalkota-Uppada                                       |
| 8       | RR           | Samalkota-Unduru                                       |

Due to the above links converging and diverging at Samalkota, the traffic in and around Samalkota is very high. Further, Chennai – Howrah railway line is crossing Samalkota and the goods transport activity at Samalkota railway junction is very high due to its close proximity to Kakinada port.

The existing alignment between km 12/4 and km 15/0 at Samalkota is very critical and upgrading of this portion along the existing alignment is almost impossible. At Samalkota, for around 2.6 km length from km 12/4 to 15/0, road has many constraints like bridge over Yeleru river, siphon structure between Yeleru river and Samalkota canal, railway bridge (ROB), level crossing and road intersections and project road is sandwiched between Samalkota canal and railway track. Table 2.9 summarises the details of critical locations along this section.

**Table 6.2: Description of Critical locations between km 12/4 and 15/0 at Samalkota**

| Location | Chainage | Description                                                                                                                                                                                                                                                                                         |
|----------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A        | Km 12/6  | Very sharp curve (90°), bridge and Aqueduct.<br>Widening at this location is not possible due to the existence of Yeleru drain and Aqueduct on LHS and Railway line on RHS.                                                                                                                         |
| B        | Km 13/0  | Major canal on LHS and Samalkota railway station on RHS (Railway line located at 3-5 m from edge of the CW). Widening on both sides not possible.                                                                                                                                                   |
| C        | Km 14/0  | The intersection leading to Peddapuram/Samalkota and ADB road is located on ROB and the ramps of this intersection are just on the edge of the carriageway of KR road. Apart from the above Sugar factory is located close to the road. Hence widening of KR road at this location is not possible. |

Due to the above constraints at Samalkota, the existing bottleneck shall become more severe as the 4 lane traffic converges to 2 lane road at location which is already very critical and congested. Hence unless the problem at Samalkota is eased out, the upgrading of project road to 4 lane at both the ends of Samalkota shall be very much underutilized.

Due to the above mentioned critical locations and high intensity of traffic along KR road, the Consultants have proposed 4 Lane bypass. The schematic of the existing alignment and the proposed bypass is presented in Fig 6.2



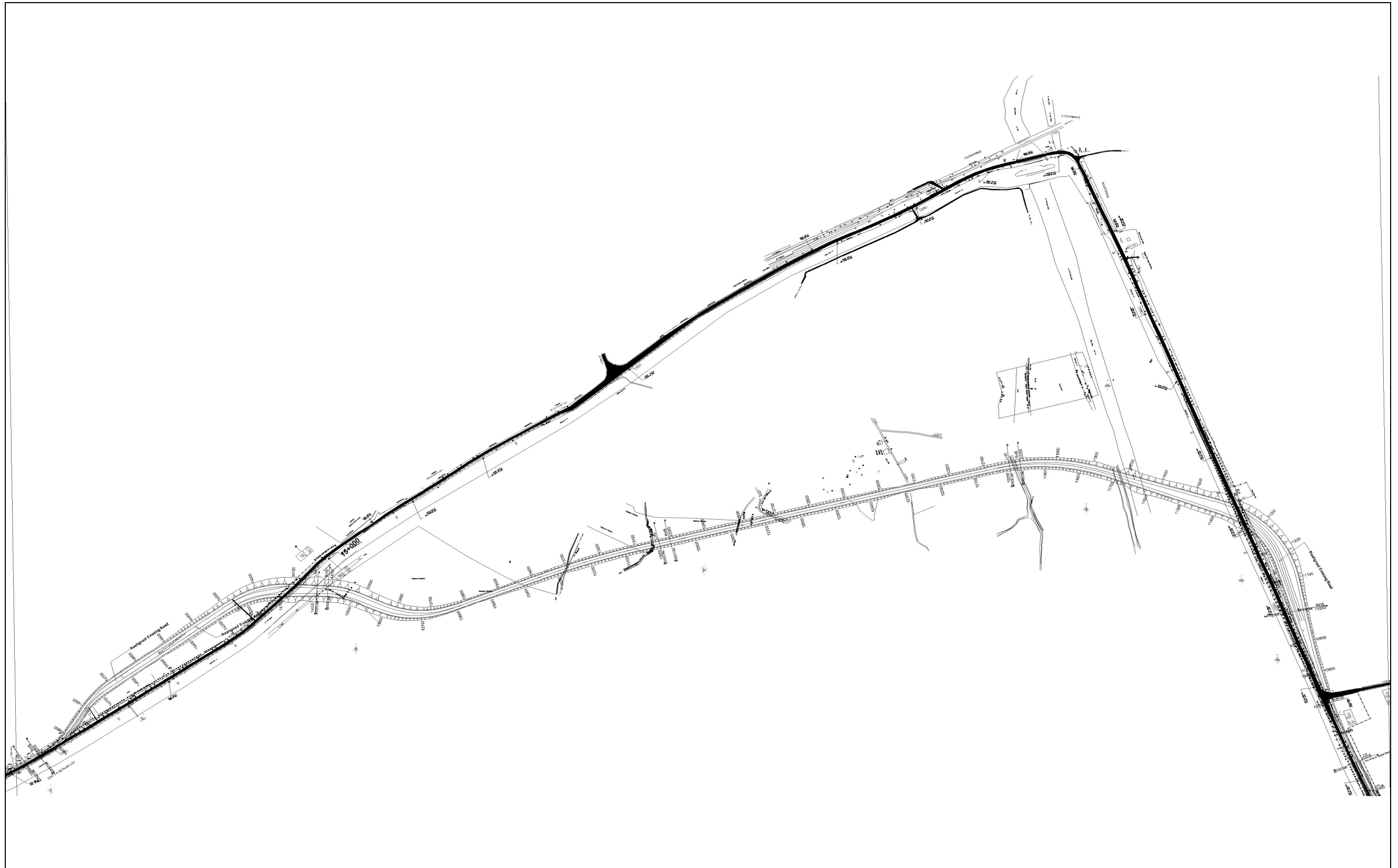


Fig 6.2: Schematic of existing alignment and bypass at Samalkota

The proposed bypass is designed as 4Lane divided carriageway and takes off with a ramp at km 11/5 and crosses the Samalkota River and travels for a total length of 3.2 km to end at km 15/5 on existing alignment. The junctions will be developed at both the ends of bypass to facilitate easy entry/exit to/from the bypass. The environmental attributes within the corridor of impact (45m wide) is presented below.

**Table 6.3 Environmental attributes along the bypass alignment**

| Sr. No. | Description         | Unit    | Quantity | Remarks                                       |
|---------|---------------------|---------|----------|-----------------------------------------------|
| 1       | Length of alignment | m       | 3100     | Shorter than existing alignment by 830m.      |
| 2       | Land Acquisition    | sqm     | 139500   | Hectare at 45m wide land width to be acquired |
| 3       | Structures          | No.s    | 0        | No residential/commercial structure exists.   |
| 4       | Canal Crossing      | Numbers | 7        | Details in <b>Table 6.2</b>                   |
| 5       | Road Crossings      | Numbers | 2        | Km 12560 & Km 13060.                          |
| 6       | No. of trees        | Number  | 66       | Details in <b>Table 6.2</b>                   |
| 7       | Electric poles      | Numbers | 1        | At Km 14.399.                                 |
| 8       | Compound wall       | Numbers | 1        | At Km 12.657.                                 |
| 9       | Field tracks        | Numbers | 4        | Details in <b>Table 6.2</b> .                 |
| 10      | Drains              | Numbers | 1        | Details in <b>Table 6.2</b> .                 |

Public consultation has been organised to know the mindset of the people about the bypass alignment. Details covered in Chapter 5: Stakeholder consultation.

**Table 6.4: Environmental Attributes presents along the Bypass alignment, Potential Impact and Mitigation Proposed**

| S.No | Chainage                                                    | Dist. From Prop. CL | Present On | Environmental Attribute | Impact        | Mitigation/ Enhancement                    |
|------|-------------------------------------------------------------|---------------------|------------|-------------------------|---------------|--------------------------------------------|
| 1    | 11600                                                       | 17.3                | LHS        | palm tree               | No Impact     |                                            |
| 2    | 11610                                                       | 11.0                | LHS        | other tree              |               |                                            |
| 3    | 11620                                                       | 21.3                | LHS        | other tree              |               |                                            |
| 4    | 11645                                                       | 19.0                | LHS        | other tree              |               |                                            |
| 5    | 11732                                                       | 17.7                | LHS        | other tree              |               |                                            |
| 6    | 11740                                                       | 15.7                | LHS        | other tree              |               |                                            |
| 7    | 11750                                                       | 10.4                | LHS        | palm tree               |               |                                            |
| 8    | 11760                                                       | 7.2                 | LHS        | palm tree               |               |                                            |
| 9    | 11768                                                       | 7.4                 | LHS        | palm tree               |               |                                            |
| 10   | 11768                                                       | 12.4                | LHS        | palm tree               |               |                                            |
| 11   | 11775                                                       | 5.9                 | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 12   | 11778                                                       | 9.9                 | LHS        | electric pole           |               |                                            |
| 13   | 11782                                                       | 8.8                 | RHS        | other tree              |               |                                            |
| 14   | 11786                                                       | 5.8                 | LHS        | other tree              | Direct Impact | Avenue plantation                          |
| 15   | 11978                                                       | 8.6                 | RHS        | neem tree               |               |                                            |
| 16   | Eluru canal crossing (At Km 12000 from R.H.S towards L.H.S) |                     |            |                         |               | Cross drainage structure will be provided. |
| 17   | Farm fencing is present from Km 12030 to 12280.             |                     |            |                         | Direct Impact | Will be relocated.                         |
| 18   | 12038                                                       | 17.0                | RHS        | palm tree               |               |                                            |
| 19   | 12038                                                       | 9.1                 | RHS        | palm tree               |               |                                            |
| 20   | 12050                                                       | 17.3                | LHS        | palm tree               |               |                                            |
| 21   | 12052                                                       | 20.0                | RHS        | mango tree              |               |                                            |
| 22   | 12058                                                       | 6.4                 | LHS        | other tree              | Direct Impact | Avenue plantation                          |
| 23   | Canal crossing (At Km 12060 from R.H.S towards L.H.S)       |                     |            |                         |               | Cross drainage structure will be provided. |
| 24   | 12068                                                       | 10.4                | LHS        | palm tree               |               |                                            |
| 25   | 12069                                                       | 5.2                 | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 26   | 12070                                                       | 2.6                 | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 27   | 12070                                                       | 2.3                 | RHS        | palm tree               | Direct Impact | Avenue plantation                          |



| S.No | Chainage | Dist. From Prop. CL                                   | Present On | Environmental Attribute | Impact        | Mitigation/ Enhancement                    |
|------|----------|-------------------------------------------------------|------------|-------------------------|---------------|--------------------------------------------|
| 29   | 12072    | 5.4                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 30   | 12134    | 10.9                                                  | RHS        | palm tree               |               |                                            |
| 31   | 12175    | 16.7                                                  | RHS        | palm tree               |               |                                            |
| 32   | 12190    | 16.4                                                  | RHS        | palm tree               |               |                                            |
| 33   | 12195    | 6.4                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 34   | 12205    | 4.8                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 35   | 12208    | 4.3                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 36   | 12212    | 3.3                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 37   | 12213    | 10.1                                                  | LHS        | palm tree               |               |                                            |
| 38   | 12213    | 13.7                                                  | LHS        | palm tree               |               |                                            |
| 39   | 12210    | 16.8                                                  | LHS        | palm tree               |               |                                            |
| 40   | 12212    | 0.1                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 41   | 12214    | 4.2                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 42   | 12212    | 7.6                                                   | RHS        | palm tree               |               |                                            |
| 43   | 12212    | 11.2                                                  | RHS        | palm tree               |               |                                            |
| 44   | 12210    | 14.5                                                  | RHS        | palm tree               |               |                                            |
|      | 12215    | 18.3                                                  | RHS        | palm tree               |               |                                            |
| 45   | 12218    | 8.4                                                   | LHS        | palm tree               |               |                                            |
| 46   | 12218    | 13.2                                                  | LHS        | palm tree               |               |                                            |
| 47   | 12218    | C/L                                                   | C/L        | palm tree               | Direct Impact | Avenue plantation                          |
| 48   | 12218    | 5.37                                                  | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 49   | 12218    | 9.8                                                   | RHS        | palm tree               |               |                                            |
| 50   | 12220    | Canal crossing (At Km 12220 from R.H.S towards L.H.S) |            |                         | Direct Impact | Cross drainage structure will be provided. |
| 51   | 12250    | Drain is present along bypass at this chainage        |            |                         | Direct Impact | Cross drainage structure will be provided. |
| 52   | 12280    | 15.3                                                  | RHS        | palm tree               |               |                                            |
| 53   | 12282    | 19.5                                                  | RHS        | palm tree               |               |                                            |
| 54   | 12285    | C/L                                                   | C/L        | palm tree               | Direct Impact | Avenue plantation                          |
| 55   | 12288    | 13.2                                                  | LHS        | palm tree               |               |                                            |
| 56   | 12282    | 12.3                                                  | LHS        | palm tree               |               |                                            |
| 57   | 12288    | 14.5                                                  | LHS        | palm tree               |               |                                            |
| 58   | 12288    | 7.4                                                   | LHS        | palm tree               |               |                                            |
| 59   | 12560    | Road crossing                                         |            |                         |               |                                            |
| 60   | 12645    | C/L                                                   | C/L        | palm tree               | Direct Impact | Avenue plantation                          |
| 61   | 12650    | Field track is present along bypass at this Chainage. |            |                         | Direct Impact | Proper connectivity will be maintained     |
| 62   | 12653    | 9.3                                                   | LHS        | electric pole           |               |                                            |
| 63   | 12657    | C/L                                                   | C/L        | compound wall           | Direct Impact | Will be reimbursed as per RAP              |
| 64   | 12672    | 14.3                                                  | RHS        | palm tree               |               |                                            |
| 65   | 12673    | 8.7                                                   | RHS        | palm tree               |               |                                            |
| 66   | 12675    | 2.9                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 67   | 12678    | 15.3                                                  | RHS        | palm tree               |               |                                            |
| 68   | 12679    | 11.1                                                  | RHS        | palm tree               |               |                                            |
| 69   | 12680    | 8.1                                                   | RHS        | palm tree               |               |                                            |
| 70   | 12682    | 19.5                                                  | RHS        | palm tree               |               |                                            |
| 71   | 12685    | 14.5                                                  | RHS        | palm tree               |               |                                            |
| 72   | 12688    | 9.8                                                   | RHS        | palm tree               |               |                                            |
| 73   | 12738    | 5.3                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 74   | 12752    | 5.2                                                   | RHS        | other tree              | Direct Impact | Avenue plantation                          |
| 75   | 12760    | 18                                                    | LHS        | coconut tree            |               |                                            |
| 76   | 12765    | C/L                                                   | C/L        | mango tree              | Direct Impact | Avenue plantation                          |
| 77   | 12775    | 14.3                                                  | RHS        | other tree              |               |                                            |
| 78   | 12792    | 3.5                                                   | RHS        | neem tree               | Direct Impact | Avenue plantation                          |
| 79   | 12795    | C/L                                                   | C/L        | neem tree               | Direct Impact | Avenue plantation                          |
| 80   | 12796    | 4.1                                                   | RHS        | other trees             | Direct Impact | Avenue plantation                          |
| 81   | 12798    | 3.4                                                   | RHS        | other trees             | Direct Impact | Avenue plantation                          |
| 82   | 12855    | 11.5                                                  | LHS        | palm tree               |               |                                            |
| 83   | 12868    | 6.6                                                   | LHS        | mango tree              | Direct Impact | Avenue plantation                          |
| 84   | 12870    | 3.2                                                   | RHS        | coconut tree            | Direct Impact | Avenue plantation                          |

| S.No | Chainage                                                                                       | Dist. From Prop. CL                                   | Present On | Environmental Attribute | Impact        | Mitigation/ Enhancement                    |
|------|------------------------------------------------------------------------------------------------|-------------------------------------------------------|------------|-------------------------|---------------|--------------------------------------------|
| 85   | 13030                                                                                          | 11.3                                                  | LHS        | palm tree               |               |                                            |
| 86   | 13035                                                                                          | 7                                                     | LHS        | palm tree               |               |                                            |
| 87   | 13040                                                                                          | 2.9                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 88   | 13040                                                                                          | 1.8                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 89   | 13045                                                                                          | 5.3                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 90   | 13049                                                                                          | 9.8                                                   | RHS        | palm tree               |               |                                            |
| 91   | 13053                                                                                          | 7.8                                                   | RHS        | other tree              |               |                                            |
| 92   | Canal crossing (At Km 13055 from R.H.S towards L.H.S)                                          |                                                       |            |                         | Direct Impact | Cross drainage structure will be provided. |
| 93   | 13060                                                                                          | Road Crossing                                         |            |                         | Direct Impact |                                            |
| 94   | 13085                                                                                          | 1.2                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 95   | 13088                                                                                          | 1.8                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 96   | 13090                                                                                          | 3.5                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 97   | 13090                                                                                          | 7.5                                                   | LHS        | palm tree               |               |                                            |
| 98   | 13091                                                                                          | 10.2                                                  | LHS        | palm tree               |               |                                            |
| 99   | 13093                                                                                          | 12                                                    | LHS        | palm tree               |               |                                            |
| 100  | 13095                                                                                          | 17.6                                                  | LHS        | palm tree               |               |                                            |
| 101  | 13097                                                                                          | 17.8                                                  | LHS        | palm tree               |               |                                            |
| 102  | 13105                                                                                          | 15.3                                                  | RHS        | coconut tree            |               |                                            |
| 103  | 13106                                                                                          | 13.5                                                  | RHS        | mango tree              |               |                                            |
| 104  | 13107                                                                                          | 13.4                                                  | RHS        | coconut tree            |               |                                            |
| 105  | 13109                                                                                          | 6.8                                                   | RHS        | mango tree              | Direct Impact | Avenue plantation                          |
| 106  | 13110                                                                                          | 8.7                                                   | RHS        | coconut tree            |               |                                            |
| 107  | 13110                                                                                          | 1.7                                                   | RHS        | mango tree              | Direct Impact | Avenue plantation                          |
| 108  | 13115                                                                                          | Field track is present along bypass at this Chainage. |            |                         | Direct Impact | Proper connectivity will be maintained.    |
| 110  | 13115                                                                                          | 1.1                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 111  | 13125                                                                                          | 1.8                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 112  | 13128                                                                                          | 3.4                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 113  | 13132                                                                                          | 5.3                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 114  | 13135                                                                                          | 7.6                                                   | RHS        | palm tree               |               |                                            |
| 115  | 13138                                                                                          | 9.9                                                   | RHS        | palm tree               |               |                                            |
| 116  | 13140                                                                                          | 12.2                                                  | RHS        | palm tree               |               |                                            |
| 117  | 13145                                                                                          | 15.3                                                  | RHS        | palm tree               |               |                                            |
| 118  | 13148                                                                                          | 19.5                                                  | RHS        | palm tree               |               |                                            |
| 119  | 13376                                                                                          | 4.9                                                   | LHS        | coconut tree            | Direct Impact | Avenue plantation                          |
| 120  | 13377                                                                                          | 3.6                                                   | LHS        | other tree              | Direct Impact | Avenue plantation                          |
| 121  | 13378                                                                                          | 2.6                                                   | LHS        | coconut tree            | Direct Impact | Avenue plantation                          |
| 122  | 13378                                                                                          | 3.2                                                   | RHS        | mango tree              | Direct Impact | Avenue plantation                          |
| 123  | 13378                                                                                          | 8.6                                                   | LHS        | mango tree              |               |                                            |
| 124  | Canal crossing (At Km 13380 from R.H.S towards L.H.S)                                          |                                                       |            |                         | Direct Impact | Cross drainage structure will be provided. |
| 125  | 13381                                                                                          | 3.0                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 126  | 13383                                                                                          | 8.0                                                   | LHS        | palm tree               |               |                                            |
| 127  | 13385                                                                                          | 1.7                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 128  | 13387                                                                                          | 3.9                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 129  | 13387                                                                                          | 5.2                                                   | LHS        | other tree              | Direct Impact | Avenue plantation                          |
| 130  | 13387                                                                                          | 9.8                                                   | LHS        | palm tree               |               |                                            |
| 131  | 13387                                                                                          | 9.7                                                   | LHS        | other tree              |               |                                            |
| 132  | 13388                                                                                          | 5.1                                                   | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 133  | 13395                                                                                          | 10.2                                                  | LHS        | palm tree               |               |                                            |
| 134  | 13395                                                                                          | 10.3                                                  | LHS        | palm tree               |               |                                            |
| 135  | 13490                                                                                          | 18.4                                                  | LHS        | palm tree               |               |                                            |
| 136  | 13400                                                                                          | 18.6                                                  | LHS        | palm tree               |               |                                            |
| 137  | 13610                                                                                          | Field track is present along bypass at this Chainage. |            |                         | Direct Impact | Proper connectivity will be maintained     |
| 138  | Canal crossing (At Km 13615 from R.H.S towards L.H.S)                                          |                                                       |            |                         | Direct Impact | Cross drainage structure will be provided. |
| 139  | Sugar plantation is found from Km 13700 to Km 14300 along the bypass alignment on either side. |                                                       |            |                         | Direct Impact |                                            |
| 140  | 14225                                                                                          | 5.5                                                   | RHS        | palm tree               | Direct Impact | Avenue plantation                          |

| S.No | Chainage                                                     | Dist. From Prop. CL                                          | Present On | Environmental Attribute | Impact        | Mitigation/ Enhancement                    |
|------|--------------------------------------------------------------|--------------------------------------------------------------|------------|-------------------------|---------------|--------------------------------------------|
| 141  | 14235                                                        | 2.2                                                          | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 142  | 14238                                                        | 13.6                                                         | LHS        | palm tree               |               |                                            |
| 143  | 14245                                                        | 15.6                                                         | LHS        | palm tree               |               |                                            |
| 144  | 14245                                                        | 18.2                                                         | LHS        | palm tree               |               |                                            |
| 145  | 14248                                                        | 18.2                                                         | LHS        | palm tree               |               |                                            |
| 146  | 14263                                                        | 15.4                                                         | LHS        | palm tree               |               |                                            |
| 147  | 14268                                                        | 12.2                                                         | LHS        | palm tree               |               |                                            |
| 148  | 14272                                                        | 8.6                                                          | LHS        | palm tree               |               |                                            |
| 149  | 14280                                                        | 2.2                                                          | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 150  | 14285                                                        | 3.6                                                          | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 151  | 14288                                                        | 2.4                                                          | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 152  | 14298                                                        | 5.2                                                          | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 153  | 14308                                                        | 0.5                                                          | RHS        | coconut tree            | Direct Impact | Avenue plantation                          |
| 154  | 14318                                                        | 6.8                                                          | LHS        | coconut tree            | Direct Impact | Avenue plantation                          |
| 155  | 14323                                                        | 1.4                                                          | RHS        | coconut tree            | Direct Impact | Avenue plantation                          |
| 156  | <b>Canal crossing (At Km 14350 from R.H.S towards L.H.S)</b> |                                                              |            |                         | Direct Impact | Cross drainage structure will be provided. |
| 157  | 14399                                                        | 6.9                                                          | RHS        | electric pole           | Direct Impact | Will be relocated.                         |
| 158  | 14414                                                        | 17.5                                                         | RHS        | palm tree               |               |                                            |
| 159  | 14418                                                        | 13.5                                                         | RHS        | palm tree               |               |                                            |
| 160  | 14425                                                        | 9.1                                                          | RHS        | palm tree               |               |                                            |
| 161  | 14430                                                        | 6.3                                                          | RHS        | palm tree               |               |                                            |
| 162  | 14430                                                        | 2.4                                                          | RHS        | other tree              | Direct Impact | Avenue plantation                          |
| 163  | 14433                                                        | 4.9                                                          | RHS        | other tree              | Direct Impact | Avenue plantation                          |
| 164  | 14437                                                        | 2.7                                                          | RHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 165  | 14442                                                        | C/L                                                          | C/L        | palm tree               | Direct Impact | Avenue plantation                          |
| 166  | 14448                                                        | 3.9                                                          | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 167  | 14455                                                        | 6.4                                                          | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 168  | 14462                                                        | 9.5                                                          | LHS        | palm tree               |               |                                            |
| 169  | 14468                                                        | 12.9                                                         | LHS        | palm tree               |               |                                            |
| 170  | 14474                                                        | 14.3                                                         | LHS        | other tree              |               |                                            |
| 171  | 14476                                                        | 14.8                                                         | LHS        | palm tree               |               |                                            |
| 172  | 14485                                                        | 17.5                                                         | LHS        | palm tree               |               |                                            |
| 173  | 14493                                                        | 19.3                                                         | LHS        | palm tree               |               |                                            |
| 174  | 14495                                                        | 20.9                                                         | LHS        | palm tree               |               |                                            |
| 175  | 14528                                                        | 10.5                                                         | LHS        | palm tree               |               |                                            |
| 176  | 14512                                                        | 16.0                                                         | LHS        | well                    |               |                                            |
| 177  | 14531                                                        | 10.8                                                         | LHS        | palm tree               |               |                                            |
| 178  | 14531                                                        | 17.3                                                         | LHS        | palm tree               |               |                                            |
| 179  | 14540                                                        | <b>Field track is present along bypass at this Chainage.</b> |            |                         | Direct Impact | Proper connectivity will be maintained     |
| 180  | 14540                                                        | 7.0                                                          | LHS        | electric pole           |               |                                            |
| 181  | 14545                                                        | 14.9                                                         | RHS        | structure               |               |                                            |
| 182  | 14550                                                        | 16.9                                                         | RHS        | structure               |               |                                            |
| 183  | 14651                                                        | 15.9                                                         | LHS        | palm tree               |               |                                            |
| 184  | 14658                                                        | 14.5                                                         | LHS        | palm tree               |               |                                            |
| 185  | 14665                                                        | 12.5                                                         | LHS        | palm tree               |               |                                            |
| 186  | 14671                                                        | 10.5                                                         | LHS        | palm tree               |               |                                            |
| 187  | 14678                                                        | 8.3                                                          | LHS        | palm tree               |               |                                            |
| 188  | 14685                                                        | 7.0                                                          | LHS        | palm tree               |               |                                            |
| 189  | 14692                                                        | 5.5                                                          | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 190  | 14698                                                        | 3.7                                                          | LHS        | palm tree               | Direct Impact | Avenue plantation                          |
| 191  | 14703                                                        | 2.7                                                          | LHS        | palm tree               | Direct Impact | Avenue plantation                          |

As the bypass alignment crosses Samalkota distributaries at 7 locations and large number of trees, utmost care has been taken in avoiding the impact on the existing Environmental attributes. The attributes under impact has been proposed for mitigation/enhancement as presented in **Table 6.5**.

**Table 6.5: Environmental Attributes along the bypass alignment.**

| SR. No. | Description         | Unit    | Quantity | Mitigation Measures and Budgeting                                                                                                                                                      |
|---------|---------------------|---------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Length of alignment | m       | 3100     | Shorter than existing alignment by 830m.                                                                                                                                               |
| 2       | Land Acquisition    | Ha      | 13.95    | <ul style="list-style-type: none"> <li>✓ Public Consultation</li> <li>✓ NGO will be hired for information dissemination</li> <li>✓ PAP will be informed 4 month in advance.</li> </ul> |
| 3       | Structures          | Numbers | 0        | Utmost care has been given in avoiding any loss to structures, as many religious structures exists along the corridor.                                                                 |
| 4       | Canal Crossing      | Numbers | 7        | Canals are crossing at 7 locations, adequate cross drainage structures has been provided for minimizing impact on existing drainage network.                                           |
| 5       | Road Crossings      | Numbers | 2        | Grade Separated crossing is provided at Km 12560 &<br>At grade crossing at km 13060.                                                                                                   |
| 6       | No. of trees        | Numbers | 66       | Avenue plantation is proposed all along the corridor.                                                                                                                                  |
| 7       | Compound wall       | Numbers | 1        | Will be reimbursed as per RAP                                                                                                                                                          |
| 8       | Field Tracks        | Numbers | 4        | ✓ Field Tracks will be relocated at desired location of the people in order to access their agricultural fields.                                                                       |
| 9       | Drains              | Numbers | 1        | Cross Drainage structure will be provided.                                                                                                                                             |
| 10      | Electric pole       | Numbers | 1        | Will be relocated as per utility shifting Plan.                                                                                                                                        |

## ***CHAPTER - 7: PROJECT IMPACTS AND ISSUES***

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## CHAPTER 7: PROJECT IMPACTS AND ISSUES

### 7.1 PROJECT IMPACTS AND ISSUES

Previous chapters of this report have dealt about-proposed improvement and existing environmental condition. Following sections summarizes impact of proposed improvement on existing environment. As mentioned in earlier section the improvement proposal envisages 2-lane roads with paved/earthen shoulder as a result direct impact zone up to towline is in the range of 12-16 meters of corridors and magnitude of indirect impact varies depending upon location of environmental receptors and type of impact. The planning of proposed project intervention point towards the impacts in the pre-construction, the construction stages and the operation stage. The subsequent sections deal with the prediction of impacts due to the project on the natural environment and socio & cultural environment.

Tables 7.1 & 7.2 below presents the general environmental impacts expected due to the proposed upgradation of the project road. Impacts have been assessed based on the information collected from the screening & scoping of environmental attributes at feasibility stage. The quanta of all the impacts on Natural Environment are discussed in details in subsequent paragraphs.

**Table 7.1 : General Impacts on Natural Environment**

| Project Activity        | Planning and Design Phase | Pre-construction Phase             |                                                                          | Construction Phase                                                                                 |                                         |                                                 |                                          |                                    | Road Operation                                                                 | Indirect effects of operation or Induced development |
|-------------------------|---------------------------|------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------|------------------------------------------|------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------|
| Env. component Affected | Land acquisition          | Removal of Structures              | Removal of trees and vegetation                                          | Earth works including quarrying                                                                    | Laying of pavement                      | Vehicle & Machine operation & maintenance       | Asphalt & crusher plants                 | Sanitation & Waste (labour campus) | Vehicle operation                                                              |                                                      |
| Air                     |                           | Dust generation during dismantling | Reduced buffering of air and noise pollution, Hotter, drier microclimate | Dust generation                                                                                    | Asphalt odour                           | Noise, dust, pollution                          | Noise, soot, odour, dust, pollution      | Odour / smoke                      | Noise, dust, pollution                                                         | other pollution                                      |
| Land                    | Loss of productive Land   | Generation of debris               | Erosion and loss of top soil                                             | Erosion and loss of top soil                                                                       |                                         | Contamination by fuel and lubricants Compaction | Contamination Compaction of soil         | Contamination from wastes          | Spill from accidents Deposition of lead                                        | Change in cropping pattern                           |
| Water                   | Loss of water sources     | Siltation due to loose earth       | Siltation due to loose earth                                             | Alteration of drainage Break in continuity of ditches Siltation, Stagnant water pools in quarries. | Reduction of ground water recharge area | Contamination by fuel and lubricants            | Contamination by asphalt leakage or fuel | Contamination from wastes Overuse  | Spill Contamination by fuel, lubricants and washing of vehicles                | Increased contamination of ground water              |
| Noise                   |                           | Noise Pollution                    | Noise Pollution due to machinery                                         | Noise Pollution                                                                                    |                                         | Noise pollution                                 | Noise Pollution                          |                                    | Noise Pollution                                                                | Noise pollution                                      |
| Flora                   |                           | Loss of Biomass                    |                                                                          | Lowered productivity Loss of ground for vegetation                                                 |                                         | Removal of vegetation                           | Lower productivity Use as fuel wood      | Felling trees for fuel             | Impact of pollution on vegetation Lowered productivity Toxicity of vegetation. |                                                      |
| Fauna                   |                           |                                    | Disturbance Habitat loss                                                 | Disturbance                                                                                        |                                         | Disturbance                                     | Disturbance                              | Poaching                           | Collision with traffic                                                         | Distorted habitat                                    |

Besides, above-mentioned impact on natural environment there will be socio-economic impacts due to disruptions on the social and economic interactions of communities. This involves effect on both the adjacent communities (mostly direct) as well as the nearby communities (mostly indirect). The various impacts have been detailed as:

- General impacts that apply to the entire Project Influence Area (PIA),



- Specific impacts on likely properties and PAPs, within the Corridor of Impact (CoI) of the Project Road.

For investigation/monitoring purpose the study area has been defined as under.

- **Corridor of Impact**, CoI: is the 20m wide strip on either side, beyond Right of Way, RoW.
- **Project Influence Area**, PIA: is the 10km area on either side along the alignment.
- **Project District**, PD: is/are the district/districts through which project road is passing.

**Table 7.2: General Impact on Social and Cultural Environment**

| Project Activity               | Planning and Design Phase                  | Pre Construction Phase |                                                                          |                                                                               | Construction Phase                              |                        |                                                                            |                                               |                                       | Operation                             |                                            |
|--------------------------------|--------------------------------------------|------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------|------------------------|----------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------|---------------------------------------|--------------------------------------------|
|                                |                                            |                        |                                                                          |                                                                               |                                                 |                        |                                                                            |                                               |                                       | Direct                                | Indirect Induced development               |
| Env. Component Affected        | Design decisions & Implementation policies | Land acquisition       | Removal of Structures                                                    | Removal of trees & vegetation                                                 | Earth works including quarrying                 | Laying of pavement     | Vehicle & machine operation & maintenance                                  | Asphalt and crusher plants                    | Labour Camps                          | Vehicle operation                     | -                                          |
| Agricultural land              | -                                          | Change in land prices  | Loss of land economic value                                              | Loss of standing crops                                                        | Loss of productive land                         | -                      | -                                                                          | Dust on agricultural land reduce productivity | -                                     | -                                     | Conversion of Agricultural Land            |
| Buildings and built structures | -                                          | -                      | Loss of structures, Debris generation, Noise and Air pollution           | -                                                                             | Noise, vibration may cause damage to structures | -                      | Noise, vibration may cause damage to structures                            | Dust accumulation on building and structure   | -                                     | Vibration and noise                   | Change in building use and characteristics |
| People and Community           | Anxiety and fear among community           | -                      | Displacement of people Psychological impact on people loss of livelihood | Loss of shade & community trees, Loss of fuel wood and fodder, Loss of income | Noise and Air pollution                         | Odour and dust         | Noise and Air pollution, Collision with pedestrians livestock and vehicles | Air and noise pollution and discomfort        | Community clashes with migrant labour | Noise pollution, Risk of accident     | Induced pollution                          |
| Cultural Assets                | -                                          | -                      | Displacement loss of structure from RoW                                  | Loss of sacred trees.                                                         | Noise, vibration may cause damage to structure  | -                      | Damage from vibration & air pollution                                      | Dust accumulation                             | -                                     | Damage from vibration & air pollution | -                                          |
| Utilities and Amenities        | -                                          | -                      | Interruption in supply                                                   | -                                                                             | -                                               | -                      | Damage to utility and amenities                                            | Dust accumulation on water bodies             | Pressure on existing amenities        | -                                     | -                                          |
| Labour's Health & Safety       | -                                          | -                      | -                                                                        | -                                                                             | Increase of stagnant water and disease          | Asphalt odour and dust | Collisions with vehicles, pedestrians & livestock                          | Impact on health due to inhale of dust        | Increase in communicable diseases     | Collisions pedestrians & livestock    | -                                          |

## 7.2 PROJECT INTERVENTION

The project intervention necessitate/demand the involvement of following resources, having major direct and indirect impact

- 🔧 Land acquisition for widening and geometric improvement.
- 🔧 Dismantling of few roadside residential and commercial structures (mostly encroachments) for proposed improvements.
- 🔧 Clearing and grubbing of vegetation & felling of existing tree for construction activity and road safety.

In addition project road upgradation will require 3M i.e. man, material and machinery for execution of construction Work.

- ✚ Construction work involves large number of manpower (both local and migrated, contractor staff).
- ✚ Construction material includes Borrow Earth, coarse and fine aggregated and water.
- ✚ Construction machinery includes Crusher Plant, BT plant, Batching Plant and Pavers, Graders, road rollers and dumpers etc.

The provisional quantification of each of the above mention resources, required for project intervention, is tabulated in table 7.3 below.

**Table 7.3: Potential Impacts and Requirement of Man, Materials & Machinery**

| Sr. No. | Description                                                              | Unit      | Quantity | Remarks                                                                   |
|---------|--------------------------------------------------------------------------|-----------|----------|---------------------------------------------------------------------------|
| 1       | Land Acquisition                                                         | Ha.       | 14.25    | Moderate Impact                                                           |
| 2       | Removal of trees                                                         | No.s      | 4162     | 596 on LHS and 3566 on RHS.                                               |
| 3       | Removal of vegetation                                                    | Ha        | 80       | Including 4.5 ha for bypass.                                              |
| 4       | <b>Man, Material and Machinery Requirement for Construction Activity</b> |           |          |                                                                           |
| 5       | Labour                                                                   | No.s      | 80       | Total No.s of man days divided by time allotted for construction activity |
| 6       | Operators and drivers                                                    | No.s      | 40       | Total Machinery working time divided by 8hours of operator working.       |
| 7       | Semi skilled labour-mate/supervisor                                      | No.s      | 15       | One mate/supervisor over 8 labours.                                       |
| 8       | Officers/incharge                                                        | No.s      | 10       |                                                                           |
| 9       | Borrow Earth                                                             | cum       | 108,000  | Requirement for embankment, subgrade shoulder etc                         |
| 10      | Fine Aggregate/Sand                                                      | cum       | 35,410   | Concrete works, screening material and in GSB                             |
| 11      | Coarse Aggregate/Metal                                                   | cum       | 20,643   | WMM, GSB, concrete and BT works                                           |
| 12      | Water                                                                    | Kilolitre | 110,000  | All construction activities and for worker use.                           |
| 13      | Crusher Plant/BT plant/Batching Plant                                    | No.s      | 1 each   | Based on the project requirement, capacity will be judged.                |
| 14      | Pavers, Grader,                                                          | No.s      | 2 each   |                                                                           |
| 15      | Dumpers                                                                  | No.s      | 30       |                                                                           |

Consultant during the design process made desperate endeavor to minimize the impact and use of resources,

### 7.2.1 Land Acquisition

The project will be required LA for upgradation of existing 2L carriageway. Besides the horizontal and vertical alignment will be improved, to standard as per IRC/MoRTH guidelines, requiring additional land width for realignment and geometric improvement. A total of **14.25 hectares** land will be acquired for project intervention however certain measures had been taken to minimize the land acquisition some of them are as follows.

- As far as possible the proposed alignment follows the existing alignment, with concentric widening.
- The land acquisition will be restricted to bare minimum required.
- Provision of protection works like retaining/toe wall is kept to confine the embankment within RoW /minimizing the width to be acquires.
- Details and location of land to be acquired is presented below

| Chainages |       | Length (m) | Proposed Improvement | Area to be acquired (sqm) |
|-----------|-------|------------|----------------------|---------------------------|
| From      | To    |            |                      |                           |
| 0.250     | 0.400 | 0.150      | Minor Realignment    | 2200                      |
| 0.400     | 0.500 | 0.100      | Minor Realignment    | 1659                      |

| Chainages        |        | Length<br>(m) | Proposed Improvement       | Area to be acquired<br>(sqm) |
|------------------|--------|---------------|----------------------------|------------------------------|
| From             | To     |               |                            |                              |
| 11.200           | 14.900 | 3.700         | <b>Samarlakota Bypass</b>  | <b>96647</b>                 |
| 15.600           | 16.000 | 0.400         | Minor Realignment          | 1670                         |
| 17.980           | 18.100 | 0.120         | Minor Realignment          | 1115                         |
| 19.700           | 19.850 | 0.150         | Minor Realignment          | 1593                         |
| 22.350           | 22.750 | 0.400         | Minor Realignment          | 1456                         |
| 24.200           | 24.320 | 0.120         | Minor Realignment          | 1270                         |
| 25.550           | 25.650 | 0.100         | Minor Realignment          | 427                          |
| 31.650           | 32.050 | 0.400         | Minor Realignment          | 8531                         |
| 34.300           | 34.600 | 0.300         | Minor Realignment          | 2948                         |
| 35.100           | 35.250 | 0.150         | Minor Realignment          | 1857                         |
| 35.500           | 35.750 | 0.250         | Minor Realignment          | 3580                         |
| 41.600           | 41.800 | 0.200         | Minor Realignment          | 1778                         |
| 42.050           | 42.200 | 0.150         | Minor Realignment          | 1055                         |
| 43.300           | 43.550 | 0.250         | Minor Realignment          | 2062                         |
| 49.900           | 50.100 | 0.200         | Minor Realignment          | 2838                         |
| 54.000           | 54.250 | 0.250         | Minor Realignment          | 3464                         |
| 57.280           | 57.520 | 0.240         | Minor Realignment          | 3130                         |
| 58.900           | 59.100 | 0.200         | Minor Realignment          | 3003                         |
| 59.350           | 59.520 | 0.170         | Minor Realignment          | 230                          |
| <b>Total (m)</b> |        | <b>8.000</b>  | Total Area in sqm          | 142513                       |
|                  |        |               | <b>Total Area in Hect.</b> | <b>14.25</b>                 |

### 7.2.2 Removal of Pavement & Road side Structures

Strategies have been adopted for the reuse of the materials from pavement scarification and removed of cross drainage structures, so that impact of disposal of these materials could be minimized, as follows:

- The excavated bituminous material from the PR will be used in paving of the gravel village roads.
- The pavement crust excavated (WBM/WMM) will be made to be reuse in the PR crust formation.
- The boulders obtained from dismantling of the masonry work will be dressed to reuse.
- Dismantled RCC structure will be crushed to adequate size to be used in village road as gravel.
- Hume pipes have limitation of its re-use because of its fixed diameter and suitability to specific conditions. Also most of the time they are not amendable.

Unlike sustainable use of pavement and sub grade materials, roadside dwelling and business unit would also be impacted and their impacted may not be transformed rather need mitigation measures. Engineering Design team in consultation with environmental and social team has minimized/ restricted land width in contiguous built-up areas. Even after such engineering efforts some of this residential and business unit required to be dismantled (partially or fully). The impact is not considered very high because these structures are in the form of katchha, kiosks or semi-permanent. (For detail analysis of impact please refers Social Assessment Report).

Strategies have been adopted for the reuse of these materials so that impact of disposal of these materials could be minimized, as follows

- Dismantled RCC structure will be crushed to adequate size to be used in village road as gravel.

- Bricks in full will be given/taken by the dwellers or will be paved in footpath and the broken/fine bricks will be used as screening material in village roads.

### 7.2.3 Removal of Trees and Vegetation

A total of 4162 (596 on LHS and 3566 on RHS) numbers of trees need to be fell, out of the total number of existing 4197 trees, however certain preventive/avoidance measures had been taken to minimize the adverse impact, as follows:

- The road alignment has been adjusted to minimize tree felling
- As part of the effort it is also considered to remove tree up to toe line only
- Alignment has been adjusted to save the Green Tunnel as well.
- In case of both side trees not having sufficient clear distance, one side trees has been saved.

Around 80.0 Ha of land area need clearing and grubbing for smooth operation of construction activity, machineries. Vegetation needs to be removed including Bypass and 2.5 ha. additional temporary LA to be done. Certain preventive measures taken includes

- Clearing and grubbing restricted to the construction land width and in no case exceeding toe line.
- Providing the embankment slopes with top soil for re-vegetation, immediately after completion of construction work in that section.

### 7.2.4 Extraction of Material for Construction Activity

Huge quantity of construction material is required for project road upgradation. The queries for (Coarse and Fine) aggregates and borrow areas for subgrade and Embankment are locally available hence NO significant direct impact is envisaged, however strict guidelines has been prepared minimize the adverse impact; some of these guidelines are as follows.

- Top soil will be excavated and preserved and reapplied for vegetation growth.
- The contractor shall prepare redevelopment plan for the quarry site and get it approved by the EMU, before the commencement of any quarry activity.
- Contractor will be responsible for adhering to the condition laid by the Director of Geology and Mining at the time of issue of quarrying permit or quarrying lease.
- The contractor shall restore all haul roads to their original state.
- Contractor will be responsible for enforcing all safety measures/guidelines detailed time to time by EMU.

### 7.2.5 Machinery and Plant for Construction Activity

Large quantum of vehicles & machinery are required for project intervention. These machineries will have its bearing on surrounding environment especially on air quality subject to emission level of machinery and Noise Environment, however strict guideline has been prepared to minimize the adverse impact, some of these guidelines are as follows:

- Construction equipment and Machineries deployed for construction will be regularly maintained and not older than 3 years.
- Vehicles /equipment will be regularly subjected for emission tests and will have valid "Pollution under Control" certificate.
- All vehicles deployed for material haulage will be spill proof.
- Water tankers with suitable sprinkling system will be deployed along the haulage road and in the work site.

### 7.2.6 Manpower for Construction Activity

Besides contractor manpower, that will be mobilized to the project site, huge number of local labour will be hired for the execution of this project. This interaction between local and migrated labour will put pressure on the existing resourced and hence adversely impact the environment. Following measures will be taken to minimize the impact.

- Contractor will be encouraged for hiring/engaging more and more local labours.
- Contractor is responsible to adhere to the safety measures indicated in the Appendix 8.6 Safety Measures during Construction activity.
- Contractor is responsible to ensure proper medical, hygiene, Sanitation and shelter facilities.
- Contractor will encourage/facilitate HIV/AIDS prevention awareness programs in construction Camp.

## 7.3 AIR ENVIRONMENT- IMPACTS

Motor vehicles have emerged as one of the major sources of air pollution especially in urban areas. Due to the proposed road improvements aimed at enhancing the efficiency of road transport system the number of vehicles on these roads will be increased over time, so impacts on ambient air environment may be significant, which needs detailed analysis.

### 7.3.1 Meteorological factors and climate

Due to the construction and operation of the project no impacts are expected to contribute to the micro-climatic and meteorological conditions of the Project affected region. The project will have a comprehensive afforestation programme and avenue plantation that includes shrub plantation in the median and RoW edge. This could provide shade and canopy to larger areas. This could reverse if any minor or negligible impacts do exists due to the widening of the project road.

### 7.3.2 Air quality - emissions

**Preconstruction Stage:** The preconstruction stage activities include site clearance, shifting of utilities, removal of trees present in the corridor of impact, transportation of man and material, construction of accommodations, construction of stock yards, installation of construction plants and construction of office buildings. Dust generation during such activities would be the predominant polluting activity during pre-construction stage and particularly so if pre-construction tasks are performed during dry weather.

The impacts due to the pre construction activity are temporary and location specific and the width of the impacts is limited. Quantification of impacts at the pre construction stage is very difficult as these are very temporary and localized.

**Construction Stage:** During the construction stage the most predominant air pollutant would be particulate matter along with various other gaseous pollutants like oxides of nitrogen, carbon monoxide, sulphur dioxide and carbon based emissions from the hot mix plants.

The air pollution during the construction phase may be considerable locally, particularly near the working zones, construction plant sites, quarries and from construction machinery and construction vehicles. The list of activities which generate air pollutants are:

- Dust generation from the construction zone during different stages of the construction such as clearing and grubbing, materials dumping, drying of materials, brushing of the surface etc.,
- Dust generation from the access roads to the soil borrow-areas, aggregate quarries construction plants and construction camp sites.
- Operation of the construction plants such as hot mix plants, Crushers, WMM plants and Concrete Batching Plants.
- Material storage, transportation and handling (loading/unloading) of different construction

materials such as sand, earth from borrow pits and aggregates.

- Asphalt odor during paving of asphalt layers.
- Odor and smoke from construction labour camp.

**Operational Stage:** During the operational stage air pollutant will be from vehicular movements on roads and dust emission from tyres.

The severity of impact of gaseous pollutants due to vehicles plying on the highway at any given time will depend upon the traffic volume, emission rates of auto exhausted pollutants and prevailing metrological condition within the project corridor. Emissions are part and parcel of overall infrastructural development process and efficiency augmentation of transport system. However, compliance with future statutory regulatory requirements with respect to emission limits, auto-technology, and vehicular fuel quality should be adequate to prevent any negative public health impacts of this project.

## 7.4 LAND ENVIRONMENT - IMPACTS

### 7.4.1 Impact on Topography

The design has not suggested any substantial change in the height of the embankments of the road except to raise in flood prone areas. These are where the bridges have to be raised. The overall topography of the area is not going to alter due to these minor changes rather there will be some positive impacts as follows:

- Raising of embankment leads to NO submergence areas, for round the year connectivity.
- More drainage structure will ease/improve the hydrology of the area, relieve the flooding situation.
- Vertical geometrics improvement leads to better sight distance, safer transportation.
- Junction design/improvement for better and efficient traffic movement.

### 7.4.2 Impact on Geology

Likely impact on the geological resources will occur from the extraction of materials (borrow of earth, granular sub base and aggregates for base courses and bridges). The boulders will be procured from the authorized suppliers and prevalent rules will be followed for borrowing of soil, sand and aggregates. Hence, the impact on general geology of the region is insignificant. At the construction sites, no blasting is proposed; therefore, there will be no added impact on the geology of the area.

### 7.4.3 Impact on Seismology

The construction and operation of the project road will not lead to any adverse impact on seismology settings of the regional environment. On the contrary, the seismic events that could occur on the region could damage the road and structures if not constructed as per the specification recommended for the seismic zone. Present upgradation will incorporate both, existing structures will be checked & complied and new structures will be designed earth quake resistant. Project road falls under **ZONE III** as per Seismic Map.



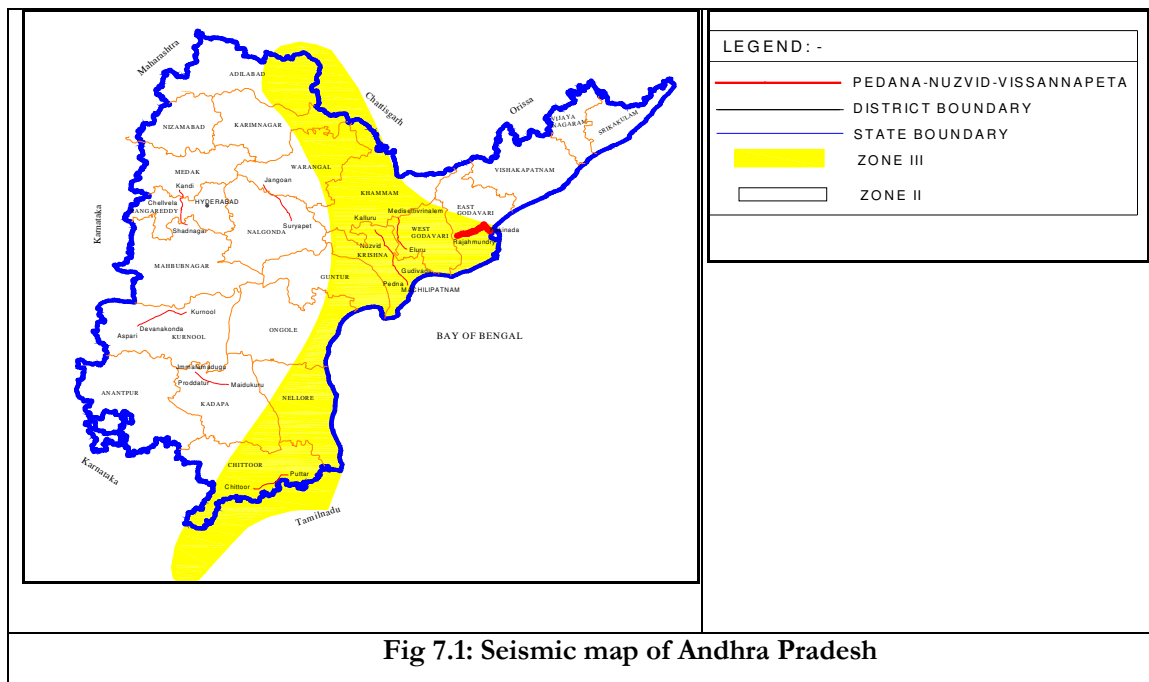


Fig 7.1: Seismic map of Andhra Pradesh

#### 7.4.4 Impact on Land

**Loss of land:** One of the major 'local' impacts due to highway project is upon the local land resources required for widening and improvement of the project road. There are stretches along the corridors where acquisition of agricultural land has been unavoidable due to non-availability of sufficient right of way to accommodate the proposed cross-sections.

**Generation of Debris:** The major source of debris generation is dismantling of existing cross drainage structures and road side residential and commercial structures.

**Soil Erosion:** Erosion of top-soil can be considered a moderate, direct and long-term negative impact resulting from the construction and maintenance of roads. The potential for soil erosion is high and pervasive during the construction stage. Starting with clearing and grubbing of trees vegetation is stripped away, exposing raw soil. The construction of new fill slopes for grading and bridge-end fills also exposes large areas to erosion, if protection methods are not implemented. Finally, during the operation or maintenance phase of highway development, erosion can continue to occur in areas not vegetated. Fills are exposed to long-term exposure to water and wind. Although soil erosion occurs sporadically on highway corridors, the sites most affected are generally bridge end fills and over-steep banks.

##### a) Road slopes and spoils

Erosion problems may occur on newly constructed slopes and fills depending on soil type, angle of slope, height of slope and climatic factors like wind (direction, speed and frequency) and rain (intensity and duration). Since slope protection methods (re-vegetation or stone pitching) form part of good engineering practice, and have been incorporated into the detailed design for the roads, erosion concerns should be minimised. However, failure to maintain soil erosion protection can reduce the security of high road embankments and add siltation to the rivers during the monsoon season.

##### b) Construction of new bridges and culverts

Along each corridor rehabilitation/reconstruction/widening of a number of major and minor bridges and culverts is planned. Construction of new bridges involves excavation of river bed and banks for the construction of the foundation and piers. If the residual spoil is not properly disposed of, increased sedimentation downstream of the bridge may take place during the monsoon. Also, the bridge-end fills require armouring to ensure gullying and slumping are minimised. For details of bridge proposed for reconstruction, refer Chapter 2: Project Description.

During the construction period some amount of drainage alteration and downstream erosion/siltation is anticipated. Some of these alterations may be because of construction of temporary traffic detours/diversion. Except for these temporary works, in almost all cases there should be an improvement in the drainage characteristics of the surrounding area due to improved design and added culvert/ditch capacity. Changes in the drainage pattern due to the raising of the road profile has not been discussed in specific cases, as the likely impact is not adverse and does not warrant mitigation (as the road design itself takes care of cross-pavement drainage). New culverts are being incorporated in the project roads not only to prevent over-topping but also to maintain equal water distribution on either side of the road. In fact, the bridges and culverts, as designed, are an automatic enhancement to the local environment (flooding, stagnation, scour, torrent run-off velocity– all would be reduced as a result of this project).

#### c) Quarries and borrow areas

The excavation of quarries and borrow pits used for obtaining soil and aggregate materials for road construction can cause direct, and indirect long-term major adverse impacts on the environment. While loss of productive soil is the most direct negative impact, other significant indirect negative impacts can also occur.

Since most of the construction materials would be available from existing quarries nearby, relatively few new borrow areas will be required.

One of the long-term residual adverse impacts of borrow pits not reclaimed is the spread of malaria. Mosquitoes breeding and multiplying in stagnant water that collects in these pits can affect humans in villages and towns close to the features.

#### 7.4.5 Contamination of Soil

**Construction Stage:** In this project contamination of the soil may take place, from the following activities at the construction zones, construction labour camps, construction plant sites and other auxiliary facilities required for the construction. Details of the activities from which the contamination can occur are presented below;

- Scarified bitumen wastes, over production of bituminous product,
- Debris generation due to dismantling of structures,
- Maintenance of the machinery and operation of the diesel generator sets on site,
- Oil Spill from the operation of the diesel pumps and diesel storage, during transportation and transfer, parking places, and diesel generator sets,
- Operation of the emulsion sprayer and laying of hot mix,
- Operation of the residential facilities for the labour and officers,
- Storage and stock yards of bitumen and emulsion,
- Excess production of hot mix and rejected materials,

**Operation Stage:** During the operation stage, soil may get contaminated with similar reasons, as mentioned above, during routine and periodical maintenance of the project road. ***The implications of accidental discharge are potentially disastrous. But, it must be emphasized that the probability of such an accident is quite low, as one of the objectives of the design is the enhancement of road safety.***

#### 7.5 WATER ENVIRONMENT –IMPACTS

Due to the proposed project there will be some direct and indirect long-term impacts on the water resources. Table below presents the major adverse impacts on the water resources and the indicators chosen to assess the impacts for the study.

| Impacts Due To Construction               | Indicators                     |
|-------------------------------------------|--------------------------------|
| Loss of water bodies                      | Area of water bodies affected  |
| Loss of other water supply sources        | Number of wells affected       |
| Alteration of drainage, run off, flooding | No. of cross drainage channels |



| Impacts Due To Construction                                                     | Indicators                                                             |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Depletion of Ground Water recharge                                              | Area rendered impervious                                               |
| Use of Water Supply for Construction                                            | Quantum of water used                                                  |
| Contamination from fuel and lubricants                                          | Nature and quantum of contaminants                                     |
| Contamination from improper sanitation and Waste Disposal in Construction Camps | Area of camp / disposal site and, proximity to water bodies / channels |

### 7.5.1 Loss of Water Bodies

The project road passes through the coastal region of Andhra Pradesh with mean annual rainfall of 1090 mm. 80% of the land is under cultivation and remaining 20% is Builtup.

#### Surface Water Bodies

Samalkota Canal is running parallel to the project road almost all along, at places the canal is very close to the project road and at Samalkota Bypass the proposed bypass crosses the Samalkota Canal 2 times. Hence due the proximity of the canal with the project road, calls for major avoidance measures to check the runoff into the canal for the degradation of the canal water. Consultant has carried out the water quality test on the existing quality of the canal water, presented in Chapter 4 baseline environment and measures are adopted for the regular checking during construction.

Provision of silt fencing, precautions during construction for spilling of waste near canal are adopted for the prevention of degradation of canal water. Temporary erosion/sedimentation and pollution control measures will be used to control the phenomenon of erosion, sedimentation and pollution that may develop during normal construction practice. Details of prevention of **SOIL EROSION AND SEDIMENTATION CONTROL** are presented in Appendix E-9. of this report.

The impact due to acquisition of the tank area or shifting of the community water tanks. Table below present the details of the existing surface water bodies and possible impacts.

**Table 7.4A: Impact on surface Water Bodies/Public Utilities**

|                   |                          |                                                                                     |                  |                                                                                       |
|-------------------|--------------------------|-------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------|
| Chainage          | 15+600                   |  | 18+950           |  |
| Structure ID      | Pond                     |                                                                                     | Pond             |                                                                                       |
| Side (Left/Right) | Right                    |                                                                                     | Right            |                                                                                       |
| Distance from ECL | 15.0                     |                                                                                     | 6.5              |                                                                                       |
| Proposed Widening | Eccentric on RHS (4.38m) |                                                                                     | Concentric       |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | No Direct Impact |                                                                                       |
| Chainage          | 24+800                   |  | 34+670           |  |
| Structure ID      | Ponds                    |                                                                                     | Pond             |                                                                                       |
| Side (Left/Right) | Right                    |                                                                                     | Right            |                                                                                       |
| Distance from ECL | 4.5                      |                                                                                     | 4.5              |                                                                                       |
| Proposed Widening | Eccentric on RHS (0.70m) |                                                                                     | Concentric       |                                                                                       |
| Impact            | Direct Impact            |                                                                                     | Direct Impact    |                                                                                       |

Out of 4 water tank locations, 2 are under direct impact and need to be partly filled/partly excavated before the construction activities starts.

#### Ground water resources

The impacts due to relocation of hand pumps, open wells, Borewells. Table below present the details of the existing ground water resources and possible impacts.

**Table 7.4B: Impacts on Ground Water Bodies**

|                   |                          |                                                                                     |                          |                                                                                       |
|-------------------|--------------------------|-------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------|
| Chainage          | 2+300                    |  | 3+000                    |  |
| Structure ID No   | Handpump                 |                                                                                     | Well                     |                                                                                       |
| Side (Left/Right) | Right                    |                                                                                     | Left                     |                                                                                       |
| Distance from ECL | 6.0                      |                                                                                     | 9.5                      |                                                                                       |
| Proposed Widening | Eccentric on RHS (3.97m) |                                                                                     | Eccentric on RHS (3.98m) |                                                                                       |
| Impact            | Direct Impact            |                                                                                     | No Direct Impact         |                                                                                       |
| Chainage          | 27+050                   |  | 39+300                   |  |

|                   |                             |                             |
|-------------------|-----------------------------|-----------------------------|
| Structure ID No   | Old Wells                   | Well                        |
| Side (Left/Right) | Right                       | Right                       |
| Distance from ECL | 25.5                        | 18.5                        |
| Proposed Widening | Eccentric on RHS<br>(3.42m) | Eccentric on RHS<br>(3.93m) |
| Impact            | No Direct Impact            | No Direct Impact            |
| Chainage          | 41+550                      | 54+200                      |
| Structure ID No   | Hand Pump                   | Hand Pump                   |
| Side (Left/Right) | Left                        | Right                       |
| Distance from ECL | 4.0                         | 7.0                         |
| Proposed Widening | Concentric                  | Concentric                  |
| Impact            | Direct Impact               | No Direct Impact            |
| Chainage          | 56+400                      | 57+100                      |
| Structure ID No   | Well                        | Well                        |
| Side (Left/Right) | Left                        | Left                        |
| Distance from ECL | 4.5                         | 13.5                        |
| Proposed Widening | Concentric                  | Concentric                  |
| Impact            | Direct Impact               | No Direct Impact            |



Out of the 8nos existing ground water sources 3nos (2 hand pumps and 1 well) are under direct impact, needs to be relocated before the start of project road.

### 7.5.2 Alteration of Cross Drainage

During the construction period some amount of drainage alteration is anticipated, due to construction of temporary traffic diversions. The need of diversion is meant for the reconstruction and new proposed bridges.

**Table 7.5: Bridges reconstruction**

| S.No | Existing Chainage | Design Chainage | Existing Span Arrangement | Proposal                                                                 | Proposed Span Arrangement |
|------|-------------------|-----------------|---------------------------|--------------------------------------------------------------------------|---------------------------|
| 1    | 1+550             | 1+515           | 5X7.8                     | Existing Bridge is retained and additional new 2 lane bridge is proposed | 5X8.00                    |
| 2    | 3+750             | 3+715           | 4X5.35                    | Reconstruction                                                           | 3X8.00                    |
| 3    | 12+100            | -               | 3X4.25                    | Not in scope,(Eliminated due to Samalkota Bypass)                        | -                         |
| 4    | 12+300            | -               | 10X6.75                   |                                                                          | -                         |
| 5    | 14+920            | -               | 3X3.60                    |                                                                          | -                         |
| 6    | 18+300            | 17+540          | 3X4.25                    | Reconstruction                                                           | 2X8.40                    |
| 7    | 20+380            | 19+565          | 3X3.60                    | Reconstruction                                                           | 2X6.00                    |
| 8    | 28+050            | 27+191          | 3X3.60                    | Reconstruction                                                           | 2X6.00                    |
| 9    | 31+050            | 30+240          | 3X3.60                    | Reconstruction                                                           | 1X10.0                    |
| 10   | 40+980            | 40+100          | 2X3.60                    | Reconstruction                                                           | 1X10.0                    |
| 11   | 43+380            | 42+532          | 3x3.00                    | Reconstruction                                                           | 1X10.4                    |
| 12   | 49+900            | 49+030          | 1X3.60                    | Reconstruction                                                           | 1X6.4                     |
| 13   | 55+700            | 54+984          | 1x6.0                     | Retained                                                                 | -                         |
| 14   | 61+400            | 60+580          | 1x10.90                   | Retained                                                                 | -                         |

| S.No | Design Chainage Left CW | Proposed Span Arrangement Left CW | Proposal         | Design Chainage Right CW | Proposed Span Arrangement Right CW | Proposal         |
|------|-------------------------|-----------------------------------|------------------|--------------------------|------------------------------------|------------------|
| 1    | 11+335                  | 1x31.5                            | New construction | 11+389                   | 2x31.5                             | New construction |
| 2    | 11+639                  | 5x21.6                            | New construction | 11+680                   | 5x21.6                             | New construction |
| 3    | 14+055                  | 1x31.5                            | New construction | 14+095                   | 1x31.5+2x35.0                      | New construction |

With these widening of minor bridges and bridging of existing causeways, there will be an improvement in the drainage characteristics of the surrounding area. And the alteration will be temporary in nature. All the diversions will be provided with adequate water way for drainage.

### 7.5.3 Run-off and Drainage

**Sediment accumulation in water bodies decreases the storage capacity for road run-off. To worsen the situation road construction activities can lead to increased run-off both, during the construction and operational stage. This can be considered a high adverse impact.**

| Sr. No. | Phase of Construction | Reason                                                                                                        |
|---------|-----------------------|---------------------------------------------------------------------------------------------------------------|
| 1       | Construction phase    | The removal of vegetation and compaction of soil can lead to increased run-off during the monsoon             |
| 2       | Operational phase     | The area of open ground lost and added impervious black top surface increases the amount and rate of run-off. |

The project involves widening of existing roads within the RoW, with addition of 2x1.5 paved shoulders to the existing pavement of 7m; provision of 2x1m earthen (98 per cent compacted) shoulders; raising the pavement and as a result widening the embankment. Overall the increase in the impervious surface would be 6m. During the operation phase increased run-off can be calculated using the formula:

*Increase in runoff per year ( $m^3$ ) = increase in runoff coefficient due to construction \* annual rainfall in the area (m) \* area of the constructed surface ( $m^2$ )*

*The appropriate run-off coefficients are: 0.95 for asphalt, 0.2 for silty and sandy soil, 0.3 for loamy soil, and 0.55 for Black cotton soil.*

Hence, for Kakinada - Rajahmundry Road

Annual average rainfall is equals to 936mm say 0.936m (statistical data book)

The increase in construction surface (BT) equals to 246,050 sqm

Increase in runoff (cum) =  $(0.95-0.2) \times 0.936 \times 246.05 \times 10^3 = 172.73 \times 10^3$  cum

Hence additional 1.73 lakh KL of additional storm water will be required to drain off.

Since soil erosion is associated with concentrated flow of water it is imperative to prevent any increased diversion of run-off into drainage channels.

### 7.5.4 Water requirement for project

The water requirement for construction depends on the climatic conditions, type of equipment, type of material available, mix design, type of construction plant and number of people working on the project. With the following assumptions the approximate water quantity required for the Project has been calculated.

- 1 8-10% of weight of soil for the embankment construction
- 2 7-8% of weight of soil for sub grade construction
- 3 5-6% of weight of GSB materials for GSB and WMM

4 150 liters/ cum for concrete

The water requirement has been assumed based on past project experiences and on the strict quality control basis. Domestic requirement of 150 liters per worker has been assumed. For this project about 100 resident workers have been considered. Details of the water requirement assessed for the project are presented in **Table 7.6**.

**Table 7.6: Requirement of Water for Proposed Construction Works**

| S. No.                            | Purpose                                          | Cum/day                         |
|-----------------------------------|--------------------------------------------------|---------------------------------|
| 1                                 | Permanent works (Total quantity in cum)          | 110,000 (2 Years) or 150 kl/day |
| 2                                 | Dust Suppression at work zone in (cum/day)       | 15                              |
| 3                                 | Curing (cum/day)                                 | 8                               |
| 4                                 | Laboratory (cum/day)                             | 5                               |
| 5                                 | Haul Roads (cum/day)                             | 15                              |
| 6                                 | Crusher (cum/day)                                | 10                              |
| 7                                 | Plant Cleaning and workshop washing in (cum/day) | 8                               |
| 8                                 | Domestic Purpose in (cum/day)                    | 15                              |
| <b>Total Requirement(cum/day)</b> |                                                  | <b>226</b>                      |

Daily water requirement for the permanent works has been calculated assuming the construction period of 2 years.

#### 7.5.5 Water Quality

##### Increased Sedimentation

Degradation of water quality due to sediment transport may occur from activities like removal of trees, removal of grass cover, excavation, stock piling of materials as part of the pre construction and construction activities. The soil type present along the project corridor consists of the sandy loam to clay loam.

The impacts due to increased sediment laden run-off will make the water more turbid. This is a significant negative impact on the water bodies supporting aquatic life. Heavier sediment may smother the algae growing in the lower strata and could completely alter the nature of the watercourse. Excessive sediment loads may also mean disruption to areas of fish breeding.

##### Contamination of Water

**Construction Stage:** The degradation of the surface and to a much less extent ground water quality can occur from pavement construction works, bridge construction works, construction plants, machinery and accommodations of workers. The sources of water pollution from the construction activities are as follows;

- Water flow from scarified bitumen materials
- From the foundation works of the bridges and culverts such as piling and excavation for open/ well foundations
- Oil spills from the Maintenance of the machinery and operation of the diesel generator sets on site.
- Oil Spill from the operation of the diesel pumps and diesel storage, transportation and transfer, parking places, and diesel generators.
- Operation of the emulsion sprayer and laying of hot mix.
- Operation of the residential facilities for the labour and officers and offices.
- Storage and stock yards of bitumen and emulsion.

Degradation of water quality is also possible due to accidental discharges into watercourses from drainage of workers camps and from spillages from vehicle parking and/or fuel and lubricant storage areas.

**Operation Stage:** During the operation stage, water may get contaminated with similar reasons, as mentioned above, during routine and periodical maintenance of the project road. *The implications of accidental discharge are potentially disastrous. But, it must be emphasized that the probability of such an accident is quite low, as one of the objectives of the design is the enhancement of road safety.*

## 7.6 NOISE ENVIRONMENT - IMPACT

Environmental noise particularly highway traffic noise, is a complex phenomenon because its intensity and characteristics vary with time depending upon the frequency as well as type of vehicles on the road.

The impacts of noise due to the project will be of temporary significance locally in the construction phase and slight increase may occur during the operation stages. Table below present the source of noise pollution and the impact categorization.

**Table 7.7: Source of noise pollution**

| Sr. No. | Phase              | Source of Noise pollution                                                                                                                                                                                                                                                                                                          | Impact categorization                                                                                                                                                                                                                                                |
|---------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Pre-construction   | <ul style="list-style-type: none"> <li>Man, material &amp; machinery movements</li> <li>establishment of labor camps onsite offices, stock yards and construction plants</li> </ul>                                                                                                                                                | <ul style="list-style-type: none"> <li>all activities will last for a short duration and also shall be localized in nature</li> </ul>                                                                                                                                |
| 2       | Construction Phase | <ul style="list-style-type: none"> <li><b>Plant Site</b> <ul style="list-style-type: none"> <li>stone crushing, asphalt production plant and batching plants, diesel generators etc</li> </ul> </li> <li><b>Work zones</b> <ul style="list-style-type: none"> <li>Community residing near to the work zones</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li><b>Plant Site:</b> Impact will be significant within 500m.</li> <li><b>Work zones:</b> Such impacts again will be of temporary nature as the construction site will go on changing with the progress of the works.</li> </ul> |
| 3       | Operation Phase    | <ul style="list-style-type: none"> <li>due to increase in traffic (due to improved facility)</li> </ul>                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>will be compensated with the uninterrupted movement of heavy and light vehicles till the facility reaches the level of service C.</li> </ul>                                                                                  |

Although the baseline day & night time noise levels monitored at various locations along the project roads are within permissible limits specified by the MoEF for residential and rural areas, still, noise is a major area of concern, especially since sensitive receptors (schools, colleges and hospitals) have been located quite close to the road. To assess the noise levels at the various sensitive receptors located along the roads, consultant carried out the noise survey through noise integrator both at inside and outside the sensitive receptor. List of the sensitive receptors located along the project road are presented in **Table 7.8**.

**Table 7.8: Sensitive Receptors with respect to Noise Pollution**

|                   |                               |                                                                                     |                         |                                                                                       |
|-------------------|-------------------------------|-------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------|
| Chainage          | 0+200                         |  | 0+600                   |  |
| Structure ID No   | PRG College (BW)              |                                                                                     | Hospital (BW)           |                                                                                       |
| Side (Left/Right) | Right                         |                                                                                     | Left                    |                                                                                       |
| Distance from ECL | 12.0                          |                                                                                     | 12.0                    |                                                                                       |
| Length x Breadth  | 50x100                        |                                                                                     | 400x200                 |                                                                                       |
| Proposed Widening | Eccentric on RHS(2.40m)       |                                                                                     | Eccentric on LHS(0.82m) |                                                                                       |
| Impact            | No Direct Impact              |                                                                                     | No Direct Impact        |                                                                                       |
| Chainage          | 0+900                         |  | 1+600                   |  |
| Structure ID No   | Rangarai Medical College (BW) |                                                                                     | Ideal College (BW)      |                                                                                       |
| Side (Left/Right) | Right                         |                                                                                     | Right                   |                                                                                       |
| Distance from ECL | 15.0                          |                                                                                     | 10.5                    |                                                                                       |
| Length x Breadth  | 200x100                       |                                                                                     | 200x400                 |                                                                                       |
| Proposed Widening | Eccentric on LHS(5.02m)       |                                                                                     | Eccentric on RHS(4.78m) |                                                                                       |
| Impact            | No Direct Impact              |                                                                                     | No Direct Impact        |                                                                                       |
| Chainage          | 5+700                         |  | 17+950                  |  |
| Structure ID No   | Saraswathi Vidya Pitam        |                                                                                     | Lyceum School           |                                                                                       |
| Side (Left/Right) | Left                          |                                                                                     | Right                   |                                                                                       |
| Distance from ECL | 23.5                          |                                                                                     | 7.0                     |                                                                                       |
| Length x Breadth  | 50x50                         |                                                                                     | 20x10                   |                                                                                       |
| Proposed Widening | Eccentric on RHS(4.46m)       |                                                                                     | Concentric              |                                                                                       |



|                          |                          |                                                                                    |                          |                                                                                      |
|--------------------------|--------------------------|------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------|
| <b>Impact</b>            | No Direct Impact         |                                                                                    | No Direct Impact         |                                                                                      |
| <b>Chainage</b>          | 30+200                   |   | 41+500                   |   |
| <b>Structure ID No</b>   | School                   |                                                                                    | Private Hospital         |                                                                                      |
| <b>Side (Left/Right)</b> | Right                    |                                                                                    | Right                    |                                                                                      |
| <b>Distance from ECL</b> | 7.5                      |                                                                                    | 7.5                      |                                                                                      |
| <b>Length x Breadth</b>  | 20x20                    |                                                                                    |                          |                                                                                      |
| <b>Proposed Widening</b> | Eccentric on RHS(8.43m)  |                                                                                    | Eccentric on RHS(0.67m)  |                                                                                      |
| <b>Impact</b>            | No Direct Impact         |                                                                                    | No Direct Impact         |                                                                                      |
| <b>Chainage</b>          | 57+300                   |   | 57+800                   |   |
| <b>Structure ID No</b>   | Veterinary Hospital (BW) |                                                                                    | Hospital (BW)            |                                                                                      |
| <b>Side (Left/Right)</b> | Right                    |                                                                                    | Left                     |                                                                                      |
| <b>Distance from ECL</b> | 4.5                      |                                                                                    | 7.5                      |                                                                                      |
| <b>Length x Breadth</b>  | 25x50                    |                                                                                    | 30x150                   |                                                                                      |
| <b>Proposed Widening</b> | Concentric               |                                                                                    | Concentric               |                                                                                      |
| <b>Impact</b>            | <b>Direct Impact</b>     |                                                                                    | No Direct Impact         |                                                                                      |
| <b>Chainage</b>          | 57+850                   |   | 60+050                   |   |
| <b>Structure ID No</b>   | School                   |                                                                                    | School (BW)              |                                                                                      |
| <b>Side (Left/Right)</b> | Left                     |                                                                                    | Right                    |                                                                                      |
| <b>Distance from ECL</b> | 23.5                     |                                                                                    | 5.0                      |                                                                                      |
| <b>Length x Breadth</b>  | 20x35                    |                                                                                    | 25x50                    |                                                                                      |
| <b>Proposed Widening</b> | Concentric               |                                                                                    | Eccentric on RHS (1.69m) |                                                                                      |
| <b>Impact</b>            | No Direct Impact         |                                                                                    | <b>Direct Impact</b>     |                                                                                      |
| <b>Chainage</b>          | 61+200                   |  | 61+350                   |  |
| <b>Structure ID No</b>   | School                   |                                                                                    | School                   |                                                                                      |
| <b>Side (Left/Right)</b> | Right                    |                                                                                    | Right                    |                                                                                      |
| <b>Distance from ECL</b> | 153.5                    |                                                                                    | 53.5                     |                                                                                      |
| <b>Length x Breadth</b>  | 20x50                    |                                                                                    |                          |                                                                                      |
| <b>Proposed Widening</b> | Eccentric on LHS (0.96m) |                                                                                    | Concentric               |                                                                                      |
| <b>Impact</b>            | No Direct Impact         |                                                                                    | No Direct Impact         |                                                                                      |

As such no direct impact is envisaged at all the 14 noise sensitive locations along the project corridor, however the boundary wall at 2 locations will be shifted for road widening. Considering their sensitivity to noise, measures have been suggested for the mitigation/enhancement of these locations in next chapter.

## 7.7 FLORA AND FAUNA - IMPACT

The major impact in this project on flora involves the removal of trees to permit construction and to provide clear zone for safety of the road users. Table below presents the major adverse impacts on the flora & fauna and the indicators chosen to assess the impacts for this study.

**Table 7.9: Impacts on flora & fauna due To Construction Activities**

| Impacts Due To Construction             | Indicators                                      |
|-----------------------------------------|-------------------------------------------------|
| Loss of forest area/wild life sanctuary | Area of forest affected                         |
| Alteration of Wild life passage         | Locations of wild life crossing of project road |
| Tree felling                            | No. of trees to be fell                         |
| vegetation                              | Area of vegetation loss                         |
| Cattle Grazing Ground                   | Area and location of grazing ground.            |

### 7.7.1 Forest Area

NO forest area is found along the project corridor.

### 7.7.2 Wild Life

NO wild life crossing is found along the project corridor.

### 7.7.3 Tree Cutting

Trees located within the toe line (bottom of formation) need to be removed for efficient construction workmanship and more importantly to prevent collision with the trees, in case of accident. Roadside trees with strong and rigid stems can pose safety hazards. Some trees obstruct clear sight distances. Others have a propensity to overturn when old and are potential safety hazards depending upon age and decay condition. All such trees that are safety hazards need to be cleared.

There will be a significant, direct impact on cutting of the roadside trees, it includes

- The loss of shade
- Loss of tree products
- Loss of birds nesting place
- Removal of roadside trees will also reduce comfort levels for slow moving traffic and pedestrians
- The removal of trees would lead to erosion and contributes to the loss of the micro-ecosystems developed on the roadside.
- Besides this trees act as noise barrier, dust absorption, air purifier etc.

A detailed tree inventory was carried out of all the existing trees within the corridor of impact, due care had been given in alignment design to reduce/minimize the loss of flora and green tunnels. As a result a total of about **4162 trees out of a total 5732 existing trees are required to be cut** for the proposed project. About 116 trees on the left side and 1965 trees on the right side are being affected due to the proposed widening. The major trees affected are Neem, Tamarind etc.

No trees will be cut beyond the proposed toe line. Cutting of trees for fuel by workers, especially near their camps is of major concern. Therefore adequate training of the workers and availability of their fuel requirements are to be ensured by the Contractor.

### 7.7.4 Removal of Vegetation

Clearing and grubbing of the area is the foremost requirement to start the construction activities in accordance with MoRTH specifications. The impact due to removal of vegetation includes

- Dust generation during windy atmosphere
- Loss of productive top soil
- Soil erosion during rainy season, may lead to water contamination.

Measures have been taken in reducing and curtailing the clearing and grubbing of excess land. Refer table 7.5 for the item wise quantitative requirement of clearing and grubbing.

### 7.7.5 Cattle Grazing

No cattle grazing grounds have been found along the corridor of Impact.

## 7.8 SOCIO-ECONOMIC ENVIRONMENT-IMPACTS

Adverse socio-economic impacts include all disruptions on the social and economic interactions of communities due to the road intervention. This involves effect on both the adjacent communities (mostly direct) as well as the nearby communities (mostly indirect). The various impacts have been detailed as:

- General impacts that apply to the entire project Influence Area,
- Specific impacts on likely properties and PAPs, within the Corridor of Impact.

Table 7.10: Impact on Social and Cultural Environment

| Project Activity               | Planning and Design Phase                  | Pre Construction Phase |                                                                          |                                                                               | Construction Phase                              |                        |                                                                            |                                               |                                       | Operation                             |                                            |
|--------------------------------|--------------------------------------------|------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------|------------------------|----------------------------------------------------------------------------|-----------------------------------------------|---------------------------------------|---------------------------------------|--------------------------------------------|
|                                |                                            |                        |                                                                          |                                                                               |                                                 |                        |                                                                            |                                               |                                       | Direct                                | Indirect Induced development               |
| Env. Component Affected        | Design decisions & Implementation policies | Land acquisition       | Removal of Structures                                                    | Removal of trees & vegetation                                                 | Earth works including quarrying                 | Laying of pavement     | Vehicle & machine operation & maintenance                                  | Asphalt and crusher plants                    | Labour Camps                          | Vehicle operation                     | -                                          |
| Agricultural land              | -                                          | Change in land prices  | Loss of land economic value                                              | Loss of standing crops                                                        | Loss of productive land                         | -                      | -                                                                          | Dust on agricultural land reduce productivity | -                                     | -                                     | Conversion of Agricultural Land            |
| Buildings and built structures | -                                          | -                      | Loss of structures, Debris generation, Noise and Air pollution           | -                                                                             | Noise, vibration may cause damage to structures | -                      | Noise, vibration may cause damage to structures                            | Dust accumulation on building and structure   | -                                     | Vibration and noise                   | Change in building use and characteristics |
| People and Community           | Anxiety and fear among community           | -                      | Displacement of people Psychological impact on people loss of livelihood | Loss of shade & community trees, Loss of fuel wood and fodder, Loss of income | Noise and Air pollution                         | Odour and dust         | Noise and Air pollution, Collision with pedestrians livestock and vehicles | Air and noise pollution and discomfort        | Community clashes with migrant labour | Noise pollution, Risk of accident     | Induced pollution                          |
| Cultural Assets                | -                                          | -                      | Displacement loss of structure from RoW                                  | Loss of sacred trees.                                                         | Noise, vibration may cause damage to structure  | -                      | Damage from vibration & air pollution                                      | Dust accumulation                             | -                                     | Damage from vibration & air pollution | -                                          |
| Utilities and Amenities        | -                                          | -                      | Interruption in supply                                                   | -                                                                             | -                                               | -                      | Damage to utility and amenities                                            | Dust accumulation on water bodies             | Pressure on existing amenities        |                                       | -                                          |
| Labour's Health & Safety       | -                                          | -                      | -                                                                        | -                                                                             | Increase of stagnant water and disease          | Asphalt odour and dust | Collisions with vehicles, pedestrians & livestock                          | Impact on health due to inhale of dust        | Increase in communicable diseases     | Collisions pedestrian & livestock     | -                                          |

### General Impacts

Engineering, environmental and socio-economic surveys, conducted during the design phase, for the generation of the baseline information, give indications of several adverse impacts in the vicinity of the alignment, which are related to common human psychology and general in nature.

#### 7.8.1 Fear of uncertainties regarding future

These normally become long lived, given the length of time, which elapses between initial surveys and commencement of construction.

Land and property owners are subjected to sufferings regarding uncertainties of the extent of loss and the nature of compensation. These involve:

- uncertainty of the amount of land/property to be acquired,
- time of acquisition and evacuation,
- extent and amount compensation,
- Provision of alternative land or job, etc.

#### 7.8.2 Inducement of Land Prices

Once the project becomes common knowledge, there may be a danger of unscrupulous speculators moving in to purchase land at what might seem to be advantageous prices, prior to the commencement of the official procedures. Such impact is more likely to occur in the case of urban



fringe areas during the design and pre-construction phase.

### 7.8.3 Inducement of Squatter Influx

Squatters may attempt to occupy land along and adjacent to the proposed alignments, in the hope of receiving compensation or some other inducements to leave when construction commences. Such squatters could cause undue pressure on local resources such as water and firewood, which could result in conflicts with those who are harvesting the resources presently.

### 7.8.4 Loss of utilities and amenities

Site clearance involves removal of various assets, utilities and amenities that are:

- Natural (trees, bushes and grasslands), and
- Physical structures (public or private assets and utilities).
- Relocation of utilities like electricity, water and telephone lines

For people dependent on the above, this constitutes economic loss for some time before these are restored to their previous status. These have been further discussed in the RAP.

### 7.8.5 Public health and safety

- a) Impacts on Public health and safety may arise during the phases of pre-construction, construction and operation phases. During the pre-construction and construction phases, dismantling of the structures for CoI clearance and road construction activities may result in the following health hazards:
  - Dismantling of properties has psychological impacts on their owners and others associated with them.
  - Debris generated on account of the above mentioned activities.
- b) Labour Camps during construction period can bring the following problems.
  - In the case of non-local labour (if so is arranged by the contractor), labour camps are set up at one or more sites adjacent to the alignment, and at some ancillary sites, like aggregate quarries. These labourers hired from outside can have clashes with the local population on account of cultural and religious differences. The influx of a large work force to an area, already hard pressed for basic services (medical services, power, water supply, etc.), can impose additional stress on these facilities.
  - If alternative fuels are not made available to the workforce, there is a likelihood that trees will be cut down for cooking or heating purposes.
  - In sanitary conditions in the labour camps might also result in impact on health of labourers as well as the local population. Transmission of diseases is also facilitated by the migration of people. During the construction phase work, crews and their dependants may bring with them a multitude of communicable diseases including sexually transmitted diseases (STDs) like AIDS. This is more so if the nature of the project requires more male-workers, who have migrated from other parts of the state or country.
- c) Allied activities during construction period may cause local disruption.
  - During road construction allied activities like quarrying and crushing operations, traffic diversions, etc., may cause disruption of social and economic life of the local population of the nearby areas.
  - Dust and noise generated in crushing and blasting operations may cause nuisance to the nearby communities.
  - Traffic jams and congestion, loss of access and other road accident risks, as a result of diversion of traffic and construction work on road.
  - There will be some impact on land during construction, limited mainly to temporary acquisition to cater to road diversion or traffic detours and establishment of labour camps.

## d) Accidents and Safety

- Although the design speeds have been kept lower in the major settlement areas, some amount of severance is expected in the rural areas. Especially where the residential area is on one side and their agricultural land and other facilities are on the other side of the highway. School children and ladies carrying pots full of water from the water sources (ponds/wells) also get exposed to this risk. In rural areas it was seen that cattle also cross the highways near the settlement.

**7.8.6 Resettlement of People**

People, displaced from their homes and agricultural lands on account of the project, shall induce additional pressures to the local resource base. These include pressure on:

- water resources in areas where availability is low,
- grazing lands and fuel-wood,
- Public services such as schools and medical facilities.
- This is critical since number of displaced persons being squatters is larger than legal **landholders formally displaced.**

**7.8.7 Land Use Changes**

Land use changes along the road corridors are anticipated. These shall bring about a change in the characteristics of the adjacent lands. There would be succession of land uses and higher return uses would displace the lower return uses. This phenomenon will occur at major intersections and in settlement areas along the project corridors. The urban fringe areas along the project roads will be subjected to ribbon development.

**7.8.8 Disturbance to the Road side Services**

Along the highway, near settlements, small shops get attracted to serve the local people as well as the highway users. A composite socio-economically inter-dependent has been developed as a consequence. The shops serve dual purpose by providing income and employment to locals as well as service to the road users. It is likely that due to implementation of the project some of the shops may get displaced. This would cause negative impact on the livelihood of people as well as loss of service to the local people and road users.

**7.8.9 Removal of encroachments and squatters**

In order to reduce the number of PAPs, land clearing shall be restricted to within the Corridor of Impact (CoI) which principally lies within the RoW. Width of the CoI varies according to the design and is narrower in the settlement areas, where the numbers of PAPs are likely to be more. However, some amount of land clearing will be essential in several of these stretches.

The potential impacts likely to arise from clearance of encroached residential areas (especially in settlements along the project corridors) may involve loss of valuable residential space to the residents. In the case of squatter settlements, displacement might lead to loss of shelter if adequate measures are not taken for their resettlement. Compensation may not be enough for the effected persons to gain access to shelter. Other impacts include disturbance to family and community life and increased distance from their workplace. In such cases the displaced persons may again resort to squatting.

The extent of loss in the case of encroached agricultural lands shall be relatively less, in comparison with residential and commercial properties. This because, the encroached lands form only a small part of the total cropped land of the farmers.

**Specific Impact**

Other socio-economic impacts involve the presence of sensitive community facilities within the Corridor of Impact such as worship places and cultural properties.

### 7.8.10 Sensitive Cultural Properties

Utmost care has been taken in finalising the horizontal and vertical alignment so that the impact on cultural properties could be better avoided than mitigated as a result, **out of the 67 nos. cultural properties existing within the Corridor of Impact 29 nos. are in direct (physical) impact**, i.e. lying within the formation width of the proposed widening/upgradation. **And also it is observed that the project highway is likely to impact khediyar nursery near km 57.3. understandably, this is a very old nursery, which supplies high quality mango and other saplings for orchards across Andhra Pradesh and adjoining states. Considering its significance, measures have been suggested for the mitigation/enhancement of this location in next chapter.**

The details of cultural properties identified within CoI that will have experience positive or negative impact is provided in the Table 7.11 below.

**Table 7.11: Details of Sensitive Cultural Properties**

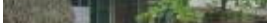
|                   |                         |                                                                                     |                          |                                                                                       |
|-------------------|-------------------------|-------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------|
| Chainage          | 1+800                   |    | 2+000                    |    |
| Structure ID No   | Venkatesh Temple        |                                                                                     | Church                   |                                                                                       |
| Side (Left/Right) | Right                   |                                                                                     | Right                    |                                                                                       |
| Distance from ECL | 4.5                     |                                                                                     | 11.0                     |                                                                                       |
| Length x Breadth  | 3x6                     |                                                                                     |                          |                                                                                       |
| Proposed Widening | Eccentric on RHS(3.43m) |                                                                                     | Eccentric on RHS(3.88m)  |                                                                                       |
| Impact            | Direct Impact           |                                                                                     | Direct Impact            |                                                                                       |
| Chainage          | 3+000                   |   | 3+300                    |   |
| Structure ID No   | Temple                  |                                                                                     | Church                   |                                                                                       |
| Side (Left/Right) | Left                    |                                                                                     | Right                    |                                                                                       |
| Distance from ECL | 9.5                     |                                                                                     | 7.0                      |                                                                                       |
| Length x Breadth  | 3X6                     |                                                                                     | 5x8                      |                                                                                       |
| Proposed Widening | Eccentric on RHS(3.98m) |                                                                                     | Eccentric on RHS(4.60m)  |                                                                                       |
| Impact            | No Direct Impact        |                                                                                     | Direct Impact            |                                                                                       |
| Chainage          | 4+100                   |  | 4+825                    |  |
| Structure ID No   | Temple                  |                                                                                     | Temple                   |                                                                                       |
| Side (Left/Right) | Left                    |                                                                                     | Right                    |                                                                                       |
| Distance from ECL | 4.5                     |                                                                                     | 7                        |                                                                                       |
| Length x Breadth  | 1x1                     |                                                                                     | 10.5                     |                                                                                       |
| Proposed Widening | Eccentric on RHS(2.47m) |                                                                                     | Eccentric on RHS(4.03m)  |                                                                                       |
| Impact            | Direct Impact           |                                                                                     | Direct Impact            |                                                                                       |
| Chainage          | 5+525                   |  | 8+700                    |  |
| Structure ID No   | Temple & Pipal Tree     |                                                                                     | Temple                   |                                                                                       |
| Side (Left/Right) | Left                    |                                                                                     | Left                     |                                                                                       |
| Distance from ECL | 5.5                     |                                                                                     | 5.5                      |                                                                                       |
| Length x Breadth  | 5x5                     |                                                                                     | 3x5                      |                                                                                       |
| Proposed Widening | Eccentric on RHS(4.04m) |                                                                                     | Eccentric on RHS(3.73)   |                                                                                       |
| Impact            | No Direct Impact        |                                                                                     | No Direct Impact         |                                                                                       |
| Chainage          | 9+175                   |  | 10+575                   |  |
| Structure ID No   | Church                  |                                                                                     | Temple Banyana&Neem Tree |                                                                                       |
| Side (Left/Right) | Right                   |                                                                                     | Right                    |                                                                                       |
| Distance from ECL | 13.5                    |                                                                                     | 4.5                      |                                                                                       |
| Length x Breadth  | 7x7                     |                                                                                     | 2x2                      |                                                                                       |
| Proposed Widening | Eccentric on RHS(4.44m) |                                                                                     | Eccentric on RHS(6.41m)  |                                                                                       |
| Impact            | No Direct Impact        |                                                                                     | Direct Impact            |                                                                                       |
| Chainage          | 16+000                  |  | 16+150                   |  |
| Structure ID No   | Temple                  |                                                                                     | Temple                   |                                                                                       |
| Side (Left/Right) | Left                    |                                                                                     | Right                    |                                                                                       |
| Distance from ECL | 6.5                     |                                                                                     | 9.5                      |                                                                                       |
| Length x Breadth  | 6x3                     |                                                                                     | 7x4                      |                                                                                       |
| Proposed Widening | Eccentric on RHS(3.95m) |                                                                                     | Eccentric on RHS(1.19m)  |                                                                                       |
| Impact            | No Direct Impact        |                                                                                     | No Direct Impact         |                                                                                       |



|                   |                         |        |  |                            |  |
|-------------------|-------------------------|--------|--|----------------------------|--|
| Chainage          |                         | 16+200 |  | 16+250                     |  |
| Structure ID No   | Temple                  |        |  | Temple                     |  |
| Side (Left/Right) | Left                    |        |  | Left                       |  |
| Distance from ECL | 5.0                     |        |  | 6.5                        |  |
| Length x Breadth  | 50x10                   |        |  | 7x10                       |  |
| Proposed Widening | Eccentric on RHS(4.86m) |        |  | Eccentric on RHS(5.07m)    |  |
| Impact            | No Direct Impact        |        |  | No Direct Impact           |  |
| Chainage          |                         | 16+450 |  | 17+550                     |  |
| Structure ID No   | Temple                  |        |  | Temple                     |  |
| Side (Left/Right) | Right                   |        |  | Left                       |  |
| Distance from ECL | 4.5                     |        |  | 6.5                        |  |
| Length x Breadth  | 3x3                     |        |  | 15x10                      |  |
| Proposed Widening | Concentric              |        |  | Concentric                 |  |
| Impact            | Direct Impact           |        |  | No Direct Impact           |  |
| Chainage          |                         | 18+000 |  | 18+050                     |  |
| Structure ID No   | Temple                  |        |  | Bathing Ghat&Ganesh Temple |  |
| Side (Left/Right) | Right                   |        |  | Left                       |  |
| Distance from ECL | 4.5                     |        |  | 6.5                        |  |
| Length x Breadth  | 7x4                     |        |  | 2x3                        |  |
| Proposed Widening | Concentric              |        |  | Concentric                 |  |
| Impact            | Direct Impact           |        |  | No Direct Impact           |  |
| Chainage          |                         | 18+600 |  | 22+100                     |  |
| Structure ID No   | Temple                  |        |  | Temple                     |  |
| Side (Left/Right) | Right                   |        |  | Left                       |  |
| Distance from ECL | 5.5                     |        |  | 6.5                        |  |
| Length x Breadth  | 7x4                     |        |  |                            |  |
| Proposed Widening | Concentric              |        |  | Concentric)                |  |
| Impact            | Direct Impact           |        |  | No Direct Impact           |  |
| Chainage          |                         | 22+650 |  | 22+700                     |  |
| Structure ID No   | Temple & Lands          |        |  | Temple                     |  |
| Side (Left/Right) | Left                    |        |  | Left                       |  |
| Distance from ECL | 8.5                     |        |  | 5.5                        |  |
| Length x Breadth  | 8x3                     |        |  | 50x10                      |  |
| Proposed Widening | Concentric              |        |  | Concentric                 |  |
| Impact            | No Direct Impact        |        |  | Direct Impact              |  |
| Chainage          |                         | 25+200 |  | 25+200                     |  |
| Structure ID No   | Temple                  |        |  | Temple                     |  |
| Side (Left/Right) | Left                    |        |  | Left                       |  |
| Distance from ECL | 6.5                     |        |  | 4.5                        |  |
| Length x Breadth  |                         |        |  |                            |  |
| Proposed Widening | Concentric              |        |  | Conventric                 |  |
| Impact            | No Direct Impact        |        |  | Direct Impact              |  |
| Chainage          |                         | 25+200 |  | 25+350                     |  |
| Structure ID No   | Temple                  |        |  | Temple                     |  |
| Side (Left/Right) | Right                   |        |  | Left                       |  |
| Distance from ECL | 5.5                     |        |  | 4.5                        |  |
| Length x Breadth  | 3x5                     |        |  |                            |  |
| Proposed Widening | Concentric              |        |  | Concentric                 |  |
| Impact            | Direct Impact           |        |  | Direct Impact              |  |
| Chainage          |                         | 28+000 |  | 30+300                     |  |
| Structure ID No   | Temple                  |        |  | Temple                     |  |
| Side (Left/Right) | Right                   |        |  | Left                       |  |
| Distance from ECL | 4.0                     |        |  | 6.5                        |  |
| Length x Breadth  | 2x2                     |        |  | 0.5x4                      |  |
| Proposed Widening | Concentric              |        |  | Concentric                 |  |

| Impact            | Direct Impact    |                                                                                     | No Direct Impact              |                                                                                       |
|-------------------|------------------|-------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|
| Chainage          | 31+800           |    | 34+670                        |    |
| Structure ID No   | Temple           |                                                                                     | Temple / Ghat                 |                                                                                       |
| Side (Left/Right) | Left             |                                                                                     | Left                          |                                                                                       |
| Distance from ECL | 6.5              |                                                                                     | 5.5                           |                                                                                       |
| Length x Breadth  | 5x5              |                                                                                     | 5x8                           |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | Direct Impact                 |                                                                                       |
| Chainage          | 35+100           |    | 35+100                        |    |
| Structure ID No   | Sai Temple       |                                                                                     | Temple                        |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Right                         |                                                                                       |
| Distance from ECL | 28.5             |                                                                                     | 8.5                           |                                                                                       |
| Length x Breadth  | 40x50            |                                                                                     | 5x3                           |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | No Direct Impact              |                                                                                       |
| Chainage          | 35+100           |    | 35+300                        |    |
| Structure ID No   | Temple           |                                                                                     | Statue Of Durga in Lion Mouth |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Right                         |                                                                                       |
| Distance from ECL | 7.5              |                                                                                     | 6.5                           |                                                                                       |
| Length x Breadth  | 2x2              |                                                                                     | 20x6                          |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | No Direct Impact              |                                                                                       |
| Chainage          | 38+100           |   | 40+050                        |   |
| Structure ID No   | Temple           |                                                                                     | Temple                        |                                                                                       |
| Side (Left/Right) | Left             |                                                                                     | Right                         |                                                                                       |
| Distance from ECL | 6.0              |                                                                                     | 5.0                           |                                                                                       |
| Length x Breadth  | 8x3              |                                                                                     | 2x2                           |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | Direct Impact                 |                                                                                       |
| Chainage          | 40+150           |  | 40+400                        |  |
| Structure ID No   | Temple           |                                                                                     | Temple                        |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Right                         |                                                                                       |
| Distance from ECL | 4.5              |                                                                                     | 5.0                           |                                                                                       |
| Length x Breadth  | 2x2              |                                                                                     | 4x4                           |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | Direct Impact    |                                                                                     | Direct Impact                 |                                                                                       |
| Chainage          | 40+400           |  | 41+200                        |  |
| Structure ID No   | Temple           |                                                                                     | Temple                        |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Left                          |                                                                                       |
| Distance from ECL | 5.0              |                                                                                     | 6.5                           |                                                                                       |
| Length x Breadth  | 20x50            |                                                                                     | 10x10                         |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | Direct Impact    |                                                                                     | No Direct Impact              |                                                                                       |
| Chainage          | 41+350           |  | 41+550                        |  |
| Structure ID No   | Temple           |                                                                                     | Temple & Hand Pump            |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Left                          |                                                                                       |
| Distance from ECL | 5.5              |                                                                                     | 4.0                           |                                                                                       |
| Length x Breadth  |                  |                                                                                     | 12x7                          |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | Direct Impact                 |                                                                                       |
| Chainage          | 41+700           |  | 42+100                        |  |
| Structure ID No   | Temple           |                                                                                     | Church                        |                                                                                       |
| Side (Left/Right) | Left             |                                                                                     | Right                         |                                                                                       |
| Distance from ECL | 5.5              |                                                                                     | 8.5                           |                                                                                       |
| Length x Breadth  | 10x10            |                                                                                     |                               |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric                    |                                                                                       |
| Impact            | Direct Impact    |                                                                                     | No Direct Impact              |                                                                                       |
| Chainage          | 44+150           |  | 44+600                        |  |



|                   |                  |                                                                                     |                  |                                                                                       |
|-------------------|------------------|-------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------|
| Structure ID No   | Temple           |                                                                                     | Big Temple       |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Right            |                                                                                       |
| Distance from ECL | 6.5              |                                                                                     | 8.5              |                                                                                       |
| Length x Breadth  | 8x4              |                                                                                     |                  |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | No Direct Impact |                                                                                       |
| Chainage          | 44+700           |    | 45+000           |    |
| Structure ID No   | Temple           |                                                                                     | Temple           |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Left             |                                                                                       |
| Distance from ECL | 6.5              |                                                                                     | 7.5              |                                                                                       |
| Length x Breadth  |                  |                                                                                     | 3x3              |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | No Direct Impact |                                                                                       |
| Chainage          | 45+350           |    | 45+900           |    |
| Structure ID No   | Temple           |                                                                                     | Chappur Temple   |                                                                                       |
| Side (Left/Right) | Left             |                                                                                     | Left             |                                                                                       |
| Distance from ECL | 10.5             |                                                                                     | 6.5              |                                                                                       |
| Length x Breadth  | 8x5              |                                                                                     | 6x3              |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | No Direct Impact |                                                                                       |
| Chainage          | 46+100           |   | 48+800           |   |
| Structure ID No   | Sai Temple       |                                                                                     | Temple (imp)     |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Right            |                                                                                       |
| Distance from ECL | 6.5              |                                                                                     | 4.0              |                                                                                       |
| Length x Breadth  |                  |                                                                                     | 13x4             |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | Direct Impact    |                                                                                       |
| Chainage          | 49+550           |  | 52+300           |                                                                                       |
| Structure ID No   | Temple           |                                                                                     | Temple           |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Right            |                                                                                       |
| Distance from ECL | 4.5              |                                                                                     | 15.0             |                                                                                       |
| Length x Breadth  | 5x5              |                                                                                     | 5x8              |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | Direct Impact    |                                                                                     | No Direct Impact |                                                                                       |
| Chainage          | 54+200           |  | 55+500           |  |
| Structure ID No   | Sai Temple       |                                                                                     | Temple           |                                                                                       |
| Side (Left/Right) | Right            |                                                                                     | Left             |                                                                                       |
| Distance from ECL | 10.5             |                                                                                     | 5.5              |                                                                                       |
| Length x Breadth  | 6x4              |                                                                                     | 10x5             |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | Direct Impact    |                                                                                     | Direct Impact    |                                                                                       |
| Chainage          | 56+800           |  | 57+050           |  |
| Structure ID No   | Temple           |                                                                                     | Temple/ island   |                                                                                       |
| Side (Left/Right) | Left             |                                                                                     | Right            |                                                                                       |
| Distance from ECL | 5.5              |                                                                                     | 2.5              |                                                                                       |
| Length x Breadth  | 8x12             |                                                                                     | 2 x 3            |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | Direct Impact    |                                                                                     | No Direct Impact |                                                                                       |
| Chainage          | 57+200           |  | 57+300           |  |
| Structure ID No   | Temple           |                                                                                     | Temple           |                                                                                       |
| Side (Left/Right) | Left             |                                                                                     | Left             |                                                                                       |
| Distance from ECL | 6.5              |                                                                                     | 8.5              |                                                                                       |
| Length x Breadth  | 50x50            |                                                                                     |                  |                                                                                       |
| Proposed Widening | Concentric       |                                                                                     | Concentric       |                                                                                       |
| Impact            | No Direct Impact |                                                                                     | No Direct Impact |                                                                                       |
| Chainage          | 57+300           |  | 58+950           |  |
| Structure ID No   | Temple BW        |                                                                                     | Temple           |                                                                                       |
| Side (Left/Right) | Left             |                                                                                     | Right            |                                                                                       |

|                   |                       |                                                                                    |                  |                                                                                     |
|-------------------|-----------------------|------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| Distance from ECL | 5.0                   |                                                                                    | 6.5              |                                                                                     |
| Length x Breadth  |                       |                                                                                    | 4x5              |                                                                                     |
| Proposed Widening | Concentric            |                                                                                    | Concentric       |                                                                                     |
| Impact            | Direct Impact         |                                                                                    | No Direct Impact |                                                                                     |
| Chainage          | 59+550                |   | 60+200           |  |
| Structure ID No   | Ganesh Temple         |                                                                                    | Temple           |                                                                                     |
| Side (Left/Right) | Right                 |                                                                                    | Right            |                                                                                     |
| Distance from ECL | 3.5                   |                                                                                    | 4.5              |                                                                                     |
| Length x Breadth  | 3x5                   |                                                                                    | 4x8              |                                                                                     |
| Proposed Widening | Concentric            |                                                                                    | Concentric       |                                                                                     |
| Impact            | Direct Impact         |                                                                                    | Direct Impact    |                                                                                     |
| Chainage          | 60+750                |   | 60+750           |                                                                                     |
| Structure ID No   | Mary Statu            |                                                                                    | Church           |                                                                                     |
| Side (Left/Right) | Right                 |                                                                                    | Right            |                                                                                     |
| Distance from ECL | 7.0                   |                                                                                    | 28.5             |                                                                                     |
| Length x Breadth  |                       |                                                                                    | 5x15             |                                                                                     |
| Proposed Widening | Concentric            |                                                                                    | Concentric       |                                                                                     |
| Impact            | No Direct Impact      |                                                                                    | No Direct Impact |                                                                                     |
| Chainage          | 61+800                |  |                  |                                                                                     |
| Structure ID No   | Park & Hanuman Temple |                                                                                    |                  |                                                                                     |
| Side (Left/Right) | Right                 |                                                                                    |                  |                                                                                     |
| Distance from ECL | 6.5                   |                                                                                    |                  |                                                                                     |
| Length x Breadth  |                       |                                                                                    |                  |                                                                                     |
| Proposed Widening | Concentric            |                                                                                    |                  |                                                                                     |
| Impact            | No Direct Impact      |                                                                                    |                  |                                                                                     |

Besides direct impact, there are many possible indirect impacts due to construction activities, as follows.

- Access to religious places would be difficult during the construction period due to the presence of working areas, consequent traffic management issues, presence of heavy equipment, machineries and numerous workers and controlled sign boards.
- Many existing signs boards and information boards will be removed for the construction work. This will make it more complicated for identifying the pilgrimage location, routes and landmarks.
- Safety issues and accidents could go high during the construction period.

## 7.9 BUS SHELTER, BUS BAYS, TRUCK LAY BYES, RESTING PLACE AND SERVICE ROAD

Concentric widening is proposed in built-up sections of the project road so as to minimize the negative impact due to upgradation/widening of the project road. The **Table 2.1** of Chapter 2 describes the settlements locations along the proposed project road. Out of the total of 19 existing pucca bus waiting sheds, 5 will be relocated, due to direct physical Impact.

**Table 7.12: List of Bus Shelters**

|                   |                          |                                                                                     |                          |                                                                                       |
|-------------------|--------------------------|-------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------|
| Chainage          | 2+800                    |  | 5+500                    |  |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters             |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Left                     |                                                                                       |
| Distance from ECL | 6.5                      |                                                                                     | 6.5                      |                                                                                       |
| Length x Breadth  |                          |                                                                                     | 5x3                      |                                                                                       |
| Proposed Widening | Eccentric on RHS (3.91m) |                                                                                     | Eccentric on RHS (4.04m) |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | No Direct Impact         |                                                                                       |
| Chainage          | 8+500                    |  | 15+750                   |  |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters             |                                                                                       |



|                   |                          |                                                                                     |                      |                                                                                       |
|-------------------|--------------------------|-------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------|
| Side (Left/Right) | Left                     |                                                                                     | Right                |                                                                                       |
| Distance from ECL | 9.5                      |                                                                                     | 7.5                  |                                                                                       |
| Length x Breadth  | 3x5                      |                                                                                     | 5x3                  |                                                                                       |
| Proposed Widening | Eccentric on RHS (4.64m) |                                                                                     | Concentric           |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | No Direct Impact     |                                                                                       |
| Chainage          | 18+600                   |    | 18+600               |    |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters         |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Right                |                                                                                       |
| Distance from ECL | 6.5                      |                                                                                     | 5.5                  |                                                                                       |
| Length x Breadth  | 15x3                     |                                                                                     | 7x5                  |                                                                                       |
| Proposed Widening | Concentric               |                                                                                     | Concentric           |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | <b>Direct Impact</b> |                                                                                       |
| Chainage          | 22+100                   |    | 25+200               |                                                                                       |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters         |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Left                 |                                                                                       |
| Distance from ECL | 6.5                      |                                                                                     | 6.5                  |                                                                                       |
| Length x Breadth  | 3.5/3.5x6.00/3.00        |                                                                                     |                      |                                                                                       |
| Proposed Widening | Concentric               |                                                                                     | Concentric           |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | No Direct Impact     |                                                                                       |
| Chainage          | 25+500                   |   | 30+300               |   |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters         |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Left                 |                                                                                       |
| Distance from ECL | 5.5                      |                                                                                     | 4.0                  |                                                                                       |
| Length x Breadth  | 10x5                     |                                                                                     |                      |                                                                                       |
| Proposed Widening | Concentric               |                                                                                     | Concentric           |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | No Direct Impact     |                                                                                       |
| Chainage          | 31+800                   |  | 34+350               |  |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters         |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Left                 |                                                                                       |
| Distance from ECL | 6.0                      |                                                                                     | 4.0                  |                                                                                       |
| Length x Breadth  | 6x4                      |                                                                                     | 15x5                 |                                                                                       |
| Proposed Widening | Concentric               |                                                                                     | Concentric           |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | <b>Direct Impact</b> |                                                                                       |
| Chainage          | 35+250                   |  | 40+150               |  |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters         |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Right                |                                                                                       |
| Distance from ECL | 6.0                      |                                                                                     | 4.5                  |                                                                                       |
| Length x Breadth  | 15x6                     |                                                                                     | 10x3                 |                                                                                       |
| Proposed Widening | Concentric               |                                                                                     | Concentric           |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | <b>Direct Impact</b> |                                                                                       |
| Chainage          | 41+500                   |  | 49+700               |  |
| Structure ID No   | Bus Shelters             |                                                                                     | Bus Shelters         |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Left                 |                                                                                       |
| Distance from ECL | 5.5                      |                                                                                     | 5.5                  |                                                                                       |
| Length x Breadth  | 10x4                     |                                                                                     | 15x5                 |                                                                                       |
| Proposed Widening | Concentric               |                                                                                     | Concentric           |                                                                                       |
| Impact            | <b>Direct Impact</b>     |                                                                                     | <b>Direct Impact</b> |                                                                                       |
| Chainage          | 57+500                   |  | 59+700               |  |
| Structure ID No   | BusStand                 |                                                                                     | Bus Shelters         |                                                                                       |
| Side (Left/Right) | Left                     |                                                                                     | Left                 |                                                                                       |
| Distance from ECL | 18.5                     |                                                                                     | 7.0                  |                                                                                       |
| Length x Breadth  |                          |                                                                                     | 4x15                 |                                                                                       |
| Proposed Widening | Concentric               |                                                                                     | Concentric           |                                                                                       |
| Impact            | No Direct Impact         |                                                                                     | No Direct Impact     |                                                                                       |
| Chainage          | 60+250                   |  |                      |                                                                                       |

|                   |                  |
|-------------------|------------------|
| Structure ID No   | Bus Shelters     |
| Side (Left/Right) | Left             |
| Distance from ECL | 5.5              |
| Length x Breadth  | 3x7s             |
| Proposed Widening | Concentric       |
| Impact            | No Direct Impact |

## ***CHAPTER - 8: IMPACT MITIGATION AND ENHANCEMENT***

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## CHAPTER 8: IMPACT MITIGATION AND ENHANCEMENT

### 8.1 MITIGATION, AVOIDANCE AND ENHANCEMENT

Prevention or avoidance of impact is better than mitigation of impact. Hence avoidance and reduction of adverse impacts approaches were adopted during the design stage through continued interaction between the design and environmental teams. This is reflected in the designs of the horizontal & vertical alignment, cross sections adopted, construction methods and construction materials. In-depth site investigations have been carried out so that sensitive environmental resources are effectively avoided, leading to the environmentally best-fit alignment option. As a result many of the trees, cultural properties, water bodies etc. have been avoided at the design stage itself, as presented in Table 8.1 below.

**Table 8.1: Environmental features saved through Avoidance measure at design stage**

| Environmental Features | Potential Impact | Under Direct Impact | Saved through alignment design |
|------------------------|------------------|---------------------|--------------------------------|
| Trees (nos.)           | 5732             | 4162                | 1570                           |
| Surface Water source   | 4                | 2                   | 2                              |
| Ground Water source    | 8                | 3                   | 5                              |
| Schools and Hospitals  | 14               | 2                   | 12                             |
| Cultural Properties    | 67               | 29                  | 38                             |
| Bus Shelters           | 19               | 5                   | 14                             |

### 8.2 AIR ENVIRONMENT – MITIGATION

Motor vehicles have emerged as one of the major sources of air pollution especially in urban areas. Due to the proposed road improvements aimed at enhancing the efficiency of road transport system the number of vehicles on these roads will be increased over time. Summary of potential impact and mitigation measures proposed is mentioned below:

| Sr. No. | Item                               | Impact                                 | Impact (Reason)                                                                                                                                                                                                                                                                   | Mitigation/Enhancement                                                                                                                                                                    |
|---------|------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Meteorological factors and climate | Marginal impact                        | Due to production and laying of hot bituminous mix.                                                                                                                                                                                                                               | <ul style="list-style-type: none"> <li>✓ Avenue plantation</li> <li>✓ Plantation in realignment sections</li> <li>✓ shrub plantation in the median/island</li> <li>✓ RoW hedge</li> </ul> |
| 2a      | Air quality - emissions            | temporary and location specific        | <ul style="list-style-type: none"> <li>▪ shifting of utilities,</li> <li>▪ removal of trees &amp; vegetation,</li> <li>▪ transportation of material</li> </ul>                                                                                                                    | Sprinkling of Water<br>Fine materials to be completely covered, during transport & stocking.                                                                                              |
|         | Pre-construction stage             | (Dust Generation)                      | <ul style="list-style-type: none"> <li>▪ installation of construction plants</li> </ul>                                                                                                                                                                                           | Plant to be installed in Downwind direction from nearby settlement.                                                                                                                       |
| 2b      | Air quality - emissions            | Moderate impact                        | <ul style="list-style-type: none"> <li>▪ clearing and grubbing</li> <li>▪ materials dumping</li> <li>▪ brushing of the surface</li> <li>▪ access roads to borrow-areas</li> <li>▪ hot mix plants, Crushers</li> <li>▪ paving of asphalt layers</li> <li>▪ Labour Camps</li> </ul> | Air pollution Norms will be enforced, Laborers will be provided mask.<br>Local people will be educated on safety and precaution.<br>on access roads, newly constructed embankment etc.    |
|         | Construction Stage                 | (Gaseous pollutants & Dust generation) |                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                           |
| 2c      | Air quality - emissions            | Moderate impact                        | air pollutants from traffic                                                                                                                                                                                                                                                       | compliance with future statutory regulatory requirements                                                                                                                                  |
|         | Operation Stage                    | (Gaseous pollutants)                   | dust emission from tyres                                                                                                                                                                                                                                                          | auto-technology, vehicular fuel quality-improvement                                                                                                                                       |
| 3       | Air quality - monitoring           | --                                     | Effectiveness / shortfall (if any)<br>Any unforeseen impact.                                                                                                                                                                                                                      | Measures will be revised & improved to mitigate/enhance environment due to any unforeseen impact.                                                                                         |

***Emissions from the vehicles are potentially disastrous. But, it must be emphasized that the air pollution from vehicle emission will reduce on completion, as one of the objectives of the design is enhancement of facility to cater traffic requirement.***

### 8.2.1 Meteorological Factors and Climate - mitigation

As indicated in previous chapter, there will not be any micro-climatic effect/impact due to increased traffic emission on the project road during construction and operation phase, if any minor impact do exists due to widening will be counter by compensatory and additional afforestation plan and avenue plantation.

### 8.2.2 Air Quality Emissions - mitigation

**Pre-construction Stage:** As indicated in previous chapter, the dust generation due to pre-construction activities is very temporary and localized, and will be efficiently countered by sprinkling of water.

**Construction Stage:** During the construction stage, there are two major sources: the first one is construction activities at working zones, which cause primarily dust emission and second are from operation of the construction plant, equipments and machinery, which causes gaseous pollutants. The specific measures include:

- Locating Plant at a significant distance from nearest human settlement in the predominant down wind direction.
- Vehicles delivering fine materials like soil and fine aggregates shall be covered to reduce spills on existing roads.
- Water will be sprayed on earthworks, temporary haulage and diversions on a regular basis.
- Batch type hot mix plants fitted with the bag filter / cyclone and scrubber will be installed for the reduction of the air pollution.
- Pollution control systems like water sprinkling and dust extractors and cover on conveyors will be installed for the crushers.
- All vehicles, equipment and machinery used for construction will be regularly maintained to ensure that the emission levels conform to the SPCB/CPCB norms.
- Air pollution monitoring plan has been delineated for construction phase separately for checking the effectiveness of the mitigation measures adopted during the construction phase of the Contract.

**Operation Stage:** As indicated in previous chapter, the air pollutant will be from vehicular movement on road and dust emission by tyres. As such the national and international bodies are quite active in controlling the air pollution through emission limit, auto technology and fuel quality which will counter the increase in air pollution due to increase in traffic during operation phase. Additional measures/proposed are below:

- Pollution resistant species, which can grow in high pollutant concentrations or even absorb pollutants, can be planted along the roadside.
- Monitoring of air pollution levels at sensitive locations shall be carried out all through the operation stage to check that the pollution levels are within standards prescribed by CPCB. A monitoring plan to this effect has been prepared for all roads separately and is presented in the individual EMPs.
- Other measures such as the reduction of vehicular emissions, ensuring vehicular maintenance and up-keep, educating drivers about driving behavior / methods that will reduce emissions are beyond the scope of the Project but will be far more effective in reducing the pollutant levels.

### 8.2.3 Air Quality Monitoring - mitigation

Apart from provision of the mitigation measures, their effectiveness and further improvement in designs to reduce the air pollution with increase in traffic shall be monitored. The monitoring plan shall be functional in construction as well as in operation stages. The frequency, duration and responsibility will be as per the **APPENDIX # E-10: ENVIRONMENTAL MONITORING**



PLAN. For location refer table 4.2: Ambient Air Quality Monitoring Locations including one at Hot mix plant site. And standard/acceptable values are given in table 4.2: National Ambient Air Quality Standards. Any value/result not within acceptable limits will be reported to engineer, for remedial measures.

### 8.3 LAND ENVIRONMENT- MITIGATION MEASURES

Land Acquisition, soil erosion and contamination of soil have emerged as major sources of land impact especially in urban areas and near by watercourses. Due to the proposed road improvements aimed at enhancing the efficiency of road transport system, which will result in economic growth in the region over time.

| Sr. No. | Item                       | Impact                                      | Impact (Reason)                                                                                                                                                                                                                              | Mitigation/Enhancement                                                                                                                                                                                                                                                                     |
|---------|----------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Change in Topography       | Marginal impact                             | Due to embankment raising.                                                                                                                                                                                                                   | Embankment rising to relieve water logging.                                                                                                                                                                                                                                                |
| 2       | Change in Geology          | Direct , long term, negative impact         | Extraction of materials (borrow earth, coarse & fine aggregates).                                                                                                                                                                            | no blasting is envisaged quarry redevelopment plan need to be enforced                                                                                                                                                                                                                     |
| 3       | Change in Seismology       | No Negative Impact                          |                                                                                                                                                                                                                                              | Cross drainage structures are checked and complied with the seismological settings of the region. (Zone)                                                                                                                                                                                   |
| 4       | Change in Land environment |                                             |                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                            |
| a       | Loss of land               | Direct, long-term negative impact           | Land Acquisition<br>Change in land use pattern                                                                                                                                                                                               | Land acquisition to be minimized with provision of Retaining walls.                                                                                                                                                                                                                        |
| b       | Generation of Debris       | Negative Impact                             | May contaminate air, water and land, if not disposed properly.                                                                                                                                                                               | disposed properly to avoid contamination                                                                                                                                                                                                                                                   |
| c       | Soil Erosion               | Moderate, direct, long-term negative impact | Road slopes and spoils<br><br>Construction of new bridges and culverts<br>Quarry and Borrow areas                                                                                                                                            | Embankment protection.<br>For Emb. ht. >3m Stone pitching,<br>Emb. ht. <3m Turfing.<br>Residual spoil need to be disposed properly.<br>Silt Fencing need to be provided.<br>Quarries need to be reclaimed                                                                                  |
| 5       | Contamination of Soil      | Direct, long term negative impact           | Scarified bitumen wastes<br><br>Oil & diesel Spills<br>Emulsion sprayer and laying of hot mix<br>Production of hot mix and rejected materials<br><br>Residential facilities for the labor and officers<br>Routine and periodical maintenance | Hazardous Wastes (Management and Handling) Rules, 1989 to be enforced.<br>Oil Interceptor will be provided for accidental spill of oil and diesel.<br>Rejected material will be layed in village roads or as directed by engineer.<br>Septic tank will be construction for waste disposal. |
| 6       | Soil quality - monitoring  |                                             | Effectiveness / shortfall (if any)<br>Any unforeseen impact                                                                                                                                                                                  | Measures will be revised & improved to mitigate/ enhance environment due to any unforeseen impact.                                                                                                                                                                                         |

***The implications of accidental discharge are potentially disastrous. But, it must be emphasized that the probability of such an accident is quite low, as one of the objectives of the design is the enhancement of road safety.***

#### 8.3.1 Change in Topography - Mitigation

As indicated in previous chapter, there is not much change in the embankment of the project road (refer table 2.13 Embankment raising sections) hence no appreciable change in topography. Pavement height will be raised only to accommodate structural layers over the existing pavement. The raising due to submergence and profile improvement is positive impacts on the local environment. The raised sections are located away from habitations. **Adequate measures have been taken so that the raising if any in settlement areas should be minimum possible and in no case exceed 0.5 m.**

#### 8.3.2 Change in Geology- Mitigation

As part of the Project preparation, the sources of quarries for the fine and coarse aggregates have been identified for use in road works and structural works, details of the quarries have been presented in Chapter 4 Baseline Environment. No new quarry has been proposed for the Project

requirements. Only existing, live, licensed quarries will be used as sources of coarse and fine aggregates. It will be ensured the aggregates procured during construction stage will be from the authorized or licensed suppliers only. In case of use of any new quarry by contractor, the instructions/procedure as detailed in **APPENDIX # E1: AGGREGATE QUARRY MANAGEMENT** will be applicable.

Selected soil borrow areas have been identified during the design stage of the Project. Most of these borrow areas are local borrow areas, and agricultural fields not in productive use, and where farmers want to lower the level of the land (reduction of energy requirement for the watering and reduce loss of water, nutrients, and fertilizer through the seepage). The detailed list of the borrow areas is given in Chapter 4 Baseline Environment. In case of taking earth from any of these borrow areas the instructions / procedure as detailed in **APPENDIX # E2: BORROW AREA MANAGEMENT**, will be applicable.

### 8.3.3 Change in Seismology - Mitigation

As indicated in previous chapter, there will be NO impact on the seismological setting of the region. Rather, as part of the project all the existing structures will be checked and constructed as per the seismological requirements of the region in conformity to the **IRC 6, 2000** guidelines. Refer Fig 7.1 Seismology map of Andhra Pradesh.

### 8.3.4 Change in Land Environment – Mitigation

**Loss of Land:** As far as possible the land acquisition has been kept to the minimum, by restricting the geometric improvement within the existing right of way. The same can be seen as only 37 hectare land is required for project road upgradation. However the land acquisition will be done at sections having width, insufficient to accommodate the approved cross-sections & geometric improvements.

**Debris Generation:** Due to the removal of structures (Residential and commercial), pavement scarification and cross drainage structures lot of debris will generate, which need to be disposed properly to avoid contamination of land and water. For safe and environmental friendly disposal of waste debris the instruction/procedure specified in **APPENDIX # E3: DEBRIS DISPOSAL SITES AND MANAGEMENT**, will be applicable.

**Soil Erosion:** could cause sever negative impact if not dealt with, in time, as this harms the environment in two ways, firstly it erodes the top soil and secondly it spoils the resource where it settles, mostly water course. Hence soil erosion cause loss of top-soil and contamination of water bodies/sources/channels.

A comprehensive instruction/procedure is prepared in **APPENDIX # E4: SITE CLEARANCE AND TREE FELLING**

#### a) Road slopes and spoils

Adequate measures have been devised for control of the soil erosion from the embankments, the slopes have been restricted to 1 vertical: 2 horizontal for most of the sections. Soil erosion through embankments is prevented and controlled by following methods.

**Alt 1. Grassing of slopes:** (for emb. ht. < 3.0m). For this purpose it is best to use locally growing grasses and bushes, as these are best adapted to the local soil, temperature and rainfall conditions. Plantation is best done just after the first pre-monsoon showers, which gives a time of 2-3 weeks for the grass to take root before the onset of monsoon. Normally, no watering of the grassed slopes is done following the planting. However, watering of the slopes may be provided if the planting is done in the non-monsoon season, or to respond to dry conditions following planting. The above methods of providing vegetation cover on embankment slopes follow provisions in IRC-56-1974, “Recommended Practice for Treatment of Embankment Slopes for Erosion Control”.

**Alt 2. Use of Pitching to Control Erosion:** (for emb. ht.  $\geq$  3.0m). Stones or bricks are hand laid on the surface and lightly tamped. The interstices between the stones are filled up with soil. Grasses may be dibbled into the soil filled spaces. As the grass grows, it develops a good binding effect on the pitching. List of stone pitching sections is provided in previous chapter.

**b) Construction of New Bridges and Culverts**

Contractor will be responsible of removing all the debris/ earth generated due to dismantling of existing structure and excavation of the foundation of cross drainage works, form the water course before the onset of monsoon.

**c) Quarries and Borrow Areas**

A comprehensive management plan for restoring of quarry and borrow area is prepared. Refer **APPENDIX # E1: AGGREGATE QUARRY MANAGEMENT & APPENDIX # E2: BORROW AREA MANAGEMENT.**

Prior to the start of the relevant construction, the Contractor shall submit to the Engineer for approval, his schedules for carrying out temporary and permanent erosion/sedimentation control works as are applicable for the items of clearing and grubbing, roadway and drainage excavation, embankment/sub-grade construction, bridges and other structures across water courses, pavement courses and shoulders. He shall also submit for approval his proposed method of erosion/sedimentation control on service road and borrow pits and his plan for disposal of waste materials. **No construction activity will start prior to approval by engineer, of the measures and method to be adopted by contractor.**

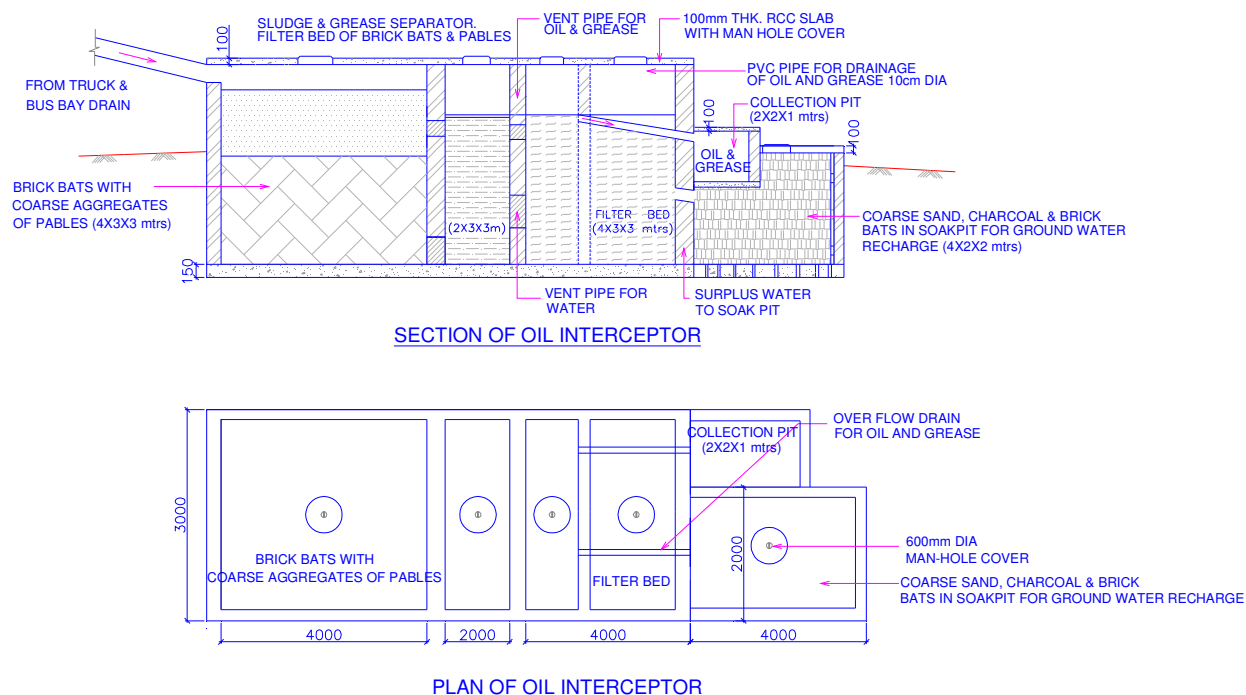
**8.3.5 Contamination of Soil: Mitigation**

Contamination of soil can spoil the soil and can also contaminate the surface as well as ground water sources. Details of the activities from which the contamination can occur are presented below;

| Potential Impact                                                                                                                                                                                                                      | Mitigation/Enhancement                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scarified bitumen wastes, Excess production of hot mix and rejected materials.                                                                                                                                                        | Scarified waste and excess/rejected hot mix, with the consent of village authority, will be used in village roads construction.                                                                                                     |
| Debris generated from dismantling of structures.                                                                                                                                                                                      | A comprehensive list of instructions/procedures has been suggested in <b>APPENDIX # E3: DEBRIS DISPOSAL SITES</b> , for contractor to adhere to for safe and environmental friendly disposal of debris.                             |
| Maintenance of the machinery and operation of the diesel generator sets on site<br>Oil Spill from the operation of the diesel pumps and diesel storage, during transportation and transfer, parking places, and diesel generator sets | The base of all machinery, generators will be paved and all the waste/spill will be drained to oil interceptor before discharging. Figure of oil interceptor is presented below.                                                    |
| Operation of the emulsion sprayer and laying of hot mix                                                                                                                                                                               | Proper demarcation of the surface to be sprayed /paved will be done to minimize the excessive spread of emulsion/hot mix.                                                                                                           |
| Operation of the residential facilities for the labour and officers                                                                                                                                                                   | The residential facilities will be provided with proper sanitation, and planned setup of construction camp. A comprehensive plan of construction camp is prepared in <b>APPENDIX # E5: SITING &amp; LAYOUT OF CONSTRUCTION CAMP</b> |
| Storage and stock yards of bitumen and emulsion                                                                                                                                                                                       | The base of bitumen/emulsion stock yard will be paved and all the waste/spill will be drained to oil interceptor before discharging. Figure of oil interceptor is presented below                                                   |

**Oil Interceptor:** Oil and grease from polluting run-off is another major concern. During construction, discharge of Oil and Grease is most likely from work shops, oil and waste oil storage areas, diesel oil pumps, vehicle parking areas from the construction camps. Vehicle/machinery and equipment maintenance and refuelling will be carried out so that spillage of fuels and lubricants do not contaminate the soil. The source is well defined and restricted. An “oil interceptor” will be provided for wash down and refuelling areas. Fuel storage will be in proper bunded areas. All spills and collected petroleum products will be disposed off in accordance with MoEF and SPCB guidelines. Fuel storage and fuelling areas will be located at least 300m from all cross drainage

structures and significant water bodies. **Drawing 8.1** provides the details of the arrangement for the oil interceptor for the removal of oil and grease.



**Drawing No. 8.1: Conceptual plan of Oil Interceptor**

### 8.3.6 Soil Quality Monitoring - mitigation

Apart from provision of the mitigation measures, their effectiveness and further improvement in designs to reduce the concentration of pollutants in the soil due to construction activity shall be monitored, as contamination of soil is directly linked with contamination of water. The monitoring plan shall be functional in construction as well as in operation stages. The frequency, duration and responsibility will be as per the **APPENDIX # E10: ENVIRONMENTAL MONITORING PLAN**. Any value/result not within acceptable limits will be reported to engineer, for remedial measures.

## 8.4 WATER ENVIRONMENT-MITIGATION MEASURES

Due to the proposed project there will be some direct and indirect long term impacts on the water resources. Table below presents the major adverse impacts on the water resources and the mitigation measures taken.

| Sr. No. | Item                          | Impact               | Impact (Reason)                                                                                | Mitigation/Enhancement                                                                                                                                                                    |
|---------|-------------------------------|----------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Loss of water Bodies          | Major, direct impact | Part or complete acquisition of source of water                                                | Land acquisition to be minimized with provision of Retaining walls.<br>Relocation of ground/surface water sources.                                                                        |
| 2       | Alteration of Cross Drainage  | Very Low Impact      | One major bridge over existing causeway.<br>Widening of minor bridges and culverts.            | widening of minor bridges and bridging of existing causeways, there will be an improvement in the drainage characteristics of the project area                                            |
| 3       | Runoff and drainage           | Direct Impact        | Siltation of water bodies<br>Reduction in ground recharge.<br><br>Increased drainage discharge | Silt fencing to be provided.<br>Recharge well to be provided to compensate the loss of pervious surface.<br>Continuous drain is provided, unlined in rural area and lined in urban areas. |
| 4       | Water requirement for project | Direct Impact        | Water requirement for construction activity.<br>Water requirement of labour.                   | Contractor needs to obtain approvals for taking adequate quantities of water from surface and ground water sources. This is required to avoid depletion of water resources                |

| Sr. No. | Item                       | Impact                | Impact (Reason)                                                                                                                                                                                                                                                                                                          | Mitigation/Enhancement                                                                                                                                                                                                                                                                     |
|---------|----------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5       | Water Quality              |                       |                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                            |
| a       | Increased sedimentation    | Direct impact         | increased sediment laden runoff alter the nature & capacity of the watercourse                                                                                                                                                                                                                                           | Silt fencing to be provided.<br>Instructions given in Appendix 8.9: Guidelines for sediment control to be enforced.                                                                                                                                                                        |
| b       | Contamination of Water     | Direct adverse impact | <ul style="list-style-type: none"> <li>Scarified bitumen wastes</li> <li>Oil &amp; diesel Spills</li> <li>Emulsion sprayer and laying of hot mix</li> <li>Production of hot mix and rejected materials</li> <li>Residential facilities for the labor and officers</li> <li>Routine and periodical maintenance</li> </ul> | Hazardous Wastes (Management and Handling) Rules, 1989 to be enforced.<br>Oil Interceptor will be provided for accidental spill of oil and diesel.<br>Rejected material will be layed in village roads or as directed by engineer.<br>Septic tank will be construction for waste disposal. |
| 6       | Water quality - monitoring |                       | Effectiveness / shortfall (if any)<br>Any unforeseen impact                                                                                                                                                                                                                                                              | Measures will be revised & improved to mitigate/enhance environment due to any unforeseen impact.                                                                                                                                                                                          |

*The implications of accidental discharge are potentially disastrous. But, it must be emphasized that the probability of such an accident is quite low, as one of the objectives of the design is the enhancement of road safety.*

#### 8.4.1 Loss of Water Bodies-Mitigation

##### 8.4.1.1 Surface Water Bodies

Samalkota Canal is running parallel to the project road almost all along, at places the canal is very close to the project road and at Samalkota Bypass the proposed bypass crosses the Samalkota Canal 7 times. Hence due the proximity of the canal with the project road, calls for major avoidance measures to check the runoff into the canal for the degradation of the canal water. Consultant has carried out the water quality test on the existing quality of the canal water, presented in Chapter 4 baseline environment and measures are adopted for the regular checking during construction and operation.

**Provision of silt fencing, precautions during construction for spilling of waste near canal are adopted for the prevention of degradation of canal water.** Temporary erosion/sedimentation and pollution control measures will be used to control the phenomenon of erosion, sedimentation and pollution that may develop during normal construction practice. Details of prevention of SOIL EROSION AND SEDIMENTATION CONTROL are presented in Appendix E-9 of this report. Mitigation measures for the remaining four Surface water bodies are present in table 8.2 below.

**Table 8.2: Surface Water Bodies-Mitigation**

| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) | Distance for PCL (in mts) | Impact           | Mitigation/Enhancement                            |
|---------|-------------------------|----------------|-------------------------|----------------------------|---------------------------|------------------|---------------------------------------------------|
| 1       | Pond                    | 15+600         | Right                   | 15                         | 10.62                     | No Direct Impact | tank platform                                     |
| 2       | Pond                    | 18+950         | Right                   | 6.5                        | 6.5                       | No Direct Impact | Protection Works                                  |
| 3       | Ponds                   | 24+800         | Right                   | 4.5                        | 4.43                      | Direct Impact    | Equivalent volume of soil will be dug./ relocated |
| 4       | Large Water Body        | 34+670         | Right                   | 4.5                        | 4.5                       | Direct Impact    | Equivalent volume of soil will be dug./ relocated |

##### 8.4.1.2 Ground water resources

Loss of other water supply sources includes; removal of private and community ground water sources like bore wells, tube wells and open wells. These losses have been covered under the utility relocation process in the pre-construction phase of the EMP. Compensatory water supply sources will be set up before the start of construction activities.

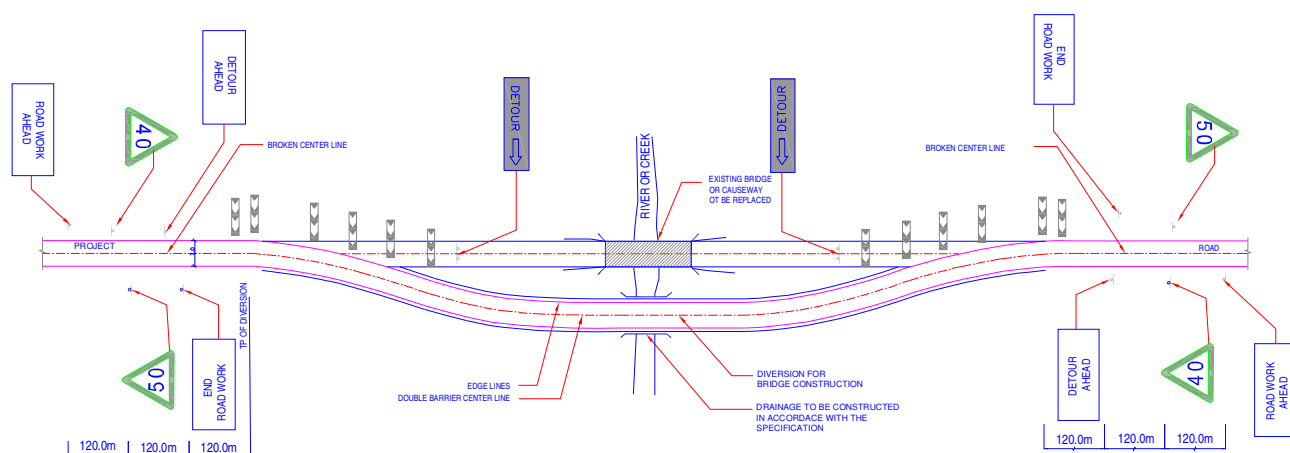
Table 8.3: Ground water Sources-mitigation

| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) | Distance for PCL (in mts) | Impact           | Mitigation/ Enhancement                |
|---------|-------------------------|----------------|-------------------------|----------------------------|---------------------------|------------------|----------------------------------------|
| 1       | Hand Pump               | 2+300          | Right                   | 6                          | 2.03                      | Direct Impact    | Will be relocated to the nearest place |
| 2       | Well                    | 3+000          | Left                    | 9.5                        | 13.48                     | No Direct Impact |                                        |
| 3       | Old Wells               | 27+050         | Right                   | 25.5                       | 22.08                     | No Direct Impact |                                        |
| 4       | Well                    | 39+300         | Right                   | 18.5                       | 14.57                     | No Direct Impact |                                        |
| 5       | Hand Pump               | 41+550         | Left                    | 4.0                        | 4.0                       | Direct Impact    | Will be relocated to the nearest place |
| 6       | Hand Pump               | 54+200         | Right                   | 7.0                        | 7.0                       | No Direct Impact |                                        |
| 7       | Well                    | 56+400         | Left                    | 4.5                        | 4.5                       | Direct Impact    | Will be relocated to the nearest place |
| 8       | Well                    | 57+100         | Left                    | 13.5                       | 13.5                      | No Direct Impact |                                        |

Any source of community water (potable or otherwise) such as open wells, ponds, tube-wells and bore wells, accidentally lost will be replaced immediately. The location and sitting of the replaced source of water will be as close as possible to the original source.

#### 8.4.2 Alteration of cross drainage

- All cross drainage structures have been designed to handle a 50-year peak flood level. A detailed hydrological study had been carried to calculate the design discharge.
- Pipe drainages will be provided for diversion roads constructed for the construction of new bridges and culverts. (refer drawing 8.2)
- Storm water from all longitudinal and Cross drainage works will be connected to the natural drainage courses.
- The contractor will remove obstructions that may cause temporary flooding of local drainage channels, during the construction phase.
- Contractor will be responsible for removal of debris generated due to the dismantling of structure and earth generated due to the excavation of foundation, from the water course before the onset of monsoon. For list of bridge under reconstruction/new proposed bridges (refer table 6.11)



Drawing No. 8.2: Conceptual Plan of Diversion at Bridge Location



**8.4.3 Runoff and drainage**

- Continuous Drain (lined/unlined) is provided throughout the project road for efficient drainage of storm water.
- Lined drain is provided at builtup sections for quick drainage of storm water.
- The increased runoff due to increased impervious (Bituminous Top) surface will be countered with increased pervious surface area through soak pits at surface and ground resources.
- The local bodies need to discourage/ stop the filling of private water bodies, ponds etc to develop commercial places and shops due to the improved roads and improved connectivity.

**8.4.4 Water Requirement for Project**

Acquisition and management of water for construction is an issue that must be addressed by the contractor. The Contractor is expected to obtain water for construction purposes that is of a high water quality.

- Contractor needs to obtain approvals for taking adequate quantities of water from surface and ground water sources. This is required to avoid depletion of water resources.
- Contractor is required to minimise wastage of water.
- Water conservation methods to be adopted during construction process to make optimum use of water, as limited water sources are available along the PR.

**8.4.5 Water Quality**

## a) Increased sedimentation

- Silt fencing will be provided.
- Desilting will be done for the existing ponds/Tanks.
- Desilting of the waste water will be ensured before the discharge of drain water into natural stream.
- Instructions given in **APPENDIX E9: SOIL EROSION AND SEDIMENTATION CONTROL** to be enforced.

**Table 8.4 List of silt fencing locations**

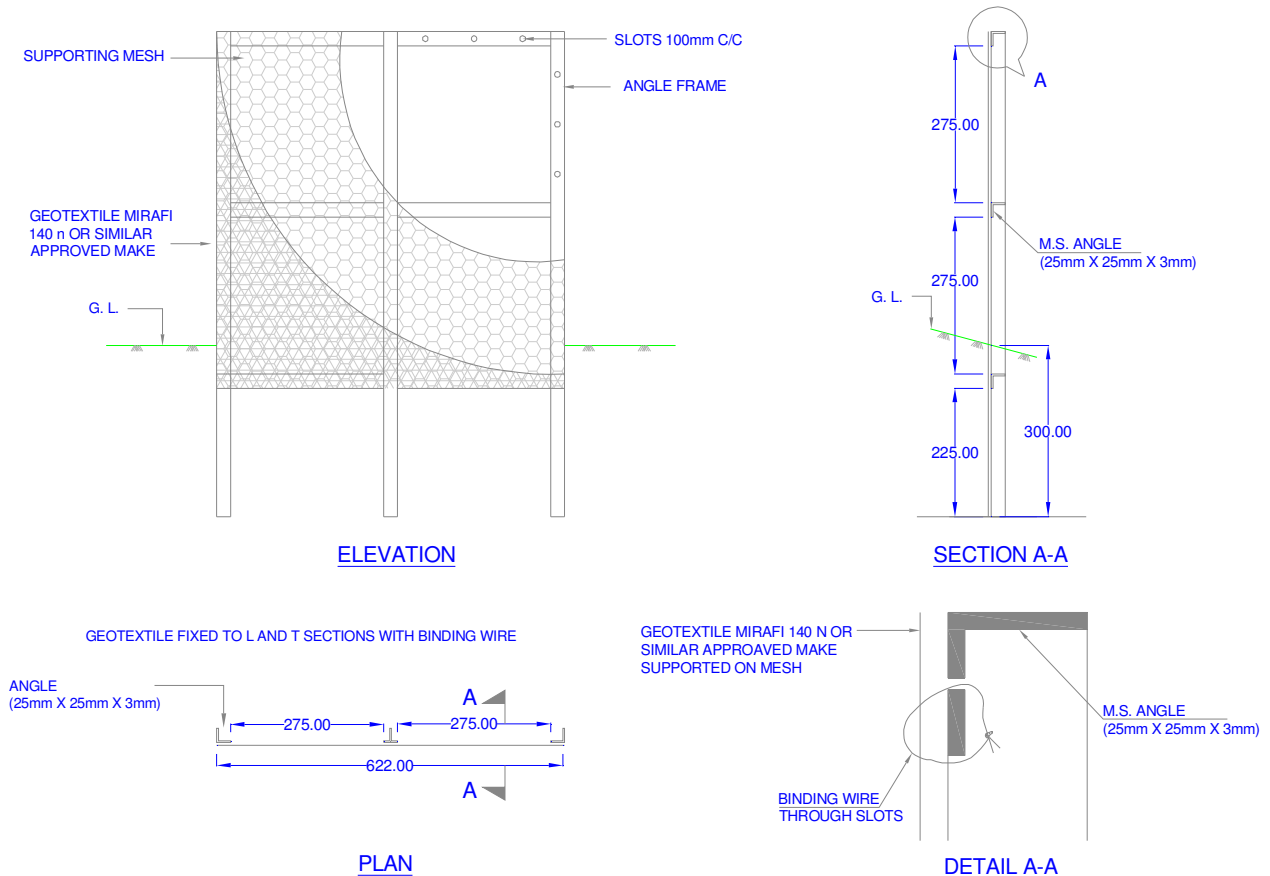
| Sl. No. | Location             | Length (approximate) | Remark                                                  |
|---------|----------------------|----------------------|---------------------------------------------------------|
| 1       | Top soil storage     | 300m                 | Around the periphery of dumping yard                    |
| 2       | Borrow area site     | 100m                 | Along the length of borrow area in down slope direction |
| 3       | Surface Water bodies | 500m                 | Equals to diameter/length or as instructed by engineer. |
| 4       | Plant Site           | 100m                 | In down slope direction of fine material                |
| 5       | canal                | 1000m                | Along the canal where ever it is required               |

## b) Contamination of water.

- Oil interceptor will be provided at plant site and truck lay byes.
- Construction work close to the streams or water bodies will be avoided during monsoon.
- The discharge standards promulgated under the Environmental Protection Act, 1986 will be strictly adhered to. All wastes arising from the project will be disposed off in a manner that is acceptable to the State Pollution Control Board (SPCB).
- All relevant provisions of the Factories Act, 1948 and the Building and other Construction Workers (regulation of Employment and Conditions of Service) Act, 1996 will be adhered to.



- Construction labourers' camps will be located at least 1000m away from the nearest habitation.
- Unless otherwise authorised by the local sanitary authority, arrangements for proper disposal of excreta by incineration at the workplace suitably approved by the local medical health or municipal authorities will be made.
- All approach roads to rivers and other surface water bodies need to be closed permanently to avoid vehicle washing and to avoid major pollution sources. This applicable to all areas including the secondary construction sites.
- Automotive service centres will be discouraged from establishing along the corridors without installing preventive measures against petroleum and oil contamination.



**Drawing No. 8.3: Conceptual plan of silt fencing.**

#### 8.4.6 Water Quality Monitoring - Mitigation

Apart from provision of the mitigation measures, their effectiveness and further improvement in designs to reduce the concentration of pollutants in the soil due to construction activity shall be monitored, as contamination of soil is directly linked with contamination of water. The monitoring plan shall be functional in construction as well as in operation stages. The frequency, duration and responsibility will be as per the **APPENDIX # E10: ENVIRONMENTAL MONITORING PLAN**. For location refer table on Ambient Water Quality Monitoring Locations including one at Hot mix plant site and standard/acceptable values are given in table on National Ambient Water Quality Standards. Any value/result not within acceptable limits will be reported to engineer, for remedial measures.

#### 8.5 NOISE ENVIRONMENT-MITIGATION MEASURES

Environmental noise particularly highway traffic noise, is a complex phenomenon because its intensity and characteristics vary with time depending upon the frequency as well as type of vehicles on the road.

| Sr. No. | Item                                     | Impact                        | Impact (Reason)                                                                                                                  | Mitigation/Enhancement                                                                                                                                                                                               |
|---------|------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Sensitive receptors                      | Direct impact                 | Increase in noise pollution                                                                                                      | Noise barrier to be provided<br>Traffic calming devices to be used.<br>NO Horn Zone sign Post.                                                                                                                       |
| 2a      | Noise Pollution (Pre-Construction Stage) | Direct impact, short duration | Man, material & machinery movements<br>Establishment of labor camps onsite offices, stock yards and construction plants          | Area specific and for short duration<br>Machinery to be checked & complied with noise pollution regulations.<br>Camps to be setup away from the settlements, in the down wind direction.                             |
| 2b      | Noise Pollution (Construction Stage)     | Marginal Impact               | stone crushing, asphalt production plant and batching plants, diesel generators etc<br>Community residing near to the work zones | Camps to be setup away from the settlements, in the down wind direction.<br>Noise pollution regulation to be monitored and enforced.<br>Temporary as the work zones will be changing with completion of construction |
| 2c      | Noise Pollution (Operation Stage)        | Marginal Impact               | due to increase in traffic (due to improved facility)                                                                            | will be compensated with the uninterrupted movement of heavy and light vehicles till the facility reaches the level of service C.                                                                                    |
| 3       | Noise Pollution Monitoring               |                               | Effectiveness / shortfall (if any)<br>Any unforeseen impact                                                                      | Measures will be revised & improved to mitigate/enhance environment due to any unforeseen impact.                                                                                                                    |

### 8.5.1 Sensitive Receptors – Mitigation

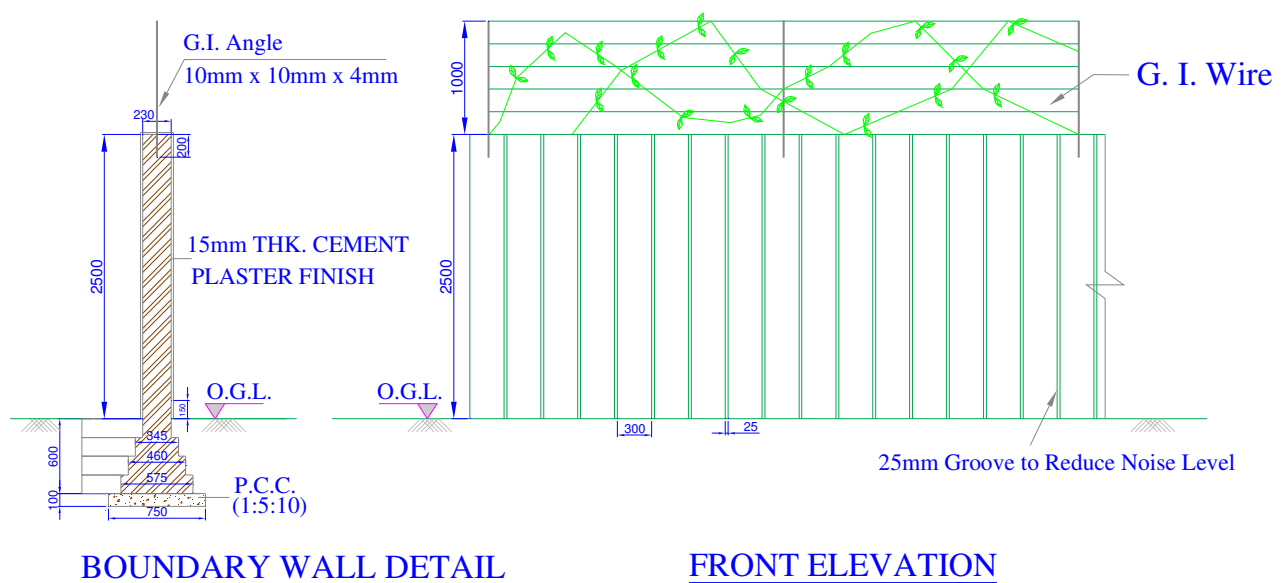
All schools, hospitals and cultural properties have been identified and those that are close to the project roads may require noise barrier to accommodate the long term impact of the improved road.. List of sensitive receptors along the project corridor is presented in table 8.5 below

**Table 8.5 List of Sensitive Receptor**

| Sl. No. | Name of Common Property               | Chainage (Km.) | Location (Left/Right) | Distance from ECL (in mts) | Distance for PCL (in mts) | Impact           | Mitigation/ Enhancement                                                                                                                                                                                                                   |
|---------|---------------------------------------|----------------|-----------------------|----------------------------|---------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | PRG College (BW)                      | 0+200          | Right                 | 12                         | 14.4                      | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) |
| 2       | Hospital (BW)                         | 0+600          | Left                  | 12                         | 11.18                     | No Direct Impact |                                                                                                                                                                                                                                           |
| 3       | Rangaraya Medical College (BW)        | 0+900          | Right                 | 15                         | 4 Lane                    | No Direct Impact |                                                                                                                                                                                                                                           |
| 4       | Ideal College Of Arts & Sciences (BW) | 1+600          | Right                 | 5.5                        | 0.72                      | No Direct Impact |                                                                                                                                                                                                                                           |
| 5       | Saraswathi Vidyapita                  | 5+700          | Right                 | 23.5                       | 27.96                     | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) |
| 6       | School                                | 17+950         | Right                 | 6                          | 6.04                      | No Direct Impact |                                                                                                                                                                                                                                           |
| 7       | School                                | 30+200         | Right                 | 5.5                        | 2.93                      | No Direct Impact |                                                                                                                                                                                                                                           |
| 8       | Private Hospital                      | 41+500         | Right                 | 5.5                        | 6.17                      | No Direct Impact | Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4)                                                                                                                                                        |
| 9       | Veterinary Hospital (BW)              | 57+300         | Right                 | 4.5                        | 4.97                      | Direct Impact    |                                                                                                                                                                                                                                           |
| 10      | Hospital (BW)                         | 57+800         | Left                  | 7.5                        | 7.92                      | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) |
| 11      | School                                | 57+850         | Left                  | 23.5                       | 23.67                     | No Direct Impact |                                                                                                                                                                                                                                           |
| 12      | School (BW)                           | 60+050         | Right                 | 5                          | 3.31                      | Direct Impact    | Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4)                                                                                                                                                        |

| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left/Right) | Distance from ECL (in mts) | Distance for PCL (in mts) | Impact           | Mitigation/ Enhancement                                                                                                                                                                                                                   |
|---------|-------------------------|----------------|-----------------------|----------------------------|---------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13      | School                  | 61+200         | Right                 | 153.5                      | 154.46                    | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) |
| 14      | School (BW)             | 61+350         | Right                 | 53.5                       | 53.58                     | No Direct Impact |                                                                                                                                                                                                                                           |

Consultant carried out a preliminary level consultation with the hospital and school authority, however the final conclusion on the dimensions of the Noise Barrier could not be reached hence consultant proposes a detailed consultation with the authorities prior to construction of Noise barrier during construction stage and accordingly works to be carried out as per the Day works items in cost estimate.



**Drawing No. 8.4: Conceptual Drawing for Noise Barrier**

### 8.5.2 Noise Pollution - Mitigation

- Noise standards will be strictly enforced for all vehicles, plants, equipment, and construction machinery. All construction equipment used for an 8-hour shift will conform to a standard of less than 90dB(A). If required, high noise producing generators such as concrete mixers, generators, graders, etc. must be provided with noise shields.
- Machinery and vehicles will be maintained regularly, with particular attention to silencers and mufflers, to keep construction noise levels to minimum.
- Workers in the vicinity of high noise levels will be provided earplugs, helmets and will be engaged in diversified activities to prevent prolonged exposure to noise levels of more than 90dB(A) per 8 hour shift.
- During construction vibratory compactors will be used sparingly within the urban areas. In case of complaints from roadside residents, the engineer will ask the site engineer to take suitable steps of restricting the work hours even further or use an alternative roller.
- Proposed tree and shrub plantations planned for avenue plantation especially close to settlements, may form an effective sound buffer during the operation stage.

- People will be convinced / educated to prevent sensitive land uses from developing up adjacent to the project corridors.

### 8.5.3 Noise Pollution Monitoring - Mitigation

Apart from provision of the mitigation measures, their effectiveness and further improvement in designs to reduce the concentration of pollutants in the soil due to construction activity shall be monitored, as contamination of soil is directly linked with contamination of water. The monitoring plan shall be functional in construction as well as in operation stages. The frequency, duration and responsibility will be as per the **APPENDIX # E10: ENVIRONMENTAL MONITORING PLAN**. For location refer table on Ambient Water Quality Monitoring Locations including one at Hot mix plant site and standard/acceptable values are given in table on National Ambient Water Quality Standards. Any value/result not within acceptable limits will be reported to engineer, for remedial measures.

## 8.6 FLORA & FAUNA - MITIGATION

The major impact in this project on flora involves the removal of trees to permit construction and to provide clear zone for safety of the road users. There is NO recorded wild life habitat in near vicinity of the project corridor.

Trees located within the clear zone need to be removed to prevent collision with the trees, in case of accident. Roadside trees with strong and rigid stems can pose safety hazards. Some trees obstruct clear sight distances. Others have a propensity to overturn when old and are potential safety hazards depending upon age and decay condition. All such trees that are safety hazards need to be cleared.

| Sr. No. | Item           | Impact        | Impact (Reason)                                                                                                         | Mitigation/Enhancement                                                                                                                                                                                                                                            |
|---------|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Forest area    | Direct Impact | Forest land to be acquired                                                                                              | Land to be diverted, application submitted to Forest Department , presented separately as <b>ANNEXURE TO MAIN REPORT</b>                                                                                                                                          |
| 2       | Wild Life      | No Impact     | No wild life habitat                                                                                                    | Nil                                                                                                                                                                                                                                                               |
| 3       | Trees Cutting  | Direct impact | Increase in soil erosion, silting of water bodies, Dust & noise pollution.<br><br>loss of shade & loss of tree products | Maximum deviation in alignment design to save the road side trees.<br>Compulsory tree plantation in the ratio of 1:2, i.e. for each tree cut, two saplings will be planted.<br>Avenue plantation along corridor.<br>Endangered Tree species will be transplanted. |
| 4       | Vegetation     | Direct Impact | Increase in soil erosion, silting of water bodies, noise pollution. Dust Pollution                                      | Clearing and grubbing will be minimized, and sprinkled with water to reduce dust pollution.<br>Exposed surface like embankment slopes will be protected with stone pitching and turfing.<br>Open land in and around plant will be vegetated.                      |
| 5       | Cattle Grazing | No Impact     | No cattle grazing found                                                                                                 | Nil                                                                                                                                                                                                                                                               |

### 8.6.1 Forest Area – Mitigation

No forest land.

### 8.6.2 Wild Life - Mitigation

NO wild life habitat/wild life crossing seen along the project corridor.

### 8.6.3 Trees – Mitigation

The mitigation and enhancement measures taken along the project corridor includes,

- Compensatory Afforestation: to compensate the felling of trees.
- Avenue Plantation
- Plantation at enhancement sites

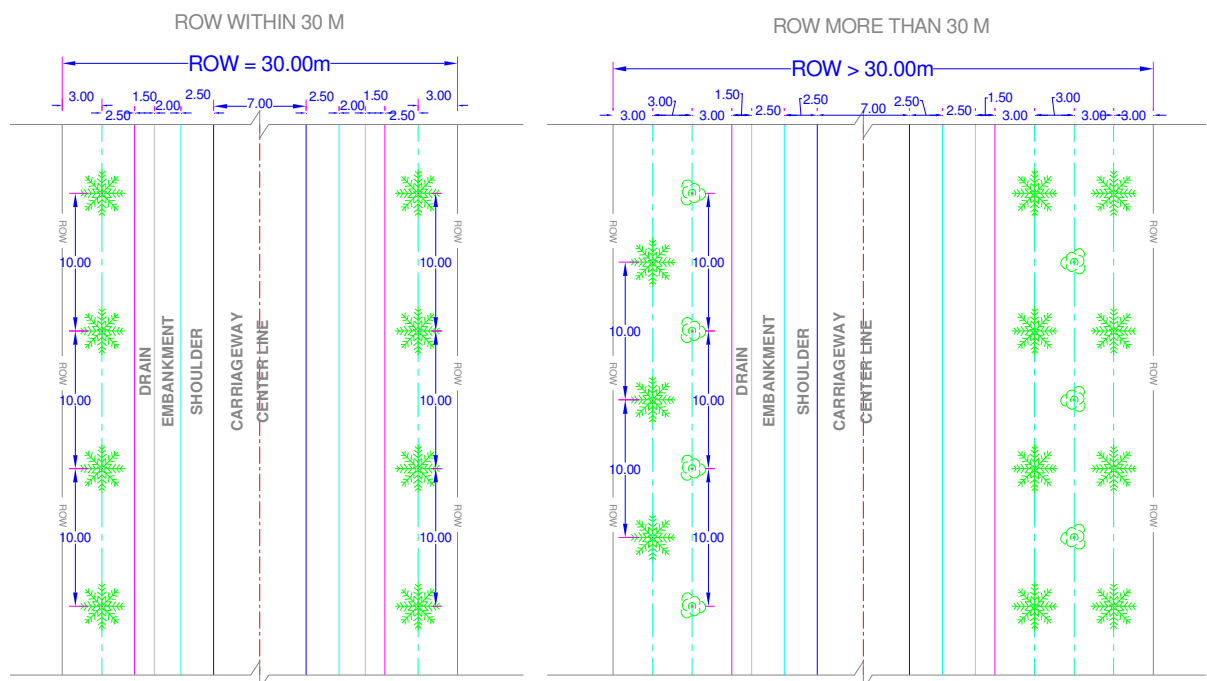
d) Landscaping at Junctions and realignment locations etc.

**Compensatory Afforestation:** Compensatory afforestation will be taken up as per the Forest (Conservation) Act, 1980.

- To minimise loss of trees, clearance of only those trees identified from the design will be removed. The felling of trees has been minimised as much as possible by having the centreline adjusted so as to necessitate felling of trees from only one side, in case of green tunnel.
- Endangered species, if found during construction, will be transplanted.
- No tree will be removed in the zone of construction without the prior approval of the DoF, through the supervising engineer.
- A total of 4162 trees need to be uprooted along the alignment because of project intervention.

**Avenue Plantation:** In addition to the compensatory afforestation, trees shall be planted all along the project corridors under avenue plantation, by the APRDC through Department of Forest. For this purpose RDC should sign a MOU with Department of Forest. Such plantations will be initiated once the construction is complete. The objective behind such plantation is to cover/ revegetate the areas within the RoW that are at presently barren. To maintain the present character of strip plantation, similar indigenous trees should be planted.

- Considering the combined compensatory afforestation by DoF and the plantation by the APRDC under avenue plantation, a total of 8450 saplings will be planted along the corridor in single row.
- Costing has been done as per forest schedule of rate, including the plantation and maintenance cost for 3 years.

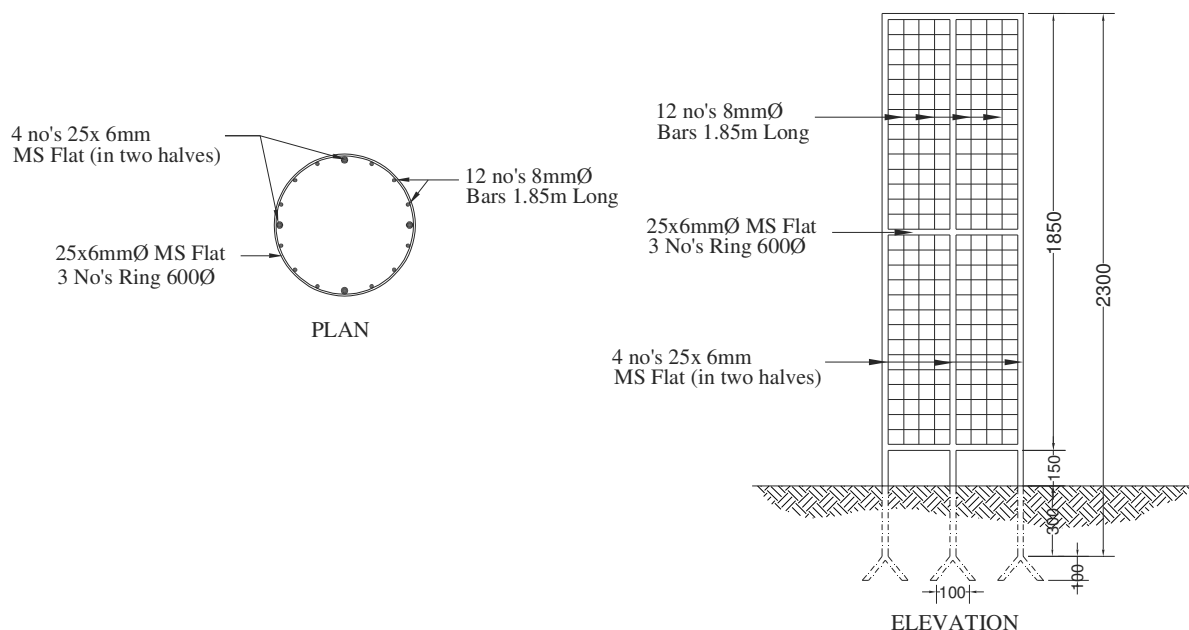


**Drawing No. 8.6: Conceptual Plan for Avenue Plantation**

**Plantation at Enhancement sites:** A number of cultural properties exist along the project corridor, which attracts large people gatherings throughout the year. Landscape design has been worked out to enhance the aesthetic beauty of selected sites. Trees like *Cassia marginata*, *Cassia fistula* and *Delonix regia* may be planted in such places. Tall growing varieties with straight stem such *Arjuna myrabalam* may be planted at selected spots to serve as effective landmarks.

- A total of 30 trees will be planted along the realignment locations as per drawings, Drawing folder.
- A total of 70 trees will be planted around cultural properties, as part of the enhancement. Refer **APPENDIX E12: MITIGATION/ENHANCEMENT OF SENSITIVE CULTURAL PROPERTIES**.

**Landscaping at Junctions and medians etc:** No trees are planted up to 50m before the intersections so as to achieve clear sight distances. For intersections, shrubs will be planted at 1.5m c/c. Beyond this point only dwarf shrubs are planted. The height of these shrubs should be such that they do not obstruct the view of traffic coming from the intersecting road. The plants recommended for intersections are *Celodendron inerme*, *Antirrhinum*, *Browallia*. The intersections where landscaping is proposed is presented in the individual EMPs. The supervision consultant shall ascertain availability of space in the junctions mentioned below and plantation shall be done in the available space.



**Drawing No. 8.7: Conceptual Plan for Tree Guard**

#### 8.6.4 Vegetation - Mitigation

- High embankment will be re-vegetated with local shrubs and grasses to prevent soil erosion from the bare earth, prior to the monsoon.

#### 8.6.5 Cattle Grazing - Mitigation

No cattle grazing seen along the corridor.

### 8.7 SOCIO-ECONOMIC ENVIRONMENT - MITIGATION

Adverse socio-economic impacts include all disruptions on the social and economic interactions of communities due to the road project. This involves effect on both the adjacent communities (mostly direct) as well as the nearby communities (mostly indirect). The various impacts have been detailed as:

- General impacts that apply to the entire corridor,
- Specific impacts on likely properties and PAPs, within the Corridor of Impact (CoI) of the project corridor.

## 8.7.1 General Impacts - Mitigation

| Sr. No. | Item                                   | Impact                     | Impact (Reason)                                                                                                                                                                                                                | Mitigation/Enhancement                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|----------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Fear of uncertainties regarding future | Direct, long Impact        | Land and property owners are subjected to sufferings regarding uncertainties of the extent of loss and the nature of compensation                                                                                              | Public participation sessions were/will be conducted in different stages of project.                                                                                                                                                                                                                                                                                                                                                                                           |
| 2       | Inducement of land prices              | Direct impact              | danger of unscrupulous speculators moving in to purchase land                                                                                                                                                                  | Market Value Assessment Committee will decide the actual cost of land                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3       | Inducement of squatter influx          | Direct impact              | Squatters may attempt to occupy adjacent land in the hope of receiving compensation undue pressure on local resources                                                                                                          | The dates of base-line socio-economic survey have been considered as cut off date for identification of project affected people PAP.                                                                                                                                                                                                                                                                                                                                           |
| 4       | Loss of utilities and amenities        | Direct Impact              | Natural (trees, bushes and grasslands), and Physical structures (public or private assets and utilities).                                                                                                                      | Relocation of utilities will be completed prior to start of project work. These have been further discussed in details in RAP                                                                                                                                                                                                                                                                                                                                                  |
| 5a      | Public Health and Safety               | High direct adverse impact | Psychological impacts on their owners and others associated with them.<br><br>Debris generated                                                                                                                                 | Advance notice as per RAP(4 months before commencement of work) will be given to the owners of the affected properties.<br>Debris, so generated will be disposed to the satisfaction of Engineer. Refer <b>Appendix E3: Debris Disposal Site</b> .<br>Monitoring of air , water ,noise and land during construction and operation phase. Refer <b>Appendix E10: Environment Monitoring Plan</b> .<br>Refer <b>Appendix E7: Environment Friendly construction Methodology</b> . |
| 5b      | Labour Camps                           | Direct Impact              | can have clashes with the local population<br>Pressure on basic facilities like medical services, power, water supply, etc<br>Transmission of communicable diseases including aids.<br>sanitary conditions in the labour camps | All contractors will be encouraged to recruit the local people as labourer at least for unskilled and semi-skilled jobs.<br><br>Hygiene and basic facilities will be ensured at labour camp to prevent the spread of disease. Refer <b>Appendix E5</b> sitting and layout of construction Camps, and <b>Appendix E6</b> Workers Safety during construction Activities.                                                                                                         |
| 5c      | Allied activities                      | Indirect Impact            | social and economic life of the local population due to quarrying and crushing operations, traffic diversions, etc<br>traffic jams and congestion, loss of access and other road accident risks<br>temporary land acquisition  | Detailed traffic control plans shall be prepared and submitted to the engineer for approval 5 days prior to commencement of work on any section of road.                                                                                                                                                                                                                                                                                                                       |
| 5d      | Accidents and Safety                   | Direct Impact              | School children<br>ladies carrying pots full of water                                                                                                                                                                          | The contractor will provide, erect and maintain barricades, including signs marking flags lights and flagmen as required by the Engineer.                                                                                                                                                                                                                                                                                                                                      |
| 6       | Resettlement of People                 | In direct impact           | Pressure on civil amenities, water sources, grazing lands, fuel wood, medical facilities etc.                                                                                                                                  | A comprehensive resettlement action plan has been prepared to improve the standard of living of the affected population                                                                                                                                                                                                                                                                                                                                                        |
| 7       | Land Use Changes                       | Indirect impact            | Succession of land uses and higher return uses would displace the lower return uses at major intersections and in settlement areas.<br>Urban fringe areas will be subjected to ribbon development.                             | Project is widening of existing of pavement, hence no major change in land use pattern is envisaged along the project road.                                                                                                                                                                                                                                                                                                                                                    |
| 8       | Disturbance to road side services      | Indirect Impact            | Some Shops may be shifted, no income from highway users hence loss of service to the local people                                                                                                                              | The cleaning of such informal establishment will be carried out as phase-wise resettlement Programme                                                                                                                                                                                                                                                                                                                                                                           |
| 9       | Removal of encroachments and           | Indirect impact            | loss of shelter disturbance to family and community life                                                                                                                                                                       | Impact will be avoided by implementation of phase-wise resettlement action                                                                                                                                                                                                                                                                                                                                                                                                     |



| Sr. No. | Item                           | Impact          | Impact (Reason)                                                                                                                       | Mitigation/Enhancement                                                                               |
|---------|--------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|         | squatters                      |                 |                                                                                                                                       | Programme Notice will be served 4 months in advance.                                                 |
| 10      | Sensitive community facilities | Indirect impact | Loss of community facilities or institutions                                                                                          | Discussed in detail in following paragraphs.                                                         |
| 11      | Host Community                 | Indirect Impact | Displaced, resettled families will put pressure on the existing scarce infrastructure and amenities available with the host community | special provision has been made in the comprehensive resettlement action plan for the host community |

#### 8.7.1.1 Fear of uncertainties regarding future

The Project is only of widening and strengthening of the existing road, and within the confines of the existing RoW, the fear of uncertainties is most likely limited to the people squatting and encroaching in the RoW. At places where the community utilities are to be affected a certain amount of anxiety will be among the people in that particular community.

To remove such fear from the people, public participation sessions were/will be conducted in different stages of project, viz. pre-design, design, pre-construction and construction.

In the pre-design stage, a comprehensive socio-economic survey was conducted to prepare base line status of the households squatting or encroached upon the RoW. During the survey each households were contacted/interviewed and they were explained about the purpose of the survey, need of the project and benefits associated with the project etc.

The second stage of public participation was in the form of village meetings, focus group discussion, individual interviews, voluntary and academic institution consultation.

The third stage of participation session will start before pre-construction. The EMU will verify and consult the individual EPs with the help of an NGO for distribution of ID card. At this stage the EPs will be explained about their entitlement and R&R framework.

All people likely to be displaced will be informed in advance through NGO by a time bound Programme about resettlement to remove fear of uncertainty.

During the construction stage the consultation process will continue to avoid any inconvenience to the community at any point of time.

#### 8.7.1.2 Inducement of Land Prices

As the project becomes common knowledge, the land prices along the corridor will increase. For realignment and geometric improvement of highway, extra additional land may be required; Market Value Assessment Committee will decide the actual cost of such land. The actual cost of land may be different from induced land cost. The MVAC have time bound Programme to calculate the actual cost of land. In calculating the actual land cost individual project affected person (PAP) & NGOs will be involved.

#### 8.7.1.3 Inducement of Squatter Influx

Once the project becomes common knowledge, people may attempt to occupy the land along the corridor in anticipation of compensation. To avoid such, the dates of base-line socio-economic survey have been considered as cut off date for identification of project affected people, who are eligible for compensation. The cut off dates will be used to establish whether a person located in the right way qualifies as a PAP for the disbursement of compensation. All the PAPs recorded during socio-economic baseline survey are eligible for compensation after verification by EMU.

#### 8.7.1.4 Loss of Utilities and Amenities

The site clearance for construction of road may result in loss or relocation of certain utilities and amenities, viz. electricity, water and telephone line etc. People dependent upon these utilities and amenities may experience inconvenience and economic loss. Though such impacts are unavoidable keeping in mind the scale of work, but every care will be taken in co-ordination with concerned

departments, to restore the facility within shortest possible time to avoid any prolonged hardship or inconvenience to the community. Similarly other utilities like water source, cattle trough etc. will be constructed or replaced at appropriate place with the consent of community prior to dismantling the existing one.

#### 8.7.1.5 Public Health and Safety

- a) During the Pre-construction and Construction Phases dismantling of the structures for CoI clearance and road construction may result in health hazards. To minimise this potential negative impact the following recommendations should be adopted:
  - To avoid the psychological impacts due to the demolition of properties on the owners and other tenants. The advance notice as per RAP will be given to the owners of the affected properties. An advance notice will be served at least four months before construction commences. For squatters needing relocation, all R&R activities will be undertaken and entitlements will be completed before construction starts.
  - Debris generated from the demolition of properties will be properly disposed of to avoid the health problems in the safeties. Earth material, if required will be dumped in borrow areas as approved by the engineer. Borrow areas will be filled to avoid health hazards from stagnant water collecting in these areas. The contractor will make all arrangements for dismantling and cleaning up of debris. Implementation will be as per the approval and direction of the engineer.
  - Instructions as given in **APPENDIX E7: ENVIRONMENT FRIENDLY CONSTRUCTION METHODOLOGY**, to be enforced
- b) During the construction period the potential negative community impacts arising from imported labour in the labour camps will be avoided as per following: -

All contractors will be encouraged to recruit the local people as labourer at least for unskilled and semi-skilled jobs. This would automatically reduce the magnitude of impact expected due to outside labour. Wherever the local labourers are not available the contractor should ensure the following provision for imported labour.

- The additional stress on the facilities like medical services, power, water supply due to a labour camp in a local area will be avoided by the contractor providing these facilities for the labourers as per the direction of the Engineer
  - In areas where wildlife resources are abundant, construction workers shall be instructed how to protect natural resources, fauna, flora and aquatic life. In such areas hunting and unauthorised fishing are prohibited.
  - In the labour camps, all temporary accommodation must be constructed and maintained in such a fashion that uncontaminated water is available for drinking, cooking and washing. The sewage system for the camp will be properly designed built and operated so that no health hazard occurs. Garbage bins will be provided in the camp and regularly emptied. The collected garbage will be disposed off in a hygienic and approved manner.
  - Instructions as given in **APPENDIX E5 SITTING AND LAYOUT OF CONSTRUCTION CAMPS**, to be enforced.
  - Instructions as given in **APPENDIX E6 WORKERS SAFETY DURING CONSTRUCTION ACTIVITIES**, to be enforced.
  - Instructions as given in **APPENDIX E8 STORAGE, HANDLING, USE AND EMERGENCY RESPONSE FOR HAZARDOUS CHEMICALS**, to be enforced.
- c) Allied activities during construction period may cause local disruption:
 

In the construction phase, there may be inconvenience to the local people as well as the highway passengers due to traffic jams and congestion, loss of access and other road accident risk as a result of construction. Detailed traffic control plans shall be prepared and submitted to the

engineer for approval 5 days prior to commencement of work on any section of road. In the preparation of the traffic control plan special consideration shall be given to the safety of pedestrians and workers at night.

d) Accidents and Safety

To avoid the accidents during construction phase, contractor shall take all necessary measures to ensure traffic safety. The contractor will provide, erect and maintain barricades, including signs marking flags lights and flagmen as required by the Engineer.

In the operation phase, traffic control measures such as speed breakers and sign boards (including speed limits) will be provided and strictly enforced in residential areas, near schools and water bodies like ponds and wells.

#### 8.7.1.6 Resettlement of People

People displaced from their home and livelihood on account of the proposed activity will be taken care in the project. A comprehensive resettlement action plan has been prepared to improve the standard of living of the affected population if not at least restore their livelihood and regaining their former standard of living (Refer Resettlement Action Plan Report).

#### 8.7.1.7 Land Use Changes

As regards land use changes, it is likely that the impact would be very minimal. Since the project is widening of road within the existing ROW, the possibility of major land acquisition is not envisaged. In design stage utmost care has been taken to keep the land acquisition at minimal for road realignment and geometric purpose. The probable impact on road-side business and trees within ROW has been reduced and avoided through design.

#### 8.7.1.8 Disturbance to Roadside Service

Along the highway, near settlements, small shops serve the local people as well as highway users. Some of these shops are within the RoW. To avoid any impact on livelihood of people dependent on roadside business, the cleaning of such informal establishment will be carried out as phase-wise resettlement Programme. Loss of livelihood of the displaced shop owners will be compensated as per assistance or entitlements in Entitlement Framework of Andhra Pradesh.

#### 8.7.1.9 Removal of Encroachments and Squatters

The impact due to removal of encroachments and squatters will be avoided by implementation of phase-wise resettlement action Programme. To avoid the severance of impact, advance notice will be given to the encroachers and squatters. The notice will be given four months prior to construction phase starts. Relocation of all such impacted persons will be as per R&R schedule.

#### 8.7.1.10 Sensitive Cultural Properties

The impacts due to project on the sensitive community facilities or institution along the project corridor such as education institution, health facilities, a number of recreational facilities and others like ponds and cultural community assets avoided in the design phase of the project. These are discussed in detail in subsequent paragraphs.

#### 8.7.1.11 Host Community

Families displaced due to the project will be resettled at suitable place. This may put pressure on the existing scarce infrastructure and amenities available with the host community. To avoid such pressure on already scarce resources, viz. water, grazing land, fuel woods, and medical facility etc. special provision has been made in the comprehensive resettlement action plan for the host community. This will be applicable where the total number of resettled people exceeds 200 or 10 percent of the host community whichever is less (Refer: Resettlement Action Plan Report).

**8.7.2 Specific Impacts – Mitigation****8.7.2.1 Sensitive Cultural Properties**

A no. of sensitive Cultural properties/assets exists within the Corridor of Impact. Table below presents the impact and mitigation/enhancement proposed at each location. (Refer **APPENDIX E12: MITIGATION/ENHANCEMENT OF CULTURAL PROPERTIES**.) The Khediyar nursery near design chainage km 57.300 is under direct impact due to realignment. This nursery is known to be famous and is functioning for quite a long time as explained in detailed in chapter 7. In view of this the oxbow land which is formed due to realignment may be transferred to the nursery as a compensation. This option may be considered.

**Table 8.6 List of Cultural Properties**

| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) | Distance for PCL (in mts) | Impact           | Mitigation |
|---------|-------------------------|----------------|-------------------------|----------------------------|---------------------------|------------------|------------|
| 1       | Venkatesh Temple        | 1+800          | Right                   | 4.5                        | 1.07                      | Direct Impact    |            |
| 2       | Church                  | 2+000          | Right                   | 11                         | 7.12                      | Direct Impact    |            |
| 3       | Temple                  | 3+000          | Left                    | 9.5                        | 13.48                     | No Direct Impact |            |
| 4       | Church                  | 3+300          | Right                   | 7                          | 2.4                       | Direct Impact    |            |
| 5       | Temple                  | 4+100          | Right                   | 4.5                        | 6.97                      | Direct Impact    |            |
| 6       | Temple                  | 4+825          | Right                   | 10.5                       | 6.48                      | Direct Impact    | Relocated  |
| 7       | Temple/Pipaltree        | 5+525          | Right                   | 5.5                        | 10                        | No Direct Impact |            |
| 8       | Temple                  | 8+700          | Right                   | 5.5                        | 9.23                      | No Direct Impact |            |
| 9       | Church                  | 9+175          | Right                   | 13.5                       | 9.06                      | No Direct Impact |            |
| 10      | Temple(Banyan&Neem)     | 10+575         | Right                   | 4.5                        | 1.98                      | Direct Impact    | Relocated  |
| 11      | Temple                  | 16+000         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 12      | Temple                  | 16+150         | Right                   | 9.5                        | 9.5                       | No Direct Impact |            |
| 13      | Temple (BW)             | 16+200         | Left                    | 5                          | 5                         | No Direct Impact |            |
| 14      | Temple                  | 16+250         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 15      | Temple                  | 16+450         | Right                   | 4.5                        | 4.5                       | Direct Impact    | Relocated  |
| 16      | Temple                  | 17+550         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 17      | Temple                  | 18+000         | Right                   | 4.5                        | 4.5                       | Direct Impact    | Relocated  |
| 18      | Ganesha Temple          | 18+050         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 19      | Temple                  | 18+600         | Right                   | 5.5                        | 5.5                       | Direct Impact    | Relocated  |
| 20      | Temple                  | 22+100         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 21      | Temple& Lands           | 22+650         | Left                    | 8.5                        | 8.5                       | No Direct Impact |            |
| 22      | Temple                  | 22+700         | Left                    | 5.5                        | 5.5                       | Direct Impact    | Relocated  |
| 23      | Temple                  | 25+200         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 24      | Temple                  | 25+200         | Left                    | 4.5                        | 4.5                       | Direct Impact    | Relocated  |
| 25      | Temple                  | 25+200         | Right                   | 4.5                        | 4.5                       | Direct Impact    | Relocated  |
| 26      | Temple                  | 25+350         | Left                    | 4                          | 4                         | Direct Impact    | Relocated  |
| 27      | Temple                  | 28+000         | Right                   | 4                          | 4                         | Direct Impact    | Relocated  |
| 28      | Temple                  | 30+300         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 29      | Temple                  | 31+800         | Left                    | 6.5                        | 6.5                       | Direct Impact    | Relocated  |
| 30      | Temple/Ghat             | 34+670         | Left                    | 5.5                        | 5.5                       | No Direct Impact |            |
| 31      | Sai Temple              | 35+100         | Right                   | 28.5                       | 28.5                      | No Direct Impact |            |

| Sl. No. | Name of Common Property         | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) | Distance for PCL (in mts) | Impact           | Mitigation |
|---------|---------------------------------|----------------|-------------------------|----------------------------|---------------------------|------------------|------------|
| 32      | Temple                          | 35+100         | Right                   | 8.5                        | 8.5                       | No Direct Impact |            |
| 33      | Temple                          | 35+100         | Right                   | 5.5                        | 5.5                       | No Direct Impact |            |
| 34      | Statue Of Lion In Mouth Goddess | 35+200         | Right                   | 5.5                        | 5.5                       | No Direct Impact |            |
| 35      | Temple                          | 38+100         | Left                    | 6                          | 6                         | No Direct Impact |            |
| 36      | Temple                          | 40+050         | Right                   | 5                          | 5                         | Direct Impact    | Relocated  |
| 37      | Temple                          | 40+150         | Right                   | 4.5                        | 4.5                       | Direct Impact    | Relocated  |
| 38      | Temple                          | 40+400         | Right                   | 5                          | 5                         | Direct Impact    | Relocated  |
| 39      | Temple                          | 40+400         | Right                   | 5                          | 5                         | Direct Impact    | Relocated  |
| 40      | Temple                          | 41+200         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 41      | Temple                          | 41+350         | Right                   | 5.5                        | 5.5                       | No Direct Impact |            |
| 42      | Temple                          | 41+550         | Left                    | 4                          | 4                         | Direct Impact    | Relocated  |
| 43      | Temple                          | 41+700         | Left                    | 5.5                        | 5.5                       | Direct Impact    | Relocated  |
| 44      | Church (BW)                     | 42+100         | Right                   | 8.5                        | 8.5                       | No Direct Impact |            |
| 45      | Temple                          | 44+150         | Right                   | 6.5                        | 6.5                       | No Direct Impact |            |
| 46      | Big Temple (BW)                 | 44+600         | Right                   | 8.5                        | 8.5                       | No Direct Impact |            |
| 47      | Temple                          | 44+700         | Right                   | 8.5                        | 8.5                       | No Direct Impact |            |
| 48      | Temple                          | 45+000         | Left                    | 7.5                        | 7.5                       | No Direct Impact |            |
| 49      | Temple                          | 45+350         | Left                    | 10.5                       | 10.5                      | No Direct Impact |            |
| 50      | Temple                          | 45+900         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 51      | Temple Saibaba                  | 46+100         | Right                   | 5.5                        | 5.5                       | No Direct Impact |            |
| 52      | Temple                          | 48+800         | Right                   | 5.5                        | 5.5                       | Direct Impact    | Relocated  |
| 53      | Temple                          | 49+550         | Right                   | 4.5                        | 4.5                       | Direct Impact    | Relocated  |
| 54      | Temple                          | 52+300         | Right                   | 15                         | 15                        | No Direct Impact |            |
| 55      | Sai Temple                      | 54+200         | Right                   | 10.5                       | 10.5                      | Direct Impact    | Relocated  |
| 56      | Temple                          | 55+500         | Left                    | 5.5                        | 5.5                       | Direct Impact    | Relocated  |
| 57      | Temple                          | 56+800         | Left                    | 5.5                        | 5.5                       | Direct Impact    | Relocated  |
| 58      | Temple                          | 57+050         | Right                   |                            |                           | No Direct Impact |            |
| 59      | Temple                          | 57+200         | Left                    | 6.5                        | 6.5                       | No Direct Impact |            |
| 60      | Temple (BW)                     | 57+300         | Left                    | 8.5                        | 8.5                       | No Direct Impact |            |
| 61      | Temple                          | 57+300         | Left                    | 5                          | 5                         | No Direct Impact |            |
| 62      | Temple                          | 58+950         | Right                   | 6.5                        | 6.5                       | Direct Impact    | Relocated  |
| 63      | Ganesh Temple                   | 59+550         | Right                   | 3.5                        | 3.5                       | Direct Impact    | Relocated  |
| 64      | Temple                          | 60+200         | Right                   | 4.5                        | 4.5                       | Direct Impact    | Relocated  |
| 65      | Mary Statue                     | 60+750         | Right                   | 7                          | 7                         | No Direct Impact |            |
| 66      | Church                          | 60+750         | Right                   | 28.5                       | 28.5                      | No Direct Impact |            |
| 67      | Park & Hanuman Temple           | 61+800         | Right                   | 6.5                        | 6.5                       | No Direct Impact |            |

Besides mitigation & enhancement, following measures will be taken for safety of the locals:

- Local people will be informed about the schedule of construction activity, so that the local people remain prepared in advance.
- The existing sign boards will not be removed but shifted to appropriate place during construction, so that people don't find problem in identifying the tourist/religious place.

- During construction proper demarcation and sign board, indicators and flag man will be deputed for safe traffic movement. (Refer drawing 8.10)

## 8.8 BUS SHELTER, BUS BAYS, TRUCK LAY BYES, RESTING PLACE AND SERVICE ROAD

As a part of road improvement, bus shelters will be constructed/repared at all built up locations. The table below presents the impact and mitigation proposed for each of the existing bus shelters.

**Table 8.7 List of Existing Bus Shelters**

| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Distance from ECL (in mts) | Distance for PCL (in mts) | Impact           | Mitigation/ Enhancement |
|---------|-------------------------|----------------|-------------------------|----------------------------|---------------------------|------------------|-------------------------|
| 1       | Bus Shelter             | 2+800          | Left                    | 6.5                        | 10.41                     | No Direct Impact |                         |
| 2       | Bus Shelter             | 5+500          | Left                    | 6.5                        | 10.54                     | No Direct Impact |                         |
| 3       | Bus Shelter             | 8+500          | Right                   | 9.5                        | 14.14                     | No Direct Impact |                         |
| 4       | Bus Shelter             | 15+750         | Right                   | 7.5                        | 3.05                      | No Direct Impact |                         |
| 5       | Bus Shelter             | 18+600         | Left                    | 6.5                        | 7.58                      | No Direct Impact |                         |
| 6       | Bus Shelter             | 18+600         | Right                   | 5.5                        | 4.42                      | Direct Impact    | Will be relocated       |
| 7       | Bus Shelter             | 22+100         | Left                    | 6.5                        | 7.74                      | No Direct Impact |                         |
| 8       | Bus shelter             | 25+200         | Left                    | 6.5                        | 8.02                      | No Direct Impact |                         |
| 9       | Bus Shelter             | 25+500         | Left                    | 5.5                        | 13.25                     | No Direct Impact |                         |
| 10      | Bus Shelter             | 30+300         | Left                    | 4                          | 9.67                      | No Direct Impact |                         |
| 11      | Bus Shelter             | 31+800         | Left                    | 6                          | 6.82                      | No Direct Impact |                         |
| 12      | Bus Shelter             | 34+350         | Left                    | 4                          | 4                         | Direct Impact    | Will be relocated       |
| 13      | Bus Shelter             | 35+250         | Left                    | 5                          | 28.56                     | No Direct Impact |                         |
| 14      | Bus Shelter             | 40+150         | Right                   | 4.5                        | 2.12                      | Direct Impact    | Will be relocated       |
| 15      | Bus Shelter             | 41+500         | Left                    | 5.5                        | 6.17                      | Direct Impact    | Will be relocated       |
| 16      | Bus Shelter             | 49+700         | left                    | 5.5                        | 5                         | Direct Impact    | Will be relocated       |
| 17      | Bus Shelter             | 57+500         | Left                    | 18.5                       | 18.5                      | No Direct Impact |                         |
| 18      | Bus Shelter             | 59+700         | Left                    | 7                          | 7.08                      | No Direct Impact |                         |
| 19      | Bus Shelter             | 60+250         | Left                    | 5.5                        | 11.05                     | No Direct Impact |                         |

(Refer Drawing No 8.9)

Besides the reconstruction/relocation of 11 bus shelters and repair & rehabilitation of all existing bus shelters, 20 no.s of new bus shelter/bus laybys are proposed, tabulated below.

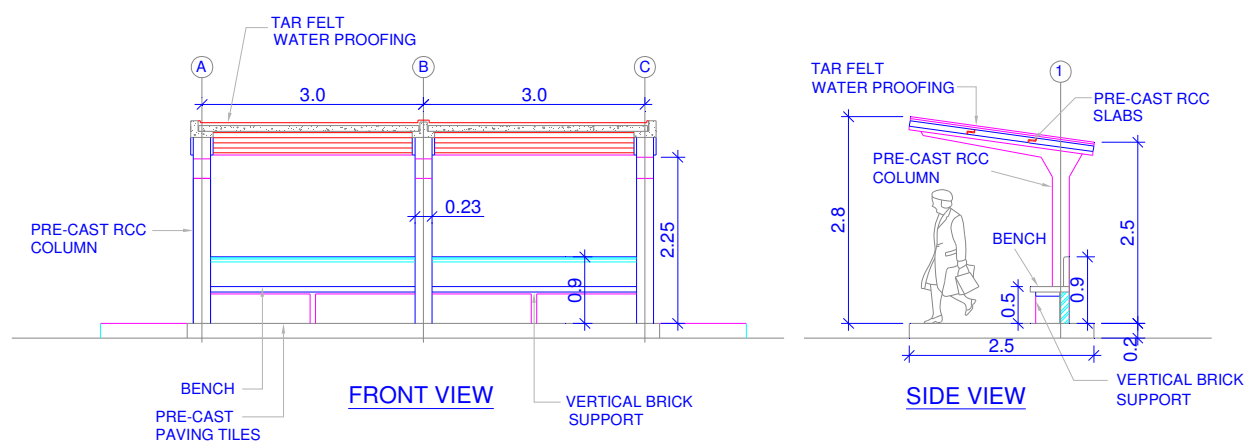
**Table 8.8 List of Proposed Bus Shelters**

| Sl.No | Description          | Location |        | Remarks                     |
|-------|----------------------|----------|--------|-----------------------------|
|       |                      | LHS      | RHS    |                             |
| 1     | Bus Shelter& Bus Bay | 0+230    |        | Kakinada                    |
| 2     | Bus Shelter& Bus Bay | 0+280    |        | Kakinada                    |
| 3     | Bus Shelter& Bus Bay |          | 0+280  | Kakinada                    |
| 4     | Bus Shelter& Bus Bay | 5+440    |        | Madhavapatnam               |
| 5     | Bus Shelter& Bus Bay |          | 5+440  | Madhavapatnam               |
| 6     | Bus Shelter& Bus Bay | 8+490    |        | Venkatakrishnarayapuram     |
| 7     | Bus Shelter& Bus Bay |          | 8+490  | Venkatakrishnarayapuram     |
| 8     | Bus Shelter& Bus Bay | 10+400   |        | Samarlakota Bypass starting |
| 9     | Bus Shelter& Bus Bay |          | 10+400 | Samarlakota Bypass starting |
| 10    | Bus Shelter& Bus Bay | 15+380   |        | Samarlakota Bypass Ending   |
| 11    | Bus Shelter& Bus Bay |          | 15+500 | Vetlapalem                  |
| 12    | Bus Shelter& Bus Bay |          | 17+730 | Gudaparti                   |
| 13    | Bus Shelter& Bus Bay |          | 21+250 | Medapadu                    |

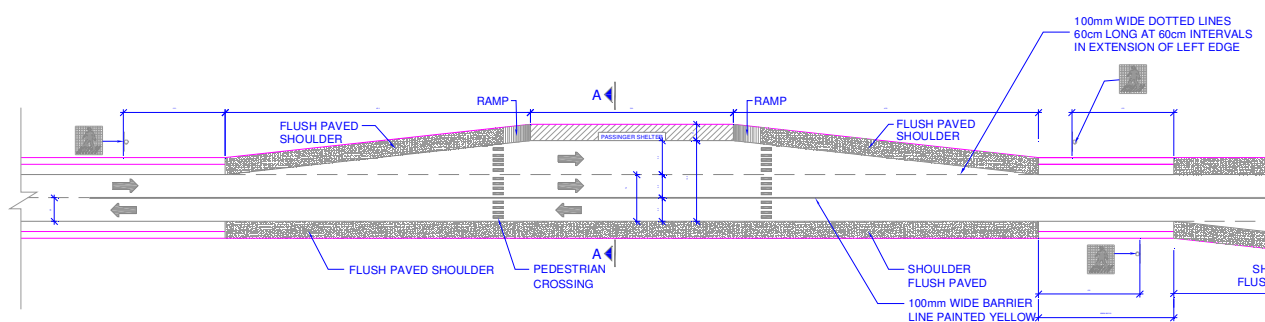


| Sl.No | Description          | Location |        | Remarks                  |
|-------|----------------------|----------|--------|--------------------------|
|       |                      | LHS      | RHS    |                          |
| 14    | Bus Shelter& Bus Bay |          | 24+740 | Peddabrahmadevam         |
| 15    | Bus Shelter& Bus Bay |          | 29+150 | Bikkavolu                |
| 16    | Bus Shelter& Bus Bay |          | 30+520 | Bikkavolu                |
| 17    | Bus Shelter& Bus Bay |          | 30+990 | Bikkavolu                |
| 18    | Bus Shelter& Bus Bay |          | 33+550 | Balabhadrapuram          |
| 19    | Bus Shelter& Bus Bay | 34+460   |        | Balabhadrapuram          |
| 20    | Bus Shelter& Bus Bay | 40+530   |        | Anaparthi                |
| 21    | Bus Shelter& Bus Bay |          | 44+700 | Dwarapudi                |
| 22    | Bus Shelter& Bus Bay | 48+580   |        | Kesavaram Rly Stn        |
| 23    | Bus Shelter& Bus Bay | 53+330   |        | Andhrapradesh paper mill |
| 24    | Bus Shelter& Bus Bay | 56+500   |        | Kadiyam                  |
| 25    | Bus Shelter& Bus Bay | 58+800   |        | Vemageri                 |

For dimensions, plan and layout refer drawing no 8.8: Conceptual plan of Bus shelter.



**Drawing No. 8.8: Conceptual Plan for Bus shelter**



**Drawing No. 8.9: Conceptual Plan for Bus Laybye**

## 8.9 AVOIDANCE OF DISRUPTION AND SAFETY RISKS DURING THE CONSTRUCTION STAGE

### 8.9.1 Disruption to the community

#### 8.9.1.1 Loss of access

At all times, the Contractor will provide safe and convenient passage for vehicles, pedestrians and livestock to and from side roads and property accesses connecting the project road. Work that affects the use of side roads and existing accesses will not be undertaken without providing adequate provisions.



The works will not interfere unnecessarily or improperly with the convenience of public or the access to, use and occupation of public or private roads, railways and any other access footpaths to or of properties whether public or private.

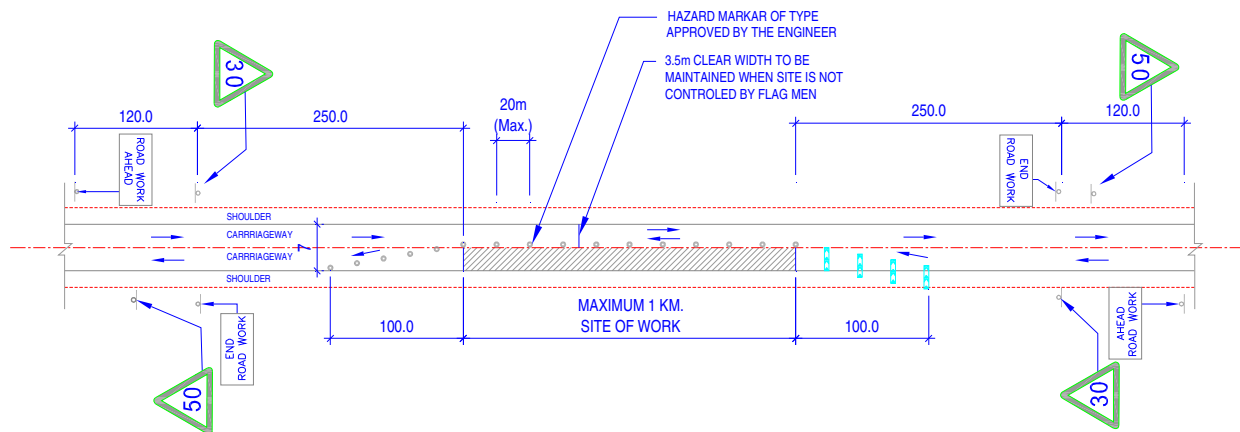
#### 8.9.1.2 Traffic Jams, Congestion and Safety

Detailed Traffic Control Plans will be prepared prior to commencement of works on any section of road. The traffic control plans will contain details of temporary diversions, details of arrangements for construction under traffic and details of traffic arrangement after cessation of work each day.

Temporary diversion (including scheme of temporary and acquisition) will be constructed with the approval of the Engineer and the EMU. Special consideration will be given in the preparation of the traffic control plan to the safety of pedestrians and workers at night.

The Contractor will ensure that the running surface is always properly maintained, particularly during the monsoon so that no disruption to the traffic flow occurs. The temporary traffic detours will be kept free of dust by frequent application of water, if necessary.

The Contractor will take all necessary measures for the safety of traffic during construction and provide, erect and maintain such barricades, including signs, markings, flags, lights and flagmen as may be required by the Engineer for the information and protection of traffic approaching or passing through the section of the highway under improvement.



**Drawing No. 8.10: Conceptual Plan for traffic Diversion**

#### 8.9.2 Safety of the workers

- REFER **APPENDIX E5**: SITTING AND LAYOUT OF CONSTRUCTION CAMP.
- REFER **APPENDIX E6**: WORKERS SAFETY DURING CONSTRUCTION ACTIVITY.
- REFER **APPENDIX E7**: ENVIRONMENT FRIENDLY CONSTRUCTION METHODOLOGY.
- REFER **APPENDIX E8**: STORAGE, HANDLING, USE AND EMERGENCY RESPONSE FOR HAZARDOUS CHEMICALS.

##### 8.9.2.1 Risk from Operations

The Contractor is required to comply with all the precautions as required for the safety of the workmen as per the International Labour Organisation (ILO) Convention No. 62 as far as those are applicable to this contract. The contractor will supply all necessary safety appliances such as safety goggles, helmets, masks, etc., to the workers and staff. The contractor has to comply with all regulation regarding safe scaffolding, ladders, working platforms, gangway, stairwells, excavations, trenches and safe means of entry and egress.

#### 8.9.2.2 Risk from Electrical Equipment

Adequate precautions will be taken to prevent danger from electrical equipment. No material or any of the sites will be so stacked or placed as to cause danger or inconvenience to any person or the public. All necessary fencing and lights will be provided to protect the public. All machines to be used in the construction will conform to the relevant Indian Standards (IS) codes, will be free from patent defect, will be kept in good working order, will be regularly inspected and properly maintained as per IS provisions and to the satisfaction of the Engineer.

#### 8.9.2.3 Risk at Hazardous Activity

All workers employed on mixing asphaltic material, cement, lime mortars, concrete etc., will be provided with protective footwear and protective goggles. Workers, who are engaged in welding works would be provided with welder's protective eye-shields. Stone-breakers will be provided with protective goggles and clothing and will be seated at sufficiently safe intervals.

The use of any herbicide or other toxic chemical will be strictly in accordance with the manufacturer's instructions. The Engineer will be given at least 6 working days notice of the proposed use of any herbicide or toxic chemical. A register of all herbicides and other toxic chemicals delivered to the site will be kept and maintained up to date by the Contractor. The register will include the trade name, physical properties and characteristics, chemical ingredients, health and safety hazard information, safe handling and storage procedures, and emergency and first aid procedures for the product.

#### 8.9.2.4 Risk of Lead Pollution

No man below the age of 18 years and no woman will be employed on the work of painting with products containing lead in any form. No paint containing lead or lead products will be used except in the form of paste or readymade paint. Face masks will be supplied for use by the workers when paint is applied in the form of spray or a surface having lead paint dry rubbed and scrapped.

#### 8.9.2.5 Risk Caused by Force' Majure

All reasonable precaution will be taken to prevent danger of the workers and the public from fire, flood, drowning, etc. All necessary steps will be taken for prompt first aid treatment of all injuries likely to be sustained during the course of work.

#### 8.9.2.6 Risk from Explosives

Except as may be provided in the contract or ordered or authorised by the Engineer, the Contractor will not use explosives. Where the use of explosives is so provided or ordered or authorised, the Contractor will comply with the requirements of the following Sub-Clauses of this Clause besides the law of the land as applicable: (also refer **Appendix E8: Storage, Handling, Use And Emergency Response For Hazordous Chemicals.**)

- The Contractor will at all times take every possible precaution and will comply with appropriate laws and regulations relating to the importation, handling, transportation, storage and use of explosives and will, at all times when engaged in blasting operations, post sufficient warning flagmen, to the full satisfaction of the Engineer.
- The Contractor will at all times make full liaison with and inform well in advance and obtain such permission as is required from all Government Authorities, public bodies and private parties whatsoever concerned or affected or likely to be concerned or affected by blasting operations.

#### 8.9.2.7 Malarial Risk

The Contractor will, at his own expense, conform to all anti-malarial instructions given to him by the Engineer, including filling up any borrow pits which may have been dug by him.

Gravid, blood laden mosquitoes can not fly very far, so they generally bite within a kilometre or so of their breeding place. Thus borrow pits and any other water bodies created during the construction

process will be situated 1 to 2km away from the human settlements. Pits dug up closer than these will be adequately drained to prevent water logging.

Similarly compensatory measures for filling up part of the water bodies situated adjacent to the project corridors will be directed towards deepening of the water bodies concerned. This way the capacity of the water body remains the same, while water surface available for breeding of mosquitoes is reduced. This will have an additional advantage of decreased evaporation losses, which will be important in water-scarce region.

#### 8.9.2.8 First Aid

At every workplace, a readily available first aid unit including an adequate supply of sterilised dressing material and appliances will be provided as per the Factory Rules. Workplaces remote and far away from regular hospitals will have indoor health units with one bed for every 250 workers. Suitable transport will be provided to facilitate take injured or ill person(s) to the nearest applicable hospital. At every workplace an ambulance room containing the prescribed equipment and nursing staff will be provided as prescribed.

#### 8.9.2.9 Potable Water

In every workplace at suitable and easily accessible places a sufficient supply of cold potable water (as per IS) will be provided and maintained. If the drinking water is obtained from an intermittent public water supply then, storage tanks will be provided. All water supply storage will be at a distance of not less than 15m from any latrine, drain or other source of pollution. Where water has to be drawn from an existing well, which is within such proximity of any latrine, drain or any other source of pollution, the well will be properly chlorinated before water is drawn from it for drinking water. All such wells will be entirely closed in and be provided with a trap door, which will be dust proof and waterproof. A reliable pump will be fitted to each covered well. The trap door will be kept locked and opened only for cleaning or inspection, which will be done at least once a month.

#### 8.9.2.10 Hygiene

The Contractor during the progress of work will provide, erect and maintain necessary (temporary) living accommodation and ancillary facilities for labour to standards and scales approved by the resident engineer. Refer **Appendix E5**: Sitting and layout of construction Camp.

There will be provided within the precincts of every workplace, latrines and urinals in an accessible place, and the accommodation, separately for each for these, as per standards set by the Building and other Construction Workers (regulation of Employment and Conditions of Service) Act, 1996. Except in workplaces provided with water-flushed latrines connected with a water borne sewage system, all latrines will be provided with dry-earth system (receptacles) which will be cleaned at least four times daily and at least twice during working hours and kept in a strict sanitary condition. Receptacles will be tarred inside and outside at least once a year. If women are employed, separate latrines and urinals, screened from those for men and marked in the vernacular will be provided. There will be adequate supply of water, close to latrines and urinals.

All temporary accommodation must be constructed and maintained in such a fashion that uncontaminated water is available for drinking, cooking and washing. The sewage system for the camp must be properly designed, built and operated so that no health hazard occurs and no pollution to the air, ground or adjacent watercourses takes place. Compliance with the relevant legislation must be strictly adhered to. Garbage bins must be provided in the camp and regularly emptied and the garbage disposed off in a hygienic manner. Construction camps are to be sited away from vulnerable people and adequate health care is to be provided for the work force.

Unless otherwise arranged for by the local sanitary authority, arrangement for proper disposal of excreta by incineration at the workplace will be made by means of a suitable incinerator approved by the local medical health or municipal authorities. Alternatively, excreta may be disposed off by putting a layer of night soils at the bottom of a permanent tank prepared for the purpose and covering it with 15 cm layer of waste or refuse and then covering it with a layer of earth for a fortnight (by then it will turn into manure).

On completion of the works, the whole of such temporary structures will be cleared away, all rubbish burnt, excreta or other disposal pits or trenches filled in and effectively sealed off and the whole of the site left clean and tidy, at the Contractor's expense, to the entire satisfaction of the Engineer.

## ***CHAPTER - 9: ENVIRONMENTAL MANAGEMENT ACTION PLAN***

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## CHAPTER - 9: ENVIRONMENTAL MANAGEMENT PLAN

Environmental Management Plan (EMP) deals with the implementation procedure of the guidelines and measures recommended to avoid, minimize and mitigate environmental impacts of the project. It also includes management of measures suggested for enhancement of the environmental quality along the highways.

The institutional arrangement made under project will look into the implementation of project as well as EMP and the various legal settings applicable to the project are briefly stated in chapter 3.

The avoidance, mitigation & enhancement measures for protection of the environment along highways have been discussed in detail in previous chapter. Although the social environmental impacts, its mitigation and management is an essential component of the EMP, this chapter excludes it for the purpose of clarity and procedural requirements. Social environmental elements have been separately dealt in separate volume namely, Resettlement and Rehabilitation Action Plan (RAP).

### 9.1 OBJECTIVE OF EMP

The EMP is a plan of action for mitigation / management / avoidance of the negative impacts of the project and enhancement of the project corridor. For each measure to be taken, its location, timeframe, implementation and overseeing / supervision responsibilities are listed. These components of the EMP have been given in **Table 9.1** which explains the environmental issues and the avoidance/ mitigation/ minimization or enhancement measures adopted and/or to be adopted during different phases of the project. It also provides the references for the suggested measures, responsible agency for its implementation/ management as well as its timeframe.

Linked activity of ROB@ Dwarapudi (ch 43/700 to 44/550) : At this locations , construction related environmental issues such as workers safety, construction dust, lack of appropriate traffic diversion and pollution to irrigation canal were already incorporated in the agreement (LS Agt.CR.No 45/2009-10 dt 18.11.2009) under safety measures and ecological balance. Hence these issues relating environmental safety aspects of the linked project are already taken care in above place.



Table 9.1: The Environmental Management Plan (EMP)

| Cl. No                                                                                                  | Environmental Issues                                                  | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Time frame        | Regulation and coordinating agency | Implementing Agency                                                           | Reference/ Bid document Clause          |                                                                               |                                                                |                                                        |                                               |                                                        |                                                      |                                                                                                         |                                     |                                                                                 |                                   |                                               |                                |                                                  |                                     |                                              |                     |                                                                                                                  |                                                   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| PRIOR TO START OF CONSTRUCTION ACTIVITY                                                                 |                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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                                                                                                                                                                                                                           |
| P.1                                                                                                     | Clearances, Approvals, Permits/NOCs etc. to be secured/complied with. | <div>List of clearance Required prior to start of construction Activity</div> <table><thead><tr><th>Type of Clearance</th><th>Applicability</th></tr></thead><tbody><tr><td>NOC and consents under Air, Water &amp; Environment Act and noise rules from SPCB</td><td>For establishment of construction camp.</td></tr><tr><td>NOC and consents under Air, Water &amp; Environment Act and noise rules from SPCB</td><td>For operating construction plant, crusher, batching plant etc.</td></tr><tr><td>Explosive License from Chief Controller of Explosives,</td><td>For storing fuel oil, lubricants, diesel etc.</td></tr><tr><td>Permission for storage of hazardous chemical from CPCB</td><td>Manufacture storage and Import of Hazardous Chemical</td></tr><tr><td>Borrow Area, approval from district collector, Consent letter, leases agreement with the owner of land.</td><td>Borrow area for excavation of earth</td></tr><tr><td>Quarry Lease Deed and Quarry License from State Department of Mines and Geology</td><td>Quarry operation (for new quarry)</td></tr><tr><td>Permission/NOC from State Ground Water board.</td><td>For extraction of ground water</td></tr><tr><td>Permission/NOC from State Irrigation department.</td><td>For taking/using river, canal water</td></tr><tr><td>Labor license from labor commissioner office</td><td>Engagement of Labor</td></tr></tbody></table> <div>Provide a copy of all necessary clearances to the CSC / EMU<br/>✓ Adhere to all clearance terms and conditions<br/>✓ Obtain written permission from private landholders to use their land for construction activities, prior to commencing any works.</div> | Type of Clearance | Applicability                      | NOC and consents under Air, Water & Environment Act and noise rules from SPCB | For establishment of construction camp. | NOC and consents under Air, Water & Environment Act and noise rules from SPCB | For operating construction plant, crusher, batching plant etc. | Explosive License from Chief Controller of Explosives, | For storing fuel oil, lubricants, diesel etc. | Permission for storage of hazardous chemical from CPCB | Manufacture storage and Import of Hazardous Chemical | Borrow Area, approval from district collector, Consent letter, leases agreement with the owner of land. | Borrow area for excavation of earth | Quarry Lease Deed and Quarry License from State Department of Mines and Geology | Quarry operation (for new quarry) | Permission/NOC from State Ground Water board. | For extraction of ground water | Permission/NOC from State Irrigation department. | For taking/using river, canal water | Labor license from labor commissioner office | Engagement of Labor | Construction stage (Prior to initiation of any work)<br><br>Time period in getting the permission is 2-3 months. | Regulatory Authorities as applicable for securing clearances.<br><br>Package Unit and PD-APRSP. | The Contractor | General Conditions of Contract<br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br><br>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| Type of Clearance                                                                                       | Applicability                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| NOC and consents under Air, Water & Environment Act and noise rules from SPCB                           | For establishment of construction camp.                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| NOC and consents under Air, Water & Environment Act and noise rules from SPCB                           | For operating construction plant, crusher, batching plant etc.        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Explosive License from Chief Controller of Explosives,                                                  | For storing fuel oil, lubricants, diesel etc.                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Permission for storage of hazardous chemical from CPCB                                                  | Manufacture storage and Import of Hazardous Chemical                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Borrow Area, approval from district collector, Consent letter, leases agreement with the owner of land. | Borrow area for excavation of earth                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Quarry Lease Deed and Quarry License from State Department of Mines and Geology                         | Quarry operation (for new quarry)                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Permission/NOC from State Ground Water board.                                                           | For extraction of ground water                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Permission/NOC from State Irrigation department.                                                        | For taking/using river, canal water                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Labor license from labor commissioner office                                                            | Engagement of Labor                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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| Cl. No                  | Environmental Issues                    | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Time frame                                              | Regulation and coordinating agency           | Implementing Agency   | Reference/ Bid document Clause                              |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
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|                         | structures within acquired Right of way | Resettlement Action Plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | construction.                                           | Officer (APRDC) & Package unit,              |                       | T ACTION PLAN                                               |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| P.3                     | Relocation of Utilities                 | All community underground and over head utilities are to be shifted as per <u>utility shifting plan</u> , prior permission has been obtained from regional offices of concerned departments like Electricity, Telecommunications, OFC, Water works etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Post design to Pre-construction                         | LAO & R&R Officer (APRDC) & Package unit,    | The Contractor        | UTILITY SHIFTING PLAN                                       |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| P.4                     | Clearing and Grubbing & Tree Felling    | <div>✓ Clearing and grubbing to be done only on the required surface &amp; just before the start of next activity on that section. In case of time gap, water should be sprinkled regularly till the start of next activity.</div> <div>✓ Tree felling permission already obtained from forest department under WALTA act.</div> <div>✓ A total of 4162 trees are to be fell, 596 on LHS and 3566 on RHS.</div>                                                                                                                                                                                                                                                                                                                                                                      | Prior to start of construction activity in that section | Environmental Officer (APRDC) & Package unit | The Contractor        | Trees to be fell marked at site.<br><br><b>Appendix E-4</b> |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| P.5                     | Relocation of drinking Water sources    | <div>Private drinking water source replaced according to RAP and public water sources replaced according to EMP.</div> <div>a) Two ponds to be dug to compensate the lost/filled volume as directed by engineer.</div> <div>b) Two Hand pumps to be relocated to nearest place.</div> <div>c) Digging of new well nearest to existing one.</div> <table><thead><tr><th>Name of Common Property</th><th>Chainage (Km.)</th><th>Location (Left/Right)</th></tr></thead><tbody><tr><td>Ponds</td><td>24+800</td><td>Right</td></tr><tr><td>Pond/Large Water Body</td><td>34+670</td><td>Right</td></tr><tr><td>Hand Pump</td><td>2+300</td><td>Right</td></tr><tr><td>Hand Pump</td><td>41+550</td><td>Left</td></tr><tr><td>Well</td><td>56+400</td><td>Left</td></tr></tbody></table> | Name of Common Property                                 | Chainage (Km.)                               | Location (Left/Right) | Ponds                                                       | 24+800 | Right | Pond/Large Water Body | 34+670 | Right | Hand Pump | 2+300 | Right | Hand Pump | 41+550 | Left | Well | 56+400 | Left | Prior to start of construction activity in that section | Environmental Officer (APRDC) & Package unit | The Contractor |  |
| Name of Common Property | Chainage (Km.)                          | Location (Left/Right)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                         |                                              |                       |                                                             |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| Ponds                   | 24+800                                  | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                         |                                              |                       |                                                             |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| Pond/Large Water Body   | 34+670                                  | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                         |                                              |                       |                                                             |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| Hand Pump               | 2+300                                   | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                         |                                              |                       |                                                             |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| Hand Pump               | 41+550                                  | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                         |                                              |                       |                                                             |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| Well                    | 56+400                                  | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                         |                                              |                       |                                                             |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |
| P.6                     | Cultural Properties                     | The main structure/boundary wall of twenty nine (29) sensitive cultural properties to be shifted, regarding 38 generic enhancement cultural properties is considered. The Khediyar nursery near design chainage km 57.300 is under direct impact due to realignment. This nursery is known to be famous and is functioning for quite a long time as explained in detailed in chapter 7. In view of this the oxbow land which is formed due to realignment may be transferred to the nursery as compensation. This option may be considered.                                                                                                                                                                                                                                          | Prior to start of construction activity in that section | Environmental Officer (APRDC) & Package unit | The Contractor        |                                                             |        |       |                       |        |       |           |       |       |           |        |      |      |        |      |                                                         |                                              |                |  |

| Cl. No                    | Environmental Issues | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Time frame                | Regulation and coordinating agency | Implementing Agency   | Reference/ Bid document Clause |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
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|                           |                      | <table><thead><tr><th>Name of Cultural Property</th><th>Chainage (Km.)</th><th>Location (Left/Right)</th></tr></thead><tbody><tr><td>Venkatesh Temple</td><td>1+800</td><td>Right</td></tr><tr><td>Church</td><td>2+000</td><td>Right</td></tr><tr><td>Church</td><td>3+300</td><td>Right</td></tr><tr><td>Temple</td><td>4+100</td><td>Right</td></tr><tr><td>Temple</td><td>4+825</td><td>Right</td></tr><tr><td>Temple(Banyan&amp;Neem)</td><td>10+575</td><td>Right</td></tr><tr><td>Temple</td><td>16+450</td><td>Right</td></tr><tr><td>Temple</td><td>18+000</td><td>Right</td></tr><tr><td>Temple</td><td>18+600</td><td>Right</td></tr><tr><td>Temple</td><td>22+700</td><td>Left</td></tr><tr><td>Temple</td><td>25+200</td><td>Left</td></tr><tr><td>Temple</td><td>25+200</td><td>Right</td></tr><tr><td>Temple</td><td>25+350</td><td>Left</td></tr><tr><td>Temple</td><td>28+000</td><td>Right</td></tr><tr><td>Temple</td><td>31+800</td><td>Left</td></tr><tr><td>Temple</td><td>40+050</td><td>Right</td></tr><tr><td>Temple</td><td>40+150</td><td>Right</td></tr><tr><td>Temple</td><td>40+400</td><td>Right</td></tr><tr><td>Temple</td><td>40+400</td><td>Right</td></tr><tr><td>Temple</td><td>41+550</td><td>Left</td></tr><tr><td>Temple</td><td>41+700</td><td>Left</td></tr><tr><td>Temple</td><td>48+800</td><td>Right</td></tr><tr><td>Temple</td><td>49+550</td><td>Right</td></tr><tr><td>Sai Temple</td><td>54+200</td><td>Right</td></tr><tr><td>Temple</td><td>55+500</td><td>Left</td></tr><tr><td>Temple</td><td>56+800</td><td>Left</td></tr><tr><td>Temple</td><td>58+950</td><td>Right</td></tr><tr><td>Ganesh Temple</td><td>59+550</td><td>Right</td></tr><tr><td>Temple</td><td>60+200</td><td>Right</td></tr></tbody></table> | Name of Cultural Property | Chainage (Km.)                     | Location (Left/Right) | Venkatesh Temple               | 1+800 | Right | Church | 2+000 | Right | Church | 3+300 | Right | Temple | 4+100 | Right | Temple | 4+825 | Right | Temple(Banyan&Neem) | 10+575 | Right | Temple | 16+450 | Right | Temple | 18+000 | Right | Temple | 18+600 | Right | Temple | 22+700 | Left | Temple | 25+200 | Left | Temple | 25+200 | Right | Temple | 25+350 | Left | Temple | 28+000 | Right | Temple | 31+800 | Left | Temple | 40+050 | Right | Temple | 40+150 | Right | Temple | 40+400 | Right | Temple | 40+400 | Right | Temple | 41+550 | Left | Temple | 41+700 | Left | Temple | 48+800 | Right | Temple | 49+550 | Right | Sai Temple | 54+200 | Right | Temple | 55+500 | Left | Temple | 56+800 | Left | Temple | 58+950 | Right | Ganesh Temple | 59+550 | Right | Temple | 60+200 | Right |  |  |  |  |
| Name of Cultural Property | Chainage (Km.)       | Location (Left/Right)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Venkatesh Temple          | 1+800                | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Church                    | 2+000                | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Church                    | 3+300                | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 4+100                | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 4+825                | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple(Banyan&Neem)       | 10+575               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 16+450               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 18+000               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 18+600               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 22+700               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 25+200               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 25+200               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 25+350               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 28+000               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 31+800               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 40+050               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 40+150               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 40+400               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 40+400               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 41+550               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 41+700               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 48+800               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 49+550               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Sai Temple                | 54+200               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 55+500               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 56+800               | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 58+950               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Ganesh Temple             | 59+550               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |
| Temple                    | 60+200               | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                    |                       |                                |       |       |        |       |       |        |       |       |        |       |       |        |       |       |                     |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |      |        |        |       |        |        |      |        |        |       |        |        |       |        |        |       |        |        |       |        |        |      |        |        |      |        |        |       |        |        |       |            |        |       |        |        |      |        |        |      |        |        |       |               |        |       |        |        |       |  |  |  |  |

| Cl. No  | Environmental Issues                                                                                | Specifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Time frame                                 | Regulation and coordinating agency | Implementing Agency   | Reference/ Bid document Clause |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |
|---------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------|-----------------------|--------------------------------|--------|-------|---|--------|------|---|--------|-------|---|--------|------|---|--------|------|---------------------------------|-------------|----------------|------------------------------|
| P.7     | Loss of existing bus stops and Waiting shed facilities                                              | <div>✓ Five existing bus shelters are to be relocated at suitable location or as directed by the engineer.</div> <table><thead><tr><th>Sl. No.</th><th>Chainage (Km.)</th><th>Location (Left/Right)</th></tr></thead><tbody><tr><td>1</td><td>18/600</td><td>Right</td></tr><tr><td>2</td><td>34+350</td><td>Left</td></tr><tr><td>3</td><td>40+150</td><td>Right</td></tr><tr><td>4</td><td>41+500</td><td>Left</td></tr><tr><td>5</td><td>49+700</td><td>left</td></tr></tbody></table> | Sl. No.                                    | Chainage (Km.)                     | Location (Left/Right) | 1                              | 18/600 | Right | 2 | 34+350 | Left | 3 | 40+150 | Right | 4 | 41+500 | Left | 5 | 49+700 | left | Prior to start of construction. | PIU – APRDC | The Contractor | Drawing No. LBG/APRDC/B S/01 |
| Sl. No. | Chainage (Km.)                                                                                      | Location (Left/Right)                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                            |                                    |                       |                                |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |
| 1       | 18/600                                                                                              | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                            |                                    |                       |                                |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |
| 2       | 34+350                                                                                              | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                            |                                    |                       |                                |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |
| 3       | 40+150                                                                                              | Right                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                            |                                    |                       |                                |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |
| 4       | 41+500                                                                                              | Left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                            |                                    |                       |                                |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |
| 5       | 49+700                                                                                              | left                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                            |                                    |                       |                                |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |
| P.8     | Environmental Management And Monitoring Facility Equipment for EMP (Meters, Vehicles and Buildings) | <div>✓ Monitoring is to be carried out regularly as per the frequency and locations mentioned in <b>Appendix E-10.</b></div>                                                                                                                                                                                                                                                                                                                                                              | During and after construction (Five Years) | SPCB, PIU - APRDC                  | The Contractor        | <b>Appendix E-10.</b>          |        |       |   |        |      |   |        |       |   |        |      |   |        |      |                                 |             |                |                              |

| Cl. No                                       | Environmental Impact & Issues                 | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Time frame                                                                                           | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause                                                                       |
|----------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------|---------------------|------------------------------------------------------------------------------------------------------|
| <b>AIR ENVIRONMENT - CONSTRUCTION PHASE</b>  |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                      |                                 |                     |                                                                                                      |
| C.1                                          | Gaseous Emissions                             | Vehicles and machinery are to be maintained so that emissions conform to National Ambient air quality standards.<br>All vehicles and machineries should obtain <u>Pollution Under Control Certificates</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Beginning with & throughout construction                                                             | PIU - APRDC, SPCB               | The Contractor      | MORTH Specification 111.1, 111.5                                                                     |
| C.2                                          | Dust Generation                               | <ul style="list-style-type: none"> <li>✓ Asphalt mixing plants should be sited over 1000 m from any communities. Mixing equipment should be well sealed, and be equipped with a dust-removal device. Operators should wear dust masks, ear protection and hard hats.</li> <li>✓ Vehicles delivering materials should be covered to reduce spills and dust blowing off the load.</li> <li>✓ Clearing and grubbing to be done, just before the start of next activity on that site. In case of time gap, water should be sprinkled regularly till the start of next activity.</li> <li>✓ Water to be sprayed during the construction phase, at mixing sites, approach roads &amp; temporary roads.</li> <li>✓ In laying sub-base, water spraying is needed to aid compaction of the material. After the compaction, water spraying should be carried out at regular intervals to prevent dust generation.</li> <li>✓ Road surface should be cleaned with air compressor and vacuum cleaners prior to the construction works. Manual labour using brooms should be avoided, if used labour to be provided masks.</li> <li>✓ Embankment slopes to be covered with turfing/stone pitching immediately after completion</li> </ul> | Beginning with & throughout construction until asphaltting is completed and side slopes are covered. | PIU – APRDC, EMU – APRDC CSC    | The Contractor      | MORTH Specification 111.1, 111.5, 111.8, 111.9, 111.10<br><br><b>Appendix E-4 &amp; Appendix E-5</b> |
| C.3                                          | Equipment Selection maintenance and operation | Construction plant and equipment will meet recognized international standards for emissions and will be maintained and operated in a manner that ensures that relevant air, noise, and discharge regulations are met.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | During construction                                                                                  | PIU – APRDC, EMU– APRDC, CSC    | The Contractor      | MORTH Spec 106, IRC: 72-1978; IRC: 90-1985, 111.5, 111.9, 111.10, 201.3                              |
| <b>LAND ENVIRONMENT - CONSTRUCTION PHASE</b> |                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                      |                                 |                     |                                                                                                      |
| C.4                                          | Soil Erosion and Sedimentation control        | Main reason of soil erosion is rains/monsoon, contractor should plan the activities so that NO bare/ loose earth surface is left out before the onset of monsoon, for minimising the soil erosion following preventive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Upon completion of construction activities at these sites.<br>During construction                    | PIU – APRDC, EMU – APRDC, CSC   | The Contractor      | MORTH Specification 306, 307, 308                                                                    |

| Cl. No | Environmental Impact & Issues               | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Time frame          | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause                                        |
|--------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------|---------------------|-----------------------------------------------------------------------|
|        |                                             | measures to be taken such as<br><ul style="list-style-type: none"> <li>✓ Embankment slopes to be covered, soon after completion</li> <li>✓ Next layer/activity to be planned, soon after completion of, clearing and grubbing, laying of embankment layer, sub grade layer, sub base layer, scarification etc.</li> <li>✓ Top soil from borrow area, Debris disposal sites; borrow area, construction site to be protected/covered for soil erosion.</li> <li>✓ Debris due to excavation of foundation, dismantling of existing cross drainage structure will be removed from the water course immediately.</li> <li>✓ Diversions for bridges, will be removed from the water course before the onset of monsoon.</li> </ul> |                     |                                 |                     | <b>Appendix E-9</b>                                                   |
| C.5    | Loss of agricultural topsoil                | All areas of cutting and all areas to be permanently covered will be stripped to a depth of 150mm and stored in stockpile.<br>Top soil will be safeguard from erosion and will be reused as follows.<br><ul style="list-style-type: none"> <li>✓ Covering all borrow areas after excavation is over.</li> <li>✓ Dressing of slopes of road embankment</li> <li>✓ Agricultural field, acquired temporarily.</li> </ul>                                                                                                                                                                                                                                                                                                        | During construction | PIU – APRDC EMU – APRDC, CSC    | The Contractor      | MORTH Specification 301.3.2, 305.3.3                                  |
| C.6    | Compaction of Soil and Damage to Vegetation | Construction vehicles should operate within the Corridor of Impact avoiding damage to soil and vegetation.<br>Diversions, access road used will be redeveloped by contractor, to the satisfaction of the owner/villagers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | During construction | PIU – APRDC EMU – APRDC, CSC    | The Contractor      | MORTH Specification 201.2                                             |
| C.7    | Contamination of soil                       | Guidelines of “Hazardous waste (management and handling) rules, 1989 will be enforced.<br><ul style="list-style-type: none"> <li>✓ Plant to be setup 500m away from surface water body.</li> <li>✓ Oil interceptor will be installed at plant site and Truck lay bye.</li> <li>✓ Septic tank will be constructed for safe disposal of waste.</li> <li>✓ Scarified Bituminous waste will be reuse for base course in cross roads and junction improvement of gravel roads.</li> </ul>                                                                                                                                                                                                                                         | During Construction | PIU – APRDC EMU – APRDC, CSC    | The Contractor      | <b>Appendix E-5 And Drawing No. LBG/APRDC/OGS/01</b>                  |
| C.8    | Borrow pits                                 | NO borrow pit will be opened without the permission of supervision consultant.<br><ul style="list-style-type: none"> <li>✓ Written approval from owner to be submitted to EMU.</li> <li>✓ Borrow pits have been identified outside the ROW. Before opening additional borrow pits, operating pits shall be closed according to IRC specification.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 | During Construction | PIU – APRDC EMU – APRDC, CSC    | The Contractor      | IRC: 10 1961<br>MORTH Spec. 111.2, 305.2.2<br><br><b>Appendix E-2</b> |

| Cl. No                                        | Environmental Impact & Issues         | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Time frame                                | Regulation/ coordinating agency                  | Implementing Agency | Reference/ Bid document Clause                                                                                 |
|-----------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------|
| C.9                                           | 1) Quarrying<br>2) Material sources   | 1. Quarrying will be carried out at approved and licensed quarries only. Copy of licenses to be submitted to the EMU.<br>2. As far as possible contractor will use the material from the approved Material sources.<br>In case on new quarry the instruction of Appendix E-1: Aggregate Quarry Management will be applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | During Construction                       | PIU – APRDC EMU – APRDC, CSC                     | The Contractor      | MORTH Specification 111.3, 302, 305.2.2<br><br><b>Appendix E-1</b>                                             |
| <b>WATER ENVIRONMENT - CONSTRUCTION PHASE</b> |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                           |                                                  |                     |                                                                                                                |
| C.10                                          | Loss of water bodies (Surface/Ground) | <ul style="list-style-type: none"> <li>✓ NO excavation from the bund of the water bodies.</li> <li>✓ NO debris disposal near, any water body.</li> <li>✓ Prior written permission from authorities for use of water for construction activity should be submitted to EMU.</li> <li>✓ Construction labours should be restricted from polluting the source or misusing the source.</li> <li>✓ Shifting of source to be completed prior to disruption of the actual source.</li> <li>✓ Source to be replaced immediately, in case of accidental loss.</li> <li>✓ Alternate measures to be taken/ensured during disrupted period.</li> <li>✓ Enhancement of Road side utilities like hand pump and water taps, providing soak pits at all hand pump locations in working condition or as directed by Engineer.</li> <li>✓ Enhancement/de-silting of existing surface water bodies as list above</li> </ul> | During Construction                       | PIU – APRDC EMU – APRDC, CSC                     | The Contractor      | MORTH Specification 111.4, 201.2, 301, 304, 306<br><br><b>Appendix E-3</b><br><br>Drawing no. LBG/APRDC/H P/01 |
| C.11                                          | Alteration of drainage                | <ul style="list-style-type: none"> <li>✓ Diversions should be constructed during dry season, with adequate drainage facility, and will be completely removed before the onset of monsoon.</li> <li>✓ Debris generated due to the excavation of foundation or due to the dismantling of existing structure should be removed from the water course.</li> <li>✓ Silt fencing has to be provided on the mouth of discharge into natural streams.</li> <li>✓ Continuous drain (lined/unlined) is provided, obstruction if any, to be removed immediately.</li> </ul>                                                                                                                                                                                                                                                                                                                                       | Whenever encountered during construction. | PIU – APRDC EMU – APRDC, CSC<br>Irrigation Dept. | The Contractor      | MORTH Specification 201.2, 301, 304, 306, 312                                                                  |
| C.12                                          | Runoff and drainage                   | ✓ Throughout monsoon uninterrupted continuous drain to be functional.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | During                                    | PIU – APRDC EMU –                                | The Contractor      |                                                                                                                |



| Cl. No  | Environmental Impact & Issues | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Time frame                                              | Regulation/ coordinating agency                     | Implementing Agency  | Reference/ Bid document Clause                                                   |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
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|         |                               | <ul style="list-style-type: none"><li>✓ Lined drain to be provided at buildup locations for quick drainage.</li><li>✓ Increased runoff due to increased impervious surface is countered through increased pervious surface area through soak pits.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Construction                                            | APRDC, CSC                                          |                      |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| C.13    | Water requirement for project | <ul style="list-style-type: none"><li>✓ Contractor has to provide list of sources (surface/ground) for approval from EMU.</li><li>✓ Prior to use of source contractor should obtain the written permission from authority, to use the water in construction activity, and submit a copy to EMU.</li><li>✓ During construction only permitted quantity (permission taken) from approved sources should be used in construction activity.</li><li>✓ Contractor to ensure optimum use of water; discourage labour from wastage of water.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | During Construction                                     | PIU – APRDC EMU – APRDC, CSC                        | The Contractor       |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| C.14    | Silting/sedimentation         | <div><ul style="list-style-type: none"><li>✓ <b>Pollution control and silt control temporary/permanent measures to be taken prior to any work in sections of project road very near (less than 100m) to the Samalkota canal.</b></li><li>✓ Measures suggested under “Soil Erosion and Sedimentation control” has to be enforced.</li><li>✓ Silt fencing to be provided.</li></ul><table><thead><tr><th>Sl. No.</th><th>Location</th><th>Length (approximate)</th><th>Remark</th></tr></thead><tbody><tr><td>1</td><td>Top soil storage</td><td>300m</td><td>Around the periphery of dumping yard</td></tr><tr><td>2</td><td>Borrow area site</td><td>100m</td><td>Along the length of borrow area in down slope direction</td></tr><tr><td>3</td><td>Surface Water bodies</td><td>500m</td><td>Equals to diameter/length or as instructed by engineer.</td></tr><tr><td>4</td><td>Plant Site</td><td>100m</td><td>In down slope direction of fine material</td></tr><tr><td>5</td><td>canal</td><td>1000m</td><td>Along the canal where ever it is required</td></tr></tbody></table><ul style="list-style-type: none"><li>✓ Construction activities should be stopped near water bodies during monsoon.</li></ul></div> | Sl. No.                                                 | Location                                            | Length (approximate) | Remark                                                                           | 1 | Top soil storage | 300m | Around the periphery of dumping yard | 2 | Borrow area site | 100m | Along the length of borrow area in down slope direction | 3 | Surface Water bodies | 500m | Equals to diameter/length or as instructed by engineer. | 4 | Plant Site | 100m | In down slope direction of fine material | 5 | canal | 1000m | Along the canal where ever it is required | Throughout construction period. | PIU – APRDC, EMU –APRDC, CSC<br>Irrigation Dept., Forest Dept. | The Contractor | MORTH Specification 111.4, 306<br><br>Drawing no. LBG/APRDC/S F/01 |
| Sl. No. | Location                      | Length (approximate)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Remark                                                  |                                                     |                      |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| 1       | Top soil storage              | 300m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Around the periphery of dumping yard                    |                                                     |                      |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| 2       | Borrow area site              | 100m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Along the length of borrow area in down slope direction |                                                     |                      |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| 3       | Surface Water bodies          | 500m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Equals to diameter/length or as instructed by engineer. |                                                     |                      |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| 4       | Plant Site                    | 100m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | In down slope direction of fine material                |                                                     |                      |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| 5       | canal                         | 1000m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Along the canal where ever it is required               |                                                     |                      |                                                                                  |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |
| C.15    | Contamination of water        | <ul style="list-style-type: none"><li>✓ Measures suggested under “Contamination of soil” has to be enforced.</li><li>✓ Construction work close to water bodies should be avoided during monsoon.</li><li>✓ Labour camps are to be located away from water bodies.</li><li>✓ Car washing/workshops near water bodies are to be avoided.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Throughout construction period.                         | PIU – APRDC, EMU –APRDC, CSC, SPCB, Irrigation Dept | The Contractor       | MORTH Specification 111.1, 111.4, 111.9, 111.13, 122, 201.2, 201.4, 304.3.3, 306 |   |                  |      |                                      |   |                  |      |                                                         |   |                      |      |                                                         |   |            |      |                                          |   |       |       |                                           |                                 |                                                                |                |                                                                    |

| Cl. No                                 | Environmental Impact & Issues         | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Time frame             | Regulation/ coordinating agency | Implementing Agency       | Reference/ Bid document Clause |                                                                                                                                                                                                                                           |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
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|                                        |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                        |                                 |                           |                                |                                                                                                                                                                                                                                           |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| NOISE ENVIRONMENT - CONSTRUCTION PHASE |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                        |                                 |                           |                                |                                                                                                                                                                                                                                           |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| C.16                                   | Noise                                 | <div><div><div>✓ Noise standard at processing sites, e.g. aggregate crushing plants, batching plant, hot mix plant are to be strictly monitored to prevent exceeding of GOI noise standards.</div><div>✓ Workers in the vicinity of strong noise to wear protectors and their working time should be limited as a safety measure.</div><div>✓ In construction sites within 150 m of sensitive receptors construction to be stopped from 22:00 to 06:00.</div><div>✓ Machinery and vehicles should be maintained to keep their noise to a minimum.</div><div>✓ Noise barrier to be constructed at noise sensitive locations. During consultation consensus could not be reached regarding the dimensions of the noise barrier, hence consultant propose consultation during the construction stage and works to be taken up as per day works.</div></div><table><tr><th>Sl. No.</th><th>Name of Common Property</th><th>Chainage (Km.)</th><th>Location (Left/ Right)</th><th>Distance from ECL (in mts)</th><th>Distance for PCL (in mts)</th><th>Impact</th><th>Mitigation/ Enhancement</th></tr><tr><td>1</td><td>PRG College (BW)</td><td>0+200</td><td>Right</td><td>12</td><td>14.4</td><td>No Direct Impact</td><td rowspan="4">Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4)</td></tr><tr><td>2</td><td>Hospital (BW)</td><td>0+600</td><td>Left</td><td>12</td><td>11.18</td><td>No Direct Impact</td></tr><tr><td>3</td><td>Rangaraya Medical College (BW)</td><td>0+900</td><td>Right</td><td>15</td><td>4 Lane</td><td>No Direct Impact</td></tr><tr><td>4</td><td>Ideal College Of Arts &amp; Sciences (BW)</td><td>1+600</td><td>Right</td><td>5.5</td><td>0.72</td><td>No Direct Impact</td></tr><tr><td>5</td><td>Saraswathi Vidyapita</td><td>5+700</td><td>Right</td><td>23.5</td><td>27.96</td><td>No Direct Impact</td><td rowspan="2">Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as</td></tr><tr><td>6</td><td>School</td><td>17+950</td><td>Right</td><td>6</td><td>6.04</td><td>No Direct</td></tr></table></div> | Sl. No.                | Name of Common Property         | Chainage (Km.)            | Location (Left/ Right)         | Distance from ECL (in mts)                                                                                                                                                                                                                | Distance for PCL (in mts) | Impact | Mitigation/ Enhancement | 1 | PRG College (BW) | 0+200 | Right | 12 | 14.4 | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) | 2 | Hospital (BW) | 0+600 | Left | 12 | 11.18 | No Direct Impact | 3 | Rangaraya Medical College (BW) | 0+900 | Right | 15 | 4 Lane | No Direct Impact | 4 | Ideal College Of Arts & Sciences (BW) | 1+600 | Right | 5.5 | 0.72 | No Direct Impact | 5 | Saraswathi Vidyapita | 5+700 | Right | 23.5 | 27.96 | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as | 6 | School | 17+950 | Right | 6 | 6.04 | No Direct | Beginning and throughout construction | PIU – APRDC<br>EMU – APRDC, CSC , SPCB | The Contractor | MORTH Specification 111, 111.5<br><br>Drawing no. LBG/APR DC/NB/01<br>LBG/APR DC/TS/01<br>LBG/APR DC/SB/01 |
| Sl. No.                                | Name of Common Property               | Chainage (Km.)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Location (Left/ Right) | Distance from ECL (in mts)      | Distance for PCL (in mts) | Impact                         | Mitigation/ Enhancement                                                                                                                                                                                                                   |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| 1                                      | PRG College (BW)                      | 0+200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Right                  | 12                              | 14.4                      | No Direct Impact               | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| 2                                      | Hospital (BW)                         | 0+600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Left                   | 12                              | 11.18                     | No Direct Impact               |                                                                                                                                                                                                                                           |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| 3                                      | Rangaraya Medical College (BW)        | 0+900                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Right                  | 15                              | 4 Lane                    | No Direct Impact               |                                                                                                                                                                                                                                           |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| 4                                      | Ideal College Of Arts & Sciences (BW) | 1+600                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Right                  | 5.5                             | 0.72                      | No Direct Impact               |                                                                                                                                                                                                                                           |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| 5                                      | Saraswathi Vidyapita                  | 5+700                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Right                  | 23.5                            | 27.96                     | No Direct Impact               | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as                               |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |
| 6                                      | School                                | 17+950                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Right                  | 6                               | 6.04                      | No Direct                      |                                                                                                                                                                                                                                           |                           |        |                         |   |                  |       |       |    |      |                  |                                                                                                                                                                                                                                           |   |               |       |      |    |       |                  |   |                                |       |       |    |        |                  |   |                                       |       |       |     |      |                  |   |                      |       |       |      |       |                  |                                                                                                                                                                                                             |   |        |        |       |   |      |           |                                       |                                        |                |                                                                                                            |

| Cl. No                                                                                                              | Environmental Impact & Issues     | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |       |        |                  |                                                                                                                                                                                                                                           | Time frame                                  | Regulation/ coordinating agency            | Implementing Agency | Reference/ Bid document Clause                                                        |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|--------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------|---------------------|---------------------------------------------------------------------------------------|
|                                                                                                                     |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |       |        |                  | Impact                                                                                                                                                                                                                                    | noise barrier (refer dwg:8.4)               |                                            |                     |                                                                                       |
| 7                                                                                                                   | School                            | 30+200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Right | 5.5   | 2.93   | No Direct Impact |                                                                                                                                                                                                                                           |                                             |                                            |                     |                                                                                       |
| 8                                                                                                                   | Private Hospital                  | 41+500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Right | 5.5   | 6.17   | No Direct Impact |                                                                                                                                                                                                                                           |                                             |                                            |                     |                                                                                       |
| 9                                                                                                                   | Veterinary Hospital (BW)          | 57+300                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Right | 4.5   | 4.97   | Direct Impact    | Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4)                                                                                                                                                        |                                             |                                            |                     |                                                                                       |
| 10                                                                                                                  | Hospital (BW)                     | 57+800                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Left  | 7.5   | 7.92   | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) |                                             |                                            |                     |                                                                                       |
| 11                                                                                                                  | School                            | 57+850                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Left  | 23.5  | 23.67  | No Direct Impact |                                                                                                                                                                                                                                           |                                             |                                            |                     |                                                                                       |
| 12                                                                                                                  | School (BW)                       | 60+050                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Right | 5     | 3.31   | Direct Impact    | Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4)                                                                                                                                                        |                                             |                                            |                     |                                                                                       |
| 13                                                                                                                  | School                            | 61+200                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Right | 153.5 | 154.46 | No Direct Impact | Provision is made for noise barrier in environmental budget. The requirement will be assessed by the environmental supervision consultant. if required Existing boundary wall will be raised and develop as noise barrier (refer dwg:8.4) |                                             |                                            |                     |                                                                                       |
| 14                                                                                                                  | School (BW)                       | 61+350                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Right | 53.5  | 53.58  | No Direct Impact |                                                                                                                                                                                                                                           |                                             |                                            |                     |                                                                                       |
| <div>✓ “HORN PROHIBITION” sign post to be erected.</div> <div>✓ Rumble strips / speed breaker to be provided.</div> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |       |        |                  |                                                                                                                                                                                                                                           |                                             |                                            |                     |                                                                                       |
| FLORA & FAUNA - CONSTRUCTION PHASE                                                                                  |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |       |        |                  |                                                                                                                                                                                                                                           |                                             |                                            |                     |                                                                                       |
| C.17                                                                                                                | Loss of trees and Avenue Planting | <div>✓ Trees felled should be cut in logs and stacked specieswise as instructed by employer.</div> <div>✓ Avenue plantation has to be taken up soon after completion of civil works. <b>Replantation is proposed, not in the ratio of the mandatory requirement, rather replantation is proposed all along the project road under avenue plantation.</b></div> <div>✓ All the realignment sections are to be enhanced with landscaping and peripheral tree plantation.</div> <div>✓ Community structures have to be enhanced with plantation.</div> <div>✓ Contractor has to ensure that no trees/branches to be fell by</div> |       |       |        |                  |                                                                                                                                                                                                                                           | After completion of construction activities | PIU – APRDC EMU – APRDC, CSC, Forest Dept. | The Contractor EMU  | MORTH Specification, 111, 111.5, 201.5, 306, 308<br><br>Drawing no. BG/APRDC/TG 3B/01 |

| Cl. No | Environmental Impact & Issues | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                             | Time frame                                        | Regulation/ coordinating agency            | Implementing Agency | Reference/ Bid document Clause                        |
|--------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------|---------------------|-------------------------------------------------------|
|        |                               | labourer for fuel, warmth during winter. Enough provision of fuel to be ensured.                                                                                                                                                                                                                                                                                            |                                                   |                                            |                     |                                                       |
| C.18   | Vegetation clearance          | <ul style="list-style-type: none"> <li>✓ Clearing and grubbing should be avoided beyond that which is directly required for construction activities.</li> <li>✓ Next activity to be planned/started immediately, to avoid dust generation and soil erosion during monsoon.</li> <li>✓ Turfing / re-vegetation to be started soon after completion of embankment.</li> </ul> | During cleaning operations<br>During construction | PIU – APRDC EMU – APRDC, CSC, Forest Dept. | The Contractor      | MORTH Specification 201.2<br><br><b>Appendix: E-4</b> |
| C.19   | Fauna                         | <ul style="list-style-type: none"> <li>✓ Construction workers must protect natural resources &amp; wild animals.</li> <li>✓ Hunting to be prohibited.</li> <li>✓ Nesting grounds &amp; migratory paths to be protected.</li> </ul>                                                                                                                                          | During construction                               | PIU–APRDC, EMU–APRDC, CSC, Forest Dept     | The Contractor      | MORTH Specification 111.1, 111.6                      |

**SOCIO-ECONOMIC ENVIRONMENT - CONSTRUCTION PHASE**

|      |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |                              |                |                                                                          |
|------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------|----------------|--------------------------------------------------------------------------|
| C.20 | General Issues – Related to Users      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | During Construction | PIU – APRDC EMU – APRDC, CSC | The Contractor | MORTH Specification 111, 111.5, 111.6, 112, 201.2, 201.3 & 201, 302, 306 |
|      | Fear of uncertainties regarding future | Public participation sessions should be conducted in different stages of project construction.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |                              |                |                                                                          |
|      | Public Health and Safety               | Debris, so generated will be disposed to the satisfaction of Engineer. Monitoring of air, water, noise and land during construction and operation phase.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |                              |                |                                                                          |
|      | Loss of access                         | At all times, the Contractor shall provide safe and convenient passage for vehicles, pedestrians and livestock to and from side roads and property accesses connecting the project road. Work that affects the use of side roads and existing accesses shall not be undertaken without providing adequate provisions to the prior satisfaction of the Engineer. The works shall not interfere unnecessarily or improperly with the convenience of public or the access to, use and occupation of public or private roads, railways and any other access footpaths to or of properties whether public or private. |                     |                              |                |                                                                          |

| Cl. No | Environmental Impact & Issues                           | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Time frame | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause |
|--------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------|---------------------|--------------------------------|
|        | Traffic Jams and Congestion                             | Detailed Traffic Management Plans shall be prepared and submitted to the Engineer for approval 5 days prior to commencement of maintenance works on any section of road. The traffic control plans shall contain details of temporary diversions, details of arrangements for construction under traffic and details of traffic arrangements after cession of work each day<br>Temporary diversion (including scheme of temporary and acquisition) will be constructed with the approval of the Engineer.<br>Special consideration shall be given in the preparation of the traffic control plan to the safety of pedestrians and workers at night<br>The contractor shall ensure that the running surface is always property maintained, particularly during the monsoon so that no disruption tot the traffic flow occurs<br>The temporary traffic detours in settlement areas shall be kept free of dust by frequent application of water. |            |                                 |                     |                                |
|        | Traffic control and safety                              | The Contractor shall take all necessary measures for the safety of traffic during construction and provide, erect and maintain such barricades, including signs, markings, flags, lights and flagmen as may be required by the Engineer for the information and protection of traffic approaching or passing through the section of the highways under improvement. The provision of traffic safety measures shall be considered incidental to work as per MoRTH specification and IRC guidelines.<br>All signs, barricades, pavement markings shall be as per MoRTH specification                                                                                                                                                                                                                                                                                                                                                            |            |                                 |                     |                                |
|        | General Issues – Related to Labours                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |            |                                 |                     |                                |
|        | Pressure on Existing Infrastructure due to labour camp. | Contractors should recruit the local people as laborers at least for unskilled and semi-skilled jobs.<br>Basic facilities should be ensured at labour camp including first aid, canteen and toilets to prevent the spread of disease and pressure on existing infrastructure.<br>Contractor is responsible for enforcing/adhering to the basic minimum requirements given in <b>Appendix E5</b> : Sitting and layout of construction camp,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |                                 |                     |                                |

Appendix E5:

| Cl. No | Environmental Impact & Issues | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Time frame | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause                                    |
|--------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------|---------------------|-------------------------------------------------------------------|
|        | Accidents and Safety          | Contractor is responsible for enforcing/adhering to the basic minimum requirements given in <b>Appendix E6</b> : Workers safety during Construction, <b>Appendix E7</b> : Environment Friendly Construction methodology and <b>Appendix E8</b> : Storage, handling use and emergency response for hazardous chemicals.<br>Environment and Safety manager will report, on the enforcement of the above mentioned appendices, to the EMU on weekly basis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |            |                                 |                     | <b>Appendix E6:</b><br><b>Appendix E7:</b><br><b>Appendix E8:</b> |
|        | Payment of Wages              | <ul style="list-style-type: none"> <li>The payment of wages should be as per the Minimum Wages Act, Department of Labour, Government of India for both male and female workers.</li> <li>Display of the minimum wages board at camps and major construction sites should be done in local languages at the construction and labour camp sites.</li> <li>Wages should be paid to the labourers only in the presence of R&amp;BD staff;</li> <li>Contractor is required to maintain register for payment of labour wages with entry of every labour working for him. Also, he has to produce it for verification if and when asked by the Engineer, EMU and/or the concerned R&amp;BD staff/Engineer's representative.</li> <li>In case there are sub-contractors, the main contractor shall be responsible for ensuring that the payment of wages to labourers is as per the Minimum Wages Act. Sub-contractor has to follow the same procedure as the main contractor.</li> </ul> |            |                                 |                     |                                                                   |

| Cl. No | Environmental Impact & Issues                  | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Time frame          | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause                                       |
|--------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------|---------------------|----------------------------------------------------------------------|
|        | Rehabilitation of labour and Construction camp | <p>At the completion of construction, all construction camp facilities shall be dismantled and removed from the site. The site shall be restored to a condition in no way inferior to the condition prior to commencement of the works. Various activities to be carried out for site rehabilitation include:</p> <ul style="list-style-type: none"> <li>Oil and fuel contaminated soil shall be removed and transported and buried in waste disposal areas.</li> <li>Soak pits, septic tanks shall be covered and effectively sealed off.</li> <li>Debris (rejected material) should be disposed off suitably</li> <li>Ramps created should be levelled.</li> <li>Underground water tank in a barren/non-agricultural land can be covered. However, in an agricultural land, the tank shall be removed.</li> <li>If the construction camp site is on an agricultural land, top soil can be spread so as to aid faster rejuvenation.</li> <li>Proper documentation of rehabilitation site is necessary. This shall include the following: <ul style="list-style-type: none"> <li>Photograph of rehabilitated site;</li> <li>Land owner consent letter for satisfaction in measures taken for rehabilitation of site;</li> <li>Undertaking from contractor; and</li> <li>Certification from Engineer in-charge of the PIC.</li> </ul> </li> </ul> <p>In cases, where the construction camps site is located on a private land holding, the contractor would still have to restore the campsite as per this guideline. The rehabilitation is mandatory and should be include in the agreement with the landowner by the contractor. Also, he would have to obtain a certificate for satisfaction from the landowner.</p> |                     |                                 |                     |                                                                      |
| C.21   | Sensitive Cultural Properties                  | <ul style="list-style-type: none"> <li>✓ A comprehensive mitigation/enhancement plan is prepared for each of the existing community properties along the project corridor, detailed in <b>Appendix E-12.</b></li> <li>✓ Precautions should be taken during construction, for accidental loss/damage of any communal property.</li> <li>✓ Any loss during construction will be the solo responsibility of contractor and the damage will be repaired immediately up to the satisfaction of people, at contractor's own cost.</li> <li>✓ Through access/identification should be maintained during construction.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | During construction | PIU – APRDC EMU – APRDC, CSC    | The Contractor      | <b>Appendix E-12.</b><br><br>Drawing no. LBG/APRDC/C P/01 through 11 |



| Cl. No                           | Environmental Impact & Issues | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Time frame                 | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
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| C.22                             | Road side amenities           | <div><div><div>✓ Bus shelter at all buildup locations, in case already existing; the same will be repaired and rehabilitated.</div><div>✓ Pedestrian crossing is provided at major settlement locations, providing zebra crossing, sign posts and speed breakers.</div><div>✓ Landscaping at Junctions</div></div><table><thead><tr><th>Sl.no</th><th>Type of Junction</th><th>Numbers/Chainage</th><th>Remarks</th></tr></thead><tbody><tr><td>1</td><td>Minor - T/ Y/ X</td><td>101 Nos</td><td>Typical Design</td></tr><tr><td>1</td><td>Major</td><td>0+000 (Design ch: 0+000)</td><td>Proposed <b>T</b> Junction</td></tr><tr><td>2</td><td>Major</td><td>0+250 (Design ch: 0+250)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>3</td><td>Major</td><td>0+600 (Design ch: 0+600)</td><td>Proposed <b>T</b> Junction</td></tr><tr><td>4</td><td>Major</td><td>1+000 (Design ch: 1+000)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>5</td><td>Major</td><td>1+480 (Design ch: 1+480)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>6</td><td>Major</td><td>10+670 (Design ch: 10+720)</td><td>Proposed <b>Y</b> Junction</td></tr><tr><td>7</td><td>Major</td><td>16+000 (Design ch: 15+000)</td><td>Proposed <b>Y</b> Junction</td></tr><tr><td>8</td><td>Major</td><td>30+200 (Design ch: 29+400)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>9</td><td>Major</td><td>35+400 (Design ch: 34+450)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>10</td><td>Major</td><td>41+600 (Design ch: 40+420)</td><td>Proposed <b>T</b> Junction</td></tr><tr><td>11</td><td>Major</td><td>45+900 (Design ch: 45+100)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>12</td><td>Major</td><td>49+600 (Design ch: 48+750)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>13</td><td>Major</td><td>56+900 (Design ch: 56+150)</td><td>Proposed <b>X</b> Junction</td></tr><tr><td>14</td><td>Major</td><td>60+100 (Design ch: 59+320)</td><td>Proposed <b>X</b> Junction</td></tr></tbody></table></div> | Sl.no                      | Type of Junction                | Numbers/Chainage    | Remarks                        | 1 | Minor - T/ Y/ X | 101 Nos | Typical Design | 1 | Major | 0+000 (Design ch: 0+000) | Proposed <b>T</b> Junction | 2 | Major | 0+250 (Design ch: 0+250) | Proposed <b>X</b> Junction | 3 | Major | 0+600 (Design ch: 0+600) | Proposed <b>T</b> Junction | 4 | Major | 1+000 (Design ch: 1+000) | Proposed <b>X</b> Junction | 5 | Major | 1+480 (Design ch: 1+480) | Proposed <b>X</b> Junction | 6 | Major | 10+670 (Design ch: 10+720) | Proposed <b>Y</b> Junction | 7 | Major | 16+000 (Design ch: 15+000) | Proposed <b>Y</b> Junction | 8 | Major | 30+200 (Design ch: 29+400) | Proposed <b>X</b> Junction | 9 | Major | 35+400 (Design ch: 34+450) | Proposed <b>X</b> Junction | 10 | Major | 41+600 (Design ch: 40+420) | Proposed <b>T</b> Junction | 11 | Major | 45+900 (Design ch: 45+100) | Proposed <b>X</b> Junction | 12 | Major | 49+600 (Design ch: 48+750) | Proposed <b>X</b> Junction | 13 | Major | 56+900 (Design ch: 56+150) | Proposed <b>X</b> Junction | 14 | Major | 60+100 (Design ch: 59+320) | Proposed <b>X</b> Junction | During construction | PIU – APRDC EMU – APRDC, CSC | The Contractor | Drawing no. LBG/APRDC/B S/01 |
| Sl.no                            | Type of Junction              | Numbers/Chainage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Remarks                    |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 1                                | Minor - T/ Y/ X               | 101 Nos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Typical Design             |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 1                                | Major                         | 0+000 (Design ch: 0+000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Proposed <b>T</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 2                                | Major                         | 0+250 (Design ch: 0+250)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 3                                | Major                         | 0+600 (Design ch: 0+600)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Proposed <b>T</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 4                                | Major                         | 1+000 (Design ch: 1+000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 5                                | Major                         | 1+480 (Design ch: 1+480)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 6                                | Major                         | 10+670 (Design ch: 10+720)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>Y</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 7                                | Major                         | 16+000 (Design ch: 15+000)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>Y</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 8                                | Major                         | 30+200 (Design ch: 29+400)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 9                                | Major                         | 35+400 (Design ch: 34+450)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 10                               | Major                         | 41+600 (Design ch: 40+420)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>T</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 11                               | Major                         | 45+900 (Design ch: 45+100)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 12                               | Major                         | 49+600 (Design ch: 48+750)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 13                               | Major                         | 56+900 (Design ch: 56+150)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| 14                               | Major                         | 60+100 (Design ch: 59+320)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Proposed <b>X</b> Junction |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |
| ROAD SAFETY - CONSTRUCTION PHASE |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |                                 |                     |                                |   |                 |         |                |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                          |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |   |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |    |       |                            |                            |                     |                              |                |                              |

| Cl. No                                    | Environmental Impact & Issues     | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Time frame                                | Regulation/ coordinating agency                      | Implementing Agency                 | Reference/ Bid document Clause |
|-------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------|-------------------------------------|--------------------------------|
| C.23                                      | Accident with hazardous materials | <p>COMPLIANCE with “Rules” as defined in Environmental (Protection) Act, 1986, including:</p> <ul style="list-style-type: none"> <li>✓ For delivery of hazardous substances, three certificates issued by transportation department are required permit license, driving license, and guarding license.</li> <li>✓ Vehicles delivering hazardous substances will be printed with standard signs.</li> <li>✓ Public security, transportation and fire fighting departments will designate a special route for these vehicles.</li> <li>✓ These vehicles can only be parked at designated parking lots.</li> <li>✓ In case of spill of hazardous materials, relevant departments will be informed at once &amp; dealt with it in accordance with spill contingency plan.</li> </ul> | During Construction                       | APRDC, State Police & Fire Station, SPCB, EMU        | The Contractor, local bodies        |                                |
| <b>AIR ENVIRONMENT – OPERATION PHASE</b>  |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |                                                      |                                     |                                |
| O.1                                       | Dust Generation                   | <ul style="list-style-type: none"> <li>✓ Dust generation due to vehicle wheel will be reduced due to increased/widened paved surface.</li> <li>✓ Avenue plantation to be maintained, casualties to be replaced.</li> <li>✓ Avenue plantation includes species having dust absorption characteristic.</li> <li>✓ Community properties and realignment locations has been proposed for peripheral plantation and landscaping.</li> <li>✓ Maintenance of roads to be ensured.</li> </ul>                                                                                                                                                                                                                                                                                             | After completion of construction activity | Forest dept. EMU                                     | Maintenance unit- APRDC Forest Dept |                                |
| O.2                                       | Air Pollution                     | <ul style="list-style-type: none"> <li>✓ With the reduction in journey time, idle engine running time air pollution will reduce.</li> <li>✓ Avenues plantation is proposed throughout the corridor, casualties to be replaced.</li> <li>✓ Avenue plantation includes species having air purifying characteristic.</li> <li>✓ Enforce Pollution Under Control (PUC) Programs. The public will be informed about the regulations on air pollution of vehicles.</li> <li>✓ Air pollution monitoring program has been devised for checking pollution level and suggesting remedial measures.</li> </ul>                                                                                                                                                                               | After completion of construction activity | EMU, SPCB Forest Dept, State Transport Dept., Police | Maintenance unit- APRDC Forest Dept |                                |
| <b>LAND ENVIRONMENT – OPERATION PHASE</b> |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |                                                      |                                     |                                |
| O.3                                       | Temporary land acquisition        | <ul style="list-style-type: none"> <li>✓ Borrow area redevelopment plan to be completed/ enforced.</li> <li>✓ All temporary acquired land for construction of diversion, transportation of material etc should be redeveloped to the</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | After completion of construction          | EMU, SPCB,                                           | Maintenance unit- APRDC             |                                |

| Cl. No                                     | Environmental Impact & Issues              | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Time frame                             | Regulation/ coordinating agency                                            | Implementing Agency                | Reference/ Bid document Clause |
|--------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------|------------------------------------|--------------------------------|
|                                            |                                            | satisfaction of owner.<br>✓ Affected productive area to be poured with top soil.                                                                                                                                                                                                                                                                                                                                                                                                           |                                        |                                                                            |                                    |                                |
| O.4                                        | Soil erosion                               | ✓ Embankment slopes to be re-vegetated, casualties to be replaced.<br>✓ Residual spoils to be disposed properly.                                                                                                                                                                                                                                                                                                                                                                           | After completion of construction       | EMU, SPCB,                                                                 | Maintenance unit-APRDC             |                                |
| O.5                                        | Soil Contamination                         | ✓ Accidental spills are potentially disastrous, but its probability is quite low as one of the objectives of this project is to enhance road safety.<br>✓ The public should be informed about the regulations on land pollution.<br>✓ Monitoring of Land pollution to be done regularly as per frequency and location mentioned in Appendix E.10 and suggesting remedial measures.                                                                                                         | After completion of construction       | EMU, SPCB, State Police, State Transport Dept.                             | Maintenance unit-APRDC             |                                |
| <b>WATER ENVIRONMENT – OPERATION PHASE</b> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                        |                                                                            |                                    |                                |
| O.6                                        | Silting/sedimentation                      | ✓ Measures suggested under “soil erosion” to be enforced.<br>✓ De-silting of existing water bodies to be taken up.<br>✓ Silt fencing to be provided.                                                                                                                                                                                                                                                                                                                                       | After completion of construction       | EMU, SPCB                                                                  | Maintenance unit-APRDC             |                                |
| O.7                                        | Contamination of water                     | ✓ Accidental spills are potentially disastrous, but its probability is quite low as one of the objective of this project is to enhance road safety.<br>✓ Discouraging local people from establishing workshops and car wash near public drinking water source.<br>✓ The public to be informed about the regulations on water pollution.<br>✓ Monitoring of water pollution to be done regularly as per frequency and location mentioned in Appendix E.10 and suggesting remedial measures. | After completion of construction       | EMU, SPCB, State Police, State Transport Dept. respective Municipal Agency | Maintenance unit-APRDC             |                                |
| O.8                                        | Maintenance of Storm Water Drainage System | ✓ The urban drainage systems should be maintained to accommodate storm water flow.<br>✓ Cleaning/removing of spoils should be ensured before/during the monsoon rains.                                                                                                                                                                                                                                                                                                                     | Especially at the start & end of rains | EMU, respective Municipal Agency                                           | Local municipal corporation        |                                |
| <b>NOISE ENVIRONMENT – OPERATION PHASE</b> |                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                        |                                                                            |                                    |                                |
| O.9                                        | Noise                                      | ✓ HORN PROHIBITED sign post will be enforced.<br>✓ Maintenance of noise barriers.<br>✓ Discouraging local people from establishing sensitive receptor near the road.                                                                                                                                                                                                                                                                                                                       | After completion of construction       | SPCB, State Police, Traffic Police, State Transport Dept.                  | Maintenance unit-APRDC Forest Dept |                                |

| Cl. No                                             | Environmental Impact & Issues       | Mitigation/Enhancement measures                                                                                                                                                                                                                                                                                                                                                      | Time frame                                      | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause            |
|----------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------------|---------------------|-------------------------------------------|
|                                                    |                                     | <ul style="list-style-type: none"> <li>✓ The public to be informed about the regulations on noise pollution.</li> <li>✓ Monitoring of noise pollution to be done regularly as per frequency and location mentioned in Appendix E.10 and suggesting remedial measures.</li> </ul>                                                                                                     |                                                 |                                 |                     |                                           |
| <b>FLORA &amp; FAUNA – OPERATION PHASE</b>         |                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                                                 |                                 |                     |                                           |
| O.10                                               | Loss of trees and Avenue Planting   | <ul style="list-style-type: none"> <li>✓ The avenue plantation should be completed, maintained and casualties to be replaced.</li> <li>✓ Discouraging local peoples from cutting tree/branches for fuel, cattle food etc.</li> <li>✓ Educating people about the usefulness of trees.</li> </ul>                                                                                      | After completion of construction                |                                 |                     |                                           |
| <b>SOCIO-ECONOMIC ENVIRONMENT– OPERATION PHASE</b> |                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                                                 |                                 |                     |                                           |
| O.11                                               | General issues                      | <ul style="list-style-type: none"> <li>✓ Public consultation to be organized after completion of construction to access the people opinion/grievances from the project intervention.</li> <li>✓ Remedial measures to mitigate the impact due to project intervention to be incorporated in the operation phase.</li> </ul>                                                           | Operation phase                                 | APRDC, EMU                      | Maintenance unit    |                                           |
| O.12                                               | Sensitive Communal Structures       | <ul style="list-style-type: none"> <li>✓ The mitigation/enhancement measures to be completed, and trees to be maintained and casualties to be replaced.</li> </ul>                                                                                                                                                                                                                   |                                                 |                                 |                     |                                           |
| <b>ROAD SAFETY ENVIRONMENT – OPERATION PHASE</b>   |                                     |                                                                                                                                                                                                                                                                                                                                                                                      |                                                 |                                 |                     |                                           |
| O.13                                               | Protection of high road embankments | <ul style="list-style-type: none"> <li>✓ Stabilization of altered (especially high emb.) embankments.</li> <li>✓ Although stone pitching is provided, vigilance to be maintained.</li> </ul>                                                                                                                                                                                         | Immediately after construction                  | APRDC                           | EMU                 | Design standard requirement               |
| O.14                                               | Safety and noise disturbance        | <ul style="list-style-type: none"> <li>✓ New buildings are prohibited within 50 m of the edge of carriageway.</li> <li>✓ No new schools and hospitals are allowed within 200 m of carriageway. Wherever required appropriate noise barrier should be constructed.</li> </ul>                                                                                                         | Throughout and after project development period | Local Bodies                    | Local Bodies        | IRC 35-1971<br>IRC 79-1981<br>IRC 93-1995 |
| O.15                                               | Accident black spot                 | <ul style="list-style-type: none"> <li>✓ Road surface to be maintained, pot holes to be filled immediately.</li> <li>✓ Regular maintenance of sign post, painting/removal of bills.</li> <li>✓ Road marking to be maintained.</li> <li>✓ People to be educated about the safety in following traffic rules.</li> <li>✓ Speed limit to be enforced at sensitive locations.</li> </ul> | During Operational stage.                       | PIU – EMU, Local Bodies         | PIU                 |                                           |

| Cl. No | Environmental Impact & Issues | Mitigation/Enhancement measures                                                                                                                                                                   | Time frame | Regulation/ coordinating agency | Implementing Agency | Reference/ Bid document Clause |
|--------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------|---------------------|--------------------------------|
|        |                               | <ul style="list-style-type: none"><li>✓ Lighting of major junctions near settlements.</li><li>✓ Mitigative /preventive measures for accident black spots, like traffic calming devises.</li></ul> |            |                                 |                     |                                |

## 9.2 ORGANIZATIONAL FRAMEWORK

The proposed project will be implemented by APRDC through its Environmental Management Unit (EMU). The EMU comprises of officers from Department of Forest, GoAP, and other environmental Engineers. The EMU will be coordinating with the field level implementing agencies such as Engineer (Supervision Consultant), Contractor and field level APSHP officials. Role and responsibilities of important officials is mentioned below.

**Table 9.2: Roles and Responsibilities of Officers**

| Officer                            | Responsibility                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chief Engineer & Managing Director | <ul style="list-style-type: none"> <li>• Overview of the project implementation</li> <li>• Ensure timely budget for the EMP.</li> <li>• Coordination with different state level committee, to obtain regulatory clearances.</li> <li>• Participate in state level meetings</li> <li>• Monthly review of the progress.</li> </ul>                                                                                                                                                                                                                                                  |
| Project Director APRSP             | <ul style="list-style-type: none"> <li>• Overall responsible for EMP implementation</li> <li>• Reporting to various stakeholders (World Bank, Regulatory bodies) on status of EMP implementation</li> <li>• Coordination with PIU Staff (EMU &amp; APSHP).</li> <li>• Responsible for obtaining regulatory Clearances</li> <li>• Review of the progress made by contractors</li> <li>• Ensure that BOQ items mentioned in EMP are executed as per Contract provisions.</li> </ul>                                                                                                 |
| Environmental Officer (PIU)        | <ul style="list-style-type: none"> <li>• Assisting Project Director in overall implementation of EMP</li> <li>• Review of periodic reports on EMP implementation and advising Project Director in taking corrective measure.</li> <li>• Conducting periodic field inspection of EMP implementation</li> <li>• Assisting Project Director to reporting various stakeholders (World Bank, Regulatory bodies) on status of EMP implementation</li> <li>• Preparing environmental training program and conducting the same for field officers and engineers of contractor.</li> </ul> |
| Engineer (Supervision Consultant)  | <ul style="list-style-type: none"> <li>• Act as an “Engineer” for supervising EMP implementation</li> <li>• Responsible for maintaining quality of EMP envisioned in detail Project Report</li> <li>• Maintaining progress reports on EMP implementation</li> <li>• Periodic reporting to PIU-APRSP about the status of EMP implementation</li> <li>• Work in close coordination with Executive Engineer (package unit) and contractor.</li> </ul>                                                                                                                                |
| Executive Engineer                 | <ul style="list-style-type: none"> <li>• Conducting need-based site inspection and preparing compliance reports and forwarding the same to the Environmental Management Unit (EMU)</li> <li>• Programming necessary training program on environmental issues.</li> </ul>                                                                                                                                                                                                                                                                                                          |
| Deputy Executive Engineer          | <ul style="list-style-type: none"> <li>• Working as site-representative of Executive Engineer (APSHP)</li> <li>• Conducting regular site inspection to all onsite and offsite works</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                    |

| Officer                              | Responsibility                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <ul style="list-style-type: none"> <li>• Maintaining records of all necessary statutory compliance, to be obtained from contractor.</li> <li>• Maintaining records of EMP implementation including photographic records</li> <li>• Attending environmental and social training programs</li> <li>• Preparing periodic reports on EMP implementation and forwarding to EE</li> </ul> |
| Environmental Engineer of Contractor | <ul style="list-style-type: none"> <li>• As detailed below</li> </ul>                                                                                                                                                                                                                                                                                                               |

***For ensuring that EMP is implemented as per provision in the document, Contractor shall nominate a qualified and experienced as Environmental Engineer from the commencement to completion of the project.***

**The responsibilities of Environmental Engineer of Contractor will include the following:**

- Directly reporting to the Project Manager of the Contractor;
- Discussing various environmental/social issues and environmental/social mitigation, enhancement and monitoring actions with all concerned directly or indirectly;
- Prepare Contractor's Checklist, traffic management plan and safety plan as part of their Work Program;
- Ensure Contractor's compliance with the ESMF stipulations and conditions of statutory bodies;
- Assisting his project manager to ensure social and environmentally sound and safe construction practices;
- Conducting periodic environmental and safety training for contractor's engineers, supervisors and workers along with sensitization on social issues that may be arising during the construction stage of the project;
- Preparing a registers for material sources, labour, pollution monitoring results (if required by APSPCB), public complaint/grievance redressal and as may be directed by the Engineer;
- Assisting the R&B on various environmental monitoring and control activities including pollution monitoring; and
- Preparing and submitting monthly/bio-monthly reports to R&B on status of implementation safeguard measures.
- Will be responsible for getting and maintaining the approvals or clearance for various departments and Environmental officer as per formats attached in **Appendix**.



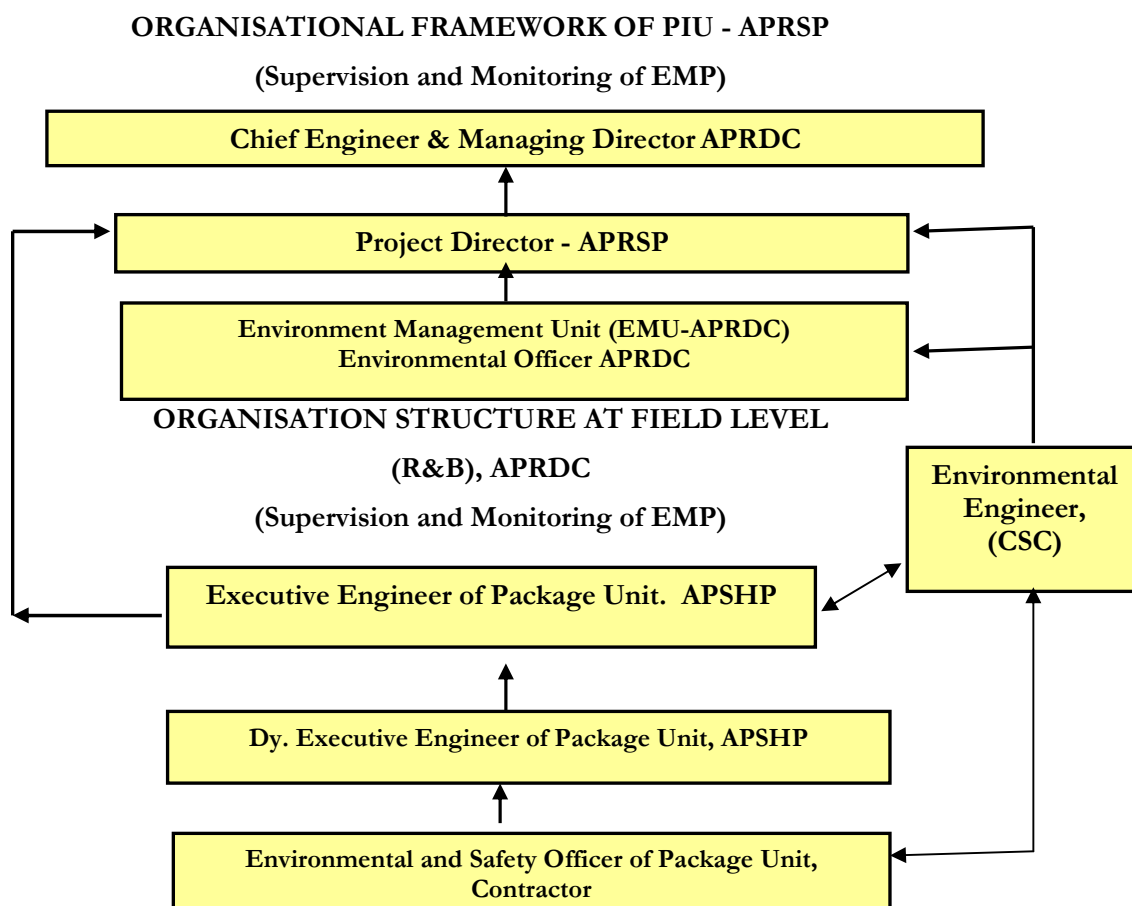


Fig. 9.1: Organisation Framework of PIU - APRSP

### 9.3 COST ESTIMATES FOR ENVIRONMENTAL MANAGEMENT

Mitigation measures proposed in the EMP will be implemented by the Contractor. The works to be undertaken by the Contractor have been quantified and the quantities included in the respective BOQ items such as earth works, slope protection, noise barriers, road safety features, and shrub plantation.

Provisional quantities have also been included for additional measures that may be identified during construction and for silt fencing which will depend on the Contractors work methods and site locations. Items and quantities have also been included for enhancement measures.

More general environmental management measures to be followed by the contractor have been included in the specifications and this EMP. These cannot be quantified and are to be included in the contract rates.

The budgetary provisions for the implementation of the environmental management plan of the Project are presented in **Table 9.3**.

Table 9.3: Cost Estimates for Environmental Management

| S. No.   | Item                          | Unit   | Rate (in INR)                                         | Quantity | Cost (in INR) | Remarks         |
|----------|-------------------------------|--------|-------------------------------------------------------|----------|---------------|-----------------|
| <b>A</b> | <b>PRE CONSTRUCTION PHASE</b> |        |                                                       |          |               |                 |
| 1        | Tree Felling Permission       | Number | 60                                                    | 4162     | 249720        | Trees to be cut |
| 2        | Utility Shifting              |        | Covered under regulatory clearance, Engineering cost. |          |               |                 |
|          | <b>Total Cost</b>             |        |                                                       |          | 249720        |                 |
| <b>B</b> | <b>CONSTRUCTION PHASE</b>     |        |                                                       |          |               |                 |

| S. No. | Item                                                             | Unit        | Rate (in INR) | Quantity | Cost (in INR) | Remarks                            |
|--------|------------------------------------------------------------------|-------------|---------------|----------|---------------|------------------------------------|
| 1      | Mitigation Measures other than Good Engineering practices        |             |               |          |               |                                    |
| 1.1    | Oil Interceptors                                                 | Number      | 5,000         | 4        | 20000         | Clause 8.3.5                       |
| 1.2    | Silt Fencing                                                     | Length, m   | 850           | 2000     | 1700000       | Clause 8.4.5                       |
| 1.3    | Soak Pits for Hand Pump/Wells                                    | Numbers     | 5000          | 8        | 40000         |                                    |
| 1.4    | Relocation of Hand Pumps                                         | Numbers     | 75,000        | 2        | 150,000       |                                    |
| 1.5    | Desilting/Deepning of Ponds                                      | Number      | 200,000       | 2        | 400,000       | Along Bypass                       |
| 1.6    | Digging of Wells                                                 | Number      | 100,000       | 1        | 100,000       | Along Bypass                       |
| 1.7    | Construction of noise barrier                                    | Sq mt       | 1200          | 700      | 840000        |                                    |
|        | Sub total                                                        |             |               |          | 32,50,000     |                                    |
| 2      | Tree Plantation and Protection                                   |             |               |          |               |                                    |
| 2.1    | Avenue Plantation including Compensatory Plantation.             |             |               |          |               |                                    |
| 2.1.1  | Plantation & maintenance of Saplings for 3 years                 | Number      | 1000          | 8,200    | 8200000       | As separate Contract               |
|        | Tree Guards                                                      |             |               |          |               |                                    |
| 2.1.2  | Brush Wood Fence                                                 | Numbers     | 180           | 5740     | 1033200       |                                    |
| 2.1.3  | Half Brick circular Tree Guard                                   | Numbers     | 974           | 1640     | 1597360       |                                    |
| 2.1.4  | Iron Tree Guard                                                  | Numbers     | 1635          | 820      | 1340700       |                                    |
| 2.2    | Plantation in Realignment/ Cultural properties Locations         |             |               |          |               |                                    |
| 2.2.1  | Plantation & maintenance of Saplings for 3 years                 | Number      | 1000          | 250      | 250000        | As separate Contract               |
|        | Tree Guards                                                      |             |               |          |               |                                    |
| 2.2.2  | Brush Wood Fence                                                 | Numbers     | 180           | 175      | 31500         |                                    |
| 2.2.3  | Half Brick circular Tree Guard                                   | Numbers     | 974           | 50       | 48700         |                                    |
| 2.2.4  | Iron Tree Guard                                                  | Numbers     | 1635          | 25       | 40875         |                                    |
|        | Sub Total                                                        |             |               |          | 12542335      |                                    |
| 3      | Landscaping                                                      |             |               |          |               |                                    |
| 3.1    | At Junction location                                             | Numbers     | 50            | 1400     | 70000         | At 1.5m for 50m                    |
| 4      | Monitoring of Environmental Attributes during Construction Phase |             |               |          |               |                                    |
| 4.1    | Air Quality                                                      |             |               |          |               |                                    |
| 4.1.1  | Monitoring of Air Quality near Hot mix plants                    | Per Samples | 5,000         | 6        | 30,000        | Three season in a year for 2 years |
| 4.1.2  | Monitoring of Air Quality at Critical Locations                  | Per Samples | 5,000         | 36       | 180,000       | Three season in a year for 2 years |
| 4.2    | Noise Levels                                                     |             |               |          |               |                                    |
| 4.2.1  | Monitoring of Noise Level at Equipment Yards                     | Per Samples | 4,000         | 6        | 24,000        | Three season in a year for 2 years |
| 4.2.2  | Monitoring of Noise Levels at Critical Locations                 | Per Samples | 4,000         | 54       | 216,000       | Three season in a year for 2 years |
| 4.3    | Water Quality                                                    |             |               |          |               |                                    |

| S. No.   | Item                                                                                                                | Unit                                   | Rate (in INR) | Quantity | Cost (in INR)    | Remarks                                   |
|----------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------|----------|------------------|-------------------------------------------|
| 4.3.1    | Monitoring of Water Quality                                                                                         | Per Samples                            | 5,000         | 48       | 240,000          | Four season in a year for 2 years         |
| 4.4      | Soil Quality                                                                                                        |                                        |               |          |                  |                                           |
| 4.4.1    | Monitoring of Soil Quality                                                                                          | Per Samples                            | 6,000         | 40       | 240,000          | Four season in a year for 2 years         |
| 4.4.2    | Additional Soil Monitoring during Spills                                                                            | Per Samples                            | 6,000         | 6        | 36,000           |                                           |
| 5        | <b>Orientation of Implementing agency staff</b>                                                                     | Per Session                            | -             | -        | -                | Deemed to be included in Consultancy fee. |
| 6        | <b>Mitigation/Enhancement Measures</b>                                                                              |                                        |               |          |                  | <b>966000</b>                             |
| 6.1      | <b>Cultural Properties (Generic Enhancement)</b>                                                                    |                                        |               |          |                  |                                           |
| 6.1.1    | Cultural Structures as per list                                                                                     | Numbers                                | 3,000         | 38       | 114,000          |                                           |
|          | Shifting of Part/total structure                                                                                    | Numbers                                | 100,000       | 29       | 2,900,000        |                                           |
| 6.3      | Surface Water Bodies/incidental                                                                                     | Numbers                                | 3,000         | 5        | 15,000           |                                           |
| 6.4      | For Incidental spaces                                                                                               | Numbers                                | 10,000        | 10       | 100,000          |                                           |
|          | Total Cost Of Environmental Measures During Construction Phase Excluding Costs Under Good Engineering Practices (B) |                                        |               |          | <b>3,129,000</b> |                                           |
| <b>C</b> | <b>GOOD ENGINEERING PRACTICES</b>                                                                                   |                                        |               |          |                  |                                           |
| 1        | Dust Suppression                                                                                                    | <b>Covered under Engineering Costs</b> |               |          |                  |                                           |
| 2        | Erosion Control Measures (Turfing / Pitching / Seeding & Mulching)                                                  |                                        |               |          |                  |                                           |
| 3        | Provision of Cross drainage & side drainage structures                                                              |                                        |               |          |                  |                                           |
| 4        | General Borrow area management and maintenance of haul roads related to borrow areas                                |                                        |               |          |                  |                                           |
| 5        | Air/noise pollution control measures in construction equipments                                                     |                                        |               |          |                  |                                           |
| 6        | Management and disposal of scarified waste bituminous material                                                      |                                        |               |          |                  |                                           |
| 7        | Provision of Informatory Signs                                                                                      |                                        |               |          |                  |                                           |
| 8        | Bus shelters                                                                                                        |                                        |               |          |                  |                                           |
| 9        | Construction of Speed Humps                                                                                         |                                        |               |          |                  |                                           |

| S. No. | Item                                                                           | Unit       | Rate (in INR)                                      | Quantity | Cost (in INR) | Remarks                            |
|--------|--------------------------------------------------------------------------------|------------|----------------------------------------------------|----------|---------------|------------------------------------|
| 10     | Cattle Crossings                                                               |            |                                                    |          |               |                                    |
| 11     | Management of quarries                                                         |            | Deemed to be included in contractor's quoted rates |          |               |                                    |
| 12     | Redevelopment of Borrow Areas                                                  |            |                                                    |          |               |                                    |
| 13     | Construction Camp Management Costs                                             |            | Will form a part of the Contractor's cost          |          |               |                                    |
| 14     | Safety measures for workers                                                    |            |                                                    |          |               |                                    |
| D      | ITEMS COVERED UNDER THE RAP BUDGET                                             |            |                                                    |          |               |                                    |
| 1      | Relocation of private Properties                                               |            | Covered under RAP Budget                           |          |               |                                    |
| 2      | Relocation of private Water points (wells, tanks, water taps and hand pumps)   |            |                                                    |          |               |                                    |
| 3      | Relocation of graveyards, statues, motor sheds                                 |            |                                                    |          |               |                                    |
| 4      | Relocation of Other Community Assets (private)                                 |            |                                                    |          |               |                                    |
| E      | OPERATION PHASE                                                                |            |                                                    |          |               |                                    |
| 1      | Monitoring of Environmental Attributes during Operation Phase                  |            |                                                    |          |               |                                    |
| 1.1    | Monitoring of Air Quality at Critical Locations                                | Per Sample |                                                    | 36       | 180000        | Two season in a year for 3 years   |
| 1.2    | Monitoring of Noise Levels at Critical Locations                               | Per Sample |                                                    | 54       | 216000        | Three season in a year for 3 years |
| 1.3    | Monitoring of Water Quality                                                    | Per Sample |                                                    | 36       | 180000        | Four season in a year for 3 years  |
| 1.4    | Monitoring of Soil Quality                                                     | Per Sample |                                                    |          |               | Four season in a year for 3 years  |
| 1.5    | Additional Soil Monitoring during Spills                                       | Per Sample |                                                    |          |               | Throughout operation phase.        |
| 1.6    | Noise mitigation measures in form of noise barrier at sensitive receptors*     | Sq. m      |                                                    |          |               |                                    |
| 2      | Information Dissemination                                                      | -          | -                                                  | -        | -             | Covered under RAP                  |
| 3      | Additional Training for Land use Management at Package Level                   | Package    |                                                    | 1        |               |                                    |
|        | Environmental Budget During Operation Phase                                    |            |                                                    |          | 576000        |                                    |
|        | Sub Total (A+B+E) including operation cost                                     |            |                                                    | INR      | 20783055      | Say                                |
|        | Sub Total (B1+B3+B4+B6)                                                        |            |                                                    | INR      | 74,15,000     |                                    |
|        | GRAND TOTAL US\$ @ INR 40.00/\$                                                |            |                                                    | US\$     | 1,42,596      |                                    |
|        | * The provision made shall be utilised in construction phase also if necessary |            |                                                    |          |               |                                    |

**9.4 BASIS OF COSTING****(A) COSTS DURING PRE-CONSTRUCTION STAGE****ITEM 1.0 TREE FELLING PERMISSION**

Tree felling permission will be obtained from Forest Department of Andhra Pradesh under WALTA ACT 2002, a fee of Rs 60 per felling tree is charged from the agency, in this case APRDC, and for tree felling cost is considered in Engineering cost.

**ITEM 2.0 UTILITY SHIFTING**

The cost of existing utility shifting is taken separately in engineering costing, under the Non civil cost Head.

**(B) COSTS DURING CONSTRUCTION PHASE****ITEM 1.0 MITIGATION MEASURES OTHER THAN GOOD ENGINEERING PRACTICES**

This section includes the costs for the mitigation measures during the construction other than those included in the Engineering Budget. It covers the following items:

**Item 1.1 Oil Interceptors**

The unit cost of Single Chamber Gravity Separation Type Oil & Grease Interceptors (Influent handling capacity 20 L/hr) as per prevalent market rates is Rs 5,000.

**Item 1.2 Silt Fencing**

The per unit length cost of silt fencing as per market rate is Rs 850.

**Item 1.3 Soak pits for hand pumps and wells.**

The unit cost of soak pit of 1.5m diameter and 1.0m deep made of brick masonry/concrete rings, filled with brick bats and pables, as per market rate of Rs 5,000.

**Item 1.4 Relocation of ground water bodies**

The unit cost of relocation of ground water sources has been considered as per market rate of Rs75,000/source.

**Item 1.5 Pond deepening/desilting**

A lump sum cost of Rs.2, 00, 000/- has been provided for deepening/desilting of pond. The unit rates for earthwork excavation from ponds have been obtained from the Schedule of Rates as Rs.62/- per cum.

**Item 1.6 Digging of Well**

A lump sum cost of Rs.100, 000/- has been provided for digging of each well. The unit rates for earthwork excavation from ponds have been obtained from the Schedule of Rates as Rs.62/- per cum.

**Item 1.7 Construction of noise barrier**

A lump sum cost of Rs.1200/- per sq.mt has been provided for each noise barrier including GI wire mesh

**ITEM 2.0 TREE PLANTATION AND PROTECTION****Item 2.1 Avenue plantation including Compensatory afforestation**

The afforestation will be done by APRDC through Forest department. The MoU will be signed between APRDC and Forest Department.

**ITEM 3.0 LANDSCAPING**

It includes the costs involved in landscaping apart from tree plantation. It is proposed to plant dwarf shrubs along the T-junctions, realignment location and hedging along Right of Way.

**Item 3.1 Along junction**

It has been estimated that the number of shrubs required is:

Considering 100no.s of shrubs at each major junction, planting at 1.5 m c/c. The unit cost of shrubs is Rs 50.

**ITEM 4.0 MONITORING OF ENVIRONMENTAL ATTRIBUTES DURING CONSTRUCTION ACTIVITY****Item 4.1 Air Quality**

The cost for continuous 24 hour monitoring for SPM, RPM, SO<sub>2</sub>, NO<sub>x</sub>, CO & Pb as per prevalent market rates is Rs 5,000/sample. (For locations and frequency refer Appendix 8.15 Monitoring plan)

**Item 4.2 Noise Level**

The cost for noise level monitoring on dB(A) scale (readings to be taken at 15 second interval for 15 minutes every hour for a total period of 24 hours) as per prevalent market rates is Rs 4,000/sample. (For locations and frequency refer Appendix 8.15 Monitoring plan)

**Item 4.3 Water Quality**

Water quality will be monitored for the following parameters:

- pH, BOD, COD, TDS, Pb, Oil & Grease and Detergents for surface water
- pH, TDS, Total Hardness, Sulphate, Chloride, Fe, Pb and Coliform count for ground water

Cost for one time monitoring of water quality as per prevalent market rates is Rs 5,000/sample. (For locations and frequency refer Appendix 8.15 Monitoring plan)

**Item 4.4 Soil Quality**

Cost for one time monitoring of soil quality for Pb, Sodium Absorption Ratio & Oil & Grease as per prevalent market rates is Rs 6,000/sample. (For locations and frequency refer Appendix 8.15 Monitoring plan) Provisions have also been included in the budget to monitor the soil quality in event of any major accident/spillage during bulk transport of hazardous material. Costs are incorporated for monitoring at such locations.

**ITEM 5.0 ORIENTATION OF IMPLEMENTING AGENCY STAFF**

This item includes the costs to be incurred for training sessions involving the PIU staff. The training programme will be attended by EMU staff –APRDC, Engineer – Supervision

**ITEM 6.0 ENHANCEMENT MEASURES**

The costs have been calculated for each enhancement design (specific & generic) based on the individual BoQ is prepared. For generic enhancements of cultural properties, water bodies and incidental spaces adequate provisions have been made in the budget. The specific enhancements proposed has formulated based on the community consultation, given as under.

**Table 9.4: Specific enhancement of Community Properties**

| Sl. No. | Name of Common Property         | Chainage (Km.) | Location (Left / Right) | Impact           | Mitigation/Enhancement |
|---------|---------------------------------|----------------|-------------------------|------------------|------------------------|
| 1       | Venkatesh Temple                | 1+800          | Right                   | Direct Impact    | Generic Enhancement    |
| 2       | Church                          | 2+000          | Right                   | Direct Impact    | Generic Enhancement    |
| 3       | Temple                          | 3+000          | Left                    | No Direct Impact | Generic Enhancement    |
| 4       | Church                          | 3+300          | Right                   | Direct Impact    | Generic Enhancement    |
| 5       | Temple                          | 4+100          | Right                   | Direct Impact    | Generic Enhancement    |
| 6       | Temple                          | 4+825          | Right                   | Direct Impact    | Relocated              |
| 7       | Temple/Pipaltree                | 5+525          | Right                   | No Direct Impact | Generic Enhancement    |
| 8       | Temple                          | 8+700          | Right                   | No Direct Impact | Generic Enhancement    |
| 9       | Church                          | 9+175          | Right                   | No Direct Impact | Generic Enhancement    |
| 10      | Temple(Banyan&Neem)             | 10+575         | Right                   | Direct Impact    | Relocated              |
| 11      | Temple                          | 16+000         | Left                    | No Direct Impact | Generic Enhancement    |
| 12      | Temple                          | 16+150         | Right                   | No Direct Impact | Generic Enhancement    |
| 13      | Temple (BW)                     | 16+200         | Left                    | No Direct Impact | Generic Enhancement    |
| 14      | Temple                          | 16+250         | Left                    | No Direct Impact | Generic Enhancement    |
| 15      | Temple                          | 16+450         | Right                   | Direct Impact    | Relocated              |
| 16      | Temple                          | 17+550         | Left                    | No Direct Impact | Generic Enhancement    |
| 17      | Temple                          | 18+000         | Right                   | Direct Impact    | Relocated              |
| 18      | Ganesha Temple                  | 18+050         | Left                    | No Direct Impact | Generic Enhancement    |
| 19      | Temple                          | 18+600         | Right                   | Direct Impact    | Relocated              |
| 20      | Temple                          | 22+100         | Left                    | No Direct Impact | Generic Enhancement    |
| 21      | Temple& Lands                   | 22+650         | Left                    | No Direct Impact | Generic Enhancement    |
| 22      | Temple                          | 22+700         | Left                    | Direct Impact    | Relocated              |
| 23      | Temple                          | 25+200         | Left                    | No Direct Impact | Generic Enhancement    |
| 24      | Temple                          | 25+200         | Left                    | Direct Impact    | Relocated              |
| 25      | Temple                          | 25+200         | Right                   | Direct Impact    | Relocated              |
| 26      | Temple                          | 25+350         | Left                    | Direct Impact    | Relocated              |
| 27      | Temple                          | 28+000         | Right                   | Direct Impact    | Relocated              |
| 28      | Temple                          | 30+300         | Left                    | No Direct Impact | Generic Enhancement    |
| 29      | Temple                          | 31+800         | Left                    | Direct Impact    | Relocated              |
| 30      | Temple/Ghat                     | 34+670         | Left                    | No Direct Impact | Generic Enhancement    |
| 31      | Sai Temple                      | 35+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 32      | Temple                          | 35+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 33      | Temple                          | 35+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 34      | Statue Of Lion In Mouth Goddess | 35+200         | Right                   | No Direct Impact | Generic Enhancement    |
| 35      | Temple                          | 38+100         | Left                    | No Direct Impact | Generic Enhancement    |
| 36      | Temple                          | 40+050         | Right                   | Direct Impact    | Relocated              |
| 37      | Temple                          | 40+150         | Right                   | Direct Impact    | Relocated              |



| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Impact           | Mitigation/Enhancement |
|---------|-------------------------|----------------|-------------------------|------------------|------------------------|
| 38      | Temple                  | 40+400         | Right                   | Direct Impact    | Relocated              |
| 39      | Temple                  | 40+400         | Right                   | Direct Impact    | Relocated              |
| 40      | Temple                  | 41+200         | Left                    | No Direct Impact | Generic Enhancement    |
| 41      | Temple                  | 41+350         | Right                   | No Direct Impact | Generic Enhancement    |
| 42      | Temple                  | 41+550         | Left                    | Direct Impact    | Relocated              |
| 43      | Temple                  | 41+700         | Left                    | Direct Impact    | Relocated              |
| 44      | Church (BW)             | 42+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 45      | Temple                  | 44+150         | Right                   | No Direct Impact | Generic Enhancement    |
| 46      | Big Temple (BW)         | 44+600         | Right                   | No Direct Impact | Generic Enhancement    |
| 47      | Temple                  | 44+700         | Right                   | No Direct Impact | Generic Enhancement    |
| 48      | Temple                  | 45+000         | Left                    | No Direct Impact | Generic Enhancement    |
| 49      | Temple                  | 45+350         | Left                    | No Direct Impact | Generic Enhancement    |
| 50      | Temple                  | 45+900         | Left                    | No Direct Impact | Generic Enhancement    |
| 51      | Temple Saibaba          | 46+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 52      | Temple                  | 48+800         | Right                   | Direct Impact    | Relocated              |
| 53      | Temple                  | 49+550         | Right                   | Direct Impact    | Relocated              |
| 54      | Temple                  | 52+300         | Right                   | No Direct Impact | Generic Enhancement    |
| 55      | Sai Temple              | 54+200         | Right                   | Direct Impact    | Relocated              |
| 56      | Temple                  | 55+500         | Left                    | Direct Impact    | Relocated              |
| 57      | Temple                  | 56+800         | Left                    | Direct Impact    | Relocated              |
| 58      | Temple                  | 57+050         | Right                   | No Direct Impact | Generic Enhancement    |
| 59      | Temple                  | 57+200         | Left                    | No Direct Impact | Generic Enhancement    |
| 60      | Temple (BW)             | 57+300         | Left                    | No Direct Impact | Generic Enhancement    |
| 61      | Temple                  | 57+300         | Left                    | No Direct Impact | Generic Enhancement    |
| 62      | Temple                  | 58+950         | Right                   | Direct Impact    | Relocated              |
| 63      | Ganesh Temple           | 59+550         | Right                   | Direct Impact    | Relocated              |
| 64      | Temple                  | 60+200         | Right                   | Direct Impact    | Relocated              |
| 65      | Mary Statue             | 60+750         | Right                   | No Direct Impact | Generic Enhancement    |
| 66      | Church                  | 60+750         | Right                   | No Direct Impact | Generic Enhancement    |
| 67      | Park & Hanuman Temple   | 61+800         | Right                   | No Direct Impact | Generic Enhancement    |

Generic Enhancement includes small repairs, white wash, small safety measures, and plantation as directed by EO.

#### **(E) COSTS DURING OPERATION STAGE**

##### **ITEM 1.0 MONITORING OF ENVIRONMENTAL ATTRIBUTES DURING OPERATION PHASE**

The rates for monitoring of quality are same as those mentioned for the construction stage. Provisions have been made in the budget to monitor each year upto 3 years from completion of construction.

##### **ITEM 2.0 INFORMATION DISSEMINATION**

This item includes the costs to be incurred for information dissemination during the implementation stage. Cost of information dissemination has been included in RAP towards the cost of NGO.

##### **ITEM 3.0 ADDITIONAL TRAINING FOR LAND USE MANAGEMENT**

It refers to the training to be imparted for land use management at package level. It is proposed to have one training session during the construction stage. The cost per session has been estimated as Rs 1,00,000. These costs are applied for all packages.

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**APPENDIX E12: MITIGATION/ENHANCEMENT OF SENSITIVE CULTURAL  
PROPERTIES**

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 1      | 0+000    | 0+000 | 10     | 62           | Bhagi     | 6                 | T2                  |
| 2      |          |       | 15     | 110          | Naredu    | 6                 | T3                  |
| 3      |          |       | 15     | 95           | Giriship  | 6                 | T3                  |
| 4      |          |       | 15     | 116          | Thurai    | 6                 | T3                  |
| 5      | 0+200    | 0+400 | 20     | 65           | Palm      | 6                 | T2                  |
| 6      |          |       | 20     | 66           | Coconut   | 6                 | T2                  |
| 7      |          |       | 20     | 69           | Raavi     | 6                 | T2                  |
| 8      | 0+400    | 0+600 | 20     | 118          | Tamarind  | 6                 | T3                  |
| 9      | 1+000    | 1+600 | 10     | 75           | Pongamia  | 6                 | T2                  |
| 10     |          |       | 10     | 96           | Pongamia  | 6                 | T3                  |
| 11     |          |       | 10     | 100          | Raavi     | 6                 | T3                  |
| 12     | 1+800    | 2+000 | 10     | 85           | Pongamia  | 6                 | T2                  |
| 13     | 2+600    | 2+800 | 10     | 106          | Pongamia  | 6                 | T3                  |
| 14     |          |       | 10     | 212          | Bhagi     | 6                 | T3                  |
| 15     | 2+800    | 3+000 | 10     | 84           | Thurai    | 6                 | T2                  |
| 16     |          |       | 10     | 86           | Pongamia  | 6                 | T2                  |
| 17     |          |       | 10     | 123          | Raavi     | 6                 | T3                  |
| 18     | 3+000    | 3+200 | 10     | 83           | Acasia    | 6                 | T2                  |
| 19     |          |       | 10     | 82           | Acasia    | 6                 | T2                  |
| 20     | 3+800    | 4+000 | 10     | 86           | Coconut   | 6                 | T2                  |
| 21     |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 22     |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 23     |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 24     |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 25     |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 26     |          |       | 10     | 72           | Coconut   | 6                 | T2                  |
| 27     |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 28     |          |       | 10     | 71           | Coconut   | 6                 | T2                  |
| 29     |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 30     |          |       | 10     | 89           | Coconut   | 6                 | T2                  |
| 31     |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 32     |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 33     |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 34     |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 35     |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 36     | 4+000    | 4+200 | 10     | 69           | Coconut   | 6                 | T2                  |
| 37     |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 38     |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 39     |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 40     |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 41     |          |       | 10     | 71           | Coconut   | 6                 | T2                  |
| 42     |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 43     |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 44     |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 45     |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 46     |          |       | 10     | 67           | Coconut   | 6                 | T2                  |
| 47     |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 48     |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 49     |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 50     |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 51     |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 52     |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 53     |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 54     |          |       | 10     | 70           | Coconut   | 6                 | T2                  |

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 55     |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 56     |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 57     |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 58     |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 59     |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 60     |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 61     |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 62     | 4+200    | 4+400 | 10     | 79           | Coconut   | 6                 | T2                  |
| 63     |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 64     |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 65     |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 66     |          |       | 10     | 72           | Coconut   | 6                 | T2                  |
| 67     |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 68     | 4+400    | 4+600 | 10     | 84           | Coconut   | 6                 | T2                  |
| 69     |          |       | 10     | 80           | Giriship  | 6                 | T2                  |
| 70     |          |       | 10     | 76           | Giriship  | 6                 | T2                  |
| 71     | 5+000    | 5+600 | 10     | 89           | Coconut   | 6                 | T2                  |
| 72     |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 73     |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 74     | 5+800    | 6+000 | 10     | 70           | Coconut   | 6                 | T2                  |
| 75     |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 76     |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 77     |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 78     |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 79     |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 80     |          |       | 10     | 63           | Coconut   | 6                 | T2                  |
| 81     |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 82     |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 83     |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 84     |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 85     |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 86     |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 87     |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 88     |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 89     |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 90     |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 91     |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 92     |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 93     |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 94     |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 95     |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 96     |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 97     |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 98     |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 99     |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 100    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 101    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 102    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 103    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 104    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 105    |          |       | 10     | 74           | Coconut   | 6                 | T2                  |
| 106    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 107    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 108    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 109    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 110    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 111    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 112    |          |       | 10     | 74           | Coconut   | 6                 | T2                  |
| 113    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 114    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 115    |          |       | 10     | 89           | Coconut   | 6                 | T2                  |
| 116    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 117    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 118    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 119    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 120    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 121    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 122    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 123    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 124    |          |       | 10     | 63           | Coconut   | 6                 | T2                  |
| 125    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 126    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 127    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 128    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 129    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 130    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 131    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 132    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 133    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 134    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 135    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 136    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 137    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 138    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 139    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 140    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 141    |          |       | 10     | 71           | Coconut   | 6                 | T2                  |
| 142    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 143    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 144    | 6+000    | 6+200 | 10     | 68           | Coconut   | 6                 | T2                  |
| 145    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 146    |          |       | 10     | 67           | Coconut   | 6                 | T2                  |
| 147    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 148    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 149    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 150    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 151    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 152    |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 153    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 154    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 155    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 156    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 157    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 158    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 159    |          |       | 10     | 60           | Coconut   | 6                 | T2                  |
| 160    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 161    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 162    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 163    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 164    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 165    |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 166    |          |       | 10     | 72           | Coconut   | 6                 | T2                  |
| 167    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 168    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 169    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 170    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 171    | 6+400    | 6+600 | 10     | 89           | Coconut   | 6                 | T2                  |
| 172    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 173    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 174    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 175    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 176    |          |       | 10     | 66           | Coconut   | 6                 | T2                  |
| 177    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 178    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 179    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 180    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 181    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 182    |          |       | 10     | 83           | Coconut   | 6                 | T2                  |
| 183    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 184    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 185    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 186    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 187    |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 188    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 189    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 190    |          |       | 10     | 72           | Coconut   | 6                 | T2                  |
| 191    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 192    |          |       | 10     | 71           | Coconut   | 6                 | T2                  |
| 193    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 194    |          |       | 10     | 89           | Coconut   | 6                 | T2                  |
| 195    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 196    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 197    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 198    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 199    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 200    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 201    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 202    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 203    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 204    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 205    |          |       | 10     | 71           | Coconut   | 6                 | T2                  |
| 206    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 207    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 208    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 209    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 210    |          |       | 10     | 67           | Coconut   | 6                 | T2                  |
| 211    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 212    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 213    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 214    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 215    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 216    |          |       | 10     | 78           | Coconut   | 6                 | T2                  |



# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 217    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 218    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 219    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 220    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 221    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 222    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 223    |          |       | 10     | 60           | Coconut   | 6                 | T2                  |
| 224    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 225    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 226    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 227    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 228    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 229    | 6+600    | 7+000 | 10     | 78           | Coconut   | 6                 | T2                  |
| 230    |          |       | 10     | 72           | Coconut   | 6                 | T2                  |
| 231    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 232    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 233    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 234    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 235    |          |       | 10     | 89           | Coconut   | 6                 | T2                  |
| 236    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 237    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 238    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 239    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 240    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 241    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 242    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 243    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 244    |          |       | 10     | 63           | Coconut   | 6                 | T2                  |
| 245    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 246    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 247    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 248    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 249    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 250    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 251    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 252    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 253    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 254    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 255    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 256    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 257    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 258    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 259    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 260    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 261    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 262    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 263    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 264    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 265    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 266    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 267    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 268    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 269    |          |       | 10     | 74           | Coconut   | 6                 | T2                  |
| 270    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 271    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 272    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 273    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 274    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 275    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 276    | 7+000    | 7+200 | 10     | 74           | Coconut   | 6                 | T2                  |
| 277    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 278    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 279    |          |       | 10     | 89           | Coconut   | 6                 | T2                  |
| 280    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 281    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 282    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 283    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 284    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 285    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 286    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 287    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 288    |          |       | 10     | 63           | Coconut   | 6                 | T2                  |
| 289    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 290    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 291    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 292    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 293    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 294    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 295    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 296    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 297    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 298    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 299    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 300    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 301    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 302    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 303    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 304    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 305    |          |       | 10     | 71           | Coconut   | 6                 | T2                  |
| 306    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 307    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 308    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 309    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 310    |          |       | 10     | 67           | Coconut   | 6                 | T2                  |
| 311    | 7+400    | 7+600 | 10     | 65           | Coconut   | 6                 | T2                  |
| 312    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 313    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 314    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 315    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 316    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 317    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 318    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 319    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 320    |          |       | 10     | 66           | Coconut   | 6                 | T2                  |
| 321    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 322    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 323    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 324    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 325    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 326    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 327    |          |       | 10     | 63           | Coconut   | 6                 | T2                  |
| 328    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 329    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 330    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 331    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 332    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 333    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 334    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 335    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 336    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 337    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 338    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 339    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 340    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 341    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 342    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 343    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 344    |          |       | 10     | 71           | Coconut   | 6                 | T2                  |
| 345    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 346    |          |       | 10     | 69           | Coconut   | 6                 | T2                  |
| 347    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 348    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 349    |          |       | 10     | 67           | Coconut   | 6                 | T2                  |
| 350    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 351    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 352    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 353    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 354    |          |       | 10     | 65           | Coconut   | 6                 | T2                  |
| 355    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 356    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 357    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 358    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 359    |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 360    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 361    | 7+600    | 7+800 | 10     | 70           | Coconut   | 6                 | T2                  |
| 362    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 363    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 364    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 365    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 366    |          |       | 10     | 60           | Coconut   | 6                 | T2                  |
| 367    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 368    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 369    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 370    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 371    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 372    |          |       | 10     | 78           | Coconut   | 6                 | T2                  |
| 373    |          |       | 10     | 72           | Coconut   | 6                 | T2                  |
| 374    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 375    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 376    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 377    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 378    |          |       | 10     | 89           | Coconut   | 6                 | T2                  |

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |       | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|-------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To    |        |              |           |                   |                     |
| 379    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 380    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 381    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 382    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 383    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 384    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 385    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 386    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 387    |          |       | 10     | 63           | Coconut   | 6                 | T2                  |
| 388    | 7+800    | 8+000 | 10     | 76           | Coconut   | 6                 | T2                  |
| 389    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 390    |          |       | 10     | 62           | Coconut   | 6                 | T2                  |
| 391    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 392    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 393    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 394    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 395    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 396    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 397    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 398    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 399    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 400    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 401    |          |       | 10     | 75           | Coconut   | 6                 | T2                  |
| 402    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 403    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 404    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 405    |          |       | 10     | 82           | Coconut   | 6                 | T2                  |
| 406    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 407    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 408    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 409    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 410    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 411    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 412    |          |       | 10     | 74           | Coconut   | 6                 | T2                  |
| 413    |          |       | 10     | 81           | Coconut   | 6                 | T2                  |
| 414    |          |       | 10     | 86           | Coconut   | 6                 | T2                  |
| 415    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 416    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 417    |          |       | 10     | 73           | Coconut   | 6                 | T2                  |
| 418    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 419    |          |       | 10     | 74           | Coconut   | 6                 | T2                  |
| 420    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 421    | 8+000    | 8+200 | 10     | 76           | Coconut   | 6                 | T2                  |
| 422    |          |       | 10     | 89           | Coconut   | 6                 | T2                  |
| 423    |          |       | 10     | 84           | Coconut   | 6                 | T2                  |
| 424    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |
| 425    |          |       | 10     | 70           | Coconut   | 6                 | T2                  |
| 426    |          |       | 10     | 80           | Coconut   | 6                 | T2                  |
| 427    |          |       | 10     | 85           | Coconut   | 6                 | T2                  |
| 428    |          |       | 10     | 79           | Coconut   | 6                 | T2                  |
| 429    |          |       | 10     | 68           | Coconut   | 6                 | T2                  |
| 430    |          |       | 10     | 64           | Coconut   | 6                 | T2                  |
| 431    |          |       | 10     | 63           | Coconut   | 6                 | T2                  |
| 432    |          |       | 10     | 76           | Coconut   | 6                 | T2                  |

**APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)**

| Sl. No | Chainage |        | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|--------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To     |        |              |           |                   |                     |
| 433    |          |        | 10     | 73           | Coconut   | 6                 | T2                  |
| 434    |          |        | 10     | 62           | Coconut   | 6                 | T2                  |
| 435    |          |        | 10     | 68           | Coconut   | 6                 | T2                  |
| 436    |          |        | 10     | 70           | Coconut   | 6                 | T2                  |
| 437    |          |        | 10     | 79           | Coconut   | 6                 | T2                  |
| 438    |          |        | 10     | 81           | Coconut   | 6                 | T2                  |
| 439    |          |        | 10     | 76           | Coconut   | 6                 | T2                  |
| 440    |          |        | 10     | 82           | Coconut   | 6                 | T2                  |
| 441    |          |        | 10     | 73           | Coconut   | 6                 | T2                  |
| 442    |          |        | 10     | 84           | Coconut   | 6                 | T2                  |
| 443    | 8+200    | 8+400  | 10     | 75           | Coconut   | 6                 | T2                  |
| 444    |          |        | 10     | 86           | Coconut   | 6                 | T2                  |
| 445    |          |        | 10     | 75           | Coconut   | 6                 | T2                  |
| 446    |          |        | 10     | 70           | Coconut   | 6                 | T2                  |
| 447    |          |        | 10     | 76           | Coconut   | 6                 | T2                  |
| 448    |          |        | 10     | 71           | Coconut   | 6                 | T2                  |
| 449    |          |        | 10     | 76           | Coconut   | 6                 | T2                  |
| 450    |          |        | 10     | 69           | Coconut   | 6                 | T2                  |
| 451    |          |        | 10     | 68           | Coconut   | 6                 | T2                  |
| 452    |          |        | 10     | 62           | Coconut   | 6                 | T2                  |
| 453    |          |        | 10     | 67           | Coconut   | 6                 | T2                  |
| 454    |          |        | 10     | 65           | Coconut   | 6                 | T2                  |
| 455    |          |        | 10     | 84           | Coconut   | 6                 | T2                  |
| 456    |          |        | 10     | 82           | Coconut   | 6                 | T2                  |
| 457    |          |        | 10     | 80           | Coconut   | 6                 | T2                  |
| 458    |          |        | 10     | 85           | Coconut   | 6                 | T2                  |
| 459    |          |        | 10     | 78           | Coconut   | 6                 | T2                  |
| 460    |          |        | 10     | 73           | Coconut   | 6                 | T2                  |
| 461    |          |        | 10     | 70           | Coconut   | 6                 | T2                  |
| 462    |          |        | 10     | 85           | Coconut   | 6                 | T2                  |
| 463    |          |        | 10     | 81           | Coconut   | 6                 | T2                  |
| 464    |          |        | 10     | 79           | Coconut   | 6                 | T2                  |
| 465    |          |        | 10     | 64           | Coconut   | 6                 | T2                  |
| 466    |          |        | 10     | 65           | Coconut   | 6                 | T2                  |
| 467    |          |        | 10     | 86           | Coconut   | 6                 | T2                  |
| 468    |          |        | 10     | 84           | Coconut   | 6                 | T2                  |
| 469    | 8+400    | 8+600  | 10     | 79           | Bhagi     | 6                 | T2                  |
| 470    |          |        | 15     | 85           | Bhagi     | 6                 | T2                  |
| 471    |          |        | 15     | 76           | Naredu    | 6                 | T2                  |
| 472    | 10+200   | 10+400 | 10     | 78           | Coconut   | 6                 | T2                  |
| 473    |          |        | 10     | 72           | Coconut   | 6                 | T2                  |
| 474    |          |        | 10     | 80           | Coconut   | 6                 | T2                  |
| 475    |          |        | 10     | 84           | Coconut   | 6                 | T2                  |
| 476    |          |        | 10     | 80           | Coconut   | 6                 | T2                  |
| 477    |          |        | 10     | 76           | Coconut   | 6                 | T2                  |
| 478    | 10+400   | 10+600 | 10     | 89           | Coconut   | 6                 | T2                  |
| 479    |          |        | 10     | 84           | Coconut   | 6                 | T2                  |
| 480    |          |        | 10     | 76           | Coconut   | 6                 | T2                  |
| 481    |          |        | 10     | 62           | Coconut   | 6                 | T2                  |
| 482    |          |        | 10     | 65           | Coconut   | 6                 | T2                  |
| 483    |          |        | 10     | 66           | Coconut   | 6                 | T2                  |
| 484    | 16+000   | 16+200 | 10     | 85           | RedThurai | 6                 | T2                  |
| 485    | 16+400   | 16+600 | 10     | 86           | Dirsena   | 6                 | T2                  |
| 486    |          |        | 10     | 78           | Dirsena   | 6                 | T2                  |

# APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |        | Height | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|--------|--------|--------------|-----------|-------------------|---------------------|
|        | From     | To     |        |              |           |                   |                     |
| 487    | 16+600   | 16+800 | 10     | 79           | Dirsena   | 6                 | T2                  |
| 488    | 16+800   | 17+000 | 10     | 76           | Dirsena   | 6                 | T2                  |
| 489    |          |        | 10     | 72           | Dirsena   | 6                 | T2                  |
| 490    | 17+000   | 17+200 | 10     | 70           | Dirsena   | 6                 | T2                  |
| 491    |          |        | 10     | 71           | Dirsena   | 6                 | T2                  |
| 492    | 18+800   | 19+000 | 10     | 70           | Thurai    | 6                 | T2                  |
| 493    | 19+000   | 19+200 | 10     | 89           | Thurai    | 6                 | T2                  |
| 494    | 19+200   | 19+400 | 10     | 80           | Dirsena   | 6                 | T2                  |
| 495    |          |        | 10     | 82           | RedThurai | 6                 | T2                  |
| 496    |          |        | 10     | 99           | RedThurai | 6                 | T3                  |
| 497    | 19+400   | 19+600 | 10     | 95           | Thurai    | 6                 | T3                  |
| 498    |          |        | 10     | 62           | Thurai    | 6                 | T2                  |
| 499    |          |        | 10     | 69           | Dirsena   | 6                 | T2                  |
| 500    |          |        | 10     | 70           | Dirsena   | 6                 | T2                  |
| 501    |          |        | 10     | 79           | Dirsena   | 6                 | T2                  |
| 502    | 19+600   | 19+800 | 10     | 73           | Dirsena   | 6                 | T2                  |
| 503    | 19+800   | 20+000 | 10     | 75           | Dirsena   | 6                 | T2                  |
| 504    |          |        | 10     | 71           | Thurai    | 6                 | T2                  |
| 505    | 20+000   | 20+200 | 10     | 76           | Dirsena   | 6                 | T2                  |
| 506    | 20+200   | 20+400 | 10     | 36           | Dirsena   | 6                 | T1                  |
| 507    |          |        | 10     | 37           | Thurai    | 6                 | T1                  |
| 508    | 20+400   | 20+600 | 10     | 33           | Dirsena   | 6                 | T1                  |
| 509    |          |        | 10     | 31           | Dirsena   | 6                 | T1                  |
| 510    |          |        | 10     | 65           | Thurai    | 6                 | T2                  |
| 511    | 20+600   | 20+800 | 10     | 53           | Dirsena   | 6                 | T1                  |
| 512    |          |        | 10     | 59           | Dirsena   | 6                 | T1                  |
| 513    |          |        | 10     | 57           | Thurai    | 6                 | T1                  |
| 514    |          |        | 10     | 54           | Kanuga    | 6                 | T1                  |
| 515    | 20+800   | 21+000 | 10     | 59           | Kanuga    | 6                 | T1                  |
| 516    |          |        | 10     | 52           | Kanuga    | 6                 | T1                  |
| 517    |          |        | 10     | 54           | Thurai    | 6                 | T1                  |
| 518    | 22+800   | 23+000 | 10     | 85           | Thurai    | 6                 | T2                  |
| 519    |          |        | 10     | 81           | Dirsena   | 6                 | T2                  |
| 520    | 23+000   | 23+200 | 10     | 58           | Dirsena   | 6                 | T1                  |
| 521    |          |        | 10     | 56           | Aggipulla | 6                 | T1                  |
| 522    | 23+200   | 23+400 | 10     | 50           | Dirsena   | 6                 | T1                  |
| 523    |          |        | 10     | 55           | Aggipulla | 6                 | T1                  |
| 524    | 23+400   | 23+600 | 10     | 84           | Aggipulla | 6                 | T2                  |
| 525    | 23+600   | 23+800 | 10     | 79           | Aggipulla | 6                 | T2                  |
| 526    | 25+000   | 25+200 | 10     | 46           | Bhagi     | 6                 | T1                  |
| 527    | 26+200   | 26+400 | 10     | 142          | Bhagi     | 6                 | T3                  |
| 528    |          |        | 10     | 158          | Bhagi     | 6                 | T3                  |
| 529    |          |        | 10     | 164          | Bhagi     | 6                 | T3                  |
| 530    | 33+400   | 33+600 | 10     | 80           | Bhagi     | 6                 | T2                  |
| 531    | 34+600   | 34+800 | 10     | 54           | Bhagi     | 6                 | T1                  |
| 532    |          |        | 10     | 50           | Bhagi     | 6                 | T1                  |
| 533    |          |        | 10     | 59           | Bhagi     | 6                 | T1                  |
| 534    |          |        | 10     | 51           | Bhagi     | 6                 | T1                  |
| 535    |          |        | 10     | 58           | Bhagi     | 6                 | T1                  |
| 536    | 35+600   | 35+800 | 10     | 97           | Bhagi     | 6                 | T3                  |
| 537    |          |        | 10     | 93           | Bhagi     | 6                 | T3                  |
| 538    |          |        | 10     | 95           | Bhagi     | 6                 | T3                  |
| 539    |          |        | 10     | 92           | Bhagi     | 6                 | T3                  |
| 540    |          |        | 10     | 99           | Bhagi     | 6                 | T3                  |

#### APPENDIX 4.1: TREE ENUMERATION : Kakinada-Rajamundry(L.H.S)

| Sl. No | Chainage |        | Height   | Girth in Cms | Tree Name | Dist. From Centre | Tree Classification |
|--------|----------|--------|----------|--------------|-----------|-------------------|---------------------|
|        | From     | To     |          |              |           |                   |                     |
| 541    |          |        | 10       | 94           | Bhagi     | 6                 | T3                  |
| 542    |          |        | 10       | 101          | Bhagi     | 6                 | T3                  |
| 543    |          |        | 10       | 152          | Bhagi     | 6                 | T3                  |
| 544    |          |        | 10       | 142          | Bhagi     | 6                 | T3                  |
| 545    |          |        | 10       | 113          | Bhagi     | 6                 | T3                  |
| 546    |          |        | 10       | 152          | Bhagi     | 6                 | T3                  |
| 547    |          |        | 10       | 142          | Bhagi     | 6                 | T3                  |
| 548    |          |        | 10       | 132          | Bhagi     | 6                 | T3                  |
| 549    |          |        | 10       | 125          | Bhagi     | 6                 | T3                  |
| 550    |          |        | 10       | 142          | Bhagi     | 6                 | T3                  |
| 551    |          |        | 10       | 152          | Bhagi     | 6                 | T3                  |
| 552    |          |        | 10       | 142          | Bhagi     | 6                 | T3                  |
| 553    | 36+800   | 37+000 | 15 above | 185          | Bhagi     | 6                 | T3                  |
| 554    |          |        | 15 above | 165          | Bhagi     | 6                 | T3                  |
| 555    |          |        | 15 above | 123          | Bhagi     | 6                 | T3                  |
| 556    |          |        | 15 above | 142          | Bhagi     | 6                 | T3                  |
| 557    |          |        | 15 above | 165          | Bhagi     | 6                 | T3                  |
| 558    |          |        | 15 above | 132          | Bhagi     | 6                 | T3                  |
| 559    | 41+000   | 41+200 | 10       | 175          | Aggipulla | 6                 | T3                  |
| 560    |          |        | 10       | 162          | Aggipulla | 6                 | T3                  |
| 561    | 43+800   | 44+000 | 10       | 46           | Bhagi     | 6                 | T1                  |
| 562    | 46+600   | 46+800 | 10       | 49           | Kanuga    | 6                 | T1                  |
| 563    |          |        | 10       | 44           | Kanuga    | 6                 | T1                  |
| 564    | 57+400   | 57+600 | 10       | 47           | Aggipulla | 6                 | T1                  |
| 565    |          |        | 10       | 40           | Aggipulla | 6                 | T1                  |
| 566    | 57+600   | 57+800 | 10       | 46           | Bhadam    | 6                 | T1                  |
| 567    |          |        | 10       | 40           | Bhadam    | 6                 | T1                  |
| 568    |          |        | 10       | 62           | Bhadam    | 6                 | T2                  |
| 569    | 59+200   | 59+400 | 10       | 100          | Bhagi     | 6                 | T3                  |
| 570    |          |        | 10       | 62           | Bhadam    | 6                 | T2                  |

NOTE: Number of Trees to be fell are considered up to 8m from CL on both RHS and LHS side =4057



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1             | 0+128           | 24            | 75                  | Neelagiri        | 2.5                      |
| 2             | 0+132           | 23            | 75                  | Neelagiri        | 2.5                      |
| 3             | 0+140           | 18            | 145                 | Ashoka           | 2.5                      |
| 4             | 0+150           | 18            | 85                  | Thurai           | 2.5                      |
| 5             | 0+180           | 15            | 125                 | Thurai           | 4.8                      |
| 6             | 1+270           | 10            | 95                  | Thurai           | 5.0                      |
| 7             | 1+273           | 10            | 96                  | Thurai           | 5.1                      |
| 8             | 1+280           | 10            | 240                 | Tamarind         | 6.2                      |
| 9             | 1+310           | 10            | 78                  | Pongamia         | 6.8                      |
| 10            | 1+440           | 10            | 120                 | Pongamia         | 3.2                      |
| 11            | 1+444           | 10            | 110                 | Pongamia         | 2.8                      |
| 12            | 1+450           | 10            | 100                 | Pongamia         | 3.8                      |
| 13            | 1+590           | 5             | 55                  | Pongamia         | 2.6                      |
| 14            | 1+596           | 6             | 67                  | Pongamia         | 4.8                      |
| 15            | 1+607           | 5             | 52                  | Pongamia         | 1.2                      |
| 16            | 1+635           | 5             | 105                 | Thurai           | 7.2                      |
| 17            | 1+642           | 5             | 85                  | Maddi            | 6.3                      |
| 18            | 1+650           | 10            | 125                 | Bhagi            | 1.5                      |
| 19            | 1+735           | 6             | 95                  | Bhagi            | 5.2                      |
| 20            | 1+745           | 7             | 95                  | Bhagi            | 6.3                      |
| 21            | 1+752           | 5             | 75                  | Pongamia         | 6.8                      |
| 22            | 1+756           | 5             | 80                  | Pongamia         | 7.0                      |
| 23            | 1+768           | 7             | 105                 | Pongamia         | 7.5                      |
| 24            | 1+775           | 8             | 110                 | Bhagi            | 7.8                      |
| 25            | 1+780           | 6             | 95                  | Pongamia         | 7.8                      |
| 26            | 1+846           | 6             | 105                 | Coconut          | 6.0                      |
| 27            | 1+885           | 7             | 75                  | Pongamia         | 5.4                      |
| 28            | 1+898           | 5             | 60                  | Neem             | 5.4                      |
| 29            | 1+930           | 18            | 450                 | Bhagi            | 1.1                      |
| 30            | 1+946           | 20            | 550                 | Bhagi            | 1.0                      |
| 31            | 1+955           | 20            | 595                 | Bhagi            | 0.8                      |
| 32            | 1+962           | 20            | 480                 | Bhagi            | 1.3                      |
| 33            | 1+978           | 15            | 320                 | Bhagi            | 1.6                      |
| 34            | 1+995           | 14            | 185                 | Bhuraga          | 1.6                      |
| 35            | 2+003           | 4             | 45                  | Jama             | 2.9                      |
| 36            | 2+007           | 6             | 52                  | Neredu           | 5.4                      |
| 37            |                 | 8             | 110                 | Coconut          | 5.4                      |
| 38            | 2+012           | 15            | 485                 | Bhagi            | 0.9                      |
| 39            | 2+029           | 12            | 220                 | Raavi            | 3.3                      |
| 40            | 2+033           | 4             | 75                  | Neem             | 3.4                      |
| 41            | 2+104           | 6             | 95                  | Thurai           | 2.3                      |
| 42            | 2+105           | 8             | 110                 | Coconut          | 5.4                      |
| 43            | 2+109           | 7             | 100                 | Coconut          | 5.4                      |
| 44            | 2+109           | 8             | 70                  | Raavi            | 2.1                      |
| 45            | 2+115           | 5             | 100                 | Coconut          | 5.4                      |
| 46            | 2+119           | 6             | 58                  | Bhagi            | 4.4                      |
| 47            | 2+125           | 6             | 60                  | Thurai           | 4.2                      |
| 48            | 2+138           | 10            | 75                  | Ashoka           | 3.3                      |
| 49            | 2+139           | 10            | 70                  | Ashoka           | 3.4                      |
| 50            | 2+140           | 10            | 70                  | Ashoka           | 3.4                      |
| 51            | 2+144           | 7             | 80                  | Maddi            | 6.2                      |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 52            | 2+158           | 20            | 105                 | Palm             | 6.5                      |
| 53            | 2+260           | 10            | 85                  | Thurai           | 4.5                      |
| 54            | 2+263           | 10            | 95                  | Pongamia         | 4.6                      |
| 55            | 2+290           | 8             | 205                 | Pongamia         | 4.6                      |
| 56            | 2+310           | 12            | 240                 | Pongamia         | 4.4                      |
| 57            | 2+324           | 10            | 88                  | Pongamia         | 5.1                      |
| 58            | 2+350           | 17            | 500                 | Bhagi            | 2.6                      |
| 59            | 2+358           | 6             | 105                 | Coconut          | 6.0                      |
| 60            | 2+368           | 6             | 105                 | Coconut          | 6.0                      |
| 61            | 2+375           | 7             | 85                  | Neredu           | 5.8                      |
| 62            | 2+379           | 5             | 75                  | Thurai           | 4.5                      |
| 63            | 2+398           | 5             | 65                  | Pongamia         | 4.5                      |
| 64            | 2+405           | 14            | 125                 | Mango            | 6.2                      |
| 65            | 2+428           | 10            | 80                  | Mango            | 4.8                      |
| 66            | 2+465           | 8             | 105                 | Thurai           | 5.6                      |
| 67            | 2+475           | 6             | 75                  | Mango            | 4.3                      |
| 68            | 2+486           | 7             | 75                  | Pongamia         | 4.3                      |
| 69            | 2+501           | 10            | 105                 | Pongamia         | 4.9                      |
| 70            | 2+530           | 12            | 120                 | Coconut          | 5.9                      |
| 71            | 2+534           | 12            | 105                 | Coconut          | 6.0                      |
| 72            | 2+542           | 10            | 145                 | Pongamia         | 6.5                      |
| 73            | 2+545           | 8             | 105                 | Coconut          | 6.0                      |
| 74            | 2+558           | 12            | 105                 | Coconut          | 6.0                      |
| 75            | 2+565           | 10            | 98                  | Coconut          | 6.1                      |
| 76            | 2+566           | 12            | 110                 | Coconut          | 7.8                      |
| 77            | 2+578           | 12            | 110                 | Pongamia         | 7.8                      |
| 78            | 2+595           | 15            | 365                 | Bhagi            | 1.5                      |
| 79            | 2+600           | 18            | 440                 | Bhagi            | 1.7                      |
| 80            | 2+610           | 10            | 105                 | Pongamia         | 6.8                      |
| 81            | 2+690           | 10            | 140                 | Pongamia         | 4.0                      |
| 82            | 2+708           | 8             | 95                  | Thurai           | 4.0                      |
| 83            | 2+730           | 12            | 520                 | Bhagi            | 3.5                      |
| 84            | 2+745           | 7             | 70                  | Neredu           | 5.6                      |
| 85            | 2+750           | 7             | 75                  | Pongamia         | 4.5                      |
| 86            | 2+764           | 10            | 190                 | Maddi            | 4.6                      |
| 87            | 2+794           | 10            | 140                 | Raavi            | 5.4                      |
| 88            | 2+796           | 10            | 105                 | Pongamia         | 6.2                      |
| 89            | 2+800           | 11            | 105                 | Pongamia         | 6.0                      |
| 90            | 2+805           | 12            | 115                 | Pongamia         | 6.2                      |
| 91            | 2+810           | 10            | 60                  | Neem             | 5.2                      |
| 92            | 2+828           | 15            | 320                 | Bhagi            | 3.4                      |
| 93            | 2+842           | 11            | 160                 | Thurai           | 4.2                      |
| 94            | 2+859           | 15            | 175                 | Pongamia         | 4.4                      |
| 95            | 2+870           | 6             | 82                  | Pongamia         | 3.1                      |
| 96            | 2+876           | 15            | 385                 | Bhagi            | 3.0                      |
| 97            | 2+884           | 18            | 75                  | Neelagiri        | 3.1                      |
| 98            | 2+896           | 15            | 220                 | Raavi            | 2.4                      |
| 99            | 2+903           | 10            | 95                  | Thurai           | 3.4                      |
| 100           | 2+903           | 8             | 105                 | Coconut          | 5.4                      |
| 101           | 2+911           | 6             | 70                  | Pongamia         | 3.3                      |
| 102           | 2+911           | 10            | 80                  | Pongamia         | 5.6                      |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 103           | 2+913           | 12            | 100                 | Coconut          | 6.8                      |
| 104           | 2+915           | 7             | 45                  | Neem             | 3.5                      |
| 105           | 2+917           | 8             | 185                 | Neredu           | 3.5                      |
| 106           | 2+925           | 8             | 125                 | Thurai           | 4.0                      |
| 107           | 2+936           | 10            | 155                 | Thurai           | 3.8                      |
| 108           | 2+943           | 8             | 95                  | Pongamia         | 6.2                      |
| 109           | 2+945           | 14            | 110                 | Coconut          | 6.1                      |
| 110           | 2+947           | 8             | 85                  | Pongamia         | 3.2                      |
| 111           | 2+960           | 8             | 120                 | Pongamia         | 3.6                      |
| 112           | 2+965           | 10            | 120                 | Pongamia         | 3.8                      |
| 113           | 2+972           | 10            | 145                 | Neredu           | 3.4                      |
| 114           | 2+995           | 8             | 125                 | Pongamia         | 3.0                      |
| 115           | 3+003           | 10            | 104                 | Coconut          | 4.6                      |
| 116           | 3+014           | 10            | 95                  | Pongamia         | 5.0                      |
| 117           | 3+017           | 8             | 110                 | Coconut          | 6.2                      |
| 118           | 3+022           | 8             | 110                 | Coconut          | 6.1                      |
| 119           | 3+035           | 6             | 65                  | Pongamia         | 3.6                      |
| 120           | 3+036           | 10            | 80                  | Neem             | 3.5                      |
| 121           | 3+040           | 8             | 80                  | Neem             | 3.8                      |
| 122           | 3+044           | 8             | 60                  | Pongamia         | 6.2                      |
| 123           | 3+050           | 10            | 105                 | Pongamia         | 6.0                      |
| 124           | 3+055           | 12            | 110                 | Neem             | 6.2                      |
| 125           | 3+074           | 10            | 105                 | Pongamia         | 4.4                      |
| 126           | 3+094           | 8             | 105                 | Coconut          | 3.4                      |
| 127           | 3+100           | 8             | 104                 | Pongamia         | 3.8                      |
| 128           | 3+101           | 8             | 110                 | Coconut          | 3.4                      |
| 129           | 3+114           | 10            | 125                 | Pongamia         | 3.1                      |
| 130           | 3+119           | 8             | 85                  | Neem             | 3.0                      |
| 131           | 3+130           | 10            | 105                 | Pongamia         | 3.1                      |
| 132           | 3+150           | 10            | 210                 | Thurai           | 4.4                      |
| 133           | 3+162           | 10            | 140                 | Pongamia         | 5.4                      |
| 134           | 3+185           | 8             | 120                 | Pongamia         | 3.4                      |
| 135           | 3+198           | 11            | 160                 | Thurai           | 4.5                      |
| 136           | 3+204           | 10            | 104                 | Pongamia         | 4.1                      |
| 137           | 3+210           | 12            | 276                 | Raavi            | 4.5                      |
| 138           | 3+216           | 8             | 75                  | Thurai           | 1.7                      |
| 139           | 3+219           | 4             | 58                  | Pongamia         | 1.9                      |
| 140           | 3+219           | 6             | 85                  | Neem             | 7.8                      |
| 141           | 3+262           | 8             | 105                 | Thurai           | 4.0                      |
| 142           | 3+270           | 6             | 115                 | Coconut          | 5.2                      |
| 143           |                 | 8             | 120                 | Coconut          | 7.2                      |
| 144           | 3+290           | 6             | 55                  | Neredu           | 4.0                      |
| 145           | 3+296           | 15            | 95                  | Neelagiri        | 4.0                      |
| 146           | 3+297           | 15            | 85                  | Neelagiri        | 4.1                      |
| 147           | 3+320           | 9             | 110                 | Pongamia         | 3.9                      |
| 148           | 3+325           | 9             | 95                  | Neem             | 5.2                      |
| 149           | 3+327           | 14            | 98                  | Neelagiri        | 4.0                      |
| 150           | 3+332           | 10            | 75                  | Bhuraga          | 3.8                      |
| 151           | 3+357           | 14            | 105                 | Coconut          | 5.4                      |
| 152           | 3+370           | 7             | 65                  | Thurai           | 4.1                      |
| 153           | 3+385           | 10            | 95                  | Thurai           | 4.4                      |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 154           | 3+387           | 10            | 116                 | Seemasinta       | 4.8                      |
| 155           | 3+394           | 9             | 97                  | Pongamia         | 3.8                      |
| 156           | 3+396           | 5             | 80                  | Pongamia         | 4.2                      |
| 157           | 3+397           | 10            | 140                 | Thurai           | 4.8                      |
| 158           | 3+397           | 10            | 110                 | Coconut          | 7.6                      |
| 159           | 3+398           | 12            | 110                 | Coconut          | 7.8                      |
| 160           | 3+400           | 10            | 110                 | Pongamia         | 3.2                      |
| 161           | 3+402           | 10            | 130                 | Thurai           | 5.4                      |
| 162           | 3+408           | 5             | 110                 | Coconut          | 4.9                      |
| 163           | 3+410           | 8             | 120                 | Coconut          | 5.4                      |
| 164           | 3+415           | 10            | 325                 | Bhagi            | 1.5                      |
| 165           | 3+422           | 18            | 100                 | Neelagiri        | 4.0                      |
| 166           | 3+425           | 10            | 145                 | Thurai           | 4.1                      |
| 167           | 3+428           | 4             | 110                 | Coconut          | 3.5                      |
| 168           | 3+442           | 14            | 560                 | Bhagi            | 3.6                      |
| 169           | 3+451           | 6             | 75                  | Bhagi            | 1.4                      |
| 170           | 3+460           | 16            | 85                  | Neelagiri        | 6.0                      |
| 171           | 3+464           | 6             | 75                  | Pongamia         | 2.6                      |
| 172           | 3+470           | 10            | 120                 | Pongamia         | 3.4                      |
| 173           | 3+480           | 5             | 110                 | Coconut          | 4.0                      |
| 174           | 3+485           | 16            | 80                  | Neelagiri        | 3.9                      |
| 175           | 3+502           | 16            | 80                  | Neelagiri        | 2.8                      |
| 176           | 3+509           | 18            | 95                  | Neelagiri        | 4.2                      |
| 177           | 3+510           | 8             | 135                 | Coconut          | 7.5                      |
| 178           | 3+580           | 20            | 120                 | Palm             | 6.2                      |
| 179           | 3+582           | 15            | 380                 | Bhagi            | 4.1                      |
| 180           | 3+600           | 10            | 134                 | Coconut          | 6.0                      |
| 181           | 3+608           | 8             | 155                 | Coconut          | 4.5                      |
| 182           | 3+615           | 10            | 160                 | Coconut          | 4.2                      |
| 183           | 3+621           | 5             | 70                  | Pongamia         | 3.8                      |
| 184           | 3+623           | 5             | 85                  | Pongamia         | 3.4                      |
| 185           | 3+630           | 6             | 85                  | Mango            | 5.4                      |
| 186           | 3+635           | 15            | 115                 | Palm(4)          | 3.2                      |
| 187           | 3+640           | 8             | 110                 | Pongamia         | 4.2                      |
| 188           | 3+642           | 8             | 95                  | Thurai           | 5.0                      |
| 189           | 3+649           | 8             | 92                  | Pongamia         | 4.7                      |
| 190           | 3+655           | 12            | 110                 | Palm             | 4.0                      |
| 191           | 3+670           | 6             | 110                 | Coconut          | 5.6                      |
| 192           | 3+673           | 7             | 70                  | Neredu           | 3.7                      |
| 193           | 3+676           | 8             | 50                  | Mango            | 4.4                      |
| 194           | 3+682           | 5             | 55                  | Pongamia         | 3.1                      |
| 195           | 3+696           | 10            | 115                 | Mango            | 6.2                      |
| 196           | 3+706           | 10            | 106                 | Neem             | 7.3                      |
| 197           | 3+802           | 18            | 110                 | Palm             | 5.0                      |
| 198           | 3+815           | 15            | 110                 | Palm,Coconut     | 3.5                      |
| 199           | 3+855           | 15            | 120                 | Palm             | 4.0                      |
| 200           | 3+870           | 10            | 95                  | Pongamia         | 7.0                      |
| 201           | 3+889           | 12            | 110                 | Palm             | 8.0                      |
| 202           | 3+894           | 10            | 130                 | Pongamia         | 6.5                      |
| 203           | 3+902           | 15            | 130                 | Coconut          | 6.5                      |
| 204           | 3+950           | 11            | 140                 | Thurai           | 7.6                      |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 205           | 3+955           | 10            | 105                 | Tamarind         | 4.0                      |
| 206           | 4+025           | 14            | 265                 | Bhagi            | 3.1                      |
| 207           | 4+035           | 10            | 185                 | Thurai           | 4.0                      |
| 208           | 4+037           | 6             | 85                  | Thurai           | 7.8                      |
| 209           | 4+050           | 10            | 106                 | Coconut          | 5.7                      |
| 210           | 4+055           | 8             | 110                 | Thurai           | 6.0                      |
| 211           | 4+068           | 15            | 104                 | Palm             | 7.8                      |
| 212           | 4+073           | 10            | 110                 | Coconut          | 5.5                      |
| 213           | 4+096           | 8             | 130                 | Bhagi            | 6.7                      |
| 214           | 4+104           | 12            | 100                 | Coconut          | 4.6                      |
| 215           | 4+140           | 10            | 104                 | Coconut          | 5.7                      |
| 216           | 4+143           | 12            | 110                 | Coconut          | 5.5                      |
| 217           | 4+158           | 10            | 95                  | Coconut          | 5.6                      |
| 218           | 4+186           | 10            | 120                 | Coconut          | 4.6                      |
| 219           | 4+189           | 11            | 120                 | Coconut          | 5.0                      |
| 220           | 4+195           | 10            | 125                 | Coconut          | 4.5                      |
| 221           | 4+250           | 15            | 125                 | Coconut          | 5.2                      |
| 222           | 4+254           | 12            | 110                 | Coconut          | 5.7                      |
| 223           | 4+257           | 20            | 95                  | Palm             | 2.7                      |
| 224           | 4+270           | 10            | 125                 | Coconut          | 6.1                      |
| 225           | 4+286           | 15            | 320                 | Bhagi            | 1.1                      |
| 226           | 4+310           | 15            | 315                 | Bhagi            | 1.8                      |
| 227           | 4+380           | 12            | 165                 | Bhuraga          | 2.9                      |
| 228           | 4+385           | 15            | 106                 | Coconut          | 3.9                      |
| 229           | 4+392           | 15            | 310                 | Bhagi            | 1.6                      |
| 230           | 4+401           | 15            | 240                 | Raavi            | 4.0                      |
| 231           | 4+407           | 10            | 130                 | Pongamia         | 4.2                      |
| 232           | 4+428           | 15            | 460                 | Bhagi            | 1.0                      |
| 233           | 4+434           | 10            | 95                  | Coconut          | 5.2                      |
| 234           | 4+450           | 10            | 110                 | Bhuraga          | 5.5                      |
| 235           | 4+474           | 16            | 285                 | Bhagi            | 1.0                      |
| 236           | 4+480           | 15            | 270                 | Bhagi            | 1.1                      |
| 237           | 4+481           | 20            | 110                 | Palm             | 1.5                      |
| 238           | 4+481           | 8             | 100                 | Coconut          | 5.2                      |
| 239           | 4+537           | 12            | 110                 | Tamarind         | 6.4                      |
| 240           | 4+548           | 10            | 70                  | Thurai           | 5.9                      |
| 241           | 4+558           | 9             | 105                 | Thurai           | 7.0                      |
| 242           | 4+583           | 10            | 95                  | Tamarind         | 7.5                      |
| 243           | 4+590           | 18            | 230                 | Bhagi            | 3.8                      |
| 244           | 4+596           | 12            | 130                 | Coconut          | 7.8                      |
| 245           | 4+610           | 10            | 102                 | Bhagi            | 7.5                      |
| 246           | 4+619           | 12            | 160                 | Neredu           | 7.8                      |
| 247           | 4+632           | 8             | 85                  | Tamarind         | 7.2                      |
| 248           | 4+675           | 15            | 310                 | Bhagi            | 2.1                      |
| 249           | 4+708           | 16            | 300                 | Bhagi            | 2.3                      |
| 250           | 4+745           | 13            | 295                 | Bhagi            | 1.5                      |
| 251           | 4+780           | 12            | 265                 | Bhagi            | 3.6                      |
| 252           | 4+791           | 15            | 310                 | Bhagi            | 1.5                      |
| 253           | 4+792           | 12            | 95                  | Palm             | 4.0                      |
| 254           | 4+807           | 14            | 290                 | Bhagi            | 1.7                      |
| 255           | 4+818           | 12            | 285                 | Bhagi            | 2.0                      |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 256           | 4+842           | 8             | 85                  | Pongamia         | 8.0                      |
| 257           | 4+865           | 6             | 80                  | Seemasinta       | 5.2                      |
| 258           | 4+980           | 7             | 96                  | Tamarind         | 0 - 6.0                  |
| 259           | 5+110           | 7             | 70                  | Tamarind         | 5.5                      |
| 260           | 5+120           | 8             | 110                 | Palm             | 6.8                      |
| 261           | 5+190           | 18            | 104                 | Neelagiri        | 7.0                      |
| 262           | 5+315           | 10            | 85                  | Thurai           | 5.4                      |
| 263           | 5+335           | 6             | 110                 | Pongamia         | 6.7                      |
| 264           | 5+346           | 7             | 88                  | Pongamia         | 4.2                      |
| 265           | 5+352           | 9             | 98                  | Raavi            | 4.2                      |
| 266           | 5+356           | 6             | 75                  | Pongamia         | 3.5                      |
| 267           | 5+360           | 7             | 70                  | Pongamia         | 4.1                      |
| 268           | 5+370           | 6             | 80                  | Pongamia         | 6.7                      |
| 269           | 5+375           | 8             | 98                  | Coconut          | 3.7                      |
| 270           | 5+377           | 6             | 84                  | Neem             | 3.8                      |
| 271           | 5+395           | 12            | 195                 | Bhagi            | 4.2                      |
| 272           | 5+400           | 10            | 135                 | Pongamia         | 4.2                      |
| 273           | 5+440           | 8             | 85                  | Pongamia         | 5.3                      |
| 274           | 5+454           | 10            | 165                 | Pongamia         | 4.3                      |
| 275           | 5+504           | 12            | 260                 | Bhagi            | 6.4                      |
| 276           | 5+522           | 12            | 325                 | Raavi            | 3.2                      |
| 277           | 5+528           | 15            | 95                  | Palm             | 4.0                      |
| 278           | 5+560           | 15            | 80                  | Palm             | 2.4                      |
| 279           | 5+592           | 18            | 110                 | Palm             | 4.8                      |
| 280           | 5+615           | 12            | 106                 | Palm             | 3.0                      |
| 281           | 5+655           | 6             | 110                 | Badige           | 3.5                      |
| 282           | 5+664           | 14            | 85                  | Palm             | 3.8                      |
| 283           | 5+705           | 6             | 120                 | Badige           | 4.5                      |
| 284           | 5+750           | 12            | 110                 | Palm             | 4.6                      |
| 285           | 5+802           | 8             | 120                 | Badige           | 5.8                      |
| 286           | 5+812           | 15            | 110                 | Palm             | 2.8                      |
| 287           | 5+830           | 8             | 140                 | Badige           | 6.0                      |
| 288           | 5+845           | 10            | 95                  | Palm             | 3.8                      |
| 289           | 5+860           | 10            | 135                 | Coconut          | 6.0                      |
| 290           | 5+908           | 12            | 106                 | Coconut          | 5.2                      |
| 291           | 5+918           | 11            | 125                 | Coconut          | 4.1                      |
| 292           | 5+999           | 5             | 70                  | Coconut          | 4.2                      |
| 293           | 6+001           | 10            | 128                 | Coconut(36)      | 4.2                      |
| 294           | 6+001           | 10            | 128                 | Palm(6)          | 5.2                      |
| 295           | 6+201           | 9             | 132                 | Coconut(21)      | 5.2                      |
| 296           | 6+201           | 9             | 132                 | Palm(12)         | 5.2                      |
| 297           | 6+401           | 10            | 122                 | Coconut(4)       | 5.3                      |
| 298           | 6+401           | 15            | 110                 | Palm(3)          | 5.3                      |
| 299           | 6+495           | 6             | 80                  | Occassic         | 5.6                      |
| 300           | 6+601           | 9             | 132                 | Coconut(23)      | 5.8                      |
| 301           | 6+801           | 10            | 128                 | Coconut(10)      | 5.6                      |
| 302           | 6+801           | 15            | 115                 | Palm(4)          | 5.6                      |
| 303           | 6+815           | 10            | 175                 | Tamarind         | 4.2                      |
| 304           | 6+986           | 10            | 315                 | Tamarind         | 2.2                      |
| 305           | 7+000           | 8             | 126                 | Coconut(11)      | 5.6                      |
| 306           | 7+000           | 8             | 126                 | Palm(2)          | 5.6                      |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 307           | 7+201           | 8             | 125                 | Coconut(12)      | 5.6                      |
| 308           | 7+415           | 8             | 115                 | Coconut(24)      | 5.8                      |
| 309           | 7+605           | 10            | 120                 | Coconut(18)      | 5.0                      |
| 310           | 7+605           | 12            | 110                 | Palm(2)          | 5.5                      |
| 311           | 7+850           | 12            | 128                 | Coconut(22)      | 4.8                      |
| 312           | 7+850           | 18            | 102                 | Palm             | 3.7                      |
| 313           | 8+000           | 8             | 108                 | Coconut(9)       | 3.8                      |
| 314           |                 | 10            | 36                  | Sunkesula        | 0 - 6.0                  |
| 315           |                 | 10            | 45                  | Sunkesula        | 0 - 6.0                  |
| 316           |                 | 15            | 58                  | Pongamia         | 0 - 6.0                  |
| 317           |                 | 15            | 59                  | Pongamia         | 0 - 6.0                  |
| 318           |                 | 15            | 62                  | Pongamia         | 0 - 6.0                  |
| 319           |                 | 15            | 65                  | Pongamia         | 0 - 6.0                  |
| 320           |                 | 15            | 66                  | Palm             | 0 - 6.0                  |
| 321           |                 | 15            | 69                  | Palm             | 0 - 6.0                  |
| 322           |                 | 15            | 75                  | Palm             | 0 - 6.0                  |
| 323           |                 | 15            | 85                  | Palm             | 0 - 6.0                  |
| 324           |                 | 15            | 84                  | Palm             | 0 - 6.0                  |
| 325           |                 | 15            | 86                  | Neem             | 0 - 6.0                  |
| 326           |                 | 15            | 83                  | Neem             | 0 - 6.0                  |
| 327           |                 | 15            | 82                  | Neem             | 0 - 6.0                  |
| 328           |                 | 15            | 86                  | Coconut          | 0 - 6.0                  |
| 329           |                 | 15            | 85                  | Coconut          | 0 - 6.0                  |
| 330           |                 | 15            | 86                  | Coconut          | 0 - 6.0                  |
| 331           |                 | 15            | 78                  | Coconut          | 0 - 6.0                  |
| 332           |                 | 15            | 79                  | Neem             | 0 - 6.0                  |
| 333           |                 | 15            | 76                  | Neem             | 0 - 6.0                  |
| 334           |                 | 15            | 72                  | Giriship         | 0 - 6.0                  |
| 335           |                 | 15            | 70                  | Giriship         | 0 - 6.0                  |
| 336           |                 | 15            | 71                  | Coconut          | 0 - 6.0                  |
| 337           |                 | 15            | 70                  | Coconut          | 0 - 6.0                  |
| 338           |                 | 15            | 89                  | Coconut          | 0 - 6.0                  |
| 339           |                 | 15            | 80                  | Giriship         | 0 - 6.0                  |
| 340           |                 | 15            | 82                  | Giriship         | 0 - 6.0                  |
| 341           |                 | 20            | 65                  | Giriship         | 0 - 6.0                  |
| 342           |                 | 20            | 64                  | Raavi            | 0 - 6.0                  |
| 343           |                 | 20            | 62                  | Raavi            | 0 - 6.0                  |
| 344           |                 | 20            | 69                  | Raavi            | 0 - 6.0                  |
| 345           |                 | 20            | 70                  | Raavi            | 0 - 6.0                  |
| 346           |                 | 20            | 79                  | Tamarind         | 0 - 6.0                  |
| 347           |                 | 20 above      | 73                  | Tamarind         | 0 - 6.0                  |
| 348           |                 | 20 above      | 75                  | Tamarind         | 0 - 6.0                  |
| 349           |                 | 20 above      | 71                  | Tamarind         | 0 - 6.0                  |
| 350           |                 | 20 above      | 76                  | Tamarind         | 0 - 6.0                  |
| 351           |                 | 20 above      | 109                 | Tamarind         | 0 - 6.0                  |
| 352           |                 | 20 above      | 108                 | Raavi            | 0 - 6.0                  |
| 353           |                 | 20 above      | 110                 | Coconut          | 0 - 6.0                  |
| 354           |                 | 20 above      | 95                  | Coconut          | 0 - 6.0                  |
| 355           |                 | 20 above      | 98                  | Raavi            | 0 - 6.0                  |
| 356           |                 | 20 above      | 128                 | Raavi            | 0 - 6.0                  |
| 357           |                 | 20 above      | 138                 | Coconut          | 0 - 6.0                  |



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 358           |                 | 20 above      | 104                 | Tamarind         | 0 - 6.0                  |
| 359           | 8+200           | 8             | 165                 | Tamarind         | 6.0                      |
| 360           |                 | 9             | 124                 | Tamarind         | 5.5                      |
| 361           |                 | 8             | 129                 | Coconut(2)       | 4.1                      |
| 362           |                 | 8             | 146                 | Tamarind         | 4.2                      |
| 363           |                 | 9             | 132                 | Tamarind         | 4.0                      |
| 364           |                 | 10            | 145                 | Tamarind         | 4.0                      |
| 365           |                 | 12            | 132                 | Coconut(6)       | 4.2                      |
| 366           |                 | 15            | 86                  | Coconut          | 0 - 6.0                  |
| 367           |                 | 15            | 84                  | Coconut          | 0 - 6.0                  |
| 368           |                 | 15            | 66                  | Coconut          | 0 - 6.0                  |
| 369           |                 | 15            | 69                  | Coconut          | 0 - 6.0                  |
| 370           |                 | 15            | 75                  | Coconut          | 0 - 6.0                  |
| 371           |                 | 15            | 85                  | Coconut          | 0 - 6.0                  |
| 372           |                 | 15            | 84                  | Coconut          | 0 - 6.0                  |
| 373           |                 | 15            | 86                  | Coconut          | 0 - 6.0                  |
| 374           |                 | 15            | 83                  | Coconut          | 0 - 6.0                  |
| 375           |                 | 15            | 82                  | Coconut          | 0 - 6.0                  |
| 376           |                 | 15            | 86                  | Coconut          | 0 - 6.0                  |
| 377           |                 | 15            | 85                  | Coconut          | 0 - 6.0                  |
| 378           |                 | 15            | 86                  | Coconut          | 0 - 6.0                  |
| 379           |                 | 15            | 78                  | Coconut          | 0 - 6.0                  |
| 380           |                 | 15            | 79                  | Coconut          | 0 - 6.0                  |
| 381           | 8+400           | 5             | 76                  | Thurai           | 4.6                      |
| 382           |                 | 16            | 143                 | Palm             | 3.4                      |
| 383           |                 | 4             | 70                  | Pongamia         | 4.1                      |
| 384           |                 | 8             | 94                  | Thurai           | 4.5                      |
| 385           |                 | 10            | 105                 | Pongamia         | 4.4                      |
| 386           |                 | 8             | 109                 | Badige           | 2.6                      |
| 387           |                 | 20            | 80                  | Coconut          | 0 - 6.0                  |
| 388           |                 | 20            | 82                  | Coconut          | 0 - 6.0                  |
| 389           |                 | 20            | 65                  | Coconut          | 0 - 6.0                  |
| 390           |                 | 20            | 64                  | Coconut          | 0 - 6.0                  |
| 391           |                 | 20            | 62                  | Coconut          | 0 - 6.0                  |
| 392           |                 | 20            | 65                  | Coconut          | 0 - 6.0                  |
| 393           |                 | 20            | 66                  | Coconut          | 0 - 6.0                  |
| 394           |                 | 20            | 69                  | Coconut          | 0 - 6.0                  |
| 395           |                 | 20            | 75                  | Coconut          | 0 - 6.0                  |
| 396           |                 | 20            | 85                  | Coconut          | 0 - 6.0                  |
| 397           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 398           |                 | 20            | 86                  | Coconut          | 0 - 6.0                  |
| 399           | 8+600           | 20 above      | 83                  | Coconut          | 0 - 6.0                  |
| 400           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 401           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 402           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 403           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 404           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 405           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 406           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 407           |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |
| 408           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 409           |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 410           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 411           |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 412           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 413           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 414           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 415           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 416           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 417           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 418           |                 | 20 above      | 63                  | Coconut          | 0 - 6.0                  |
| 419           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 420           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 421           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 422           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 423           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 424           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 425           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 426           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 427           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 428           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 429           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 430           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 431           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 432           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 433           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 434           | 8+800           | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 435           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 436           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 437           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 438           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 439           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 440           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 441           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 442           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 443           |                 | 20 above      | 74                  | Coconut          | 0 - 6.0                  |
| 444           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 445           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 446           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 447           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 448           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 449           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 450           |                 | 20 above      | 74                  | Coconut          | 0 - 6.0                  |
| 451           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 452           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 453           |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 454           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 455           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 456           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 457           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 458           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 459           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 460           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 461           |                 | 20 above      | 63                  | Coconut          | 0 - 6.0                  |
| 462           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 463           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 464           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 465           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 466           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 467           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 468           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 469           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 470           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 471           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 472           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 473           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 474           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 475           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 476           |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 477           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 478           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 479           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 480           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 481           |                 | 20 above      | 67                  | Coconut          | 0 - 6.0                  |
| 482           | 9+200           | 20 above      | 65                  | Coconut          | 0 - 6.0                  |
| 483           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 484           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 485           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 486           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 487           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 488           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 489           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 490           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 491           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 492           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 493           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 494           |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 495           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 496           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 497           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 498           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 499           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 500           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 501           |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |
| 502           | 9+400           | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 503           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 504           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 505           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 506           |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 507           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 508           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 509           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 510           |                 | 20 above      | 65                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 511           |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 512           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 513           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 514           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 515           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 516           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 517           |                 | 20 above      | 83                  | Coconut          | 0 - 6.0                  |
| 518           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 519           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 520           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 521           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 522           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 523           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 524           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 525           |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |
| 526           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 527           |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 528           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 529           | 9+600           | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 530           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 531           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 532           |                 | 20 above      | 65                  | Coconut          | 0 - 6.0                  |
| 533           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 534           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 535           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 536           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 537           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 538           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 539           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 540           |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 541           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 542           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 543           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 544           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 545           |                 | 20 above      | 67                  | Coconut          | 0 - 6.0                  |
| 546           |                 | 20 above      | 65                  | Coconut          | 0 - 6.0                  |
| 547           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 548           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 549           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 550           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 551           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 552           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 553           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 554           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 555           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 556           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 557           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 558           |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 559           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 560           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 561           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 562           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 563           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 564           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 565           |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |
| 566           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 567           |                 | 20 above      | 63                  | Coconut          | 0 - 6.0                  |
| 568           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 569           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 570           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 571           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 572           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 573           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 574           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 575           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 576           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 577           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 578           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 579           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 580           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 581           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 582           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 583           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 584           | 9+800           | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 585           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 586           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 587           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 588           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 589           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 590           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 591           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 592           |                 | 20 above      | 74                  | Coconut          | 0 - 6.0                  |
| 593           |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 594           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 595           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 596           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 597           |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 598           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 599           |                 | 20 above      | 74                  | Coconut          | 0 - 6.0                  |
| 600           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 601           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 602           |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 603           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 604           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 605           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 606           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 607           | 10+000          | 20            | 85                  | Coconut          | 0 - 6.0                  |
| 608           |                 | 20            | 79                  | Coconut          | 0 - 6.0                  |
| 609           |                 | 20            | 68                  | Coconut          | 0 - 6.0                  |
| 610           |                 | 20            | 64                  | Coconut          | 0 - 6.0                  |
| 611           |                 | 20            | 63                  | Coconut          | 0 - 6.0                  |
| 612           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 613           |                 | 20            | 73                  | Coconut          | 0 - 6.0                  |
| 614           |                 | 20            | 62                  | Coconut          | 0 - 6.0                  |
| 615           |                 | 20            | 68                  | Coconut          | 0 - 6.0                  |
| 616           |                 | 20            | 70                  | Coconut          | 0 - 6.0                  |
| 617           |                 | 20            | 79                  | Coconut          | 0 - 6.0                  |
| 618           |                 | 20            | 81                  | Coconut          | 0 - 6.0                  |
| 619           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 620           |                 | 20            | 82                  | Coconut          | 0 - 6.0                  |
| 621           |                 | 20            | 73                  | Coconut          | 0 - 6.0                  |
| 622           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 623           |                 | 20            | 75                  | Coconut          | 0 - 6.0                  |
| 624           |                 | 20            | 86                  | Coconut          | 0 - 6.0                  |
| 625           |                 | 20            | 75                  | Coconut          | 0 - 6.0                  |
| 626           |                 | 20            | 70                  | Coconut          | 0 - 6.0                  |
| 627           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 628           |                 | 20            | 71                  | Coconut          | 0 - 6.0                  |
| 629           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 630           |                 | 20            | 69                  | Coconut          | 0 - 6.0                  |
| 631           |                 | 20            | 68                  | Coconut          | 0 - 6.0                  |
| 632           |                 | 20            | 62                  | Coconut          | 0 - 6.0                  |
| 633           |                 | 20            | 67                  | Coconut          | 0 - 6.0                  |
| 634           |                 | 20            | 65                  | Coconut          | 0 - 6.0                  |
| 635           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 636           |                 | 20            | 82                  | Coconut          | 0 - 6.0                  |
| 637           |                 | 20            | 65                  | Coconut          | 0 - 6.0                  |
| 638           |                 | 20            | 69                  | Coconut          | 0 - 6.0                  |
| 639           | 10+200          | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 640           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 641           |                 | 20            | 85                  | Coconut          | 0 - 6.0                  |
| 642           |                 | 20            | 73                  | Coconut          | 0 - 6.0                  |
| 643           |                 | 20            | 66                  | Coconut          | 0 - 6.0                  |
| 644           |                 | 20            | 69                  | Coconut          | 0 - 6.0                  |
| 645           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 646           |                 | 20            | 85                  | Coconut          | 0 - 6.0                  |
| 647           |                 | 20            | 79                  | Coconut          | 0 - 6.0                  |
| 648           |                 | 20            | 68                  | Coconut          | 0 - 6.0                  |
| 649           |                 | 20            | 64                  | Coconut          | 0 - 6.0                  |
| 650           |                 | 20            | 63                  | Coconut          | 0 - 6.0                  |
| 651           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 652           |                 | 20            | 73                  | Coconut          | 0 - 6.0                  |
| 653           |                 | 20            | 62                  | Coconut          | 0 - 6.0                  |
| 654           |                 | 20            | 68                  | Coconut          | 0 - 6.0                  |
| 655           |                 | 20            | 70                  | Coconut          | 0 - 6.0                  |
| 656           |                 | 20            | 79                  | Coconut          | 0 - 6.0                  |
| 657           |                 | 20            | 81                  | Coconut          | 0 - 6.0                  |
| 658           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 659           |                 | 20            | 82                  | Coconut          | 0 - 6.0                  |
| 660           |                 | 20            | 73                  | Coconut          | 0 - 6.0                  |
| 661           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 662           |                 | 20            | 75                  | Coconut          | 0 - 6.0                  |
| 663           |                 | 20            | 86                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 664           |                 | 20            | 75                  | Coconut          | 0 - 6.0                  |
| 665           |                 | 20            | 70                  | Coconut          | 0 - 6.0                  |
| 666           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 667           |                 | 20            | 82                  | Coconut          | 0 - 6.0                  |
| 668           |                 | 20            | 80                  | Coconut          | 0 - 6.0                  |
| 669           |                 | 20            | 85                  | Coconut          | 0 - 6.0                  |
| 670           |                 | 20            | 78                  | Coconut          | 0 - 6.0                  |
| 671           |                 | 20            | 73                  | Coconut          | 0 - 6.0                  |
| 672           |                 | 20            | 70                  | Coconut          | 0 - 6.0                  |
| 673           |                 | 20            | 85                  | Coconut          | 0 - 6.0                  |
| 674           |                 | 20            | 81                  | Coconut          | 0 - 6.0                  |
| 675           |                 | 20            | 79                  | Coconut          | 0 - 6.0                  |
| 676           |                 | 20            | 64                  | Coconut          | 0 - 6.0                  |
| 677           | 10+400          | 20            | 62                  | Coconut          | 0 - 6.0                  |
| 678           |                 | 20            | 86                  | Coconut          | 0 - 6.0                  |
| 679           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 680           |                 | 20            | 79                  | Coconut          | 0 - 6.0                  |
| 681           |                 | 20            | 85                  | Coconut          | 0 - 6.0                  |
| 682           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 683           |                 | 20            | 78                  | Coconut          | 0 - 6.0                  |
| 684           |                 | 20            | 72                  | Coconut          | 0 - 6.0                  |
| 685           |                 | 20            | 80                  | Coconut          | 0 - 6.0                  |
| 686           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 687           |                 | 20            | 80                  | Coconut          | 0 - 6.0                  |
| 688           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 689           |                 | 20            | 89                  | Coconut          | 0 - 6.0                  |
| 690           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 691           |                 | 20            | 76                  | Coconut          | 0 - 6.0                  |
| 692           |                 | 20            | 70                  | Thurai           | 0 - 6.0                  |
| 693           |                 | 20            | 80                  | Thurai           | 0 - 6.0                  |
| 694           |                 | 20            | 85                  | Thurai           | 0 - 6.0                  |
| 695           |                 | 20            | 79                  | Thurai           | 0 - 6.0                  |
| 696           |                 | 20            | 68                  | Thurai           | 0 - 6.0                  |
| 697           |                 | 20            | 64                  | Thurai           | 0 - 6.0                  |
| 698           |                 | 20            | 63                  | Thurai           | 0 - 6.0                  |
| 699           |                 | 20            | 76                  | Thurai           | 0 - 6.0                  |
| 700           |                 | 20            | 73                  | Thurai           | 0 - 6.0                  |
| 701           |                 | 20            | 62                  | Mango            | 0 - 6.0                  |
| 702           |                 | 20            | 68                  | Mango            | 0 - 6.0                  |
| 703           |                 | 20            | 70                  | Mango            | 0 - 6.0                  |
| 704           |                 | 20            | 79                  | Mango            | 0 - 6.0                  |
| 705           |                 | 20            | 81                  | Mango            | 0 - 6.0                  |
| 706           |                 | 20            | 76                  | Mango            | 0 - 6.0                  |
| 707           |                 | 20            | 82                  | Mango            | 0 - 6.0                  |
| 708           |                 | 20            | 73                  | Mango            | 0 - 6.0                  |
| 709           |                 | 20            | 84                  | Mango            | 0 - 6.0                  |
| 710           |                 | 20            | 75                  | Mango            | 0 - 6.0                  |
| 711           |                 | 20            | 86                  | Mango            | 0 - 6.0                  |
| 712           |                 | 20            | 75                  | Mango            | 0 - 6.0                  |
| 713           |                 | 20            | 70                  | Mango            | 0 - 6.0                  |
| 714           |                 | 20            | 76                  | Mango            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 715           |                 | 20            | 84                  | Mango            | 0 - 6.0                  |
| 716           |                 | 20            | 82                  | Raavi            | 0 - 6.0                  |
| 717           |                 | 15            | 110                 | Raavi            | 0 - 6.0                  |
| 718           |                 | 15            | 99                  | Raavi            | 0 - 6.0                  |
| 719           |                 | 15            | 91                  | Raavi            | 0 - 6.0                  |
| 720           |                 | 15            | 93                  | Ashoka           | 0 - 6.0                  |
| 721           |                 | 15            | 102                 | Ashoka           | 0 - 6.0                  |
| 722           |                 | 15            | 116                 | Ashoka           | 0 - 6.0                  |
| 723           | 10+600          | 15            | 74                  | Coconut          | 0 - 6.0                  |
| 724           |                 | 15            | 81                  | Coconut          | 0 - 6.0                  |
| 725           |                 | 15            | 86                  | Coconut          | 0 - 6.0                  |
| 726           |                 | 15            | 76                  | Coconut          | 0 - 6.0                  |
| 727           |                 | 15            | 84                  | Coconut          | 0 - 6.0                  |
| 728           |                 | 15            | 73                  | Coconut          | 0 - 6.0                  |
| 729           |                 | 15            | 85                  | Coconut          | 0 - 6.0                  |
| 730           |                 | 15            | 74                  | Coconut          | 0 - 6.0                  |
| 731           |                 | 15            | 80                  | Coconut          | 0 - 6.0                  |
| 732           |                 | 15            | 76                  | Coconut          | 0 - 6.0                  |
| 733           |                 | 20            | 89                  | Coconut          | 0 - 6.0                  |
| 734           |                 | 20            | 84                  | Coconut          | 0 - 6.0                  |
| 735           |                 | 20            | 76                  | Bhagi            | 0 - 6.0                  |
| 736           |                 | 20            | 70                  | Bhagi            | 0 - 6.0                  |
| 737           |                 | 20            | 80                  | Bhagi            | 0 - 6.0                  |
| 738           |                 | 20            | 85                  | Bhagi            | 0 - 6.0                  |
| 739           |                 | 20            | 79                  | Bhagi            | 0 - 6.0                  |
| 740           |                 | 20            | 68                  | Mango            | 0 - 6.0                  |
| 741           |                 | 20            | 64                  | Mango            | 0 - 6.0                  |
| 742           |                 | 20            | 63                  | Mango            | 0 - 6.0                  |
| 743           |                 | 20            | 76                  | Mango            | 0 - 6.0                  |
| 744           |                 | 20            | 73                  | Mango            | 0 - 6.0                  |
| 745           |                 | 20            | 62                  | Mango            | 0 - 6.0                  |
| 746           |                 | 20            | 68                  | Mango            | 0 - 6.0                  |
| 747           |                 | 20            | 70                  | Mango            | 0 - 6.0                  |
| 748           |                 | 20            | 79                  | Bhagi            | 0 - 6.0                  |
| 749           |                 | 20            | 81                  | Bhagi            | 0 - 6.0                  |
| 750           |                 | 20            | 76                  | Bhagi            | 0 - 6.0                  |
| 751           |                 | 20            | 82                  | Bhagi            | 0 - 6.0                  |
| 752           |                 | 20            | 108                 | Bhagi            | 0 - 6.0                  |
| 753           |                 | 20            | 102                 | Bhagi            | 0 - 6.0                  |
| 754           | 16+000          | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 755           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 756           |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 757           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 758           |                 | 20 above      | 65                  | Coconut          | 0 - 6.0                  |
| 759           |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 760           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 761           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 762           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 763           |                 | 20 above      | 83                  | Coconut          | 0 - 6.0                  |
| 764           |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 765           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 766           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 767           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 768           |                 | 20 above      | 81                  | Thurai           | 0 - 6.0                  |
| 769           |                 | 20 above      | 79                  | Thurai           | 0 - 6.0                  |
| 770           |                 | 20 above      | 85                  | Thurai           | 0 - 6.0                  |
| 771           |                 | 20 above      | 76                  | Thurai           | 0 - 6.0                  |
| 772           |                 | 20 above      | 81                  | Thurai           | 0 - 6.0                  |
| 773           |                 | 20 above      | 76                  | Thurai           | 0 - 6.0                  |
| 774           |                 | 20 above      | 81                  | Thurai           | 0 - 6.0                  |
| 775           | 16+200          | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 776           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 777           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 778           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 779           |                 | 20 above      | 65                  | Coconut          | 0 - 6.0                  |
| 780           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 781           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 782           |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 783           |                 | 20 above      | 65                  | Coconut          | 0 - 6.0                  |
| 784           |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 785           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 786           |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 787           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 788           |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 789           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 790           |                 | 20 above      | 83                  | Coconut          | 0 - 6.0                  |
| 791           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 792           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 793           |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 794           |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 795           |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 796           |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 797           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 798           |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |
| 799           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 800           |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 801           |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 802           |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 803           |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 804           | 16+400          | 20 above      | 82                  | Bhagi            | 0 - 6.0                  |
| 805           |                 | 20 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 806           |                 | 20 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 807           |                 | 20 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 808           |                 | 20 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 809           |                 | 20 above      | 147                 | Bhagi            | 0 - 6.0                  |
| 810           |                 | 20 above      | 168                 | Bhagi            | 0 - 6.0                  |
| 811           |                 | 20 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 812           |                 | 20 above      | 175                 | Bhagi            | 0 - 6.0                  |
| 813           |                 | 20 above      | 159                 | Bhagi            | 0 - 6.0                  |
| 814           | 16+600          | 15 above      | 146                 | Bhagi            | 0 - 6.0                  |
| 815           |                 | 15 above      | 135                 | Bhagi            | 0 - 6.0                  |
| 816           |                 | 15 above      | 145                 | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 817           |                 | 15 above      | 128                 | Bhagi            | 0 - 6.0                  |
| 818           |                 | 15 above      | 67                  | Bhagi            | 0 - 6.0                  |
| 819           |                 | 15 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 820           |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 821           |                 | 15 above      | 82                  | Bhagi            | 0 - 6.0                  |
| 822           |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 823           |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 824           |                 | 15 above      | 78                  | Bhagi            | 0 - 6.0                  |
| 825           |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 826           |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 827           |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 828           |                 | 15 above      | 81                  | Bhagi            | 0 - 6.0                  |
| 829           |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 830           |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 831           |                 | 15 above      | 66                  | Bhagi            | 0 - 6.0                  |
| 832           |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 833           |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 834           |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 835           |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 836           |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 837           |                 | 15 above      | 78                  | Bhagi            | 0 - 6.0                  |
| 838           | 16+800          | 15 above      | 72                  | Bhagi            | 0 - 6.0                  |
| 839           |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 840           |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 841           |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 842           |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 843           |                 | 15 above      | 89                  | Bhagi            | 0 - 6.0                  |
| 844           |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 845           |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 846           |                 | 15 above      | 70                  | Sunkesula        | 0 - 6.0                  |
| 847           |                 | 15 above      | 80                  | Sunkesula        | 0 - 6.0                  |
| 848           |                 | 15 above      | 85                  | Sunkesula        | 0 - 6.0                  |
| 849           |                 | 15 above      | 79                  | Sunkesula        | 0 - 6.0                  |
| 850           |                 | 15 above      | 68                  | Sunkesula        | 0 - 6.0                  |
| 851           |                 | 15 above      | 64                  | Sunkesula        | 0 - 6.0                  |
| 852           |                 | 15 above      | 63                  | Sunkesula        | 0 - 6.0                  |
| 853           | 17+000          | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 854           |                 | 10            | 73                  | Coconut          | 0 - 6.0                  |
| 855           |                 | 10            | 62                  | Coconut          | 0 - 6.0                  |
| 856           |                 | 10            | 68                  | Coconut          | 0 - 6.0                  |
| 857           |                 | 10            | 70                  | Coconut          | 0 - 6.0                  |
| 858           |                 | 10            | 79                  | Coconut          | 0 - 6.0                  |
| 859           |                 | 10            | 81                  | Coconut          | 0 - 6.0                  |
| 860           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 861           |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 862           |                 | 20 above      | 73                  | Thurai           | 0 - 6.0                  |
| 863           |                 | 20 above      | 84                  | Thurai           | 0 - 6.0                  |
| 864           |                 | 20 above      | 75                  | Thurai           | 0 - 6.0                  |
| 865           |                 | 20 above      | 86                  | Thurai           | 0 - 6.0                  |
| 866           |                 | 20 above      | 75                  | Thurai           | 0 - 6.0                  |
| 867           |                 | 20 above      | 70                  | Thurai           | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 868           |                 | 20 above      | 76                  | Thurai           | 0 - 6.0                  |
| 869           | 17+200          | 20 above      | 84                  | Thurai           | 0 - 6.0                  |
| 870           |                 | 20 above      | 82                  | Thurai           | 0 - 6.0                  |
| 871           |                 | 20 above      | 80                  | Thurai           | 0 - 6.0                  |
| 872           |                 | 20 above      | 70                  | Thurai           | 0 - 6.0                  |
| 873           |                 | 20 above      | 76                  | Thurai           | 0 - 6.0                  |
| 874           |                 | 15 above      | 81                  | Thurai           | 0 - 6.0                  |
| 875           |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 876           |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 877           |                 | 15 above      | 74                  | Coconut          | 0 - 6.0                  |
| 878           |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |
| 879           |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 880           |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 881           |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 882           |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 883           |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 884           |                 | 15 above      | 74                  | Coconut          | 0 - 6.0                  |
| 885           |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 886           |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 887           |                 | 15 above      | 89                  | Coconut          | 0 - 6.0                  |
| 888           |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 889           |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 890           |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 891           |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 892           |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 893           |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 894           | 17+400          | 15 above      | 68                  | Coconut          | 0 - 6.0                  |
| 895           |                 | 15 above      | 64                  | Coconut          | 0 - 6.0                  |
| 896           |                 | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 897           |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 898           |                 | 20 above      | 73                  | Thurai           | 0 - 6.0                  |
| 899           |                 | 20 above      | 62                  | Thurai           | 0 - 6.0                  |
| 900           |                 | 20 above      | 68                  | Thurai           | 0 - 6.0                  |
| 901           |                 | 20 above      | 70                  | Thurai           | 0 - 6.0                  |
| 902           |                 | 20 above      | 79                  | Thurai           | 0 - 6.0                  |
| 903           |                 | 20 above      | 81                  | Thurai           | 0 - 6.0                  |
| 904           |                 | 20 above      | 76                  | Thurai           | 0 - 6.0                  |
| 905           |                 | 20 above      | 82                  | Thurai           | 0 - 6.0                  |
| 906           | 17+600          | 10            | 73                  | Thurai           | 0 - 6.0                  |
| 907           |                 | 10            | 84                  | Thurai           | 0 - 6.0                  |
| 908           |                 | 10            | 75                  | Thurai           | 0 - 6.0                  |
| 909           |                 | 10            | 86                  | Coconut          | 0 - 6.0                  |
| 910           |                 | 10            | 75                  | Coconut          | 0 - 6.0                  |
| 911           |                 | 10            | 70                  | Coconut          | 0 - 6.0                  |
| 912           |                 | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 913           |                 | 10            | 71                  | Coconut          | 0 - 6.0                  |
| 914           |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 915           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 916           |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 917           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 918           |                 | 20 above      | 136                 | Pongamia         | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 919           |                 | 20 above      | 185                 | Pongamia         | 0 - 6.0                  |
| 920           |                 | 20 above      | 134                 | Pongamia         | 0 - 6.0                  |
| 921           |                 | 20 above      | 134                 | Pongamia         | 0 - 6.0                  |
| 922           |                 | 20 above      | 134                 | Pongamia         | 0 - 6.0                  |
| 923           | 18+000          | 15 above      | 85                  | Pongamia         | 0 - 6.0                  |
| 924           |                 | 15 above      | 78                  | Pongamia         | 0 - 6.0                  |
| 925           |                 | 15 above      | 73                  | Pongamia         | 0 - 6.0                  |
| 926           |                 | 15 above      | 70                  | Pongamia         | 0 - 6.0                  |
| 927           |                 | 15 above      | 85                  | Pongamia         | 0 - 6.0                  |
| 928           |                 | 15 above      | 81                  | Almond           | 0 - 6.0                  |
| 929           | 18+200          | 10            | 79                  | Raavi            | 0 - 6.0                  |
| 930           |                 | 10            | 64                  | Raavi            | 0 - 6.0                  |
| 931           |                 | 10            | 66                  | Raavi            | 0 - 6.0                  |
| 932           |                 | 10            | 86                  | Raavi            | 0 - 6.0                  |
| 933           |                 | 10            | 84                  | Thurai           | 0 - 6.0                  |
| 934           |                 | 10            | 79                  | Thurai           | 0 - 6.0                  |
| 935           |                 | 15 above      | 85                  | Thurai           | 0 - 6.0                  |
| 936           |                 | 15 above      | 76                  | Thurai           | 0 - 6.0                  |
| 937           |                 | 15 above      | 69                  | Thurai           | 0 - 6.0                  |
| 938           |                 | 20 above      | 134                 | Thurai           | 0 - 6.0                  |
| 939           |                 | 20 above      | 154                 | Thurai           | 0 - 6.0                  |
| 940           |                 | 20 above      | 134                 | Pongamia         | 0 - 6.0                  |
| 941           |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 942           |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 943           |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 944           |                 | 15 above      | 168                 | Bhagi            | 0 - 6.0                  |
| 945           |                 | 15 above      | 101                 | Bhagi            | 0 - 6.0                  |
| 946           | 18+600          | 15 above      | 134                 | Neem             | 0 - 6.0                  |
| 947           |                 | 15 above      | 134                 | Neem             | 0 - 6.0                  |
| 948           |                 | 15 above      | 134                 | Neredu           | 0 - 6.0                  |
| 949           |                 | 15 above      | 156                 | Neredu           | 0 - 6.0                  |
| 950           |                 | 15 above      | 108                 | Neredu           | 0 - 6.0                  |
| 951           |                 | 15 above      | 109                 | Neredu           | 0 - 6.0                  |
| 952           |                 | 15 above      | 134                 | Neredu           | 0 - 6.0                  |
| 953           |                 | 15 above      | 134                 | Neredu           | 0 - 6.0                  |
| 954           |                 | 15 above      | 136                 | Neredu           | 0 - 6.0                  |
| 955           |                 | 15 above      | 101                 | Coconut          | 0 - 6.0                  |
| 956           |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 957           |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 958           | 18+800          | 20 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 959           |                 | 20 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 960           |                 | 20 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 961           |                 | 20 above      | 75                  | Bhagi            | 0 - 6.0                  |
| 962           |                 | 20 above      | 71                  | Bhagi            | 0 - 6.0                  |
| 963           |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 964           |                 | 20 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 965           |                 | 20 above      | 68                  | Bhagi            | 0 - 6.0                  |
| 966           |                 | 20 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 967           |                 | 20 above      | 67                  | Bhagi            | 0 - 6.0                  |
| 968           |                 | 20 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 969           |                 | 20 above      | 84                  | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 970           |                 | 20 above      | 111                 | Coconut          | 0 - 6.0                  |
| 971           |                 | 20 above      | 198                 | Coconut          | 0 - 6.0                  |
| 972           |                 | 20 above      | 114                 | Coconut          | 0 - 6.0                  |
| 973           |                 | 20 above      | 157                 | Coconut          | 0 - 6.0                  |
| 974           |                 | 20 above      | 139                 | Coconut          | 0 - 6.0                  |
| 975           |                 | 20 above      | 175                 | Coconut          | 0 - 6.0                  |
| 976           |                 | 20 above      | 185                 | Coconut          | 0 - 6.0                  |
| 977           |                 | 20 above      | 142                 | Coconut          | 0 - 6.0                  |
| 978           |                 | 20 above      | 153                 | Coconut          | 0 - 6.0                  |
| 979           |                 | 20 above      | 145                 | Coconut          | 0 - 6.0                  |
| 980           |                 | 20 above      | 136                 | Coconut          | 0 - 6.0                  |
| 981           |                 | 20 above      | 175                 | Coconut          | 0 - 6.0                  |
| 982           |                 | 20 above      | 135                 | Coconut          | 0 - 6.0                  |
| 983           | 19+000          | 15 above      | 145                 | Coconut          | 0 - 6.0                  |
| 984           |                 | 15 above      | 147                 | Coconut          | 0 - 6.0                  |
| 985           |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 986           |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 987           |                 | 15 above      | 143                 | Coconut          | 0 - 6.0                  |
| 988           |                 | 15 above      | 146                 | Coconut          | 0 - 6.0                  |
| 989           |                 | 15 above      | 184                 | Coconut          | 0 - 6.0                  |
| 990           |                 | 15 above      | 135                 | Coconut          | 0 - 6.0                  |
| 991           |                 | 15 above      | 143                 | Coconut          | 0 - 6.0                  |
| 992           |                 | 15 above      | 94                  | Coconut          | 0 - 6.0                  |
| 993           |                 | 15 above      | 93                  | Coconut          | 0 - 6.0                  |
| 994           |                 | 15 above      | 94                  | Coconut          | 0 - 6.0                  |
| 995           |                 | 15 above      | 99                  | Coconut          | 0 - 6.0                  |
| 996           |                 | 15 above      | 91                  | Coconut          | 0 - 6.0                  |
| 997           |                 | 15 above      | 94                  | Coconut          | 0 - 6.0                  |
| 998           |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 999           |                 | 15 above      | 135                 | Coconut          | 0 - 6.0                  |
| 1000          |                 | 15 above      | 144                 | Coconut          | 0 - 6.0                  |
| 1001          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 1002          |                 | 15 above      | 135                 | Coconut          | 0 - 6.0                  |
| 1003          |                 | 20 above      | 101                 | Coconut          | 0 - 6.0                  |
| 1004          |                 | 20 above      | 111                 | Coconut          | 0 - 6.0                  |
| 1005          |                 | 20 above      | 125                 | Coconut          | 0 - 6.0                  |
| 1006          |                 | 20 above      | 142                 | Coconut          | 0 - 6.0                  |
| 1007          |                 | 20 above      | 135                 | Coconut          | 0 - 6.0                  |
| 1008          |                 | 20 above      | 101                 | Coconut          | 0 - 6.0                  |
| 1009          | 19+200          | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1010          |                 | 20 above      | 74                  | Coconut          | 0 - 6.0                  |
| 1011          |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 1012          |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1013          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1014          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1015          |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1016          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1017          |                 | 20 above      | 74                  | Coconut          | 0 - 6.0                  |
| 1018          |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1019          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1020          |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1021          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1022          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1023          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1024          |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1025          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1026          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1027          |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 1028          |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 1029          |                 | 20 above      | 63                  | Coconut          | 0 - 6.0                  |
| 1030          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1031          |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1032          |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 1033          |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 1034          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1035          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1036          |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 1037          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1038          |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1039          |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1040          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1041          |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 1042          |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1043          |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 1044          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1045          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1046          |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 1047          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1048          |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 1049          |                 | 15 above      | 68                  | Sunkesula        | 0 - 6.0                  |
| 1050          |                 | 15 above      | 62                  | Sunkesula        | 0 - 6.0                  |
| 1051          |                 | 15 above      | 67                  | Sunkesula        | 0 - 6.0                  |
| 1052          |                 | 15 above      | 65                  | Sunkesula        | 0 - 6.0                  |
| 1053          |                 | 15 above      | 84                  | Sunkesula        | 0 - 6.0                  |
| 1054          | 19+400          | 10            | 82                  | Sunkesula        | 0 - 6.0                  |
| 1055          |                 | 10            | 80                  | Sunkesula        | 0 - 6.0                  |
| 1056          |                 | 10            | 85                  | Sunkesula        | 0 - 6.0                  |
| 1057          |                 | 10            | 78                  | Sunkesula        | 0 - 6.0                  |
| 1058          |                 | 20 above      | 73                  | Sunkesula        | 0 - 6.0                  |
| 1059          |                 | 20 above      | 70                  | Sunkesula        | 0 - 6.0                  |
| 1060          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1061          |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 1062          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1063          |                 | 20 above      | 64                  | Coconut          | 0 - 6.0                  |
| 1064          |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 1065          |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1066          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1067          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1068          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1069          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1070          | 19+600          | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 1071          |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |

#### APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 1072   |          | 20 above | 80           | Coconut   | 0 - 6.0           |
| 1073   |          | 20 above | 84           | Coconut   | 0 - 6.0           |
| 1074   |          | 20 above | 80           | Coconut   | 0 - 6.0           |
| 1075   |          | 20 above | 76           | Coconut   | 0 - 6.0           |
| 1076   |          | 20 above | 89           | Coconut   | 0 - 6.0           |
| 1077   |          | 20 above | 84           | Coconut   | 0 - 6.0           |
| 1078   |          | 20 above | 76           | Coconut   | 0 - 6.0           |
| 1079   |          | 20 above | 62           | Coconut   | 0 - 6.0           |
| 1080   |          | 20 above | 65           | Coconut   | 0 - 6.0           |
| 1081   |          | 20 above | 66           | Coconut   | 0 - 6.0           |
| 1082   |          | 20 above | 69           | Coconut   | 0 - 6.0           |
| 1083   |          | 20 above | 75           | Coconut   | 0 - 6.0           |
| 1084   |          | 15 above | 59           | Sunkesula | 0 - 6.0           |
| 1085   |          | 15 above | 58           | Sunkesula | 0 - 6.0           |
| 1086   |          | 15 above | 50           | Sunkesula | 0 - 6.0           |
| 1087   |          | 15 above | 54           | Sunkesula | 0 - 6.0           |
| 1088   | 19+800   | 15 above | 82           | Sunkesula | 0 - 6.0           |
| 1089   |          | 15 above | 86           | Sunkesula | 0 - 6.0           |
| 1090   |          | 15 above | 85           | Sunkesula | 0 - 6.0           |
| 1091   |          | 15 above | 86           | Sunkesula | 0 - 6.0           |
| 1092   |          | 15 above | 78           | Sunkesula | 0 - 6.0           |
| 1093   |          | 15 above | 79           | Sunkesula | 0 - 6.0           |
| 1094   |          | 15 above | 76           | Sunkesula | 0 - 6.0           |
| 1095   |          | 15 above | 72           | Sunkesula | 0 - 6.0           |
| 1096   |          | 15 above | 70           | Sunkesula | 0 - 6.0           |
| 1097   |          | 15 above | 71           | Sunkesula | 0 - 6.0           |
| 1098   |          | 15 above | 70           | Sunkesula | 0 - 6.0           |
| 1099   |          | 15 above | 89           | Sunkesula | 0 - 6.0           |
| 1100   |          | 15 above | 80           | Sunkesula | 0 - 6.0           |
| 1101   |          | 15 above | 82           | Sunkesula | 0 - 6.0           |
| 1102   |          | 15 above | 65           | Sunkesula | 0 - 6.0           |
| 1103   |          | 10       | 64           | Sunkesula | 0 - 6.0           |
| 1104   |          | 10       | 62           | Sunkesula | 0 - 6.0           |
| 1105   |          | 10       | 69           | Sunkesula | 0 - 6.0           |
| 1106   |          | 10       | 70           | Sunkesula | 0 - 6.0           |
| 1107   |          | 10       | 79           | Sunkesula | 0 - 6.0           |
| 1108   |          | 10       | 73           | Sunkesula | 0 - 6.0           |
| 1109   |          | 10       | 75           | Sunkesula | 0 - 6.0           |
| 1110   |          | 10       | 71           | Sunkesula | 0 - 6.0           |
| 1111   |          | 10       | 76           | Sunkesula | 0 - 6.0           |
| 1112   |          | 10       | 69           | Sunkesula | 0 - 6.0           |
| 1113   |          | 10       | 68           | Sunkesula | 0 - 6.0           |
| 1114   |          | 10       | 62           | Sunkesula | 0 - 6.0           |
| 1115   |          | 10       | 67           | Sunkesula | 0 - 6.0           |
| 1116   |          | 10       | 65           | Sunkesula | 0 - 6.0           |
| 1117   |          | 10       | 84           | Sunkesula | 0 - 6.0           |
| 1118   |          | 10       | 82           | Palm      | 0 - 6.0           |
| 1119   |          | 10       | 80           | Palm      | 0 - 6.0           |
| 1120   | 20+000   | 10       | 85           | Coconut   | 0 - 6.0           |
| 1121   |          | 10       | 78           | Coconut   | 0 - 6.0           |
| 1122   |          | 10       | 73           | Coconut   | 0 - 6.0           |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1123          |                 | 10            | 70                  | Coconut          | 0 - 6.0                  |
| 1124          |                 | 10            | 85                  | Coconut          | 0 - 6.0                  |
| 1125          |                 | 10            | 81                  | Coconut          | 0 - 6.0                  |
| 1126          |                 | 10            | 79                  | Coconut          | 0 - 6.0                  |
| 1127          |                 | 10            | 64                  | Coconut          | 0 - 6.0                  |
| 1128          |                 | 10            | 66                  | Bhagi            | 0 - 6.0                  |
| 1129          |                 | 10            | 86                  | Bhagi            | 0 - 6.0                  |
| 1130          |                 | 15 above      | 111                 | Bhagi            | 0 - 6.0                  |
| 1131          |                 | 15 above      | 157                 | Bhagi            | 0 - 6.0                  |
| 1132          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 1133          | 20+200          | 10            | 153                 | Bhagi            | 0 - 6.0                  |
| 1134          |                 | 10            | 142                 | Bhagi            | 0 - 6.0                  |
| 1135          |                 | 10            | 165                 | Bhagi            | 0 - 6.0                  |
| 1136          |                 | 10            | 132                 | Bhagi            | 0 - 6.0                  |
| 1137          |                 | 10            | 142                 | Bhagi            | 0 - 6.0                  |
| 1138          |                 | 10            | 154                 | Bhagi            | 0 - 6.0                  |
| 1139          |                 | 10            | 123                 | Bhagi            | 0 - 6.0                  |
| 1140          |                 | 10            | 101                 | Bhagi            | 0 - 6.0                  |
| 1141          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1142          |                 | 10            | 73                  | Bhagi            | 0 - 6.0                  |
| 1143          |                 | 10            | 62                  | Bhagi            | 0 - 6.0                  |
| 1144          |                 | 10            | 68                  | Bhagi            | 0 - 6.0                  |
| 1145          |                 | 10            | 70                  | Bhagi            | 0 - 6.0                  |
| 1146          |                 | 10            | 79                  | Bhagi            | 0 - 6.0                  |
| 1147          |                 | 10            | 81                  | Bhagi            | 0 - 6.0                  |
| 1148          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1149          |                 | 10            | 82                  | Bhagi            | 0 - 6.0                  |
| 1150          |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 1151          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1152          |                 | 15 above      | 75                  | Bhagi            | 0 - 6.0                  |
| 1153          |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 1154          |                 | 15 above      | 75                  | Bhagi            | 0 - 6.0                  |
| 1155          |                 | 15 above      | 70                  | Kanuga           | 0 - 6.0                  |
| 1156          |                 | 15 above      | 76                  | Kanuga           | 0 - 6.0                  |
| 1157          |                 | 15 above      | 84                  | Kanuga           | 0 - 6.0                  |
| 1158          |                 | 15 above      | 82                  | Kanuga           | 0 - 6.0                  |
| 1159          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1160          |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1161          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1162          |                 | 20 above      | 44                  | Coconut          | 0 - 6.0                  |
| 1163          |                 | 20 above      | 46                  | Coconut          | 0 - 6.0                  |
| 1164          |                 | 20 above      | 47                  | Coconut          | 0 - 6.0                  |
| 1165          |                 | 20 above      | 38                  | Coconut          | 0 - 6.0                  |
| 1166          |                 | 20 above      | 33                  | Coconut          | 0 - 6.0                  |
| 1167          |                 | 20 above      | 46                  | Coconut          | 0 - 6.0                  |
| 1168          |                 | 20 above      | 48                  | Coconut          | 0 - 6.0                  |
| 1169          |                 | 20 above      | 55                  | Coconut          | 0 - 6.0                  |
| 1170          | 20+400          | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 1171          |                 | 10            | 55                  | Coconut          | 0 - 6.0                  |
| 1172          |                 | 10            | 111                 | Coconut          | 0 - 6.0                  |
| 1173          |                 | 10            | 140                 | Coconut          | 0 - 6.0                  |



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1174          |                 | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 1175          |                 | 10            | 89                  | Coconut          | 0 - 6.0                  |
| 1176          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1177          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1178          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1179          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 1180          |                 | 20 above      | 85                  | Raavi            | 0 - 6.0                  |
| 1181          | 20+600          | 10            | 79                  | Bhagi            | 0 - 6.0                  |
| 1182          |                 | 10            | 68                  | Bhagi            | 0 - 6.0                  |
| 1183          |                 | 10            | 64                  | Bhagi            | 0 - 6.0                  |
| 1184          |                 | 10            | 63                  | Bhagi            | 0 - 6.0                  |
| 1185          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1186          |                 | 10            | 73                  | Bhagi            | 0 - 6.0                  |
| 1187          |                 | 10            | 62                  | Bhagi            | 0 - 6.0                  |
| 1188          |                 | 10            | 47                  | Bhagi            | 0 - 6.0                  |
| 1189          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 1190          |                 | 10            | 40                  | Bhagi            | 0 - 6.0                  |
| 1191          |                 | 15 above      | 38                  | Bhagi            | 0 - 6.0                  |
| 1192          |                 | 15 above      | 31                  | Bhagi            | 0 - 6.0                  |
| 1193          |                 | 15 above      | 39                  | Bhagi            | 0 - 6.0                  |
| 1194          |                 | 15 above      | 37                  | Bhagi            | 0 - 6.0                  |
| 1195          |                 | 15 above      | 33                  | Bhagi            | 0 - 6.0                  |
| 1196          |                 | 15 above      | 46                  | Bhagi            | 0 - 6.0                  |
| 1197          |                 | 20 above      | 164                 | Coconut          | 0 - 6.0                  |
| 1198          |                 | 20 above      | 136                 | Coconut          | 0 - 6.0                  |
| 1199          |                 | 20 above      | 132                 | Coconut          | 0 - 6.0                  |
| 1200          |                 | 20 above      | 142                 | Coconut          | 0 - 6.0                  |
| 1201          |                 | 20 above      | 101                 | Coconut          | 0 - 6.0                  |
| 1202          | 20+800          | 10            | 76                  | Thumma           | 0 - 6.0                  |
| 1203          |                 | 10            | 69                  | Thumma           | 0 - 6.0                  |
| 1204          |                 | 20 above      | 68                  | Thumma           | 0 - 6.0                  |
| 1205          |                 | 20 above      | 62                  | Thumma           | 0 - 6.0                  |
| 1206          |                 | 20 above      | 67                  | Thumma           | 0 - 6.0                  |
| 1207          |                 | 20 above      | 65                  | Thumma           | 0 - 6.0                  |
| 1208          |                 | 20 above      | 84                  | Kanuga           | 0 - 6.0                  |
| 1209          |                 | 20 above      | 82                  | Kanuga           | 0 - 6.0                  |
| 1210          |                 | 15 above      | 65                  | Kanuga           | 0 - 6.0                  |
| 1211          |                 | 15 above      | 69                  | Kanuga           | 0 - 6.0                  |
| 1212          |                 | 15 above      | 76                  | Kanuga           | 0 - 6.0                  |
| 1213          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1214          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1215          |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 1216          |                 | 15 above      | 66                  | Bhagi            | 0 - 6.0                  |
| 1217          |                 | 15 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 1218          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1219          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1220          |                 | 15 above      | 103                 | Bhagi            | 0 - 6.0                  |
| 1221          |                 | 15 above      | 106                 | Bhagi            | 0 - 6.0                  |
| 1222          |                 | 15 above      | 141                 | Bhagi            | 0 - 6.0                  |
| 1223          |                 | 15 above      | 141                 | Bhagi            | 0 - 6.0                  |
| 1224          |                 | 15 above      | 140                 | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1225          |                 | 15 above      | 141                 | Bhagi            | 0 - 6.0                  |
| 1226          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 1227          |                 | 15 above      | 164                 | Bhagi            | 0 - 6.0                  |
| 1228          |                 | 15 above      | 164                 | Bhagi            | 0 - 6.0                  |
| 1229          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 1230          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1231          | 21+000          | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1232          |                 | 10            | 82                  | Bhagi            | 0 - 6.0                  |
| 1233          |                 | 10            | 73                  | Bhagi            | 0 - 6.0                  |
| 1234          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 1235          |                 | 10            | 75                  | Bhagi            | 0 - 6.0                  |
| 1236          |                 | 10            | 86                  | Bhagi            | 0 - 6.0                  |
| 1237          |                 | 10            | 75                  | Bhagi            | 0 - 6.0                  |
| 1238          |                 | 10            | 70                  | Bhagi            | 0 - 6.0                  |
| 1239          |                 | 10            | 38                  | Bhagi            | 0 - 6.0                  |
| 1240          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 1241          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 1242          |                 | 10            | 54                  | Bhagi            | 0 - 6.0                  |
| 1243          |                 | 10            | 52                  | Bhagi            | 0 - 6.0                  |
| 1244          |                 | 10            | 56                  | Bhagi            | 0 - 6.0                  |
| 1245          |                 | 10            | 50                  | Coconut          | 0 - 6.0                  |
| 1246          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 1247          |                 | 10            | 43                  | Coconut          | 0 - 6.0                  |
| 1248          |                 | 10            | 41                  | Coconut          | 0 - 6.0                  |
| 1249          |                 | 10            | 48                  | Coconut          | 0 - 6.0                  |
| 1250          |                 | 10            | 37                  | Coconut          | 0 - 6.0                  |
| 1251          |                 | 10            | 32                  | Coconut          | 0 - 6.0                  |
| 1252          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 1253          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 1254          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 1255          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 1256          |                 | 10            | 55                  | Coconut          | 0 - 6.0                  |
| 1257          |                 | 10            | 33                  | Coconut          | 0 - 6.0                  |
| 1258          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 1259          |                 | 10            | 57                  | Coconut          | 0 - 6.0                  |
| 1260          |                 | 10            | 52                  | Coconut          | 0 - 6.0                  |
| 1261          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 1262          | 21+200          | 10            | 31                  | Thati            | 0 - 6.0                  |
| 1263          |                 | 10            | 52                  | Thati            | 0 - 6.0                  |
| 1264          |                 | 10            | 43                  | Thati            | 0 - 6.0                  |
| 1265          |                 | 10            | 51                  | Thati            | 0 - 6.0                  |
| 1266          |                 | 10            | 53                  | Thati            | 0 - 6.0                  |
| 1267          |                 | 10            | 59                  | Thati            | 0 - 6.0                  |
| 1268          |                 | 10            | 57                  | Thati            | 0 - 6.0                  |
| 1269          |                 | 10            | 50                  | Mango            | 0 - 6.0                  |
| 1270          |                 | 10            | 46                  | Mango            | 0 - 6.0                  |
| 1271          |                 | 10            | 40                  | Bhagi            | 0 - 6.0                  |
| 1272          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 1273          |                 | 10            | 57                  | Bhagi            | 0 - 6.0                  |
| 1274          |                 | 10            | 50                  | Bhagi            | 0 - 6.0                  |
| 1275          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1276          |                 | 10            | 52                  | Bhagi            | 0 - 6.0                  |
| 1277          |                 | 10            | 53                  | Bhagi            | 0 - 6.0                  |
| 1278          |                 | 10            | 32                  | Bhagi            | 0 - 6.0                  |
| 1279          |                 | 10            | 39                  | Bhagi            | 0 - 6.0                  |
| 1280          |                 | 10            | 31                  | Bhagi            | 0 - 6.0                  |
| 1281          |                 | 10            | 38                  | Bhagi            | 0 - 6.0                  |
| 1282          |                 | 10            | 54                  | Coconut          | 0 - 6.0                  |
| 1283          |                 | 10            | 59                  | Coconut          | 0 - 6.0                  |
| 1284          |                 | 10            | 50                  | Coconut          | 0 - 6.0                  |
| 1285          |                 | 10            | 57                  | Coconut          | 0 - 6.0                  |
| 1286          |                 | 10            | 50                  | Coconut          | 0 - 6.0                  |
| 1287          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 1288          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 1289          |                 | 10            | 52                  | Coconut          | 0 - 6.0                  |
| 1290          |                 | 10            | 56                  | Coconut          | 0 - 6.0                  |
| 1291          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 1292          |                 | 10            | 50                  | Coconut          | 0 - 6.0                  |
| 1293          |                 | 10            | 56                  | Coconut          | 0 - 6.0                  |
| 1294          |                 | 10            | 36                  | Coconut          | 0 - 6.0                  |
| 1295          |                 | 10            | 34                  | Coconut          | 0 - 6.0                  |
| 1296          |                 | 15 above      | 36                  | Coconut          | 0 - 6.0                  |
| 1297          |                 | 15 above      | 75                  | Coconut          | 0 - 6.0                  |
| 1298          |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1299          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1300          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1301          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1302          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1303          |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1304          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1305          |                 | 20 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1306          |                 | 20 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 1307          |                 | 20 above      | 81                  | Bhagi            | 0 - 6.0                  |
| 1308          |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1309          |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1310          |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1311          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1312          |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 1313          |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1314          |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 1315          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1316          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1317          |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 1318          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1319          |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 1320          |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 1321          |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 1322          |                 | 20 above      | 67                  | Coconut          | 0 - 6.0                  |
| 1323          |                 | 20 above      | 65                  | Coconut          | 0 - 6.0                  |
| 1324          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1325          |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1326          |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1327          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1328          |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 1329          |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1330          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1331          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1332          |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 1333          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1334          |                 | 20 above      | 38                  | Coconut          | 0 - 6.0                  |
| 1335          |                 | 20 above      | 49                  | Coconut          | 0 - 6.0                  |
| 1336          |                 | 20 above      | 48                  | Coconut          | 0 - 6.0                  |
| 1337          |                 | 20 above      | 41                  | Coconut          | 0 - 6.0                  |
| 1338          |                 | 20 above      | 40                  | Coconut          | 0 - 6.0                  |
| 1339          |                 | 20 above      | 46                  | Coconut          | 0 - 6.0                  |
| 1340          |                 | 20 above      | 33                  | Coconut          | 0 - 6.0                  |
| 1341          |                 | 20 above      | 38                  | Coconut          | 0 - 6.0                  |
| 1342          | 21+600          | 20 above      | 39                  | Bhagi            | 0 - 6.0                  |
| 1343          |                 | 20 above      | 42                  | Bhagi            | 0 - 6.0                  |
| 1344          |                 | 20 above      | 51                  | Bhagi            | 0 - 6.0                  |
| 1345          |                 | 20 above      | 56                  | Bhagi            | 0 - 6.0                  |
| 1346          |                 | 20 above      | 57                  | Bhagi            | 0 - 6.0                  |
| 1347          |                 | 20 above      | 52                  | Bhagi            | 0 - 6.0                  |
| 1348          |                 | 20 above      | 51                  | Bhagi            | 0 - 6.0                  |
| 1349          |                 | 20 above      | 34                  | Bhagi            | 0 - 6.0                  |
| 1350          |                 | 20 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 1351          |                 | 20 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 1352          |                 | 20 above      | 66                  | Coconut          | 0 - 6.0                  |
| 1353          |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 1354          |                 | 20 above      | 75                  | Coconut          | 0 - 6.0                  |
| 1355          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1356          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1357          |                 | 20 above      | 83                  | Coconut          | 0 - 6.0                  |
| 1358          |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1359          |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1360          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1361          |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1362          |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 1363          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1364          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1365          |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |
| 1366          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1367          |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 1368          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1369          |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 1370          |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1371          |                 | 20 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1372          |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 1373          |                 | 20 above      | 69                  | Coconut          | 0 - 6.0                  |
| 1374          |                 | 10            | 81                  | Coconut          | 0 - 6.0                  |
| 1375          |                 | 15 above      | 36                  | Coconut          | 0 - 6.0                  |
| 1376          |                 | 15 above      | 31                  | Coconut          | 0 - 6.0                  |
| 1377          |                 | 15 above      | 39                  | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1378          |                 | 15 above      | 38                  | Bhagi            | 0 - 6.0                  |
| 1379          |                 | 15 above      | 41                  | Raavi            | 0 - 6.0                  |
| 1380          |                 | 15 above      | 136                 | Teak             | 0 - 6.0                  |
| 1381          |                 | 15 above      | 125                 | Teak             | 0 - 6.0                  |
| 1382          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 1383          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1384          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1385          |                 | 10            | 70                  | Coconut          | 0 - 6.0                  |
| 1386          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1387          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1388          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1389          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1390          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1391          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1392          |                 | 15 above      | 75                  | Coconut          | 0 - 6.0                  |
| 1393          |                 | 15 above      | 66                  | Coconut          | 0 - 6.0                  |
| 1394          |                 | 15 above      | 69                  | Coconut          | 0 - 6.0                  |
| 1395          | 22+200          | 15 above      | 84                  | Neem             | 0 - 6.0                  |
| 1396          |                 | 15 above      | 85                  | Kanuga           | 0 - 6.0                  |
| 1397          |                 | 15 above      | 79                  | Kanuga           | 0 - 6.0                  |
| 1398          |                 | 10            | 65                  | Coconut          | 0 - 6.0                  |
| 1399          |                 | 10            | 84                  | Coconut          | 0 - 6.0                  |
| 1400          | 22+400          | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 1401          | 22+600          | 15 above      | 33                  | Coconut          | 0 - 6.0                  |
| 1402          |                 | 15 above      | 37                  | Coconut          | 0 - 6.0                  |
| 1403          |                 | 15 above      | 46                  | Coconut          | 0 - 6.0                  |
| 1404          |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1405          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1406          |                 | 20 above      | 89                  | Neem             | 0 - 6.0                  |
| 1407          | 22+800          | 10            | 80                  | Bhagi            | 0 - 6.0                  |
| 1408          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1409          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 1410          |                 | 10            | 75                  | Bhagi            | 0 - 6.0                  |
| 1411          |                 | 10            | 86                  | Coconut          | 0 - 6.0                  |
| 1412          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1413          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 1414          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1415          |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |
| 1416          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1417          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1418          |                 | 15 above      | 74                  | Coconut          | 0 - 6.0                  |
| 1419          |                 | 15 above      | 48                  | Coconut          | 0 - 6.0                  |
| 1420          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |
| 1421          |                 | 15 above      | 58                  | Coconut          | 0 - 6.0                  |
| 1422          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 1423          |                 | 15 above      | 164                 | Coconut          | 0 - 6.0                  |
| 1424          | 23+000          | 15 above      | 163                 | Bhagi            | 0 - 6.0                  |
| 1425          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 1426          |                 | 10            | 79                  | Bhagi            | 0 - 6.0                  |
| 1427          |                 | 10            | 85                  | Bhagi            | 0 - 6.0                  |
| 1428          |                 | 10            | 76                  | Coconut          | 0 - 6.0                  |

#### APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 1429   |          | 10       | 78           | Coconut   | 0 - 6.0           |
| 1430   |          | 10       | 62           | Coconut   | 0 - 6.0           |
| 1431   |          | 10       | 65           | Coconut   | 0 - 6.0           |
| 1432   |          | 10       | 66           | Coconut   | 0 - 6.0           |
| 1433   |          | 10       | 69           | Coconut   | 0 - 6.0           |
| 1434   |          | 10       | 75           | Coconut   | 0 - 6.0           |
| 1435   |          | 10       | 85           | Coconut   | 0 - 6.0           |
| 1436   |          | 10       | 83           | Coconut   | 0 - 6.0           |
| 1437   | 23+200   | 10       | 82           | Bhagi     | 0 - 6.0           |
| 1438   |          | 10       | 86           | Bhagi     | 0 - 6.0           |
| 1439   |          | 10       | 85           | Bhagi     | 0 - 6.0           |
| 1440   |          | 10       | 79           | Bhagi     | 0 - 6.0           |
| 1441   |          | 10       | 70           | Bhagi     | 0 - 6.0           |
| 1442   |          | 10       | 89           | Bhagi     | 0 - 6.0           |
| 1443   |          | 10       | 80           | Bhagi     | 0 - 6.0           |
| 1444   |          | 10       | 82           | Bhagi     | 0 - 6.0           |
| 1445   |          | 10       | 62           | Bhagi     | 0 - 6.0           |
| 1446   |          | 10       | 69           | Bhagi     | 0 - 6.0           |
| 1447   |          | 10       | 70           | Bhagi     | 0 - 6.0           |
| 1448   |          | 10       | 79           | Bhagi     | 0 - 6.0           |
| 1449   |          | 10       | 70           | Bhagi     | 0 - 6.0           |
| 1450   |          | 10       | 81           | Bhagi     | 0 - 6.0           |
| 1451   |          | 10       | 81           | Bhagi     | 0 - 6.0           |
| 1452   | 23+400   | 10       | 51           | Mango     | 0 - 6.0           |
| 1453   |          | 10       | 56           | Mango     | 0 - 6.0           |
| 1454   |          | 10       | 50           | Bhagi     | 0 - 6.0           |
| 1455   |          | 10       | 58           | Bhagi     | 0 - 6.0           |
| 1456   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 1457   |          | 10       | 33           | Bhagi     | 0 - 6.0           |
| 1458   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 1459   |          | 10       | 37           | Bhagi     | 0 - 6.0           |
| 1460   |          | 10       | 55           | Coconut   | 0 - 6.0           |
| 1461   |          | 10       | 51           | Coconut   | 0 - 6.0           |
| 1462   |          | 10       | 59           | Coconut   | 0 - 6.0           |
| 1463   |          | 10       | 58           | Coconut   | 0 - 6.0           |
| 1464   |          | 10       | 51           | Coconut   | 0 - 6.0           |
| 1465   |          | 10       | 53           | Coconut   | 0 - 6.0           |
| 1466   |          | 15 above | 62           | Coconut   | 0 - 6.0           |
| 1467   |          | 15 above | 65           | Coconut   | 0 - 6.0           |
| 1468   |          | 15 above | 66           | Coconut   | 0 - 6.0           |
| 1469   |          | 15 above | 69           | Coconut   | 0 - 6.0           |
| 1470   |          | 15 above | 75           | Coconut   | 0 - 6.0           |
| 1471   |          | 15 above | 86           | Coconut   | 0 - 6.0           |
| 1472   |          | 15 above | 83           | Coconut   | 0 - 6.0           |
| 1473   |          | 15 above | 66           | Coconut   | 0 - 6.0           |
| 1474   |          | 15 above | 78           | Coconut   | 0 - 6.0           |
| 1475   |          | 15 above | 64           | Coconut   | 0 - 6.0           |
| 1476   | 23+600   | 15 above | 154          | Bhagi     | 0 - 6.0           |
| 1477   |          | 15 above | 57           | Bhagi     | 0 - 6.0           |
| 1478   |          | 20 above | 46           | Bhagi     | 0 - 6.0           |
| 1479   |          | 10       | 51           | Bhagi     | 0 - 6.0           |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1480          |                 | 10            | 50                  | Bhagi            | 0 - 6.0                  |
| 1481          |                 | 10            | 38                  | Bhagi            | 0 - 6.0                  |
| 1482          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 1483          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 1484          |                 | 10            | 51                  | Bhagi            | 0 - 6.0                  |
| 1485          |                 | 10            | 57                  | Bhagi            | 0 - 6.0                  |
| 1486          |                 | 10            | 51                  | Bhagi            | 0 - 6.0                  |
| 1487          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 1488          |                 | 10            | 55                  | Bhagi            | 0 - 6.0                  |
| 1489          | 23+800          | 10            | 36                  | Bhagi            | 0 - 6.0                  |
| 1490          |                 | 10            | 78                  | Thumma           | 0 - 6.0                  |
| 1491          | 24+000          | 15 above      | 64                  | Thumma           | 0 - 6.0                  |
| 1492          |                 | 15 above      | 83                  | Thumma           | 0 - 6.0                  |
| 1493          |                 | 15 above      | 82                  | Tamarind         | 0 - 6.0                  |
| 1494          |                 | 15 above      | 86                  | Tamarind         | 0 - 6.0                  |
| 1495          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1496          |                 | 20 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1497          |                 | 20 above      | 110                 | Bhagi            | 0 - 6.0                  |
| 1498          |                 | 20 above      | 147                 | Bhagi            | 0 - 6.0                  |
| 1499          | 24+200          | 10            | 123                 | Neredu           | 0 - 6.0                  |
| 1500          |                 | 10            | 153                 | Thumma           | 0 - 6.0                  |
| 1501          |                 | 10            | 125                 | Thumma           | 0 - 6.0                  |
| 1502          |                 | 10            | 71                  | Bhagi            | 0 - 6.0                  |
| 1503          |                 | 10            | 70                  | Bhagi            | 0 - 6.0                  |
| 1504          |                 | 10            | 89                  | Bhagi            | 0 - 6.0                  |
| 1505          |                 | 10            | 80                  | Bhagi            | 0 - 6.0                  |
| 1506          |                 | 10            | 82                  | Bhagi            | 0 - 6.0                  |
| 1507          |                 | 10            | 65                  | Bhagi            | 0 - 6.0                  |
| 1508          |                 | 10            | 64                  | Bhagi            | 0 - 6.0                  |
| 1509          |                 | 20 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 1510          |                 | 20 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 1511          |                 | 20 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1512          |                 | 20 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 1513          | 24+400          | 20 above      | 73                  | Tamarind         | 0 - 6.0                  |
| 1514          |                 | 20 above      | 75                  | Tamarind         | 0 - 6.0                  |
| 1515          |                 | 20 above      | 71                  | Bhagi            | 0 - 6.0                  |
| 1516          |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1517          |                 | 20 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 1518          |                 | 20 above      | 68                  | Bhagi            | 0 - 6.0                  |
| 1519          |                 | 20 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 1520          |                 | 15 above      | 67                  | Bhagi            | 0 - 6.0                  |
| 1521          |                 | 15 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 1522          |                 | 15 above      | 130                 | Bhagi            | 0 - 6.0                  |
| 1523          |                 | 15 above      | 120                 | Bhagi            | 0 - 6.0                  |
| 1524          |                 | 15 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 1525          |                 | 15 above      | 120                 | Bhagi            | 0 - 6.0                  |
| 1526          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1527          |                 | 15 above      | 121                 | Bhagi            | 0 - 6.0                  |
| 1528          |                 | 15 above      | 131                 | Bhagi            | 0 - 6.0                  |
| 1529          |                 | 15 above      | 132                 | Bhagi            | 0 - 6.0                  |
| 1530          | 24+600          | 10            | 132                 | Thumma           | 0 - 6.0                  |

# APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 1531   |          | 10       | 135          | Thumma    | 0 - 6.0           |
| 1532   |          | 10       | 110          | Coconut   | 0 - 6.0           |
| 1533   |          | 10       | 126          | Coconut   | 0 - 6.0           |
| 1534   |          | 10       | 120          | Coconut   | 0 - 6.0           |
| 1535   |          | 20 above | 84           | Coconut   | 0 - 6.0           |
| 1536   |          | 20 above | 79           | Coconut   | 0 - 6.0           |
| 1537   |          | 20 above | 85           | Coconut   | 0 - 6.0           |
| 1538   |          | 20 above | 76           | Coconut   | 0 - 6.0           |
| 1539   |          | 20 above | 78           | Bhagi     | 0 - 6.0           |
| 1540   |          | 15 above | 72           | Bhagi     | 0 - 6.0           |
| 1541   |          | 15 above | 80           | Bhagi     | 0 - 6.0           |
| 1542   |          | 15 above | 84           | Bhagi     | 0 - 6.0           |
| 1543   |          | 15 above | 80           | Bhagi     | 0 - 6.0           |
| 1544   |          | 15 above | 76           | Bhagi     | 0 - 6.0           |
| 1545   |          | 15 above | 89           | Bhagi     | 0 - 6.0           |
| 1546   |          | 15 above | 84           | Bhagi     | 0 - 6.0           |
| 1547   |          | 15 above | 76           | Bhagi     | 0 - 6.0           |
| 1548   |          | 15 above | 70           | Bhagi     | 0 - 6.0           |
| 1549   | 24+800   | 10       | 80           | Bhagi     | 0 - 6.0           |
| 1550   |          | 10       | 85           | Bhagi     | 0 - 6.0           |
| 1551   |          | 10       | 79           | Bhagi     | 0 - 6.0           |
| 1552   |          | 10       | 68           | Bhagi     | 0 - 6.0           |
| 1553   |          | 10       | 64           | Bhagi     | 0 - 6.0           |
| 1554   |          | 10       | 63           | Bhagi     | 0 - 6.0           |
| 1555   |          | 10       | 76           | Bhagi     | 0 - 6.0           |
| 1556   |          | 10       | 73           | Bhagi     | 0 - 6.0           |
| 1557   |          | 10       | 62           | Coconut   | 0 - 6.0           |
| 1558   |          | 10       | 68           | Coconut   | 0 - 6.0           |
| 1559   |          | 10       | 125          | Coconut   | 0 - 6.0           |
| 1560   |          | 10       | 126          | Coconut   | 0 - 6.0           |
| 1561   |          | 10       | 123          | Coconut   | 0 - 6.0           |
| 1562   |          | 10       | 120          | Coconut   | 0 - 6.0           |
| 1563   |          | 10       | 130          | Coconut   | 0 - 6.0           |
| 1564   | 25+000   | 10       | 73           | Bhagi     | 0 - 6.0           |
| 1565   |          | 10       | 84           | Bhagi     | 0 - 6.0           |
| 1566   |          | 10       | 75           | Bhagi     | 0 - 6.0           |
| 1567   |          | 10       | 86           | Bhagi     | 0 - 6.0           |
| 1568   |          | 10       | 75           | Bhagi     | 0 - 6.0           |
| 1569   |          | 10       | 70           | Bhagi     | 0 - 6.0           |
| 1570   |          | 15 above | 76           | Bhagi     | 0 - 6.0           |
| 1571   |          | 15 above | 140          | Bhagi     | 0 - 6.0           |
| 1572   |          | 15 above | 151          | Bhagi     | 0 - 6.0           |
| 1573   |          | 15 above | 124          | Bhagi     | 0 - 6.0           |
| 1574   | 25+200   | 15 above | 120          | Bhagi     | 0 - 6.0           |
| 1575   |          | 15 above | 37           | Bhagi     | 0 - 6.0           |
| 1576   |          | 15 above | 54           | Bhagi     | 0 - 6.0           |
| 1577   |          | 15 above | 86           | Bhagi     | 0 - 6.0           |
| 1578   |          | 15 above | 73           | Bhagi     | 0 - 6.0           |
| 1579   |          | 15 above | 74           | Bhagi     | 0 - 6.0           |
| 1580   |          | 15 above | 81           | Bhagi     | 0 - 6.0           |
| 1581   |          | 15 above | 86           | Bhagi     | 0 - 6.0           |



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1582          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1583          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1584          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1585          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1586          |                 | 15 above      | 74                  | Coconut          | 0 - 6.0                  |
| 1587          |                 | 10            | 80                  | Coconut          | 0 - 6.0                  |
| 1588          |                 | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 1589          |                 | 10            | 89                  | Coconut          | 0 - 6.0                  |
| 1590          |                 | 10            | 84                  | Coconut          | 0 - 6.0                  |
| 1591          | 25+400          | 10            | 70                  | Bhagi            | 0 - 6.0                  |
| 1592          |                 | 10            | 80                  | Bhagi            | 0 - 6.0                  |
| 1593          |                 | 10            | 85                  | Bhagi            | 0 - 6.0                  |
| 1594          |                 | 10            | 79                  | Bhagi            | 0 - 6.0                  |
| 1595          |                 | 15 above      | 154                 | Bhagi            | 0 - 6.0                  |
| 1596          |                 | 15 above      | 135                 | Bhagi            | 0 - 6.0                  |
| 1597          |                 | 15 above      | 135                 | Bhagi            | 0 - 6.0                  |
| 1598          |                 | 15 above      | 126                 | Bhagi            | 0 - 6.0                  |
| 1599          |                 | 15 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 1600          | 25+600          | 15 above      | 128                 | Bhagi            | 0 - 6.0                  |
| 1601          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 1602          |                 | 15 above      | 135                 | Bhagi            | 0 - 6.0                  |
| 1603          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1604          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 1605          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1606          |                 | 15 above      | 142                 | Bhagi            | 0 - 6.0                  |
| 1607          |                 | 20 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 1608          |                 | 20 above      | 143                 | Bhagi            | 0 - 6.0                  |
| 1609          |                 | 20 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 1610          |                 | 20 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1611          |                 | 20 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 1612          |                 | 20 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1613          |                 | 20 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 1614          |                 | 10            | 148                 | Bhagi            | 0 - 6.0                  |
| 1615          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 1616          |                 | 10            | 52                  | Bhagi            | 0 - 6.0                  |
| 1617          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 1618          |                 | 10            | 42                  | Bhagi            | 0 - 6.0                  |
| 1619          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 1620          |                 | 10            | 65                  | Bhagi            | 0 - 6.0                  |
| 1621          |                 | 10            | 84                  | Coconut          | 0 - 6.0                  |
| 1622          |                 | 10            | 82                  | Coconut          | 0 - 6.0                  |
| 1623          |                 | 10            | 80                  | Coconut          | 0 - 6.0                  |
| 1624          |                 | 10            | 81                  | Coconut          | 0 - 6.0                  |
| 1625          |                 | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 1626          |                 | 10            | 75                  | Coconut          | 0 - 6.0                  |
| 1627          |                 | 10            | 70                  | Coconut          | 0 - 6.0                  |
| 1628          |                 | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 1629          |                 | 10            | 71                  | Coconut          | 0 - 6.0                  |
| 1630          |                 | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 1631          |                 | 10            | 69                  | Coconut          | 0 - 6.0                  |
| 1632          |                 | 10            | 68                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1633          |                 | 10            | 62                  | Coconut          | 0 - 6.0                  |
| 1634          |                 | 10            | 67                  | Coconut          | 0 - 6.0                  |
| 1635          |                 | 15 above      | 43                  | Coconut          | 0 - 6.0                  |
| 1636          |                 | 15 above      | 47                  | Coconut          | 0 - 6.0                  |
| 1637          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |
| 1638          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 1639          |                 | 15 above      | 48                  | Bhagi            | 0 - 6.0                  |
| 1640          |                 | 15 above      | 43                  | Bhagi            | 0 - 6.0                  |
| 1641          |                 | 15 above      | 48                  | Bhagi            | 0 - 6.0                  |
| 1642          |                 | 15 above      | 146                 | Bhagi            | 0 - 6.0                  |
| 1643          |                 | 15 above      | 144                 | Bhagi            | 0 - 6.0                  |
| 1644          |                 | 15 above      | 134                 | Bhagi            | 0 - 6.0                  |
| 1645          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1646          |                 | 15 above      | 129                 | Bhagi            | 0 - 6.0                  |
| 1647          | 26+000          | 15 above      | 41                  | Thati            | 0 - 6.0                  |
| 1648          |                 | 15 above      | 86                  | Thati            | 0 - 6.0                  |
| 1649          |                 | 15 above      | 84                  | Kanuga           | 0 - 6.0                  |
| 1650          |                 | 15 above      | 79                  | Kanuga           | 0 - 6.0                  |
| 1651          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1652          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1653          |                 | 15 above      | 78                  | Bhagi            | 0 - 6.0                  |
| 1654          |                 | 15 above      | 72                  | Bhagi            | 0 - 6.0                  |
| 1655          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 1656          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1657          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 1658          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1659          |                 | 15 above      | 89                  | Coconut          | 0 - 6.0                  |
| 1660          |                 | 20 above      | 128                 | Coconut          | 0 - 6.0                  |
| 1661          |                 | 20 above      | 132                 | Coconut          | 0 - 6.0                  |
| 1662          |                 | 20 above      | 124                 | Coconut          | 0 - 6.0                  |
| 1663          |                 | 20 above      | 136                 | Coconut          | 0 - 6.0                  |
| 1664          |                 | 10            | 185                 | Coconut          | 0 - 6.0                  |
| 1665          |                 | 10            | 136                 | Coconut          | 0 - 6.0                  |
| 1666          |                 | 10            | 125                 | Coconut          | 0 - 6.0                  |
| 1667          |                 | 10            | 142                 | Coconut          | 0 - 6.0                  |
| 1668          |                 | 10            | 112                 | Coconut          | 0 - 6.0                  |
| 1669          |                 | 10            | 126                 | Coconut          | 0 - 6.0                  |
| 1670          |                 | 10            | 125                 | Coconut          | 0 - 6.0                  |
| 1671          |                 | 10            | 136                 | Coconut          | 0 - 6.0                  |
| 1672          |                 | 10            | 128                 | Coconut          | 0 - 6.0                  |
| 1673          |                 | 10            | 127                 | Coconut          | 0 - 6.0                  |
| 1674          |                 | 10            | 129                 | Coconut          | 0 - 6.0                  |
| 1675          | 26+200          | 10            | 122                 | Bhagi            | 0 - 6.0                  |
| 1676          |                 | 10            | 125                 | Bhagi            | 0 - 6.0                  |
| 1677          |                 | 10            | 136                 | Bhagi            | 0 - 6.0                  |
| 1678          |                 | 10            | 134                 | Bhagi            | 0 - 6.0                  |
| 1679          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1680          |                 | 15 above      | 71                  | Bhagi            | 0 - 6.0                  |
| 1681          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1682          |                 | 15 above      | 89                  | Bhagi            | 0 - 6.0                  |
| 1683          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1684          |                 | 15 above      | 82                  | Bhagi            | 0 - 6.0                  |
| 1685          |                 | 15 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 1686          |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 1687          |                 | 15 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 1688          |                 | 15 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 1689          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1690          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 1691          |                 | 20 above      | 75                  | Bhagi            | 0 - 6.0                  |
| 1692          |                 | 20 above      | 71                  | Bhagi            | 0 - 6.0                  |
| 1693          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1694          |                 | 15 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 1695          |                 | 15 above      | 68                  | Bhagi            | 0 - 6.0                  |
| 1696          |                 | 15 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 1697          |                 | 15 above      | 67                  | Bhagi            | 0 - 6.0                  |
| 1698          |                 | 15 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 1699          |                 | 15 above      | 135                 | Bhagi            | 0 - 6.0                  |
| 1700          |                 | 15 above      | 156                 | Bhagi            | 0 - 6.0                  |
| 1701          |                 | 15 above      | 157                 | Bhagi            | 0 - 6.0                  |
| 1702          |                 | 15 above      | 152                 | Bhagi            | 0 - 6.0                  |
| 1703          | 26+600          | 15 above      | 155                 | Bhagi            | 0 - 6.0                  |
| 1704          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 1705          |                 | 15 above      | 138                 | Bhagi            | 0 - 6.0                  |
| 1706          |                 | 15 above      | 145                 | Bhagi            | 0 - 6.0                  |
| 1707          |                 | 15 above      | 139                 | Bhagi            | 0 - 6.0                  |
| 1708          |                 | 15 above      | 142                 | Bhagi            | 0 - 6.0                  |
| 1709          |                 | 15 above      | 123                 | Bhagi            | 0 - 6.0                  |
| 1710          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 1711          |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 1712          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1713          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 1714          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1715          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1716          |                 | 10            | 78                  | Bhagi            | 0 - 6.0                  |
| 1717          |                 | 10            | 72                  | Bhagi            | 0 - 6.0                  |
| 1718          |                 | 10            | 80                  | Bhagi            | 0 - 6.0                  |
| 1719          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 1720          |                 | 10            | 80                  | Bhagi            | 0 - 6.0                  |
| 1721          | 26+800          | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1722          |                 | 10            | 89                  | Bhagi            | 0 - 6.0                  |
| 1723          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 1724          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1725          |                 | 10            | 70                  | Bhagi            | 0 - 6.0                  |
| 1726          |                 | 10            | 80                  | Bhagi            | 0 - 6.0                  |
| 1727          |                 | 10            | 85                  | Bhagi            | 0 - 6.0                  |
| 1728          |                 | 10            | 79                  | Bhagi            | 0 - 6.0                  |
| 1729          |                 | 10            | 68                  | Coconut          | 0 - 6.0                  |
| 1730          |                 | 10            | 64                  | Coconut          | 0 - 6.0                  |
| 1731          |                 | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 1732          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1733          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1734          |                 | 15 above      | 62                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1735          |                 | 15 above      | 128                 | Coconut          | 0 - 6.0                  |
| 1736          |                 | 15 above      | 126                 | Coconut          | 0 - 6.0                  |
| 1737          |                 | 15 above      | 122                 | Coconut          | 0 - 6.0                  |
| 1738          |                 | 15 above      | 124                 | Coconut          | 0 - 6.0                  |
| 1739          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 1740          | 27+000          | 10            | 122                 | Bhagi            | 0 - 6.0                  |
| 1741          |                 | 10            | 145                 | Bhagi            | 0 - 6.0                  |
| 1742          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 1743          |                 | 20 above      | 75                  | Bhagi            | 0 - 6.0                  |
| 1744          |                 | 20 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 1745          |                 | 20 above      | 75                  | Bhagi            | 0 - 6.0                  |
| 1746          |                 | 20 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1747          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1748          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1749          |                 | 15 above      | 82                  | Bhagi            | 0 - 6.0                  |
| 1750          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 1751          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1752          | 27+200          | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1753          |                 | 10            | 81                  | Bhagi            | 0 - 6.0                  |
| 1754          |                 | 10            | 86                  | Bhagi            | 0 - 6.0                  |
| 1755          |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 1756          |                 | 15 above      | 74                  | Bhagi            | 0 - 6.0                  |
| 1757          |                 | 15 above      | 156                 | Bhagi            | 0 - 6.0                  |
| 1758          |                 | 15 above      | 123                 | Bhagi            | 0 - 6.0                  |
| 1759          |                 | 15 above      | 152                 | Bhagi            | 0 - 6.0                  |
| 1760          |                 | 20 above      | 146                 | Bhagi            | 0 - 6.0                  |
| 1761          |                 | 20 above      | 158                 | Bhagi            | 0 - 6.0                  |
| 1762          |                 | 20 above      | 169                 | Bhagi            | 0 - 6.0                  |
| 1763          |                 | 20 above      | 154                 | Bhagi            | 0 - 6.0                  |
| 1764          |                 | 20 above      | 49                  | Bhagi            | 0 - 6.0                  |
| 1765          |                 | 20 above      | 47                  | Bhagi            | 0 - 6.0                  |
| 1766          | 27+400          | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 1767          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 1768          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 1769          |                 | 10            | 70                  | Bhagi            | 0 - 6.0                  |
| 1770          |                 | 10            | 80                  | Bhagi            | 0 - 6.0                  |
| 1771          |                 | 10            | 85                  | Bhagi            | 0 - 6.0                  |
| 1772          |                 | 10            | 79                  | Coconut          | 0 - 6.0                  |
| 1773          |                 | 10            | 68                  | Coconut          | 0 - 6.0                  |
| 1774          |                 | 15 above      | 64                  | Coconut          | 0 - 6.0                  |
| 1775          |                 | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 1776          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1777          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1778          |                 | 15 above      | 165                 | Coconut          | 0 - 6.0                  |
| 1779          |                 | 15 above      | 126                 | Coconut          | 0 - 6.0                  |
| 1780          |                 | 15 above      | 133                 | Coconut          | 0 - 6.0                  |
| 1781          |                 | 15 above      | 155                 | Coconut          | 0 - 6.0                  |
| 1782          |                 | 20 above      | 166                 | Coconut          | 0 - 6.0                  |
| 1783          | 27+600          | 15 above      | 177                 | Coconut          | 0 - 6.0                  |
| 1784          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 1785          |                 | 15 above      | 144                 | Coconut          | 0 - 6.0                  |

#### APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 1786   |          | 15 above | 165          | Coconut   | 0 - 6.0           |
| 1787   |          | 15 above | 136          | Coconut   | 0 - 6.0           |
| 1788   |          | 20 above | 186          | Coconut   | 0 - 6.0           |
| 1789   |          | 20 above | 128          | Coconut   | 0 - 6.0           |
| 1790   |          | 20 above | 136          | Coconut   | 0 - 6.0           |
| 1791   |          | 20 above | 142          | Coconut   | 0 - 6.0           |
| 1792   |          | 20 above | 136          | Coconut   | 0 - 6.0           |
| 1793   |          | 20 above | 43           | Coconut   | 0 - 6.0           |
| 1794   |          | 20 above | 69           | Coconut   | 0 - 6.0           |
| 1795   |          | 20 above | 68           | Coconut   | 0 - 6.0           |
| 1796   |          | 20 above | 62           | Coconut   | 0 - 6.0           |
| 1797   |          | 20 above | 67           | Coconut   | 0 - 6.0           |
| 1798   |          | 20 above | 65           | Coconut   | 0 - 6.0           |
| 1799   |          | 20 above | 84           | Bhagi     | 0 - 6.0           |
| 1800   |          | 20 above | 82           | Bhagi     | 0 - 6.0           |
| 1801   |          | 20 above | 65           | Bhagi     | 0 - 6.0           |
| 1802   |          | 20 above | 69           | Bhagi     | 0 - 6.0           |
| 1803   |          | 20 above | 76           | Bhagi     | 0 - 6.0           |
| 1804   |          | 20 above | 84           | Bhagi     | 0 - 6.0           |
| 1805   |          | 20 above | 68           | Bhagi     | 0 - 6.0           |
| 1806   |          | 20 above | 64           | Bhagi     | 0 - 6.0           |
| 1807   | 27+800   | 10       | 63           | Bhagi     | 0 - 6.0           |
| 1808   |          | 10       | 76           | Bhagi     | 0 - 6.0           |
| 1809   |          | 10       | 73           | Bhagi     | 0 - 6.0           |
| 1810   |          | 10       | 62           | Bhagi     | 0 - 6.0           |
| 1811   |          | 10       | 68           | Bhagi     | 0 - 6.0           |
| 1812   |          | 10       | 70           | Bhagi     | 0 - 6.0           |
| 1813   |          | 10       | 79           | Bhagi     | 0 - 6.0           |
| 1814   |          | 10       | 81           | Bhagi     | 0 - 6.0           |
| 1815   |          | 15 above | 76           | Coconut   | 0 - 6.0           |
| 1816   |          | 15 above | 82           | Coconut   | 0 - 6.0           |
| 1817   |          | 15 above | 73           | Coconut   | 0 - 6.0           |
| 1818   |          | 15 above | 84           | Coconut   | 0 - 6.0           |
| 1819   |          | 15 above | 75           | Coconut   | 0 - 6.0           |
| 1820   |          | 15 above | 115          | Coconut   | 0 - 6.0           |
| 1821   |          | 15 above | 49           | Coconut   | 0 - 6.0           |
| 1822   |          | 15 above | 48           | Coconut   | 0 - 6.0           |
| 1823   |          | 15 above | 43           | Coconut   | 0 - 6.0           |
| 1824   |          | 15 above | 48           | Coconut   | 0 - 6.0           |
| 1825   |          | 15 above | 43           | Coconut   | 0 - 6.0           |
| 1826   |          | 15 above | 49           | Coconut   | 0 - 6.0           |
| 1827   |          | 20 above | 46           | Coconut   | 0 - 6.0           |
| 1828   |          | 20 above | 48           | Coconut   | 0 - 6.0           |
| 1829   |          | 20 above | 41           | Coconut   | 0 - 6.0           |
| 1830   |          | 20 above | 46           | Coconut   | 0 - 6.0           |
| 1831   |          | 20 above | 44           | Coconut   | 0 - 6.0           |
| 1832   |          | 20 above | 49           | Coconut   | 0 - 6.0           |
| 1833   | 28+000   | 10       | 117          | Tamarind  | 0 - 6.0           |
| 1834   |          | 10       | 65           | Tamarind  | 0 - 6.0           |
| 1835   |          | 20 above | 84           | Bhagi     | 0 - 6.0           |
| 1836   |          | 20 above | 82           | Bhagi     | 0 - 6.0           |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1837          |                 | 20 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 1838          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1839          |                 | 15 above      | 78                  | Bhagi            | 0 - 6.0                  |
| 1840          |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 1841          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 1842          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1843          |                 | 15 above      | 81                  | Bhagi            | 0 - 6.0                  |
| 1844          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 1845          |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 1846          |                 | 15 above      | 66                  | Bhagi            | 0 - 6.0                  |
| 1847          |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 1848          | 28+200          | 10            | 84                  | Tamarind         | 0 - 6.0                  |
| 1849          |                 | 20 above      | 79                  | Tamarind         | 0 - 6.0                  |
| 1850          |                 | 20 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 1851          |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1852          |                 | 20 above      | 78                  | Bhagi            | 0 - 6.0                  |
| 1853          |                 | 15 above      | 72                  | Bhagi            | 0 - 6.0                  |
| 1854          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 1855          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1856          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 1857          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1858          |                 | 15 above      | 119                 | Bhagi            | 0 - 6.0                  |
| 1859          | 28+400          | 10            | 60                  | Bhagi            | 0 - 6.0                  |
| 1860          |                 | 15 above      | 56                  | Bhagi            | 0 - 6.0                  |
| 1861          |                 | 15 above      | 58                  | Bhagi            | 0 - 6.0                  |
| 1862          |                 | 15 above      | 51                  | Bhagi            | 0 - 6.0                  |
| 1863          |                 | 15 above      | 180                 | Bhagi            | 0 - 6.0                  |
| 1864          |                 | 15 above      | 166                 | Tamarind         | 0 - 6.0                  |
| 1865          |                 | 15 above      | 136                 | Tamarind         | 0 - 6.0                  |
| 1866          |                 | 20 above      | 144                 | Tamarind         | 0 - 6.0                  |
| 1867          |                 | 20 above      | 136                 | Coconut          | 0 - 6.0                  |
| 1868          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1869          |                 | 20 above      | 73                  | Coconut          | 0 - 6.0                  |
| 1870          |                 | 20 above      | 62                  | Coconut          | 0 - 6.0                  |
| 1871          |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 1872          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1873          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1874          |                 | 20 above      | 81                  | Coconut          | 0 - 6.0                  |
| 1875          |                 | 20 above      | 50                  | Coconut          | 0 - 6.0                  |
| 1876          |                 | 20 above      | 57                  | Coconut          | 0 - 6.0                  |
| 1877          |                 | 20 above      | 133                 | Coconut          | 0 - 6.0                  |
| 1878          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1879          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1880          |                 | 15 above      | 81                  | Kanuga           | 0 - 6.0                  |
| 1881          |                 | 15 above      | 86                  | Kanuga           | 0 - 6.0                  |
| 1882          |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 1883          |                 | 15 above      | 74                  | Bhagi            | 0 - 6.0                  |
| 1884          |                 | 15 above      | 81                  | Bhagi            | 0 - 6.0                  |
| 1885          |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 1886          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1887          | 28+800          | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |

#### APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 1888   |          | 15 above | 73           | Bhagi     | 0 - 6.0           |
| 1889   |          | 15 above | 85           | Bhagi     | 0 - 6.0           |
| 1890   |          | 15 above | 74           | Bhagi     | 0 - 6.0           |
| 1891   |          | 20 above | 80           | Bhagi     | 0 - 6.0           |
| 1892   |          | 20 above | 59           | Coconut   | 0 - 6.0           |
| 1893   |          | 20 above | 55           | Coconut   | 0 - 6.0           |
| 1894   |          | 20 above | 133          | Coconut   | 0 - 6.0           |
| 1895   |          | 20 above | 133          | Coconut   | 0 - 6.0           |
| 1896   |          | 20 above | 133          | Coconut   | 0 - 6.0           |
| 1897   |          | 20 above | 156          | Coconut   | 0 - 6.0           |
| 1898   |          | 20 above | 156          | Coconut   | 0 - 6.0           |
| 1899   | 29+000   | 20 above | 68           | Tamarind  | 0 - 6.0           |
| 1900   |          | 20 above | 64           | Tamarind  | 0 - 6.0           |
| 1901   |          | 20 above | 63           | Coconut   | 0 - 6.0           |
| 1902   |          | 20 above | 76           | Coconut   | 0 - 6.0           |
| 1903   |          | 20 above | 73           | Coconut   | 0 - 6.0           |
| 1904   |          | 20 above | 62           | Coconut   | 0 - 6.0           |
| 1905   |          | 20 above | 68           | Coconut   | 0 - 6.0           |
| 1906   |          | 20 above | 70           | Coconut   | 0 - 6.0           |
| 1907   |          | 20 above | 156          | Coconut   | 0 - 6.0           |
| 1908   |          | 20 above | 156          | Coconut   | 0 - 6.0           |
| 1909   |          | 15 above | 82           | Coconut   | 0 - 6.0           |
| 1910   |          | 15 above | 86           | Coconut   | 0 - 6.0           |
| 1911   |          | 15 above | 85           | Coconut   | 0 - 6.0           |
| 1912   | 29+400   | 10       | 86           | Coconut   | 0 - 6.0           |
| 1913   |          | 20 above | 79           | Bhagi     | 0 - 6.0           |
| 1914   |          | 15 above | 81           | Coconut   | 0 - 6.0           |
| 1915   |          | 15 above | 73           | Coconut   | 0 - 6.0           |
| 1916   |          | 15 above | 84           | Coconut   | 0 - 6.0           |
| 1917   |          | 15 above | 86           | Coconut   | 0 - 6.0           |
| 1918   |          | 15 above | 76           | Coconut   | 0 - 6.0           |
| 1919   |          | 15 above | 81           | Coconut   | 0 - 6.0           |
| 1920   |          | 15 above | 86           | Coconut   | 0 - 6.0           |
| 1921   |          | 15 above | 73           | Coconut   | 0 - 6.0           |
| 1922   |          | 15 above | 74           | Coconut   | 0 - 6.0           |
| 1923   | 29+800   | 15 above | 81           | Coconut   | 0 - 6.0           |
| 1924   |          | 15 above | 86           | Coconut   | 0 - 6.0           |
| 1925   |          | 15 above | 76           | Coconut   | 0 - 6.0           |
| 1926   |          | 15 above | 76           | Coconut   | 0 - 6.0           |
| 1927   |          | 15 above | 82           | Coconut   | 0 - 6.0           |
| 1928   |          | 15 above | 73           | Coconut   | 0 - 6.0           |
| 1929   |          | 15 above | 84           | Coconut   | 0 - 6.0           |
| 1930   |          | 15 above | 75           | Coconut   | 0 - 6.0           |
| 1931   |          | 15 above | 66           | Coconut   | 0 - 6.0           |
| 1932   |          | 15 above | 69           | Coconut   | 0 - 6.0           |
| 1933   |          | 15 above | 185          | Coconut   | 0 - 6.0           |
| 1934   |          | 15 above | 123          | Coconut   | 0 - 6.0           |
| 1935   |          | 15 above | 152          | Coconut   | 0 - 6.0           |
| 1936   |          | 15 above | 165          | Coconut   | 0 - 6.0           |
| 1937   |          | 15 above | 142          | Coconut   | 0 - 6.0           |
| 1938   | 30+200   | 15 above | 135          | Coconut   | 0 - 6.0           |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1939          |                 | 15 above      | 162                 | Coconut          | 0 - 6.0                  |
| 1940          |                 | 15 above      | 154                 | Coconut          | 0 - 6.0                  |
| 1941          |                 | 15 above      | 125                 | Thati            | 0 - 6.0                  |
| 1942          |                 | 20 above      | 74                  | Bhagi            | 0 - 6.0                  |
| 1943          |                 | 20 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1944          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1945          |                 | 20 above      | 89                  | Coconut          | 0 - 6.0                  |
| 1946          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1947          |                 | 20 above      | 68                  | Coconut          | 0 - 6.0                  |
| 1948          |                 | 15 above      | 64                  | Coconut          | 0 - 6.0                  |
| 1949          |                 | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 1950          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 1951          |                 | 15 above      | 173                 | Coconut          | 0 - 6.0                  |
| 1952          |                 | 15 above      | 162                 | Coconut          | 0 - 6.0                  |
| 1953          |                 | 20 above      | 86                  | Coconut          | 0 - 6.0                  |
| 1954          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1955          |                 | 20 above      | 79                  | Coconut          | 0 - 6.0                  |
| 1956          |                 | 20 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1957          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1958          |                 | 20 above      | 78                  | Coconut          | 0 - 6.0                  |
| 1959          |                 | 20 above      | 72                  | Coconut          | 0 - 6.0                  |
| 1960          |                 | 20 above      | 84                  | Coconut          | 0 - 6.0                  |
| 1961          | 31+200          | 20 above      | 51                  | Bhagi            | 0 - 6.0                  |
| 1962          |                 | 20 above      | 53                  | Bhagi            | 0 - 6.0                  |
| 1963          |                 | 20 above      | 50                  | Bhagi            | 0 - 6.0                  |
| 1964          |                 | 20 above      | 152                 | Bhagi            | 0 - 6.0                  |
| 1965          |                 | 20 above      | 143                 | Bhagi            | 0 - 6.0                  |
| 1966          |                 | 20 above      | 165                 | Bhagi            | 0 - 6.0                  |
| 1967          |                 | 20 above      | 169                 | Bhagi            | 0 - 6.0                  |
| 1968          |                 | 15 above      | 190                 | Bhagi            | 0 - 6.0                  |
| 1969          |                 | 15 above      | 197                 | Coconut          | 0 - 6.0                  |
| 1970          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 1971          |                 | 15 above      | 74                  | Coconut          | 0 - 6.0                  |
| 1972          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 1973          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1974          |                 | 15 above      | 73                  | Neem             | 0 - 6.0                  |
| 1975          |                 | 20 above      | 86                  | Neem             | 0 - 6.0                  |
| 1976          |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1977          |                 | 20 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 1978          |                 | 20 above      | 57                  | Bhagi            | 0 - 6.0                  |
| 1979          |                 | 20 above      | 59                  | Bhagi            | 0 - 6.0                  |
| 1980          |                 | 20 above      | 190                 | Bhagi            | 0 - 6.0                  |
| 1981          |                 | 20 above      | 199                 | Bhagi            | 0 - 6.0                  |
| 1982          |                 | 20 above      | 162                 | Bhagi            | 0 - 6.0                  |
| 1983          | 31+600          | 10            | 143                 | Bhagi            | 0 - 6.0                  |
| 1984          |                 | 10            | 54                  | Bhagi            | 0 - 6.0                  |
| 1985          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 1986          |                 | 10            | 59                  | Bhagi            | 0 - 6.0                  |
| 1987          | 31+800          | 10            | 76                  | Coconut          | 0 - 6.0                  |
| 1988          |                 | 10            | 82                  | Coconut          | 0 - 6.0                  |
| 1989          |                 | 10            | 73                  | Coconut          | 0 - 6.0                  |



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 1990          |                 | 10            | 185                 | Coconut          | 0 - 6.0                  |
| 1991          |                 | 10            | 165                 | Coconut          | 0 - 6.0                  |
| 1992          |                 | 10            | 86                  | Coconut          | 0 - 6.0                  |
| 1993          |                 | 10            | 75                  | Coconut          | 0 - 6.0                  |
| 1994          |                 | 20 above      | 70                  | Coconut          | 0 - 6.0                  |
| 1995          |                 | 20 above      | 76                  | Coconut          | 0 - 6.0                  |
| 1996          |                 | 20 above      | 71                  | Coconut          | 0 - 6.0                  |
| 1997          |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 1998          |                 | 20 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 1999          |                 | 15 above      | 68                  | Bhagi            | 0 - 6.0                  |
| 2000          |                 | 15 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 2001          |                 | 15 above      | 67                  | Bhagi            | 0 - 6.0                  |
| 2002          |                 | 15 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 2003          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 2004          |                 | 15 above      | 82                  | Marri            | 0 - 6.0                  |
| 2005          |                 | 15 above      | 80                  | Marri            | 0 - 6.0                  |
| 2006          |                 | 15 above      | 85                  | Marri            | 0 - 6.0                  |
| 2007          |                 | 15 above      | 78                  | Marri            | 0 - 6.0                  |
| 2008          |                 | 15 above      | 73                  | Marri            | 0 - 6.0                  |
| 2009          | 32+000          | 15 above      | 70                  | Kanuga           | 0 - 6.0                  |
| 2010          |                 | 15 above      | 85                  | Kanuga           | 0 - 6.0                  |
| 2011          |                 | 15 above      | 81                  | Bhagi            | 0 - 6.0                  |
| 2012          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 2013          |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 2014          |                 | 15 above      | 61                  | Bhagi            | 0 - 6.0                  |
| 2015          |                 | 15 above      | 86                  | Bhagi            | 0 - 6.0                  |
| 2016          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 2017          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 2018          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2019          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2020          |                 | 15 above      | 78                  | Coconut          | 0 - 6.0                  |
| 2021          |                 | 15 above      | 71                  | Thati            | 0 - 6.0                  |
| 2022          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 2023          |                 | 15 above      | 89                  | Bhagi            | 0 - 6.0                  |
| 2024          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 2025          |                 | 15 above      | 82                  | Bhagi            | 0 - 6.0                  |
| 2026          |                 | 15 above      | 65                  | Bhagi            | 0 - 6.0                  |
| 2027          |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 2028          |                 | 15 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 2029          | 32+400          | 10            | 39                  | Kanuga           | 0 - 6.0                  |
| 2030          |                 | 10            | 37                  | Kanuga           | 0 - 6.0                  |
| 2031          |                 | 10            | 146                 | Thati            | 0 - 6.0                  |
| 2032          |                 | 10            | 185                 | Thati            | 0 - 6.0                  |
| 2033          |                 | 10            | 124                 | Coconut          | 0 - 6.0                  |
| 2034          |                 | 10            | 136                 | Coconut          | 0 - 6.0                  |
| 2035          |                 | 10            | 152                 | Coconut          | 0 - 6.0                  |
| 2036          |                 | 20 above      | 175                 | Coconut          | 0 - 6.0                  |
| 2037          |                 | 15 above      | 188                 | Coconut          | 0 - 6.0                  |
| 2038          |                 | 15 above      | 154                 | Coconut          | 0 - 6.0                  |
| 2039          |                 | 15 above      | 67                  | Coconut          | 0 - 6.0                  |
| 2040          |                 | 15 above      | 65                  | Coconut          | 0 - 6.0                  |

# APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 2041   |          | 15 above | 84           | Bhagi     | 0 - 6.0           |
| 2042   |          | 15 above | 82           | Bhagi     | 0 - 6.0           |
| 2043   |          | 10       | 35           | Coconut   | 0 - 6.0           |
| 2044   |          | 10       | 120          | Coconut   | 0 - 6.0           |
| 2045   |          | 20 above | 142          | Coconut   | 0 - 6.0           |
| 2046   |          | 20 above | 153          | Coconut   | 0 - 6.0           |
| 2047   |          | 15 above | 142          | Coconut   | 0 - 6.0           |
| 2048   |          | 15 above | 72           | Coconut   | 0 - 6.0           |
| 2049   |          | 15 above | 80           | Bhagi     | 0 - 6.0           |
| 2050   |          | 15 above | 84           | Bhagi     | 0 - 6.0           |
| 2051   |          | 15 above | 80           | Bhagi     | 0 - 6.0           |
| 2052   |          | 15 above | 76           | Bhagi     | 0 - 6.0           |
| 2053   |          | 15 above | 89           | Bhagi     | 0 - 6.0           |
| 2054   |          | 15 above | 84           | Bhagi     | 0 - 6.0           |
| 2055   |          | 15 above | 76           | Bhagi     | 0 - 6.0           |
| 2056   |          | 15 above | 73           | Bhagi     | 0 - 6.0           |
| 2057   |          | 15 above | 62           | Bhagi     | 0 - 6.0           |
| 2058   | 32+800   | 10       | 34           | Coconut   | 0 - 6.0           |
| 2059   |          | 10       | 38           | Coconut   | 0 - 6.0           |
| 2060   |          | 10       | 32           | Coconut   | 0 - 6.0           |
| 2061   |          | 10       | 35           | Coconut   | 0 - 6.0           |
| 2062   |          | 20 above | 38           | Coconut   | 0 - 6.0           |
| 2063   |          | 20 above | 45           | Coconut   | 0 - 6.0           |
| 2064   |          | 20 above | 146          | Coconut   | 0 - 6.0           |
| 2065   |          | 20 above | 185          | Coconut   | 0 - 6.0           |
| 2066   |          | 20 above | 135          | Thati     | 0 - 6.0           |
| 2067   |          | 20 above | 142          | Thati     | 0 - 6.0           |
| 2068   |          | 15 above | 75           | Tamarind  | 0 - 6.0           |
| 2069   |          | 15 above | 70           | Tamarind  | 0 - 6.0           |
| 2070   |          | 15 above | 76           | Tamarind  | 0 - 6.0           |
| 2071   |          | 15 above | 84           | Bhagi     | 0 - 6.0           |
| 2072   |          | 15 above | 82           | Bhagi     | 0 - 6.0           |
| 2073   |          | 15 above | 80           | Bhagi     | 0 - 6.0           |
| 2074   |          | 15 above | 70           | Bhagi     | 0 - 6.0           |
| 2075   |          | 15 above | 76           | Bhagi     | 0 - 6.0           |
| 2076   |          | 15 above | 81           | Bhagi     | 0 - 6.0           |
| 2077   |          | 15 above | 86           | Bhagi     | 0 - 6.0           |
| 2078   |          | 15 above | 73           | Bhagi     | 0 - 6.0           |
| 2079   |          | 15 above | 74           | Bhagi     | 0 - 6.0           |
| 2080   |          | 15 above | 81           | Bhagi     | 0 - 6.0           |
| 2081   |          | 15 above | 86           | Bhagi     | 0 - 6.0           |
| 2082   |          | 15 above | 76           | Bhagi     | 0 - 6.0           |
| 2083   |          | 10       | 84           | Bhagi     | 0 - 6.0           |
| 2084   |          | 20 above | 76           | Bhagi     | 0 - 6.0           |
| 2085   |          | 20 above | 70           | Bhagi     | 0 - 6.0           |
| 2086   |          | 20 above | 80           | Bhagi     | 0 - 6.0           |
| 2087   |          | 15 above | 85           | Bhagi     | 0 - 6.0           |
| 2088   |          | 15 above | 79           | Bhagi     | 0 - 6.0           |
| 2089   |          | 15 above | 68           | Bhagi     | 0 - 6.0           |
| 2090   |          | 15 above | 64           | Bhagi     | 0 - 6.0           |
| 2091   |          | 15 above | 63           | Bhagi     | 0 - 6.0           |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2092          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 2093          |                 | 15 above      | 46                  | Bhagi            | 0 - 6.0                  |
| 2094          |                 | 15 above      | 45                  | Bhagi            | 0 - 6.0                  |
| 2095          |                 | 15 above      | 49                  | Bhagi            | 0 - 6.0                  |
| 2096          | 33+200          | 10            | 48                  | Ashoka           | 0 - 6.0                  |
| 2097          |                 | 10            | 44                  | Ashoka           | 0 - 6.0                  |
| 2098          |                 | 10            | 44                  | Ashoka           | 0 - 6.0                  |
| 2099          |                 | 15 above      | 44                  | Ashoka           | 0 - 6.0                  |
| 2100          |                 | 15 above      | 110                 | Ashoka           | 0 - 6.0                  |
| 2101          |                 | 15 above      | 185                 | Ashoka           | 0 - 6.0                  |
| 2102          |                 | 15 above      | 165                 | Ashoka           | 0 - 6.0                  |
| 2103          |                 | 15 above      | 144                 | Ashoka           | 0 - 6.0                  |
| 2104          |                 | 15 above      | 144                 | Ashoka           | 0 - 6.0                  |
| 2105          |                 | 15 above      | 144                 | Bhagi            | 0 - 6.0                  |
| 2106          |                 | 15 above      | 155                 | Bhagi            | 0 - 6.0                  |
| 2107          |                 | 15 above      | 146                 | Bhagi            | 0 - 6.0                  |
| 2108          |                 | 20 above      | 144                 | Bhagi            | 0 - 6.0                  |
| 2109          |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 2110          |                 | 20 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 2111          |                 | 20 above      | 68                  | Bhagi            | 0 - 6.0                  |
| 2112          |                 | 20 above      | 62                  | Bhagi            | 0 - 6.0                  |
| 2113          |                 | 20 above      | 67                  | Bhagi            | 0 - 6.0                  |
| 2114          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2115          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 2116          |                 | 15 above      | 68                  | Bhagi            | 0 - 6.0                  |
| 2117          |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 2118          |                 | 15 above      | 63                  | Bhagi            | 0 - 6.0                  |
| 2119          |                 | 20 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 2120          |                 | 20 above      | 49                  | Bhagi            | 0 - 6.0                  |
| 2121          |                 | 20 above      | 48                  | Bhagi            | 0 - 6.0                  |
| 2122          |                 | 20 above      | 56                  | Bhagi            | 0 - 6.0                  |
| 2123          |                 | 20 above      | 144                 | Bhagi            | 0 - 6.0                  |
| 2124          | 33+600          | 10            | 57                  | Bhagi            | 0 - 6.0                  |
| 2125          |                 | 10            | 55                  | Bhagi            | 0 - 6.0                  |
| 2126          |                 | 10            | 144                 | Bhagi            | 0 - 6.0                  |
| 2127          |                 | 10            | 82                  | Bhagi            | 0 - 6.0                  |
| 2128          |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 2129          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 2130          |                 | 20 above      | 75                  | Bhagi            | 0 - 6.0                  |
| 2131          | 33+800          | 10            | 86                  | Bhagi            | 0 - 6.0                  |
| 2132          |                 | 10            | 75                  | Bhagi            | 0 - 6.0                  |
| 2133          |                 | 10            | 70                  | Bhagi            | 0 - 6.0                  |
| 2134          |                 | 10            | 55                  | Bhagi            | 0 - 6.0                  |
| 2135          |                 | 10            | 50                  | Bhagi            | 0 - 6.0                  |
| 2136          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 2137          |                 | 10            | 55                  | Bhagi            | 0 - 6.0                  |
| 2138          |                 | 10            | 53                  | Bhagi            | 0 - 6.0                  |
| 2139          | 34+000          | 10            | 51                  | Neem             | 0 - 6.0                  |
| 2140          |                 | 10            | 59                  | Bhagi            | 0 - 6.0                  |
| 2141          |                 | 10            | 54                  | Marri            | 0 - 6.0                  |
| 2142          |                 | 10            | 59                  | Giriship         | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2143          |                 | 10            | 58                  | Giriship         | 0 - 6.0                  |
| 2144          |                 | 10            | 54                  | Giriship         | 0 - 6.0                  |
| 2145          |                 | 10            | 52                  | Giriship         | 0 - 6.0                  |
| 2146          |                 | 10            | 53                  | Giriship         | 0 - 6.0                  |
| 2147          |                 | 10            | 51                  | Giriship         | 0 - 6.0                  |
| 2148          |                 | 10            | 58                  | Giriship         | 0 - 6.0                  |
| 2149          |                 | 15 above      | 144                 | Giriship         | 0 - 6.0                  |
| 2150          | 34+200          | 10            | 57                  | Giriship         | 0 - 6.0                  |
| 2151          |                 | 10            | 59                  | Giriship         | 0 - 6.0                  |
| 2152          |                 | 10            | 51                  | Giriship         | 0 - 6.0                  |
| 2153          |                 | 10            | 56                  | Giriship         | 0 - 6.0                  |
| 2154          |                 | 10            | 51                  | Giriship         | 0 - 6.0                  |
| 2155          |                 | 10            | 55                  | Giriship         | 0 - 6.0                  |
| 2156          | 34+400          | 15 above      | 59                  | Bhagi            | 0 - 6.0                  |
| 2157          |                 | 15 above      | 57                  | Giriship         | 0 - 6.0                  |
| 2158          |                 | 15 above      | 51                  | Giriship         | 0 - 6.0                  |
| 2159          |                 | 15 above      | 50                  | Giriship         | 0 - 6.0                  |
| 2160          |                 | 15 above      | 58                  | Giriship         | 0 - 6.0                  |
| 2161          |                 | 15 above      | 54                  | Giriship         | 0 - 6.0                  |
| 2162          |                 | 15 above      | 59                  | Giriship         | 0 - 6.0                  |
| 2163          |                 | 15 above      | 58                  | Neem             | 0 - 6.0                  |
| 2164          |                 | 15 above      | 54                  | Coconut          | 0 - 6.0                  |
| 2165          |                 | 15 above      | 56                  | Coconut          | 0 - 6.0                  |
| 2166          |                 | 15 above      | 51                  | Coconut          | 0 - 6.0                  |
| 2167          |                 | 15 above      | 58                  | Coconut          | 0 - 6.0                  |
| 2168          |                 | 15 above      | 55                  | Coconut          | 0 - 6.0                  |
| 2169          |                 | 15 above      | 54                  | Coconut          | 0 - 6.0                  |
| 2170          |                 | 15 above      | 39                  | Coconut          | 0 - 6.0                  |
| 2171          |                 | 15 above      | 34                  | Coconut          | 0 - 6.0                  |
| 2172          |                 | 15 above      | 38                  | Coconut          | 0 - 6.0                  |
| 2173          |                 | 15 above      | 39                  | Coconut          | 0 - 6.0                  |
| 2174          |                 | 15 above      | 34                  | Coconut          | 0 - 6.0                  |
| 2175          |                 | 15 above      | 144                 | Coconut          | 0 - 6.0                  |
| 2176          | 34+600          | 15 above      | 33                  | Coconut          | 0 - 6.0                  |
| 2177          |                 | 15 above      | 38                  | Coconut          | 0 - 6.0                  |
| 2178          |                 | 15 above      | 34                  | Coconut          | 0 - 6.0                  |
| 2179          |                 | 10            | 39                  | Giriship         | 0 - 6.0                  |
| 2180          | 34+800          | 10            | 36                  | Giriship         | 0 - 6.0                  |
| 2181          |                 | 10            | 31                  | Giriship         | 0 - 6.0                  |
| 2182          |                 | 10            | 38                  | Giriship         | 0 - 6.0                  |
| 2183          |                 | 15 above      | 144                 | Bhagi            | 0 - 6.0                  |
| 2184          |                 | 15 above      | 152                 | Neem             | 0 - 6.0                  |
| 2185          | 35+000          | 10            | 136                 | Giriship         | 0 - 6.0                  |
| 2186          |                 | 10            | 185                 | Giriship         | 0 - 6.0                  |
| 2187          |                 | 10            | 31                  | Giriship         | 0 - 6.0                  |
| 2188          |                 | 10            | 39                  | Giriship         | 0 - 6.0                  |
| 2189          |                 | 10            | 31                  | Giriship         | 0 - 6.0                  |
| 2190          |                 | 10            | 35                  | Giriship         | 0 - 6.0                  |
| 2191          |                 | 10            | 36                  | Giriship         | 0 - 6.0                  |
| 2192          |                 | 10            | 35                  | Giriship         | 0 - 6.0                  |
| 2193          |                 | 10            | 35                  | Giriship         | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2194          |                 | 10            | 38                  | Giriship         | 0 - 6.0                  |
| 2195          |                 | 15 above      | 39                  | Giriship         | 0 - 6.0                  |
| 2196          |                 | 15 above      | 39                  | Giriship         | 0 - 6.0                  |
| 2197          |                 | 15 above      | 34                  | Giriship         | 0 - 6.0                  |
| 2198          |                 | 15 above      | 38                  | Giriship         | 0 - 6.0                  |
| 2199          |                 | 15 above      | 46                  | Giriship         | 0 - 6.0                  |
| 2200          | 35+200          | 15 above      | 164                 | Bhagi            | 0 - 6.0                  |
| 2201          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2202          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2203          |                 | 15 above      | 136                 | Neem             | 0 - 6.0                  |
| 2204          |                 | 10            | 86                  | Giriship         | 0 - 6.0                  |
| 2205          |                 | 10            | 76                  | Giriship         | 0 - 6.0                  |
| 2206          |                 | 10            | 84                  | Giriship         | 0 - 6.0                  |
| 2207          |                 | 10            | 73                  | Giriship         | 0 - 6.0                  |
| 2208          |                 | 10            | 85                  | Giriship         | 0 - 6.0                  |
| 2209          |                 | 10            | 74                  | Giriship         | 0 - 6.0                  |
| 2210          |                 | 10            | 80                  | Giriship         | 0 - 6.0                  |
| 2211          |                 | 10            | 76                  | Marri            | 0 - 6.0                  |
| 2212          |                 | 10            | 89                  | Coconut          | 0 - 6.0                  |
| 2213          | 35+400          | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 2214          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 2215          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 2216          |                 | 15 above      | 80                  | Neem             | 0 - 6.0                  |
| 2217          |                 | 15 above      | 85                  | Neem             | 0 - 6.0                  |
| 2218          |                 | 15 above      | 79                  | Kanuga           | 0 - 6.0                  |
| 2219          |                 | 15 above      | 68                  | Coconut          | 0 - 6.0                  |
| 2220          |                 | 15 above      | 64                  | Coconut          | 0 - 6.0                  |
| 2221          |                 | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 2222          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2223          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2224          |                 | 15 above      | 62                  | Coconut          | 0 - 6.0                  |
| 2225          |                 | 15 above      | 68                  | Coconut          | 0 - 6.0                  |
| 2226          |                 | 15 above      | 115                 | Coconut          | 0 - 6.0                  |
| 2227          |                 | 15 above      | 116                 | Coconut          | 0 - 6.0                  |
| 2228          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2229          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2230          |                 | 15 above      | 145                 | Coconut          | 0 - 6.0                  |
| 2231          |                 | 15 above      | 152                 | Coconut          | 0 - 6.0                  |
| 2232          |                 | 15 above      | 185                 | Coconut          | 0 - 6.0                  |
| 2233          |                 | 15 above      | 146                 | Coconut          | 0 - 6.0                  |
| 2234          |                 | 15 above      | 125                 | Giriship         | 0 - 6.0                  |
| 2235          |                 | 15 above      | 136                 | Giriship         | 0 - 6.0                  |
| 2236          |                 | 15 above      | 185                 | Giriship         | 0 - 6.0                  |
| 2237          |                 | 15 above      | 156                 | Giriship         | 0 - 6.0                  |
| 2238          |                 | 15 above      | 175                 | Giriship         | 0 - 6.0                  |
| 2239          |                 | 15 above      | 165                 | Giriship         | 0 - 6.0                  |
| 2240          |                 | 15 above      | 142                 | Giriship         | 0 - 6.0                  |
| 2241          |                 | 15 above      | 136                 | Giriship         | 0 - 6.0                  |
| 2242          |                 | 15 above      | 152                 | Giriship         | 0 - 6.0                  |
| 2243          |                 | 15 above      | 132                 | Giriship         | 0 - 6.0                  |
| 2244          |                 | 15 above      | 152                 | Giriship         | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2245          |                 | 15 above      | 144                 | Giriship         | 0 - 6.0                  |
| 2246          |                 | 15 above      | 133                 | Giriship         | 0 - 6.0                  |
| 2247          |                 | 15 above      | 165                 | Giriship         | 0 - 6.0                  |
| 2248          |                 | 15 above      | 122                 | Giriship         | 0 - 6.0                  |
| 2249          |                 | 15 above      | 136                 | Giriship         | 0 - 6.0                  |
| 2250          |                 | 15 above      | 115                 | Giriship         | 0 - 6.0                  |
| 2251          | 35+600          | 10            | 165                 | Giriship         | 0 - 6.0                  |
| 2252          |                 | 10            | 142                 | Giriship         | 0 - 6.0                  |
| 2253          |                 | 10            | 136                 | Giriship         | 0 - 6.0                  |
| 2254          |                 | 10            | 152                 | Giriship         | 0 - 6.0                  |
| 2255          |                 | 10            | 132                 | Giriship         | 0 - 6.0                  |
| 2256          |                 | 10            | 152                 | Giriship         | 0 - 6.0                  |
| 2257          |                 | 10            | 144                 | Giriship         | 0 - 6.0                  |
| 2258          |                 | 10            | 133                 | Giriship         | 0 - 6.0                  |
| 2259          |                 | 10            | 165                 | Giriship         | 0 - 6.0                  |
| 2260          |                 | 10            | 122                 | Giriship         | 0 - 6.0                  |
| 2261          |                 | 10            | 135                 | Bhagi            | 0 - 6.0                  |
| 2262          |                 | 10            | 114                 | Bhagi            | 0 - 6.0                  |
| 2263          |                 | 10            | 165                 | Bhagi            | 0 - 6.0                  |
| 2264          |                 | 10            | 185                 | Bhagi            | 0 - 6.0                  |
| 2265          |                 | 10            | 145                 | Thurai           | 0 - 6.0                  |
| 2266          |                 | 10            | 146                 | Thurai           | 0 - 6.0                  |
| 2267          |                 | 10            | 136                 | Thurai           | 0 - 6.0                  |
| 2268          |                 | 15 above      | 70                  | Bhagi            | 0 - 6.0                  |
| 2269          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 2270          |                 | 15 above      | 73                  | Bhagi            | 0 - 6.0                  |
| 2271          |                 | 15 above      | 75                  | Giriship         | 0 - 6.0                  |
| 2272          |                 | 15 above      | 71                  | Giriship         | 0 - 6.0                  |
| 2273          |                 | 15 above      | 76                  | Giriship         | 0 - 6.0                  |
| 2274          |                 | 15 above      | 65                  | Panasa           | 0 - 6.0                  |
| 2275          |                 | 15 above      | 85                  | Thurai           | 0 - 6.0                  |
| 2276          |                 | 15 above      | 58                  | Thurai           | 0 - 6.0                  |
| 2277          |                 | 15 above      | 48                  | Thurai           | 0 - 6.0                  |
| 2278          |                 | 15 above      | 41                  | Thurai           | 0 - 6.0                  |
| 2279          |                 | 15 above      | 92                  | Thurai           | 0 - 6.0                  |
| 2280          |                 | 15 above      | 99                  | Thurai           | 0 - 6.0                  |
| 2281          |                 | 15 above      | 94                  | Thurai           | 0 - 6.0                  |
| 2282          |                 | 15 above      | 95                  | Thurai           | 0 - 6.0                  |
| 2283          |                 | 15 above      | 99                  | Thurai           | 0 - 6.0                  |
| 2284          |                 | 15 above      | 102                 | Thurai           | 0 - 6.0                  |
| 2285          |                 | 15 above      | 105                 | Thurai           | 0 - 6.0                  |
| 2286          |                 | 15 above      | 184                 | Thurai           | 0 - 6.0                  |
| 2287          |                 | 15 above      | 136                 | Thurai           | 0 - 6.0                  |
| 2288          | 36+000          | 15 above      | 152                 | Coconut          | 0 - 6.0                  |
| 2289          |                 | 15 above      | 175                 | Coconut          | 0 - 6.0                  |
| 2290          |                 | 15 above      | 162                 | Coconut          | 0 - 6.0                  |
| 2291          |                 | 15 above      | 135                 | Coconut          | 0 - 6.0                  |
| 2292          |                 | 15 above      | 142                 | Coconut          | 0 - 6.0                  |
| 2293          | 36+400          | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 2294          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2295          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2296          |                 | 15 above      | 75                  | Coconut          | 0 - 6.0                  |
| 2297          |                 | 15 above      | 86                  | Palm             | 0 - 6.0                  |
| 2298          |                 | 15 above      | 84                  | Palm             | 0 - 6.0                  |
| 2299          |                 | 15 above      | 82                  | Palm             | 0 - 6.0                  |
| 2300          |                 | 15 above      | 76                  | Thurai           | 0 - 6.0                  |
| 2301          | 36+600          | 15 above      | 46                  | Coconut          | 0 - 6.0                  |
| 2302          |                 | 15 above      | 46                  | Coconut          | 0 - 6.0                  |
| 2303          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |
| 2304          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 2305          |                 | 15 above      | 48                  | Coconut          | 0 - 6.0                  |
| 2306          |                 | 15 above      | 42                  | Coconut          | 0 - 6.0                  |
| 2307          |                 | 15 above      | 43                  | Coconut          | 0 - 6.0                  |
| 2308          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 2309          |                 | 15 above      | 47                  | Coconut          | 0 - 6.0                  |
| 2310          |                 | 15 above      | 43                  | Coconut          | 0 - 6.0                  |
| 2311          |                 | 15 above      | 89                  | Coconut          | 0 - 6.0                  |
| 2312          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 2313          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 2314          |                 | 15 above      | 62                  | Coconut          | 0 - 6.0                  |
| 2315          |                 | 15 above      | 69                  | Coconut          | 0 - 6.0                  |
| 2316          | 36+800          | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 2317          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2318          |                 | 15 above      | 71                  | Coconut          | 0 - 6.0                  |
| 2319          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2320          |                 | 15 above      | 69                  | Coconut          | 0 - 6.0                  |
| 2321          | 37+000          | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2322          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 2323          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 2324          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2325          |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |
| 2326          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2327          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2328          |                 | 15 above      | 74                  | Coconut          | 0 - 6.0                  |
| 2329          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2330          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2331          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2332          |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |
| 2333          |                 | 15 above      | 66                  | Coconut          | 0 - 6.0                  |
| 2334          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2335          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 2336          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 2337          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2338          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2339          |                 | 15 above      | 78                  | Coconut          | 0 - 6.0                  |
| 2340          |                 | 15 above      | 62                  | Coconut          | 0 - 6.0                  |
| 2341          |                 | 15 above      | 65                  | Palm             | 0 - 6.0                  |
| 2342          |                 | 15 above      | 66                  | Palm             | 0 - 6.0                  |
| 2343          | 37+200          | 10            | 79                  | Coconut          | 0 - 6.0                  |
| 2344          |                 | 10            | 70                  | Coconut          | 0 - 6.0                  |
| 2345          |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |
| 2346          |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2347          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2348          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2349          |                 | 15 above      | 152                 | Coconut          | 0 - 6.0                  |
| 2350          |                 | 15 above      | 144                 | Coconut          | 0 - 6.0                  |
| 2351          |                 | 15 above      | 165                 | Coconut          | 0 - 6.0                  |
| 2352          |                 | 15 above      | 185                 | Coconut          | 0 - 6.0                  |
| 2353          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2354          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2355          |                 | 15 above      | 142                 | Coconut          | 0 - 6.0                  |
| 2356          |                 | 15 above      | 177                 | Coconut          | 0 - 6.0                  |
| 2357          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2358          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2359          |                 | 15 above      | 142                 | Coconut          | 0 - 6.0                  |
| 2360          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2361          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2362          |                 | 15 above      | 152                 | Coconut          | 0 - 6.0                  |
| 2363          | 37+400          | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 2364          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 2365          |                 | 15 above      | 48                  | Coconut          | 0 - 6.0                  |
| 2366          |                 | 15 above      | 46                  | Coconut          | 0 - 6.0                  |
| 2367          |                 | 15 above      | 41                  | Coconut          | 0 - 6.0                  |
| 2368          |                 | 15 above      | 40                  | Coconut          | 0 - 6.0                  |
| 2369          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |
| 2370          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 2371          |                 | 15 above      | 40                  | Coconut          | 0 - 6.0                  |
| 2372          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 2373          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 2374          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2375          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2376          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 2377          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 2378          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2379          | 37+600          | 10            | 84                  | Kanuga           | 0 - 6.0                  |
| 2380          |                 | 10            | 80                  | Thurai           | 0 - 6.0                  |
| 2381          |                 | 10            | 116                 | Bhagi            | 0 - 6.0                  |
| 2382          | 37+800          | 10            | 70                  | Coconut          | 0 - 6.0                  |
| 2383          |                 | 10            | 76                  | Kanuga           | 0 - 6.0                  |
| 2384          |                 | 10            | 71                  | Kanuga           | 0 - 6.0                  |
| 2385          |                 | 10            | 71                  | Kanuga           | 0 - 6.0                  |
| 2386          |                 | 10            | 76                  | Bhagi            | 0 - 6.0                  |
| 2387          | 38+000          | 10            | 47                  | Kanuga           | 0 - 6.0                  |
| 2388          |                 | 10            | 49                  | Kanuga           | 0 - 6.0                  |
| 2389          |                 | 10            | 73                  | Kanuga           | 0 - 6.0                  |
| 2390          |                 | 10            | 70                  | Thurai           | 0 - 6.0                  |
| 2391          |                 | 10            | 85                  | Thurai           | 0 - 6.0                  |
| 2392          |                 | 15 above      | 81                  | Thurai           | 0 - 6.0                  |
| 2393          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 2394          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2395          |                 | 15 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 2396          |                 | 15 above      | 113                 | Bhagi            | 0 - 6.0                  |
| 2397          |                 | 10            | 142                 | Coconut          | 0 - 6.0                  |



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2398          |                 | 10            | 136                 | Coconut          | 0 - 6.0                  |
| 2399          |                 | 15 above      | 125                 | Giriship         | 0 - 6.0                  |
| 2400          |                 | 15 above      | 185                 | Palm             | 0 - 6.0                  |
| 2401          |                 | 15 above      | 165                 | Palm             | 0 - 6.0                  |
| 2402          |                 | 15 above      | 195                 | Kanuga           | 0 - 6.0                  |
| 2403          |                 | 15 above      | 175                 | Kanuga           | 0 - 6.0                  |
| 2404          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 2405          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 2406          |                 | 15 above      | 79                  | Giriship         | 0 - 6.0                  |
| 2407          | 38+400          | 15 above      | 68                  | Giriship         | 0 - 6.0                  |
| 2408          |                 | 15 above      | 64                  | Giriship         | 0 - 6.0                  |
| 2409          |                 | 15 above      | 63                  | Kanuga           | 0 - 6.0                  |
| 2410          |                 | 15 above      | 76                  | Kanuga           | 0 - 6.0                  |
| 2411          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2412          |                 | 15 above      | 62                  | Thurai           | 0 - 6.0                  |
| 2413          |                 | 15 above      | 68                  | Thurai           | 0 - 6.0                  |
| 2414          |                 | 10            | 70                  | Thurai           | 0 - 6.0                  |
| 2415          |                 | 10            | 79                  | Thurai           | 0 - 6.0                  |
| 2416          |                 | 10            | 81                  | Thurai           | 0 - 6.0                  |
| 2417          |                 | 10            | 76                  | Thurai           | 0 - 6.0                  |
| 2418          |                 | 10            | 82                  | Thurai           | 0 - 6.0                  |
| 2419          |                 | 10            | 73                  | Thurai           | 0 - 6.0                  |
| 2420          |                 | 10            | 46                  | Thurai           | 0 - 6.0                  |
| 2421          |                 | 10            | 56                  | Thurai           | 0 - 6.0                  |
| 2422          |                 | 15 above      | 175                 | Bhagi            | 0 - 6.0                  |
| 2423          |                 | 15 above      | 80                  | Bhagi            | 0 - 6.0                  |
| 2424          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 2425          |                 | 15 above      | 71                  | Mango            | 0 - 6.0                  |
| 2426          |                 | 15 above      | 54                  | Mango            | 0 - 6.0                  |
| 2427          | 38+800          | 15 above      | 51                  | Coconut          | 0 - 6.0                  |
| 2428          |                 | 15 above      | 59                  | Coconut          | 0 - 6.0                  |
| 2429          |                 | 15 above      | 58                  | Neredu           | 0 - 6.0                  |
| 2430          |                 | 15 above      | 51                  | Neredu           | 0 - 6.0                  |
| 2431          |                 | 15 above      | 52                  | Palm             | 0 - 6.0                  |
| 2432          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2433          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2434          |                 | 15 above      | 62                  | Coconut          | 0 - 6.0                  |
| 2435          |                 | 15 above      | 68                  | Coconut          | 0 - 6.0                  |
| 2436          |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 2437          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 2438          |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |
| 2439          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2440          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 2441          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2442          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 2443          |                 | 15 above      | 75                  | Coconut          | 0 - 6.0                  |
| 2444          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2445          | 39+200          | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2446          |                 | 15 above      | 69                  | Coconut          | 0 - 6.0                  |
| 2447          |                 | 15 above      | 68                  | Coconut          | 0 - 6.0                  |
| 2448          |                 | 15 above      | 62                  | Coconut          | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2449          |                 | 15 above      | 67                  | Coconut          | 0 - 6.0                  |
| 2450          |                 | 15 above      | 65                  | Coconut          | 0 - 6.0                  |
| 2451          |                 | 15 above      | 84                  | Coconut          | 0 - 6.0                  |
| 2452          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 2453          |                 | 15 above      | 135                 | Coconut          | 0 - 6.0                  |
| 2454          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2455          |                 | 15 above      | 54                  | Palm             | 0 - 6.0                  |
| 2456          |                 | 15 above      | 55                  | Thurai           | 0 - 6.0                  |
| 2457          |                 | 15 above      | 58                  | Thurai           | 0 - 6.0                  |
| 2458          |                 | 15 above      | 59                  | Thurai           | 0 - 6.0                  |
| 2459          |                 | 15 above      | 53                  | Thurai           | 0 - 6.0                  |
| 2460          |                 | 15 above      | 58                  | Thurai           | 0 - 6.0                  |
| 2461          |                 | 15 above      | 54                  | Thurai           | 0 - 6.0                  |
| 2462          |                 | 15 above      | 38                  | Ragi             | 0 - 6.0                  |
| 2463          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2464          |                 | 15 above      | 86                  | Coconut          | 0 - 6.0                  |
| 2465          |                 | 15 above      | 78                  | Coconut          | 0 - 6.0                  |
| 2466          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 2467          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2468          |                 | 15 above      | 72                  | Coconut          | 0 - 6.0                  |
| 2469          |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 2470          |                 | 15 above      | 71                  | Coconut          | 0 - 6.0                  |
| 2471          |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 2472          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 2473          |                 | 15 above      | 85                  | Coconut          | 0 - 6.0                  |
| 2474          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 2475          | 39+600          | 15 above      | 68                  | Coconut          | 0 - 6.0                  |
| 2476          |                 | 15 above      | 64                  | Coconut          | 0 - 6.0                  |
| 2477          |                 | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 2478          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2479          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2480          |                 | 15 above      | 62                  | Palm             | 0 - 6.0                  |
| 2481          |                 | 15 above      | 68                  | Palm             | 0 - 6.0                  |
| 2482          |                 | 15 above      | 70                  | Palm             | 0 - 6.0                  |
| 2483          |                 | 15 above      | 79                  | Palm             | 0 - 6.0                  |
| 2484          |                 | 15 above      | 81                  | Palm             | 0 - 6.0                  |
| 2485          |                 | 15 above      | 76                  | Palm             | 0 - 6.0                  |
| 2486          |                 | 15 above      | 158                 | Palm             | 0 - 6.0                  |
| 2487          |                 | 15 above      | 165                 | Bhagi            | 0 - 6.0                  |
| 2488          |                 | 15 above      | 123                 | Bhagi            | 0 - 6.0                  |
| 2489          |                 | 15 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 2490          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2491          |                 | 15 above      | 158                 | Bhagi            | 0 - 6.0                  |
| 2492          |                 | 15 above      | 165                 | Bhagi            | 0 - 6.0                  |
| 2493          |                 | 15 above      | 155                 | Bhagi            | 0 - 6.0                  |
| 2494          |                 | 10            | 34                  | Bhagi            | 0 - 6.0                  |
| 2495          |                 | 15 above      | 145                 | Coconut          | 0 - 6.0                  |
| 2496          |                 | 15 above      | 122                 | Coconut          | 0 - 6.0                  |
| 2497          |                 | 15 above      | 135                 | Coconut          | 0 - 6.0                  |
| 2498          |                 | 15 above      | 112                 | Coconut          | 0 - 6.0                  |
| 2499          | 40+000          | 15 above      | 110                 | Raavi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2500          |                 | 15 above      | 145                 | Raavi            | 0 - 6.0                  |
| 2501          |                 | 15 above      | 145                 | Neem             | 0 - 6.0                  |
| 2502          |                 | 15 above      | 72                  | Thurai           | 0 - 6.0                  |
| 2503          | 40+400          | 10            | 34                  | Bhagi            | 0 - 6.0                  |
| 2504          |                 | 10            | 39                  | Bhagi            | 0 - 6.0                  |
| 2505          |                 | 10            | 125                 | Bhagi            | 0 - 6.0                  |
| 2506          |                 | 15 above      | 65                  | Giriship         | 0 - 6.0                  |
| 2507          |                 | 15 above      | 33                  | Giriship         | 0 - 6.0                  |
| 2508          |                 | 15 above      | 37                  | Giriship         | 0 - 6.0                  |
| 2509          |                 | 15 above      | 38                  | Giriship         | 0 - 6.0                  |
| 2510          |                 | 15 above      | 31                  | Giriship         | 0 - 6.0                  |
| 2511          |                 | 15 above      | 39                  | Giriship         | 0 - 6.0                  |
| 2512          | 40+800          | 15 above      | 38                  | Raavi            | 0 - 6.0                  |
| 2513          | 41+000          | 15 above      | 175                 | Giriship         | 0 - 6.0                  |
| 2514          |                 | 15 above      | 72                  | Thurai           | 0 - 6.0                  |
| 2515          | 41+400          | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 2516          | 42+200          | 15 above      | 155                 | Neredu           | 0 - 6.0                  |
| 2517          |                 | 15 above      | 165                 | Neredu           | 0 - 6.0                  |
| 2518          | 42+400          | 15 above      | 185                 | Neredu           | 0 - 6.0                  |
| 2519          |                 | 15 above      | 192                 | Bhagi            | 0 - 6.0                  |
| 2520          |                 | 15 above      | 142                 | Bhagi            | 0 - 6.0                  |
| 2521          |                 | 15 above      | 146                 | Bhagi            | 0 - 6.0                  |
| 2522          |                 | 15 above      | 178                 | Bhagi            | 0 - 6.0                  |
| 2523          |                 | 10            | 175                 | Bhagi            | 0 - 6.0                  |
| 2524          |                 | 10            | 175                 | Bhagi            | 0 - 6.0                  |
| 2525          |                 | 10            | 155                 | Tamarind         | 0 - 6.0                  |
| 2526          |                 | 10            | 125                 | Tamarind         | 0 - 6.0                  |
| 2527          |                 | 10            | 136                 | Tamarind         | 0 - 6.0                  |
| 2528          |                 | 10            | 124                 | Neem             | 0 - 6.0                  |
| 2529          |                 | 10            | 114                 | Neem             | 0 - 6.0                  |
| 2530          |                 | 10            | 113                 | Giriship         | 0 - 6.0                  |
| 2531          |                 | 10            | 125                 | Giriship         | 0 - 6.0                  |
| 2532          |                 | 10            | 136                 | Giriship         | 0 - 6.0                  |
| 2533          |                 | 15 above      | 185                 | Giriship         | 0 - 6.0                  |
| 2534          |                 | 15 above      | 165                 | Coconut          | 0 - 6.0                  |
| 2535          |                 | 15 above      | 175                 | Coconut          | 0 - 6.0                  |
| 2536          | 42+600          | 15 above      | 152                 | Bhagi            | 0 - 6.0                  |
| 2537          |                 | 15 above      | 163                 | Bhagi            | 0 - 6.0                  |
| 2538          |                 | 15 above      | 145                 | Bhagi            | 0 - 6.0                  |
| 2539          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2540          |                 | 15 above      | 122                 | Bhagi            | 0 - 6.0                  |
| 2541          |                 | 15 above      | 156                 | Bhagi            | 0 - 6.0                  |
| 2542          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2543          |                 | 15 above      | 128                 | Bhagi            | 0 - 6.0                  |
| 2544          |                 | 15 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 2545          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2546          |                 | 15 above      | 185                 | Bhagi            | 0 - 6.0                  |
| 2547          | 42+800          | 10            | 124                 | Bhagi            | 0 - 6.0                  |
| 2548          |                 | 10            | 123                 | Bhagi            | 0 - 6.0                  |
| 2549          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2550          |                 | 15 above      | 162                 | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2551          |                 | 15 above      | 58                  | Bhagi            | 0 - 6.0                  |
| 2552          |                 | 10            | 145                 | Thurai           | 0 - 6.0                  |
| 2553          |                 | 10            | 185                 | Thurai           | 0 - 6.0                  |
| 2554          |                 | 15 above      | 196                 | Bhagi            | 0 - 6.0                  |
| 2555          |                 | 15 above      | 174                 | Bhagi            | 0 - 6.0                  |
| 2556          |                 | 15 above      | 154                 | Bhagi            | 0 - 6.0                  |
| 2557          |                 | 15 above      | 126                 | Kanuga           | 0 - 6.0                  |
| 2558          |                 | 15 above      | 128                 | Kanuga           | 0 - 6.0                  |
| 2559          |                 | 15 above      | 122                 | Kanuga           | 0 - 6.0                  |
| 2560          | 43+200          | 15 above      | 146                 | Thurai           | 0 - 6.0                  |
| 2561          | 43+400          | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 2562          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 2563          |                 | 15 above      | 85                  | Mango            | 0 - 6.0                  |
| 2564          |                 | 15 above      | 78                  | Mango            | 0 - 6.0                  |
| 2565          |                 | 10            | 73                  | Bhagi            | 0 - 6.0                  |
| 2566          |                 | 10            | 55                  | Bhagi            | 0 - 6.0                  |
| 2567          |                 | 10            | 50                  | Bhagi            | 0 - 6.0                  |
| 2568          |                 | 10            | 59                  | Thurai           | 0 - 6.0                  |
| 2569          |                 | 10            | 154                 | Thurai           | 0 - 6.0                  |
| 2570          |                 | 10            | 145                 | Thurai           | 0 - 6.0                  |
| 2571          |                 | 10            | 165                 | Thurai           | 0 - 6.0                  |
| 2572          |                 | 10            | 162                 | Thurai           | 0 - 6.0                  |
| 2573          |                 | 10            | 115                 | Thurai           | 0 - 6.0                  |
| 2574          | 43+600          | 10            | 115                 | Coconut          | 0 - 6.0                  |
| 2575          |                 | 10            | 57                  | Palm             | 0 - 6.0                  |
| 2576          |                 | 10            | 50                  | Bhagi            | 0 - 6.0                  |
| 2577          |                 | 15 above      | 141                 | Bhagi            | 0 - 6.0                  |
| 2578          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2579          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2580          | 44+000          | 15 above      | 185                 | Bhagi            | 0 - 6.0                  |
| 2581          |                 | 15 above      | 145                 | Bhagi            | 0 - 6.0                  |
| 2582          |                 | 15 above      | 162                 | Bhagi            | 0 - 6.0                  |
| 2583          |                 | 15 above      | 142                 | Bhagi            | 0 - 6.0                  |
| 2584          |                 | 15 above      | 196                 | Bhagi            | 0 - 6.0                  |
| 2585          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 2586          |                 | 15 above      | 76                  | Ashoka           | 0 - 6.0                  |
| 2587          |                 | 15 above      | 72                  | Ashoka           | 0 - 6.0                  |
| 2588          |                 | 10            | 70                  | Ashoka           | 0 - 6.0                  |
| 2589          |                 | 10            | 71                  | Neem             | 0 - 6.0                  |
| 2590          |                 | 15 above      | 82                  | Coconut          | 0 - 6.0                  |
| 2591          |                 | 15 above      | 185                 | Coconut          | 0 - 6.0                  |
| 2592          |                 | 15 above      | 175                 | Coconut          | 0 - 6.0                  |
| 2593          |                 | 15 above      | 144                 | Coconut          | 0 - 6.0                  |
| 2594          |                 | 15 above      | 165                 | Coconut          | 0 - 6.0                  |
| 2595          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2596          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2597          |                 | 10            | 124                 | Coconut          | 0 - 6.0                  |
| 2598          |                 | 10            | 185                 | Coconut          | 0 - 6.0                  |
| 2599          |                 | 10            | 165                 | Coconut          | 0 - 6.0                  |
| 2600          |                 | 10            | 174                 | Coconut          | 0 - 6.0                  |
| 2601          | 44+400          | 10            | 175                 | Udaga            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2602          |                 | 10            | 195                 | Neredu           | 0 - 6.0                  |
| 2603          |                 | 15 above      | 79                  | Bhagi            | 0 - 6.0                  |
| 2604          |                 | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 2605          |                 | 15 above      | 76                  | Thurai           | 0 - 6.0                  |
| 2606          | 44+600          | 15 above      | 78                  | Bhagi            | 0 - 6.0                  |
| 2607          |                 | 15 above      | 72                  | Bhagi            | 0 - 6.0                  |
| 2608          |                 | 15 above      | 145                 | Bhagi            | 0 - 6.0                  |
| 2609          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2610          | 44+800          | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2611          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2612          |                 | 15 above      | 123                 | Bhagi            | 0 - 6.0                  |
| 2613          |                 | 10            | 142                 | Bhagi            | 0 - 6.0                  |
| 2614          |                 | 10            | 136                 | Bhagi            | 0 - 6.0                  |
| 2615          |                 | 10            | 125                 | Bhagi            | 0 - 6.0                  |
| 2616          |                 | 10            | 175                 | Bhagi            | 0 - 6.0                  |
| 2617          | 45+200          | 10            | 142                 | Kanuga           | 0 - 6.0                  |
| 2618          |                 | 10            | 136                 | Kanuga           | 0 - 6.0                  |
| 2619          |                 | 10            | 125                 | Kanuga           | 0 - 6.0                  |
| 2620          |                 | 10            | 185                 | Kanuga           | 0 - 6.0                  |
| 2621          |                 | 10            | 165                 | Kanuga           | 0 - 6.0                  |
| 2622          |                 | 15 above      | 152                 | Kanuga           | 0 - 6.0                  |
| 2623          |                 | 10            | 145                 | Kanuga           | 0 - 6.0                  |
| 2624          |                 | 10            | 136                 | Almond           | 0 - 6.0                  |
| 2625          | 45+600          | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2626          |                 | 15 above      | 142                 | Bhagi            | 0 - 6.0                  |
| 2627          |                 | 15 above      | 165                 | Thurai           | 0 - 6.0                  |
| 2628          |                 | 15 above      | 59                  | Bhagi            | 0 - 6.0                  |
| 2629          |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 2630          | 46+200          | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 2631          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2632          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2633          |                 | 10            | 62                  | Coconut          | 0 - 6.0                  |
| 2634          |                 | 10            | 51                  | Coconut          | 0 - 6.0                  |
| 2635          |                 | 10            | 57                  | Coconut          | 0 - 6.0                  |
| 2636          |                 | 15 above      | 52                  | Neredu           | 0 - 6.0                  |
| 2637          |                 | 15 above      | 165                 | Bhagi            | 0 - 6.0                  |
| 2638          |                 | 15 above      | 136                 | Kanuga           | 0 - 6.0                  |
| 2639          |                 | 15 above      | 136                 | Kanuga           | 0 - 6.0                  |
| 2640          |                 | 15 above      | 185                 | Kanuga           | 0 - 6.0                  |
| 2641          |                 | 15 above      | 142                 | Kanuga           | 0 - 6.0                  |
| 2642          |                 | 15 above      | 136                 | Kanuga           | 0 - 6.0                  |
| 2643          |                 | 15 above      | 125                 | Kanuga           | 0 - 6.0                  |
| 2644          |                 | 15 above      | 175                 | Kanuga           | 0 - 6.0                  |
| 2645          |                 | 15 above      | 142                 | Kanuga           | 0 - 6.0                  |
| 2646          |                 | 15 above      | 136                 | Kanuga           | 0 - 6.0                  |
| 2647          | 46+800          | 15 above      | 125                 | Kanuga           | 0 - 6.0                  |
| 2648          |                 | 15 above      | 185                 | Kanuga           | 0 - 6.0                  |
| 2649          |                 | 15 above      | 165                 | Kanuga           | 0 - 6.0                  |
| 2650          |                 | 15 above      | 169                 | Kanuga           | 0 - 6.0                  |
| 2651          |                 | 15 above      | 165                 | Kanuga           | 0 - 6.0                  |
| 2652          |                 | 15 above      | 136                 | Kanuga           | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2653          |                 | 15 above      | 136                 | Kanuga           | 0 - 6.0                  |
| 2654          |                 | 15 above      | 185                 | Kanuga           | 0 - 6.0                  |
| 2655          |                 | 15 above      | 142                 | Mango            | 0 - 6.0                  |
| 2656          |                 | 15 above      | 136                 | Neem             | 0 - 6.0                  |
| 2657          |                 | 15 above      | 125                 | Mango            | 0 - 6.0                  |
| 2658          |                 | 15 above      | 175                 | Coconut          | 0 - 6.0                  |
| 2659          | 47+000          | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 2660          |                 | 10            | 37                  | Coconut          | 0 - 6.0                  |
| 2661          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2662          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2663          |                 | 10            | 34                  | Coconut          | 0 - 6.0                  |
| 2664          |                 | 10            | 35                  | Coconut          | 0 - 6.0                  |
| 2665          |                 | 10            | 35                  | Coconut          | 0 - 6.0                  |
| 2666          |                 | 10            | 33                  | Coconut          | 0 - 6.0                  |
| 2667          |                 | 10            | 32                  | Coconut          | 0 - 6.0                  |
| 2668          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2669          |                 | 10            | 32                  | Coconut          | 0 - 6.0                  |
| 2670          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 2671          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2672          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2673          |                 | 10            | 34                  | Coconut          | 0 - 6.0                  |
| 2674          |                 | 15 above      | 39                  | Coconut          | 0 - 6.0                  |
| 2675          |                 | 15 above      | 38                  | Coconut          | 0 - 6.0                  |
| 2676          |                 | 15 above      | 31                  | Coconut          | 0 - 6.0                  |
| 2677          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2678          | 47+200          | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2679          |                 | 15 above      | 80                  | Coconut          | 0 - 6.0                  |
| 2680          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2681          |                 | 15 above      | 89                  | Coconut          | 0 - 6.0                  |
| 2682          |                 | 20 above      | 39                  | Coconut          | 0 - 6.0                  |
| 2683          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2684          |                 | 10            | 37                  | Coconut          | 0 - 6.0                  |
| 2685          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 2686          |                 | 10            | 37                  | Coconut          | 0 - 6.0                  |
| 2687          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 2688          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2689          |                 | 10            | 48                  | Pongamia         | 0 - 6.0                  |
| 2690          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2691          |                 | 10            | 36                  | Pongamia         | 0 - 6.0                  |
| 2692          |                 | 10            | 49                  | Pongamia         | 0 - 6.0                  |
| 2693          |                 | 10            | 48                  | Pongamia         | 0 - 6.0                  |
| 2694          |                 | 10            | 41                  | Bhagi            | 0 - 6.0                  |
| 2695          |                 | 10            | 44                  | Bhagi            | 0 - 6.0                  |
| 2696          |                 | 10            | 40                  | Bhagi            | 0 - 6.0                  |
| 2697          |                 | 10            | 48                  | Tamarind         | 0 - 6.0                  |
| 2698          | 47+400          | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 2699          |                 | 10            | 40                  | Bhagi            | 0 - 6.0                  |
| 2700          |                 | 10            | 44                  | Bhagi            | 0 - 6.0                  |
| 2701          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2702          |                 | 10            | 47                  | Pongamia         | 0 - 6.0                  |
| 2703          |                 | 10            | 86                  | Pongamia         | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2704          |                 | 10            | 75                  | Pongamia         | 0 - 6.0                  |
| 2705          |                 | 10            | 70                  | Raavi            | 0 - 6.0                  |
| 2706          |                 | 10            | 142                 | Neredu           | 0 - 6.0                  |
| 2707          |                 | 10            | 125                 | Neredu           | 0 - 6.0                  |
| 2708          |                 | 10            | 136                 | Neredu           | 0 - 6.0                  |
| 2709          |                 | 10            | 142                 | Neredu           | 0 - 6.0                  |
| 2710          | 47+600          | 10            | 110                 | Neredu           | 0 - 6.0                  |
| 2711          |                 | 10            | 154                 | Neredu           | 0 - 6.0                  |
| 2712          |                 | 10            | 81                  | Neredu           | 0 - 6.0                  |
| 2713          |                 | 10            | 86                  | Neredu           | 0 - 6.0                  |
| 2714          |                 | 10            | 73                  | Neredu           | 0 - 6.0                  |
| 2715          |                 | 10            | 46                  | Neredu           | 0 - 6.0                  |
| 2716          |                 | 10            | 55                  | Pongamia         | 0 - 6.0                  |
| 2717          |                 | 10            | 35                  | Pongamia         | 0 - 6.0                  |
| 2718          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2719          |                 | 10            | 49                  | Pongamia         | 0 - 6.0                  |
| 2720          |                 | 15 above      | 42                  | Pongamia         | 0 - 6.0                  |
| 2721          |                 | 15 above      | 38                  | Pongamia         | 0 - 6.0                  |
| 2722          | 47+800          | 10            | 34                  | Neredu           | 0 - 6.0                  |
| 2723          |                 | 10            | 33                  | Neredu           | 0 - 6.0                  |
| 2724          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2725          | 48+000          | 10            | 50                  | Bhagi            | 0 - 6.0                  |
| 2726          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 2727          |                 | 10            | 44                  | Bhagi            | 0 - 6.0                  |
| 2728          |                 | 10            | 36                  | Bhagi            | 0 - 6.0                  |
| 2729          |                 | 10            | 39                  | Bhagi            | 0 - 6.0                  |
| 2730          | 48+200          | 10            | 136                 | Bhagi            | 0 - 6.0                  |
| 2731          |                 | 10            | 124                 | Bhagi            | 0 - 6.0                  |
| 2732          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2733          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2734          |                 | 15 above      | 142                 | Bhagi            | 0 - 6.0                  |
| 2735          |                 | 15 above      | 185                 | Bhagi            | 0 - 6.0                  |
| 2736          |                 | 15 above      | 162                 | Bhagi            | 0 - 6.0                  |
| 2737          |                 | 15 above      | 142                 | Pongamia         | 0 - 6.0                  |
| 2738          | 48+400          | 10            | 55                  | Pongamia         | 0 - 6.0                  |
| 2739          |                 | 10            | 35                  | Pongamia         | 0 - 6.0                  |
| 2740          |                 | 10            | 38                  | Pongamia         | 0 - 6.0                  |
| 2741          |                 | 10            | 44                  | Bhagi            | 0 - 6.0                  |
| 2742          |                 | 10            | 39                  | Bhagi            | 0 - 6.0                  |
| 2743          |                 | 10            | 31                  | Bhagi            | 0 - 6.0                  |
| 2744          |                 | 10            | 59                  | Thurai           | 0 - 6.0                  |
| 2745          |                 | 10            | 51                  | Thurai           | 0 - 6.0                  |
| 2746          |                 | 10            | 46                  | Thurai           | 0 - 6.0                  |
| 2747          |                 | 10            | 49                  | Thurai           | 0 - 6.0                  |
| 2748          | 48+600          | 10            | 44                  | Thurai           | 0 - 6.0                  |
| 2749          |                 | 10            | 38                  | Thurai           | 0 - 6.0                  |
| 2750          |                 | 10            | 31                  | Pongamia         | 0 - 6.0                  |
| 2751          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2752          |                 | 10            | 39                  | Pongamia         | 0 - 6.0                  |
| 2753          |                 | 10            | 55                  | Pongamia         | 0 - 6.0                  |
| 2754          |                 | 10            | 49                  | Pongamia         | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2755          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2756          |                 | 10            | 35                  | Tamarind         | 0 - 6.0                  |
| 2757          |                 | 10            | 38                  | Tamarind         | 0 - 6.0                  |
| 2758          |                 | 10            | 81                  | Tamarind         | 0 - 6.0                  |
| 2759          | 48+800          | 15 above      | 56                  | Palm             | 0 - 6.0                  |
| 2760          |                 | 10            | 50                  | Thurai           | 0 - 6.0                  |
| 2761          |                 | 10            | 58                  | Bhagi            | 0 - 6.0                  |
| 2762          |                 | 10            | 51                  | Pongamia         | 0 - 6.0                  |
| 2763          |                 | 10            | 51                  | Pongamia         | 0 - 6.0                  |
| 2764          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2765          |                 | 10            | 49                  | Pongamia         | 0 - 6.0                  |
| 2766          |                 | 10            | 44                  | Pongamia         | 0 - 6.0                  |
| 2767          |                 | 10            | 38                  | Pongamia         | 0 - 6.0                  |
| 2768          |                 | 10            | 31                  | Pongamia         | 0 - 6.0                  |
| 2769          |                 | 10            | 46                  | Pongamia         | 0 - 6.0                  |
| 2770          |                 | 10            | 39                  | Pongamia         | 0 - 6.0                  |
| 2771          | 49+000          | 10            | 55                  | Bhagi            | 0 - 6.0                  |
| 2772          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 2773          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 2774          |                 | 10            | 35                  | Thurai           | 0 - 6.0                  |
| 2775          |                 | 10            | 38                  | Raavi            | 0 - 6.0                  |
| 2776          |                 | 10            | 50                  | Raavi            | 0 - 6.0                  |
| 2777          | 49+200          | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 2778          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 2779          | 49+400          | 10            | 42                  | Bhagi            | 0 - 6.0                  |
| 2780          |                 | 10            | 38                  | Bhagi            | 0 - 6.0                  |
| 2781          |                 | 10            | 34                  | Bhagi            | 0 - 6.0                  |
| 2782          | 49+600          | 10            | 35                  | Bhagi            | 0 - 6.0                  |
| 2783          |                 | 10            | 38                  | Thurai           | 0 - 6.0                  |
| 2784          |                 | 10            | 44                  | Thurai           | 0 - 6.0                  |
| 2785          | 49+800          | 10            | 65                  | Bhagi            | 0 - 6.0                  |
| 2786          |                 | 10            | 69                  | Bhagi            | 0 - 6.0                  |
| 2787          |                 | 10            | 68                  | Bhagi            | 0 - 6.0                  |
| 2788          | 50+000          | 10            | 66                  | Bhagi            | 0 - 6.0                  |
| 2789          |                 | 10            | 64                  | Bhagi            | 0 - 6.0                  |
| 2790          |                 | 10            | 75                  | Bhagi            | 0 - 6.0                  |
| 2791          |                 | 10            | 71                  | Bhagi            | 0 - 6.0                  |
| 2792          |                 | 10            | 136                 | Bhagi            | 0 - 6.0                  |
| 2793          |                 | 10            | 128                 | Bhagi            | 0 - 6.0                  |
| 2794          |                 | 10            | 124                 | Bhagi            | 0 - 6.0                  |
| 2795          |                 | 10            | 125                 | Bhagi            | 0 - 6.0                  |
| 2796          |                 | 10            | 136                 | Bhagi            | 0 - 6.0                  |
| 2797          |                 | 10            | 124                 | Bhagi            | 0 - 6.0                  |
| 2798          |                 | 15 above      | 185                 | Bhagi            | 0 - 6.0                  |
| 2799          |                 | 15 above      | 165                 | Bhagi            | 0 - 6.0                  |
| 2800          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |
| 2801          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2802          |                 | 15 above      | 124                 | Bhagi            | 0 - 6.0                  |
| 2803          | 50+200          | 15 above      | 152                 | Bhagi            | 0 - 6.0                  |
| 2804          |                 | 15 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2805          |                 | 15 above      | 125                 | Bhagi            | 0 - 6.0                  |



# APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 2806   |          | 15 above | 142          | Bhagi     | 0 - 6.0           |
| 2807   |          | 15 above | 185          | Bhagi     | 0 - 6.0           |
| 2808   |          | 15 above | 162          | Bhagi     | 0 - 6.0           |
| 2809   |          | 10       | 79           | Bhagi     | 0 - 6.0           |
| 2810   |          | 10       | 70           | Bhagi     | 0 - 6.0           |
| 2811   |          | 10       | 73           | Bhagi     | 0 - 6.0           |
| 2812   |          | 10       | 72           | Bhagi     | 0 - 6.0           |
| 2813   | 50+400   | 10       | 79           | Bhagi     | 0 - 6.0           |
| 2814   |          | 10       | 76           | Bhagi     | 0 - 6.0           |
| 2815   |          | 10       | 72           | Bhagi     | 0 - 6.0           |
| 2816   |          | 10       | 79           | Bhagi     | 0 - 6.0           |
| 2817   |          | 10       | 80           | Bhagi     | 0 - 6.0           |
| 2818   |          | 10       | 32           | Bhagi     | 0 - 6.0           |
| 2819   |          | 10       | 142          | Bhagi     | 0 - 6.0           |
| 2820   |          | 15 above | 136          | Neredu    | 0 - 6.0           |
| 2821   | 50+600   | 15 above | 39           | Neredu    | 0 - 6.0           |
| 2822   |          | 10       | 35           | Bhagi     | 0 - 6.0           |
| 2823   |          | 10       | 36           | Bhagi     | 0 - 6.0           |
| 2824   |          | 10       | 35           | Bhagi     | 0 - 6.0           |
| 2825   |          | 10       | 45           | Bhagi     | 0 - 6.0           |
| 2826   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 2827   |          | 10       | 47           | Bhagi     | 0 - 6.0           |
| 2828   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 2829   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 2830   |          | 10       | 58           | Bhagi     | 0 - 6.0           |
| 2831   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 2832   | 50+800   | 10       | 53           | Bhagi     | 0 - 6.0           |
| 2833   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 2834   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 2835   |          | 10       | 55           | Bhagi     | 0 - 6.0           |
| 2836   |          | 10       | 50           | Bhagi     | 0 - 6.0           |
| 2837   | 51+000   | 10       | 58           | Bhagi     | 0 - 6.0           |
| 2838   |          | 10       | 54           | Bhagi     | 0 - 6.0           |
| 2839   |          | 10       | 59           | Bhagi     | 0 - 6.0           |
| 2840   |          | 10       | 54           | Bhagi     | 0 - 6.0           |
| 2841   |          | 10       | 54           | Bhagi     | 0 - 6.0           |
| 2842   |          | 10       | 59           | Bhagi     | 0 - 6.0           |
| 2843   |          | 10       | 55           | Neredu    | 0 - 6.0           |
| 2844   | 51+200   | 10       | 51           | Raavi     | 0 - 6.0           |
| 2845   |          | 10       | 57           | Bhagi     | 0 - 6.0           |
| 2846   |          | 10       | 52           | Bhagi     | 0 - 6.0           |
| 2847   |          | 15 above | 82           | Bhagi     | 0 - 6.0           |
| 2848   |          | 15 above | 65           | Bhagi     | 0 - 6.0           |
| 2849   |          | 15 above | 69           | Bhagi     | 0 - 6.0           |
| 2850   |          | 20 above | 76           | Bhagi     | 0 - 6.0           |
| 2851   |          | 20 above | 125          | Bhagi     | 0 - 6.0           |
| 2852   |          | 10       | 58           | Pongamia  | 0 - 6.0           |
| 2853   |          | 10       | 54           | Pongamia  | 0 - 6.0           |
| 2854   |          | 10       | 35           | Neem      | 0 - 6.0           |
| 2855   |          | 20 above | 39           | Bhagi     | 0 - 6.0           |
| 2856   |          | 15 above | 79           | Thurai    | 0 - 6.0           |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2857          |                 | 15 above      | 68                  | Bhagi            | 0 - 6.0                  |
| 2858          |                 | 15 above      | 64                  | Bhagi            | 0 - 6.0                  |
| 2859          |                 | 15 above      | 63                  | Coconut          | 0 - 6.0                  |
| 2860          |                 | 15 above      | 76                  | Coconut          | 0 - 6.0                  |
| 2861          |                 | 15 above      | 73                  | Coconut          | 0 - 6.0                  |
| 2862          |                 | 15 above      | 62                  | Coconut          | 0 - 6.0                  |
| 2863          |                 | 15 above      | 68                  | Coconut          | 0 - 6.0                  |
| 2864          |                 | 15 above      | 70                  | Coconut          | 0 - 6.0                  |
| 2865          |                 | 15 above      | 79                  | Coconut          | 0 - 6.0                  |
| 2866          |                 | 15 above      | 81                  | Coconut          | 0 - 6.0                  |
| 2867          | 51+600          | 20 above      | 76                  | Palm             | 0 - 6.0                  |
| 2868          |                 | 10            | 82                  | Velaga           | 0 - 6.0                  |
| 2869          |                 | 10            | 73                  | Thurai           | 0 - 6.0                  |
| 2870          |                 | 10            | 84                  | Bhagi            | 0 - 6.0                  |
| 2871          |                 | 10            | 75                  | Bhagi            | 0 - 6.0                  |
| 2872          |                 | 10            | 86                  | Bhagi            | 0 - 6.0                  |
| 2873          |                 | 10            | 75                  | Bhagi            | 0 - 6.0                  |
| 2874          |                 | 10            | 70                  | Pongamia         | 0 - 6.0                  |
| 2875          |                 | 10            | 76                  | Pongamia         | 0 - 6.0                  |
| 2876          |                 | 15 above      | 71                  | Pongamia         | 0 - 6.0                  |
| 2877          |                 | 15 above      | 76                  | Pongamia         | 0 - 6.0                  |
| 2878          |                 | 15 above      | 69                  | Coconut          | 0 - 6.0                  |
| 2879          |                 | 15 above      | 38                  | Coconut          | 0 - 6.0                  |
| 2880          |                 | 15 above      | 34                  | Coconut          | 0 - 6.0                  |
| 2881          |                 | 15 above      | 39                  | Coconut          | 0 - 6.0                  |
| 2882          |                 | 15 above      | 38                  | Coconut          | 0 - 6.0                  |
| 2883          |                 | 15 above      | 37                  | Coconut          | 0 - 6.0                  |
| 2884          |                 | 15 above      | 136                 | Coconut          | 0 - 6.0                  |
| 2885          |                 | 15 above      | 125                 | Coconut          | 0 - 6.0                  |
| 2886          | 51+800          | 10            | 40                  | Coconut          | 0 - 6.0                  |
| 2887          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 2888          |                 | 10            | 41                  | Coconut          | 0 - 6.0                  |
| 2889          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 2890          |                 | 15 above      | 42                  | Bhagi            | 0 - 6.0                  |
| 2891          |                 | 15 above      | 49                  | Bhagi            | 0 - 6.0                  |
| 2892          |                 | 15 above      | 48                  | Bhagi            | 0 - 6.0                  |
| 2893          |                 | 15 above      | 47                  | Raavi            | 0 - 6.0                  |
| 2894          | 52+000          | 15 above      | 135                 | Neem             | 0 - 6.0                  |
| 2895          |                 | 15 above      | 49                  | Neem             | 0 - 6.0                  |
| 2896          |                 | 10            | 40                  | Thurai           | 0 - 6.0                  |
| 2897          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 2898          |                 | 10            | 55                  | Coconut          | 0 - 6.0                  |
| 2899          |                 | 10            | 50                  | Coconut          | 0 - 6.0                  |
| 2900          |                 | 10            | 58                  | Coconut          | 0 - 6.0                  |
| 2901          |                 | 10            | 54                  | Coconut          | 0 - 6.0                  |
| 2902          |                 | 10            | 59                  | Coconut          | 0 - 6.0                  |
| 2903          |                 | 10            | 51                  | Bhagi            | 0 - 6.0                  |
| 2904          |                 | 10            | 48                  | Bhagi            | 0 - 6.0                  |
| 2905          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 2906          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 2907          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 2908          |                 | 10            | 41                  | Bhagi            | 0 - 6.0                  |
| 2909          |                 | 10            | 48                  | Bhagi            | 0 - 6.0                  |
| 2910          |                 | 20 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 2911          |                 | 20 above      | 125                 | Raavi            | 0 - 6.0                  |
| 2912          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |
| 2913          |                 | 15 above      | 46                  | Coconut          | 0 - 6.0                  |
| 2914          |                 | 15 above      | 41                  | Coconut          | 0 - 6.0                  |
| 2915          |                 | 15 above      | 41                  | Coconut          | 0 - 6.0                  |
| 2916          |                 | 15 above      | 48                  | Coconut          | 0 - 6.0                  |
| 2917          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 2918          |                 | 15 above      | 46                  | Coconut          | 0 - 6.0                  |
| 2919          |                 | 15 above      | 42                  | Coconut          | 0 - 6.0                  |
| 2920          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |
| 2921          |                 | 15 above      | 40                  | Coconut          | 0 - 6.0                  |
| 2922          |                 | 20 above      | 46                  | Palm             | 0 - 6.0                  |
| 2923          |                 | 10            | 47                  | Coconut          | 0 - 6.0                  |
| 2924          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 2925          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 2926          |                 | 10            | 40                  | Coconut          | 0 - 6.0                  |
| 2927          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 2928          |                 | 10            | 40                  | Coconut          | 0 - 6.0                  |
| 2929          |                 | 10            | 35                  | Coconut          | 0 - 6.0                  |
| 2930          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 2931          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2932          |                 | 10            | 37                  | Coconut          | 0 - 6.0                  |
| 2933          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 2934          | 52+600          | 10            | 38                  | Bhagi            | 0 - 6.0                  |
| 2935          |                 | 10            | 31                  | Palm             | 0 - 6.0                  |
| 2936          |                 | 10            | 36                  | Coconut          | 0 - 6.0                  |
| 2937          |                 | 10            | 33                  | Coconut          | 0 - 6.0                  |
| 2938          |                 | 10            | 36                  | Coconut          | 0 - 6.0                  |
| 2939          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2940          |                 | 10            | 36                  | Coconut          | 0 - 6.0                  |
| 2941          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 2942          | 52+800          | 10            | 31                  | Bhagi            | 0 - 6.0                  |
| 2943          | 53+000          | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 2944          |                 | 10            | 39                  | Bhagi            | 0 - 6.0                  |
| 2945          |                 | 10            | 35                  | Bhagi            | 0 - 6.0                  |
| 2946          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 2947          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 2948          |                 | 10            | 31                  | Coconut          | 0 - 6.0                  |
| 2949          |                 | 10            | 59                  | Thurai           | 0 - 6.0                  |
| 2950          |                 | 10            | 51                  | Thurai           | 0 - 6.0                  |
| 2951          |                 | 10            | 46                  | Thurai           | 0 - 6.0                  |
| 2952          |                 | 10            | 49                  | Thurai           | 0 - 6.0                  |
| 2953          | 53+200          | 10            | 44                  | Bhagi            | 0 - 6.0                  |
| 2954          |                 | 10            | 38                  | Bhagi            | 0 - 6.0                  |
| 2955          |                 | 10            | 31                  | Bhagi            | 0 - 6.0                  |
| 2956          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 2957          |                 | 10            | 39                  | Bhagi            | 0 - 6.0                  |
| 2958          |                 | 10            | 55                  | Bhagi            | 0 - 6.0                  |

# APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 2959   | 53+400   | 10       | 49           | Bhagi     | 0 - 6.0           |
| 2960   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 2961   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 2962   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 2963   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 2964   |          | 15 above | 49           | Bhagi     | 0 - 6.0           |
| 2965   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 2966   | 53+600   | 10       | 38           | Bhagi     | 0 - 6.0           |
| 2967   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 2968   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 2969   |          | 10       | 34           | Bhagi     | 0 - 6.0           |
| 2970   |          | 10       | 35           | Bhagi     | 0 - 6.0           |
| 2971   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 2972   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 2973   |          | 10       | 58           | Bhagi     | 0 - 6.0           |
| 2974   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 2975   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 2976   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 2977   | 53+800   | 15 above | 38           | Coconut   | 0 - 6.0           |
| 2978   |          | 15 above | 31           | Bhagi     | 0 - 6.0           |
| 2979   |          | 15 above | 46           | Bhagi     | 0 - 6.0           |
| 2980   |          | 10       | 39           | Bhagi     | 0 - 6.0           |
| 2981   |          | 10       | 55           | Bhagi     | 0 - 6.0           |
| 2982   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 2983   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 2984   |          | 10       | 35           | Bhagi     | 0 - 6.0           |
| 2985   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 2986   |          | 10       | 50           | Bhagi     | 0 - 6.0           |
| 2987   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 2988   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 2989   |          | 10       | 42           | Bhagi     | 0 - 6.0           |
| 2990   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 2991   |          | 10       | 39           | Bhagi     | 0 - 6.0           |
| 2992   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 2993   | 54+000   | 10       | 59           | Bhagi     | 0 - 6.0           |
| 2994   |          | 10       | 51           | Bhagi     | 0 - 6.0           |
| 2995   |          | 10       | 46           | Palm      | 0 - 6.0           |
| 2996   |          | 10       | 49           | Palm      | 0 - 6.0           |
| 2997   | 54+200   | 10       | 81           | Palm      | 0 - 6.0           |
| 2998   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 2999   |          | 10       | 31           | Coconut   | 0 - 6.0           |
| 3000   |          | 10       | 38           | Coconut   | 0 - 6.0           |
| 3001   |          | 10       | 56           | Coconut   | 0 - 6.0           |
| 3002   |          | 10       | 50           | Coconut   | 0 - 6.0           |
| 3003   |          | 10       | 34           | Coconut   | 0 - 6.0           |
| 3004   |          | 10       | 35           | Coconut   | 0 - 6.0           |
| 3005   |          | 10       | 38           | Coconut   | 0 - 6.0           |
| 3006   |          | 10       | 44           | Coconut   | 0 - 6.0           |
| 3007   |          | 10       | 58           | Coconut   | 0 - 6.0           |
| 3008   |          | 10       | 46           | Coconut   | 0 - 6.0           |
| 3009   |          | 10       | 49           | Coconut   | 0 - 6.0           |

#### APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 3010   |          | 10       | 44           | Coconut   | 0 - 6.0           |
| 3011   |          | 10       | 38           | Coconut   | 0 - 6.0           |
| 3012   |          | 10       | 31           | Coconut   | 0 - 6.0           |
| 3013   |          | 10       | 46           | Coconut   | 0 - 6.0           |
| 3014   |          | 10       | 39           | Coconut   | 0 - 6.0           |
| 3015   | 54+400   | 15 above | 124          | Bhagi     | 0 - 6.0           |
| 3016   | 54+600   | 10       | 125          | Mango     | 0 - 6.0           |
| 3017   |          | 10       | 136          | Coconut   | 0 - 6.0           |
| 3018   |          | 10       | 152          | Bhagi     | 0 - 6.0           |
| 3019   |          | 10       | 142          | Bhagi     | 0 - 6.0           |
| 3020   |          | 10       | 163          | Bhagi     | 0 - 6.0           |
| 3021   |          | 15 above | 46           | Bhagi     | 0 - 6.0           |
| 3022   |          | 15 above | 49           | Bhagi     | 0 - 6.0           |
| 3023   | 54+800   | 15 above | 42           | Raavi     | 0 - 6.0           |
| 3024   |          | 10       | 142          | Bhagi     | 0 - 6.0           |
| 3025   |          | 10       | 136          | Bhagi     | 0 - 6.0           |
| 3026   |          | 10       | 152          | Bhagi     | 0 - 6.0           |
| 3027   |          | 10       | 142          | Bhagi     | 0 - 6.0           |
| 3028   |          | 10       | 136          | Bhagi     | 0 - 6.0           |
| 3029   |          | 10       | 152          | Bhagi     | 0 - 6.0           |
| 3030   | 55+000   | 15 above | 80           | Tamarind  | 0 - 6.0           |
| 3031   |          | 15 above | 89           | Tamarind  | 0 - 6.0           |
| 3032   |          | 15 above | 81           | Tamarind  | 0 - 6.0           |
| 3033   |          | 15 above | 38           | Tamarind  | 0 - 6.0           |
| 3034   |          | 15 above | 56           | Tamarind  | 0 - 6.0           |
| 3035   |          | 15 above | 50           | Pongamia  | 0 - 6.0           |
| 3036   |          | 10       | 34           | Pongamia  | 0 - 6.0           |
| 3037   |          | 10       | 35           | Pongamia  | 0 - 6.0           |
| 3038   |          | 10       | 185          | Neem      | 0 - 6.0           |
| 3039   |          | 10       | 165          | Neem      | 0 - 6.0           |
| 3040   |          | 10       | 175          | Neem      | 0 - 6.0           |
| 3041   |          | 10       | 145          | Neem      | 0 - 6.0           |
| 3042   |          | 10       | 135          | Neem      | 0 - 6.0           |
| 3043   |          | 10       | 125          | Neem      | 0 - 6.0           |
| 3044   |          | 10       | 196          | Neem      | 0 - 6.0           |
| 3045   |          | 10       | 145          | Thurai    | 0 - 6.0           |
| 3046   |          | 10       | 155          | Thurai    | 0 - 6.0           |
| 3047   |          | 10       | 126          | Coconut   | 0 - 6.0           |
| 3048   | 55+200   | 15 above | 88           | Neem      | 0 - 6.0           |
| 3049   |          | 10       | 75           | Neem      | 0 - 6.0           |
| 3050   |          | 10       | 42           | Neem      | 0 - 6.0           |
| 3051   |          | 10       | 38           | Neem      | 0 - 6.0           |
| 3052   |          | 10       | 56           | Coconut   | 0 - 6.0           |
| 3053   |          | 10       | 50           | Coconut   | 0 - 6.0           |
| 3054   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3055   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 3056   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 3057   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 3058   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 3059   |          | 10       | 46           | Raavi     | 0 - 6.0           |
| 3060   |          | 10       | 34           | Raavi     | 0 - 6.0           |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 3061          | 55+400          | 15 above      | 67                  | Bhagi            | 0 - 6.0                  |
| 3062          |                 | 15 above      | 69                  | Bhagi            | 0 - 6.0                  |
| 3063          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |
| 3064          |                 | 15 above      | 58                  | Coconut          | 0 - 6.0                  |
| 3065          |                 | 15 above      | 46                  | Coconut          | 0 - 6.0                  |
| 3066          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 3067          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 3068          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 3069          |                 | 10            | 31                  | Coconut          | 0 - 6.0                  |
| 3070          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3071          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 3072          |                 | 10            | 55                  | Coconut          | 0 - 6.0                  |
| 3073          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 3074          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3075          | 55+600          | 15 above      | 35                  | Bhagi            | 0 - 6.0                  |
| 3076          |                 | 15 above      | 38                  | Bhagi            | 0 - 6.0                  |
| 3077          |                 | 15 above      | 50                  | Bhagi            | 0 - 6.0                  |
| 3078          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 3079          |                 | 10            | 49                  | Bhagi            | 0 - 6.0                  |
| 3080          |                 | 10            | 42                  | Bhagi            | 0 - 6.0                  |
| 3081          |                 | 10            | 38                  | Bhagi            | 0 - 6.0                  |
| 3082          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 3083          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 3084          |                 | 10            | 55                  | Coconut          | 0 - 6.0                  |
| 3085          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 3086          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3087          |                 | 10            | 51                  | Coconut          | 0 - 6.0                  |
| 3088          |                 | 10            | 51                  | Coconut          | 0 - 6.0                  |
| 3089          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3090          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 3091          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 3092          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 3093          | 55+800          | 15 above      | 155                 | Bhagi            | 0 - 6.0                  |
| 3094          | 56+000          | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3095          |                 | 10            | 34                  | Coconut          | 0 - 6.0                  |
| 3096          |                 | 10            | 35                  | Coconut          | 0 - 6.0                  |
| 3097          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 3098          |                 | 10            | 44                  | Coconut          | 0 - 6.0                  |
| 3099          |                 | 10            | 58                  | Coconut          | 0 - 6.0                  |
| 3100          | 56+400          | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3101          |                 | 10            | 49                  | Acasia           | 0 - 6.0                  |
| 3102          |                 | 10            | 44                  | Acasia           | 0 - 6.0                  |
| 3103          |                 | 10            | 38                  | Neem             | 0 - 6.0                  |
| 3104          | 56+600          | 15 above      | 31                  | Thurai           | 0 - 6.0                  |
| 3105          |                 | 15 above      | 46                  | Thurai           | 0 - 6.0                  |
| 3106          |                 | 10            | 39                  | Bhagi            | 0 - 6.0                  |
| 3107          |                 | 10            | 55                  | Coconut          | 0 - 6.0                  |
| 3108          |                 | 10            | 35                  | Coconut          | 0 - 6.0                  |
| 3109          | 57+000          | 10            | 42                  | Palm             | 0 - 6.0                  |
| 3110          |                 | 10            | 38                  | Ashoka           | 0 - 6.0                  |
| 3111          |                 | 10            | 56                  | Thurai           | 0 - 6.0                  |

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| <b>Sl. No</b> | <b>Chainage</b> | <b>Height</b> | <b>Girth in Cms</b> | <b>Tree Name</b> | <b>Dist. From Centre</b> |
|---------------|-----------------|---------------|---------------------|------------------|--------------------------|
| 3112          | 57+200          | 15 above      | 61                  | Thurai           | 0 - 6.0                  |
| 3113          |                 | 10            | 46                  | Thurai           | 0 - 6.0                  |
| 3114          |                 | 10            | 49                  | Palm             | 0 - 6.0                  |
| 3115          |                 | 10            | 44                  | Palm             | 0 - 6.0                  |
| 3116          |                 | 10            | 38                  | Palm             | 0 - 6.0                  |
| 3117          |                 | 10            | 31                  | Thurai           | 0 - 6.0                  |
| 3118          |                 | 10            | 46                  | Bhagi            | 0 - 6.0                  |
| 3119          | 57+400          | 15 above      | 82                  | Tamarind         | 0 - 6.0                  |
| 3120          |                 | 15 above      | 89                  | Bhagi            | 0 - 6.0                  |
| 3121          |                 | 15 above      | 84                  | Bhagi            | 0 - 6.0                  |
| 3122          |                 | 20 above      | 136                 | Bhagi            | 0 - 6.0                  |
| 3123          |                 | 10            | 125                 | Bhagi            | 0 - 6.0                  |
| 3124          |                 | 10            | 124                 | Bhagi            | 0 - 6.0                  |
| 3125          |                 | 10            | 38                  | Neem             | 0 - 6.0                  |
| 3126          |                 | 10            | 31                  | Neem             | 0 - 6.0                  |
| 3127          |                 | 10            | 46                  | Neem             | 0 - 6.0                  |
| 3128          |                 | 10            | 44                  | Neem             | 0 - 6.0                  |
| 3129          |                 | 10            | 58                  | Ashoka           | 0 - 6.0                  |
| 3130          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3131          | 57+600          | 15 above      | 85                  | Bhagi            | 0 - 6.0                  |
| 3132          |                 | 15 above      | 76                  | Bhagi            | 0 - 6.0                  |
| 3133          |                 | 15 above      | 38                  | Bhagi            | 0 - 6.0                  |
| 3134          |                 | 15 above      | 31                  | Raavi            | 0 - 6.0                  |
| 3135          |                 | 15 above      | 46                  | Tamarind         | 0 - 6.0                  |
| 3136          |                 | 15 above      | 39                  | Coconut          | 0 - 6.0                  |
| 3137          |                 | 15 above      | 55                  | Coconut          | 0 - 6.0                  |
| 3138          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 3139          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3140          |                 | 10            | 35                  | Coconut          | 0 - 6.0                  |
| 3141          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 3142          |                 | 10            | 50                  | Coconut          | 0 - 6.0                  |
| 3143          |                 | 10            | 46                  | Coconut          | 0 - 6.0                  |
| 3144          |                 | 10            | 49                  | Coconut          | 0 - 6.0                  |
| 3145          |                 | 10            | 42                  | Coconut          | 0 - 6.0                  |
| 3146          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 3147          |                 | 10            | 39                  | Coconut          | 0 - 6.0                  |
| 3148          |                 | 10            | 31                  | Coconut          | 0 - 6.0                  |
| 3149          |                 | 10            | 59                  | Coconut          | 0 - 6.0                  |
| 3150          |                 | 10            | 42                  | Coconut          | 0 - 6.0                  |
| 3151          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 3152          |                 | 10            | 56                  | Coconut          | 0 - 6.0                  |
| 3153          |                 | 10            | 50                  | Coconut          | 0 - 6.0                  |
| 3154          |                 | 10            | 34                  | Coconut          | 0 - 6.0                  |
| 3155          |                 | 10            | 35                  | Coconut          | 0 - 6.0                  |
| 3156          |                 | 10            | 42                  | Coconut          | 0 - 6.0                  |
| 3157          |                 | 10            | 38                  | Coconut          | 0 - 6.0                  |
| 3158          | 57+800          | 15 above      | 77                  | Neredu           | 0 - 6.0                  |
| 3159          |                 | 15 above      | 50                  | Palm             | 0 - 6.0                  |
| 3160          |                 | 15 above      | 46                  | Neem             | 0 - 6.0                  |
| 3161          |                 | 15 above      | 49                  | Coconut          | 0 - 6.0                  |
| 3162          |                 | 15 above      | 44                  | Coconut          | 0 - 6.0                  |

# APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 3163   |          | 10       | 38           | Coconut   | 0 - 6.0           |
| 3164   |          | 10       | 31           | Coconut   | 0 - 6.0           |
| 3165   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3166   |          | 10       | 34           | Bhagi     | 0 - 6.0           |
| 3167   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 3168   |          | 10       | 58           | Bhagi     | 0 - 6.0           |
| 3169   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3170   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 3171   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 3172   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 3173   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 3174   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3175   | 58+000   | 15 above | 76           | Oka       | 0 - 6.0           |
| 3176   |          | 15 above | 69           | Ashoka    | 0 - 6.0           |
| 3177   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3178   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 3179   |          | 10       | 58           | Bhagi     | 0 - 6.0           |
| 3180   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3181   |          | 10       | 38           | Palm      | 0 - 6.0           |
| 3182   |          | 10       | 31           | Coconut   | 0 - 6.0           |
| 3183   |          | 10       | 46           | Coconut   | 0 - 6.0           |
| 3184   |          | 10       | 39           | Coconut   | 0 - 6.0           |
| 3185   |          | 10       | 55           | Coconut   | 0 - 6.0           |
| 3186   |          | 10       | 49           | Coconut   | 0 - 6.0           |
| 3187   |          | 10       | 46           | Coconut   | 0 - 6.0           |
| 3188   |          | 10       | 35           | Coconut   | 0 - 6.0           |
| 3189   | 58+400   | 15 above | 64           | Mango     | 0 - 6.0           |
| 3190   |          | 15 above | 50           | Palm      | 0 - 6.0           |
| 3191   |          | 15 above | 46           | Palm      | 0 - 6.0           |
| 3192   |          | 10       | 49           | Palm      | 0 - 6.0           |
| 3193   |          | 10       | 42           | Palm      | 0 - 6.0           |
| 3194   |          | 10       | 38           | Neredu    | 0 - 6.0           |
| 3195   |          | 10       | 31           | Mothuga   | 0 - 6.0           |
| 3196   |          | 10       | 46           | Mothuga   | 0 - 6.0           |
| 3197   |          | 10       | 39           | Bhagi     | 0 - 6.0           |
| 3198   |          | 10       | 55           | Bhagi     | 0 - 6.0           |
| 3199   |          | 10       | 35           | Bhagi     | 0 - 6.0           |
| 3200   | 58+600   | 15 above | 68           | Palm      | 0 - 6.0           |
| 3201   |          | 15 above | 75           | Palm      | 0 - 6.0           |
| 3202   |          | 10       | 56           | Palm      | 0 - 6.0           |
| 3203   |          | 10       | 46           | Palm      | 0 - 6.0           |
| 3204   |          | 10       | 49           | Almond    | 0 - 6.0           |
| 3205   |          | 10       | 44           | Raavi     | 0 - 6.0           |
| 3206   |          | 10       | 38           | Coconut   | 0 - 6.0           |
| 3207   |          | 10       | 31           | Coconut   | 0 - 6.0           |
| 3208   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3209   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 3210   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 3211   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3212   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 3213   |          | 10       | 58           | Bhagi     | 0 - 6.0           |



**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| Sl. No | Chainage | Height   | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|----------|--------------|-----------|-------------------|
| 3214   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3215   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 3216   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 3217   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3218   |          | 10       | 39           | Bhagi     | 0 - 6.0           |
| 3219   |          | 10       | 55           | Bhagi     | 0 - 6.0           |
| 3220   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 3221   |          | 10       | 42           | Bhagi     | 0 - 6.0           |
| 3222   | 58+800   | 10       | 38           | Bhagi     | 0 - 6.0           |
| 3223   |          | 10       | 56           | Bhagi     | 0 - 6.0           |
| 3224   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3225   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 3226   |          | 10       | 44           | Bhagi     | 0 - 6.0           |
| 3227   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 3228   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 3229   | 59+000   | 15 above | 85           | Palm      | 0 - 6.0           |
| 3230   |          | 10       | 38           | Bhagi     | 0 - 6.0           |
| 3231   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 3232   | 59+200   | 15 above | 185          | Pongamia  | 0 - 6.0           |
| 3233   |          | 15 above | 135          | Pongamia  | 0 - 6.0           |
| 3234   |          | 15 above | 124          | Pongamia  | 0 - 6.0           |
| 3235   |          | 15 above | 75           | Pongamia  | 0 - 6.0           |
| 3236   |          | 15 above | 74           | Coconut   | 0 - 6.0           |
| 3237   |          | 10       | 76           | Thurai    | 0 - 6.0           |
| 3238   |          | 10       | 77           | Neredu    | 0 - 6.0           |
| 3239   |          | 10       | 79           | Bhagi     | 0 - 6.0           |
| 3240   |          | 10       | 55           | Bhagi     | 0 - 6.0           |
| 3241   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 3242   | 59+400   | 15 above | 80           | Thurai    | 0 - 6.0           |
| 3243   |          | 20 above | 163          | Thurai    | 0 - 6.0           |
| 3244   |          | 10       | 125          | Pongamia  | 0 - 6.0           |
| 3245   |          | 10       | 50           | Pongamia  | 0 - 6.0           |
| 3246   |          | 10       | 46           | Pongamia  | 0 - 6.0           |
| 3247   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 3248   |          | 10       | 42           | Bhagi     | 0 - 6.0           |
| 3249   | 59+600   | 15 above | 147          | Neem      | 0 - 6.0           |
| 3250   |          | 15 above | 142          | Neem      | 0 - 6.0           |
| 3251   |          | 20 above | 162          | Coconut   | 0 - 6.0           |
| 3252   |          | 20 above | 142          | Coconut   | 0 - 6.0           |
| 3253   |          | 10       | 88           | Coconut   | 0 - 6.0           |
| 3254   |          | 10       | 31           | Bhagi     | 0 - 6.0           |
| 3255   |          | 10       | 55           | Bhagi     | 0 - 6.0           |
| 3256   |          | 10       | 49           | Bhagi     | 0 - 6.0           |
| 3257   | 59+800   | 15 above | 81           | Giriship  | 0 - 6.0           |
| 3258   |          | 10       | 46           | Bhagi     | 0 - 6.0           |
| 3259   |          | 10       | 58           | Bhagi     | 0 - 6.0           |
| 3260   |          | 10       | 71           | Bhagi     | 1 - 6.0           |

at junctions and bypass tree enumeration is 105(refer chapter no 6)

**APPENDIX 4.1: TREE ENUMERATION:Kakinada-Rajamundry**

| Sl. No | Chainage | Height | Girth in Cms | Tree Name | Dist. From Centre |
|--------|----------|--------|--------------|-----------|-------------------|
|--------|----------|--------|--------------|-----------|-------------------|

total trees to be enumerated 4057+105=4162

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## APPENDIX E.1: AGGREGATE QUARRY MANAGEMENT

The Contractor will finalize the locations from the list given by DPR Consultant's for procuring materials. The Contractor shall establish a new quarry only with the prior consent of the EO only in cases when: (i) Lead from existing quarries is uneconomical and (ii) Alternative material sources are not available. The Contractor shall prepare a Redevelopment Plan for the quarry site and get it approved by the EO/Supervision Consultant.

The construction schedule and operations plans to be submitted to the EO prior to commencement of work shall contain a detailed work plan for procuring materials that includes procurement, transportation and storage of quarry materials.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operation & redevelopment plan (if a new quarry is opened).....                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>• Photograph of the quarry site prior to commencement</li> <li>• The quarry boundaries as well as location of the materials deposits, working equipments, stockpiling, access roads and final shape of the pit.</li> <li>• Drainage and erosion control measures at site.</li> <li>• Safety Measures during quarry operation.</li> <li>• Design for redevelopment of exhaust site.</li> </ul> <p><i>Option-A: Revegetating the quarry to merge with surrounding landscape:</i> This is done by conserving and reapplying the topsoil for the vegetative growth.</p> <p><i>Option-B: Developing exhausted quarries as water bodies:</i> The pit shall be reshaped and developed into pond, for harvesting rainwater. This option shall only be considered where the location of quarry is at the lowest point, i.e. surrounding areas/natural drainage slopes towards it.</p> |

### CONSTRUCTION STAGE

Development of site: To minimize the adverse impact during excavation of material following measures are need to be undertaken:

- Adequate drainage system shall be provided to prevent the flooding of the excavated area
- At the stockpiling locations, the Contractor shall construct sediment barriers to prevent the erosion of excavated material due to runoff
- Construction of offices, laboratory, workshop and rest places shall be done in the up-wind of the plant to minimize the adverse impact due to dust and noise.
- The access road to the plant shall be constructed taking into consideration location of units and also slope of the ground to regulate the vehicle movement within the plant.
- Incase of storage of blasting material, all precautions shall be taken as per The Explosive Rules, 1983.

### QUARRY OPERATIONS INCLUDING SAFETY

- Overburden shall be removed and disposed inline with Guidelines for Debris Disposal Site and management giving in Annexure-8.3
- During excavation, slopes shall be flatter than 20 degrees to prevent their sliding. Incases where quarry strata are good and where chances of sliding are less this restriction can be ignored.
- Incase of blasting, procedure and safety measures shall be taken as per The Explosive Rules, 1983
- The contractor shall ensure that all workers related safety measures shall be done as per guidelines for Workers and Safety attached as Annexure-8.8
- The Contractor shall ensure maintenance of crushers regularly as per manufacturer's recommendation.

Topsoil will be excavated and preserved during transportation of the material measures shall be taken to minimize the generation of dust and prevent accidents.

The EO and the Supervision Consultant shall review the quarry site for the management measures during quarry operation, including the compliance to pollution norms.

### **POST CONSTRUCTION STAGE**

The Contractor shall restore all haul roads constructed for transporting the material from the quarries to construction site to their original state.

The EO and the Supervision Consultant shall be entrusted the responsibility of reviewing the quarry site for the progress of implementation of Redevelopment Plan. These shall include the following two cases;

- Redevelopment of quarries opened by the Contractor for the project
- Redevelopment of existing quarries operated by other agencies

In the first case, the Contractor shall be responsible for the Redevelopment Plan prior to completion after five years, during the defect liability period. The EO shall be responsible for reviewing this case of redevelopment prior to the issuing the defect liability certificate.

In the second case, the redevelopment of exhaust quarry shall be the responsibility of the agency providing the permit to ensure the implementation of Redevelopment Plan.

**Case I-** For each aggregate-cum-quarry sand source, the plan should contain a separate section. If the quarry is an existing one and is managed directly by the Contractor, the plan should contain the following:

| Sr. No. | ITEM                                                                                                                         | UNIT | DETAILS | REMARKS BY SC/EO, IF ANY |
|---------|------------------------------------------------------------------------------------------------------------------------------|------|---------|--------------------------|
| 1.      | NAME / IDENTITY OF THE LOCATION                                                                                              |      |         |                          |
| 2.      | NEAREST PROJECT ROAD CHAINAGE.                                                                                               |      |         |                          |
| 3.      | NAME OF THE OWNER                                                                                                            |      |         |                          |
| 4.      | AREA INVOLVED                                                                                                                | CUM  |         |                          |
| 5.      | ARRANGEMENT WITH THE OWNER (AGREEMENT WITH LAND OWNER SHOULD BE ATTACHED AS AN ANNEXURE)                                     |      |         |                          |
| 6.      | QUANTITY OF MATERIAL TO BE WITHDRAWN VIS-À-VIS THE MATERIAL AVAILABLE                                                        | CUM  |         |                          |
| 7.      | MACHINERY & EQUIPMENT TO BE USED                                                                                             |      |         |                          |
| 8.      | COPY OF THE CONSENTS TO ESTABLISH AND OPERATE SHOULD BE ATTACHED AS AN ANNEXURE.                                             |      |         |                          |
| 9.      | COPY OF THE LICENCE FROM MINING & GEOLOGY, POLICE & FIRE DEPT.                                                               |      |         |                          |
| 10.     | CONDITIONS LAID DOWN IN THE CLEARANCES / LICENCES AND PLANS TO ENSURE COMPLIANCE                                             |      |         |                          |
| 11.     | PHOTOGRAPHS OF THE QUARRY PRIOR TO COMMENCING OPERATIONS.                                                                    |      |         |                          |
| 12.     | ACCESS ROADS CONDITION AND PROPOSED MAINTENANCE                                                                              |      |         |                          |
| 13.     | TREE-CUTTING REQUIRED, IF ANY, ALONG WITH COMPENSATION MEASURES                                                              | NO.S |         |                          |
| 14.     | INFORMATION ON WHETHER OR NOT THE QUARRY WILL BE CLOSED UNDER THIS PROJECT. IF YES, THE PROPOSED CLOSURE & RESTORATION PLAN. |      |         |                          |
| 15.     | SKETCH OF THE LAYOUT OF THE QUARRY                                                                                           |      |         |                          |
| 16.     | DESCRIPTION OF THE OPERATING PRACTICES TO BE ADOPTED.                                                                        |      |         |                          |

Attach Photograph of Proposed Site, Location Map, and Agreement with land owner

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| REMARKS |
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NAME .....

DESIGNATION .....

CONTRACTOR

NAME .....

CONSTRUCTION SUPERVISION  
CONSULTANT

NAME .....

EXECUTIVE ENGINEER

**Case II** -If the quarry is an existing one and is managed directly by a sub-contractor from whom the Contractor is sourcing the materials, the plan should contain the following:

| Sr. No. | ITEM                                                                                                                                                                                                                                         | UNIT | DETAILS | REMARKS BY SC/EO, IF ANY |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------|--------------------------|
| 1.      | NAME / IDENTITY OF THE LOCATION                                                                                                                                                                                                              |      |         |                          |
| 2.      | NEAREST PROJECT ROAD CHAINAGE.                                                                                                                                                                                                               |      |         |                          |
| 3.      | NAME OF THE OWNER                                                                                                                                                                                                                            |      |         |                          |
| 4.      | AREA INVOLVED                                                                                                                                                                                                                                |      |         |                          |
| 5.      | ARRANGEMENT WITH THE OWNER                                                                                                                                                                                                                   |      |         |                          |
| 6.      | ARRANGEMENT WITH THE SUB-CONTRACTOR (AGREEMENT WITH SUB-CONTRACTOR SHOULD BE ATTACHED AS AN ANNEXURE AND SHOULD NECESSARILY REQUIRE THE ADOPTION OF GOOD QUARRY MANAGEMENT PRACTICES – A DESCRIPTION OF THE REQUIREMENTS SHOULD BE INCLUDED) |      |         |                          |
| 7.      | QUANTITY OF MATERIAL TO BE WITHDRAWN VIS-À-VIS THE MATERIAL AVAILABLE                                                                                                                                                                        |      |         |                          |
| 8.      | MACHINERY & EQUIPMENT TO BE USED                                                                                                                                                                                                             |      |         |                          |
| 9.      | DRAINAGE PLANS                                                                                                                                                                                                                               |      |         |                          |
| 10.     | TOP SOIL MANAGEMENT                                                                                                                                                                                                                          |      |         |                          |
| 11.     | DESCRIPTION OF THE OPERATING PRACTICES                                                                                                                                                                                                       |      |         |                          |
| 12.     | HEALTH FACILITIES                                                                                                                                                                                                                            |      |         |                          |
| 13.     | SAFETY PROVISIONS MADE INCLUDING FIRE PROTECTION SYSTEMS AND THE AVAILABILITY OF DIFFERENT PERSONAL PROTECTIVE EQUIPMENT SUCH AS HELMETS, EAR PLUGS / MUFFS AND FACE MASKS                                                                   |      |         |                          |
| 14.     | COPY OF THE SUB-CONTRACTOR LICENCES FROM PCB, MINING & GEOLOGY, POLICE & FIRE DEPT.                                                                                                                                                          |      |         |                          |
| 15.     | CONDITIONS LAID DOWN IN THE CLEARANCES / LICENCES                                                                                                                                                                                            |      |         |                          |
| 16.     | MONITORING PLANS FOR AIR QUALITY                                                                                                                                                                                                             |      |         |                          |
| 17.     | INFORMATION ON WHETHER OR NOT THE QUARRY WILL BE CLOSED UNDER THIS PROJECT. IF YES, THE PROPOSED CLOSURE & RESTORATION PLAN.                                                                                                                 |      |         |                          |
| 18.     | SKETCH OF THE LAYOUT OF THE QUARRY                                                                                                                                                                                                           |      |         |                          |
| 19.     | PHOTOGRAPHS OF THE QUARRY BEFORE MATERIAL SOURCING FOR THE PROJECT                                                                                                                                                                           |      |         |                          |

Attach Photograph of Proposed Site, Location Map, and Agreement with land owner

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NAME .....

DESIGNATION .....

CONTRACTOR

NAME .....

CONSTRUCTION SUPERVISION  
CONSULTANT

NAME .....

EXECUTIVE ENGINEER



**Case III** - If the quarry is a new one and is managed directly by the Contractor from whom the contractor is sourcing the materials, the plan should contain the following:

| Sr. No. | ITEM                                                                                                                                                                                                                                             | UNIT     | DETAILS | REMARKS BY SC/EO, IF ANY |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------|--------------------------|
| 1.      | NAME / IDENTITY OF THE LOCATION                                                                                                                                                                                                                  |          |         |                          |
| 2.      | NEAREST PROJECT ROAD CHAINAGE.                                                                                                                                                                                                                   |          |         |                          |
| 3.      | NAME OF THE OWNER                                                                                                                                                                                                                                |          |         |                          |
| 4.      | AREA INVOLVED                                                                                                                                                                                                                                    | SQ<br>M  |         |                          |
| 5.      | EXISTING LAND USE (VERIFICATION REQUIRED FROM LAND RECORDS WITH REVENUE DEPARTMENT)                                                                                                                                                              |          |         |                          |
| 6.      | LAND USE OF THE AREA SURROUNDING THE PROPOSED SITE INCLUDING A MAP                                                                                                                                                                               |          |         |                          |
| 7.      | ACCESS ROADS – EXISTING CONDITIONS, PROPOSED DEVELOPMENT AND MAINTENANCE                                                                                                                                                                         |          |         |                          |
| 8.      | TREE-CUTTING AND VEGETATION CLEARANCE REQUIRED, IF ANY, ALONG WITH COMPENSATION MEASURES                                                                                                                                                         | NO<br>.S |         |                          |
| 9.      | ARRANGEMENT WITH THE OWNER (AGREEMENT SHOULD NECESSARILY INCLUDE COMMITMENT OF THE CONTRACTOR TO ADOPT GOOD QUARRY MANAGEMENT PRACTICES – A DESCRIPTION OF THE REQUIREMENTS SHOULD BE INCLUDED - AND SHOULD INDICATE PROPOSED RESTORATION PLANS) |          |         |                          |
| 10.     | QUANTITY OF MATERIAL TO BE WITHDRAWN VIS-À-VIS THE MATERIAL AVAILABLE                                                                                                                                                                            | CU<br>M  |         |                          |
| 11.     | PARTICULAR AREAS TO BE QUARRIED SHOULD BE CLEARLY IDENTIFIED                                                                                                                                                                                     |          |         |                          |
| 12.     | PRE-ESTABLISHMENT ACTIVITIES, E.G. PURCHASE / LEASE OF NEARBY PROPERTIES                                                                                                                                                                         |          |         |                          |
| 13.     | MACHINERY & EQUIPMENT TO BE USED                                                                                                                                                                                                                 |          |         |                          |
| 14.     | DRAINAGE PLANS                                                                                                                                                                                                                                   |          |         |                          |
| 15.     | TOP SOIL MANAGEMENT                                                                                                                                                                                                                              |          |         |                          |
| 16.     | DESCRIPTION OF THE OPERATING PRACTICES                                                                                                                                                                                                           |          |         |                          |
| 17.     | HEALTH FACILITIES                                                                                                                                                                                                                                |          |         |                          |
| 18.     | SAFETY PROVISIONS MADE INCLUDING FIRE PROTECTION SYSTEMS AND THE AVAILABILITY OF DIFFERENT PERSONAL PROTECTIVE EQUIPMENT SUCH AS HELMETS, EAR PLUGS / MUFFS AND FACE MASKS                                                                       |          |         |                          |
| 19.     | MONITORING PLANS FOR AIR QUALITY                                                                                                                                                                                                                 |          |         |                          |
| 20.     | COPY OF THE SUB-CONTRACTOR LICENSES FROM PCB, MINING & GEOLOGY, POLICE & FIRE DEPT.                                                                                                                                                              |          |         |                          |
| 21.     | CONDITIONS LAID DOWN IN THE CLEARANCES / LICENCES                                                                                                                                                                                                |          |         |                          |
| 22.     | INFORMATION ON WHETHER OR NOT THE QUARRY WILL BE CLOSED UNDER THIS PROJECT. IF YES, THE PROPOSED CLOSURE & RESTORATION PLAN.                                                                                                                     |          |         |                          |
| 23.     | SKETCH OF THE LAYOUT OF THE QUARRY                                                                                                                                                                                                               |          |         |                          |
| 24.     | CONCERNS OF THE LOCAL PEOPLE LIVING IN THE IMMEDIATE / NEAR VICINITY SHOULD BE IDENTIFIED AND APPROPRIATE MEASURES SHOULD BE REFLECTED                                                                                                           |          |         |                          |
| 25.     | TREE-CUTTING REQUIRED, IF ANY, ALONG WITH COMPENSATION MEASURES                                                                                                                                                                                  |          |         |                          |
| 26.     | PHOTOGRAPHS OF THE QUARRY BEFORE THE PROJECT                                                                                                                                                                                                     |          |         |                          |

Attach Photograph of Proposed Site, Location Map, and Agreement with land owner

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NAME .....  
DESIGNATION .....  
CONTRACTOR

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SIGNATURE .....

NAME .....

CONSTRUCTION SUPERVISION  
CONSULTANT

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## APPENDIX E.2: BORROW AREAS MANAGEMENT

Borrow areas will be finalized either from the list of locations recommended by DPR consultants or new areas identified by contractor. The finalization of locations identified by DPR consultant or identified by contractor depends upon the formal agreement between landowners and contractor and its suitability from civil engineering as well as environmental consideration. Meeting the guidelines/notifications as stipulated from time to time by the Ministry of Environment and Forests, Government of India, and local bodies, as applicable shall be the sole responsibility of the contractor.

Besides this certain precautions have to be taken to restrict unauthorized borrowing by the contractor. No borrow area shall be opened without permission of the Engineer/EO. The engineer in addition to the established practices, rules and regulation will also consider following criteria before approving the Borrow areas.

To avoid any embankment slippage, the borrow areas will not be dug continuously, and the size and shape of borrow pits will be decided by the Engineer. Redevelopment of the borrow areas to mitigate the impacts will be the responsibility of the contractor. The contractor shall evolve site-specific redevelopment plans for each borrow area location, which shall be implemented after the approval of the Supervision Consultant.

To ensure that the spills, which might result from the transport of borrow and quarry materials do not impact the settlements, it will be ensured that the excavation and carrying of earth will be done during day-time only. The unpaved surfaces used for the haulage of borrow materials will be maintained properly.

Borrowing of earth shall be carried out at locations recommended as follows:

**Non-Cultivable Lands:** Borrowing of earth will be carried out upto a depth of 2.0 m from the existing ground level. Borrowing of earth shall not be done continuously. Ridges of not less than 8m width shall be left at intervals not exceeding 300 m. Small drains shall be cut through the ridges, if necessary, to facilitate drainage. Borrow pits shall have slopes not steeper than 1 vertical in 4 horizontal.

**Productive Lands:** Borrowing of earth shall be avoided on productive lands. However, in the event of borrowing from productive lands, under circumstances as described above, topsoil shall be preserved in stockpiles. The conservation of topsoil shall be carried out as described in section of this chapter. At such locations, the depth of borrow pits shall not exceed 45 cm and it may be dug out to a depth of not more than 30 cm after stripping the 15 cm top soil aside.

**Elevated Lands:** At locations where private owners desire their fields to be levelled, the borrowing shall be done to a depth of not more than 2 m or up to the level of surrounding fields.

**Borrow pits along Roadside:** Borrow pits shall be located 5m away from the toe of the embankment. Depth of the pit should be such that the bottom of the pit shall not fall within an imaginary line of slope 1 vertical to 4 horizontal projected from the edge of the final section of the bank. Borrow pits should not be dug continuously. Ridges of not less than 8 m width should be left at intervals not exceeding 300 m. Small drains should be cut through the ridges to facilitate drainage.

**Borrow pits on the riverside:** The borrow pit should be located not less than 15m from the toe of the bank, distance depending on the magnitude and duration of flood to be withstood.

**Community / Private Ponds:** Borrowing can be carried out at locations, where the private owners (or in some cases, the community) desire to develop lands (mostly low-lying areas) for pisciculture purposes and for use as fishponds.

**Borrow Areas near Settlements:** Borrow pit location shall be located at least 1.0 km from villages and settlements. If unavoidable, they should not be dug for more than 30 cm and should be drained.

After identification of borrow areas based on guidelines. Contractor will fill reporting format as under and submit the same for approval to the “Engineer” Once approved the contractor will adhere to the recommendation for borrow area to the satisfaction of Engineer.

- (1) In no case the depth of borrow area should exceed 2m from the existing ground level.
- (2) Borrow pits slope should be maintained, no steeper than 1 Vertical: 2 Horizontal.
- (3) Water pooling to be avoided/managed so that NO disease spread due to water stagnation.

- (4) Precautionary measures as the covering of vehicles will be taken to avoid spillage during transportation of borrow area.
- (5) The unpaved surfaces used for the haulage of borrow materials should be maintained properly for dust suppression.
- (6) Haulage of material to embankments or other areas of fill shall proceed only when sufficient spreading and compaction facility is operating at the place of deposition, to minimize dust pollution.
- (7) During rains appropriate measures to be taken to minimize soil erosion, silt fencing to be provided as directed by Engineer/EO.

The Contractor will keep record of photographs of various stages i.e., before using materials from the location (pre-project), for the period borrowing activities (construction Phase) and after rehabilitation (post development), to ascertain the pre and post borrowing status of the area.

**Format EM1: Reporting for Borrow Areas****(To be Submitted by Contractor for taking consent for opening of Borrow area)**

Construction Stage Report: Date \_\_\_\_ Month\_\_\_\_ Year\_\_\_\_

Site Layout of Borrow Area and Proposed Borrow Area Redevelopment Plan to be attached with format

Format to be submitted before target date as (decided by PIU) for establishing Borrow Areas

Borrow Area No. **BA**

Location of Borrow Area (Km\_\_\_\_ Package \_\_\_\_)

| Sl. No | Item                                                             | Unit                                     | Details | Remarks by SC/EO, if any |
|--------|------------------------------------------------------------------|------------------------------------------|---------|--------------------------|
| 1      | Details of Borrow Area                                           |                                          |         |                          |
| a      | Date of Borrow Area becoming operational dd/mm/yy                |                                          |         |                          |
| b      | Current Landuse                                                  |                                          |         |                          |
| c      | Distance from Nearest Settlement                                 | Km                                       |         |                          |
| d      | No of settlements within 200m of Haul Road                       | No.                                      |         |                          |
| e      | No of settlements within 500m of Borrow Area                     | No.                                      |         |                          |
| f      | Total Capacity                                                   | cum                                      |         |                          |
| g      | No of Trees with girth more than 0.3 m                           | No.                                      |         |                          |
| h      | Length of Haul Road                                              | km                                       |         |                          |
| i      | Width of Haul road                                               | m                                        |         |                          |
| j      | Type of Haul Road                                                | metal/dirt                               |         |                          |
| k      | Size of Borrow Area                                              | sqkm                                     |         |                          |
| l      | Area of Borrow Area                                              | km x km                                  |         |                          |
| m      | Quantity Available                                               | cum                                      |         |                          |
| n      | Distance of Nearest Water Source                                 | Type/Size/Capacity/Present Use/Ownership |         |                          |
| o      | Quantity of top soil removed                                     | cum                                      |         |                          |
| p      | Detail of storage of topsoil                                     |                                          |         |                          |
| q      | Daily/occasional use of the Borrow Area by the community, if any | -                                        |         |                          |
| r      | Probable reuse of Borrow pit-ask community                       | -                                        |         |                          |
| s      | Drainage channels/slope/characteristics of the area              | -                                        |         |                          |
| 2      | Enhancement Elements                                             |                                          |         |                          |
| a      | Quantity of top soil removed                                     | sq.m                                     |         |                          |
| b      | Detail of storage of topsoil                                     | sq.m                                     |         |                          |
| c      | Adjoining land use/Natural elements                              |                                          |         |                          |
| d      | Near by catchment for storing water                              |                                          |         |                          |
| e      | Erosion Control Programme                                        |                                          |         |                          |
| f      | Preventive measures for                                          |                                          |         |                          |
| i      | Leaching                                                         |                                          |         |                          |
| ii     | Mosquito Breeding                                                |                                          |         |                          |
| iii    | Water run-off/contamination                                      |                                          |         |                          |
| iv     | Any other environmental degradation                              |                                          |         |                          |
| 3      | Details of workforce                                             |                                          |         |                          |

| Sl. No | Item                                          | Unit | Details | Remarks by SC/EO, if any |
|--------|-----------------------------------------------|------|---------|--------------------------|
| a      | Total No of Labourers                         | No.  |         |                          |
| b      | Total no of Male Workers                      | No.  |         |                          |
| c      | No of Male Workers below 18 years of age      | No.  |         |                          |
| d      | Total No of Female Workers                    | No.  |         |                          |
| e      | No of Female workers below 18 years of age    | No.  |         |                          |
| 4      | Details of redevelopment, Plan to be enclosed |      |         |                          |

Attach Photograph of Proposed Site, Location Map, and Agreement with land owner

|                              |             |
|------------------------------|-------------|
| REHABILITATION PLAN MEASURES |             |
| LOCATION 1:                  | LOCATION 4: |
| LOCATION 2:                  | LOCATION 5: |
| LOCATION 3:                  | LOCATION 6: |

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| REMARKS |
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DESIGNATION .....

CONTRACTOR

CONSTRUCTION SUPERVISION  
CONSULTANT

EXECUTIVE ENGINEER

### APPENDIX E.3: DEBRIS DISPOSAL SITES AND MANAGEMENT

The locations of Disposal sites have to be selected such that:

- Productive land to be avoided and available wasteland to be given preference.
- Disposal sites to be located at least 1000m away from sensitive locations like Settlements, Water body, notified forest areas, Sanctuaries or any other sensitive locations.
- Should be located in the downwind side of nearest settlement locations.
- Disposal sites do not contaminate any water sources, rivers etc for this, site should be located away from water body, disposal site should be lined properly to prevent infiltration of water.
- Public perception about the location of debris disposal site has to be obtained before finalizing the location.
- Permission from the Villager/local community is to be obtained in writing by contractor for finalising the disposal site identified.
- The Plan must be approved by EO/Supervision Consultant and PIU- APRDC.

#### PRECAUTIONS TO BE ADOPTED DURING DISPOSAL OF DEBRIS / WASTE MATERIAL

The contractor shall take the following precautions while disposing off the waste material

- During the site clearance and disposal of debris, the contractor will take full care to ensure that public or private properties are not damaged/affected, there is no dwellings below the dumpsite and that the traffic is not interrupted.
- Contractor will dispose off debris only to the identified places or at other places only with prior permission of Engineer/EO.
- In the event of any spoil or debris from the sites being deposited on any adjacent land, the contractor will immediately remove all such spoil debris and restore the affected area to its original state to the satisfaction of the Engineer/EO.
- The contractor will at all times ensure that the entire existing drains within and adjacent to the site are kept safe and free from any debris.
- Contractor will utilize effective water sprays during the delivery and handling of materials when dust is likely to be created and to dampen stored materials during dry and windy weather.
- Materials having the potential to produce dust will not be loaded to a level higher than the side and tail boards and will be covered with a tarpaulin in good condition.
- Care should always be taken to maintain the hydrological flow in the area.

#### REHABILITATION OF DISPOSAL SITES

The dumpsites filled only upto the ground level could be rehabilitated as per guidelines below and to be decided by the engineer and the supervision consultant

- The dumpsites have to be suitably rehabilitated by planting local species of shrubs and other plants. Local species of trees has also to be planted so that the landscape is coherent and is in harmony with its various components.
- In cases where a dumpsite is near to the local village community settlements, it could be converted into a play field by spreading the dump material evenly on the ground. Such playground could be made coherent with the landscape by planting trees all along the periphery of the playground.
- Material excavated for foundation of bridge works should not be dumped in the water course; if same has to be refilled then precaution has to be taken so that the excavated material should not be carried away by flowing/rainy water, thereby silting the water course.
- Care should always be taken to maintain the hydrological flow in the area.

**Format EM1: Selection of dump site locations**

From \_\_\_\_\_ To \_\_\_\_\_

(Give chainage and nearest settlements from both ends)

| Criteria on which information for each site is to be collected      |  |  |  | Site 1 | Site 2 | Site 3 | Site 4 |
|---------------------------------------------------------------------|--|--|--|--------|--------|--------|--------|
| Area covered (m <sup>2</sup> )                                      |  |  |  |        |        |        |        |
| Total Material that can be dumped within the site (m <sup>3</sup> ) |  |  |  |        |        |        |        |
| Depth to which dumping is feasible (m)                              |  |  |  |        |        |        |        |
| Distance of nearest watercourse (m)                                 |  |  |  |        |        |        |        |
| Nearest Settlement (m)                                              |  |  |  |        |        |        |        |
| Date/s of Community Consultation/s                                  |  |  |  |        |        |        |        |
| Whether the community is agreeable to siting of dumping site (Y/N)  |  |  |  |        |        |        |        |
| Date of Permission from Village Council President(VCP)              |  |  |  |        |        |        |        |
| Proposed future use of the Site                                     |  |  |  |        |        |        |        |

**Selected Site (tick any one column only)**

Certified that the above information is correct to the best of my knowledge and belief.

(Contractor)

Verified:

Date:

**Recommendation on the suitability of the site**

Signed:

Date:

Name &amp; Designation:

Decision Taken (tick one):

Approved/Not Approved

Signed:

Date:

**Name and Designation of Deciding Authority****Enclosures**

(Tick as appropriate)

- 1 Maps of each location
- 2 Photographs
  - a Each dumping location
  - b Each community consultation
- 3 Photocopies of permissions from VCPs

**Name and Designation of Verifier:**

1.



## APPENDIX E.4: SITE CLEARANCE AND TREE FELLING

### 1. VEGETATION CLEARANCE

Vegetation clearance shall comprise uprooting of vegetation, grass, brushwood, shrubs, stumps, trees and saplings of girth upto 30 cm. measured at a height of one meter above the ground level. Where only clearance of grass is involved it shall be measured and paid for separately. The procedure/ steps involved for uprooting, skating and felling trees are described below.

#### 1.1 Uprooting of Vegetation

- The roots of trees and saplings shall be removed to a depth of 60 cm. Below ground level or 30 cm. below formation level or 15 cm below sub grade level, whichever is lower.
- All holes or hollows formed due to removal of roots shall be filled up with earth rammed and levelled.
- Trees, shrubs, poles, fences, signs, monuments, pipe lines, cables etc. within or adjacent to the area, which are not required to be disturbed during vegetation clearance shall be properly protected by the contractor at his own cost.

#### Staking and Disposal

- All useful materials obtained from clearing and grubbing operation shall be staked in the manner as directed by the Engineer.
- Trunks and branches of trees shall be cleared of limbs and tops stacked properly at the places indicated by the Engineer- in – charge. These materials shall be the property of the Government.
- All unserviceable materials are disposed off in such a manner that there is no livelihood of getting mixed up with the materials meant for construction.

#### Felling Trees

- *Marking of trees:* Trees, above 30 cm. Girth (measured at a height of one meter above ground level) to be cut, shall be approved by the Engineer-in-charge and then marked at the site.
- *Felling of trees:* Felling of trees shall include taking out roots up to 60 cm. below ground level or 30 cm. below formation level or 15 cm. below sub-grade level, whichever is lower.
- *Filling:* All excavations below general ground level arising out of removal of trees, stumps etc. shall be filled with suitable material in 20 cm. layers and compacted thoroughly so that the surface at these points conform to the surrounding area.
- *Sizing:* The trunks and branches of trees shall be cleared of limbs and tops and cut into suitable pieces as directed by the Engineer-in-charge.
- *Staking:* The serviceable materials shall be staked in the manner as directed by the Environmental specialist of Supervision Consultants/Engineer-in-charge.
- *Disposal:* The material, which cannot be used or auctioned shall be removed from the area and disposed off as per the directions of the Engineer-in-charge. Unsuitable waste materials should not get mixed with construction material during disposal.

Format EM4: Tree Felling

| S. No | Links | Physical Target |        |                 |                     | Completion Target |                                      | Reason for Delay if any |
|-------|-------|-----------------|--------|-----------------|---------------------|-------------------|--------------------------------------|-------------------------|
|       |       | Total           | Target | Target Achieved | % of task completed | Target Date       | Date of Completion if task completed |                         |
|       |       | Unit            |        |                 |                     |                   |                                      |                         |
| 1     |       | nos             |        |                 |                     |                   |                                      |                         |
| 2     |       | nos             |        |                 |                     |                   |                                      |                         |
| 3     |       | nos             |        |                 |                     |                   |                                      |                         |
| 4     |       | nos             |        |                 |                     |                   |                                      |                         |

(Signed)

Assistant Conservator of Forests

## APPENDIX E.5: SITING & LAYOUT OF CONSTRUCTION CAMP

### (A) SITING

The contractor based on the following guidelines shall identify the location of the construction site. The construction site shall be located:

- The construction camps will be located at least 1000m away from major settlements in downwind direction. The living accommodation and ancillary facilities for labour shall be erected and maintained to standards and scales approved by the resident engineer.
- A minimum 500m away from surface water course or body.
- A minimum 1000m away from Wild life Sanctuary/Ecologically sensitive areas.
- Should not be located in cultivable land unless unavoidable.
- All sites used for camps must be adequately drained.
- The camps must be located such that the drainage from and through the camps will not endanger any domestic or public water supply.
- All sites must be graded, ditched and rendered free from depressions such that water may get stagnant and become a nuisance.

### (B) LAYOUT

- A conceptual layout of a typical construction site has been presented in Figure below. The Contractor during the progress of work will provide, erect and maintain necessary (temporary) living accommodation and ancillary facilities for labour to standards and scales approved by the engineer. All temporary accommodation must be constructed and maintained in such a fashion that uncontaminated water is available for drinking, cooking and washing. Safe drinking water should be provided to the dwellers of the construction camps. Adequate washing and bathing places shall be provided, and kept in clean and drained condition. Construction camps are to be sited away from vulnerable people and adequate health care is to be provided for the work force.

**Sanitation Facilities:** Construction camps shall be provided sanitary latrines and urinals. Sewerage drains should be provided for the flow of used water outside the camp. Drains and ditches should be treated with bleaching powder on a regular basis. The sewage system for the camp must be properly designed, built and operated so that no health hazard occurs and no pollution to the air, ground or adjacent watercourses takes place. Compliance with the relevant legislation must be strictly adhered to. Garbage bins must be provided in the camp and regularly emptied and the garbage disposed off in a hygienic manner

**Shelter at Workplace:** At every workplace, there shall be provided free of cost, four suitable shelters, two for meals and two others for rest, separately for use of men and women labourers. The height of shelter shall not be less than 3m from floor level to lowest part of the roof. Sheds shall be kept clean and the space provided shall be on the basis of at least 0.5m<sup>2</sup> per head. Besides the rest area/dwelling place structure have following minimum features:

- Very Tough, Durable and capable to resist wind pressure up to 200km/hour
- Effective acoustics and Insulated
- Water, fire and earthquake resistant/proof.
- Secured with grill and locking devices.

- **Canteen Facilities:** A cooked food canteen on a moderate scale shall be provided for the benefit of workers wherever it is considered necessary. The contractor shall conform generally to sanitary requirements of local medical, health and municipal authorities and at all times adopt such precautions as may be necessary to prevent soil pollution of the site.

**First aid facilities:** Medical facilities shall be provided to the labour at the construction camp. Visits of doctor shall be arranged twice a month wherein routine checkups would be conducted for women and children. A separate room for medical checkups and keeping of first aid facilities should be built. Workplaces remote and far away from regular hospitals will have indoor health units with one bed for every 250 workers. Suitable transport will be provided to facilitate take injured or ill person(s) to the nearest approachable hospital. The site medical room should display awareness posters on safety facilitation hygiene and HIV/AIDS awareness. The medical office should also distribute condoms at regular intervals to labourers.

- First Aid Box will be provided at every construction campsite and under the charge of a responsible person who shall always be readily available during working hours. He shall be adequately trained in administering first aid-treatment. Formal arrangement shall be prescribed to carry injured person or person suddenly taken ill to the nearest hospital. The first aid box shall contain the following.
  - 6 small sterilized dressings
  - 3 medium size sterilized dressings
  - 3 large size sterilized dressings
  - 3 large sterilized burns dressings
  - 1 (30 ml) bottle containing 2 % alcoholic solution of iodine
  - 1 (30 ml) bottle containing salvolatile
  - 1 snakebite lancet
  - 1 (30 gms) bottle of potassium permanganate crystals
  - 1 pair scissors
  - Ointment for burns
  - A bottle of suitable surgical antiseptic solution
- In case, the number of labour exceeds 50, the items in the first aid box shall be doubled.

**Health Care Facilities:** Health problems of the workers should be taken care of by providing basic health care facilities through health centres temporarily set up for the construction camp. The health centre should have at least a doctor, nurses, duty staff, medicines and minimum medical facilities to tackle first-aid requirements or minor accidental cases, linkage with nearest higher order hospital to refer patients of major illnesses or critical cases.

- The health centre should have MCW (Mother and Child Welfare) units for treating mothers and children in the camp. Apart from this, the health centre should provide with regular vaccinations required for children.

**Day Crèche Facilities:** At every construction site, provision of a day crèche shall be worked out so as to enable women to leave behind their children. At construction sites where 20 or more women are ordinarily employed, there shall be provided at least a hut for use of children under the age of 6 years belonging to such women. Huts shall not be constructed to a standard lower than that of thatched roof, mud walls and floor with wooden planks spread over mud floor and covered with matting. Huts shall be provided with suitable and sufficient openings for light and ventilation. There shall be adequate provision of sweepers to keep the places clean. There shall be two maidservants (or aayas) in the satisfaction of local medical, health, municipal or cantonment authorities. Where the number of women workers is more than 25 but less than 50, the contractor shall provide with at least one hut and one maidservant to look after the children of women workers. Size of crèches shall vary according to the number of women workers employed.

**Rehabilitation of labor and construction camp :** At the completion of construction, all construction camp facilities shall be dismantled and removed from the site. The site shall be restored to a condition in no way inferior to the condition prior to commencement of the works. Various activities to be carried out for site rehabilitation include:

- Oil and fuel contaminated soil shall be removed and transported and buried in waste disposal areas.
- Soak pits, septic tanks shall be covered and effectively sealed off.
- Debris (rejected material) should be disposed off suitably (Refer Chapter 10 on “Waste Management and Debris Disposal”).
- Ramps created should be levelled.
- Underground water tank in a barren/non-agricultural land can be covered. However, in an agricultural land, the tank shall be removed.
- If the construction camp site is on an agricultural land, top soil can be spread so as to aid faster rejuvenation.
- Proper documentation of rehabilitation site is necessary. This shall include the following:
  - Photograph of rehabilitated site;
  - Land owner consent letter for satisfaction in measures taken for rehabilitation of site;
  - Undertaking from contractor; and
  - Certification from Engineer in-charge of the PIC.

In cases, where the construction camps site is located on a private land holding, the contractor would still have to restore the campsite as per this guideline. The rehabilitation is mandatory and should be include in the agreement with the landowner by the contractor. Also, he would have to obtain a certificate for satisfaction from the landowner.

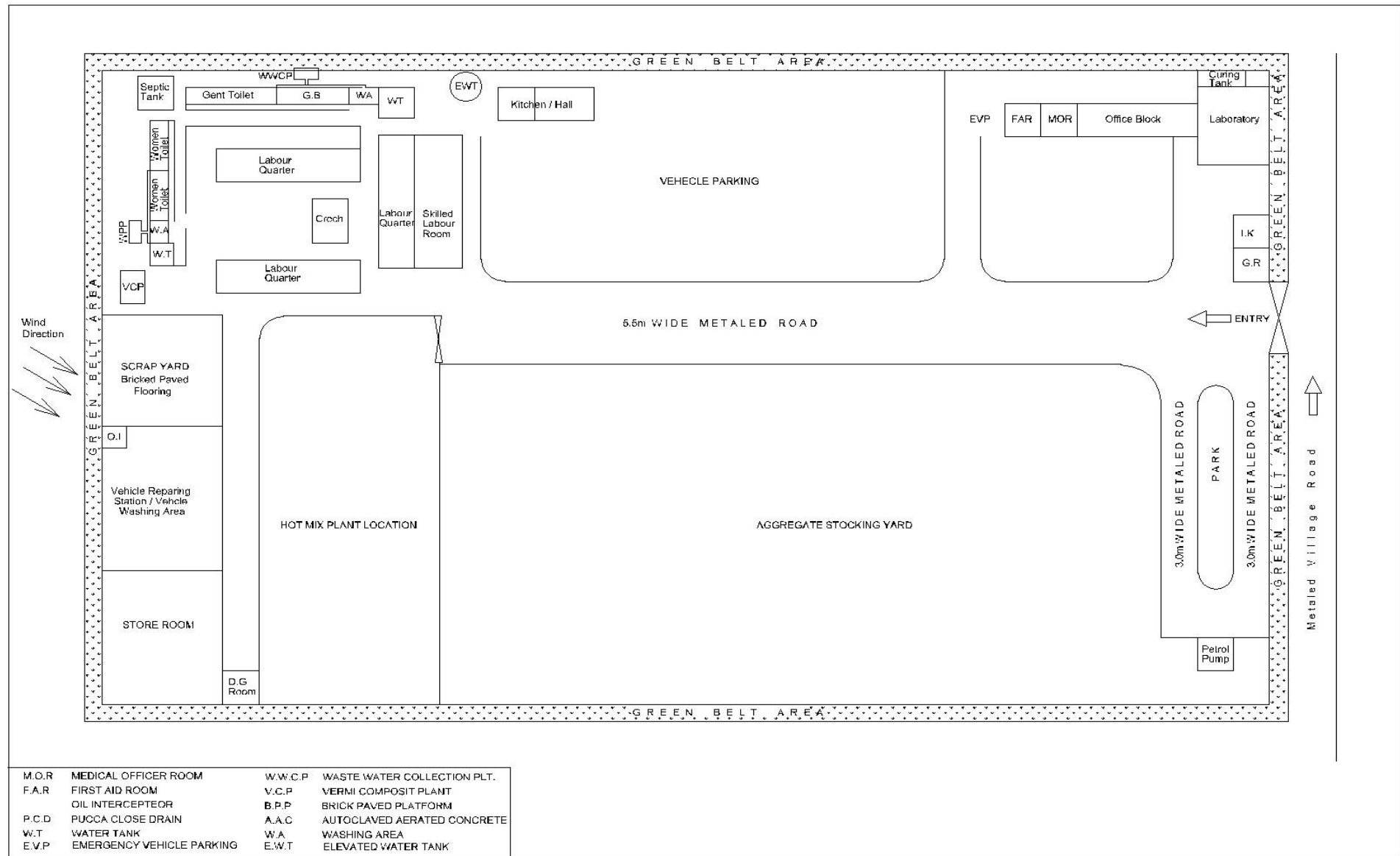


Figure-1: Typical Layout of Construction Camp

## **APPENDIX E.6: WORKERS SAFETY DURING CONSTRUCTION ACTIVITIES**

### **SAFE LAYOUT IN THE CONSTRUCTION PLANT, CAMP AND QUARRY AREAS**

1. Arrange border to perimeter fencing
2. Ensure good visibility and safe access at site entrances
3. Provide adequate warning signs at the entrance and exit where necessary
4. Provide adequate space/area for loading and unloading, storage of materials, plant and machinery
5. Display emergency procedure and statutory notices at conspicuous location
6. Consider welfare facilities required
7. Provide areas for dumping garbage and other waste materials, and also arrange for their regular clearance.
8. Arrange storage, transport and use of fuel, other flammable materials and explosives in line with the license requirements to be obtained from appropriate authorities
9. Plan emergency assembly points, fire escape routes and locate fire-fighting equipment
10. Provide access roads and plant movement areas within the site.
11. Ensure the availability of first aid facilities and display notices at the various works to show the location of these facilities
12. Provide proper drainage and sewage & drainage facilities

### **HOUSE KEEPING PRACTICES**

1. Maintain washrooms and canteens clean
2. Keep all walkways clear and unobstructed at all times
3. Ensure that spillages of oil and greasy
4. Stack raw materials and finished products clear of walkways or inside roads
5. Do not leave tools on the floor or in any location where they can be easily dislodged
6. Keep windows and light fitting clean
7. Maintain the workplace floors dry and in a non-slippery condition
8. Provide and maintain proper drainage system to prevent water ponding
9. Use metal bins for oily and greasy rags and store all flammable materials in appropriate bins, racks or cabinets. Ensure that the metal bins for storing oily and grease rags should be covered with lids.
10. Ensure that protruding nails in boards or walls are moved or bent over so that they do not constitute a hazard to people
11. Make sure that hazardous/dangerous chemicals are kept in the goods stores with the appropriate labeling, display of the material-safety-data-sheet (MSDS) and other precautionary measures
12. Display 'no smoking' signs in areas with high fire risks, e.g. paint stores, wood working area and others

### **TREE FELLING**

- Use hard hats during tree felling
- Ensure tools such as the axes are in good condition
- Determine proper foot and body position when using the axe. Do not cut above your head
- Wear appropriate foot protection
- Carry a first aid kit to the site
- Determine possible hazards in the area, e.g. electrical or telephone or other utility lines
- Prior to felling, determine the safest direction for the fall
- Determine the proper hinge size before directing the tree fall.

## **NOISE HAZARDS AND ITS CONTROL**

1. Note that indications of noise levels are:
  - You have to shout to be heard;
  - Your hearing is dulled just after work;
  - You get head noises or ringing in the ears after work;
  - You have difficulty hearing people while others are talking
2. Use sound level meters to measure. If the sound level exceeds 85 dB(A), then preventive measures should be taken
3. Make personnel aware of noisy areas by using suitable warning signs and insisting that ear protectors should necessarily be worn.
4. Reduce noise at source by improved maintenance, replacing noisy machines, screening with noise absorbing material, making changes to the process/equipment, controlling machine speeds, ensuring that two noise-generating machines are not running at the same time, using cutting oils and hydraulic breakers.
5. Appoint a competent person to carryout a detailed noise assessment of the site, designate ear protection zone, and give instructions on the necessary precautionary measures to be observed by site personnel, including the use of suitable type of ear protections.
6. Wear and maintain ear muffs and ear plugs as required
7. In construction or repair work, noise should be kept to a low-level bearing in mind the disturbance to local residents.

## **ROAD WORKS**

1. The use of signage is most important to caution the road users of possible unsafe conditions due to the road works.
2. Use the appropriate signage devices as required by the site conditions/situation. The devices include regulatory signs, delineators, barricades, cones, pavement markings, lanterns and traffic control lights.
3. In using signs, make sure that they are (i) simple, easy-to-understand and convey only one message, (ii) luminescent and with reflective properties, and (iii) broad, prominent and of appropriate size.
4. In using barricades, make sure that you keep traffic away from work areas and you guide the drivers to keep along a safe, alternative path.
5. Ensure that proper personal protective equipment (PPE) is provided to all the workers.
6. Cover existing road signs and install new ones at appropriate locations taking into account the distances that would be required and reaction times.
7. Plan layout and traffic management so that hazard are not created.
8. Deploy flagmen, who control traffic at the work areas. The flag should be 600mm x 600mm fastened to a 1m length staff.
9. Flagmen should wear reflective safety vests along with hard hats
10. If required, use wireless devices for flagmen to co-ordinate from either ends of the road, where works are being carried out.

## **ELECTRICAL HAZARDS IN CONSTRUCTION AREAS**

1. Treat all wires as live wires
2. Never touch dangling wires, but report them to your manager
3. Unless you are a qualified electrician, do not attempt electrical repairs
4. Never use electrical equipment if you hands are wet or you are standing in water
5. If electrical equipment is sparking or smoking, turn the power off and report the condition to your supervisor
6. Never use electrical wires that have physical damage
7. Never allow equipment or traffic to run over electrical wires.

**USE AND STORAGE OF GAS/LPG**

1. Store filled gas/LPG cylinder in the open area, i.e. outside of the building
2. Transport, store, use and secure cylinders in upright position
3. Ensure proper ventilation at the ground level in locations where gas/LPG is in use
4. Avoid physical damage to the cylinders
5. Never weld or cut on or near the cylinders
6. Store empty cylinders secured and upright
7. Make sure that the cylinder is closed immediately after use
8. Investigate immediately if there is the smell of LPG or gas
9. Never use destenched gas/LPG on site.
10. Make sure that there is no other unrelated fire in the vicinity of the cylinder

**OPERATION OF EXCAVATORS**

1. Ensure that excavators are operated by authorized persons who have been adequately trained.
2. Prevent unauthorized movement or use of the excavators
3. Check regularly and maintain the machine thoroughly
4. Ensure that all relevant information, including those related to instruction, training, supervision and safe system of work are provided to the operators.
5. Ensure that the operation and maintenance manuals, manufacturer's specifications, inspection and maintenance log books are provided for the use of the mechanics, service engineers or other safety personnel during periodic maintenance, inspection and examination.
6. During tipping or running alongside the trenches, excavators must be provided with stop blocks.
7. Excavators must be rested on firm ground during operation
8. Avoid operating the machine too close to an overhang, deep ditch or hole and be alert to potential carving edges, falling rocks and slides, rough terrain and obstacles.
9. Locate and identify underground services by checking with all utility companies before excavations.
10. Ensure that all excavations are supervised by experienced and competent persons.
11. When reversing or in case the operator's view is restricted, adequate supervision and signaling should be provided.
12. Ensure that the type and capacity of the excavator are properly chosen for the intended purposes and site conditions. Never use a machine for any purposes other than it is designed for.
13. Check and report for excessive wear and any breakage of the bucket, blade, edge, tooth and other working tools of the excavator.
14. Check that all linkages/hinges are properly lubricated and ensure that the linkage pins are secured. Never use improper linkage pins.
15. Never dismount or mount a moving machine
16. Work only with adequate ventilation and lighting
17. Ensure that the protective front screen of the driving cabin is fixed in position during excavations to avoid eye injury to the operator.
18. Ensure switch-off of the unattended vehicle.

**OPERATION OF TRUCKS AND DUMPERS**

1. Ensure that only trained, authorized and licensed drivers operate the vehicles
2. Enlist the help of another worker before reversing the vehicle
3. Switch-off the engine of an unattended vehicle

4. Lower the tipping bodies when the machine is unattended, but if it is necessary to leave them in the raised position they should be blocked to prevent their fall.
5. Wear safety boots or shoes to avoid injuries during loading and unloading.
6. Carryout periodic servicing to the manufacturer's requirements. All records of maintenance and repairs should be in writing or kept on site.
7. Keep the vehicle tidy and the cabin free from tools and material, which might obstruct the controls.
8. Keep to speed limits.
9. No passenger should be carried on a dumper except the driver
10. Never drive the vehicle across a slope
11. Provide stop blocks when the vehicle is tipping into or running alongside excavations
12. Do not overload the vehicle.
13. Carry only well secured loads
14. Park only on level ground, in neutral with the parking brake applied
15. Never mount or dismount from a moving vehicle

### **GAS WELDING**

1. Use the following personal protective equipment during welding
  - Face or hand shield fitted with filters
  - Goggles, particularly when chipping slag
  - Gloves long enough to protect wrists and forearms against heats, sparks, molten metal and radiation
  - High-top boots to prevent sparks from entering footwear.
2. Screen off the work area with sturdy opaque or translucent materials because glare can cause eye injury.
3. Key for opening the acetylene cylinder valve must be on the valve stem while the cylinder is in use so that the cylinder valve may be immediately shut-off in an emergency.
4. Ventilate the workplace using air blowers and exhaust fans to remove poisonous fumes and gases that are given off during welding
5. Take precautions against flying sparks and hot slag where welding is being done near flammable materials and check the area before leaving.
6. Do not weld material degreased with solvents until completely dry.
7. Do not use gas cylinders for supporting work or as rollers
8. Do not use oil grease on oxygen cylinder fittings
9. Do not use cylinders with damaged valves.
10. Do not use too much force if valves are stuck.
11. Replace valve caps after use
12. Search for leaks in equipment by using a solution of soapy water.
13. Shut the cylinder valve if acetylene from a cylinder catches fire at the valve or regulator due to leakage at a connection.
14. Treat all gas cylinders as "full" unless you are sure otherwise.
15. Never attempt to transfer acetylene from one cylinder to another or attempt to refill an acetylene cylinder.
16. Place portable fire extinguishers near the welding area
17. Secure all cylinders against accidental displacement.
18. Always lift gas cylinders. Do not slide them along the ground or drop them from trucks.
19. Keep gas cylinders in vertical position both in storage and when in use



20. Keep the work place dry, secure, free from combustible materials and obstruction.
21. Store the acetylene and oxygen cylinders separately, and in a proper store.
22. Keep the gas cylinders from source of heat, flammable materials, corrosive chemicals and fumes.

### **MANUAL HANDLING AND LIFTING**

1. Use mechanical equipment in place of manual handling as far as possible.
2. Assess the manpower required to handle or lift the load safely and arrange the manpower accordingly.
3. In handling hazardous materials, the workers shall be informed of the hazards and safety precautions.
4. All relevant persons shall be trained in the proper methods of lifting and carrying.
5. Where team work is required, select the persons whose ages and physical builds are compatible for teaming up. Coordinate the actions of the team members by giving necessary instructions.
6. Always lighten or suitably shape the load for manual handling as far as possible. Keep a look out for splinters, sharp edges, loose banding and nails.
7. Clear path of obstruction and tripping hazards.
8. Stack and secure goods safely on trucks, otherwise they fall off and injure passers-by.
9. Use personal protective equipment such as gloves, safety shoes, etc.
10. Adopt the following procedure when you lift a load:
11. Stand close to the object. Have a firm footing with feet spread on either side of the road.
12. Bend the knees and keep your back as straight as you can
13. Grasp object firmly. Be sure grip will not slip
14. Breathe in and throw the shoulder backwards.
15. Straighten the legs, continuing to keep the back as straight as you can.
16. Hold object firmly close to the body
17. Always lift smoothly. Avoid jerky motions. Turn with feet instead of twisting the back.

### **HANDLING CHEMICALS AND HAZARDOUS SUBSTANCES**

1. Always substitute hazardous chemicals with harmless or less hazardous ones wherever possible.
2. Enclose the process using chemicals or provide other engineering controls such as local exhaust ventilation, a fume cupboard or a safety cabinet.
3. Exercise great care in the storage and use of chemicals because they may be explosive, poisonous, corrosive or combustible.
4. Separate different chemicals physically
5. Store chemicals classified as dangerous goods in a properly constructed and approved goods store. Keep proper records of all chemicals and hazardous substances delivered, stored and used on site.
6. Consider unknown substances and liquids as dangerous until proven otherwise.
7. All containers should be clearly labeled to indicate contents. Never use a wrongly labeled container for chemicals.
8. Prohibit smoking in the vicinity of dangerous chemicals
9. Ensure that you are wearing the correct personal protective equipment before you handle chemicals
10. Maintain the Material Safety Data Sheet of all chemicals for reference on safety precautions to be taken and the use of suitable PPE.
11. When opening containers, hold a rag over the cap or lid, as some volatile liquids tend to spurt up when released.
12. Wash before you eat and do not eat at the work place.
13. If the skin is splashed with a chemical, rinse it immediately with plenty of clean water. Eye should be flushed thoroughly with water followed by immediate medical attention.
14. Eye fountain, emergency shower and breathing apparatus should be available in the vicinity of the workplace.

15. Safety instructions for handling emergency situations should be displayed prominently at both the storage and use locations.

**FIRST AID**

1. Provide first aid boxes at every site
2. Ensure that training on the use of the first aid box is provided to a handful of staff working in the site.
3. Display the list of persons who are trained on providing first aid.
4. Ensure that every first aid box is marked plainly “First Aid” in English and local language.
5. The responsible person or first aider should replenish the contents of the first aid box as necessary.

**PERSONAL PROTECTIVE EQUIPMENT****General**

1. Consider the provision of personal protective equipment only after all measures for removing or controlling safety hazards have been provided reasonably impractical.
2. Ensure that sufficient personal protective equipment are provided and that they are readily available for every person who may need to use them.
3. The management should ensure that all persons make full and proper use of the personal protective equipment provided.
4. Provide instruction and training in the proper use and care of any specific protective equipment where necessary
5. Do not willfully misuse, interfere with or ill-treat any protective clothing and equipment provided.
6. Ensure that the personal protective equipment are in good condition. Report immediately any damage to the management for replacement. Always keep the personal protective equipment as clean as possible.

**Eye protection**

1. Issue eye protection equipment where there is a foreseeable risk of eye injury
2. Ensure an adequate supply of goggles/shields is available.
3. Keep the goggles clean and make sure they are good fit.
4. Do not watch welding operations unless your eyes are protected from the damaging effect of flash.

**Head Protection**

1. No person shall enter a construction site unless he is wearing a suitable safety helmet
2. Wear a safety helmet:
  - When there is the risk of being hit by falling objects
  - While on or near a construction site
  - During adverse weather conditions
  - When in any area designated as a “hard hat” area.
3. Provide identification labels to all helmets in some way to prevent random exchange among wearers, with one helmet exclusive to each person.
4. Inspect helmets for cracks or sign of impact or rough treatment before each usage. Destroy, remove and replace all worn, defective or damaged helmets.

**Hearing Protection**

1. Provide ear plugs or ear muffs as required. Use re-usable ear plugs when the reduction required (15-25 dBA) is not excessive. Use ear muffs where a large attenuation of upto 40 dBA is demanded.
2. Do not use dry cotton wool for hearing protection because it cannot provide any.
3. Provide disposable ear plugs for infrequent visitors and ensure that they are never re-used.
4. Provide re-usable ear plugs for those who need to work continuously for a long period in a high noise area.
5. Use ear muffs with replaceable ear cushions because they deteriorate with age or may be damaged in use.
6. Avoid wearing spectacles with ear muffs.
7. Use soap and water or the recommended solvent for cleaning ear muffs.

8. Provide ear muffs for those who may need to get in and out of a high noise area frequently.

### **RESPIRATORY PROTECTIVE EQUIPMENT**

1. Wear suitable respirable for protection when there is a potential for small particles entering the lungs, e.g. emptying of cement bags.
2. Ensure that the exhalators can provide adequate protection.
3. Provide training to all persons using the respirators for their correct fitting, use, limitations and symptoms of exposure.
4. Clean and inspect all respirators before and after use.
5. Store respirators properly when not in use.

### **Safety Footwear**

1. Wear suitable footwear for work
2. Use safety footwear on site or in other dangerous areas
3. Wear suitable safety shoes or ankle boots when working anywhere where there is high risk of foot injuries from slippery or uneven ground, sharp objects, falling objects, etc.
4. All safety footwear, including safety shoes, ankle boots and rubber boots, should be fitted with steel toe caps.
5. Avoid wearing flip flops, high heeled shoes, slippers, light sport shoes in situations where there is a risk of foot injury.
6. Keep shoe lace knots tight.

### **Hand Protection**

1. Wear suitable gloves for selected activities such as welding & cutting and manual handling of materials & equipment.
2. Do not wear gloves where there is a risk of them becoming entangled in moving parts of machinery
3. Wash hands properly with disinfectant soap and clean water before drinking, eating or smoking. Wash hands immediately after each operation on site when the situation warrants.

### **FIRE PREVENTION, FIGHTING AND EQUIPMENT**

#### **Before fire breaks out**

1. Store flammable material in proper areas having adequate fire protection systems.
2. Display sufficient warning signs.
3. Train selected personnel to use these fire extinguishers
4. Inspect fire extinguishers regularly and replace as necessary
5. Fire escape route should be kept clear at all times and clearly indicated.
6. Know the escape route and assembly point.
7. Display escape route maps prominently on each floor
8. Carry out fire drill regularly. Designate fire officers
9. Install fire alarm wherever required and test regularly.
10. Provide sufficient exit signs at prominent locations for directing people to the escape staircases and routes.

#### **When fire breaks out.**

1. Alert all persons
2. Put off the fire with appropriate fire extinguishers only when you are sure that you are safe to do so.
3. Escape if you are in danger through the fire escape route to assembly point
4. Fire officers to carry out head count at the assembly point.

#### **Incident and accident investigations**

1. Carry out the investigation as quickly as possible.
2. Conduct interviews with as many witnesses as necessary

3. Do not rely on any one sole source of evidence
4. Use the following tools:  
Checklists for obtaining basic and typical information for accidents
  - Notebook
  - Tape records
  - Camera
  - Measuring tape
  - Special equipment for the particular investigation
5. Obtain answers to the following questions:
  - When did the accident occur?
  - Where did it occur?
  - Who was injured and what was damaged?
  - What caused the accident?
  - Why did it occur?
  - How could it have been prevented?
  - How can a recurrence be prevented?
6. Prepare a short but sufficient investigation report that contains the following:
  - A summary of what had happened
  - A summary of events prior to the accident
  - Information gathered during the investigation
  - Details of witnesses
  - Information on injury or loss sustained
  - Conclusions and possible causes of the accident
  - Recommendations to prevent recurrence
  - Supporting materials (photos, diagrams, etc.)

#### WORKERS SAFETY DURING CONSTRUCTION

| Sl No. | Stage and Nature of Construction Hazard                                                                                                  | Safety measures expected to be taken by the Contractors and Site Engineers                                                                                                                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Excavation in soft loose & slushy soil above 2.00 m depth sliding of earth or collapsing of sides.                                       | The Excavation beyond 1.5 m to 2.00 m to be done in steps of minimum 500 mm offsets as shown in Clause 2.18.2(b) and also planking and strutting should be done as in Clause 2.19.1.                                                                                                                              |
| 2      | Excavation in slippery area (water logged) – The labour may fall or machinery on site may slip.                                          | Try to dewater the area and spread minimum 150 mm thick sand layer to avoid slipping                                                                                                                                                                                                                              |
| 3      | Excavation in Rock where chiseling is involved – The fall of hammer may injure the hand, small rock pieces may injure the eyes and legs. | For hammer work, only experienced and skilled labour should be employed. Chisel should not be allowed to be held by hand, while hammering but chisel holding clamp should be provided. The labour should be provided with goggles and leg cover to protect eyes and legs, from injuries due to small rock pieces. |

| Sl No. | Stage and Nature of Construction Hazard                                                                                                               | Safety measures expected to be taken by the Contractors and Site Engineers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4      | Excavation in Rock where blasting is involved - Careless handling may lead to injury to main worker or a passer by.                                   | The work of blasting should be entrusted to only experienced persons. Provide sufficient length of fuse to give ample margin of time from the time of lighting to the time of explosion. A danger zone at least 180m diameter is to be flagged off 10 minutes before actual firing. All workmen should be sent away from danger zone except the firing man, who should be provided with a whistle.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5      | Excavation for drain across road or manhole adjacent to a road – chances of a passer by falling into the excavated portion                            | The area should be well barricaded & a red lamp provided at night. A watchman should be deputed to prevent any movement of persons, or vehicles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6      | During Excavation or some times even while concreting – Snake bites or Scorpion stings –                                                              | In places where the movement of snakes are more the contractor should provide the labour with gum boots, gloves etc. and also make snake antidotes available on site. A particular care that has to be taken on such site is to always keep a vehicle available on site to rush the patient to a doctor. This applies to snake stinged patients as well.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7      | Centring (form-work) and scaffolding – Form-work collapse while concreting or just before concreting especially when wooden ballies are used.         | Many a times ballies joined together give way due to weak joint. Hence the use of joined ballies should be restricted. Only 2 joined ballies out of 8 ballies should be allowed. In case of double staging for a Slab at a height, utmost care should be taken to see that the top balli rests on the bottom balli. A particular care that should be taken during each concreting operating of slabs and beams is that, one carpenter and two helpers with spare ballies, nails etc. should be deputed below the slab/beam that is being concreted to watch any disturbance in the supports of the form-work below during concreting and in case of any doubt the concreting should be stopped immediately and the form work strengthened. Never allow bricks below a balli to make up the required height. This is most dangerous. |
| 8      | Form-work for beams and slabs: The bottom of beam collapses and many a times brings down the slab as well, injuring the labour and supervision staff. | This case is noticed when slender ballies are used without bracing. In fact, no concreting should be allowed without bracing at 300 mm above ground, and at mid way, in normal beams & slabs. The bracings should be for the support of beams as well as slabs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 9      | Form-work for sides of a slab–The labour just rests his foot on the plank and looses balance and falls resulting a fatal accident.                    | This is noticed when the carpenter fixes the side shuttering of a slab with a plank just tied by binding wire to the steel reinforcements and by wooden pieces nailed in wall and plank. This is so weak a portion that with little pressure the plank gives way.<br>Hence side shuttering should be done with a direct balli support from ground or floor, and the practice of tying planks with binding wire to the steel reinforcement should be totally avoided. A temporary railing along the periphery of slab will guard the life of labour and supervision staff.                                                                                                                                                                                                                                                           |
| 10     | Form-work for beams and slabs–Opening the form- work–Accident due to fall of materials during removing the forms.                                     | In fact, this is a most dangerous work. One should be very careful while form-work is removed. Only trained carpenters should be deputed for the work. A safe resting place outside the area of slab as a temporary measure should be constructed from where the Slab can be removed safely. Removal of form-work during night should not be permitted under any circumstances.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Sl No. | Stage and Nature of Construction Hazard                                                                                                                                                                                            | Safety measures expected to be taken by the Contractors and Site Engineers                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11     | Scaffolding–Fall of work-man, Supervision Staff, Standing on Chalis not tied properly or tied only at one end. (Chalis mainly made of Bamboos).                                                                                    | This is a very common negligence on the part of labour who do scaffolding work. The Chalis on which they work either span over its complete length or is tied loosely and many a times at one end only. Hence, care must be taken that the Chali do not span over the full length but some middle support should be provided and also the same is tied properly on both ends.                                                                                                                          |
| 12     | Ladders–Balli or bamboo ladders – The horizontal member breaks and the person falls. Some times the top face just rests on wall and the whole ladder tilts causing an accident.                                                    | The ladders should be strong enough to bear the weight of a labour with materials on head. As far as possible a hand rail should be provided at one end. The horizontal member should be preferably fixed with bolt & nuts or strong nails. When the ladder is placed across a wall the top portion should be tied firmly to a strong support so that the ladder does not move laterally.                                                                                                              |
| 13     | Column Reinforcements – Column reinforcements mainly in independent footings collapses – Injury to persons working nearby.                                                                                                         | The tendency of bar-benders is to tie the vertical steel with coir rope or 8 mm steel rods as ties on all four sides of the column reinforcement. This method of supporting the column reinforcements results in a weak support. Hence, the column reinforcements should be supported by strong ballies on all four sides of reinforcements and as far as possible a combined platform should be constructed out of ballies over which the reinforcements can be supported.                            |
| 14     | Concreting chajjas – When chajjas are concreted with out care and on opening the form-work the chajja would collapse, causing injury to labour on top or bottom of chajja.                                                         | While concreting chajjas care must be taken that the labour do not stand on the reinforcement and disturb the position. Separate scaffolding must be tied over which the labour can stand and work without disturbing the reinforcements. The main reason is in chajja the steel is placed on top face but if the labour stands on the steel, it will bend and come to bottom face and hence the chajja will fall when form-work is removed, thus, causing injury to labour working on top, or bottom. |
| 15     | Dismantling–Dismantled materials may fall on passer by or the person engaged in dismantling work may fall due to slipping. The dismantled materials may fall on persons working below.                                             | When work of demolition is to be taken up the area should be closed for all outsiders. No one should be allowed up to 50 m. from the place of demolition. The workers engaged in demolition should be asked to wear safety belts. Helmets must be worn by all the workers engaged in dismantling work. The place should be strictly guarded at night with red lights at prominent places, and watchman should be posted.                                                                               |
| 16     | Electric-Connections/Cables etc. –High tension/L.T. Electric wire passing near the slab structure- while bending, lifting or tying reinforcements the bar benders may sustain the Electric Shock, causing fatal injury.            | The work in such places, should not be allowed to the workers themselves, but in such position the work must be executed under the strict supervision of a responsible Foreman or a Supervisor.                                                                                                                                                                                                                                                                                                        |
| 17     | Electric Connections/Cables etc. –Cables below ground may get punctured during excavation & thus electrocute the labour working. Similarly when concreting is in progress the punctured cable may prone to be fatal to the labour. | Before taking up the work all available drawings should be studied, local enquiry to be made to know the position of cables and work in such area should be got executed under strict supervision of an experienced Foreman or a Supervisor.                                                                                                                                                                                                                                                           |

| Sl No. | Stage and Nature of Construction Hazard                                                                                                                                                                                                                                                                                                                          | Safety measures expected to be taken by the Contractors and Site Engineers                                                                                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18     | Electric Connections/Cables etc. – Temporary Electric lines near damp walls, near joinery stretched on a considerable length – There is every chance that the wire may get cut due to usage and may develop short circuits/leakages etc. and may electrocute the person touching the wire accidentally.                                                          | The Electric wires should be maintained by an electrician who should regularly check up the insulation of wires especially placed near steel items & damp areas.<br>ring should be supported properly.<br>As far as possible a good quality wire should be used which may not get damaged easily.                   |
| 19     | Electric and gas welding work – Drilling, polishing work – Done by temporary cables used on a number of works – Due to the fact that the wires are old & when they come in contact with water even in the process of curing the surrounding area may get affected due to leakage in the electric current thus causing damage to the workers & supervision staff. | All wiring works to be inspected by experienced electrician. All wires to be properly insulated and fixed at height on temporary poles. No welding work should be permitted near damp area. The welders to be provided with welder's goggles & gloves. As far as possible machine in good condition should be used. |
| 20     | Construction Machinery & Lifts – Concrete Mixers – Safety precautions. A mixer with hopper tried to be operated by an helper could not release brake in time thus causing injury to the person near hopper- some times fatal one.                                                                                                                                | The Mixers with hopper should be operated by an experienced mixer operator and such mixers should not be allowed to be handled by a helper or a labour.                                                                                                                                                             |
| 21     | Construction Machinery & Lifts - Lifts - Safety precautions.<br>(1) The lift pit if left unguarded the children of workers may fall in the pit resulting in fatal accident                                                                                                                                                                                       | (1) A brick protection wall of minimum 1.00 m height should be constructed around the Lift Pit, thus, preventing the children going near the pit. A special care should be taken to see that the children are not allowed to come near the machinery.                                                               |
|        | (2) The manually operated brakes of the lift failed or the communication between the labour at the top and the liftman failed and thus, the lift was not controlled and resulted in fatal accident.                                                                                                                                                              | (2) The condition of the lift must be maintained properly. The lift operator should be well trained. The labour receiving the bucket at top should be smart and active enough to convey the message of stopping & releasing the lift-to-lift operator properly.                                                     |
| 22     | Water Storage Tank for general use & curing - chances of children of workers falling in the tank with fatal accident.                                                                                                                                                                                                                                            | The water tanks constructed on site should be protected by at least 1.00 m high walls on four sides, so that the children do not fall.                                                                                                                                                                              |
| 23     | Misuse of lift by labour and some times supervision staff The lifts that are meant for lifting materials used by labour to go to upper floors – The labour thus traveling many a times get injured.                                                                                                                                                              | No person should be allowed to go to upper floors by lifts that are mainly meant for conveying the building materials. Fatal accidents have taken place due to above action of workers.                                                                                                                             |
| 24     | Site Cleaning–Cleaning top floors of buildings – Upper portion of any structure – Throwing waste materials broken concrete pieces, brick bats, sand etc. straightway from top to ground injuring person below or even a passerby.                                                                                                                                | This dangerous practice should not be allowed at all. The materials should be brought to the ground with the help of lift or the use of rope over pulley with a bucket, thus bringing down materials safely.                                                                                                        |
| 25     | Bar bending work-Helpers of bar benders to follow short cut method, throw surplus steel pieces from top floors to ground and may cause fatal injuries.                                                                                                                                                                                                           | This is a very bad practice. The helpers should bring the rods to ground with the help of lift or rope & pulley.                                                                                                                                                                                                    |

## APPENDIX E.7: ENVIRONMENT FRIENDLY CONSTRUCTION METHODOLOGY

The contractor shall be deemed to have acquainted himself with the requirements of all the current statutes, ordinances, by-laws, rules and regulations or their instruments having the force of law including without limitation those relating to protection of the environment, health and safety, importation of labour, demolition of houses, protection of environment and procurement, transportation, storage and use of explosives, etc.

### 1. PROTECTION OF ENVIRONMENT

- (i) The contractor will take all necessary measures and precautions and ensure that the execution of the works and all associated operations on site or offsite are carried out in conformity with statutory and regulatory environmental requirements including those prescribed in EMP.
- (ii) The contractor will take all measures and precautions to avoid any nuisance or disturbance to inhabitants arising from the execution of works.
- (iii) All liquid waste products arising on the sites will be collected and disposed of at a location on or off the sites and in a manner that will not cause either nuisance or pollution.
- (iv) The contractor will at all times ensure that all existing water courses and drains within and adjacent to the site are kept safe and free from any contamination.
- (v) The contractor will submit details of his temporary drainage work system (including all surface channels, sediment traps, washing basins and discharge pits) to the Project Implementation Unit –APRDC / Environment Officer for approval prior to commencing work on its construction.
- (vi) The contractor will arrange all the equipment in good condition to minimize dust, gaseous or other air-borne emissions and carry out the works in such a manner as to minimize adverse impact on air.
- (vii) Any vehicle with an open load-carrying area used for transporting potentially dust-producing material will have properly fitted side and tailboards. Materials having the potential to produce dust will not be loaded to a level higher than the side and tail boards and will be covered with a clean tarpaulin in good condition.
- (viii) The contractor will take all necessary measures to ensure that the operation of all mechanical equipment and condition processes on and off the site will not cause any unnecessary or excessive noise, taking into account applicable environmental requirements.
- (ix) The contractor will take necessary measures to maintain all plant and equipment in good condition.
- (x) Where the execution of the works requires temporary closure of road to traffic, the contractor will provide and maintain temporary traffic diversions subject to the approval of the EO/Engineer.
- (xi) Where the execution of the works requires single-lane operation on public road the contractor will provide and maintain all necessary barriers, warning signs and traffic control signals to the satisfaction of the EO/Engineer.
- (xii) Wherever traffic diversions, warning signs, traffic control signals, barriers and the like are required, the contractor will install them to the satisfaction of EO/Engineer prior to commencing the work, in that area.
- (xiii) Contractor will install asphalt plants and other machineries away from the populated areas as per laid down regulations.
- (xiv) Permit for felling of trees will be obtained from the forest department before the execution of any work.
- (xv) Trees and plants going to be uprooted will be duly compensated and maintained up to 3 years.
- (xvi) Mist sprays should be provided at appropriate places for preventing dust pollution during handling and stockpiling of stones and loose earth.
- (xvii) Over Burden (OB) waste dumps shall be sprayed with water, as they are the major source of air borne particulate matter.
- (xviii) OB waste dumps shall be reclaimed / afforested to bind the loose soil and to prevent soil erosion. The frequency of sprinkling should be fixed as per the seasonal requirement and in consultation with engineer.
- (xix) Regular water spraying on haulage roads during transportation of construction material by water sprinklers. The frequency of sprinkling should be fixed as per the seasonal requirements in consultation with engineer.
- (xx) Transfer point for transporting construction material shall be provided with appropriate hoods/ chutes to prevent dust emissions.



- (xxi) Dumping of construction material should be from an optimum height (preferably not too high), so as to reduce the dust blow.
- (xxii) Innovative approaches of using improvised machinery designs, with in-built mechanism to reduce sound emission.
- (xxiii) Procurement of drill loaders, dumpers and other equipment with noise proof system in operator's cabin.
- (xxiv) Confining the equipment with heavy noise emissions in soundproof cabins, so that noise is not transmitted to other areas.
- (xxv) Regular and proper maintenance of noise generating machinery including the transport vehicles to maintain noise levels.
- (xxvi) Provisions should be made for noise absorbing pads at foundations of vibrating equipments to reduce noise emissions.

## 2. QUARRY OPERATIONS

The Contractor shall obtain materials from quarries only after the consent of the Forest Department or other concerned authorities and in consultation with the EO/Engineer. The quarry operations shall be undertaken within the purview of the rules and regulations in force and instructions as mentioned in **Appendix E.1: Aggregate Quarry Management** and **Appendix E.2: Borrow Area Management**.

## 3. PREVENTION OF WATER COURSES FROM SOIL EROSION AND SEDIMENTATION SILTATION

The Contractor shall apply following mitigation measures to prevent sedimentation and pollution of watercourses.

- To prevent increased siltation, if need be existing bridges maybe widened downstream side of the water body;
- Cement and coal ash should be stacked together, fenced by bricks or earth wall, and kept away from water, to prevent leachate formation and contamination of surface and ground water;
- If need be, slope of the embankments leading to water bodies should be modified and rechannelised to prevent entry of contaminants into the water body;
- During construction silt fencing (consists of geo-textile with extremely small size supported by wire-mesh mounted on a panel made up of angle frame) could be used along the road at all canals and rivers to prevent sediments from the construction site to enter into the watercourses.

## 4. POLLUTION FROM HOT-MIX PLANTS AND BATCHING PLANTS

Bituminous hot-mix plants and concrete batching plants shall be located sufficiently away from habitation, agricultural operations. The Contractor shall take every precaution to reduce the levels of noise, vibration, dust and emissions from his plants and shall be fully responsible for any claims for damages caused to the owners of property, fields and residents in the vicinity.

## 5. ARRANGEMENT FOR TRAFFIC DURING CONSTRUCTION

The Contractor shall at all times carry out work on the road in a manner creating least interference to the flow of traffic with the satisfactory execution. For all works involving improvements to the existing state highway, the Contractor shall, in accordance with the directives of the SE, provide and maintain, during execution of the work, a passage for traffic either along a part of the existing carriageway under improvement, or along a temporary diversion constructed close to the state highway. The Contractor shall take prior approval of the SE regarding traffic arrangements during construction.

## 6. TRAFFIC SAFETY AND CONTROL

- (i) Where subject to the approval of the Engineer the execution of the works requires temporary closure of road to traffic use, the Contractor shall provide and maintain temporary traffic diversions. The diversion shall generally consist of 200 mm thickness of gravel 4.5 meters wide laid directly upon natural ground and where any additional earthworks are required for this purpose that will be provided under the appropriate payment items.
- (ii) Where the execution of the works requires single-lane operation on public road, the Contractor shall provide and maintain all necessary barriers, warning signs and traffic control signals to the approval of the Engineer.
- (iii) With the exception of temporary traffic arrangements or diversions required within the first 4 weeks of the Contract, the Contractor shall submit details of his proposals to the Engineer for approval not less than 4 weeks

prior to the temporary arrangement or diversion being required. Details of temporary arrangements or diversions for approval as soon as possible after the date of the Letter of Acceptance.

- (iv) The color, configuration, size and location of all traffic signs shall be in accordance with the code of practice for road sign. In the absence of any detail or for any missing details, the signs shall be provided as directed by the Supervising Engineer (SE).
- (v) The Contractor shall take all necessary measures for the safety of traffic during construction and provide, erect and maintain such barricades, including signs, marking, flags, lights and flagmen as may be required by the Engineer for the information and protection of traffic approaching or passing through the section of the road under improvement. Before taking up any construction, an agreed phased programme for the diversion of traffic or closer of traffic on the road shall be drawn up in consultation with the SE.
- (vi) At the points where traffic is to deviate from its normal path (whether on temporary diversion or part width of the carriageway) the lane width path for traffic shall be clearly marked with the aid of pavement markings, painted drums or a similar device to the directions of the SE. At night, the passage shall be delineated with lanterns or other suitable light source.
- (vii) One-way traffic operation shall be established whenever the traffic is to be passed over part of the carriageway inadequate for two-lane traffic. This shall be done with the help of temporary traffic signals or flagmen kept positioned on opposite sides during all hours. For regulation of traffic, the flagmen shall be equipped with red and green flags and lanterns / lights.
- (viii) On both sides, suitable regulatory / warnings signs as approved by the PIU-APRDC shall be installed for the guidance of road users. On each approach, at least two signs shall be put up, one close to the point where transition of carriageway begins and the other 120 m away. The signs shall be of design and of reflectory type, if so directed by the Engineer.
- (ix) Upon completion of the works for which the temporary traffic arrangements or diversions have been made, the Contractor shall remove all temporary installations and signs and reinstate all affected roads and other structures or installations to the conditions that existed before the work started, as directed by the Engineer.

## 7. HEALTH AND SAFETY

The contractor shall take all measures and precautions necessary to ensure the health, safety and welfare of all persons entitled to be on the site. Such precautions shall include those that, in the opinion of the Engineer, are reasonable to prevent unauthorized entry upon the site and to protect members of the public from any activities under the control of the contractor. The contractor's responsibilities shall include but not be limited to:

- (i) The provision and maintenance of the Contractor's Equipment in a safe working condition and the adoption of methods of work that are safe and without risks to the health of any person entitled to be on the site.
- (ii) The execution of suitable arrangements for ensuring safety and absence of risks to health in connection with the use, handling, storage, transport and disposal of articles and substances,
- (iii) The provision of lighting, including standby facilities in the event of failure that, in the opinion of the Engineer, is adequate to ensure the safe execution of any works that are to be carried out at night.
- (iv) The provision of protective clothing and safety equipment, with such personnel and equipment and such information, instruction, training and supervision as are necessary to ensure the health and safety at work of all persons employed on or entering on the site in connection with the works, including the Engineer's supervisory staff, all in accordance with the laws.
- (v) Near towns, villages and all frequented places, trenches and foundation pits shall be securely fenced provided with proper caution signs and marked with lights at night to avoid accidents. Contractor shall take adequate protective measures to see that the excavation operations do not affect or damage adjoining structures.
- (vi) The contractor shall not use or generate any materials in the works, which are hazardous to the health of persons, animals or vegetation. Where it is necessary to use some substances, which can cause injury to the health of workers, the Contractor shall provide protective clothing or appliances to his workers.
- (vii) The contractor will take all measures necessary to safeguard the health; safety and welfare of all persons entitled to be on site and will ensure that works are carried out in a safe and efficient manner.
- (viii) The contractor will provide, and ensure the utilization of appropriate safety equipment for all workmen and staff employed directly or indirectly by the contractor. Such safety equipment will include but not be limited to the safety helmets, goggles and other eye protectors, hearing protectors, safety harnesses, safety equipment for working over water, rescue equipment, fire extinguishers and first-aid equipment. The personnel working at vulnerable locations at site will wear safety helmets and strong footwear.

- (ix) The contractor will provide an adequate number of latrines and other sanitary arrangements at areas of the site where work is in progress and ensure that they are regularly cleaned and maintained in a hygienic condition.

#### **8. FIRST AID**

- (i) The provision and maintenance of suitably equipped and staffed first aid stations throughout the extent of the works to the satisfaction of the Engineer. The contractor shall allow in his prices and be responsible for the costs of all such site welfare arrangements and requirements.
- (ii) Injuries might occur during the construction period. It is therefore pertinent to provide first aid facilities for all the construction workers. At construction camps and at all workplaces first aid equipment and nursing staff must be provided. Since many of the workplaces may be far away from regular hospitals, an indoor health unit having one bed facility every 150 workers needs to be provided.
- (iii) Adequate transport facilities for moving the injured persons to the nearest hospital must also be provided in ready to move condition.
- (iv) The first-aid units apart from an adequate supply of sterilized dressing material should contain other necessary appliances as per the factory rules.

#### **9. MAINTENANCE**

- (i) All buildings, rooms and equipment and the grounds surrounding them shall be maintained in a clean and operable condition and be protected from rubbish accumulation.
- (ii) Each structure made available for occupancy shall be of sound construction, shall assure adequate protection against weather, and shall include essential facilities to permit maintenance in a clean and operable condition. Comfort and safety of occupants shall be provided for by adequate heating, lighting, ventilation or insulation when necessary to reduce excessive heat.
- (iii) Each structure made available for occupancy shall comply with the requirements of the Uniform Building Code. This shall not apply to tent camps.

#### **10. MAINTENANCE OF DIVERSIONS AND TRAFFIC CONTROL DEVICES**

Signs, lights, barriers and other traffic control devices, as well as the riding surface of diversion shall be maintained in a satisfactory condition till such time they are required as directed by the EO/Engineer. The temporary traveled way shall be kept free of dust by frequent applications of water, if necessary.

## **APPENDIX E.8: STORAGE, HANDLING, USE AND EMERGENCY RESPONSE FOR HAZARDOUS CHEMICALS**

### **A1. REFUELING/MAINTENANCE PROCEDURE**

- Truck or suitable containers will bring in all fuel and fluids. There will be no storage of fuel, oil or fluids within 100m (or 50m) of the permanent water line.
- Prior to re-fueling or maintenance, drip pans and containment pans will be placed under the equipment. Absorbent blankets may also be required to be placed under the equipment and hoses where there is a possibility of spillage to occur.
- All used oils or fluids will be properly contained and transported to appropriately licensed (authorized) disposal facilities;
- Following re-fueling and maintenance, the absorbent blankets (if any) and spill pans will be picked up and the fuel truck or container moved outside of the 100m (or 50m) wide area.

#### **Emergency Spill Procedure**

Should a spill occur, either through spillage or equipment failure, the applicable emergency spill procedure outlined in sections A-2 to A-4 must be followed.

### **A2. SPILL PROCEDURE (INSIDE THE STREAM)**

In the case of a spill, overflow or release fluid into the stream waterway (whether water is flowing during the spill or not), do what is practical and safely possible to control the situation, then get help.

- **Stop the flow**
  - Stop the release into the stream waterway
  - Shutdown equipment
  - Close valves and pumps
  - Plug hoses
- **Remove Ignition Sources**
  - Shut off vehicles and other engines
  - Do not allow tiger torches, vehicles, smoking or other sources of ignition near the area. Keep a fire extinguisher on hand but keep it a safe distance away from the potential ignition source (if a fire starts, the extinguisher must be easily accessible).
- **Contract the environmental Officer and initiate Emergency Response**
  - Notify the site supervisor and the Contractor's Environmental Officer as soon as possible
  - The Environmental Officer will review the situation and decide if Emergency Services like Fire Brigade are required
  - Appropriate parties to be notified of the spill are
    - The contractor's Project Manager
    - The Engineer through his designated Environmental Officer
    - The Client
    - Regulatory Agencies like Pollution Control Board, Municipal Authorities, as applicable.
    - Site Safety Officer
- **Cleanup and Disposal**
  - Emergency Services will be engaged for the containment, cleanup and disposal of contamination release into the environment

- **Reporting**
  - The contractor's Environmental Officer will document the event and submit reports to the EO/Engineer, the Client and appropriate regulatory agencies like the Pollution Control Board (s).
- **Procedure Review**
  - The Engineer will review the report, determine if changes are required to procedures and recommend implementation of all required changes....

### **A3. SPILL PROCEDURE (ON LAND)**

In the case of a spill, overflow or release fluid onto land, do what is practical and safety possible to control the situation, then get help.

- **Stop the flow**
  - Stop the release into the water body
  - Shut down equipment
  - Close valves and pumps
  - Plug hoses
- **Remove Ignition Sources**
  - Shut off vehicles and other engines
  - Do not allow tiger torches, vehicles, smoking or other sources of ignition near the area. Keep a fire extinguisher on hand but keep it a safe distance away from the potential ignition sources (if a fire starts the extinguisher must be easily accessible).
- **Contain the Spill**
  - Dike around the spill to contain the material
  - Spread absorbent or place a spill blanket on the spill
  - Enlist the help of personnel on site
  - Notify your supervisor as soon as possible
- **Notification**
  - Appropriate parties to be notified of the spill are:
    - The Contractor's Project Manager
    - The Engineer through his designated Environmental Officer
    - The Client
    - Regulatory Agencies like Pollution Control Board, Municipal Authorities, as applicable
    - Site Safety Coordinator
- **Cleanup and Disposal**
  - The Engineer's Environmental Officer will ensure that a proper cleanup and disposal method is determined.
- **Reporting**
  - The Contractor's Environmental Officer will document the event and submit reports to the Engineer, the Client and appropriate regulatory agencies like the Pollution Control Board (s).
- **Procedure Review**
  - The Engineer will review the report, determine if changes are required to procedures are recommend implementation of all required changes.

### **A4. SPILL PROCEDURE (WITHIN PONDS)**

In the case of a spill, overflow or release fluid due to equipment or hose failure, do what is practical and safely possible to control the situation, then get help

- **Stop the flow**
  - Stop the release
  - Shut down equipment
  - Close valves and pumps
  - Plug hoses
- **Remove Ignition Sources**
  - Shut off vehicles and other engines
  - Do not allow tiger torches, vehicles, smoking or other sources of ignition near the area. Keep a fire extinguisher on hand but keep it a safe distance away from the potential ignition sources (if a fire starts the extinguisher must be easily accessible).
- **Contain the Spill**
  - Stop any pumps that may be moving the water from the area where the spill occurred
  - Enlist the help of personnel on site
  - Notify your supervisor as soon as possible
- **Notification**
  - Appropriate parties to be notified or the spill are:
    - The Contractor's Project Manager
    - The Engineer through his designated Environmental Officer
    - The Client
    - Regulatory Agencies like Pollution Control Board, Municipal Authorities, as applicable
    - Site Safety Coordinator
- **Cleanup and Disposal**
  - The Engineer's Environmental Officer will ensure that a proper cleanup and disposal method is determined. Absorbent pads will soak up the spilled material. The pads will be contained and removed from site for disposal at a licensed (authorized) facility.
- **Reporting**
  - The Contractor's Environmental Officer will document the event and submit reports to the Engineer, the Client and appropriate regulatory agencies like the Pollution Control Board (s)
- **Procedure Review**
  - The Engineer will review the report, determine if changes are required to procedures and recommend implementation of all required changes.

## **APPENDIX E.9: SOIL EROSION AND SEDIMENTATION CONTROL**

All materials shall meet commercial grade standards and shall be approved by the Engineer before being used in the work.

### **CONSTRUCTION STAGE**

Prior to the start of the relevant construction, the Contractor shall submit to the Engineer for approval, his schedules for carrying out temporary and permanent erosion/sedimentation control works as are applicable for the items of clearing and grubbing, roadway and drainage excavation, embankment/sub-grade construction, bridges and other structures across water courses, pavement courses and shoulders. He shall also submit for approval his proposed method of erosion/sedimentation control on service road and borrow pits and his plan for disposal of waste materials. Work shall not be started until the erosion/sedimentation control schedules and methods of operations for the applicable construction have been approved by the Engineer.

The surface area of erodible earth material exposed by clearing and grubbing, excavation, borrow and fill operations shall be limited to the extent practicable. The Contractor may be directed to provide immediate control measures to prevent soil erosion and sedimentation that will adversely affect construction operations, damage adjacent properties, or cause contamination of nearby streams or other watercourses. Such work may involve the construction of temporary berms, dikes, sediment basins, slope drains and use of temporary mulches, fabrics, mats, seeding, or other control devices or methods as necessary to control erosion and sedimentation.

The Contractor shall be required to incorporate all permanent erosion and sedimentation control features into the project at the earliest practicable time as outlined in his accepted schedule to minimize the need for temporary erosion and sedimentation control measures.

Temporary erosion/sedimentation and pollution control measures will be used to control the phenomenon of erosion, sedimentation and pollution that may develop during normal construction practices, but may neither be foreseen during design stage nor associated with permanent control features on the Project.

Where erosion or sedimentation is likely to be a problem, clearing and grubbing operations should be so scheduled and performed that grading operations and permanent erosion or sedimentation control features can follow immediately thereafter if the project conditions permit; otherwise temporary erosion or sedimentation control measures may be required between successive construction stages. Under no conditions shall a large surface area of erodible earth material be exposed at one time by clearing and grubbing or excavation without prior approval of the EO/Engineer.

The Engineer may limit the area of excavation, borrow and embankment operations in progress, commensurate with the Contractor's capability and progress in keeping the finish grading, mulching, seeding and other such permanent erosion, sedimentation and pollution control measures, in accordance with the accepted schedule.

Temporary erosion is sometimes caused due to the Contractor's negligence, carelessness or failure to install permanent controls. Sedimentation and pollution control measures then become necessary as a part of the work as scheduled or ordered by the Engineer, and these shall be carried out at the Contractor's own expense. Temporary erosion, sedimentation and pollution control work required, which is not attributed to the Contractor's negligence, carelessness or failure to install permanent controls, will be performed as ordered by the EO/Engineer.

Temporary erosion, sedimentation and pollution control may include construction work outside the right of way where such work is necessary as a result of road construction such as borrow pit operations, service roads and equipment storage sites.

The temporary erosion, sedimentation and pollution control features installed by the Contractor shall be maintained by him till these are needed, unless otherwise agreed by the Engineer.

## APPENDIX E.10: ENVIRONMENTAL MONITORING PROGRAM

### 1.1 Environmental Monitoring Plan

- The monitoring programme is devised to ensure that the envisaged purpose of the project is achieved and results in the desired benefit to the target population. To ensure the effective implementation of the EMP, it is essential that an effective monitoring programme be designed and carried out. Broad objectives of the monitoring programme are:
  - To evaluate the performance of mitigation measures proposed in the EMP
  - To suggest improvements in the management plans, if required
  - To satisfy the statutory and community obligations
- The monitoring programme contains monitoring plan for all performance indicators, reporting formats and necessary budgetary provisions. Monitoring plan for performance indicators and reporting system is presented in the following sections.

#### 1.1.1 Performance Indicators

- Physical, biological and environmental management components identified as of particular significance in affecting the environment at critical locations have been suggested as Performance Indicators (PIs). The Performance Indicators shall be evaluated under three heads as:
  - Environmental condition indicators to determine efficacy of environmental management measures in control of air, noise, water and soil pollution;
  - Environmental management indicators to determine compliance with the suggested environmental management measures
  - Operational performance indicators have also been devised to determine efficacy and utility of the mitigation/enhancement designs proposed

The Performance Indicators and monitoring plans prepared for *Project Implementation* are presented in **Table 1.1**.

**Table 1.1: Performance Indicators for Project Implementation**

| Sl. No. | Indicator                                              | Details                                                                                                                                                                              | Stage            | Responsibility                         |
|---------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------|
| A       | Environmental Condition Indicators and Monitoring Plan |                                                                                                                                                                                      |                  |                                        |
| 1       | Air Quality                                            | The parameters to be monitored, frequency and duration of monitoring as well as the locations to be monitored will be as per the Monitoring Plan prepared ( <i>Refer Table 1.2</i> ) | Pre Construction | PU through approved monitoring agency  |
|         |                                                        |                                                                                                                                                                                      | Construction     |                                        |
|         |                                                        |                                                                                                                                                                                      | Operation        | R&B through approved monitoring agency |
| 2       | Noise Levels                                           |                                                                                                                                                                                      | Pre Construction | PU through approved monitoring agency  |
|         |                                                        |                                                                                                                                                                                      | Construction     |                                        |
|         |                                                        |                                                                                                                                                                                      | Operation        | R&B through approved monitoring agency |
| 3       | Water Quality                                          |                                                                                                                                                                                      | Pre Construction | PU through approved monitoring agency  |
|         |                                                        |                                                                                                                                                                                      | Construction     |                                        |
|         |                                                        |                                                                                                                                                                                      | Operation        | R&B through approved monitoring agency |
| 4       | Soil Quality                                           |                                                                                                                                                                                      | Pre Construction | PU through approved monitoring agency. |
|         |                                                        |                                                                                                                                                                                      | Construction     |                                        |
|         |                                                        |                                                                                                                                                                                      | Operation        | R&B through approved monitoring agency |



| Sl. No. | Indicator                                               | Details                                                                                                                                                                                                                       | Stage            | Responsibility                                                 |
|---------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------|
| B       | Environmental Management Indicators and Monitoring Plan |                                                                                                                                                                                                                               |                  |                                                                |
| 1       | Construction Camps                                      | Location of construction camps have to be identified and parameters indicative of environment in the area has to be reported                                                                                                  | Pre-construction | PU                                                             |
| 2       | Borrow Areas                                            | Location of borrow areas have to be identified and parameters indicative of environment in the area has to be reported. <i>Refer Guidelines 7.2 (Guidelines for Borrow Areas Management) for the identified Borrow Areas.</i> | Pre-construction | PU                                                             |
| 3       | Tree Cutting                                            | Progress of tree removal marked for cutting is to be reported                                                                                                                                                                 | Pre-construction | Forest Department to PU                                        |
| 4       | Tree Plantation                                         | Progress of measures suggested as part of the Strategy is to be reported                                                                                                                                                      | Construction     | Forest Department                                              |
| C       | Management & Operational Performance Indicators         |                                                                                                                                                                                                                               |                  |                                                                |
| 1       | Survival Rate of Trees                                  | The number of trees surviving during each visit will be compared with the number of saplings planted                                                                                                                          | Operation        | Forest Department/R&B                                          |
| 2       | Status Regarding Rehabilitation of Borrow Areas         | The PU will undertake site visits to determine how many borrow areas have been rehabilitated in line with the landowner's request and to their full satisfaction.                                                             | Operation        | The R&B/APRDC will be responsible for a period of three years. |
| 3       | Soil Erosion                                            | Visual monitoring and operation inspection of embankments will be carried out once in three months.                                                                                                                           | Operation        | The R&B/APRDC will be responsible for a period of three years. |

### 1.1.2 Monitoring Parameters and Standards

- The Environmental monitoring of the parameters involved and the threshold limits specified are discussed below:

#### Ambient Air Quality Monitoring (AAQM)

The air quality parameters viz: Sulphur Dioxide (SO<sub>2</sub>), Oxides of Nitrogen (NO<sub>x</sub>), Carbon Monoxide (CO), Hydro-Carbons (HC), Suspended Particulate Matter (SPM), and Respirable Particulate Matter (RPM) shall be regularly monitored at identified locations from the start of the construction activity. The air quality parameters shall be monitored in accordance with the National Ambient Air Quality Standards as given in **Table 1.2A**. The duration and the pollution parameters to be monitored and the responsible institutional arrangements are detailed out in the Environmental Monitoring Plan **Table 1.3** and specific details in chapter on baseline environment.

**Table 1.2A National Standard of Air**

| Pollutant                             | Time Weighted Average | Sensitive Area                                | Industrial Area                                | Residential, Rural & Other Areas               | Method of Measurement                                                            |
|---------------------------------------|-----------------------|-----------------------------------------------|------------------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------|
| Sulphur Dioxide (SO <sub>2</sub> )    | Annual*<br>24 hours** | 15 µg/m <sup>3</sup><br>30 µg/m <sup>3</sup>  | 80 µg/m <sup>3</sup><br>120 µg/m <sup>3</sup>  | 60 µg/m <sup>3</sup><br>80 µg/m <sup>3</sup>   | Improved West and Gaeke Method<br>Ultraviolet Fluorescence                       |
| Oxides of Nitrogen (NO <sub>x</sub> ) | Annual*<br>24 hours** | 15 µg/m <sup>3</sup><br>30 µg/m <sup>3</sup>  | 80 µg/m <sup>3</sup><br>120 µg/m <sup>3</sup>  | 60 µg/m <sup>3</sup><br>80 µg/m <sup>3</sup>   | Jacob & Hochheiser Modified method<br>Gas phase Chemiluminescence                |
| Suspended Particulate Matter (SPM)    | Annual*<br>24 hours** | 70 µg/m <sup>3</sup><br>100 µg/m <sup>3</sup> | 360 µg/m <sup>3</sup><br>500 µg/m <sup>3</sup> | 140 µg/m <sup>3</sup><br>200 µg/m <sup>3</sup> | High Volume Sampler (Average flow rate not less than 1.1 m <sup>3</sup> /minute) |
| Respirable                            | Annual*               | 50 µg/m <sup>3</sup>                          | 120 µg/m <sup>3</sup>                          | 60 µg/m <sup>3</sup>                           | Respirable Particulate                                                           |

| Pollutant                                     | Time Weighted Average | Sensitive Area                                             | Industrial Area                                          | Residential, Rural & Other Areas                           | Method of Measurement                                               |
|-----------------------------------------------|-----------------------|------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Particulate Matter (RPM)<br>Size < 10 $\mu$ m | 24 hours**            | 75 $\mu$ g/m <sup>3</sup>                                  | 150 $\mu$ g/m <sup>3</sup>                               | 100 $\mu$ g/m <sup>3</sup>                                 | Matter Sampler                                                      |
| Lead (Pb)                                     | Annual*<br>24 hours** | 0.50 $\mu$ g/m <sup>3</sup><br>0.75 $\mu$ g/m <sup>3</sup> | 1.0 $\mu$ g/m <sup>3</sup><br>1.5 $\mu$ g/m <sup>3</sup> | 0.75 $\mu$ g/m <sup>3</sup><br>1.00 $\mu$ g/m <sup>3</sup> | AAS Method after sampling using EPM 2000 or equivalent filter paper |
| Carbon Monoxide (CO)                          | 8 hours**<br>1 hour   | 1.0 mg/m <sup>3</sup><br>2.0 mg/m <sup>3</sup>             | 5.0 mg/m <sup>3</sup><br>10.0 mg/m <sup>3</sup>          | 2.0 mg/m <sup>3</sup><br>4.0 mg/m <sup>3</sup>             | Non-dispersive infrared Spectroscopy                                |

\* Annual arithmetic mean of minimum 104 measurements in a year taken twice a week 24 hourly at uniform interval

\*\*24 hourly/8 hourly values should be met 98% of the time in a year. However, 2% of the time, it may exceed but not on two consecutive days.

Source: Central Pollution Control Board, 1997 Gazette Notification dated 4/94, Part II Sec 3 (ii)

### Noise Quality Monitoring

The noise levels shall be monitored at already designated locations in accordance with the Ambient Noise Quality standards given in **Table 1.2B**. The duration and the noise pollution parameters to be monitored and the responsible institutional arrangements are detailed in the Environmental Monitoring Plan **Table 1.3** and specific details in chapter on baseline environment.

**Table 1.2B National Standard of Noise**

| Area Code | Category of Zones | Limits of Leq in dB(A) Day* | Night* |
|-----------|-------------------|-----------------------------|--------|
| A         | Industrial        | 75                          | 70     |
| B         | Commercial        | 65                          | 55     |
| C         | Residential       | 55                          | 45     |
| D         | Silence Zone **   | 50                          | 40     |

\* Daytime shall mean from 6.00am to 10.00 pm and Night shall mean from 10.00 pm to 6.00 am

\*\* Silence zone is defined as area up to 100 meters around premises of hospitals, educational institutions and courts. Use of vehicles horns, loud speakers and bursting of cracking are banned in these zones.

### Water Quality Monitoring

Water quality parameters such as pH, BOD, COD, DO coliform count, total suspended solids, total dissolved solids, Iron, etc. shall be monitored at all identified locations during the construction stage as per standards prescribed by Central Pollution Control Board and Indian Standard Drinking water specifications IS 10500, 1991, presented in **Table 1.2C**. The duration and the pollution parameters to be monitored and the responsible institutional arrangements are detailed out in the Environmental Monitoring Plan **Table 1.3** and specific details in chapter on baseline environment.

**Table 1.2C: National Standard of Water**

| Sr. No | Parameters             | IS:2296 (Class C) | IS:10500   | Method Adopted                            |
|--------|------------------------|-------------------|------------|-------------------------------------------|
| 1      | PH                     | 6.5-8.5           | 6.5-8.5    | pH meter                                  |
| 2      | BOD (3 days 27°C)      | 3.0               | NS         | DO-Azide modification of Winkler's method |
| 3      | Temperature (°C)       | NS                | NS         | Thermometer                               |
| 4      | Dissolved oxygen       | 4                 | NS         | Azide Modification of Winkler's method    |
| 5      | Color (Hazen)          | 300               | NS         | Visual Comparison method                  |
| 6      | Fluorides (F)          | 1.5               | 1.0 (1.5)  | SPANDS method                             |
| 7      | Chlorides (Cl)         | 600               | 250(1000)  | Argentometric Titration                   |
| 8      | Total Dissolved Solids | 1500              | 500 (2000) | Gravimetric Analysis                      |

| Sr. No | Parameters                   | IS:2296 (Class C) | IS:10500     | Method Adopted                       |
|--------|------------------------------|-------------------|--------------|--------------------------------------|
| 9      | Sulphates (SO <sub>4</sub> ) | 400               | 200 (400)    | Barium Chloride method               |
| 10     | Iron (Fe)                    | 50                | 0.3 (1.0)    | Phenanthroline method                |
| 11     | Oil and Grease               | 0.1               | NS           | Partition – Gravimetric method       |
| 12     | Nitrates                     | 50                | 45 (100)     | Chromotropic acid                    |
| 13     | Chromium (Cr <sup>6+</sup> ) | 0.05              | 0.05         | Atomic Absorption Spectrophotometry  |
| 14     | Cadmium (Cd)                 | 0.01              | 0.01         | Atomic Absorption Spectrophotometry  |
| 15     | Lead (Pb)                    | 0.1               | 0.05         | Atomic Absorption Spectrophotometry  |
| 16     | Copper (Cu)                  | 1.5               | 0.05 (1.5)   | Atomic Absorption Spectrophotometry  |
| 17     | Cyanide (CN)                 | 0.05              | 0.05         | Chloramine-T-method                  |
| 18     | Selenium (Se)                | 0.05              | 0.01         | Atomic Absorption Spectrophotometry  |
| 19     | Arsenic (As)                 | 0.2               | 0.05         | Atomic Absorption Spectrophotometry  |
| 20     | Phenols                      | 0.005             | 0.001(0.002) | Spectrophotometer                    |
| 21     | Detergents                   | 1.0               | 0.2 (1.0)    | Spectrophotometer                    |
| 22     | DDT                          | Absent            | Absent       | Spectrophotometer                    |
| 23     | Total Coliform (MPN/100 ml)  | 5000              | NS           | Multiple Tube Fermentation Technique |

NS: Not specified; Brackets ([ ]) indicates extended limits. All the values in mg/l if otherwise mentioned

### 1.1.3 Monitoring Plans for Environment Condition

For each of the environmental components, the monitoring plan specifies the parameters to be monitored; location of the monitoring sites; frequency and duration of monitoring. The monitoring plan also specifies the applicable standards, implementation and supervising responsibilities. The monitoring plan for the various environmental condition indicators of the project in construction and operation stages is presented in Table 1.3.

Monitoring plan does not include the requirement of arising out of Regulation Provision such as obtaining NOC/ consent for plant site operation.

Table 1.3: Environmental Monitoring Plan

| Attribute              | Project Stage   | Parameter                                                                                                                           | Special Guidance                                                                                                                                                                                                             | Standards                                                                                              | Frequency                               | Duration                                | Location                                                         | Implementation           |
|------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------|------------------------------------------------------------------|--------------------------|
| <b>Air</b>             | Construction    | CO, NO <sub>x</sub> , SPM, RPM, and SO <sub>2</sub>                                                                                 | High volume sampler to be located 50 m from the plant in the downwind direction. Use method specified by CPCB for analysis                                                                                                   | Air (prevention and Control of Pollution) Rules, CPCB, 1994                                            | Three seasons per year                  | 24 hours Sampling                       | Along the road (refer table no 4.2)<br>Hot mix / batching plant  | Contractor/<br>EMU-APRSP |
|                        | Operation       |                                                                                                                                     |                                                                                                                                                                                                                              |                                                                                                        | Two seasons in a year for three years   |                                         | Along the road (refer table no 4.2)                              | Contractor/<br>EMU-APRSP |
| <b>Water</b>           | Construction    | All essential characteristics and some of desirable characteristics as decided by the Environmental Specialist of the CSC and APRDC | Grab sample collected from source and analyse as per Standard Methods for Examination of Water and Wastewater                                                                                                                | Indian Standards for Inland Surface Waters (IS: 2296, 1982) and for Drinking Water (IS : 10500 - 1991) | Four seasons per year                   | Grab Sampling                           | Along the road (refer table no 4.6)                              | Contractor/<br>EMU-APRSP |
|                        | Operation       |                                                                                                                                     |                                                                                                                                                                                                                              |                                                                                                        | four seasons for three years            |                                         | Surface water sources                                            | Contractor/<br>EMU-APRSP |
| <b>Noise</b>           | Construction    | Noise levels on dB (A) scale                                                                                                        | Equivalent noise levels using an integrated noise level meter kept at a distance of 15 from edge of pavement<br>Equivalent noise levels using an integrated noise level meter kept at a distance of 15 from edge of pavement | MoEF Noise Rules, 2000                                                                                 | Three seasons per year                  | Leq in dB(A) of day time and night time | Along the road (refer table no 4.10)<br>Hot mix / batching plant | Contractor/<br>EMU-APRSP |
|                        | Operation       |                                                                                                                                     |                                                                                                                                                                                                                              |                                                                                                        | Three seasons per year for three years. |                                         | Along the road (refer table no 4.10)                             | Contractor/<br>EMU-APRSP |
| <b>Soil</b>            | Construction    | Monitoring of Pb, SAR and Oil & Grease                                                                                              | Sample of soil collected to acidified and analysed using absorption spectrophotometer                                                                                                                                        | Threshold for each contaminant set by IRIS database of USEPA until national standards are promulgated  | Four seasons per year                   | Grab Sampling                           | Along the road (refer table no 4.4)<br>Hot mix / batching plant  | Contractor/<br>EMU-APRSP |
|                        | Operation       |                                                                                                                                     |                                                                                                                                                                                                                              |                                                                                                        | four seasons for three years            |                                         | Along the road (refer table no 4.4)                              | Contractor/<br>EMU-APRSP |
| <b>Borrow area</b>     | Construction    | As per Guidelines                                                                                                                   | Visual Observation                                                                                                                                                                                                           | -                                                                                                      | Once in a month                         | -                                       | Borrow area location                                             | Contractor/<br>EMU-APRSP |
| <b>Tree plantation</b> | Operation stage | As per Rehabilitation Plan                                                                                                          |                                                                                                                                                                                                                              |                                                                                                        | Quarterly                               | -                                       | Areas where plantation is being done                             | Contractor/<br>EMU-APRSP |

## ENVIRONMENTAL MONITORING LOCATIONS

In addition of the critical locations selected during design stage, the environmental monitoring will also be done at the construction camp site and any other plant site during construction stage. List of critical locations for caring out monitoring is presented in **chapter 4: Baseline environment**.

### Locations of the Air Samples

| Sl. No.                                              |                 |                                                   | 1              | 2                   | 3              | 4         | 5               | 6              |
|------------------------------------------------------|-----------------|---------------------------------------------------|----------------|---------------------|----------------|-----------|-----------------|----------------|
| Name of location                                     |                 | National Ambient Air Quality Standards (24 Hours) | Kakinada       | Venkatakrishnapuram | Samarlakota    | Medapeta  | Balabadhrapuram | Anaparthi      |
| Chainage                                             |                 |                                                   | Km 1/250       | Km 8/550            | Km 15/000      | km 22/300 | Km 34/600       | Km 42/000      |
| Land use                                             |                 |                                                   | Junction point | Built Up            | Junction point | Built Up  | Built Up        | Junction point |
| Location code                                        |                 |                                                   | AQ-1           | AQ-2                | AQ-3           | AQ-4      | AQ-5            | AQ-6           |
| Date of monitoring                                   |                 |                                                   |                |                     |                |           |                 |                |
| Pollutants concentration in $\mu\text{g}/\text{m}^3$ | RPM             |                                                   |                |                     |                |           |                 |                |
|                                                      | SPM             |                                                   |                |                     |                |           |                 |                |
|                                                      | SO <sub>2</sub> |                                                   |                |                     |                |           |                 |                |
|                                                      | NO <sub>2</sub> |                                                   |                |                     |                |           |                 |                |
|                                                      | Pb              |                                                   |                |                     |                |           |                 |                |
| mg/m <sup>3</sup>                                    | CO              |                                                   |                |                     |                |           |                 |                |
| PPM                                                  | HC              |                                                   |                |                     |                |           |                 |                |

### Locations of the Soil Samples

| S. N. | Location Code | Description of the Monitoring location | Chainage  | Type of Land |
|-------|---------------|----------------------------------------|-----------|--------------|
| 1.    | S1            | Venkatakrishnapuram                    | Km 8/550  | Open land    |
| 2.    | S2            | Vetlapalem                             | Km 17/800 | Open land    |
| 3.    | S3            | Bikkavolu                              | Km 34/150 | Open land    |
| 4.    | S4            | Dwarapudi                              | Km 44/800 | Open land    |
| 5.    | S5            | Vemagiri                               | Km 59/400 | Open land    |

### Water Quality Monitoring Locations

| Location Code | Location            | Chainage  | Name of Water body    | Type of Water Body |
|---------------|---------------------|-----------|-----------------------|--------------------|
| GW1           | Kakinada            | Km 1/250  | Ground Water          | Hand Pump          |
| GW2           | Venkatakrishnapuram | Km 8/550  | Ground Water          | Hand Pump          |
| SW 1          | Samalakota Canal    | Km 15/000 | Surface Water         | Canal              |
| GW3           | Medapeta            | km 22/300 | Ground Water          | Hand Pump          |
| SW 2          | Dwarapudi           | Km 51/500 | Drinking Water source | Water Tank         |
| GW4           | Kadiyam             | Km 57/750 | Ground Water          | Hand Pump          |

### Noise Quality Monitoring Locations

| Sl. No. | Location code | Location (Junction Points) | Chainage  | Leq value dB (A) |            |
|---------|---------------|----------------------------|-----------|------------------|------------|
|         |               |                            |           | Day time         | Night time |
| 1.      | NQ - 1        | Kakinada                   | Km 1/250  |                  |            |
| 2.      | NQ - 2        | Venkatakrishnapuram        | Km 8/550  |                  |            |
| 3.      | NQ - 3        | Samarlakota                | Km 15/000 |                  |            |
| 4.      | NQ - 4        | Medapeta                   | km 22/300 |                  |            |
| 5.      | NQ - 5        | Bikkavolu                  | Km 34/050 |                  |            |
| 6.      | NQ - 6        | Anaparthi                  | Km 42/000 |                  |            |
| 7.      | NQ - 7        | Dwarapudi                  | Km 51/500 |                  |            |
| 8.      | NQ - 8        | Kadiyam                    | Km 57/750 |                  |            |
| 9.      | NQ - 9        | Vemagiri                   | Km 59/400 |                  |            |

| Sl. No. | Location code | Location (Junction Points) | Chainage | Leq value dB (A) |            |
|---------|---------------|----------------------------|----------|------------------|------------|
|         |               |                            |          | Day time         | Night time |
|         |               |                            |          |                  |            |

### REPORTING PROCEDURES

Mitigation and enhancement measures adopted in the final design have been identified in the contract documents and Bill of quantities so that performance and completion is effective. The periodic site visits of the EO/Engineer/EE of the PU will keep a record of progress as well as the site-specific EMP implementation records. The frequent meeting of the EO/Engineer with the Contractors will ensure any information and communication gap with regard to the Construction phase environmental management at construction site; labour and construction camps quarry and borrow areas etc. It is necessary that the EO/Engineer should visit the sites for evolving a concept for the Environmental Management with regard to the siting of various construction requirements. The various reporting guidelines and arrangements are presented in TABLE 1.4. DESIRED MONITORING AND REPORTING PROCESS AND RESPONSIBILITIES.

Table 1.4: Desired Monitoring and Reporting Process and Responsibilities

| Format No.                   | Item                                                  | Stage                        | Contractor                                    | CSC Consultant    |                                   | PIU-APRDC                           |                                   | Reporting from APRDC to World Bank) |
|------------------------------|-------------------------------------------------------|------------------------------|-----------------------------------------------|-------------------|-----------------------------------|-------------------------------------|-----------------------------------|-------------------------------------|
|                              |                                                       |                              | Implementation and Reporting to TA Consultant | Supervision       | Reporting to CU, APRSP            | Oversee Field Compliance Monitoring | Reporting to EMU, APRSP           |                                     |
| P1                           | Identification for disposal locations                 | Pre -construction            | One Time                                      | One Time          | One Time                          | One Time                            | One Time                          | One Time                            |
| P2                           | Setting up of Construction Camp                       | Pre- construction            | One Time                                      | One Time          | One Time                          | One Time                            | One Time                          | One Time                            |
| P3                           | Establishment of Borrow areas                         | Pre- construction            | Monthly                                       | Monthly           | Quarterly                         | Quarterly                           | Quarterly                         | Quarterly                           |
| P4                           | Establishment of HMP/ BMP                             | Before start of construction | One Time                                      | One Time          | One Time                          | One Time                            | One Time                          | One Time                            |
| P5                           | Road Safety and Traffic Management                    | Pre- construction            | Monthly                                       | Monthly           | Monthly                           | Monthly                             | Quarterly                         | Quarterly                           |
| P6                           | Arrangement for Temporary Land                        | Pre- Construction            | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Quarterly                         | Quarterly                           |
| P7                           | Pollution Monitoring                                  | Pre- Construction            | Quarterly                                     | During Monitoring | Immediately on receipt of results | Quarterly                           | Quarterly                         | Quarterly                           |
| P8                           | Tree cutting/Stump Removal                            | During construction period   | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Quarterly                         | Quarterly                           |
| P9                           | Identification of Source of water for Construction    | Pre- construction            | One Time                                      | One Time          | One Time                          | One Time                            | One Time                          | One Time                            |
| C1                           | Details of earth work                                 | During Construction period   | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Quarterly                         | Quarterly                           |
| C2                           | Details of Hot Mix Plant                              | During Construction period   | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Quarterly                         | Quarterly                           |
| C3                           | Details of landfill locations/                        | During Construction period   | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Quarterly                         | Quarterly                           |
| C4                           | Details of Machinery in Operations                    | During Construction period   | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Quarterly                         | Quarterly                           |
| C5                           | Redevelopment of borrow areas                         | During construction period   | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Half Yearly                       | Quarterly                           |
| C6                           | Safety Check List                                     | During construction period   | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Half Yearly                       | Quarterly                           |
| C7                           | Accident Report                                       | During construction period   | After Accident                                | After Accident    | Immediately on receipt of report  | Quarterly                           | Half Yearly                       | Quarterly                           |
| C8                           | Pollution Monitoring                                  | During construction period   | Quarterly                                     | During Monitoring | Quarterly                         | Quarterly                           | Quarterly                         | Quarterly                           |
| C9                           | Enhancement Measures                                  | During Construction          | Monthly                                       | Monthly           | Monthly                           | Quarterly                           | Quarterly                         | Quarterly                           |
| C10                          | Restoration of Construction Sites                     | Immediate after Construction | One Time                                      | One Time          | One Time                          | One Time                            | One Time                          | Quarterly                           |
| O1                           | Pollution Monitoring                                  | During Operation             | -                                             | -                 | -                                 | Quarterly                           | Quarterly                         | Quarterly                           |
| O2                           | Monitoring of culvert opening and longitudinal drains | During Operation             | -                                             | -                 | -                                 | Twice a year Pre and Post monsoon   | Twice a year Pre and Post monsoon | -                                   |
| FORMS TO BE FILLED BY PU/CSC |                                                       |                              |                                               |                   |                                   |                                     |                                   |                                     |
| PU1                          | Form for keeping records of Consent                   | Construction                 | Quarterly                                     |                   | Quarterly                         | Quarterly                           | Half Yearly                       | -                                   |

| Format No. | Item                                                            | Stage        | Contractor                                    | CSC Consultant |                        | PIU-APRDC                           |                         | Reporting from APRDC to World Bank) |
|------------|-----------------------------------------------------------------|--------------|-----------------------------------------------|----------------|------------------------|-------------------------------------|-------------------------|-------------------------------------|
|            |                                                                 |              | Implementation and Reporting to TA Consultant | Supervision    | Reporting to CU, APRSP | Oversee Field Compliance Monitoring | Reporting to EMU, APRSP |                                     |
|            | obtained by contractor (To be filled by Supervision Consultant) |              |                                               |                |                        |                                     |                         |                                     |
| PU2        | Checklist for Environment Inspection                            | -            |                                               | -              | -                      | -                                   | -                       | -                                   |
| PU3        | Summary Sheet (To be filled by PU)                              | Construction |                                               | -              | -                      | Quarterly                           | Half Yearly             | -                                   |

The reporting formats are presented in the end of the EMP Document



## P1: IDENTIFICATION OF DISPOSAL LOCATIONS

Name of Corridor \_\_\_\_\_

(Give chainages and nearest settlements from both ends)

| Sl. No. | Criteria on which information for each site is to be collected      | Site 1 | Site 2 | Site 3 | Site 4 |
|---------|---------------------------------------------------------------------|--------|--------|--------|--------|
| 1       | Existing Land Use                                                   |        |        |        |        |
| 2       | Area covered (m <sup>2</sup> )                                      |        |        |        |        |
| 3       | Total Material that can be dumped within the site (m <sup>3</sup> ) |        |        |        |        |
| 4       | Depth to which dumping is feasible (m)                              |        |        |        |        |
| 5       | Distance of nearest watercourse (m)                                 |        |        |        |        |
| 6       | Nearest Settlement (m)                                              |        |        |        |        |
| 7       | Date/s of Community Consultation/s                                  |        |        |        |        |
| 8       | Whether the community is agreeable to siting of dumping site (Y/N)  |        |        |        |        |
| 9       | Date of Permission from Villager/local community                    |        |        |        |        |
| 10      | Proposed future use of the Site                                     |        |        |        |        |
| 11      | Selected Site (tick any one column only)                            |        |        |        |        |

Enclosures (Tick as appropriate)

1. Map of each location
2. Photographs
  - a. Each Disposal location
  - b. Each community consultation
3. Photo copy of Agreement

Remark

### Submitted

Signature.....

Name.....

Designation: Environmental Engineer  
Contractor

### Checked

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision Consultant

### Approved

Signature.....

Name.....

Designation: Executive Engineer - APRDC  
Client

## P2: SETTING –UP CONSTRUCTION CAMP AND STORAGE AREA

Name of Corridor \_\_\_\_\_

Construction Stage Report: Date \_\_\_\_\_ Month \_\_\_\_\_ Year \_\_\_\_\_

(Site Layout of Construction camp and working drawings of dwelling units with allied facilities to be attached with format)

Format to be submitted before target date of establishing camps

Location of Camp \_\_\_\_\_

| Sl. No.   | Item                                                         | Unit                                         | Details | Remarks by CMU if any |
|-----------|--------------------------------------------------------------|----------------------------------------------|---------|-----------------------|
| <b>1.</b> | <b>Detail of item camp</b>                                   |                                              |         |                       |
| a.        | Size of Camp                                                 | m x m                                        |         |                       |
| b.        | Area of Camp                                                 | Sq.m                                         |         |                       |
| c.        | Distance from Nearest Settlement                             |                                              |         |                       |
| d.        | Distance from Nearest Water Source                           | Type/Size/Capacity/<br>present Use/Ownership |         |                       |
|           | Date of camp being operational dd/mm/yy                      |                                              |         |                       |
|           | Present land use                                             |                                              |         |                       |
|           | No of trees with girth > 0.3m.                               |                                              |         |                       |
| e.        | Details of Storage area (Availability of impervious surface) | Mxm                                          |         |                       |
| f.        | Availability of separate waste disposal from storage area    | Cum                                          |         |                       |
| <b>2.</b> | <b>Details of toposoil stacking</b>                          |                                              |         |                       |
| a.        | Quantity of top soil removed                                 | sq.m                                         |         |                       |
| b.        | Detail of storage of topsoil                                 | Describe stacking arrangement                |         |                       |
| <b>3.</b> | <b>Details of workforce</b>                                  |                                              |         |                       |
| a.        | Total No of Labourers                                        | Nos                                          |         |                       |
| b.        | Total no of Male Workers                                     | Nos                                          |         |                       |
| c.        | No of Male Workers below 18 years of age                     | Nos                                          |         |                       |
| d.        | Total No of Female Workers                                   | Nos                                          |         |                       |
| e.        | No of Female workers below 18 years of age                   | Nos                                          |         |                       |
| f.        | No of children                                               | Nos                                          |         |                       |
| <b>4.</b> | <b>Details of dwelling units</b>                             |                                              |         |                       |
| a.        | No of dwellings/huts                                         | Nos                                          |         |                       |
| b.        | Minimum Size of Dwelling                                     | Mxm                                          |         |                       |
| c.        | No. of openings per dwelling                                 | Nos                                          |         |                       |
| d.        | Minimum size of opening                                      | Mxm                                          |         |                       |
| e.        | Walls                                                        | specifications                               |         |                       |
| f.        | Roofing                                                      | specifications                               |         |                       |
| g.        | Flooring                                                     | specifications                               |         |                       |
| h.        | Drinking Water Tank                                          | specifications                               |         |                       |
| i.        | Capacity of Drinking water Tank                              | Cum                                          |         |                       |
| j.        | Size of Drinking Water Tank                                  | Mxm                                          |         |                       |
| k.        | Total no of WC                                               | Nos                                          |         |                       |
| l.        | No of Wcs for female workers                                 | Nos                                          |         |                       |
| m.        | Minimum Size of WC                                           | Mxm                                          |         |                       |

| Sl. No. | Item                                                          | Unit   | Details | Remarks by CMU if any |
|---------|---------------------------------------------------------------|--------|---------|-----------------------|
| n.      | Total No of Bathrooms for female workers                      | Nos    |         |                       |
| o.      | Size of septic tank for WC/Baths                              | Mxm    |         |                       |
| p.      | Capacity of Water Tank for WCs/ Bathrooms and general purpose |        |         |                       |
| q.      | Fencing around camp                                           | Y/N    |         |                       |
| 5       | Details of facilities                                         |        |         |                       |
| a.      | Availability of security guard 24 hrs a day                   | Yes/No |         |                       |
| b.      | Details of First Aid Facility                                 | Yes/No |         |                       |
| c.      | Availability of Day Care Centre                               | Yes/No |         |                       |
| d.      | Availability of dust bins (capacity 60 ltr)                   | Nos    |         |                       |

Remark

**Submitted**

Signature.....

Name.....

Designation: Environmental Engineer  
Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC  
Client

**P3: ESTABLISHMENT OF BORROW AREAS**  
(To be submitted by Contractor for taking consent to open Borrow area)

Name of Corridor\_\_\_\_\_

| Sl No. | Location        |               |                  |                      | Area (m <sup>2</sup> ) | Quantity of Available Material | Type of Material | Distance from nearest Water Course (m) | Distance from nearest Settlement | Land Use |       | No. of Trees to be Affected | Approved by EO (Y/N) | Remark |
|--------|-----------------|---------------|------------------|----------------------|------------------------|--------------------------------|------------------|----------------------------------------|----------------------------------|----------|-------|-----------------------------|----------------------|--------|
|        | Name of Village | Chainage (km) | Side (LHS / RHS) | Haul road length (m) |                        |                                |                  |                                        |                                  | Before   | After |                             |                      |        |
|        |                 |               |                  |                      |                        |                                |                  |                                        |                                  |          |       |                             |                      |        |
|        |                 |               |                  |                      |                        |                                |                  |                                        |                                  |          |       |                             |                      |        |
|        |                 |               |                  |                      |                        |                                |                  |                                        |                                  |          |       |                             |                      |        |
|        |                 |               |                  |                      |                        |                                |                  |                                        |                                  |          |       |                             |                      |        |

Attach Photograph of Proposed Site, Location Map, Agreement

Rehabilitation Plan Measures

Location 1:

Location 2:

Remarks

**Submitted**

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer

Construction Supervision Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client

#### P4: ESTABLISHMENT OF HMP / BMP

Name of Corridor\_\_\_\_\_

| SI No. | Location        |               |                  |                      | Area (m <sup>2</sup> ) | Distance from nearest Water Course (m) | Distance from nearest Settlement | Existing Land Use | Prevalent Wind Direction | Weather in Down Wind Direction (Y/N) | Approved by EO (Yes / No) | Remarks |
|--------|-----------------|---------------|------------------|----------------------|------------------------|----------------------------------------|----------------------------------|-------------------|--------------------------|--------------------------------------|---------------------------|---------|
|        | Name of Village | Chainage (km) | Side (LHS / RHS) | Haul road length (m) |                        |                                        |                                  |                   |                          |                                      |                           |         |
|        |                 |               |                  |                      |                        |                                        |                                  |                   |                          |                                      |                           |         |
|        |                 |               |                  |                      |                        |                                        |                                  |                   |                          |                                      |                           |         |
|        |                 |               |                  |                      |                        |                                        |                                  |                   |                          |                                      |                           |         |
|        |                 |               |                  |                      |                        |                                        |                                  |                   |                          |                                      |                           |         |

1. Attach Photograph of Proposed Site

|        |
|--------|
| Remark |
|--------|

#### Submitted

Signature.....

Name.....

Designation: Environmental Engineer  
Contractor

#### Checked

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision Consultant

#### Approved

Signature.....

Name.....

Designation: Executive Engineer - APRDC  
Client

### P5: ROAD SAFETY REPORTING FORMATS

Name of Corridor \_\_\_\_\_

One time reporting before commencement construction in the Construction Zone

Sketch construction zone showing all sub zones and location of signs, etc.to be attached with format (Reporting by Contractor to APRDC)

Format on Acquisition of Temporary diversions to be attached with format

Construction stage: Monthly Report – Date .....Month.....Year.....

DIVERSION NO.\_\_\_\_Location ( km\_\_\_\_)

| Sl. No.                                    | Item                                                                           | Unit | Compliance | Remarks |
|--------------------------------------------|--------------------------------------------------------------------------------|------|------------|---------|
| <b>Details of Construction Zone</b>        |                                                                                |      |            |         |
| 1.                                         | Length of Construction Zone                                                    | km   |            |         |
| 2.                                         | Distance between this and next construction zone                               | km   |            |         |
| 3.                                         | Length of transition sub zone (should be min 50 for a speed of 50 km / hr)     |      |            |         |
| 4.                                         | Length of work sub zone in urban stretch (should be <2 km)                     | km   |            |         |
| 5.                                         | Length of work sub zone in rural stretch (5-10 km)                             | km   |            |         |
| 6.                                         | Distance between two work sub zones                                            |      |            |         |
| <b>Signage's in Construction Zones</b>     |                                                                                |      |            |         |
| 1.                                         | Sign saying 'Men at Work' 1 km ahead of transition sub zone                    | Y/N  |            |         |
| 2.                                         | Supplementary sign saying diversion 1 km provided                              | Y/N  |            |         |
| 3.                                         | Sign saying 'Road Closed ahead' provided                                       | Y/N  |            |         |
| 4.                                         | Compulsory Right Turn / Left sign provided                                     | Y/N  |            |         |
| 5.                                         | Detour sign placed                                                             |      |            |         |
| 6.                                         | Sharp deviation sign placed at end of advance warning sub zone                 | Y/N  |            |         |
| <b>Signage in Transition Sub Work Zone</b> |                                                                                |      |            |         |
| 1.                                         | Signage saying 'Keep Right / Left' provided                                    | Y/N  |            |         |
| 2.                                         | Delineators placed along length of transition                                  | Y/N  |            |         |
| <b>Signage in work sub zone</b>            |                                                                                |      |            |         |
| 1.                                         | Hazard Marker placed where railing for CD structure on diversion starts        | Y/N  |            |         |
| 2.                                         | Barricade on either side of work sub zone                                      | Y/N  |            |         |
| <b>Signage in Termination sub zone</b>     |                                                                                |      |            |         |
| 1.                                         | Sign for indication of end of work zone 120 m from end of termination sub zone | Y/N  |            |         |
| <b>Road Delineator</b>                     |                                                                                |      |            |         |
| 1.                                         | Roadway indicators provided                                                    |      |            |         |
| 2.                                         | Hazard Makers provided                                                         |      |            |         |
| 3.                                         | Object Makers Provided                                                         |      |            |         |

Remark \_\_\_\_\_

#### Submitted

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

#### Checked

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

#### Approved

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client

**P6: ARRANEMENT FOR TEMPORARY LAND**

Reporting by Contractor to APRDC

Name of Corridor\_\_\_\_\_

Construction stage: quarterly Report – Date: \_\_\_\_\_Month\_\_\_\_\_Year\_\_\_\_\_

(Site Layout of all locations to be attached with format)

| Sl. No. | Item                    | Target date for Establish ment | Date of Establis hment | Location        |               |                  |                        |                      | Present Land use | Size (mxm) | Existing Trees >30 cm girth | Dist. From nearest settlement | Dist. From nearest water source | Site approved or not (Y/N) | Remarks by CMU (PRBDB), if any |
|---------|-------------------------|--------------------------------|------------------------|-----------------|---------------|------------------|------------------------|----------------------|------------------|------------|-----------------------------|-------------------------------|---------------------------------|----------------------------|--------------------------------|
|         |                         |                                |                        | Name of Village | Chainage (km) | Side (LHS / RHS) | Area (m <sup>2</sup> ) | Haul road length (m) |                  |            |                             |                               |                                 |                            |                                |
| 1.      | Borrow Areas            |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
|         | BA1                     |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
| 2.      | Workers Camps           |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
|         | WC1                     |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
| 3.      | Site for Batching Plant |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
|         | BP1                     |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
| 4.      | Site for Hot Mix Plant  |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
|         | HMP1                    |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
| 5.      | Stock Yard              |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |
|         | SY1                     |                                |                        |                 |               |                  |                        |                      |                  |            |                             |                               |                                 |                            |                                |

Remarks

**Submitted**

Signature.....

Name.....

Designation: Environmental Engineer  
Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC  
Client

## P7: POLLUTION MONITORING

Name of Corridor \_\_\_\_\_

Construction Stage: Report – Date: \_\_\_\_\_ Month \_\_\_\_\_ Year \_\_\_\_\_

Mitigation measures suggested in last report complied or Not.....  
if not reason thereof.....

(Location at which monitoring to be conducted as per EMP)

| S. No.                     | Chainage (km) | Details of locations | Duration of monitoring | Instruments used | Completion | Standards                                                                           | Results                                                                             | Reasons for exceeding standards | Mitigation Measures suggested | Type of area (Residential / Industrial / Commercial) | Remarks |
|----------------------------|---------------|----------------------|------------------------|------------------|------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------------------------------------------------|---------|
| <b>1. Air Monitoring</b>   |               |                      |                        |                  |            |                                                                                     |                                                                                     |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            | SPM<br>RSPM<br>HC<br>SOx<br>NOx                                                     | SPM<br>RSPM<br>HC<br>SOx<br>NOx                                                     |                                 |                               |                                                      |         |
| <b>2. Water Monitoring</b> |               |                      |                        |                  |            |                                                                                     |                                                                                     |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            | pH<br>TSS<br>TDS<br>Turbidity<br>Hardness<br>Coliform<br>BOD<br>COD<br>Oil & Grease | pH<br>TSS<br>TDS<br>Turbidity<br>Hardness<br>Coliform<br>BOD<br>COD<br>Oil & Grease |                                 |                               |                                                      |         |
| <b>3. Soil Monitoring</b>  |               |                      |                        |                  |            |                                                                                     |                                                                                     |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            | pH<br>Organic Matter<br>Alkalinity<br>Conductivity                                  | pH<br>Organic Matter<br>Alkalinity<br>Conductivity                                  |                                 |                               |                                                      |         |



| S. No.                     | Chainage (km) | Details of locations | Duration of monitoring | Instruments used | Completion | Standards                                              | Results                                                | Reasons for exceeding standards | Mitigation Measures suggested | Type of area (Residential / Industrial / Commercial) | Remarks |
|----------------------------|---------------|----------------------|------------------------|------------------|------------|--------------------------------------------------------|--------------------------------------------------------|---------------------------------|-------------------------------|------------------------------------------------------|---------|
|                            |               |                      |                        |                  |            | Water holding capacity Pb                              | Water holding capacity Pb                              |                                 |                               |                                                      |         |
| <b>4. Noise Monitoring</b> |               |                      |                        |                  |            |                                                        |                                                        |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            | L day equivalent<br>L night equivalent<br>L equivalent | L day equivalent<br>L night equivalent<br>L equivalent |                                 |                               |                                                      |         |

Remark

**Submitted**

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client

**P8: Tree cutting / Stump Removal**

Name of the Road:

Date: \_\_\_\_\_ Month \_\_\_\_\_ Year \_\_\_\_\_

| Sl. No. | Section<br>(Km) | Distance<br>from<br>edge of<br>existing<br>road | Identification<br>Number<br>Marked in<br>Field | Species |                   | Girth<br>(cm) | Dry /<br>Green | Average<br>Height<br>(m) | Wt. of<br>Tree/Stump | Remarks |
|---------|-----------------|-------------------------------------------------|------------------------------------------------|---------|-------------------|---------------|----------------|--------------------------|----------------------|---------|
|         |                 |                                                 |                                                | Local   | Botanical<br>Name |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |
|         |                 |                                                 |                                                |         |                   |               |                |                          |                      |         |

Remark

**Submitted**

Signature.....

Name.....

Designation: Environmental  
Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer -  
APRDC

Client

**P9: IDENTIFICATION OF SOURCE OF WATER FOR CONSTRUCTION**

Name of Corridor\_\_\_\_\_

Construction Stage: Monthly Report – Date.      Month.....Year

| S. No. | Source (Name) | Location /Ch. | Distance from Road | Permission Required | Remarks |
|--------|---------------|---------------|--------------------|---------------------|---------|
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |
|        |               |               |                    |                     |         |

Remark

**Submitted**

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client

## C1: DETAILS OF EARTHWORK

(To be filled by the Contractor)

Name of Corridor \_\_\_\_\_

Monthly Report for Each Borrow Area under use

Reporting Month.....

Date of submission.....

### Location of Borrow Area under use

|    | Name of Village | Chainage (km) | Side (LHS / RHS) | Haul road length (m) |
|----|-----------------|---------------|------------------|----------------------|
| I  |                 |               |                  |                      |
| II |                 |               |                  |                      |

(Show on a Sketch Plan clearly indicating distance and approach roads)

|  |
|--|
|  |
|--|

### 2. Details of Borrow Areas

|      |                                                                                                                                                                                                                                                                                  |  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 2.1  | Capacity of the Borrow Area                                                                                                                                                                                                                                                      |  |
| 2.2  | Percentage of the capacity exhausted                                                                                                                                                                                                                                             |  |
| 2.3  | Total quantity of the Earth Excavated (in cum)                                                                                                                                                                                                                                   |  |
| 2.4  | Quantity of Top Soil removed from the Borrow Areas                                                                                                                                                                                                                               |  |
| 2.5  | Location of Top Soil stored removed                                                                                                                                                                                                                                              |  |
| 2.6  | Quantity of Top Soil stored at the beginning of the month                                                                                                                                                                                                                        |  |
| 2.7  | Quantity of Top Soil utilized at the end of the month                                                                                                                                                                                                                            |  |
| 2.8  | Location (s) where Top Soil has been utilized (Specify on a location plan)                                                                                                                                                                                                       |  |
| 2.9  | Quantity of earthwork excavation from existing road                                                                                                                                                                                                                              |  |
| 2.10 | Total quantity of earthwork reused in cum. (5%)                                                                                                                                                                                                                                  |  |
| 2.11 | Location disposal (if other than sites) (Specify clearly on a location plan)                                                                                                                                                                                                     |  |
| 2.12 | Quantity of earthwork re-used in fill operation                                                                                                                                                                                                                                  |  |
| 2.13 | Location of borrow areas in disuse / exhausted                                                                                                                                                                                                                                   |  |
| 2.14 | Outline a rehabilitation plan for each of the exhausted borrow areas with special reference to Erosion Protection Measures. Also, submit at separate detailed rehabilitation plan for exhausted borrow areas for approval supported adequately with layouts, plans and drawings. |  |

|        |
|--------|
| Remark |
|--------|

Submitted

Checked

Approved

The Louis Berger Group. Inc, USA

Signature.....

Name.....

Designation: Environmental  
Engineer

Contractor

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

Signature.....

Name.....

Designation: Executive Engineer -  
APRDC

Client

## C2: DETAILS OF HOT MIX PLANT

(To be filled by Contractor)

Name of Corridor.....

Monthly Report for Each Hot Mix Plant

Reporting Month.....

Date of Submission.....

### 1. Environment Features of the surrounding area

|     |                                                                                   |  |
|-----|-----------------------------------------------------------------------------------|--|
| 1.1 | Name and location of Hot Mix Plant (w.r.t. PWD km ch.)                            |  |
| 1.2 | Wind direction                                                                    |  |
| 1.3 | Name (s), distance population and type of settlements in a 1.5 km radius of site. |  |

### 2. Draw Sketch Plan of HMP clearly indicating distance and approach roads.

|  |
|--|
|  |
|--|

### 3. Details of HMP and Mitigation Measures taken

|     |                     |  |
|-----|---------------------|--|
| 3.1 | Installed Capacity  |  |
| 3.2 | Average Utilization |  |
| 3.3 | Make                |  |
| 3.4 | Model               |  |
| 3.5 | Last Serviced       |  |

### 4. Explain Air Pollution Control Measures taken at the HMP site

|  |
|--|
|  |
|--|

### 5. Explain Noise Pollution Control Measures taken at the HMP site

|  |
|--|
|  |
|--|

Remark

**Submitted**

Signature.....

Name.....

Designation: Environmental  
Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant**Approved**

Signature.....

Name.....

Designation: Executive Engineer -  
APRDC

Client

**C3: DETAILS OF LAND FILL OPERATIONS**

(To be filled by Contractor)

Name of Corridor.....

Monthly Report for Each Package

Reporting Month.....

Date of Submission.....

**1. Environment Features of the surrounding area**

| 1.1    | Location of each land fill site (Provide sketch Map below) | Name of Village | Chainage (km) | Side (LHS/RHS) | Haul road length (m) |
|--------|------------------------------------------------------------|-----------------|---------------|----------------|----------------------|
|        | I                                                          |                 |               |                |                      |
|        | II                                                         |                 |               |                |                      |
| 1.2    | Capacity of each land fill site                            |                 |               |                |                      |
| 1.3    | Safety measure taken at land fill site (s)                 |                 |               |                |                      |
| 1.     |                                                            |                 |               |                |                      |
| 2.     |                                                            |                 |               |                |                      |
| 3.     |                                                            |                 |               |                |                      |
| 4.     |                                                            |                 |               |                |                      |
| 5.     |                                                            |                 |               |                |                      |
| Remark |                                                            |                 |               |                |                      |

**Submitted**

Signature.....

Name.....

Designation: Environmental  
Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant**Approved**

Signature.....

Name.....

Designation: Executive Engineer -  
APRDC

Client

## C4: DETAILS OF MACHINERY IN OPERATION

(To be filled by Contractor)

Name of Corridor \_\_\_\_\_

Monthly Report for Each Package

Reporting Month.....

Date of Submission.....

### 1. Details of Machinery Operation

|      |                                                                                         |                                             |
|------|-----------------------------------------------------------------------------------------|---------------------------------------------|
| 1.1  | Total machinery in operation (Nos.)                                                     |                                             |
| 1.2  | Number of pavers                                                                        |                                             |
| 1.3  | Number of rollers                                                                       |                                             |
| 1.4  | Number of excavators                                                                    |                                             |
| 1.5  | Number of graders                                                                       | Nos.                                        |
| 1.6  | Number of dumpers                                                                       | Nos.                                        |
| 1.7  | No. of workshops with repairs facility (furnish location and type of facility provided) | Workshop on Location      Facility Provided |
| 1.8  | Number of vehicles in repair at each location                                           |                                             |
| 1.9  | Number of oil interceptor provided in each repair / fuelling site                       |                                             |
| 1.10 | Total quantity of oil and-wastes recovered in each interceptor during last month.       | Oil waste      Liters, Kg.                  |
| 1.11 | Details of waste disposal. (Whether Sold/ Disposed)                                     |                                             |

Remark

#### Submitted

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

#### Checked

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

#### Approved

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client

**C5: Redevelopment of Borrow Areas**  
(To be filled by Contractor)

Name of Corridor \_\_\_\_\_

Construction stage: Monthly Report – Date: \_\_\_\_\_ Month \_\_\_\_\_ Year \_\_\_\_\_

Drawing for Redevelopment to be attached for each Borrow Area, (photographs of sites before use & after rehabilitation to be attached)

| Sl. No. | Borrow Area No. | Borrow Area Location |               |                  |                        |                      | Land use | Rehabilitation Measures | Date of approval of Rehabilitation | Date of Handing Over to Owner | Remarks |
|---------|-----------------|----------------------|---------------|------------------|------------------------|----------------------|----------|-------------------------|------------------------------------|-------------------------------|---------|
|         |                 | Name of Village      | Chainage (km) | Side (LHS / RHS) | Area (M <sup>2</sup> ) | Haul road length (m) |          |                         |                                    |                               |         |
|         |                 |                      |               |                  |                        |                      |          |                         |                                    |                               |         |
|         |                 |                      |               |                  |                        |                      |          |                         |                                    |                               |         |
|         |                 |                      |               |                  |                        |                      |          |                         |                                    |                               |         |

Remark

**Submitted**

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client



### C6: SAFETY CHECK LIST.

(To be filled by Contractor)

1. Name of Contractor : \_\_\_\_\_
2. Representation : \_\_\_\_\_
3. Name of Safety Officer : \_\_\_\_\_
4. Date of Inspection : \_\_\_\_\_

|                                  | Location 1 ..... |   |   | Location 2 ..... |   |   | Location 3 ..... |   |   | Remark |
|----------------------------------|------------------|---|---|------------------|---|---|------------------|---|---|--------|
|                                  | A                | B | C | A                | B | C | A                | B | C |        |
| Adequate at time of Inspection   |                  |   |   |                  |   |   |                  |   |   |        |
| Needs Improvement                |                  |   |   |                  |   |   |                  |   |   |        |
| Needs Immediate Attention        |                  |   |   |                  |   |   |                  |   |   |        |
| General                          |                  |   |   |                  |   |   |                  |   |   |        |
| House keeping                    |                  |   |   |                  |   |   |                  |   |   |        |
| Stacking of Material             |                  |   |   |                  |   |   |                  |   |   |        |
| Passageway                       |                  |   |   |                  |   |   |                  |   |   |        |
| Lighting                         |                  |   |   |                  |   |   |                  |   |   |        |
| Ventilation                      |                  |   |   |                  |   |   |                  |   |   |        |
| Others                           |                  |   |   |                  |   |   |                  |   |   |        |
| Electrical                       |                  |   |   |                  |   |   |                  |   |   |        |
| Switches                         |                  |   |   |                  |   |   |                  |   |   |        |
| Wirings                          |                  |   |   |                  |   |   |                  |   |   |        |
| Fixed Installation               |                  |   |   |                  |   |   |                  |   |   |        |
| Portable Lighting                |                  |   |   |                  |   |   |                  |   |   |        |
| Portable Tool                    |                  |   |   |                  |   |   |                  |   |   |        |
| Welding Machine                  |                  |   |   |                  |   |   |                  |   |   |        |
| Others                           |                  |   |   |                  |   |   |                  |   |   |        |
| Fire Prevention                  |                  |   |   |                  |   |   |                  |   |   |        |
| Fire Fighting Appliance          |                  |   |   |                  |   |   |                  |   |   |        |
| Dangerous Goods Store            |                  |   |   |                  |   |   |                  |   |   |        |
| Gas Welding Cylinders            |                  |   |   |                  |   |   |                  |   |   |        |
| Others                           |                  |   |   |                  |   |   |                  |   |   |        |
| Others                           |                  |   |   |                  |   |   |                  |   |   |        |
| Dust Control                     |                  |   |   |                  |   |   |                  |   |   |        |
| Noise Control                    |                  |   |   |                  |   |   |                  |   |   |        |
| First Aid Equipment              |                  |   |   |                  |   |   |                  |   |   |        |
| Washing Facility                 |                  |   |   |                  |   |   |                  |   |   |        |
| Latrine                          |                  |   |   |                  |   |   |                  |   |   |        |
| Canteen                          |                  |   |   |                  |   |   |                  |   |   |        |
| Provision of Personal Protective |                  |   |   |                  |   |   |                  |   |   |        |
| Helmet                           |                  |   |   |                  |   |   |                  |   |   |        |
| Eye Protector                    |                  |   |   |                  |   |   |                  |   |   |        |
| Ear Protector                    |                  |   |   |                  |   |   |                  |   |   |        |
| Respirator                       |                  |   |   |                  |   |   |                  |   |   |        |
| Safety Shoes                     |                  |   |   |                  |   |   |                  |   |   |        |
| Safety Belts                     |                  |   |   |                  |   |   |                  |   |   |        |
| Others                           |                  |   |   |                  |   |   |                  |   |   |        |

Remark

#### Submitted

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

#### Checked

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

#### Approved

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client

## C7: ACCIDENT REPORT

(To be completed by the Safety Officer of contractor on occurrence of injury)

### Type of Accident

|         |                                     |         |                                                |
|---------|-------------------------------------|---------|------------------------------------------------|
| D01 ( ) | Fall of person from a height        | D11 ( ) | Explosion                                      |
| D02 ( ) | Slip, trip or fall on same level    | D12 ( ) | Fire                                           |
| D03 ( ) | Struck against fixed objects        | D13 ( ) | Contact with hot or corrosive substances       |
| D04 ( ) | Struck by flying or falling objects | D14 ( ) | Contact with poisonous gas or toxic substances |
| D05 ( ) | Struck by moving objects            | D15 ( ) | Contact with electric current                  |
| D06 ( ) | Struck / caught by cable            | D16 ( ) | Hand tool accident                             |
| D07 ( ) | Stepping on nail etc.               | D17 ( ) | Vehicle / Mobile plant accident                |
| D08 ( ) | Handling without machinery          | D18 ( ) | Machinery operation accident                   |
| D09 ( ) | Crushing / burying                  | D19 ( ) | Other (please specify)                         |
| D10 ( ) | Drowning or asphyxiation            |         |                                                |

#### Agent Involved in Accident

|         |                                             |         |                                                                       |
|---------|---------------------------------------------|---------|-----------------------------------------------------------------------|
| E01 ( ) | Machinery                                   | E11 ( ) | Excavation / underground working                                      |
| E02 ( ) | Portable power appliance                    | E12 ( ) | Floor, ground, stairs or any working, surface                         |
| E03 ( ) | Vehicle or associated equipment / machinery | E13 ( ) | Ladder                                                                |
| E04 ( ) | Material being handled, used or stored      | E14 ( ) | Scaffolding/gondola                                                   |
| E05 ( ) | Gas, vapour, dust, fume or oxygen           | E15 ( ) | Construction formwork, shuttering and falsework                       |
| E06 ( ) | Hand tools                                  | E16 ( ) | Electricity supply cable, wiring switchboard and associated equipment |
| E07 ( ) | Floor edge                                  | E17 ( ) | Nail, splinter or chipping                                            |
| E08 ( ) | Floor opening                               | E18 ( ) | Other (Please specify)                                                |
| E09 ( ) | Left shaft                                  | E19 ( ) |                                                                       |
| E10 ( ) | Stair edge                                  |         |                                                                       |

#### Unsafe Action Relevant to the Accident

|         |                                          |         |                                                    |
|---------|------------------------------------------|---------|----------------------------------------------------|
| F01 ( ) | Operating without authority              | F11 ( ) | Failure to use eye protector                       |
| F02 ( ) | Failure to secure objects                | F12 ( ) | Failure to use respirator                          |
| F03 ( ) | Making safety devices inoperative        | F13 ( ) | Failure to use proper clothing                     |
| F04 ( ) | Working on moving or dangerous equipment | F14 ( ) | Failure to use warn others or given proper signals |
| F05 ( ) | Using un-safety equipment                | F15 ( ) | Horseplay                                          |
| F06 ( ) | Adopting unsafe position or posture      | F16 ( ) | No unsafe action                                   |
| F07 ( ) | Operating or working at unsafe speed     | F17 ( ) | Others (please specify)                            |
| F08 ( ) | Unsafe loading, Placing, mixing etc.     | F18 ( ) |                                                    |
| F09 ( ) | Failure to use helmet                    | F19 ( ) |                                                    |
| F10 ( ) | Failure to use proper footwear           |         |                                                    |

|         |                           |         |                                        |
|---------|---------------------------|---------|----------------------------------------|
| G01 ( ) | No Protective gear        | G08 ( ) | Unsafe layout of job, traffic etc.     |
| G02 ( ) | Defective protective gear | G09 ( ) | Unsafe process of job methods          |
| G03 ( ) | Improper dress / footwear | G10 ( ) | Poor housekeeping                      |
| G04 ( ) | Improper guarding         | G11 ( ) | Lack of warning system                 |
| G05 ( ) | Improper ventilation      | G12 ( ) | Defective tool, machinery or materials |

|                                          |                             |         |                              |
|------------------------------------------|-----------------------------|---------|------------------------------|
| G06 ( )                                  | Improper illumination       | G13 ( ) | No unsafe condition          |
| G07 ( )                                  | Improper procedure          | G14 ( ) | Others (please specify)      |
| Personal Factor Relevant to the Accident |                             |         |                              |
| H01 ( )                                  | Incorrect attitude / motive | H04 ( ) | Unsafe act by another person |
| H02 ( )                                  | Lack of knowledge or skill  | H05 ( ) | No unsafe personal factor    |
| H03 ( )                                  | Physical defects            | H06 ( ) | Other ( please specify)      |
| Remark                                   |                             |         |                              |

**Submitted**

Signature.....  
Name.....  
Designation: Environmental  
Engineer  
  
Contractor

**Checked**

Signature.....  
Name.....  
Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....  
Name.....  
Designation: Executive Engineer -  
APRDC  
  
Client

**Part –II – To be completed Upon Finalization of Employee's Compensation Claim**

- 101( ) No permanent in capacity  
102( ) Less than 5% in capacity  
103( ) More than 5% in capacity  
104( ) Fatal

**Submitted**

Signature.....  
Name.....  
Designation: Environmental  
Engineer  
  
Contractor

**Checked**

Signature.....  
Name.....  
Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....  
Name.....  
Designation: Executive Engineer -  
APRDC  
  
Client

### C8: POLLUTION MONITORING

Name of Corridor \_\_\_\_\_

Construction Stage: Report – Date: \_\_\_\_\_ Month \_\_\_\_\_ Year \_\_\_\_\_

Mitigation measures suggested in last report complied or Not.....

If not reason thereof.....

(Location at which monitoring to be conducted as per EMP)

| Sl. No.                    | Chainage (km) | Details of locations | Duration of monitoring | Instruments used | Completion | Standards                                                                           | Results                                                                             | Reasons for exceeding standards | Mitigation Measures suggested | Type of area (Residential / Industrial / Commercial) | Remarks |
|----------------------------|---------------|----------------------|------------------------|------------------|------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------------------------------------------------|---------|
| <b>1. Air Monitoring</b>   |               |                      |                        |                  |            |                                                                                     |                                                                                     |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            | SPM<br>RSPM<br>HC<br>SO <sub>x</sub><br>NO <sub>x</sub>                             | SPM<br>RSPM<br>HC<br>SO <sub>x</sub><br>NO <sub>x</sub>                             |                                 |                               |                                                      |         |
| <b>2. Water Monitoring</b> |               |                      |                        |                  |            |                                                                                     |                                                                                     |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            | pH<br>TSS<br>TDS<br>Turbidity<br>Hardness<br>Coliform<br>ROD<br>COD<br>Oil & Grease | pH<br>TSS<br>TDS<br>Turbidity<br>Hardness<br>Coliform<br>ROD<br>COD<br>Oil & Grease |                                 |                               |                                                      |         |
| <b>3. Soil Monitoring</b>  |               |                      |                        |                  |            |                                                                                     |                                                                                     |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            | pH                                                                                  | pH                                                                                  |                                 |                               |                                                      |         |

| Sl. No.                    | Chainage (km) | Details of locations | Duration of monitoring | Instruments used | Completion |  |  | Standards                                                                    | Results                                                                      | Reasons for exceeding standards | Mitigation Measures suggested | Type of area (Residential / Industrial / Commercial) | Remarks |
|----------------------------|---------------|----------------------|------------------------|------------------|------------|--|--|------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------------------------------------------------|---------|
|                            |               |                      |                        |                  |            |  |  | Organic Matter<br>Alkalinity<br>Conductivity<br>Water holding capacity<br>Pb | Organic Matter<br>Alkalinity<br>Conductivity<br>Water holding capacity<br>Pb |                                 |                               |                                                      |         |
| <b>4. Noise Monitoring</b> |               |                      |                        |                  |            |  |  |                                                                              |                                                                              |                                 |                               |                                                      |         |
|                            |               |                      |                        |                  |            |  |  | L day equivalent<br>L night equivalent<br>L equivalent                       | L day equivalent<br>L night equivalent<br>L equivalent                       |                                 |                               |                                                      |         |

Remark

**Submitted**

Signature.....  
 Name.....  
 Designation: Environmental Engineer  
 Contractor

**Checked**

Signature.....  
 Name.....  
 Designation: R.E / Environmental Engineer  
 Construction Supervision  
 Consultant

**Approved**

Signature.....  
 Name.....  
 Designation: Executive Engineer - APRDC  
 Client

## C9: ENHANCEMENT MEASURES

(To be filled by Contractor)

Name of Corridor\_\_\_\_\_

[illegible]

Remarks

## Submitted

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer -  
APRDC

Client

### **C10: RESTORATION OF CONSTRUCTION SITES**

(To be filled by Contractor)

Name of Corridor\_\_\_\_\_

(Reporting by Contractor to APRDC)

Construction stage: Monthly Report – Date.....Month.....Year.....

| Sl. No. | Contract Package | Labor Camp |   | Construction Camp |   | Plant Site |   | Borrow areas |   | Disposal Locations |   | Top Soil  |          |
|---------|------------------|------------|---|-------------------|---|------------|---|--------------|---|--------------------|---|-----------|----------|
|         |                  | O          | R | O                 | R | O          | R | O            | R | O                  | R | Preserved | Restored |
|         |                  |            |   |                   |   |            |   |              |   |                    |   |           |          |
|         |                  |            |   |                   |   |            |   |              |   |                    |   |           |          |

Remark

#### **Submitted**

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

#### **Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

#### **Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client

## O 1. POLLUTION MONITORING

Name of Corridor \_\_\_\_\_

Construction Stage: Report – Date: \_\_\_\_\_ Month \_\_\_\_\_ Year \_\_\_\_\_

Mitigation measures suggested in last report complied or Not.....

If not reason thereof.....

(Location at which monitoring to be conducted as per EMP)

| S. No. | Chainage (km) | Details of locations | Duration of monitoring | Instruments used | Standards                                                                           | Results                                                                             | Reasons for exceeding standards | Mitigation Measures suggested | Type of area (Residential / Industrial / Commercial) | Remarks |
|--------|---------------|----------------------|------------------------|------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------|-------------------------------|------------------------------------------------------|---------|
|        |               |                      |                        |                  | SPM<br>RSPM<br>HC<br>SOx<br>NOx                                                     | SPM<br>RSPM<br>HC<br>SOx<br>NOx                                                     |                                 |                               |                                                      |         |
|        |               |                      |                        |                  | pH<br>TSS<br>TDS<br>Turbidity<br>Hardness<br>Coliform<br>BOD<br>COD<br>Oil & Grease | pH<br>TSS<br>TDS<br>Turbidity<br>Hardness<br>Coliform<br>BOD<br>COD<br>Oil & Grease |                                 |                               |                                                      |         |
|        |               |                      |                        |                  | pH<br>Organic Matter<br>Alkalinity<br>Conductivity<br>Water holding capacity<br>Pb  | pH<br>Organic Matter<br>Alkalinity<br>Conductivity<br>Water holding capacity<br>Pb  |                                 |                               |                                                      |         |
|        |               |                      |                        |                  | L day equivalent<br>L night equivalent<br>L equivalent                              | L day equivalent<br>L night equivalent<br>L equivalent                              |                                 |                               |                                                      |         |

**Remark**

**Submitted**

Signature.....

Name.....

Designation: Environmental Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant

**Approved**

Signature.....

Name.....

Designation: Executive Engineer - APRDC

Client



**O 2. CLEANING OF CULVERT OPENINGS AND LONGITUDINAL DRAINS**

(To be filled by contractor)

Name of Corridor\_\_\_\_\_

Construction Stage: Report – Date: \_\_\_\_\_Month\_\_\_\_\_Year

| Sl. No.              | Structural No | Pre monsoon | Date | Post monsoon | Date |
|----------------------|---------------|-------------|------|--------------|------|
| Name of the Corridor |               |             |      |              |      |
| 1                    |               |             |      |              |      |
| 2                    |               |             |      |              |      |
| 3                    |               |             |      |              |      |
| 4                    |               |             |      |              |      |
| 5                    |               |             |      |              |      |
| 6                    |               |             |      |              |      |
| 7                    |               |             |      |              |      |
| 8                    |               |             |      |              |      |
| 9                    |               |             |      |              |      |
| 10                   |               |             |      |              |      |
| 11                   |               |             |      |              |      |
| 12                   |               |             |      |              |      |
| 13                   |               |             |      |              |      |
| 14                   |               |             |      |              |      |

Remark

**Submitted**

Signature.....

Name.....

Designation: Environmental  
Engineer

Contractor

**Checked**

Signature.....

Name.....

Designation: R.E / Environmental Engineer  
Construction Supervision  
Consultant**Approved**

Signature.....

Name.....

Designation: Executive Engineer -  
APRDC

Client

**PU 1: FORMAT FOR KEEPING RECORDS OF CONSENT OBTAINED BY CONTRACTOR**

Name of Corridor\_\_\_\_\_

Construction Stage: Report – Date: \_\_\_\_\_Month\_\_\_\_\_Year

| Sl.<br>No. | Contractor's<br>Name | Clearance | Applicable<br>Acts | Agencies | Obtained<br>on | Valid upto | Remarks |
|------------|----------------------|-----------|--------------------|----------|----------------|------------|---------|
|            | NAME OF CORRIDOR     |           |                    |          |                |            |         |
|            |                      |           |                    |          |                |            |         |
|            |                      |           |                    |          |                |            |         |
|            |                      |           |                    |          |                |            |         |
|            |                      |           |                    |          |                |            |         |
|            |                      |           |                    |          |                |            |         |

Remark

## PU 2: CHECKLIST FOR ENVIRONMENT INSPECTION

(Points / Issues to be covered)

Name of Road \_\_\_\_\_

Date of Inspection \_\_\_\_\_

| Sl. No. | ESMP Measures                                                                                                              |
|---------|----------------------------------------------------------------------------------------------------------------------------|
| 1       | Provision of a personnel accountable for implementation of ESMP / Safety Measures with Contractor                          |
| 2       | Consent of PCB to Establish HMP                                                                                            |
| 3       | Consent of PCB to operate HMP                                                                                              |
| 4       | Compliance of PCB Conditions for HMP installation and operation                                                            |
| 5       | Whether compliance reported through monthly Progress report to Divisional Office of Executive Engineer                     |
| 6       | PUC taken for all Construction vehicles                                                                                    |
| 7       | Concrete platform with trap under bitumen boiler, Fuel Tank for HMP and generator set provided or not                      |
| 8       | Precautions to prevent contamination of soil by emulsion, Bitumen, oil and lubricant taken while storing                   |
| 9       | Providing cover to fine construction material & bituminous mix during transportation                                       |
| 10      | Borrow areas:                                                                                                              |
|         | a) Borrow areas approved by Department                                                                                     |
|         | b) Existing land was used                                                                                                  |
|         | c) Nos Opened                                                                                                              |
|         | d) Available Quantity                                                                                                      |
|         | e) Utilized Quantity                                                                                                       |
|         | f) Balance Quantity                                                                                                        |
| 11      | Spoil and debris disposal:                                                                                                 |
|         | a) Present status of land                                                                                                  |
|         | b) Closure and completion plan                                                                                             |
| 12      | Site specific traffic Safety management Plan:                                                                              |
|         | a) Contractor installed the warning / regulatory Traffic signs at the construction site                                    |
|         | b) The arrangement adequate                                                                                                |
| 13      | Safety equipment i.e helmet, gloves, gumboot, mask, earplugs etc. provided to workers                                      |
| 14      | Health Facility at camp and work site i.e. First Aid kit & suitable vehicle for conveyance in case of emergency / accident |
| 15      | Permit for Procuring River sand                                                                                            |
| 16      | Licence from Department of mines for quarrying                                                                             |
| 17      | Consent to establish / operation of crusher                                                                                |
| 18      | Provision of labour camp with sanitation & potable water                                                                   |
| 19      | Fire precautions at Hot Mix Plant and site Office                                                                          |
| 20      | Air and noise monitoring done in camp site                                                                                 |
| 21      | Whether any cultural property is being impacted                                                                            |
| 22      | Status of drainage provision in camp area                                                                                  |
| 23      | General House Keeping                                                                                                      |

Remark

**PU3: CHECKLIST FOR ENVIRONMENT INSPECTION**  
(Points / Issues to be covered)

Name of Corridor\_\_\_\_\_

Month\_\_\_\_\_

Date\_\_\_\_\_

| SI No.    | Description                                                                   | Remarks |
|-----------|-------------------------------------------------------------------------------|---------|
|           | No. of sites identified                                                       |         |
|           | Approved                                                                      |         |
|           | Opened                                                                        |         |
|           | Material available                                                            |         |
|           | Material obtained                                                             |         |
|           | No. of sites Rehabilitated                                                    |         |
| <b>8</b>  | <b>Disposal Locations</b>                                                     |         |
|           | No. of sites identified                                                       |         |
|           | Approved                                                                      |         |
|           | Opened                                                                        |         |
|           | Amount of Waste disposed                                                      |         |
|           | Type of waste disposed                                                        |         |
|           | No. of sites Rehabilitated                                                    |         |
| <b>9</b>  | <b>Road Safety</b>                                                            |         |
|           | Road Safety norms followed as per guidelines, SP-55 and approved Traffic plan |         |
| <b>10</b> | <b>Cleaning of Culvert/ drains</b>                                            |         |
|           | No. of culverts/ drains                                                       |         |
|           | Nos Cleaned                                                                   |         |
| <b>11</b> | <b>Trees</b>                                                                  |         |
|           | No of trees marked for cutting in field                                       |         |
|           | No of trees cut                                                               |         |
|           | No of trees to be Planted                                                     |         |
|           | Trees Planted                                                                 |         |
| <b>12</b> | <b>Haul Roads</b>                                                             |         |
|           | Adequacy of maintenance of Haul Road Network                                  |         |

Remark

## APPENDIX E.11: TRAINING PROGRAMME ON ENVIRONMENTAL ASPECTS

A comprehensive training programme has been planned for the project by PCC & PIU intended to address all components of the project. The programme has been intended for all Contractors, Supervision Consultants and the Package Unit -APSH. As and when found necessary PIU in consultation with PCC will select appropriate modules for the training of contractors and for the training of engineers responsible for supervision and maintenance work. List of appropriate training modules and their time frames is discussed in subsequent paragraphs..

### Training Module Details

The training components may be broadly divided into the following categories:

- Principles and policies for (natural and social) environmental mitigation in development projects;
- Legal and institutional aspects; project mandates;
- Probable (natural and social) environmental impacts and losses in road strengthening and widening projects;
- The EMAP in APSHP consisting of
  - i) the construction stage environmental concerns;
  - ii) the environmental designs and implementation plans;
  - iii) the project entitlement framework;
  - iv) types and aspects of vulnerability of the EPs;
  - v) counselling and grievance redressal methods and mechanisms and
  - vi) financial control mechanisms;
- Monitoring, evaluation and reporting methods and mechanisms and,
- Inter-sectoral and inter-agency collaboration, etc.

**Table 1.1 - Training Programme for the EMU**

| Training Sessions              | Training Components                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module I<br>Initiation         | <ul style="list-style-type: none"> <li>• Principles and policies for (natural and social) environmental mitigation in development projects;</li> <li>• Legal and institutional aspects; project mandates including the WB operational guidelines;</li> <li>• Introduction to the designs, implementation schedule for APSHP.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Module II<br>Project Specifics | <ul style="list-style-type: none"> <li>• Probable (natural and social) environmental impacts and losses in road strengthening and widening projects;</li> <li>• Basic features of the EMAP and the RAP in APSHP.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Module III<br>EMAP             | <ul style="list-style-type: none"> <li>• Construction stage environmental concerns including hot-spot mitigation matrices</li> <li>• Environmental designs and implementation plans; Location-wise and class-wise;</li> <li>• Project entitlement framework including group, individual or community entitlements; the verification procedure; conditions/circumstances under which additional persons may be considered affected/entitled;</li> <li>• Vulnerability of the EPs; special measures or additional supports proposed in the entitlement framework to assist vulnerable groups to protect their livelihood;</li> <li>• Counselling for each category of entitled persons, households or groups; method and scope of counselling; Completion and distribution of entitlement photo identity cards;</li> <li>• grievance redressal methods and mechanisms; market value assessment method and mechanisms proposed in APSHP;</li> <li>• Financial control mechanisms including disbursement of compensation/assistance payments in a</li> </ul> |

| Training Sessions                      | Training Components                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | transparent method.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Module IV<br>Construction stage issues | <ul style="list-style-type: none"> <li>• Laws and other statutes associated with the proposed project such as the Labour Laws, the various pollution control acts, Environmental (Protection) Act, Mining Act, Hazardous Materials (Handling) Act, Forest (Conservation) Act, Land Acquisition Act, Draft National Rehabilitation Policy, etc;</li> <li>• Efficient construction activity monitoring; compliance monitoring;</li> <li>• Environmental clauses in APSHP contract documents and their implications.</li> </ul>                                                                                    |
| Module V<br>Operation stage issues     | <ul style="list-style-type: none"> <li>• Monitoring requirements; monitoring techniques;</li> <li>• Environmental evaluation techniques;</li> <li>• Reporting requirements and mechanisms for APSHP.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 |
| Module VI<br>Long term issues          | <ul style="list-style-type: none"> <li>• environmental surveys including ambient air, noise, biological and water quality surveys;</li> <li>• social surveys, survey questionnaire design, sampling techniques;</li> <li>• data storage, analysis and retrieval;</li> <li>• contract documents and incorporation of environmental clauses;</li> <li>• community consultation and participatory technology generation methods;</li> <li>• legal or statutory requirements for environmental, social and forestry clearance of road projects and,</li> <li>• contingency planning and management, etc.</li> </ul> |

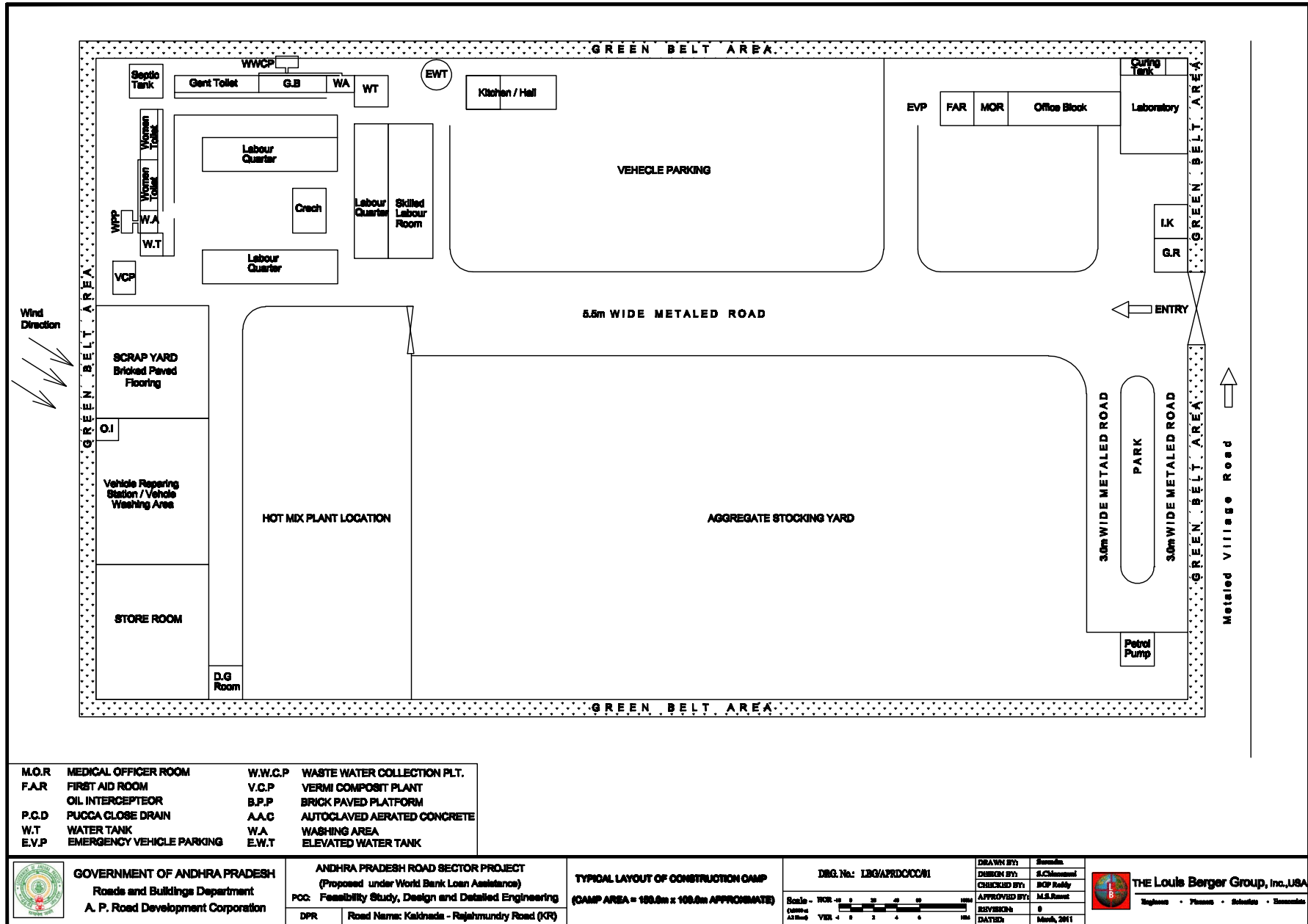
The above are the major areas of training and discussion and is not an exclusive list. The additional areas of training, in order to develop long term capacity in the EMU will also be arranged, as desired/planed.

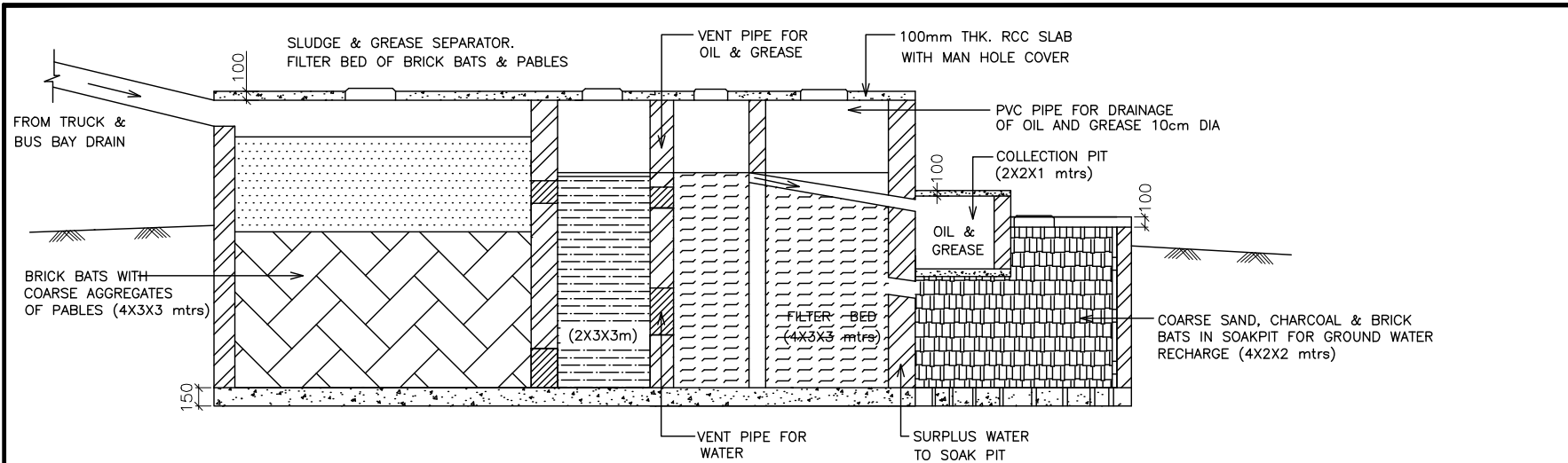
## APPENDIX E.12: MITIGATION ENHANCEMENT OF CULTURAL PROPERTIES

| Sl. No. | Name of Common Property         | Chainage (Km.) | Location (Left / Right) | Impact           | Mitigation/Enhancement |
|---------|---------------------------------|----------------|-------------------------|------------------|------------------------|
| 1       | Venkatesh Temple                | 1+800          | Right                   | Direct Impact    | Generic Enhancement    |
| 2       | Church                          | 2+000          | Right                   | Direct Impact    | Generic Enhancement    |
| 3       | Temple                          | 3+000          | Left                    | No Direct Impact | Generic Enhancement    |
| 4       | Church                          | 3+300          | Right                   | Direct Impact    | Generic Enhancement    |
| 5       | Temple                          | 4+100          | Right                   | Direct Impact    | Generic Enhancement    |
| 6       | Temple                          | 4+825          | Right                   | Direct Impact    | Relocated              |
| 7       | Temple/Pipaltree                | 5+525          | Right                   | No Direct Impact | Generic Enhancement    |
| 8       | Temple                          | 8+700          | Right                   | No Direct Impact | Generic Enhancement    |
| 9       | Church                          | 9+175          | Right                   | No Direct Impact | Generic Enhancement    |
| 10      | Temple(Banyan&Neem)             | 10+575         | Right                   | Direct Impact    | Relocated              |
| 11      | Temple                          | 16+000         | Left                    | No Direct Impact | Generic Enhancement    |
| 12      | Temple                          | 16+150         | Right                   | No Direct Impact | Generic Enhancement    |
| 13      | Temple (BW)                     | 16+200         | Left                    | No Direct Impact | Generic Enhancement    |
| 14      | Temple                          | 16+250         | Left                    | No Direct Impact | Generic Enhancement    |
| 15      | Temple                          | 16+450         | Right                   | Direct Impact    | Relocated              |
| 16      | Temple                          | 17+550         | Left                    | No Direct Impact | Generic Enhancement    |
| 17      | Temple                          | 18+000         | Right                   | Direct Impact    | Relocated              |
| 18      | Ganesha Temple                  | 18+050         | Left                    | No Direct Impact | Generic Enhancement    |
| 19      | Temple                          | 18+600         | Right                   | Direct Impact    | Relocated              |
| 20      | Temple                          | 22+100         | Left                    | No Direct Impact | Generic Enhancement    |
| 21      | Temple& Lands                   | 22+650         | Left                    | No Direct Impact | Generic Enhancement    |
| 22      | Temple                          | 22+700         | Left                    | Direct Impact    | Relocated              |
| 23      | Temple                          | 25+200         | Left                    | No Direct Impact | Generic Enhancement    |
| 24      | Temple                          | 25+200         | Left                    | Direct Impact    | Relocated              |
| 25      | Temple                          | 25+200         | Right                   | Direct Impact    | Relocated              |
| 26      | Temple                          | 25+350         | Left                    | Direct Impact    | Relocated              |
| 27      | Temple                          | 28+000         | Right                   | Direct Impact    | Relocated              |
| 28      | Temple                          | 30+300         | Left                    | No Direct Impact | Generic Enhancement    |
| 29      | Temple                          | 31+800         | Left                    | Direct Impact    | Relocated              |
| 30      | Temple/Ghat                     | 34+670         | Left                    | No Direct Impact | Generic Enhancement    |
| 31      | Sai Temple                      | 35+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 32      | Temple                          | 35+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 33      | Temple                          | 35+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 34      | Statue Of Lion In Mouth Goddess | 35+200         | Right                   | No Direct Impact | Generic Enhancement    |
| 35      | Temple                          | 38+100         | Left                    | No Direct Impact | Generic Enhancement    |
| 36      | Temple                          | 40+050         | Right                   | Direct Impact    | Relocated              |
| 37      | Temple                          | 40+150         | Right                   | Direct Impact    | Relocated              |
| 38      | Temple                          | 40+400         | Right                   | Direct Impact    | Relocated              |
| 39      | Temple                          | 40+400         | Right                   | Direct Impact    | Relocated              |
| 40      | Temple                          | 41+200         | Left                    | No Direct Impact | Generic Enhancement    |

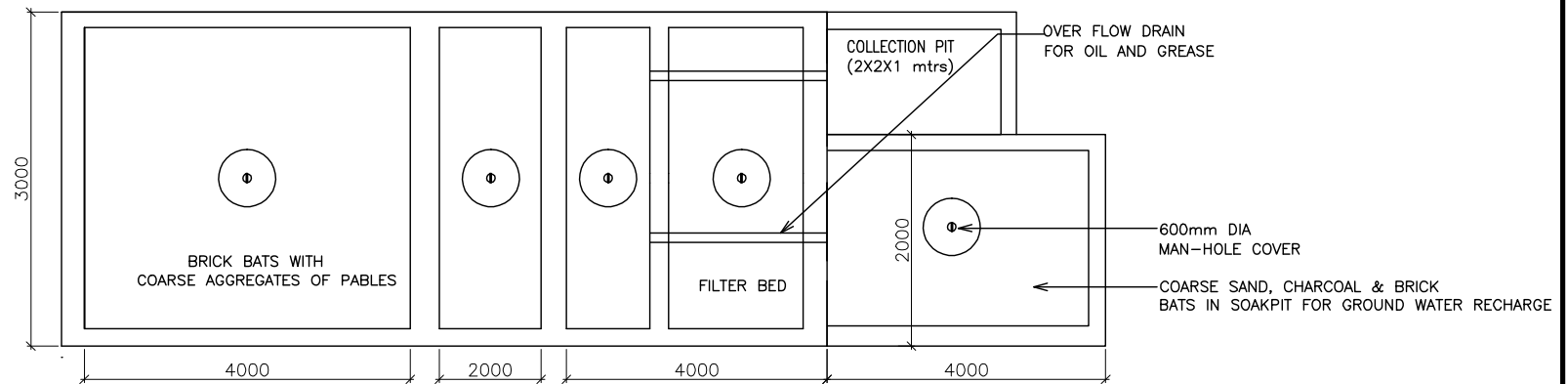
| Sl. No. | Name of Common Property | Chainage (Km.) | Location (Left / Right) | Impact           | Mitigation/Enhancement |
|---------|-------------------------|----------------|-------------------------|------------------|------------------------|
| 41      | Temple                  | 41+350         | Right                   | No Direct Impact | Generic Enhancement    |
| 42      | Temple                  | 41+550         | Left                    | Direct Impact    | Relocated              |
| 43      | Temple                  | 41+700         | Left                    | Direct Impact    | Relocated              |
| 44      | Church (BW)             | 42+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 45      | Temple                  | 44+150         | Right                   | No Direct Impact | Generic Enhancement    |
| 46      | Big Temple (BW)         | 44+600         | Right                   | No Direct Impact | Generic Enhancement    |
| 47      | Temple                  | 44+700         | Right                   | No Direct Impact | Generic Enhancement    |
| 48      | Temple                  | 45+000         | Left                    | No Direct Impact | Generic Enhancement    |
| 49      | Temple                  | 45+350         | Left                    | No Direct Impact | Generic Enhancement    |
| 50      | Temple                  | 45+900         | Left                    | No Direct Impact | Generic Enhancement    |
| 51      | Temple Saibaba          | 46+100         | Right                   | No Direct Impact | Generic Enhancement    |
| 52      | Temple                  | 48+800         | Right                   | Direct Impact    | Relocated              |
| 53      | Temple                  | 49+550         | Right                   | Direct Impact    | Relocated              |
| 54      | Temple                  | 52+300         | Right                   | No Direct Impact | Generic Enhancement    |
| 55      | Sai Temple              | 54+200         | Right                   | Direct Impact    | Relocated              |
| 56      | Temple                  | 55+500         | Left                    | Direct Impact    | Relocated              |
| 57      | Temple                  | 56+800         | Left                    | Direct Impact    | Relocated              |
| 58      | Temple                  | 57+050         | Right                   | No Direct Impact | Generic Enhancement    |
| 59      | Temple                  | 57+200         | Left                    | No Direct Impact | Generic Enhancement    |
| 60      | Temple (BW)             | 57+300         | Left                    | No Direct Impact | Generic Enhancement    |
| 61      | Temple                  | 57+300         | Left                    | No Direct Impact | Generic Enhancement    |
| 62      | Temple                  | 58+950         | Right                   | Direct Impact    | Relocated              |
| 63      | Ganesh Temple           | 59+550         | Right                   | Direct Impact    | Relocated              |
| 64      | Temple                  | 60+200         | Right                   | Direct Impact    | Relocated              |
| 65      | Mary Statue             | 60+750         | Right                   | No Direct Impact | Generic Enhancement    |
| 66      | Church                  | 60+750         | Right                   | No Direct Impact | Generic Enhancement    |
| 67      | Park & Hanuman Temple   | 61+800         | Right                   | No Direct Impact | Generic Enhancement    |







**SECTION OF OIL INTERCEPTOR**



**PLAN OF OIL INTERCEPTOR**



GOVERNMENT OF ANDHRA PRADESH  
Roads and Buildings Department  
A. P. Road Development Corporation

ANDHRA PRADESH ROAD SECTOR PROJECT  
(Proposed under World Bank Loan Assistance)  
POC: Feasibility Study, Design and Detailed Engineering  
DPR Road Name: Kalknada - Rajahmundry Road (KR)

CONCEPTUAL DESIGN FOR  
SEPERATION OF OIL, GREASE & SLUDGE

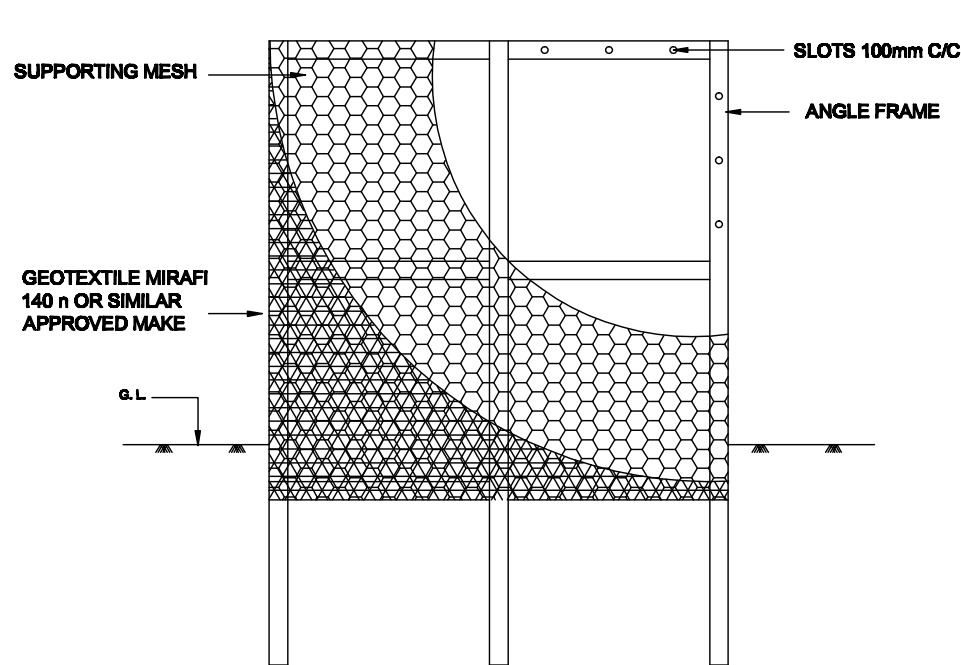
DRG. No.: LBG/APRDC/OGS/01  
Scale - 1:100  
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|              |              |
|--------------|--------------|
| DRAWN BY:    | S. Chinnam   |
| DESIGN BY:   | S. Chinnam   |
| CHECKED BY:  | S. Chinnam   |
| APPROVED BY: | M. A. Ramesh |
| REVISION:    | 0            |
| DATE:        | March 2011   |

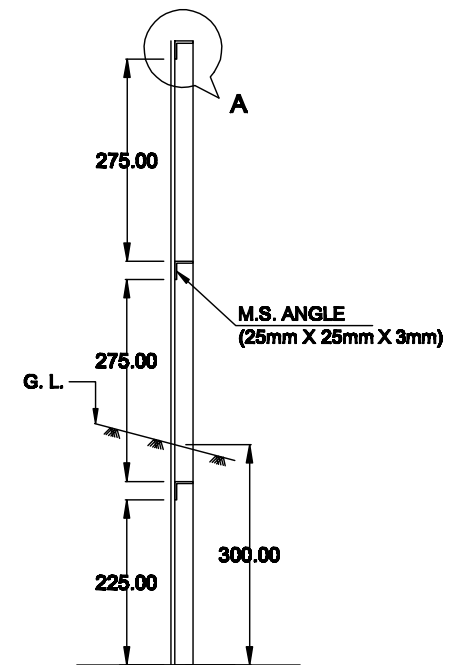


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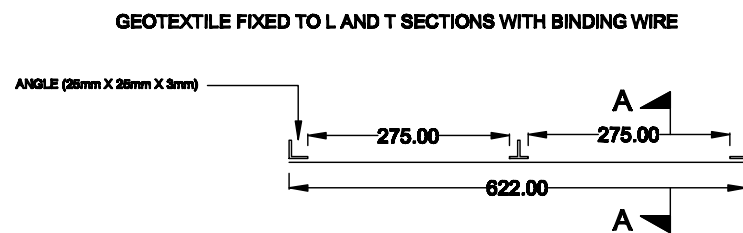
Engineering • Planning • Consulting • Construction



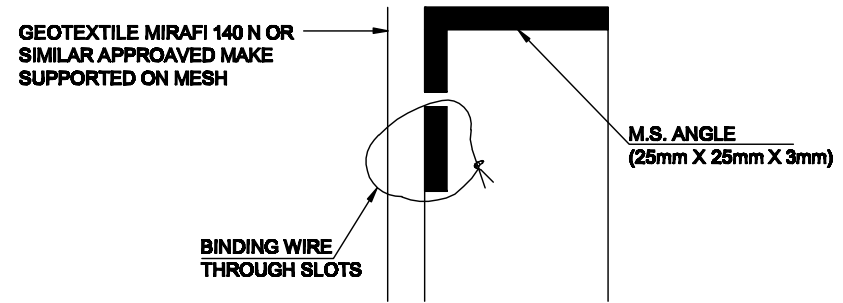
**ELEVATION**



**SECTION A-A**

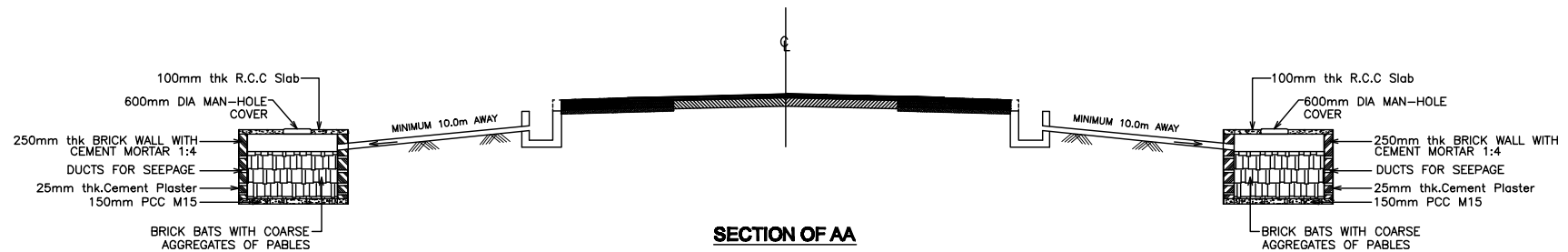
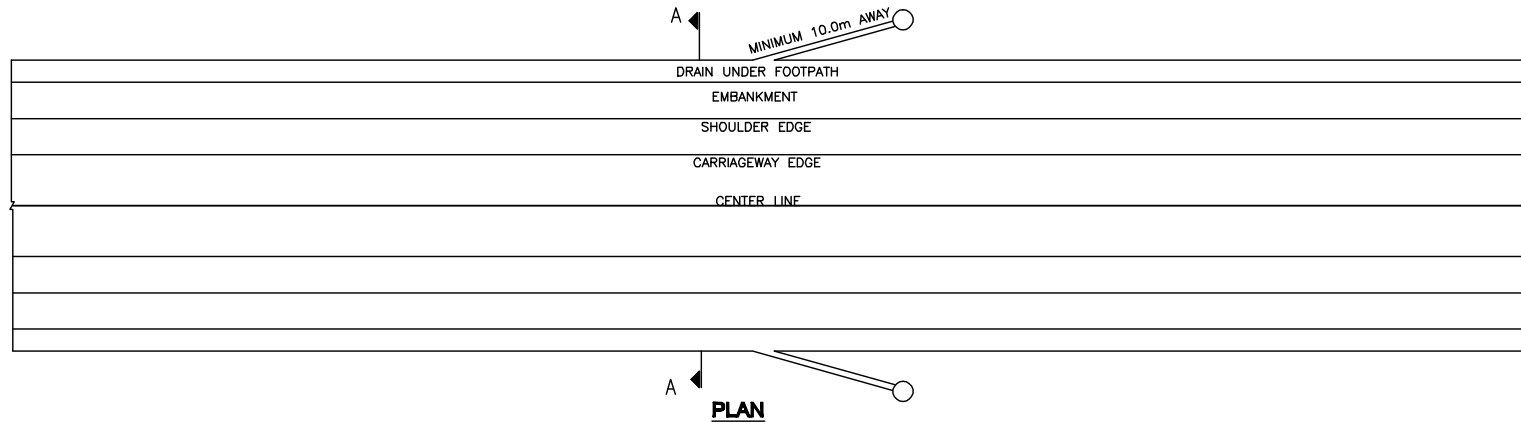


**PLAN**



**DETAIL A-A**

|                                                                                                                                                                                                   |                                                                                                                                                                                                           |                     |                                                                                                                                                                                                      |                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>GOVERNMENT OF ANDHRA PRADESH<br/>Roads and Buildings Department<br/>A. P. Road Development Corporation</p> | <p>ANDHRA PRADESH ROAD SECTOR PROJECT<br/>(Proposed under World Bank Loan Assistance)<br/>PCC: Feasibility Study, Design and Detailed Engineering<br/>DPR Road Name: Kalknada - Rajahmundry Road (KR)</p> | <p>SILT FENCING</p> | <p>DRG. No: LRG/ARDC/01</p> <p>Scale - 1:500</p> <p>DATE: 10/01/2011</p> <p>DESIGN BY: S.Chandrasekhar<br/>CHECKED BY: BOP Reddy<br/>APPROVED BY: M.S.Ram<br/>REVISION: 0<br/>DATED: March, 2011</p> |  <p>THE Louis Berger Group, Inc., USA<br/>Engineers • Planners • Scientists • Executives</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



GOVERNMENT OF ANDHRA PRADESH  
Roads and Buildings Department  
A. P. Road Development Corporation

ANDHRA PRADESH ROAD SECTOR PROJECT  
(Proposed under World Bank Loan Assistance)  
PCC: Feasibility Study, Design and Detailed Engineering  
DPR Road Name: Kaknada - Rajahmundry Road (KR)

### CONCEPTUAL DESIGN OF LINED DRAIN WITH RECHARGE PIT

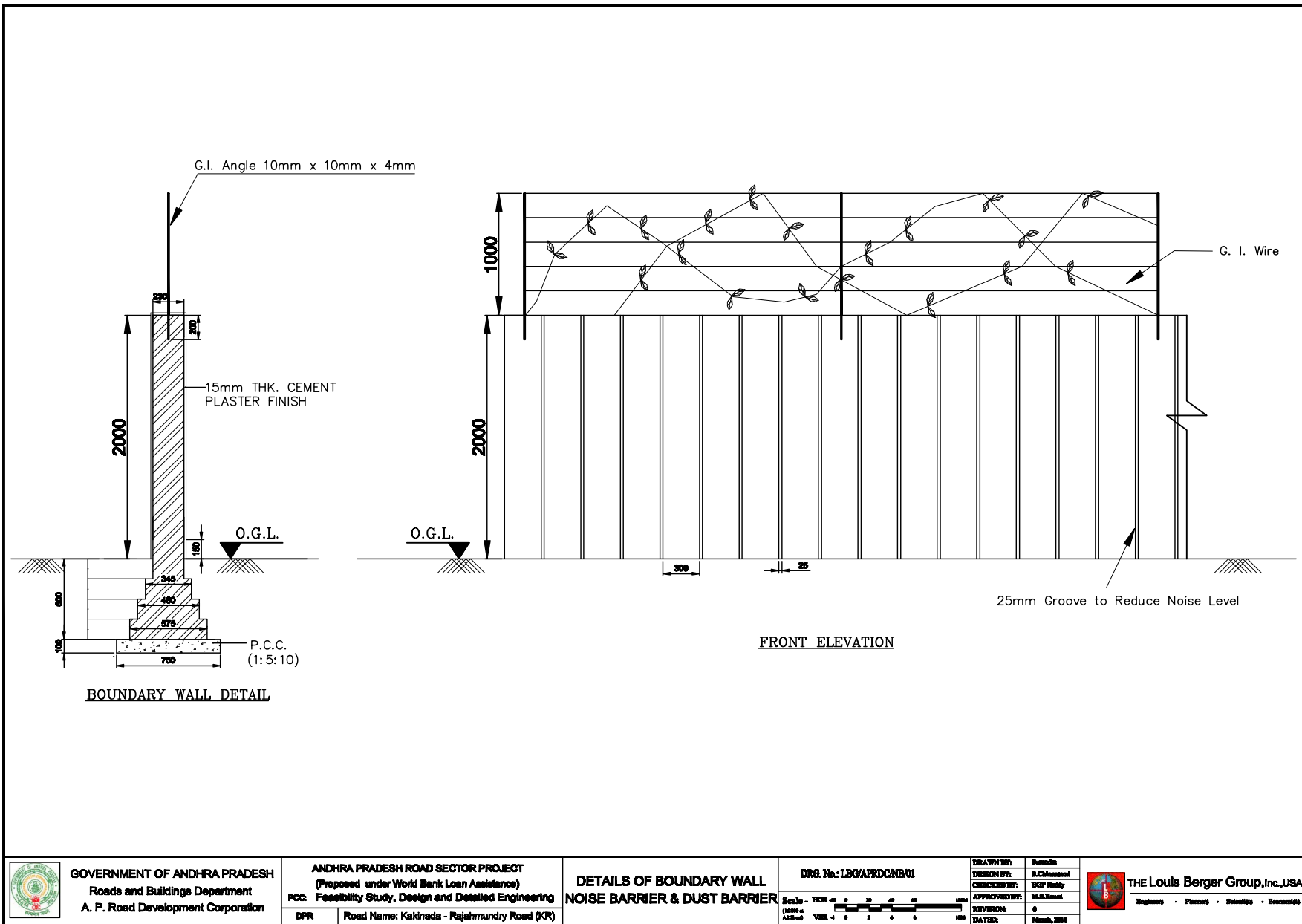
DRG. No: LB04KRD000

Scale - HOR - 0 20 40 60 mms  
VER - 0 2 4 6 cm

DRAWN BY: S. Chinnappa  
CHECKED BY: RGP Study  
APPROVED BY: M.S. Ravi  
REVISION: 0  
DATE: March, 2011



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CAUTIONARY ROAD SIGN



CROSS ROAD  
C-1



SIDE ROAD-RIGHT  
C-2



T INTERSECTION  
C-3



ROUNDABOUT  
C-4



STAGGERED INTERSECTIONS  
C-5



MAJOR ROAD  
C-13



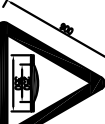
T<sup>+</sup> INTERSECTIONS  
C-6



START OF DUAL  
CARRIAGEWAY  
C-14



END OF DUAL  
CARRIAGEWAY  
C-15



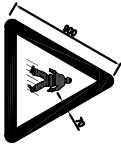
SPEED BREAKER  
C-16



RUNBLE STRIP  
C-17



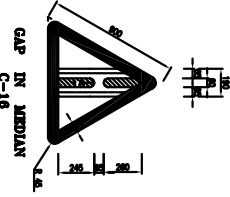
LEFT HAND  
CURVE



PEDESTRIAN  
CROSSING  
C-14



MEN AT WORK  
C-15



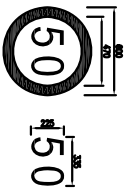
GAP IN MEDIAN  
C-16



STOP  
I-1  
FOR CENTRAL ISLAND STOP SIGN  
SIZE SHALL BE OF 600



HORN PROHIBITED  
I-2

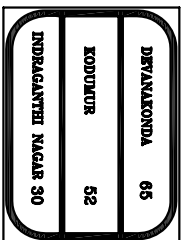


SPEED LIMIT  
I-3



COMPULSORY  
KEEP LEFT  
I-4

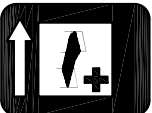
INFORMATORY ROAD SIGN



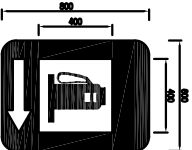
REASSURANCE SIGN  
I-1



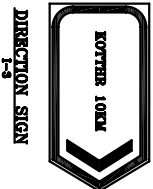
BUS STOP  
I-2



HOSPITAL  
I-3



PETROL PUMP  
I-4



DIRECTION SIGN  
I-5



GOVERNMENT OF ANDHRA PRADESH  
Roads and Buildings Department  
A. P. Road Development Corporation

ANDHRA PRADESH ROAD SECTOR PROJECT  
(Proposed under World Bank Loan Assistance)  
PCC: Feasibility Study, Design and Detailed Engineering  
DPR: Road Name: Kakinada - Rajahmundry Road (KR)

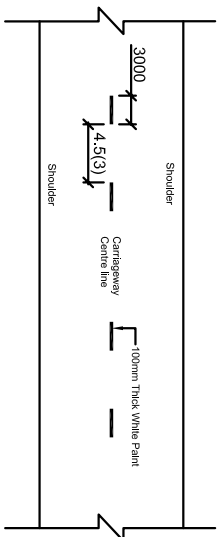
DETAILS OF  
TRAFFIC SIGN

DRG. No.: LBG/ARPD/TS/001  
Scale: HORIZONTAL: 1:2000, VERTICAL: 1:1000  
DESIGN BY: S.Chennamm  
CHECKED BY: B.P. Reddy  
APPROVED BY: M.S. Kumar  
REVISION: 0  
DATE: March 2011

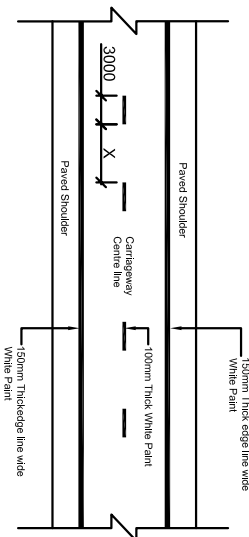


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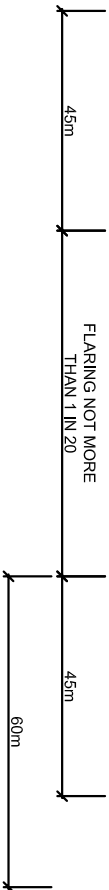
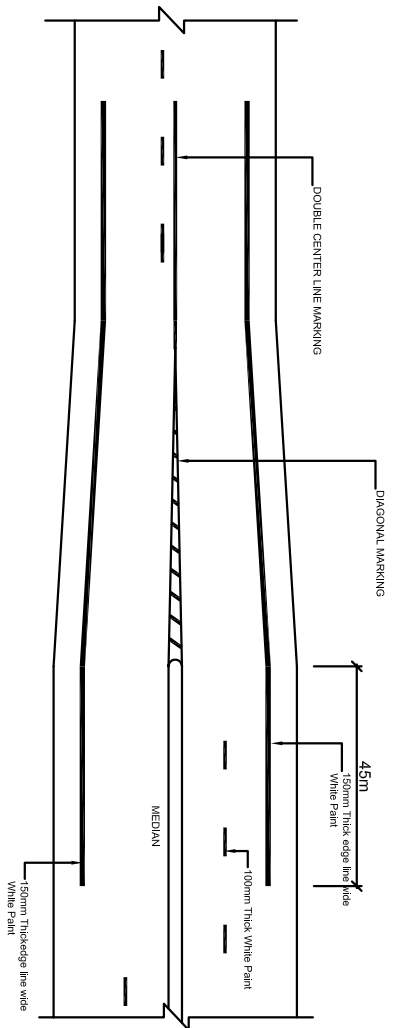




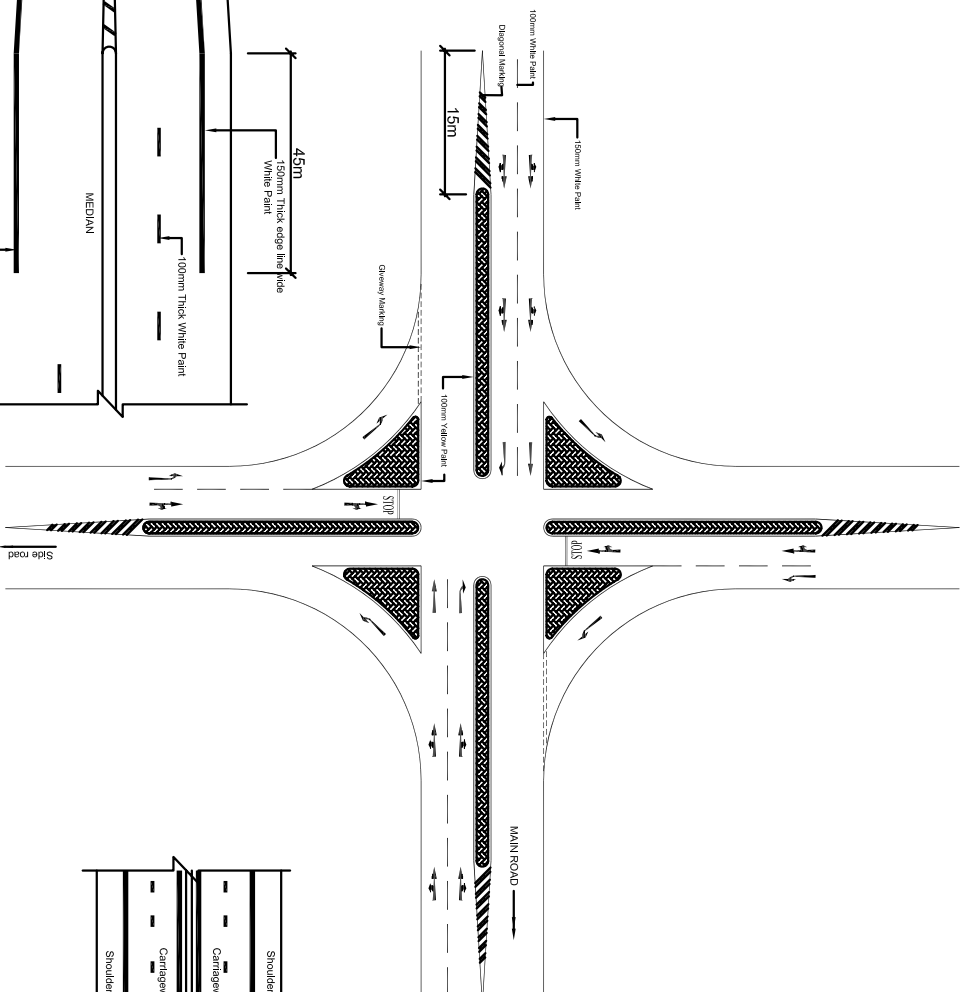
**TWO LANE MARKING WITHOUT PAVED SHOULDER**  
**LAYOUT OF EDGE & LANE STRIPS (NORMAL)**



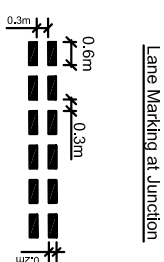
**TWO LANE MARKING WITH PAVED SHOULDER**



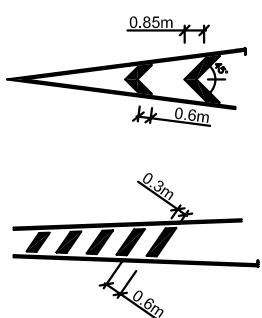
**MARKING ON TRANSITION 2-LANE UNDIVIDED TO 4-L DIVIDED CARRIAGEWAY**



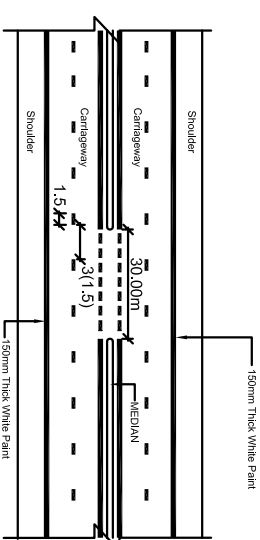
**Road Marking at Junctions**



**Lane Marking at Junction**



**Details of Chevron Marking**  
**Details of Diagonal Marking**



**MEDIAN OPENING**  
**Continuous No Overtaking Center Line**

- NOTES:-**
1. x - 6.0m ON STRAIGHT REACHES
  2. x - 3.0m ON CURVES AND APPROACHES TO INTERSECTION
  3. EDGE LINE MARKING SHALL BE PROVIDED WHERE THE SHOULDER IS PAVED, IT SHOULD NOT BE CARRIED ACROSS THE MOUTHS OF SIDE ROADS.



GOVERNMENT OF ANDHRA PRADESH  
Roads and Buildings Department  
A. P. Road Development Corporation

ANDHRA PRADESH ROAD SECTOR PROJECT  
(Proposed under World Bank Loan Assistance)  
PCC: Feasibility Study, Design and Detailed Engineering  
DPR: Road Name: Kakinada - Rajahmundry Road (KR)

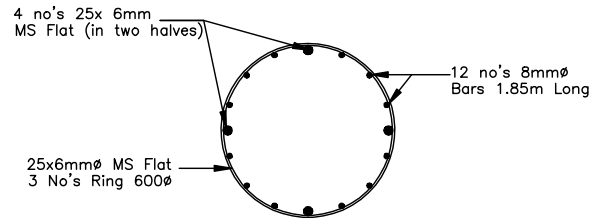
## ROAD MARKING

|                                     |                         |
|-------------------------------------|-------------------------|
| DRG. No.: LBG/APRDC/RM/01           | DESIGN BY: S.Channamma  |
| Scale: HQR - 1:1000<br>VLR - 1:2000 | CHECKED BY: B.P. Reddy  |
| 0 20 40 60 80 100                   | APPROVED BY: M.S. Kumar |
| 0 2 4 6                             | REVISION: 0             |
|                                     | DATED: March 2011       |

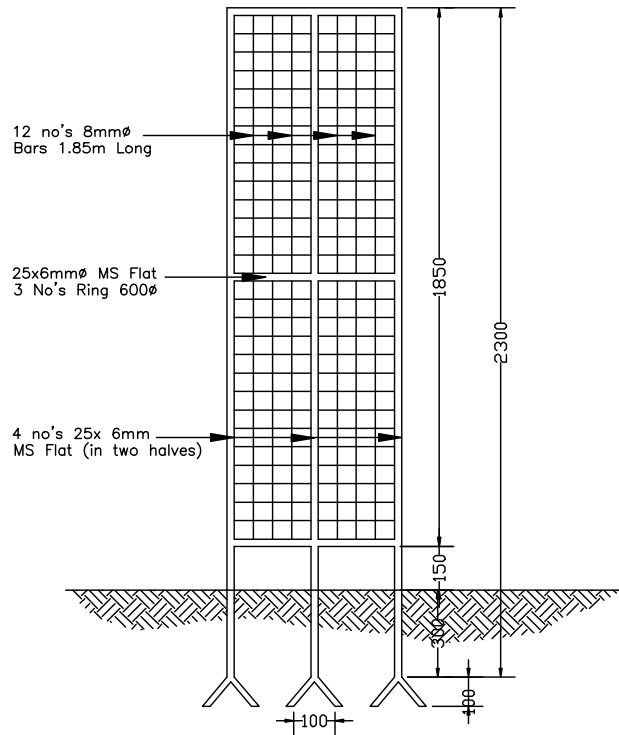


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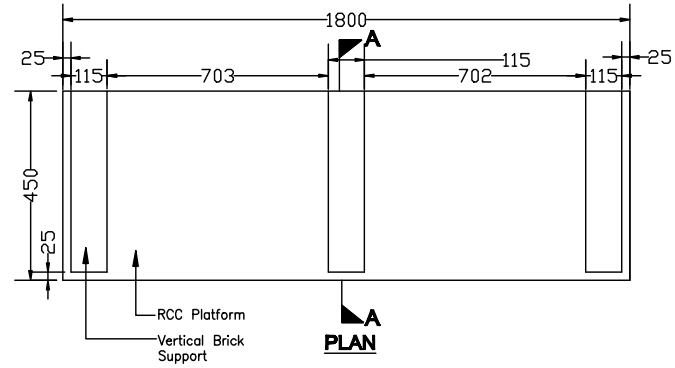




**PLAN**

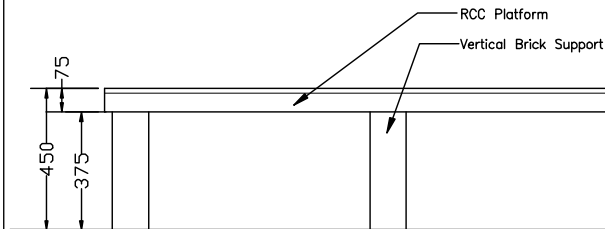


**ELEVATION**

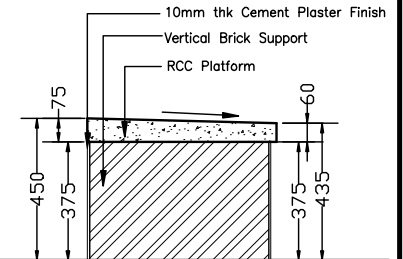


**Note:-**

1. All Dimensions are in mm.



**ELEVATION**



**SECTION AT AA**



**GOVERNMENT OF ANDHRA PRADESH**  
Roads and Buildings Department  
A. P. Road Development Corporation

**ANDHRA PRADESH ROAD SECTOR PROJECT**  
(Proposed under World Bank Loan Assistance)  
POC: Feasibility Study, Design and Detailed Engineering  
DPR Road Name: Kaldnada - Rajahmundry Road (KR)

**CONCEPTUAL DESIGN FOR ORNAMENTAL  
TREE GUARD WITH MS BARS AND  
ENHANCEMENT OF SITTING BENCHES**

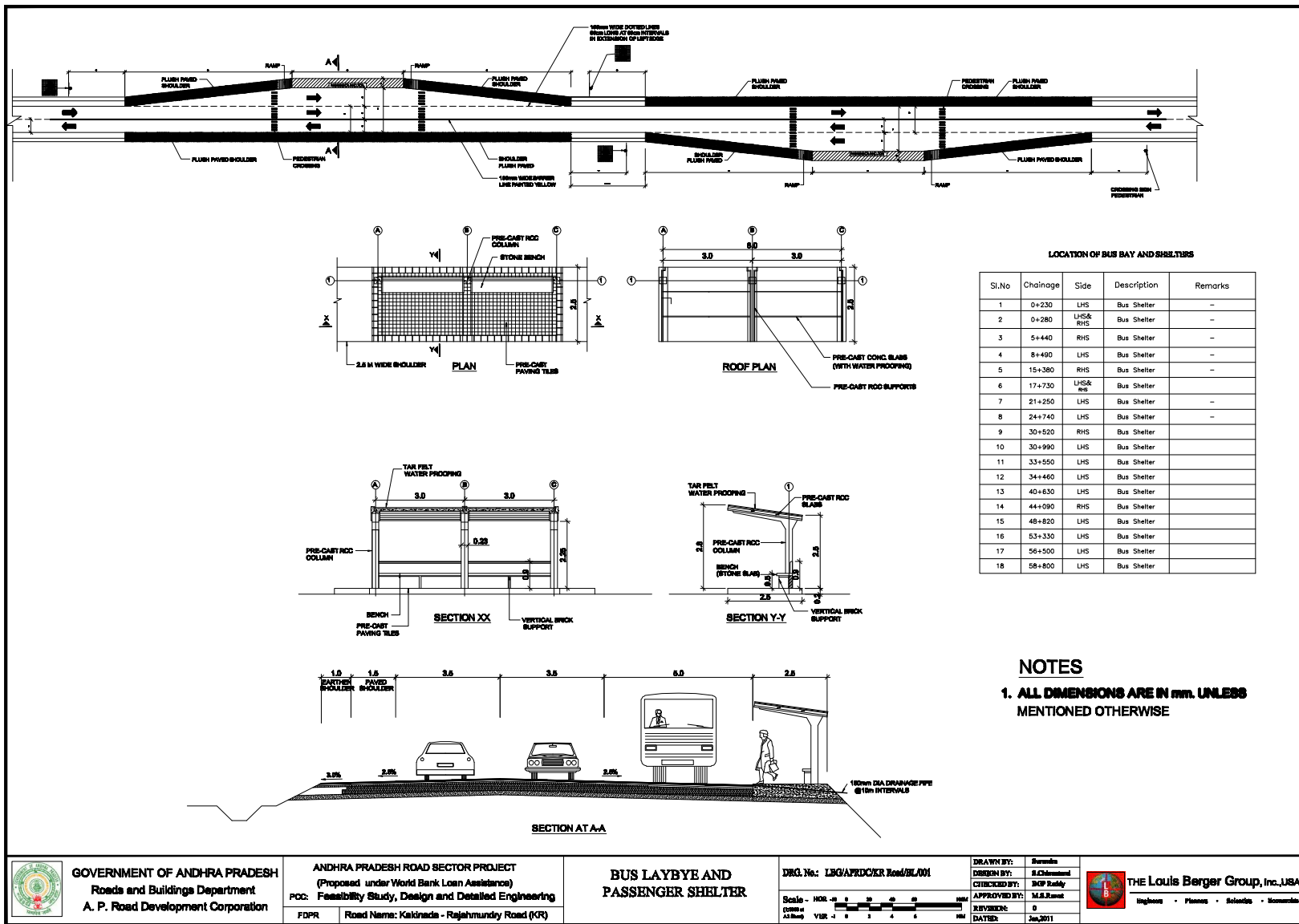
**DRG. No: LBOR/20070-0001**

Scale - 1:100  
VER. 4 0 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34 36 38 40 42 44 46 48 50 52 54 56 58 60 62 64 66 68 70 72 74 76 78 80 82 84 86 88 90 92 94 96 98 100

**DRAWN BY:** S. Chandra  
**DESIGN BY:** S. Chandra  
**CHECKED BY:** J. P. Reddy  
**APPROVED BY:** M. S. Ramani  
**REVISIONS:** 0  
**DATED:** March, 2011



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GOVERNMENT OF ANDHRA PRADESH  
Roads and Buildings Department  
A. P. Road Development Corporation

ANDHRA PRADESH ROAD SECTOR PROJECT  
(Proposed under World Bank Loan Assistance)  
Feasibility Study, Design and Detailed Engineering  
POC:   
FDR: Road Name: Kakkinada - Rajahmundry Road (GR)

### BUS LAYBY AND PASSENGER SHELTER

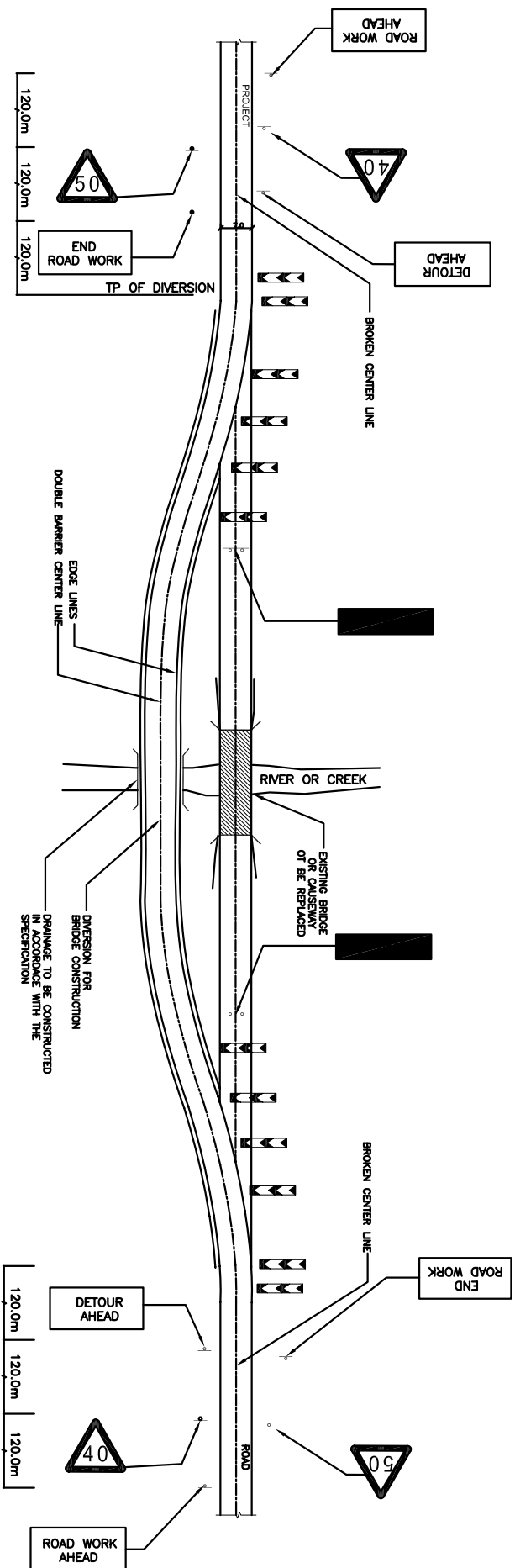
DRG. No.: LBG/AP/DOCR/Road/BL/001

Scale - HOR. - 1:100  
VERT. - 1:10  
DATE: 30/06/2011

DRAWN BY:   
DESIGN BY:   
CHECKED BY:   
APPROVED BY:   
REVIEWED:   
DATE: 30/06/2011



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GOVERNMENT OF ANDHRA PRADESH  
Roads and Buildings Department  
A. P. Road Development Corporation

ANDHRA PRADESH ROAD SECTOR PROJECT  
(Proposed under World Bank Loan Assistance)  
PCC: Feasibility Study, Design and Detailed Engineering  
DPR: Road Name: Kakinda - Rajahmundry Road (KR)

### TEMPORARY TRAFFIC CONTROL AT DIVERSION

DRG. No.: LBG/APRD/TTTC/01  
Scale: HCR - 1:1000  
VTR - 1:1000  
VTR - 1:1000

|              |              |
|--------------|--------------|
| DESIGN BY:   | DESIGNED BY: |
| CHECKED BY:  | CHECKED BY:  |
| APPROVED BY: | APPROVED BY: |
| REVISION:    | REVISION:    |
| DATE:        | DATE:        |

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